

Systematic Literature Review

Introduction

1. Who is who?
2. What is systematic literature review?
3. How do I do a systematic literature review?
4. How to present a systematic literature review?
5. Conclusion.

Introduction

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What is Systematic Literature Review?

Systematic Literature Review:

Is a review of a clearly formulated question(s) that uses systematic and explicit methods to **identify, select and critically appraise relevant research** and to **collect and analyse** data from the studies that are included in the review.

(Cochrane Collaboration, 2014).

Why Systematic Literature Review?

SLR aims to address the problem being researched by **identifying**, **critically evaluating** and **integrating** the findings of all relevant **high-quality individual studies**.

Systematic Literature Review?

A good SR should achieve the following;

1. Establish to what extent existing research has progressed towards clarifying a problem,
2. Identify relations, contradictions, gaps and inconsistencies in the literature, and explore reasons.
3. Comment on, evaluate, extend, or develop theory.
4. Provide implications for practice and policy.
5. Describe directions for future research.

(Baumeister & Leary, 1997; Bem, 1995; Cooper, 2003)

Systematic Literature Review?

A good SLR is characterised by being;

- **Objective:** *Explains criteria for inclusion & exclusion of findings.*
- **Systematic:** *Methodical & organised.*
- **Transparent:** *Clear judgement on inclusion & exclusion.*
- **Replicable:** *Allows for update and further review at a later stage.*

SLR should adopt a JUDGE AND JURY approach, rather than JUDGE AND LAWYER approach.

How do I do a SLR?

Key stages in conducting a SLR.

1. Scoping,
2. Planning,
3. Identification/ searching,
4. Screening,
5. Eligibility.

Scoping

Formulate your research questions

Example:

1. What can be done to improve the currently used workload framework?
2. How best can we implement a real-time system to detect and prevent fraud?

This should help you to be focused and narrow your literature search.

Planning

Break your research questions down into individual concepts to create search terms.

- Identify key terms/ ideas from the RQ, *(eg: workload frameworks)*

Formulate preliminary inclusion & exclusion criteria

- Based on your RQ *(eg: only real-time systems are considered)*

Make a record of the details of your searches and results obtained.

Identification /Searching

- Select databases that are relevant to your topic.
- Consider using Boolean search operators (NOT, AND, OR).
- Consider searching by publication year.
- Consider additional searches on alternative sources to ensure diversity.
- Consider published works.

Screening

- Read titles and abstracts of identified articles.
- Export references of selected articles to a citation manager (EndNote, RefWorks, Ms Word, etc.)

Eligibility

- Sift the full-text version of potentially eligible articles and extract relevant information.
- Add the included material onto your rough SLR document.
- Paraphrase and reference.
- Use turnitin to check your plagiarism level.

How to Present a Systematic Review

- Provide a theoretical and empirical background to the literature, explaining key terms, definitions and concepts;
- Describe the characteristics of included studies.
- Critically evaluate and integrate the results in an **unbiased and systematic way**, including stating whether the results are heterogeneous and discussing possible reasons.
- Balance conflict resolution:

(by identifying inconsistencies in study results & points of contention in theories, conceptualizations and methods in the literature)

Cont...

- Summarize and discuss the findings and conclusions of the review in a balanced and impartial way.
- Explicitly and intuitively link your conclusions to the evidence reviewed.
- Discuss the strengths and limitations of the literature articles.
- Establish to what extent existing research has progressed towards clarifying a particular problem.
- Comment on, evaluate, extend, or propose a new theory.
- Draw conclusions and make recommendations for practice

Common Errors with literature reviews

Common Errors

- **List-like writing that lacks synthesis**
- Relying on material that is likely to be out-of-date
- Exclusion of landmark studies
- Lack of a clear organisational structure
- Not discriminating between relevant and irrelevant materials
- **Not being sufficiently critical**

List-like Writing that Lacks Synthesis

Resulting from failing to integrate arguments into the review.



DO NOT SUMMARISE YOUR READINGS.

Example 1:

During the past decade, many researchers have shown interest in computer security. Their findings are reported here. Smith (2004) developed a model investigation on *(the next paragraph describes Smith's model)*... Jones (2005) showed that... *(the following paragraph says what Jones asserted)*... Harry (2008) makes a clear distinction between... *(the following paragraph states what Harry says)*.

Example 2:

This study brings together work in two areas of computer security. According to Robbins et al (1992) the early theories of computer security were concerned with unauthorised access.

Current thinking however does not discount these theories, but simply builds upon them to include contemporary approaches. Scholl (2007), argue that there are three basic approaches to security. He proposed that these approaches can be seen as an elaboration of Peter (2014)'s original ideas...

Not being sufficiently critical

Note that, the purpose of a literature review:

1. Is not just to summarise what is currently known about a topic.
2. Is to provide a detailed justification for your research.

That is, ultimately it is in the form of an argument and this argument is developed by pointing out "holes" or "Gaps" in the reviewed sources.

The End.

Thank you for attending, hope you
learnt something.

How to Develop: Problem Statement, Topic, Objectives and Research Questions, Hypothesis

AN

A Seminar presentation

BY

Dr. Ambrose Azeta

Department of Software Engineering

Namibia University of Science and Technology

27 February, 2023

PRESENTATION OUTLINE

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1. Introduction
2. Areas of Research in ICT
- 3. Computer Science Research**
 - 3.1 Formulation of Problem statement
 - 3.2 Research title
 - 3.3 Research Objectives
 - 3.4 Research Sub Objectives
 - 3.5 Research Questions and Hypothesis
 - 3.6 Research Methodology
 - 3.7 Computer Science Dissertation/thesis (sample outline)
 - 3.8 Computer Science publication/papers (sample outline)
- 4. Information Systems Research**
 - 4.1 Research Aim
 - 4.2 Research Objectives
 - 4.3 Mapping of Research Questions, Hypothesis and Objectives
 - 4.4 Confirmation of Research questions and Hypothesis
 - 4.5 Conclusion

2. AREAS OF RESEARCH IN ICT

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- Computer Science
- Information Systems
- **Software Engineering**
- Cyber Security,
ETC
- My main research area is Software Engineering/
Software Development
- SE is applied in all domains

3. SOFTWARE ENGINEERING RESEARCH

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- Model driven Engineering (MDE)
- Design using Unified Modelling Language (UML)
- System testing/Automated
- System Evaluation
- System maintenance and monitoring
- Automated software architecture and debugging
- Software defect prediction using machine learning

Definition of Concepts

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What is Software Engineering (SE) ?

- Software engineering is the process of applying engineering principles to analyze the needs of a specific user and then design, build and test software applications which will meet those needs.
- Software engineering goes beyond the fundamentals of programming to create more complex software systems that can serve as critical systems for businesses and organizations.
- Software engineers need to understand how to write software and how software and hardware interact.

What is Computer-aided software engineering (CASE)

- *Computer-aided software engineering (CASE)* uses programs to assist in the creation and maintenance of other computer programs.
- Programming has historically been performed by (human) programmers or teams; CASE adds software to the programming team.

Three (3) ways of finding a Problem Statement

- 1) Existing manual/traditional system is not working well
- (2) Existing automated/computerized system exist but giving problem.
- (3) A New system. No existing system, rather the project has not been done before in the community.

Sample Problem Statement

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SN	Project title	Sample problem statement
1	Development of Fake News and Advertisement Detection System	Individuals and groups have obligated themselves the task of spreading falsified information to other users with the aim of gaining some money, change the public's opinion on the integrity of reputable persons and organizations, potentially alter the results of an election or some other sinister reason.
2	Development of A Web-Based Medical Appointment System For Cove University	With the high number of appointments that are scheduled in clinics and hospitals every day, it is almost impossible for the clinic to achieve its goals. Definitely, there will be some dissatisfaction on both clinic and students. The clinic has to deal with urgent or emergency cases, absentees or no-shows among scheduled cases which may complicate the work of the receptionist who is booking these appointments manually through walk-ins.

3.1 FORMULATION OF PROBLEM (GAPS)

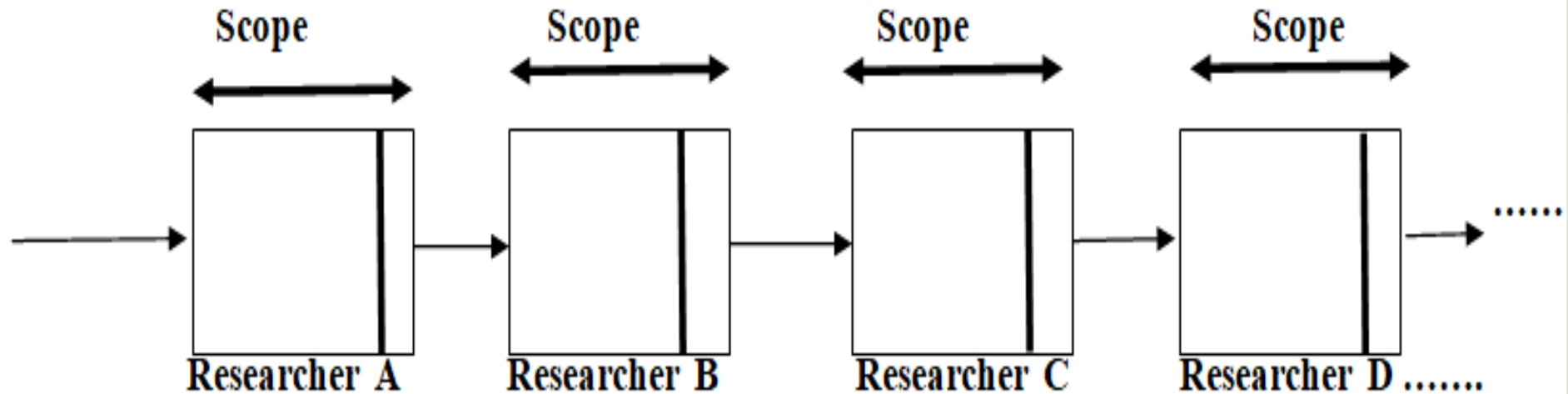
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- ➡ State the problem you want to solve
- ➡ Cite the publication/papers that contains the limitations of a study or a problem that you want to solve, and could not be solved in the current research
- ➡ Gaps are normally discovered through systematic review

1. INTRODUCTION

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RESEARCH CHAIN



Research is continuous (Research Advancements)

Some questions to address:

Which work are you extending?

What problem are you trying to solve?

What is the societal benefit of your research?

Etc

3.2 RESEARCH TOPIC

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- Must contain: a technique/methodology (**Using** or **a**), domain (in), system or platform(**of** or **for**),

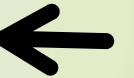
Examples

- Development of a model **for** result management system **in** E-Education **using** DevOps OR
- Development of **a** DevOps-based model **for** result management system **in** E-Education using DevOps ←
- Development of **a** result management system **in** E-Education using DevOps

3.3 RESEARCH OBJECTIVE

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- The objective of this study is to develop a devops-based model for result management system in e-education



3.4 RESEARCH SUB OBJECTIVES

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The Sub **objectives** of this study are **To**:

- **To** carry out requirement elicitation of result management system
- **To** design a DevOps model for result management system
- **To** implement a result management system based on DevOps as proof of concept
- **To** evaluate the implemented system and benchmark with existing system

3.5 RESEARCH QUESTIONS AND HYPOTHESIS

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Research Questions

- How can a DevOps model be developed for result management system?
- How can a result management system based on DevOps be evaluated.

Hypothesis

- **H1:** The development of a DevOps model for result management system is realistic
- **H1:** A result management system based on DevOps has a high usability evaluation compared to other models.

3.6 RESEARCH METHODOLOGY

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- Tools/techniques such as software tools
- Research design/instruments such as questionnaire

3.7 CHAPTER OUTLINE OF DISSERTATION/THESIS

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A Sample Thesis outline:

- ➡ Ch1-Introduction
- ➡ Ch2-Literature review
- ➡ Ch3-Research methodology
- ➡ Ch4-Results and Discussion
- ➡ Ch5-Conclusion

3.8 SECTION OUTLINE OF PUBLICATION/PAPERS

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A Sample paper outline:

- ➡ Section1-Introduction
- ➡ Section2-Literature review
- ➡ Section3-Research methodology (System Design)
- ➡ Section4-Result and Discussion (System Implementation and Evaluation)
- ➡ Section5-Conclusion

4 INFORMATION SYSTEM RESEARCH

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Research Title: *IMPACT OF ICT ON STUDENT ACADEMIC ACHIEVEMENT*

Formulation of Research Questions and Hypothesis

- **Research Question 1:** *To what extent does teaching with the aid of ICTs impact on student academic achievement?*
- **Hypothesis H1:** *Teaching with the aid of ICTs has a positive impact on student academic achievement.*
- **Research Question 2:** *To what extent does course student engagement impact on their academic achievement?*
- **Hypothesis H2:** *Student engagement has a positive impact on their academic achievement.*
- **Research Question 3:** *To what extent does student use of ICTs impact on academic achievement?*
- **Hypothesis H3:** *Student use of ICTs has a positive impact on academic achievement.*

4.1 RESEARCH AIM

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The aim of this research is to explore the impact of ICT on student academic achievements

4.2 RESEARCH OBJECTIVES

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The objectives are as follows:

- ➡ To determine if teaching with the aid of ICTs impact on student academic achievement
- ➡ To determine the extent to which course student engagement impact on student academic achievement
- ➡ To determine the extent to which student use of ICTs impact on student academic achievement

4.3 MAPPING OF RESEARCH QUESTIONS, HYPOTHESIS AND RESEARCH OBJECTIVES

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SN	Research Questions (RQ)	Hypothesis (H)	Research Objectives (RO)
1	<i>To what extent does teaching with the aid of ICTs impact on student academic achievement</i>	<i>Teaching with the aid of ICTs has a positive impact on student academic achievement.</i>	<i>To determine if teaching with the aid of ICTs impact on student academic achievement</i>
2	<i>To what extent does course student engagement impact on their academic achievement?</i>	<i>Student engagement as a positive impact on their academic achievement</i>	<i>To determine the extent to which course student engagement impact on student academic achievement</i>
3	<i>To what extent does student use of ICTs impact on academic achievement?</i>	<i>Student use of ICTs has a positive impact on academic achievement</i>	<i>To determine the extent to which student use of ICTs impact on student academic achievement</i>

4.4 CONFIRMATION OF RESEARCH QUESTIONS AND HYPOTHESIS

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RQ1: *To what extent does teaching with the aid of ICTs impact on student academic achievement?*

Construct B: Teaching with the aid of ICTs returned a strong positive correlation and prediction (2nd in weight(β) out of 5 predictors) of *student academic achievement*.

H1: *Teaching with the aid of ICTs has a positive impact on student academic achievement.*

H1 was supported since a significant positive relationship was established with model 1 & 2. (Unstd β = 5.954 & 5.848, $p < 0.01$). The finding is in agreement with the results of Bai, et al. (2016); Rugutt and Chemosit (2005.); Ilomaki (2008); Abuet al., (2011); Oye et al., (2012) and Angeli&Tsaggari (2016).

4.4 CONFIRMATION OF RESEARCH QUESTIONS AND HYPOTHESIS (CON'T)

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RQ2: *To what extent does course student engagement impact on student academic achievement?*

Construct C: Student engagement, *had* a weak positive correlation with, and prediction (3rd in weight (β) out of 4 predictors) of Class Marks.

H2: *Student engagement has a positive impact on student academic achievement.*

H2 however, was supported since a significant positive relationship was established for model 1 & 2 (Unstd β = 1.086 & 1.140, $p < 0.05$). The finding is in line with Gunuc's (2014) conclusions.

5. PUBLICATION OF RESEARCH OUTPUT

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Publishers of good Journal Outlets

- Nature
- Springer Nature
- Elsevier
- Sage
- Emerald
- Wiley
- University-based Journals
- Etc

Conference outlets

- ACM
- IEEE
- Springer
- Etc

Index Outlets/Databases

- Scopus
- Web of Science/ISI
- University databases/
repositories
- Etc

5. CONCLUSION

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This presentations has no doubt achieved its objectives of highlighting the research methodology in Computer Science and related areas

THANK YOU

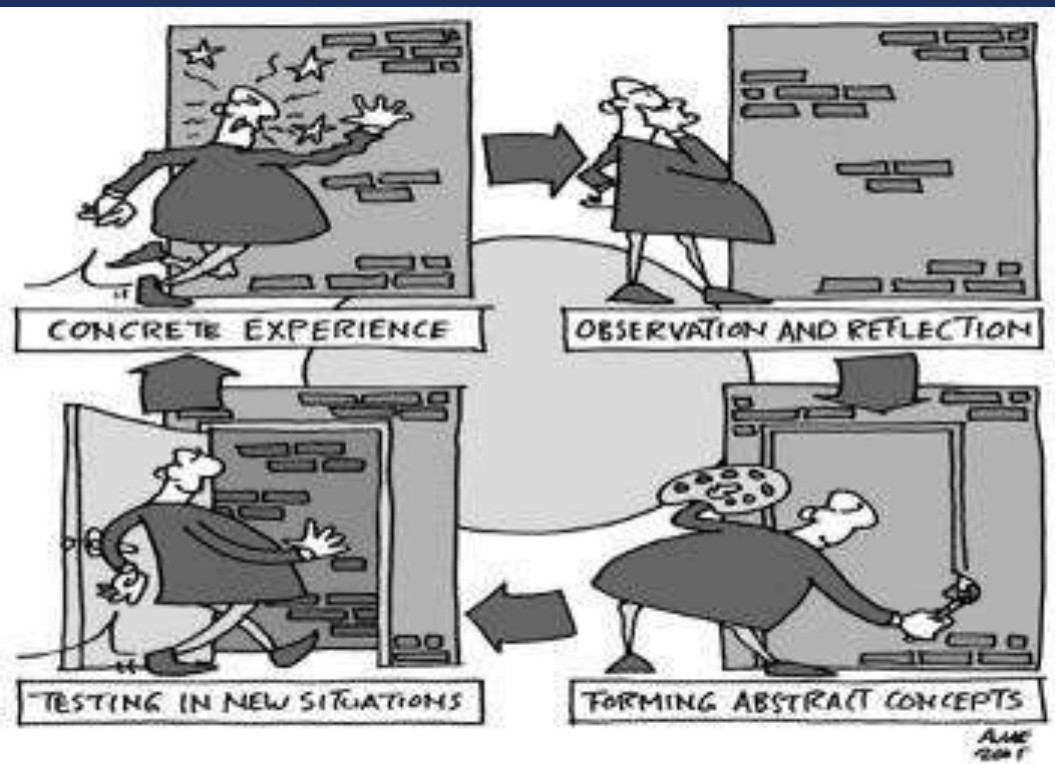


NAMIBIA UNIVERSITY
OF SCIENCE AND TECHNOLOGY

Faculty of Computing & Informatics

Action Research (AR) & Design-based Research

Towards relevant SW development research



Heike
Winschiers-Theophilus

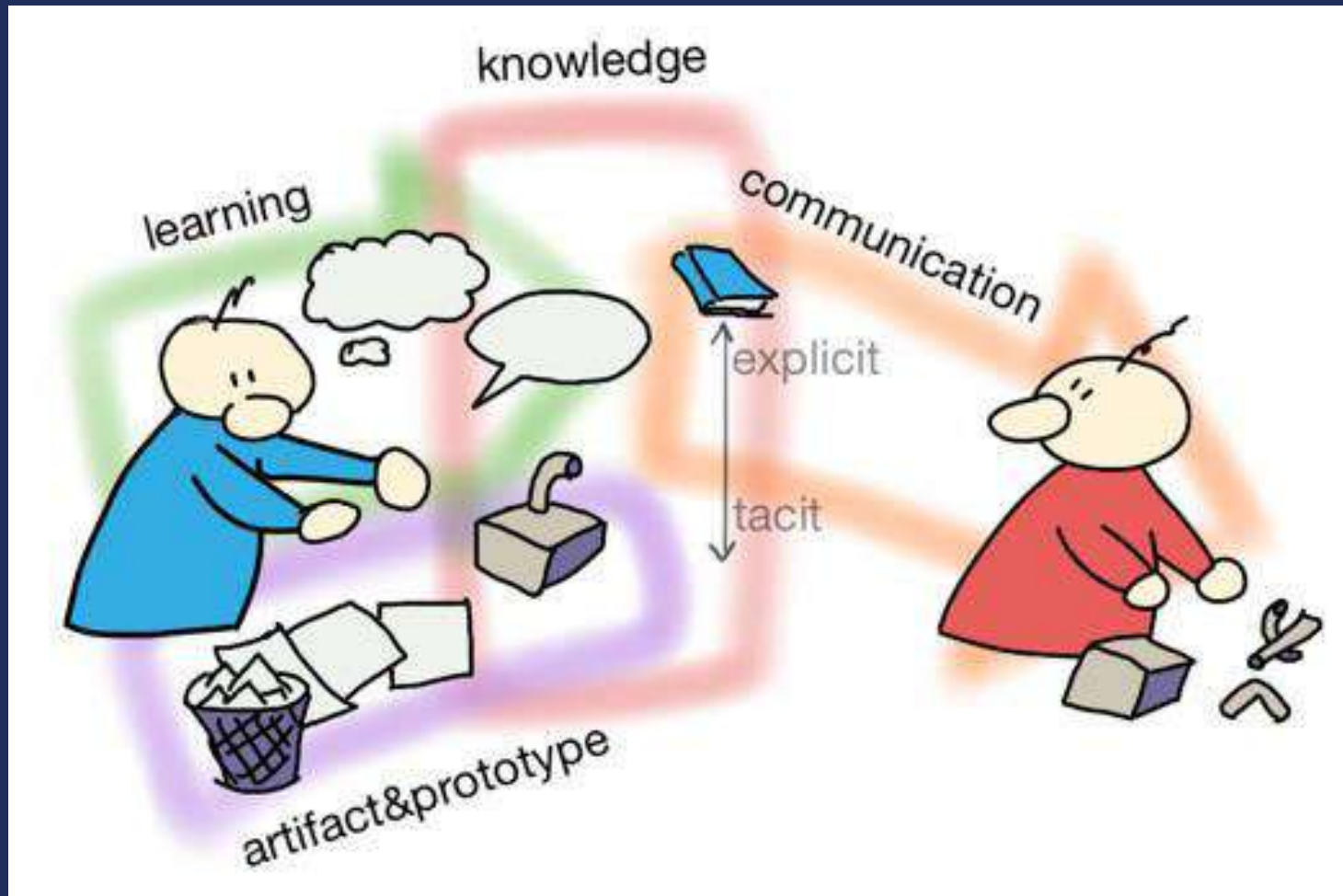
Terminology

	Working Definition	Key Associated Terms
Doing Research	Work done with the intention to produce knowledge for use by others	Question, hypothesis, theory, investigation, interpretation, generalization, validation, discovery
Doing Design	Work done with the intention to produce a feasible solution to improve a given situation	Idea and concept generation, synthesis, development, integration, discovery, prototyping, invention, implementation, realization
Artifact	Object (often material) created during a design process	Sketch, blueprint, brief, specifications, vision, proposal, recommendation, business plan
Prototype	Artifact used in research that can realize the (inter)action that is studied	Implementation, realization, test , exploration, solution, proof of concept, construction
Knowledge	Understanding about the world that can be communicated to others	Theory, book, publication, expertise
Design Practice	The ways in which design professionals conduct their work	Brief, contract, client, stakeholder, studio
Experiment	Specific: a piece of controlled, hypothesis-testing research; General: an explorative confrontation with real-world situations	Specific: hypothesis, statistics, (in)dependent variables; General: trial, intervention, exploration



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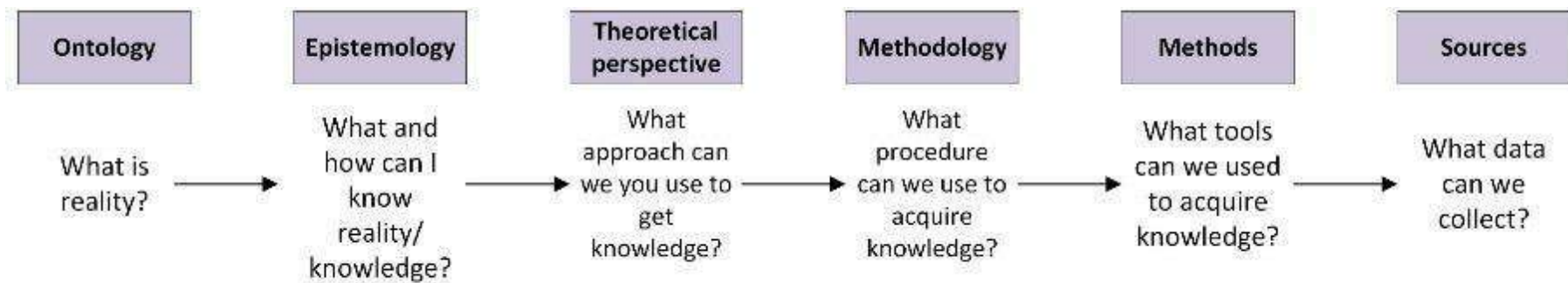
Faculty of Computing & Informatics





Paradigm	Ontology <i>What is reality?</i>	Epistemology <i>How can I know reality?</i>	Theoretical Perspective <i>Which approach do you use to know something?</i>	Methodology <i>How do you go about finding out?</i>	Method <i>What techniques do you use to find out?</i>
Positivism	There is a single reality or truth (more realist).	Reality can be measured and hence the focus is on reliable and valid tools to obtain that.	Positivism Post-positivism	Experimental research Survey research	Usually quantitative, could include: Sampling Measurement and scaling Statistical analysis Questionnaire Focus group Interview
Constructivist / Interpretive	There is no single reality or truth. Reality is created by individuals in groups (less realist).	Therefore, reality needs to be interpreted. It is used to discover the underlying meaning of events and activities.	Interpretivism (reality needs to be interpreted) • Phenomenology • Symbolic interactionism • Hermeneutics Critical Inquiry Feminism	Ethnography Grounded Theory Phenomenological research Heuristic inquiry Action Research Discourse Analysis Feminist Standpoint research etc	Usually qualitative, could include: Qualitative interview Observation Participant Non participant Case study Life history Narrative Theme identification etc
Pragmatism	Reality is constantly renegotiated, debated, interpreted in light of its usefulness in new unpredictable situations.	The best method is one that solves problems. Finding out is the means, change is the underlying aim.	Deweyan pragmatism <i>Research through design</i>	Mixed methods Design-based research Action research	Combination of any of the above and more, such as data mining expert review, usability testing, physical prototype
Subjectivism	Reality is what we perceive to be real	All knowledge is purely a matter of perspective.	Postmodernism Structuralism Post-structuralism	Discourse theory Archaeology Genealogy Deconstruction etc.	Autoethnography Semiotics Literary analysis Pastiche Intertextuality etc.
Critical	Realities are socially constructed entities that are under constant internal influence.	Reality and knowledge is both socially constructed and influenced by power relations from within society	Marxism Queer theory feminism	critical discourse analysis, critical ethnography action research ideology critique	Ideological review Civil actions open-ended interviews, focus groups, open-ended questionnaires, open-ended observations, and journals.

AR and Design-based research a Pragmatic Approach



Adapted from Hay (2002) pg. 64 and Crotty (1998)

Pragmatism	Reality is constantly renegotiated, debated, interpreted in light of its usefulness in new unpredictable situations.	The best method is one that solves problems. Finding out is the means, change is the underlying aim.	Deweyan pragmatism <i>Research through design</i>	Mixed methods Design-based research Action research	Combination of any of the above and more, such as data mining expert review, usability testing, physical prototype
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Value-laden/
subjective

Dependability
co-constructed

Learning
through
change

Open-ended/
iterative

Triangulation

Action Research

Motivations

- To make academic research relevant, researchers should try out their theories with practitioners in real situations and real organizations
- The emphasis is more on what practitioners do than on what they say they do

Key Assumptions

- Social settings cannot be reduced for study, and
- Actions brings understanding

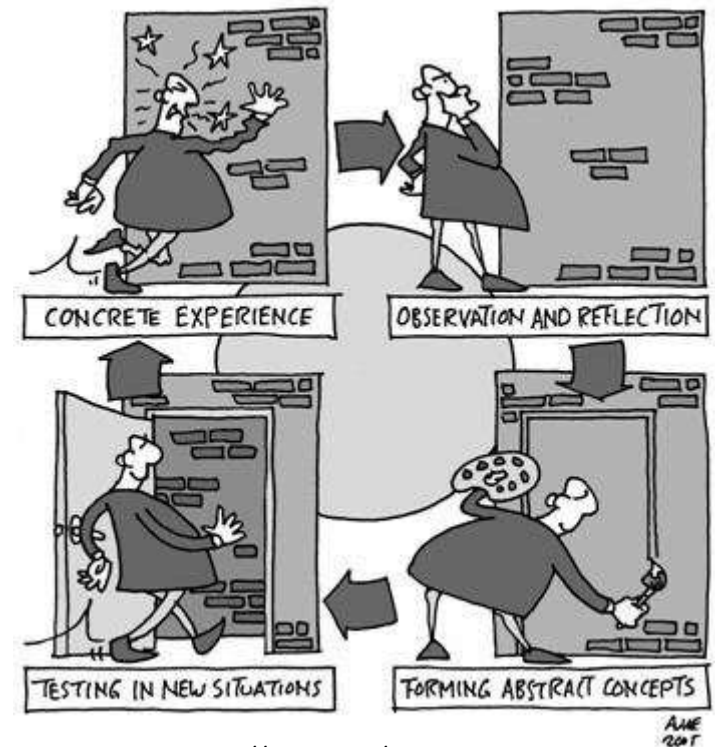
AR characteristics

- Based on **real life problems**
- Taking actions, aiming for change
- Knowledge creation (Learning) come from evaluation and reflection of the actions
- **Collaboration** with participants, democratic, interdisciplinary
- Researcher is immersed
- Situational, subjective
- Transferable rather than generalizable
- Interpretative, theory building

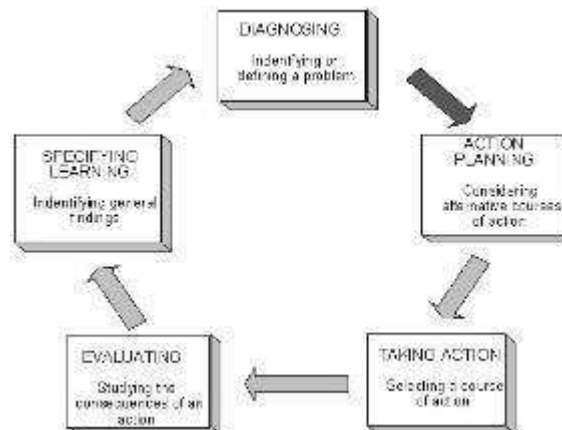
A cyclical approach



<https://edresearch.nmsu.edu/cro-research-publications-2/action-research-initiatives/ar/>

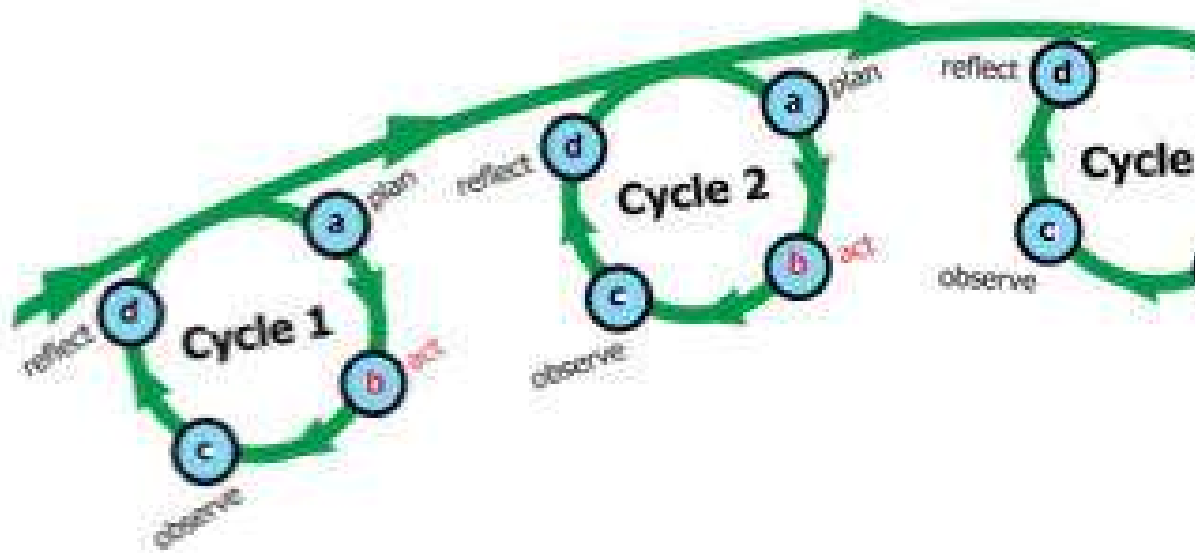


http://loop2.org/?page_id=175

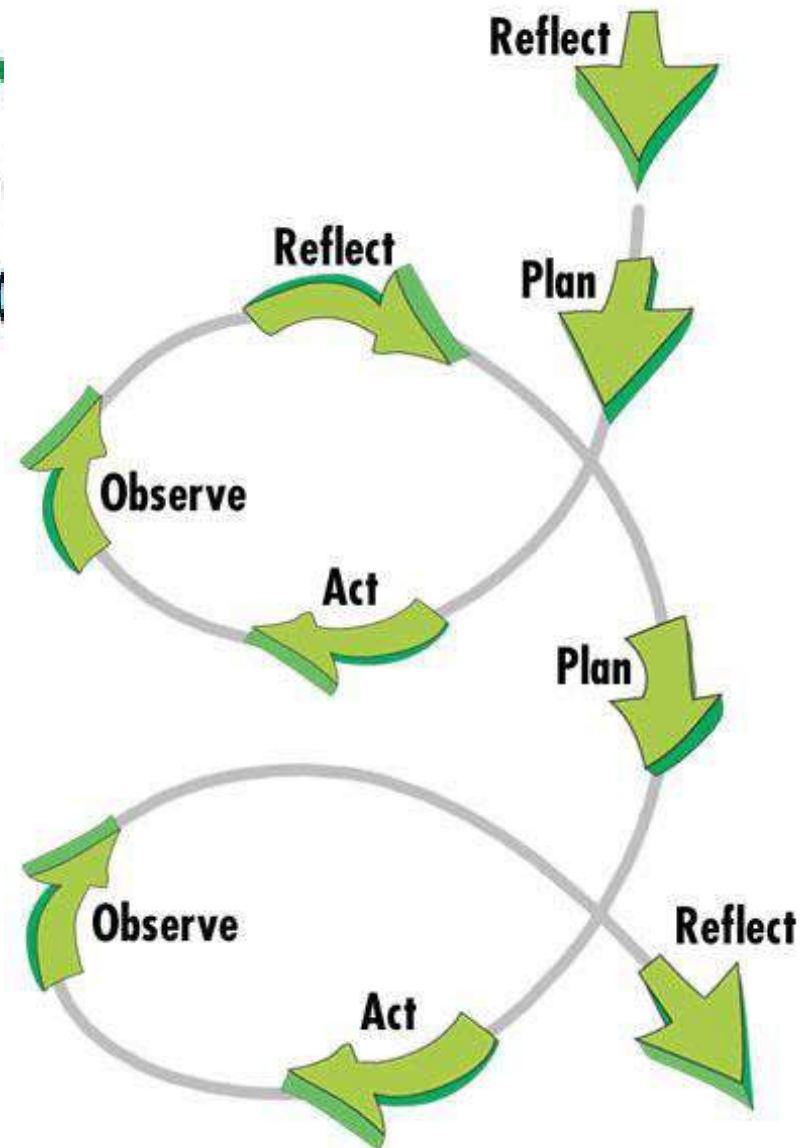


<http://www.vkmaheshwari.com/WP/?p=402>

An iterative approach



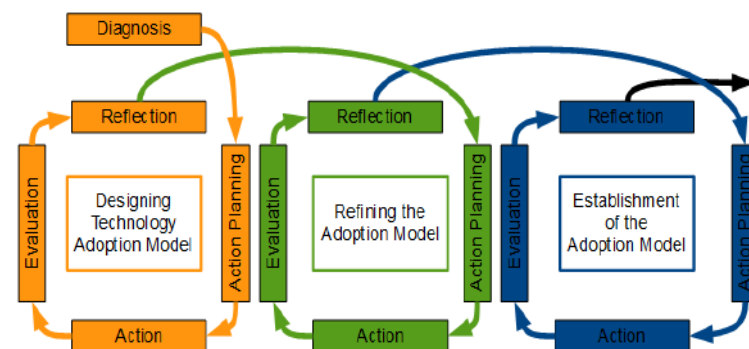
<http://robertsonandchang.com/services/learning-labs-action-research-for-innovation-and-excellence>



Developing a Technology adoption model for youth services in an informal settlement

Master Thesis by Asnath Kambunga

Cycle	Phase	Question/objective	Method	Finding/results
Cycle 1 Design Technology Adoption strategy	Diagnosis 1 (D1)	Which are the current technologies known to the Havana Youth? How are the youth currently adopting those technologies?	1. Questionnaire 2. Focus groups 3. Digital services expo workshop	Few technologies known, Yet no adoption, extracted factors affecting adoption in Havana
	Action Planning 1 (P1)	How is the rest of the world in similar setting as Havana adopting technologies?	1. Systematic literature review	Extracted factors affecting adoption Created strategy Created adoption time schedule
	Action 1 (A1)	How should new technologies be introduced in Havana to ensure better adoption?	1. Introduction to the adoption strategy workshop session 2. Training on the technology usage 3. Joint design of adoption campaign;	The campaign materials for deployment were not clear, youth were not committed
	Evaluation 1 (E1)	What were the action taken and their outcomes?	1. Synthesise results from AP1 and A1	Technology not adopted as the youth seems not to understand their roles and responsibilities, nor the system.
	Reflection 1 (R1)	What are the lesson learned to be considered in AP2	Reflect upon the action and its outcome. Summarise the lesson learned	Lesson learned to be carried over to AP2
Cycle 2 Refine Technology Adoption strategy	Action Planning 2 (AP2)	What are the elements that contribute to the ambassadors' understanding of their role in the technology adoption campaign?	Dedicated commitment session, Campaign material refinement Content training	Commitment session Campaigns materials redesign workshop, training plan
	Action 2 (A2)	How should the ambassadors understand their role in the technology adoption campaign to lead to better adoption of the technology?	1. Dedicated commitment session 2. Campaign materials refinement workshop 3. Content training	Youth set adoption/ deployment goals, better understanding of the system, partial execution of the adoption plan
	Evaluation 2 (E2)	What do the youth think of the whole adoption process?	Questionnaires	Youth not motivated
	Reflection 2 (R2)	What are the lesson learned to be considered in AP3	1. Summarise the lesson learned	Lesson learned to be carried over to AP3
Cycle 3 Establishment of the Technology Adoption strategy	Action Planning 3 (AP3)	What are the elements that will improve the technology adoption strategy	1. Scrutinize results from E2	Developed a refined adoption strategy, Developed a deployment schedule
	Action 3 (A3)	What are the best methods to implement the proposed adoption strategy in AP3?	1. Workshops 2. Meetings 3. Training	System deployed, motivated youth
	Evaluation 3 (E3)	How effective was the adoption strategy?	1. Questionnaires 2. Discussion among researchers	Committed ambassadors, Successful deployment
	Reflection 3 (R3)	What are the lesson learned to be considered in the development of a technology adoption model	1. Summarise the lesson learned	Social influence and capability plays a role in adoption



Kambunga, A. P., Winschiers-Theophilus, H., & Goagoses, N. (2018, December). Re-conceptualizing technology adoption in informal settlements based on a Namibian application. In *Proceedings of the Second African Conference for Human Computer Interaction: Thriving Communities* (pp. 1-10)

Data Collection Methods

- Semi-structured interviews
- Participant observations
- Analysis of documents
- Use of checklists: data registers, analysis tools, and health workers
- Software prototyping process and logs
- Group discussions and Training workshops
- Video/still pictures
- Analysis of press media reports

Data Analysis and Presentation

Interviews, Observations, Questionnaires, and site documents work together to support the research claims.

Empirical materials are presented in

- Descriptive statistics (quantitative data)
 - Model and measurement instrument for evaluating user satisfaction (7-point scale)
- Qualitative excerpts of encoded user reactions
- Software evaluation via criteria such as reliability and usability
- Logs and Screen shots of programs

Secondary Sources of Data

- Documents from the field
- Photos and videos

Validity of AR

Values

- Credibility
- Transferability
- Dependability
- Confirmability

Means

- Prolonged engagement, verbatim principle, multiple perspectives, workability
- Share context knowledge
- Transparency in description
- Present enough evidences

Hayes, G.R. 2011. The relationship of action research to human-computer interaction. ACM Trans. Comput. Hum. Interact. 18, 3, Article 15 (July 2011), 20 pages.
DOI=10.1145/1993060.1993065 <http://doi.acm.org/10.1145/1993060.1993065>

Ethics in AR

- Conventional ones +
- Participants: from subject to collaborator
- Relationship
- Sustainability of change
- Choice of action

Design-based research

“The designing act of creating prototypes is in itself a potential generator of knowledge (if only its insights do not disappear into the prototype, but are fed back into the disciplinary and cross-disciplinary platforms that can fit these insights into the growth of theory).”

—(Stappers 2007)

<https://www.interaction-design.org/literature/book/the-encyclopedia-of-human-computer-interaction-2nd-ed/research-through-design>

“we intend the term design research (sic) to mean an intention to produce knowledge and not the work to more immediately inform the development of a commercial product”. Zimmerman, Forlizzi, and Evenson (2007, p. 494)

RtD is defined as “a research approach that employs methods and processes from design practice [sic] as a legitimate method of inquiry” Zimmerman, Stolterman, and Forlizzi (2010)

Zimmerman, J., Forlizzi, J., & Evenson, S. (2007, April). Research through design as a method for interaction design research in HCI. In *Proceedings of the SIGCHI conference on Human factors in computing systems* (pp. 493-502)

Zimmerman, J., Stolterman, E., & Forlizzi, J. (2010, August). An analysis and critique of Research through Design: towards a formalization of a research approach. In *proceedings of the 8th ACM conference on designing interactive systems* (pp. 310-319)

Research through Design

“ the investigation of knowledge through purposeful design. ... the study of design and the process of knowledge production that occurs through the act of design”

Faste, T., & Faste, H. (2012, August). Demystifying “design research”: Design is not research, research is design. In *IDSA education symposium* (Vol. 2012, p. 15)

USER INTERFACE ACTIONS FOR THE RESTORATION OF EYE-TRACKED ATTENTION LOSS

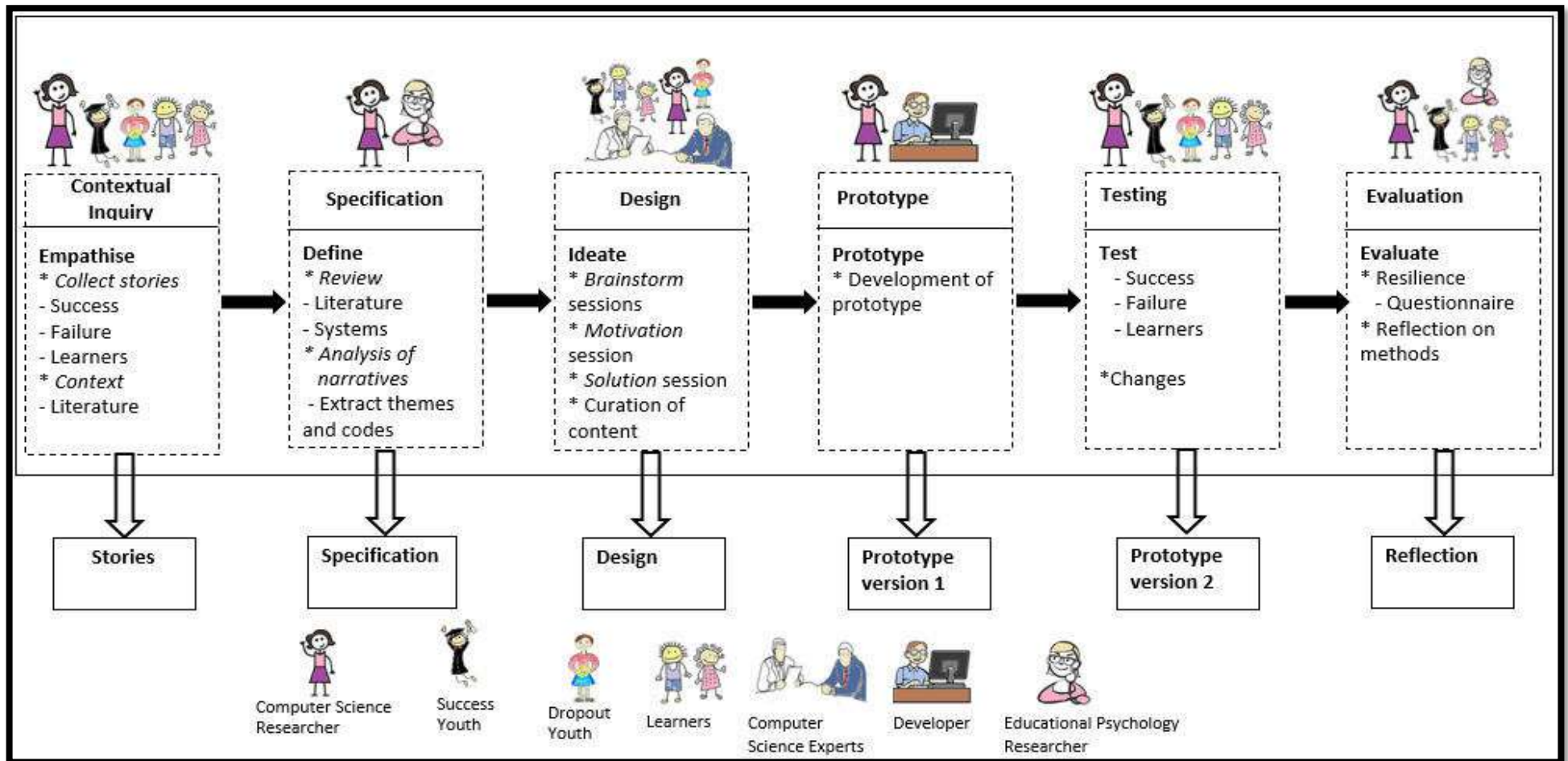
Honors thesis by Baptista Jacinto Joao

A Designed Experiment



RtD process

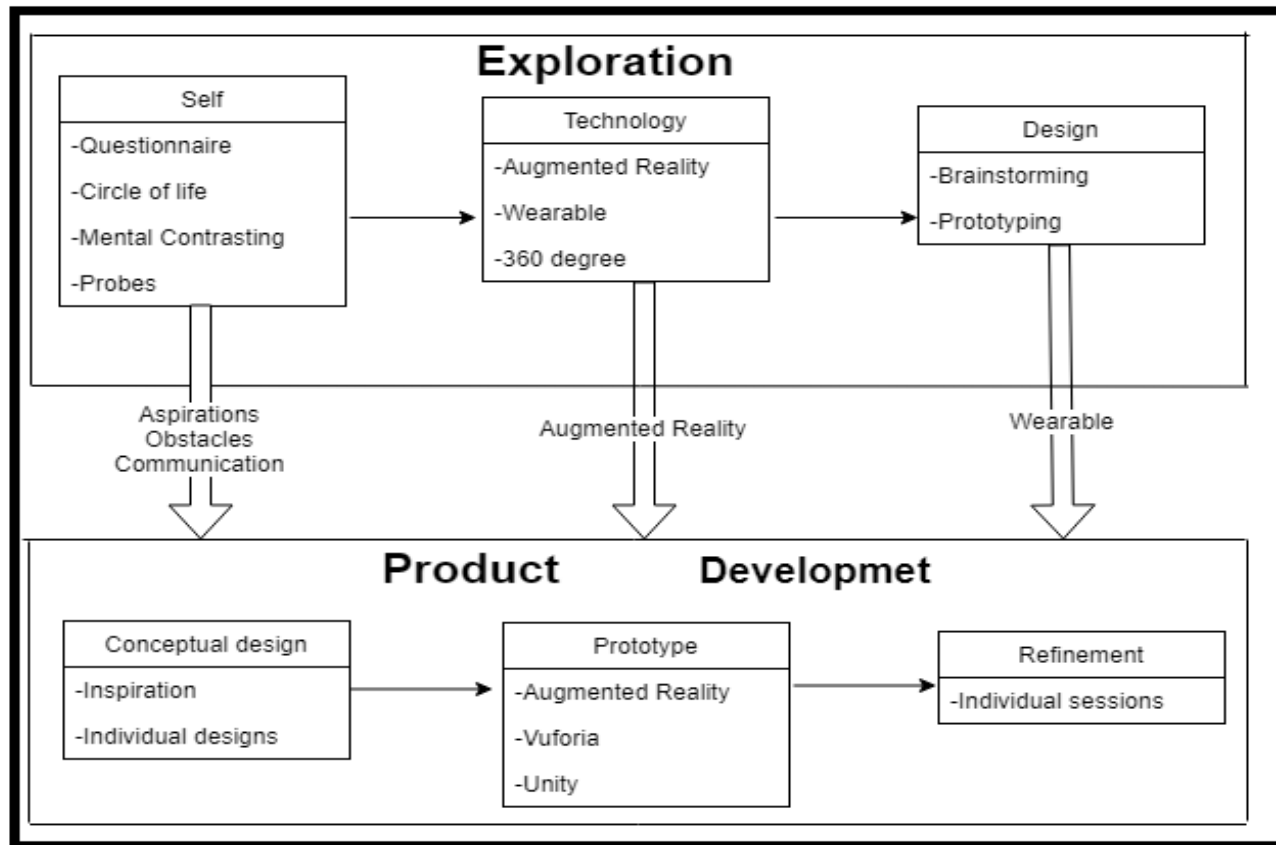
Technology to increase resilience to decrease school drop out



Kays, R. (2020) DEVELOPING TECHNOLOGY TO STRENGTHEN RESILIENCE IN SAN CHILDREN TO REDUCE SCHOOL DROPOUTS, Master Thesis

RtD process

Co-design of personal AR wearables to enhance social well-being



Kauhondamwa, M., Winschiers-Theophilus, M., Kapembe, S., Costa, H., Guxab, J., Kamati, I., & Afrikaner, H. (2018). Co-creating personal augmented reality accessories to enhance social well-being of urban San youth. *In Proceedings of the Second African Conference for Human Computer Interaction: Thriving Communities*



PAMIBIA UNIVERSITY
OF SCIENCE AND TECHNOLOGY

Faculty of Computing & Informatics





Problem Statement: A Comprehensive Guide

Prof. Jude Osakwe

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01 Understanding the Problem Statement

Purpose & Importance

Clearly Define Problem

The problem statement should immediately identify the core issue being addressed. A well-defined problem statement leaves no ambiguity about the specific challenge at hand.

Justify Need for Solution

It should strongly argue why solving the problem is crucial and significant. Emphasize the importance of finding a resolution and highlight the potential benefits of doing so.

Focus Project Efforts

The problem statement should provide a sharp, concise target for all project activities. By setting a clear objective, it helps streamline efforts and avoid scope creep.

Reference Point for Alignment

It should act as a constant point of reference throughout the project lifecycle to ensure the team remains focused on the primary goal. This helps maintain alignment and direction.

Key Elements Overview

Ideal Situation

1

Describe the desired state or how things should ideally function absent the identified problem. This sets the context for understanding the gap.

2

Current Reality

Clearly articulate the problem itself by detailing the current undesirable situation. Describe the issue as it exists and provide a contrast to the ideal.

Consequences

3

Explain the negative impacts and implications of the problem if left unresolved. Emphasize the repercussions and their effects on stakeholders.

4

Suggestion (Optional)

Optionally, offer a high-level suggestion of a potential direction for solving the problem. This provides a preliminary idea of what might be done.



02 Defining the "Ideal Situation"

Vision of the Desired State

Paint a Picture of Success

Create a vivid depiction of how things will operate when the problem is resolved. Describe the scenario where objectives are met and improvements are realized.

Remove the Problem

Imagine a scenario where the identified issue no longer exists and highlight the resulting improvements. Focus on operational excellence and smooth processes.

Future Vision

Illustrate a future state that underscores the benefits of having addressed and corrected the problem. Emphasize the value of long-term positive outcomes.

Specificity is Key



01

Concrete Language

Use precise, unambiguous language to clearly articulate the desired outcome. Avoid broad or abstract terms that can be misinterpreted.

02

Avoid Vague Generalities

Steer clear of general statements; instead, provide specific details about what success looks like. Focus on quantifiable and observable improvements.

03

Customer Example

"Customers should complete online purchases within 3 minutes with no errors." This makes it easy to understand and measure the desired improvement in customer experience

Establishing a Baseline

01

Quantify Ideal Situation

Use quantifiable metrics to define the ideal state. This helps establish a clear, measurable target for improvement.

02

Metrics and Benchmarks

Define specific benchmarks and key performance indicators (KPIs) that will be used to track progress. Include metrics such as time, cost, and satisfaction levels.

03

Measuring Future Progress

Explain how these metrics will be used to assess progress and demonstrate the impact of implemented solutions. Ensure there is a method for ongoing tracking.



03 Assessing the "Current Reality"

Articulating the Problem

Deviating from the Ideal

Explain how the current situation deviates from the envisioned ideal state. Clearly show the gap that needs to be bridged.

Heart of Problem Statement

Highlight that this description is at the core of the problem statement. This is where the actual problem is defined and brought to light.

Current Obstacles

Describe the specific obstacles or challenges that are preventing the achievement of the ideal state. Understand why the ideal state is not currently attained.

Competitive Advantage

Data-Driven Approach

Base the description of the current reality on tangible data and evidence, not personal opinions. Use factual information to underpin the problem statement.

Avoid Opinions, Use Facts

Refrain from including subjective viewpoints or interpretations in the description of the problem. Maintain a neutral, fact-based perspective.

Statistics and Examples

Support the problem description with specific statistics, examples, and relevant data points. Include tangible information that provides a clear view of the issue.

Focusing on Symptoms



01

Observable Effects

Concentrate on the observable effects or symptoms of the underlying issue. Describe the tangible outcomes rather than speculating on root causes.

02

Postpone Root Cause Analysis

Defer any attempts to identify the root cause of the problem at this stage. Focus solely on describing the symptoms that are being observed.

03

Current Performance Problems

Highlight the performance issues or deficiencies that are directly related to the identified problem. Describe the current state of poor performance.

Providing Context



Background Information

Include relevant background information that provides context for the problem. This can include details about the organization, industry, or market.

Organizational Factors

Explain organizational factors or internal dynamics that are contributing to the problem. Understand how internal processes or structures impact the situation.

Target Audience Details

Provide details about the target audience or stakeholders who are affected by the problem. Describe who is impacted and how they are influenced by the issue.



04 Analyzing the "Consequences"

Articulating the Impact



Negative Consequences

Clearly describe the negative consequences resulting from the problem. Highlight the adverse effects and detrimental outcomes.



Who is Affected? How?

Identify the specific individuals, groups, or stakeholders that are affected, and explain how they are impacted. Understand the breadth and depth of the problem's reach.



Pain points

Showcase specific problems that are painful or difficult to endure. Understand and describe problems from key stakeholders' perspectives

Quantifying the Impact

01 Numbers Make it Compelling

Emphasize the importance of quantifying the impact of the problem whenever possible. Use numerical data to make the consequences more compelling and understandable.

02 Examples of Quantification

- "Reduced online sales conversion by 15%" or "Customer service costs increased by \$20,000 per month" are tangible examples of quantification.

03 Demonstrate Actual Costs

Quantify the financial, operational, or reputational costs associated with the problem. Show the direct financial burden and indirect expenses incurred.



Long-Term Implications



Immediate vs. Future

Consider both the immediate consequences and the potential long-term implications of the problem. Understand the short-term and long-term effects.



Effects of Inaction

Describe the effects of not addressing the problem, including potential risks and missed opportunities. Highlight the dangers of inaction and complacency.



Negative Brand Impact

Emphasize how and how long unchecked problems will impact customer loyalty and brand image



Organizational Goals

Hindered Progress

Explain how the problem is impeding the organization's ability to achieve its targets.
Demonstrate the impact on performance and productivity.

01

Strategic Objectives

Connect the consequences of the problem to the organization's overall strategic objectives. Show how the problem hinders progress toward key goals.

02

03

Goal Achievements Impact

Showcase specific impacts to goals and their achievements based on the effects of problem





05 Suggesting a Direction ("What Could Be")

Outlining Potential Solutions



High Level Solutions

Summarize potential high level solutions to give context to possible ways of addressing the problem



Focus and Motivation

Use the suggestion to provide focus and motivation for the problem-solving effort. Inspire the team and provide a sense of direction.



General Statement of Action

Provide a general statement about the actions needed to address the problem. Outline the broad steps that should be taken.



Desired Outcome



01

Result of Solving Problem

Clearly state the desired result of solving the problem. Describe the ideal outcome that should be achieved.

02

Focus on Benefits

Emphasize the benefits that will be realized once the problem is resolved. Highlight the positive impacts and improvements.

03

Measurable Outcomes

Use numeric and measurable outcomes to demonstrate the effects of addressing a specific problem

Keeping it Broad



01

Avoid Specificity

Avoid being too specific or prescriptive about the solution at this stage. Maintain flexibility and encourage innovative thinking.

02

Open to Brainstorming

Encourage brainstorming and exploration of different approaches to solving the problem. Foster a creative and collaborative environment.

03

Solution Agnostic

Remain solution-agnostic and avoid predetermining the specific solution that should be implemented. Encourage open-mindedness and flexibility.

Reiterate Benefits

01

Positive Impact Summary

Summarize the positive impact that will result from successful implementation of a solution. Reinforce the value of resolving the problem.

02

Implementing a Solution

Showcase the opportunities provided by the problem that should be addressed when implementing a solution

03

Highlight Improvements

Describe the improvements that can be anticipated once the solution is in place. Focus on tangible benefits and positive changes.



06 Template & Examples

Putting It All Together

“ **Ideal, Reality, Consequences**

Utilize the problem statement structure template: "Ideally, [desired state]; However, [current problem]; As a result, [negative consequences]." Ensure all sections flow together logically.

“ **Problem Statement Structure**

Follow the logical structure of Ideal state, Current Reality, and defined Consequences to make a clear and compelling case for change and to provide a clear understanding.

“ **Filling In-Depth Template**

Review and refine the comprehensive problem statement template to ensure all dimensions of the problem are captured in detail, setting the stage for effective solutions.



Customer Service Example



01

Online Retailer Scenario

Imagine an online retail business aiming to improve its poor customer service experience. Customer satisfaction is down, returns are up, and response times are slow.

02

Specific Examples of Template use

Ideal: "Customers can easily resolve inquiries in under 3 minutes."
Reality: "Average resolution time is 10 minutes." Consequence: "Customer churn increased by 15%."

03

Customer Satisfaction Metrics

Monitor metrics such as Net Promoter Score (NPS), customer service resolution times, and customer feedback scores to measure the impact of improvements.

•Thank you for



Research Methodology

Simulations

Peter Gallert

Department of Computer Science
School of Information Technology
Faculty of Computing and Informatics
Namibia University of Science and Technology

20 April 2023



Outline

- 1 Introduction
- 2 Example 1: Packet Tracer
- 3 Example 2: ns-3



Brain in a jar



Figure: “Brain in a vat”. Picture by Alexander Wivel, CC-BY-SA

- If you were just a brain in a jar
- and had no sensory organs whatsoever
- with some computer to give input
- but no way to experience reality
- There would be no way to figure that out! (contrary to *The Matrix* Trilogy)
- A.k.a a **simulation**

What is a simulation?

- A **simulation** is a simplified imitation of a real-world process over time
- Simulations use a **model** of perceived reality
- Some characteristics are deemed important, others are not
- By using the simulation (the model), we **abstract from** all unimportant details
- Example: flight simulator
 - **Important:** pilot input, weather, airport, airplane type, mechanics, emergencies, ...
 - **Unimportant:** pilot mental health, airplane specimen, passengers, ticket price, ...
- Different from an **emulation** which has **no model** and attempts **no abstraction**, for instance VMWare



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Why simulations?

1 Money:

- Cisco "pod" (3 routers, 3 switches, and accessories): 100,000 N\$
- Cisco Packet Tracer: 0 N\$
- And that's still reasonably cheap! There are simulations for power grids, satellite launches, and the global financial system

2 Safety: How do you learn to

- Land a Boeing 747?
- Transplant a human heart?
- React to a pandemic?
- Reanimate an accident victim?
- Extinguish a fire in a residential building?

3 Convenience:

- Book lab 5, travel to campus, do a networking lab,
- Or: Fill in a set of forms, get HoD approval, take a Cisco pod home,
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Simulations in science

- Empirical scientific method: **experiment**
- Experiment is only valid if it is **repeatable**
- Many factors hinder repeatability:
 - The person / team running the experiment
 - Environmental factors: ambient temperature, time of day / year
 - Equipment manufacturing faults and deviations
 - Detail of the documentation
- A **simulation** helps suppress outside factors (abstraction)
- Also simplifies documentation
- Some simulations are **better than the real thing!**



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Packet Tracer



- **Packet Tracer** is a network simulator for teaching
- Developed by Cisco Systems, a leading networking hardware manufacturer
- **Real-time simulator** for routers, switches, firewalls, cell phones, access points, et cetera
- GUI, drop devices on screen, connect, configure
- Simulation mode: "follow" a packet on its way through the network
- Simple download, all operating systems, free of charge (account creation at www.netacad.com required)



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Main screen

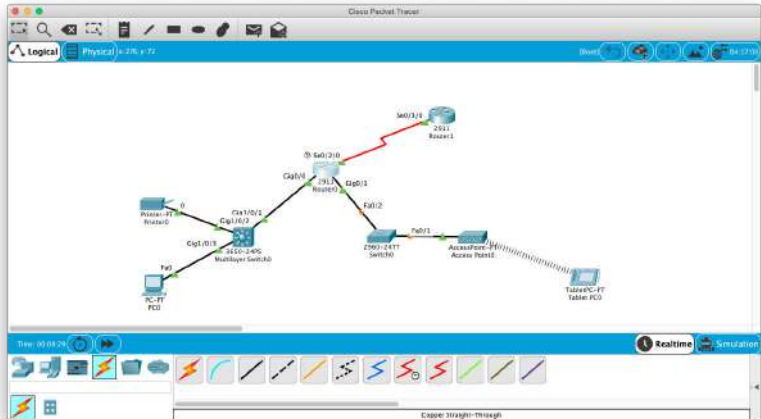


Figure: Packet Tracer screenshot with some connected devices. Fair use. 

Packet Tracer properties

- **Modelled:** Device configuration (hardware and software), protocol behaviour and performance
- **Not modelled:** Space considerations, reduced feature set
- Only selected commands and hardware options—those taught in, and sold to, the Cisco Academies



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Packet Tracer hardware configuration



Figure: Packet Tracer configuration window: hardware. Fair use.

Packet Tracer software configuration



Figure: Packet Tracer configuration window: Cisco IOS command line interface. Fair use.



Advantages, disadvantages

- **Good for teaching computer networking:**

- 1 Point-and-click, easy to learn
- 2 Realistic simulation of hardware installation and configuration mistakes
- 3 Runs on all major operating systems, even on Android
- 4 Accompanied by an online course (CCNA) to teach network device configuration

- **Not suitable for performance estimates:**

- 1 Timers independent of host performance
- 2 Illogical result, for instance for Quality of Service impact

- **Proprietary and vendor-dependent:**

- 1 Non-Cisco devices cannot be simulated
- 2 Many "fantasy devices" that do not exist in this form



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ns-3



- **Network simulator (ns)** is a network simulator for research
- Currently in version 3, **ns-3**
- Developed by its user community worldwide
- **Discrete event simulator**, calculating **events**, their **effects**, **duration**, and **output**
- **Input:** C++ source code
- **Output:** (all optional)
 - .pcap packet capture file for display with Wireshark
 - .tr bare text trace file of chronological events
 - .xml graphic for display with offline animator NetAnim
 - Command line messages



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Command line

```

peter@debian: ~/work/ns-3-allinone/ns-3-dev
File Edit View Search Terminal Help
peter@debian:~/work/ns-3-allinone/ns-3-dev$ ./waf --run stest
Waf: Entering directory `/home/peter/work/ns-3-allinone/ns-3-dev/build'
[2809/2857] Compiling scratch/stest.cc
[2818/2857] Linking build/scratch/stest
Waf: Leaving directory `/home/peter/work/ns-3-allinone/ns-3-dev/build'
Build commands will be stored in build/compile_commands.json
'build' finished successfully (5.047s)
Total Bytes Received: 680448
peter@debian:~/work/ns-3-allinone/ns-3-dev$ ./waf --run stest
Waf: Entering directory `/home/peter/work/ns-3-allinone/ns-3-dev/build'
[2810/2857] Compiling scratch/stest.cc
[2818/2857] Linking build/scratch/stest
Waf: Leaving directory `/home/peter/work/ns-3-allinone/ns-3-dev/build'
Build commands will be stored in build/compile_commands.json
'build' finished successfully (2.570s)
At time +2s client sent 1024 bytes to 10.1.2.4 port 55
At time +2.01799s server received 1024 bytes from 10.1.3.3 port 49153
At time +2.01799s server sent 1024 bytes to 10.1.3.3 port 49153
At time +2.03371s client received 1024 bytes from 10.1.2.4 port 55
At time +3s client sent 1024 bytes to 10.1.2.4 port 55
At time +3.00529s server received 1024 bytes from 10.1.3.3 port 49153
At time +3.00529s server sent 1024 bytes to 10.1.3.3 port 49153
At time +3.01058s client received 1024 bytes from 10.1.2.4 port 55
peter@debian:~/work/ns-3-allinone/ns-3-dev$

```

Figure: Command line screenshot of ns-3, with simulation output. Own work



ns-3 properties

- **Modelled:** Protocol behaviour and performance
- **Not modelled:** most device configuration steps, hardware restrictions
- With ns-3, all devices **behave exactly as they should:**
 - Implementing all aspects of the simulated protocols
 - Delivering exactly the performance as determined by standards and configuration
 - Only runtime differs
- But many default values make little sense, for instance Ethernet with 4.2 GBps and no delay



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ns-3 C++ script

```

// Step 4: Define global declarations
GlobalValue::Bind ("ChecksumEnabled", BooleanValue (true));
// Calculate FCSs

// Step 5: Create nodes and node containers
NodeContainer routers;
routers.Create (2);

NodeContainer whkAP;
whkAP.Add (routers.Get (0)); // attach AP to router Windhoek
NodeContainer whkSTA;
whkSTA.Create (12);

NodeContainer rehoLAN;
rehoLAN.Add (routers.Get (1)); // attach switch to router Rehoboth
rehoLAN.Create (8);

// Step 6: Create connections
PointToPointHelper whkreho; // Define parameters of the Windhoek-Rehoboth WAN
whkreho.SetDeviceAttribute ("DataRate", StringValue ("5Mbps"));
whkreho.SetChannelAttribute ("Delay", StringValue ("2ns"));

CsmaHelper rehoswitch; // Define parameters of the Rehoboth LAN
rehoswitch.SetChannelAttribute ("DataRate", StringValue ("100Mbps"));
rehoswitch.SetChannelAttribute ("Delay", TimeValue (NanoSeconds (6560)));

Ssid whkssid = Ssid ("Windhoek WiFi");

```

Figure: C++ source code to produce a simulation. Own work

Advantages, disadvantages

- **Ideal for scientific experiments:**

- 1 The same C++ script will produce the same output, on any computer. Only runtime can differ
- 2 Control variables do not change
- 3 Easy control of the dependent variable, easy re-run with different values. For instance, same data transfer with 802.11a, 11n, 11ax
- 4 Manufacturing faults or marketing tricks irrelevant

- **Steep learning curve:**

- 1 Requires Linux, and only Linux
- 2 Tricky installation, lots of prerequisites, must be compiled on target machine
- 3 Lots of networking knowledge needed: Any logical mistake in the C++ script breaks compilation
- 4 Some understanding of C++ required—classes, inheritance, screen output, and general syntax



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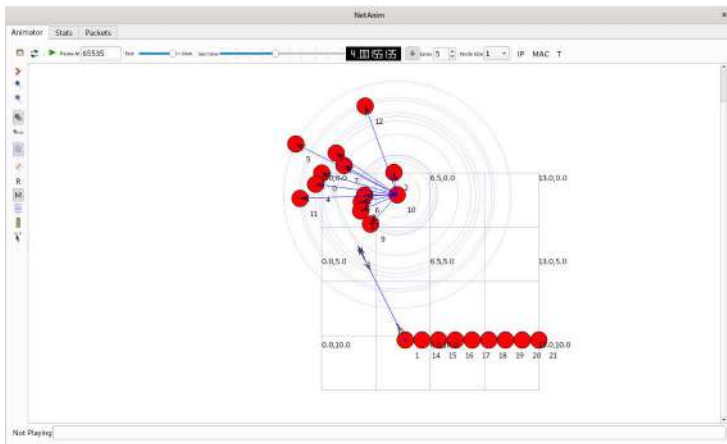


Figure: Graphical display of an ns-3 simulation with NetAnim. Own work



End of Presentation

Any questions?





PAMIBIA UNIVERSITY
OF SCIENCE AND TECHNOLOGY

Faculty of Computing and Informatics

School of Computer Science

Literature Review

Presented by: Dr A. M. Gamundani

Senior Lecturer : Cyber Security



**You are what you do,
not what you say you'll do**

Agenda

- Getting you to think about the purpose of a literature review
- Getting you to think about what your own literature review will look like
- Critical thinking, reading and writing
- Planning and structuring options



**Review the
Available
Literature**

**Publish
Findings**

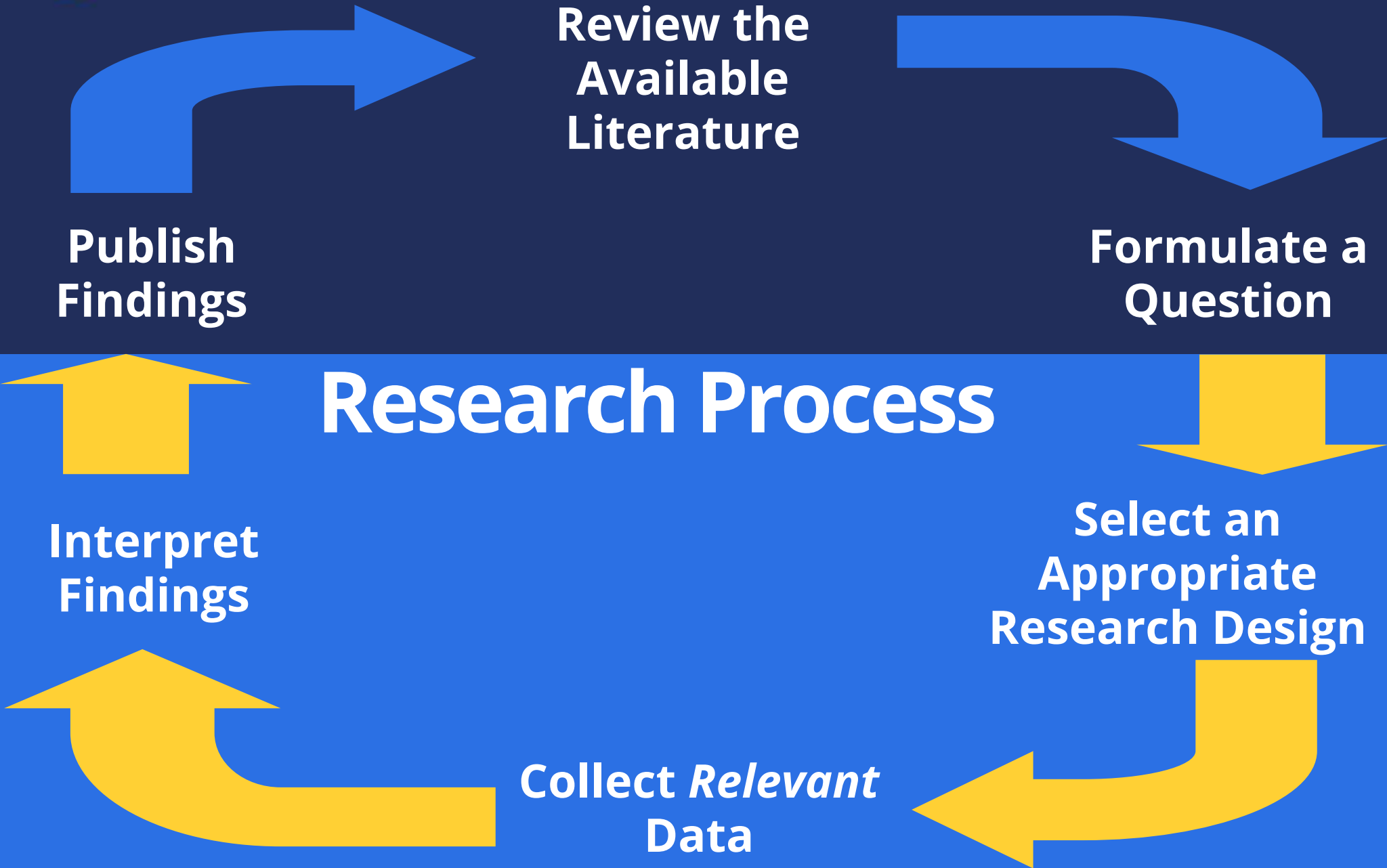
**Formulate a
Question**

Research Process

**Interpret
Findings**

**Select an
Appropriate
Research Design**

**Collect *Relevant*
Data**





What is a literature review (for)?

- When thinking about what a literature review *is*, we need to think about what it is *for*.
- What is the purpose of a literature review?
- *Activity 1:- What is literature review in your own understanding?*



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What is a literature review (for)?

Establishes the terms and context.

- *How else will you define exactly what you're looking at and where its limits are?*

Presents a survey of preceding literature on the topic.

- *How else will you know what's been done already?*

Explores ways that others have solved similar questions/problems.

- *How else will you select an appropriate methodology and approach?*

Outlines the relationship of these texts to each other.

- *How else will you know what the different perspectives and debates are, and where you are coming from?*

Evaluates the quality and relevance of the literature.

- *How else will you be able to build on or reject it?*

Establishes the gaps or inadequacies.

- *How else will you justify your own contribution?*

Demonstrates your scholarly rigor.

- *How else can I have faith in your conclusions?*

Literature Review

A critical analysis of existing research in your field - highlights both the strengths and weaknesses of existing research

Allows you to gain a critical understanding of your field

Opportunity to think about what has been done in your field; opportunity to think about the similarities, patterns, trends and also differences across the existing research

By identifying strengths and weakness, you will be able to think about what has not/needs to be done in your field

The gap in the literature is your justification for your research

MORE THAN JUST A CHAPTER...



- A literature review is a process as well as an outcome!
- **Literature review as an outcome:** appears in the final draft of your thesis as part of your introduction or as a separate chapter.
- **Literature review as a process:** critical engagement (thinking, reading and writing) with relevant research on your topic. It is a crucial and formative stage of your thesis journey.





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CRITICAL THINKING LEVELS



Undergraduate level:

backing your points up with appropriate, authoritative sources



Masters Level:

demonstrating that you understand how knowledge is created



PhD Level:

demonstrating that you can create new, original knowledge



Professional academic:

peer review, building a coherent body of work of your own and assuring others' work





READING CRITICALLY

THINK ABOUT:

What were the research aims of the paper/book?

1

Is the research aim achieved?
If so, how did they do it?

2

Are there any problems with
their methodology?

3

Was it a strong or a weak
research model?

4

How will this research help with
your own research?

5

What can you take from it?
What needs to be avoided?
What are you doing differently?



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Critical thinking (a three-year-old's guide)

Why?

How does that
work?

What's that
made of?

But X says...

How do you
know?

What's that
for?

What does
that mean?

Says who?

What happens
if...

So what?



What critical thinking can mean in terms of the literature review

- Understanding research on its own terms – testing its viability
- Understanding research in relation to other arguments
- Critiquing research in relation to what you want to do



Starting to think about your own literature review

Who are the key players in my field? This could be anything from academics, governing bodies, schools of thought etc. (Sources!)

What are the main ideas/debates in my field?

How have these ideas changed over time?

What are some of the problems with these ideas/debates?
Is there a problem with the methodology?

What are you going to do differently?



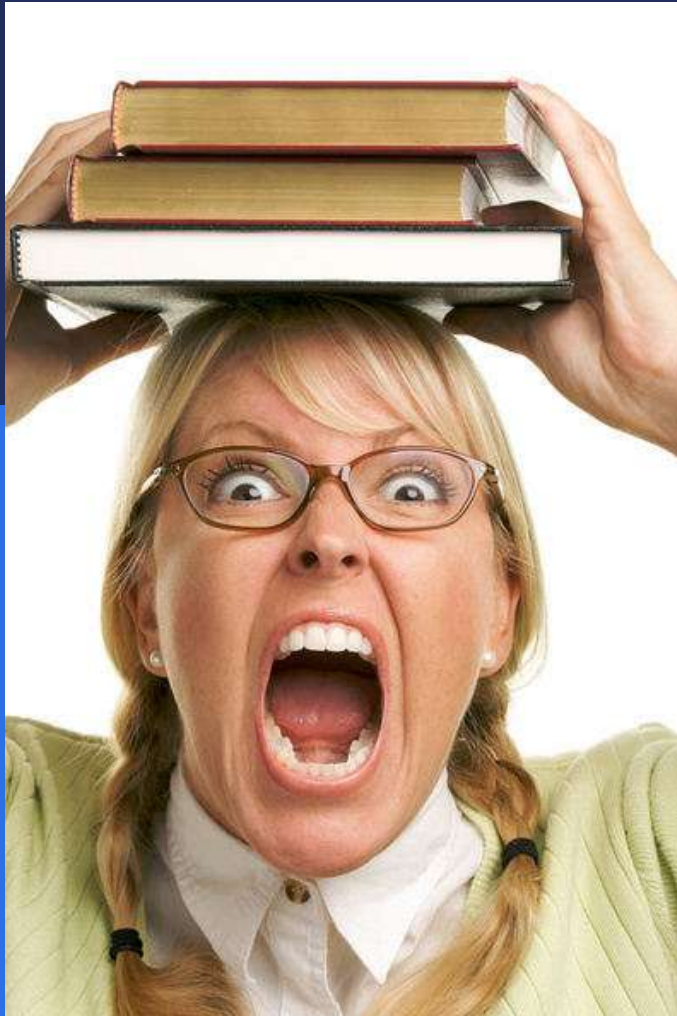
Key players and sources

First stage of the literature review is to identify the key people in your field and collate all relevant sources about your topic.

Ask yourself:

- What research and theory is there on my topic?
- What are the key sources (books, articles) on my topic?
- Who are the main theorists and researchers in this area?
- How has the topic/problem been investigated over time?

Where to start?!



YOU DON'T NEED TO READ
EVERYTHING – you can't!

- You don't need to read every text
- You don't need to read every word



(" Artificial Intelligence" OR "AI"
OR " Machine Learning" OR
"ML") AND (" Ethics" OR "
Governance")



Main ideas/debates

Once you have the relevant sources you can begin to think about what the key ideas, debates, methodologies etc. are in your field.

You can also think about how these ideas have changed over time.

Ask yourself:

- How has the topic or problem been defined?
- Are there any trends and patterns across the literature?
- What methodological assumptions and approaches have been used?
- What are the agreements and disagreements between theorists on my topic?

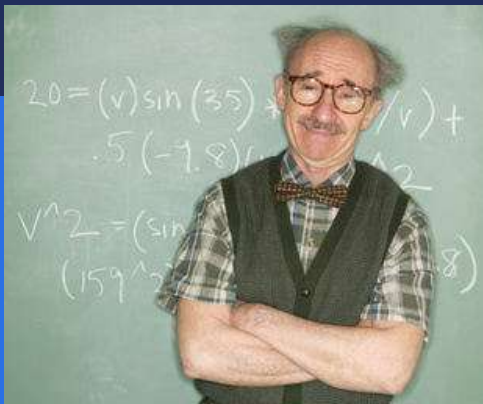


Organising your material: Identifying a debate

Scholar X



Scholar Y



- disagrees with
- agrees with (*school of thought?*)
- builds on the conclusions of
- confirms the findings of
- has reservations about



Thinking critically

When identifying the key ideas, themes and methodologies in your field, it is important to think critically about them

This will allow you to identify a 'gap' in the literature

Ask yourself:

- What are the strengths and weaknesses of these debates?
- What evidence is lacking, inconclusive or limited?
- What will you add to the topic? What will you do differently?



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- What's your story?



Map your story (literature review):

- What is your topic?
- Who are the key people in your field?
- What are the key resources?
- What are the key ideas in your field?
- What methodologies have been used?
- What are some of the strengths and weaknesses of existing research?
- What will your contribution be?
- How will it be different?

(NB: If you can't answer some of these questions, make a note of this. It will come in handy later!)



Planning is about organising the structure of your literature review (your story will help with this!)

How will you organise the information?

- Chronologically?
- Thematically?
- By trends/approaches/techniques?
- Major debates/controversies?
- Probably a combination of these



You can:

- Jot down ideas
- Move them around
- Create links
- Put text on nodes
- Swap between 'Diagram' and 'Outline'
- Transfer to Word



Writing the review

Start with an overview

Decide on organising principles (themes, trends, methodology, chronology, controversies – usually a combination of some of these)

Use headings for the different sections of the review

Provide summative signposts of where your argument is leading

Summarise your review/highlight 'gap' in research



Tips for clear writing:

- Clear introduction: overview of topic, aim of review and structure
- Clear paragraph structure
- Make sure the subject of your sentence is clear
- Don't assume knowledge
- Make sure key terminology and difficult ideas are always explained thoroughly (ask your yourself: does it make sense?)
- Be objective and balanced
- Use signposts to orientate the reader



Where appropriate, begin sections and paragraphs with a statement which synthesises or analyses, rather than just describes

Use signposting words to demonstrate how texts relate to each other and also what you think of them

- Eg. However, yet, moreover, indeed, similarly etc



Paragraphs and flow

Paragraph:

- Topic sentence
- Discussion of topic
- Closing sentence

Thematic and grammatical links

- Logical progression from one paragraph to the next
- Demonstrate links in your language



Use of citations in the literature review

Two types of citations:

Integral: The author's name appears in the sentence.

- Example (author-date system): Lillis (2001) argues that both tutors and students often lack explicit knowledge of the conventions governing the construction of academic texts.

Non-integral: The author's name appears outside sentence.

- Example: Both tutors and students often lack explicit knowledge of the conventions governing the construction of academic texts (Lillis, 2001).



- **Whose voice is dominant - the writer's or the original author's?**
- The moon is made of cheese (Brie 1999).
- Brie (1999) argues that the moon is made of cheese.
- As Brie (1999) points out, the moon is made of cheese.
- According to Brie (1999), the moon is made of cheese.
However,
- Brie (1999) argues out that the moon is made of cheese.
However,



Learning from models

Look at samples of good theses in your field

Read reviews in prestigious journals in your field



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The End of Session

All the best in your research!!



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Faculty of Computing and Informatics

2022

RESEARCH REPORT

Proud hosts for the The International Conference on Information Systems
and Emerging Technologies (ICSET) and International Conference on
Data Science, Machine Learning and Artificial Intelligence (DSMLAI); 23 -
25 November 2022



**Dr Suama L Hamunyela
(Acting Associate Dean:
Research and Innovation
(2022))**

I am thrilled to present to you the 2022 Faculty of Computing and Informatics (FCI) research outputs report. 2022 was a year of re-envisioning and reconstructing research and innovation steps after a hard-hitting pandemic. Despite the aftermath feelings experienced this year, the FCI research outputs continues to rise. We witnessed graduations of thirty (32) Masters, and a total of sixty-five (65) peer reviewed publications. The statistics are a confirmation that FCI is set towards achieving its vision of being the leading enabler of societal digital transformation.

At FCI, research outputs translate to tangible products and research publications. The research publications are a mirror reflection of work done on the ground and at times an output of scientific investigations geared towards shaping new knowledge and understanding. This year, FCI rigorously ensured comprehensive knowledge creation within the auspices of enabling technologies for the 4IR national mandates. The concepts of Artificial Intelligence, Human Computer Interaction (HCI), Analytics, Blockchain, Cybersecurity and others were investigated. Peer review being the mechanism for ensuring quality, is deemed fundamental towards benchmarking the quality of the research outputs. FCI research outputs emergences from local, regional, and international research platforms.

Still the Faculty has a lot to do to accelerate in innovations and partnerships spheres. With the current level of enthusiasm and passion, I am confident that FCI will in the coming year thrive in all aspects of research, innovations, and partnerships.

Here are a few 2023 events to consider:

- careedconference@uws.ac.uk
- <https://chi2023.acm.org/>
- <https://indore.hosting.acm.org/acmwir2023/>
- ICISSET2023 and
- AfriChi 2023

A handwritten signature in black ink, appearing to read 'Suama L Hamunyela'.

Dr Suama L Hamunyela

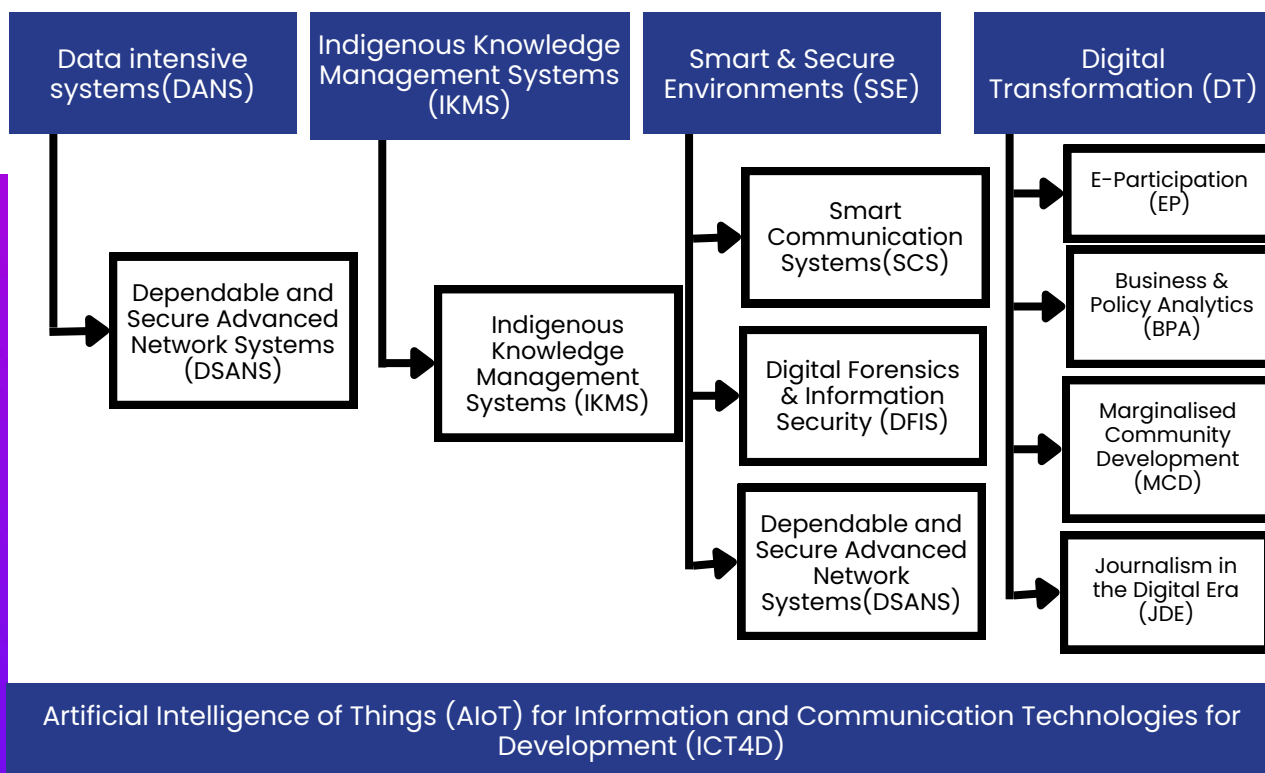
Acting Associate Dean: Research and Innovation (2022)

02 Overview of FCI Research and Innovations

The Faculty of Computing and Informatics continues to excel in research, innovations and partnership at the Namibia University of Science and Technology (NUST). The Faculty has nine (9) active research clusters with several projects across the breadth of the Faculty's four (4) research niche areas: Digital Transformation, Smart and Secure Environments, Indigenous Knowledge in Digital World, and Big Data: Infrastructure & Analytics.

The faculty holds three prestigious awards of UNESCO Research Chairs, one in Digital Technologies for Gender and the other in Smart and Secure Environment covering areas of Smart Communication, Big Data: Infrastructure and Analytics and Cyber Security and a third prestigious award of UNESCO Research Chairs, in Technology Co-design with Indigenous and Marginalized Communities was secured. Staff accountable for research activities including supervision and publications are three (2) Full professors, three (5) Associate Professors and eleven (15) Senior Lecturers.

In the 2022 academic year, the FCI graduated, eight (8) Master of Computer Science and one (9) Master of Informatics students and fifteen (15) Master of Journalism and Media Technology. FCI recorded a total of 7 Journal publications and 37 conference papers, a total of six (6) peer reviewed journal publications, and 4 book chapters. The Faculty's publication output soared with support of sponsors such as the Google DARA who provided financial support, NCRST, SASSCAL and other international partners such as EU. The Computing discipline uses a conference-publishing model more compared to the journal model, and thus the high impact outlets are peer-reviewed conferences. Faculty members served as reviewers of prestigious journals and conferences such as: 2022 DSMLAI, ICISSET, CHI 2022, IEEE Transactions on Industrial Informatics, Journal: African Journal of Science, Technology, Innovation and Development and Global Humanitarian Technology Conference (GHTC) 2021.



FCI Research Clusters

03 Research activities

The three flag ship research activities under the Faculty of Computing and Informatics are in: -

- Indigenous Knowledge in Digital World;
- Cyber Security and Digital Forensics;
- Emerging Technologies (Data Science and AI).

04 Brief profiles of key pacesetters



Prof. Fungai Bhunu Shava: is an Associate Professor in Cybersecurity within the Faculty of Computing and Informatics, Acting Executive Dean, and the leader of the Digital Forensics and Information Security Research Cluster. She is the founding and former chairperson of the Namibia National Cyber Security Competition and the coordinator of Africa Region to International Cybersecurity Competition (AFRICC). She supervises postgraduates and manages a few research projects in the areas of Information Security and Digital Forensics. She has published more than 50 papers in reputable journals and conference proceedings in her fields of specialisation namely Human Computer Interaction, User Experience, Network Security, Information Security in under-serviced communities.

Her work focuses on the secure use of ICTs as enabling tools for socio-economic development in developing economies and societies. She is also an International Visitors Leadership Program (IVLP) alumni and an Application Security Specialist with IBM Security AppScan v8.7. She is also responsible for other projects such as the Child Online Protection, Cisco Academy, and Instructor Training Centre at NUST. She has reviewed conference and journal papers in her areas of specialisation. She serves on the National cybersecurity stakeholder committees including the SADC Cyber-Infrastructure Technical Experts meetings, Africa to International Cybersecurity Competition and the International Cybersecurity Challenge. She has co-organised and spearheaded: Child Online Protection hackathons; WiSci STEAM Camp 2019 in collaboration with US Embassy and GirlUp; TechCamp Namibia: Cybersecurity for Everyone in collaboration with the USA government and Tech and Innovation Bazaars in collaboration with MTC. She participated in the Federal Department of Foreign Affairs, Switzerland (Geneva Dialogue on Responsible Behaviour in Cyberspace) as well as United Nations Institute for Disarmament Research (UNIDR) Cyber Stability Conferences. She actively contributes to Safety and Security Expo; Women in Computing Conferences, Google Developer's Circle, Namibia Internet Governance Forum (NamiIGF), Child online Protection Taskforce, Namibia; IEEE and ACM chapters in Namibia.



Dr Attlee M. Gamundani is a Senior Lecturer in the Cybersecurity Department. He is the Research lead for the Artificial Intelligence of Things (AIoT) for Information and Communication Technologies for Development (ICT4D) cluster. He is an alumni of the United Nations University (UNU) Institute in Macau where he completed his PostDoc in 2021 focusing on AI Ethics and Governance. Dr. Gamundani is serving as the country (Namibia) representative for Responsible AI Network-Africa (RAIN-Africa).

He was appointed as an Advisory Board Member and Operations - Fourth Industrial Revolution (4IR) committee for the Pan African Information Communication Technology Association (PAICTA) in 2020. Dr Gamundani's long CV consists among other publications, over 17 Journal publications, as well as over 50 Conference publications covering IoT Security, Cloud Computing Security, ICT4D, AI Security and Data Security. He delivered keynote addresses on security issues at national and international forums as well as co-authored junior level ICT Teacher's Textbooks (Grade 4, 5, 6 and 7) in Namibia. He was elevated to the Senior Member IEEE (SMIEEE) status as testimony of his wide contributions to the computer science domain. He is also a Check Point Certified Security Administrator (CCSA).

Among his main community engagement activities is the annual Namibia National Cyber Security Competition (NNCSC) which he currently chairs. He serves on the Africa Region to International Cybersecurity Competition (AFRICC) and the International Cybersecurity Competition committees.

Among other funded projects Dr Gamundani is locally executing as Co-PI or PI are the Lacuna Project (US\$48 000); Smart Parking project (N\$200 000- Funded by MTC); Making Windhoek Safer project (N\$56 000 - Funded by MTC). In 2016 he worked with the DFIS team on the Child Online Protection Research Project funded by UNICEF (US\$30 000).

Quarter 1 - 2022

- IoT Smart Gardening project (GIPE, Indonesia);
- IoT Smart Parking Project - CoW(MTC Funded);
- Visiting Faculty - Ashesi University, Ghana (Data Structures & Algorithms);
- Postgraduate Research Seminar Facilitator (Universität Leipzig, Germany).

Quarter 2 - 2022

- Personal Health Info System (GIPE, Peru);
- Cybersecurity Awareness Radio Shows (5 Shows at PAICTA & 1 Energy 100 FM -8th NNCSC Competition Overview);
- GIPE Spring School (Germany);
- Jury Member (ICC, Greece)
- 4IR AI Expo;
- 4IR RID Stream Report.

Quarter 4 - 2022

- Smart City Hunt (IoT Design);
- FinTech Digital Security Forum;
- TN Innovation Week (Hackathon);
- International Cybersecurity Webinar Hosting;
- Cybersecurity Capacity Building :The Global Constellation Annual Conference.

Quarter 3 - 2022

- Deep Learning Indaba X (Hackathon);
- IoT Innovation Challenge (Hackathon);
- NAMDEB AR/VR Challenge(Hackathon);
- SARFAI Ministerial round table Program Planning;
- SARFAI Forum Invited Speaker.
- Postgraduate Research Seminar - UNAM & Botho University , Lesotho.

DR GAMUNDANI'S 2022 RESEARCH PROFILE MILESTONES

Prof Jude Osakwe is an Associate Professor in the Department of Informatics. He is a member The Nigeria Computer Society and a member of the Digital transformation in the Department of Informatics. His research areas of interests are ICT in Education, Business Analytics and ICT for Development. Professor Osakwe is involved in various activities including being a Local Chair of the International Conference on Information Systems and Emerging Technology 2022 (ICISSET). Together with the FCI team, he sourced conference material worth 2500USD, and 40 000ND sponsorship for the ICISSET conference which will publish in the SSRN Elsevier proceedings. Graduated 3 Masters in Informatics, two (2) Journal articles Presentation ten (10) conference papers in international conference. Professor Osakwe is also a reviewer in Scientific Africa (Elsevier)



Dr Lameck Amugongo is a passionate Data Scientist. He is the DARA Big Data project lead at FCI. The DARA Big Data Project provides intensive data science skills training to students from the 8 African countries who partnered to build the mid-frequency instrument of the Square Kilometre Array (SKA) telescope in Southern Africa.

Dr Amugongo holds a PhD in Cancer Science from Manchester University. His research involves investigating image-based data mining methods to analyse how lung tumors change during treatment. For the year 2022, Dr Amugongo presented at the following conferences.

- Deep Learning Indaba 2022, Tunis (Tunisia).
- UNESCO Southern Africa sub-regional forum on Artificial intelligence (SARFAI) 2022 (Windhoek).
- Black in AI workshop co-located at Neural Information Processing Systems, New Orleans (Louisiana, US).

Together with other data scientists in the Faculty, Dr Amugongo was awarded a grant (£ 8000) through DBD/OAD Infrastructure call and a DARA Big Data grant (£ 3500).

The energetic Dr Amugongo organised the following workshops at NUST, HTTPS:

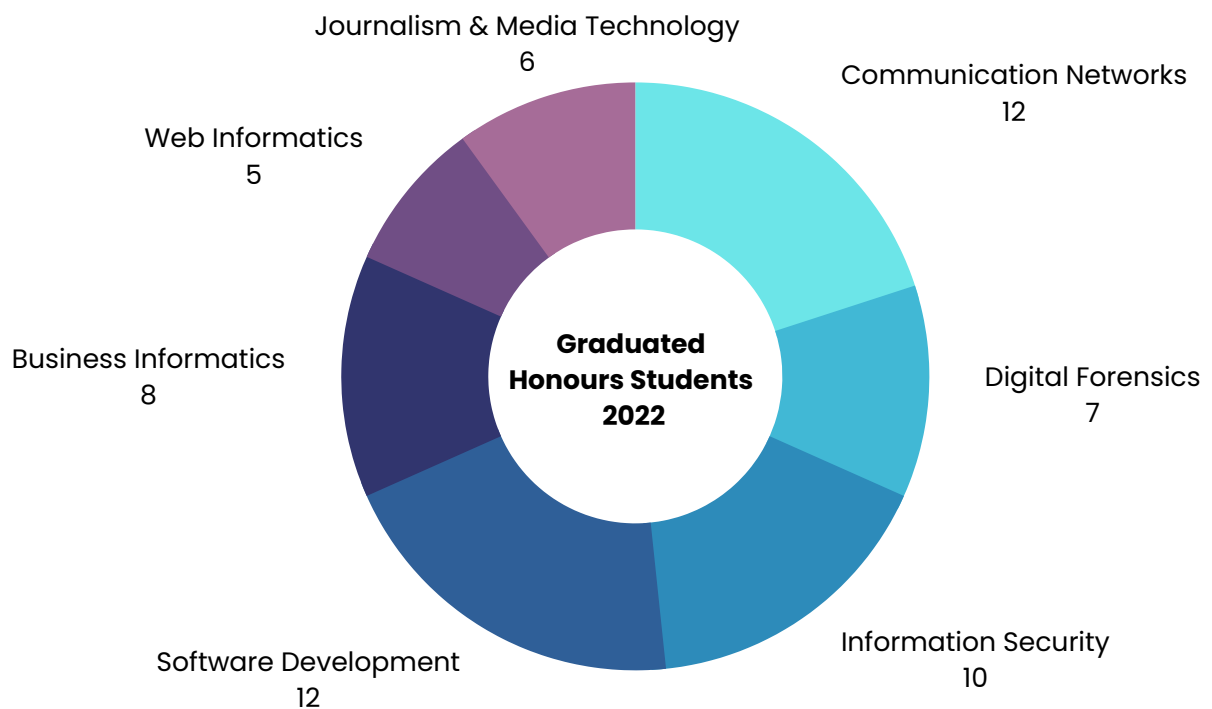
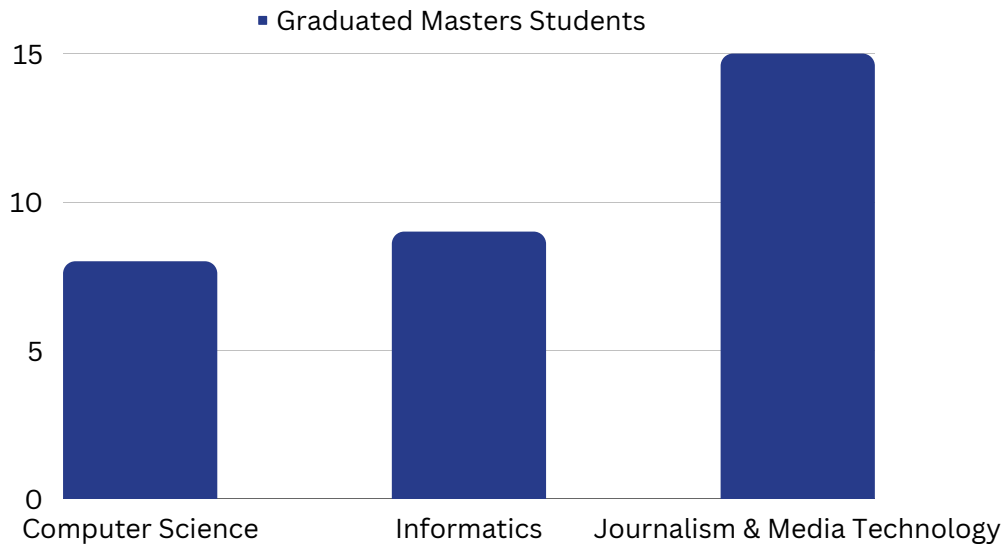
- NUST-DARA Data science school 2022.
- NAMDEB VR/AR Hackathon 2022 and
- Indaba X Namibia 2022.

He is currently working on EU Horizon funded projects with collaborators at the Institute for Ethics in Artificial Intelligence at Technical University of Munich.

05 FCI Research Output (Publications) Jan 2022 – Nov 2022



06 FCI Research Output (Graduations) Jan 2022 – Nov 2022



6 graduated with Postgraduate Certificates

07 Journal Articles

1. Ahmed Farouk, Shahid Mumtaz, Dharm Singh Jat, Heena Rathore, Zhe Li (Eds) (2022). Special Issue Computation and Modelling of Economic and Financial Complex Systems Discrete Dynamics in Nature and Society, vol. 2022. <https://www.hindawi.com/journals/ddns/si/705941/page/2/>
2. Pietari Keskinen, Heike Winschiers-Theophilus, Shilumbe Chivuno-Kuria, Anna Müller, Marko Nieminen: Digital microwork as a livelihood strategy in a Namibian informal settlement. *Electron. J. Inf. Syst. Dev. Ctries.* 88(1) (2022)
3. Heike Winschiers-Theophilus, Naska Goagoses, Erkki Rötönen, Tariq Zaman: Pushing political, cultural, and geographical boundaries: Distributed co-design with children from Namibia, Malaysia and Finland. *Int. J. Child Comput. Interact.* 31: 100439 (2022)
4. Heike Winschiers-Theophilus: Bridging worlds: indigenous knowledge in the digital world. *JCDL* 2022: 1
5. Blessing Guembe, Ambrose Azeta, Sanjay Misra, Victor Osamor, Luis Fernandez-Sanz and Vera Pospelova (2022), "The emerging Threat of AI-driven Attacks: A Review". *An International Journal of Applied Artificial Intelligence*. Vol 36 Issue 1. Published by Taylor & Francis Ltd. DOI10.1080/08839514.2022.2037254
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08 Book Chapters

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3. Idemudia, E., Iyamu, T., Ndayizigamiye, P., Shaanika, I. (Eds.). (2021) *Using Information Technology Advancements to Adapt to Global Pandemics* (pp. 66–113). IGI Global
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9. Osakwe J., Iyaloo N. W., Tarzoor T., Iyawa G., Ujakpa M. (2022). Determinants of Information Systems Resources for Business Organisations' Competitive Advantage: A Resource-Based View Approach. Presented at IST-Africa 2022. May 2022., www.IST-Africa.org/Conference2022.
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20. Emilie Maria Nybo Arendtthorp, Kasper Rodil, Heike Winschiers-Theophilus, Christof Magoath: Overcoming Legacy Bias: Re-Designing Gesture Interactions in Virtual Reality With a San Community in Namibia. CHI 2022: 555:1-555:18
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22. Nicolas Pope, Sebastian Hahta, Erkki Rötönen, Naska Goagoses, Jason Mendes, Erkki Sutinen, Heike Winschiers-Theophilus: Remote Presence: Live Holograms for a Social Classroom. IMX 2022: 249-252

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27. Nakasole, N., Chitauro, M., & Hamunyela, S. Analysing ICT initiatives towards Smart Policing to Assist African Law Enforcement in Combating Cybercrime. SETIT, Genoa-Italy & Sfax-Tunisia 28-30 May 2022. <http://www.setit.rnu.tn/Final-Version2022/33.pdf>
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35. Gerson, N., & Bhunu Shava, F. (2022). Designing A Security Model for Ehealth System in Namibia 70, International Conference on Information Systems and Emerging Technologies, (ICISSET 2022), 23- 25 November, Windhoek, Namibia.
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38. Kavikairua J., Bhunu Shava, F. (2022). Algorithm to Impact Children’s Online Behaviour and Raise their Cyber Security Awareness. International Conference on Artificial Intelligence, Big Data, Computing and Data Communication Systems (icABCD), 2022, pp. 1-7, doi: 10.1109/icABCD54961.2022.9856017.
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10 Books Authored/Co-Authored

1. Luhach A.K., Jat D.S., Bin Ghazali K.H., Gao XZ., Lingras P. (eds) (2022). Advanced Informatics for Computing Research. ICAICR 2021. Communications in Computer and Information Science, CCIS, volume 1575). Springer Nature Switzerland AG 2022. <https://doi.org/10.1007/978-3-031-09469-9>
2. Poonia, K. R., Singh, V., Jat, D.S., Diván, M. S., Khan, S. M. (2022). Proceedings of Third International Conference on Sustainable Computing, SUSCOM 2021 Book Title Proceedings of Third International Conference on Sustainable Computing: SUSCOM 2021, Series ISSN 2194-5357. <https://link.springer.com/book/9789811645372>.
3. Nagar, A.K, Jat, D.S., Marín-Raventós, G., Mishra, D.K. (Eds.2022), Intelligent Sustainable Systems, Selected Papers of Worlds4 2021, Volume 1, Springer. <https://link.springer.com/book/10.1007/978-981-16-6309-3>
4. Nagar, A.K, Jat, D.S., Marín-Raventós, G., Mishra, D.K. (Eds.2022), Intelligent Sustainable Systems, Selected Papers of Worlds4 2021, Volume 2, Springer. <https://doi.org/10.1007/978-981-16-6369-7>

11 Conference Presentations

1. Ajai Mishra, Dharm Singh Jat, and Durgesh Kumar Mishra; *An Experimental Study of Machine Learning Algorithms for Predicting Startup Success* 6th World Conference on Smart Trends in Systems Security and Sustainability organised by Global knowledge research foundation 24-27 August 2022
2. Arpit Jain, Dharm Singh Jat, *Edge Enabled Prognosis using Cognitive Ensemble Machine Learning Model for Time Constraint Applications*. 6th World Conference on Smart Trends in Systems Security and Sustainability organised by Global knowledge research foundation 24-27 August 2022.
3. Attlee M. Gamundani; *Building Responsible AI for Africa by defining the Socio-technical Constructs*. Southern Africa sub-Regional Forum on Artificial Intelligence, 7-9 September 2022, Safari Court Hotel and Conference Centre, Windhoek, Namibia
4. Lameck Amugongo; *AI Ethics from an African Context – A Case Study in the Health Sector*. Southern Africa sub-Regional Forum on Artificial Intelligence, 7-9 September 2022, Safari Court Hotel and Conference Centre, Windhoek, Namibia
5. Jose Quenum; *Session 1 Moderator – Status of AI development and utilisation in Southern Africa*. Southern Africa sub-Regional Forum on Artificial Intelligence, 7-9 September 2022, Safari Court Hotel and Conference Centre, Windhoek, Namibia
6. Attlee M. Gamundani and Shakela Viktoria; *Distilling the encounters towards building a Namibian cybersecurity awareness platform*. Global Constellation Annual Conference: Linking Cybersecurity Capacity Research to development in Africa, 2-3 November 2022, Park Inn by Radisson, Newlands, Cape Town.

7. Gabriel T. Nhinda and Fungai Bhunu Shava; Using co-design to craft cybersecurity secure practices for rural communities in Africa: A case study from Northern Namibia. Global Constellation Annual Conference: Linking Cybersecurity Capacity Research to development in Africa, 2-3 November 2022, Park Inn by Radisson, Newlands, Cape Town.

8. Nobert R. Jere , Sonwabo Mdwaba and Attlee M. Gamundani ; Co-designing a cybersecurity citizen centric framework for South Africa. Global Constellation Annual Conference: Linking Cybersecurity Capacity Research to development in Africa, 2-3 November 2022, Park Inn by Radisson, Newlands, Cape Town.

9. Nobert R. Jere, Attlee M. Gamundani and Sonwabo Mdwaba; Cybersecurity a national pandemic: Who is safer from cyber-attacks? Global Constellation Annual Conference: Linking Cybersecurity Capacity Research to development in Africa, 2-3 November 2022, Park Inn by Radisson, Newlands, Cape Town.

10. Kanyama Maria N, Fungai Bhunu Shava, Attlee M. Gamundani and Andreas Hartman; Machine learning anomaly detection techniques applicable to smart water metering networks- A systematic review. Integrated Water Resource Management for Sustainable Development in Eastern and Southern Africa, 23rd WaterNet/WARFSA/GWP-SA Symposium, Rustenburg, South Africa, 19th -21st October 2022.

11. Manfred Meyer, Katja Becker, Attlee M. Gamundani and Colin Stanley; A Global Intercultural Project Experience (GIPE): Reflections on Combining Online and On-site Project-Based Learning across four Continents. International Conference on Transformative Teaching: Focus on Pedagogy 2022, 15-17 November 2022. A virtual conference: Florida State University, University of Dundee, Zayed University.



Smart City Hunt - November 2022 (NUST, MTC, NCRST, City of Windhoek and UNDP)

12 Others (Keynotes, Talks, Magazines, e.t.c)

1. Prof Dharm Singh Jat, Session Chair and speaker, 7th International Conference on Information & Communication Technology (ICICT2022), 21-24 February 2022, London, UK
2. Prof Dharm Singh Jat invited talk on Smart Computing and Communication: Enabling future IoTs, International Conference on "Empowering Smart Future through Scientific Development and Technology organized by The Research Committee & Department of Computer Applications (MCA), Patna Women's College (Autonomous), Patna University, 5TH MAY 2022.
3. Prof Dharm Singh Jat contributed as Resource Person in AICTE Training and Learning (ATAL) Academy Online Elementary FDP on "Security in Blockchain and Cyber-Physical Systems Design" from 17/01/2022 to 21/01/2022 at Government Engineering College, Rajkot.
4. Prof Dharm Singh Jat sharing knowledge as a Speaker for the Region 10 IEEE Computer Society Mega DVP Program on the topic "Smart Computing and Communication: Enabling Future Internet of Things" on May 14th, 2022
5. Prof Dharm Singh Jat, Inaugural Speaker, 6th World Conference on Smart Trends in Systems Security and Sustainability organised by Global knowledge research foundation 24-27 August 2022.
6. Prof Fungai Bhunu Shava - Empowering Women and Girls in Science - promoting gender equity in science for women and girls PAICTA. (https://www.youtube.com/watch?v=G64-Cpkh_o&t=29s)
7. Dr Attlee M. Gamundani -Tech Power Talk – Cybersecurity – February 15, 2022 (radio 78.5 Pan-African Voice) Radio Show (<https://www.youtube.com/watch?v=pIOfnlZgKEs>)
8. Dr Attlee M. Gamundani -Tech Power Talk – Cybersecurity – March 8, 2022 (radio 78.5 Pan-African Voice) Radio Show (<https://www.youtube.com/watch?v=hbF5JqsUOUo>)
9. Dr Attlee M. Gamundani -Tech Power Talk – Cybersecurity – April 5, 2022 (radio 78.5 Pan-African Voice) Radio Show (<https://www.youtube.com/watch?v=y-eNLz3pdQw>)
10. Dr Attlee M. Gamundani -Tech Power Talk – Cybersecurity – May 17, 2022 (radio 78.5 Pan-African Voice) Radio Show (<https://www.youtube.com/watch?v=FcuQ2tFBwGU>)
11. Hickok, M., Dorsey, C., O'Brien, T., Baur, D., Ingram, K., Chauhan, C., & Gamundani, A. (2022). Case Study: The Distilling of a Biased Algorithmic Decision System through a Business Lens. Available at SSRN. (<https://ssrn.com/abstract=4019672>)



13 Postgraduate Thesis Outputs 2022

Master of Computer Science (NQF-9) – [8]

Graduate Name	Thesis Title
MUSA Gabriel P	Efficient Hybrid Deep Reinforcement Learning Mechanism for Distributed Denial of Service Attack Detection in Software Defined Networks
GERSON Naemi	Designing a Security Model for Ehealth System in Namibia
JONAS Josua	Multi-Cloud Serverless Function Composition using a Hierarchical Task Network Planner
KORUHAMA Kahuhu A	A Genealogy Tool to Account for the Dual Descent Kinship among Ovaherero Descents
MATSA Lonnie S	Design of an Intrusion Detection System for Distributed Denial of Service Attacks in Software Defined Networks Using Hybrid Deep Learning
SHEHU Yakubu	An Optimized Neural-Network Based Algorithm for Semi-Automated Identification and Quantification of Wildlife on High Resolution Drone
SHINEDIMA Gabriel N	Development of an Interactive World Map Game Co-Designed by Primary School Learners in Namibia
SIBANDA Blessing K	Mobile Application for Plant Leaf Disease Classification in a Resource-Constrained Environment



TN Mobile Innovation Week



UNESCO, Southern Africa sub-Regional Forum on AI

Master of Informatics (NQF-9) - [9]

Graduate Name	Thesis Title
GABRIEL Emeritha N	A Conceptual model for the adoption of an online program recommender system for the Namibian Higher Educational Institutions.
MARTHIN Alfriede N	A Model for Teaching Physical Science in Namibian High Schools through the Use of ICT Platforms
NAHOLE Lukas	A Strategy for Consumer Adoption and Usage of Point-of-Sale E-Payment System in Namibia Payment System in Namibia.
KUBERSKY Brandon R	Machine Learning Model to Analyse Cattle Movement with GPS-Enabled Device Tracking in Namibia-Hardap Communal Lands
SHIPULWA Fillipus S	An Automated Adaptive Risk Management System for Public Financial Information Systems: A Case Study of the Ministry of Finance in Namibia
NAKASOLE Naftali	Developing a Smart Policing Model to Assist Namibian Law Enforcement Agencies in Combating Cybercrimes
NABOTH Rosalia N N	Developing an Enterprise Resource Planning System Conversion Model: A Case Study of the Namibia Power Corporation (PTY)
NAWA Eva-Lisa T	Developing A Cybersecurity Framework for the Banking Sector of Namibia
SHALUMBU Iyaloo S	Towards the Development of an Adaptive Strategic Framework for Digital Transformation. A Case Study of Namibia Statistics Agency

Master of Journalism and Media Technology (NQF-9) - [15]

Graduate Name	Thesis Title
NTEMWA Angela K	An exploration into the perceptions of the public relations profession among selected corporate organisations in Windhoek, Namibia. NUST
MUSHAYAVANHU Kudakwashe	Framing of the COVID-19 pandemic in Namibian National Newspapers: An analysis of The Namibian, Namibian Sun and New Era. NUST
MATSI Fransina A	Investigating the effectiveness of NAMCOL's corporate communication strategy in changing brand perceptions among selected stakeholders in Windhoek, Namibia. NUST
KOOPER Immanuel	Is Language a Barrier to Public's Participation in the Law-making Process?
MATUNDU Marina	A Reception Analysis of Omurari FM's Okaruveze Kovaremane Radio Programme for People with Disabilities
TJITUKA Floyd P	The Role of Oral Media and Radio in Promoting and Preserving Indigenous Knowledge Systems: A Case Study of Otjiherero-Speaking Communities in Kunene Region
MATTY Fidel	Examining stakeholders' views on the implications of delayed enactments of the access to information bill on media reporting in Namibia
MUNDODZI Faith	A Reception Analysis of Selected Cartoon Messages on COVID-19 by Upper Primary School Learners at Highlands Christian School in Windhoek
SHIHEPO Timoteus N	An investigation into how the Namibian print media industry is coping with contemporary economic conditions
ZVIYITA Itai	An examination of ethical issues confronting the operations of selected hybrid media organisations in the digital age in Namibia

MARE Hilary	Investigating the Implications of COVID-19 on the Sustainability of the Print Media in Namibia: A Case Study of the Namibian, Confidente and New Era
KAPENDA Jairus J J	Assessing the use of social media in the Public Relations Practices of the Government Institution Pension Fund (GIPF) of Namibia
HAMBUREE Belinda	An Evaluation of Print Media Campaigns for Promoting Road Safety in Namibia from January to December 2019
IITA Ndamonako A	Investigating Stakeholder Engagement in Public Relations Practices of two Namibian Universities. (The International University of Management and Namibia University of Science and Technology
ADETONA David A	An investigation into how newspapers adhere to copyright regulations in Namibia



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**NAMIBIA UNIVERSITY
OF SCIENCE AND TECHNOLOGY**

Faculty Name: *Computing and Informatics*

Name of Department: *Computer Science*

COURSE OUTLINE FOR RESEARCH METHODOLOGY(RIT812S)

STATEMENT ABOUT ACADEMIC HONESTY AND INTEGRITY

All staff and students of the Namibia University of Science and Technology (NUST), upon signing their employment contracts and registration forms, commit themselves to abide by the policies and rules of the institution. The core activity of NUST is learning and in this respect, academic honesty and integrity are very important to ensure that learning is valid, reliable, and credible.

NUST, therefore, does not condone any form of academic dishonesty, including plagiarism and cheating on tests and assessments, amongst other such practices. NUST requires students to always do their own assignments and to produce their own academic work, unless given a group assignment.

Academic Dishonesty includes, but is not limited to:

- Using the ideas, words, works or inventions of someone else as if it is your own work.
- Using the direct words of someone else without quotation marks, even if it is referenced.
- Copying from writings (books, articles, webpages, other students' assignments, etc.), published or unpublished, without referencing.
- Syndication of a piece of work, all or part of an assignment, by a group of students, unless the assignment was a legitimate group assignment.
- The borrowing and use of another person's assignment, with or without their knowledge or permission.
- Infringing copyright, including documents, copied or cut and pasted from the internet.
- Asking someone else to prepare an assignment for you or to write or sit an assessment for you, whether this is against payment or not.
- Re-submitting work done already for another course or programme as new work, so-called self-plagiarism.
- Bringing notes into an examination or test venue, regardless of whether the notes were used to copy or not.
- Receiving any outside assistance in any form or shape during an examination or test.

All forms of academic dishonesty are viewed as misconduct under NUST Student Rules and Regulations. Students who make themselves guilty of academic dishonesty will be brought before a Disciplinary Committee and may be suspended from studying for a certain time or may be expelled. All students who are found guilty of academic dishonesty shall have an appropriate endorsement on their academic record, which will never be erased.

Campus activities that involve physical contact, whether in a meeting, laboratory, assessment, tutorial/lecture will be held under strict COVID-19 National Health and Safety Protocols. Students not adhering to such National Regulations, i.e. wearing masks correctly to cover both the mouth and nose, social distancing of at least 1.5m, hand sanitising and refraining from campus activities when experiencing COVID-19 symptoms, will not be allowed into a venue.

Course Information

Course Code and Title:	RIT812S, Research Methodology
Department(s):	Computer Science, Informatics, Cyber security and Software Engineering
Programmes:	Bachelor of Computer Science Honours (Communications Networks) Bachelor of Computer Science Honours (Software Development) Bachelor of Computer Science Honours (Information Security) Bachelor of Computer Science Honours (Digital Forensics) Bachelor of Informatics Honours (Web Informatics) Bachelor of Informatics Honours (Business Informatics)
Notational Hours:	150 hours (Contact: 45 hours; Directed self-learning and Self-Directed learning: 65 hours; Assessment: 40 hours)
NQF Level and Credit:	NQF level 8, 15 credits
Semester Offered:	1
Pre-Requisites:	None
Course Equivalencies:	Research Methodology (RIT811S)
Course Objectives:	The course aims to equip students with essential research skills in order to plan and execute a scientific or technical research project, focusing on applied research in Computing and Informatics areas of her/his own interest. Students will get the needed conceptual, writing, and design skills to develop and present a comprehensive research project proposal.

Course Delivery Methods:

The course will be facilitated through the following learning activities:

- One and a half hours, twice per week, of ONLINE contact with students.
- Introduction of course topics and progress reports assignments via Zoom or Microsoft Teams.
- Online lectures to be delivered through Microsoft Teams
- Lecturers and students to interact through online platforms: Emails, Microsoft Teams and WhatsApp.

Lectures: The theoretical foundations will be presented through online platform(Ms Teams). Students are encouraged to embrace online learning and partake in online teaching that will be taking place on **Tuesday**

and Thursday at the time as allocated on the courses timetable. The students are expected to deepen their knowledge and understanding of the covered topics through guided self-study.

Course format:

Contact: 45 hours (Online discussions between students and lecturer(s))

Directed self-learning and Self-Directed learning: 65 hours (out-of-class study time, reading prescribed material, organisational overhead, organising and conducting team meetings, online consultations with a lecturer and fellow students, collecting prescribed material, etc)

Assessment: 40 hours (All online: Proposal Presentation, Final Thesis Submission)

Course Resources:

There are several facilities that students can use as a course resource at NUST and at home.

- a) **NUST level:** Prospectus; The Library; student services especially the Writing Centre; Social Worker; Centre for Teaching and Learning; Internet is available the whole day for research.
- b) **Course Level:** Course Outline; Electronic learning resources on E-Learning (Moodle) materials will be developed.

Quality Assurance: Moderation of assessments will be done in accordance with NUST's general rules and guidelines on moderation.

Student Support and Learning Resources:

Institutional support is provided through the NUST Library, the Department of the Students Services, and the Student Representative Council. Faculty support is provided through new-student orientation activities and the FCI student representatives. Departmental and course-based support is provided through supervised practicals, customized course material and through interactions with lecturers, Head of Department, and the Deputy Head of Department.

Effective Start Date: Monday, 12 February 2025

Course Coordinating Team Information:

Full Name: Prof Jude Osakwe
Email: josakwe@nust.na
Office phone: 0612072952
Office location: Polyheights 406

Office hours:

08H00 am – 16H30 pm (consultations via emails)

Consultation hours: All lecturers (involved in the teachings) are available for consultation with students.

Consultation times can be found on the doors of lecturers' offices. You are welcome to make an appointment with the lecturer via email.

Lecturers Information:

(if a specific topic was not clear, please contact the respective lecturer for the topic via email)

Table 1: List of Lecturers who shall facilitate sessions for this course via online- Semester 1, 2024.

#	Full Name(s)	E-mail Address	Office Phone #	Office Location
1.	Prof Dharm SINGH	dsingh@nust.na	207 2991	PolyHeights, 2 nd Floor, Office 202
2.	Prof Heike WINSCHIERS-THEOPHILUS	hwinschiers@nust.na	207 2168	FCI House
3.	Prof Guy Zodi LUSILAO	gzodi@nust.na	207 2774	FCI House
4.	Dr Suama HAMUNYELA	slhamunyela@nust.na	207 2922	FCI House
5.	Prof Jose QUENUM	jqenum@nust.na	207 2235	Science & Tech Building
6.	Prof Fungai BHUNU SHAVA	fbshava@nust.na	207 2510	FCI House
7.	Prof Samuel AKINSOLA	sakinsola@nust.na	207 2704	PolyHeights, 4 th Floor, Office 401
8.	Prof Jude OSAKWE	josakwe@nust.na	207 2952	PolyHeights, 4 th Floor, Office 401
9.	Prof Gloria E. IYAWA	giyawa@nust.na	207 2760	FCI House
10.	Dr Irja SHAANIKA	ishaanika@nust.na	207 2620	FCI House
11.	Dr Colin STANLEY	cstanley@nust.na	207 2752	FCI House
12.	Dr Maravanyika MUNYARADZI	mmaravanyika@nust.na	207 2263	FCI House
13.	Dr Mercy CHITAURO	mchitauro@nust.na	207 2039	FCI House
14.	Dr Ambrose AZETA	aazeta@nust.na	207 2673	FCI House
15.	Dr Richard MALIWATU	rmaliwatu@nust.na	207 2206	FCI Housse

Possible Supervisors:

Possible supervisors and their areas of interests are listed below:

NOTE: Each supervisor on average can supervise only 4 students, on a first come first serve basis, based on their area of interests and specialisation.

Table 2 List of supervisors and their areas of interests

Full Name(s)	E-Mail/TEL	Area of Specialization/Research Interest
Prof Dharm SINGH JAT	dsingh@nust.na +264 61 207 2991	Multimedia communications, wireless technologies, mobile communication systems, and video transmission over wired/wireless networks, Network security, Internet of things and Smart and Secure Environment
Prof Heike WINSCHIERS-THEOPHILUS	hwinschiers@nust.na +264 61 207 2168	Participatory and Interaction Design, HCI, intangible Digital Cultural Heritage, and Indigenous Knowledge Preservation, Youth and rural development (Community-based co-design with indigenous and marginalised communities Cluster)
Prof Guy-Alain ZODI LUSILAO	gzodi@nust.na +264 61 207 2774	Multimedia Wireless Networks, Cloud Networking, Performance Analysis in wired and wireless Networks, Traffic Engineering, Wireless Sensor Networks, Network Security, Fog Networking and Internet of Things
Prof Jose QUENUM	jquenum@nust.na +264 61 207 2235	Big Data Infrastructure, Process Coordination/Messaging, Programming Languages and Concurrency Control, Virtual Machine Monitor for Big Data Infrastructure, Data Structures for Video Indexing
Prof Fungai BHUNU SHAVA	fbshava@nust.na +264 61 207 2510	Data communication/ computer networking security/ User experience in Security/ Human Computer Interaction Security and Cyber Security
Prof Suama L. HAMUNYELA	slhamunyela@nust.na +264 61 207 2922	E-governance/ICT4D, E-health and digital health, process management
Prof Jude OSAKWE	josakwe@nust.na +264 61 207 2952	ICT in Education, Business Information Systems, EGovernment, Data Governance/Data Analytics.
Prof Samuel AKINSOLA	sakinsola@nust.na +264 61 207 2704	Business Information Systems, Knowledge Support/Knowledge Management, Cluster: Health Informatics

Prof Gloria IYAWA	giyawa@nust.na +264 61 207 2760	Digital Health, Health Informatics and E-Government.
Dr Irja SHAANIKA	ishaanika@nust.na +264 61 207 2620	Area of Healthcare, Enterprise systems, Business Processes, Enterprise Architecture
Dr Munyaradzi MARAVANYIKA	mmaravanyika@nust.na +264 61 207 2263	ICT in education, environmental and wildlife management, intelligent systems, AI systems
Dr Mercy CHITAURO	mchitauro@nust.na +264 61 207 2039	Critical Infrastructure security, Network Security, IT law
Dr Atlee GAMUNDANI	agamundani@nust.na +264 61 207 2099	Internet of Things (IoT) Security, Cloud Computing Security, Smart and Secure environments, Data Security.
Dr Ambrose AZETA	aazeta@nust.na +264 61 207 2673	Software Engineering
Dr Richard MALIWATU	rmaliwatu@nust.na +264 61 207 2206	ICT service delivery to rural and underserved communities, Data communication/computer networking, data processing and analytics, Data mining, Machine Learning.
Mr Sebastian Mukumbira	smukumbira@nust.na +264 61 207 2895	Participatory and Interaction Design, Indigenous Knowledge Preservation, Rangeland Monitoring (Community-based codesign with pastoralists), Spatial Data Infrastructures
Mr Isaac Nyamu	inhamu@nust.na +264 61 207 2074	Digital Forensics, Database Security and Data Protection, Cloud Computing (Security), Cyber Security, Software Development with an emphasis on security, Systems Administration
Mr Admire KACHEPA	akachepa@nust.na +264 61 207 2476	Business processes their analysis and management; Implementation of ERPs in organisations; Implementation of Information Systems
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Mr Julius SILAA	jsilaa@nust.na +264 61 207 2037	IT Security, Bitcoins
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Ms Vicky SHAKELA	vshakela@nust.na +264 61 207 2856	Cyber Security and Communication Networks
Ms Shilumbe CHIVUNO-KURIA	schivuno@nust.na +264 61 207 2057	Human Computer Interactions, Co-designing technological services with unemployed youth.
Mr. Shadreck CHITAURO	schitauro@nust.na +264 61 207 2689	Cloud Computing, Network Security and E-Learning Systems Implementations.

It is the student responsibility to ensure his/her progress reports are signed by the supervisor and submitted ONTIME.. All submissions (progress reports, presentations, proposal documents) should be submitted online and not through the lecturer emails. LATE SUBMISSIONS WILL NOT BE ACCEPTED. Students are encouraged to be in touch with their supervisors, discuss with their supervisors content, topics and informations shared in class. Communicate to your supervisors your submission due dates on time and not the week the specific activity is due. Please note your supervisors are dealing with other matters in the Department hence its your responsibility to keep them up to date with your progress. ALL SUBMISSION DUE DATES ARE STATED IN THE COURSE OUTLINE

Student Readiness:

Technology & Equipment Readiness:

This course will only utilise already existing NUST equipment (on top of the Class Group WhatsApp).

Student Commitments and Contact Times:

- a) **Class Attendance:** Attend all online-classes, and be on time, and make appropriate notes. If a student fails to attend an online class, it is their responsibility to catch-up on what was covered. Students have access to the recommended books from the library. If any special presentations are used in class, it will be made available (text, audio, video) and do not need to be copied by the student.
- b) **Self-Directed Study:** After every class, and before the start of the next class, sort, complete, and annotate own lecture notes, and select further reading in case any particular item was not understood on the first attempt. Students must spend their notational hours revising, practicing or/and completing homework for the course. Students should take the reading list (prescribed and recommended reading seriously. Read all prescribed material to the extent that the student could write a one-page essay, summarising the content, without looking at the material again.

- c) **Communications:** Maintain active communication with your group lecturer and respective supervisors. Students are expected to use established channels of communication and platform (EMAILS, WHATSAPP) to solve course-related issues on time. The end of the semester is not the best time to fix issues. Use the available time to deal with matters at hand.

Recommended Reading:

Grønmo, S. (2019). *Social research methods: Qualitative, quantitative and mixed methods approaches*. Sage.

Gravetter, F. J., & Forzano, L. A. B. (2018). *Research methods for the behavioral sciences*. Cengage Learning.

Bell, E., Bryman, A., & Harley, B. (2018). *Business research methods*. Oxford university press.

Hansen, A., & Machin, D. (2018). *Media and communication research methods*. Macmillan International Higher Education.

Student Learning:

Specific Learning Outcomes:

After completing the course students will, through a written research proposal, show evidence of their ability to:

- Evaluate and apply the types, principles and procedures of research in Computing and Informatics disciplines.
- Select appropriate methods for system requirements, data gathering and analysis of sets of data in research;
- Synthesise, interpret and integrate the results and draw conclusions from data;
- Apply the basics of publishing and adhere to recommended referencing styles, e.g. APA guidelines;
- Communicate research process planning effectively.

Comprehensive Learning Outcome:

Produce and present a comprehensive research proposal.

Course Schedule:

Semester Week	Specific Date	Topic	Activity/Deliverables	
1	11-13 February	Setting Research Methodology Class Tone	• Course outline • Expectations • Process of supervision • Proposal template, • Process around proposal & • Deadlines Research Ethics	
2	18-20 February		• Ethical Considerations • University Research Policy • Paraphrasing /Plagiarism/Turnitin tool	
3	25 February 2025	Types of Scientific Research	• What is research • Types of Research	
4	27 February 2025	Cluster Presentations (I)	Data-intensive Systems (DANS): • Dependable (and secure) Advanced Network Systems • Infrastructure & Analytics (BDIA)(Big Data)	
		Cluster Presentations (II)	Smart & Secure Environment (SSE) • Smart Communication Systems • Digital Forensics and Information Security • Dependable and Secure Advanced Network Systems(DSANS)	

		Cluster Presentations (III)	Digital Transformation • E-Participation • Business and Policy Analytics(BPA) • Marginalised Community Development (MCD) • Journalism in the Digital Era(JDE)	
		Cluster Presentations (IV)	Indigenous Knowledge Management Systems(IKMS) • Indigenous Knowledge Management Systems- • Marginalised Community Development	
4	27 February 2024	Research Problem (I)	Formulation of: • Problem Statement • Topic • Aim • Research objectives • Research Questions & • Hypothesis	
5	04 March 2025	Research Problem (II)	Formulation of: • Problem Statement • Topic • Aim • Research objectives • Research Questions & Hypothesis	
6	6 March 2024	Data sources and APA Referencing (I)	APA Citations and referencing Searching for Information, credible information source, access to various NUST Library databases/portals (ACM, IEEE, etc.).	

7	11 March 2025	Data sources and APA Referencing (II)	Referencing tools • Medley Referencing manager • Zetero	
8	13 March 2025	Proposal Introduction Writing (I)	Introduction • What is proposal Introduction • How to write your proposal Introduction	
9	18 March 2025	Proposal Introduction Writing (II)	Introduction • What is proposal Introduction • How to write your proposal Introduction	

10	25 March 2025	Writing Literature review (I)	Literature Review • What is literature Review? • Why Literature Review? • How to write your study literature?	
11	27 March 2025	Writing Literature review (II)	• Writing Literature review /Organising your literature review • How to write a Literature Review • And questions on literature review	
	31 March – 04 April Institutional Holiday			
12	08 April 2025	Methodology (I)	• Qualitative Approach (Inductive) • Quantitative Approach (Deductive) • Mixed Approach (Abductive Approach)	
13	11 April 2025	Methodology (II)	Methodology SW Development (Action Research, Design Research, Waterfall, Agile, Prototyping, Spiral, XP, etc)	
	15 April 2025	Methodology (II)	Research Designs (Approach to planning) • Case study • Experimental • Theoretical	
14	17 April 2025	Methodology (IV)	Systems and Communication Networks • Modelling and Simulation Approach • Simulation Tools	

	Friday 18 April Until Monday 21 April 2024 Easter Weekend			
15	22 April 2025	Methodology (V)	Data Collection • Before data collection • Participants and stakeholders privacy • Data collection techniques’ and tools • Data Analysis	
16	24 April 2025	write up (I)	• Delineations and limitations • Assumptions • Risk/ feasibility analysis and ethical considerations • Chapter Outlines	
17	29 April 2024	Analysis and write up (II)	Developing your study models/frameworks	
18	1 st May 2024	Monday, Workers Day Observed, 1 st May		
19	04 May 2024	Cassinga Day , 4 th May		
20	06 May	• Evaluating your models/frameworks • Discussing & Recommendation Conclusions		
21	Graduation Ceremony 09 May			
22	13 May 2024	Consultation on Proposals Ethical	Questions/Answering Session Clearance Application Documents	
23	09 May 2024	Proposal Presentation Etiquettes	• Presentation outlines • Presentation checklist/Structure • How to Present your presentation	
23 May 2025 -- Deadline for submissions of Proposal Presentations (ppt)				
04- 06 June 2025 -- All RIT Proposal Presentations				
10 June 2025 - All RIT Final Proposal Submission				

Important Dates:

NOTE: The following dates are subject to change based on the needs of the students at the lecturer's prerogative. Students will be notified ahead of time of any changes.

Exact Date to be announced

Proposal Presentations Exact Date to be announced

Final Proposal Submission Assessment and Evaluation:

Continuous Assessment will apply, consisting of:

- Completion of milestones(Progress reports).
- Research proposal presentation.
- Final research proposal on a topic chosen by the student and approved by the supervisor or a senior academic.

Assessments	Weight
Proposal Online Presentations (Evaluated by: Supervisor & Assessor)	40
Final Written Proposal (Evaluated by : Supervisor & Moderators)	60
Total:	100%

Minimum pass required for this course: 50%

Course Policies:**General Academic Policies:**

It is the student's responsibility to be familiar with and adhere to the NUST's Policies. These Policies can be found in the NUST Prospectus or online at www.polytechnic.edu.na/prospectus.

Submission of reports, presentations and final proposal:

All Late Submissions will not be accepted after the due date and time and thus marked as 0%.

Failure to Pay Fees:

A student who fails to pay his/her fees may not be allowed to write the examination and if allowed, the results will be withheld until all outstanding fees are paid in full.

Important Student Services at NUST

There are a variety of services that you can use at NUST. These services are to your advantage – Use them!!! They include the following:

- Student Counselling and Career Development - Dean of Students Office.
- Writing centre and student academic problems – Centre for Teaching & Learning (CTL)
- Campus Health and Wellness Centre (CHWC) - Dean of Student's office/ NUST Clinic

Authorisation:

This course is authorised for use by:

Head of Department

Date

ACKNOWLEDGEMENT BY STUDENT

(To be completed by all students on the course, detached from the course outline and kept on record in the Department)

I, _____, student Number _____, hereby acknowledge that I have received the course outline for Research Methodology (RIT812S) in semester 1 (2024), and that I have familiarised myself with its content, in particular the statement about academic honesty and integrity. I agree to abide by the Policies and arrangements spelt out in this course outline.

Signature of student

Date

Setting a research question, aim and objective

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Author guidelines
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Abstract

Aim To describe the development of a research question, aim and objective.

Background The first steps of any study are developing the research question, aim and objective. Subsequent steps develop from these and they govern the researchers' choice of population, setting, data to be collected and time period for the study. Clear, succinctly posed research questions, aims and objectives are essential if studies are to be successful.

Discussion Researchers developing their research questions, aims and objectives generally experience difficulties. They are often overwhelmed trying to convert what they see as a relevant issue from practice

into research. This necessitates engaging with the relevant published literature and knowledgeable people.

Conclusion This paper identifies the issues to be considered when developing a research question, aim and objective. Understanding these considerations will enable researchers to effectively present their research question, aim and objective.

Implications for practice To conduct successful studies, researchers should develop clear research questions, aims and objectives.

Keywords research question, research aim, research objective, study development, nursing research, novice researchers

Introduction

DEFINING THE research question is a particularly significant step in research as it narrows the research aim and objective down to specific areas the study will address (Creswell 2014, Johnson and Christensen 2014). Research questions are vital as they guide the choice of methodology, methods, sample, sample size, data collection instrument and data analysis techniques (Lipowski 2008).

Selecting and constructing a good research question is one of any study's most difficult aspects (Blaikie 2007, 2009, Aslam and Emmanuel 2010). Despite this, there is little direction on creating good research questions (Lipowski 2008). They do not appear spontaneously and the ability to pose good questions is a skill that can be refined and developed with direction and collaboration from colleagues and supervisors (Thabane *et al* 2009). Importantly, research questions are not static: they change, adapt or are refined when

researchers work through the different steps of the design cycle, reviewing the literature, integrating theory and developing a conceptual framework (Hennink *et al* 2011).

Novice researchers are often overwhelmed and bewildered when trying to convert what they see as relevant issues from practice into a study. This difficulty arises as they progress their work and they feel they know what they need to answer.

Only by consulting a supervisor and/or other knowledgeable people about the research and critical evaluation of the relevant published literature can researchers develop research questions, aims and objectives. Developing a research question is an interactive and inductive endeavour that takes place over time.

A clear research question is vital and sets the scene for development of a clear research aim and objective that are inextricably linked (Martindale and Taylor 2014).

Table 1 Types of quantitative research question

Type	Question
Descriptive	- Seeks to quantify responses to one or more variables - Often begins with 'What is...?' or 'What are...?'
Comparative	- Seeks to compare two or more groups for some outcome variable - Often uses words such as 'compare' - When it involves two groups it can be written as: 'What is the difference in (dependent variable) between (group 1) and (group 2)?' - Can be extended to three or more groups by replacing the word 'between' with 'among' - Can be causal in nature, such as the effect of X on Y - Such causal questions are implicitly comparative in nature when a comparison is made between a group where X is involved and a group where X is not involved
Relationship	- Is concerned with trends between or among two or more variables - Often uses words such as 'relate', 'relationship', 'association' and 'trend' - Relationship questions involving two variables usually can be written using: 'What is the relationship between (independent variable) and (dependent variable) among (population)?' - Can be extended to three or more variables by replacing the word 'between' with 'among'

(Onwuegbuzie and Leech 2006)

Researchers should:

- Develop a research problem that matches your interests, background, training and publications.
- Base research on current evidence.
- The research question should logically present each step in a progression from what is known to important issues that are not.
- Do not try to be 'research trendy'. Do not follow a trend or issue with little evidence from the literature or little idea of how it will add to the knowledge base.
- Avoid topics. Look at clearly defined research problems instead.
- Stay focused. You will find many things of interest along the path, but you should ask yourself: 'Is this related to what I want to uncover or just a point of interest?'

Research questions

Research questions can be developed from theoretical knowledge, previous research or experience, or a practical need at work (Parahoo 2014). They have numerous roles, such as identifying the importance of the research, providing a framework, direction and cohesion,

and setting the limits by revealing the boundaries of the study and highlighting the type of data to be collected (White 2009).

Quantitative research questions These tend to be precise and can be categorised as 'descriptive', 'comparative' or 'relationship' (Table 1). Words such as 'do', 'does', 'is' or 'are' should be avoided as they invite 'yes'/'no' responses.

Appropriate questions should highlight the population, dependent variables and design (historical, descriptive, correlational, experimental, causal-comparative, quasi-experimental), and the link between the research question and the design (Onwuegbuzie and Leech 2006, Kloda and Bartlett 2013). In comparative/relationship research questions, independent variables should also be highlighted.

Qualitative research questions are flexible, adaptable and non-directional (Creswell 2013). They seek to determine or discover a process, or define experiences. They are normally used to try to understand particular educational, familial or social processes or experiences that happen in a specific location and/or context (Marshall and Rossman 2011). They usually describe and address 'what' and 'how' questions, avoiding words such as 'affect', 'influence', 'compare' and 'relate'.

Qualitative research questions These can represent broad or central areas of research or specific areas or sub-categories (Creswell 2013). Sub-questions can address important concerns and difficulties to be resolved or describe the context. Like quantitative questions, qualitative questions link to research design, including historical/narrative, case study, ethnography, phenomenology, grounded theory and autoethnography.

Generally, these questions are non-directional and use words that state that the study will: 'discover' (grounded theory), 'explain' or 'seek to understand' (ethnography), 'explore a process' (case study) or 'describe the experiences' (phenomenology).

Researchers can often find developing qualitative questions difficult (Agee 2009). To help researchers think about the different types and purposes of their studies, constructs for question types are presented in Table 2. Comparative questions are often overlooked in qualitative research. Onwuegbuzie and Leech (2007) described four measures that compare:

1. Participants pairwise (pairwise sampling designs).
2. Two or more subgroups (subgroup sampling designs).

3. Two or more members of the same subgroup, wherein one or more members represents a sub-sample of the full sample (nested sampling designs).
4. Two or more subgroups that are extracted from different levels of a study (multilevel sampling designs).

Mixed-methods studies Researchers conducting these studies can choose to develop separate quantitative questions and qualitative questions or develop a mixed methods question depending on the study (Creswell 2009). Separate quantitative and qualitative questions are appropriate when the approaches, not the mixed methods or integrative component of the study, are the focus (Tashakkori and Teddlie 2010). This identifies the significance of the qualitative and quantitative phases of the study and their collective power (Creswell and Plano Clarke 2011).

The development of a mixed methods question should reflect the procedures or the content and not include separate quantitative and qualitative questions (Creswell 2009, Tashakkori and Teddlie 2010). This suggests that there is some integration or linking between the study's quantitative and qualitative phases (Tashakkori and Teddlie 2010, Creswell and Plano Clark 2011).

Overall, good research questions are the basis of good research as they enable researchers to identify what they want to know and to work from knowing less towards knowing more (Payne and Payne 2004). Identifying what you want to know more about is essential. Researchers can be vague at the beginning of a study about what they want to know and unclear questions can lead to an unfocused project (White 2009). Researchers should aim to write clear, articulated questions that enable them to find the answers in a focused, clear way.

Criteria One difficulty in writing good research questions is that there are potentially an unlimited number of them. Without a clear, focused research question, it is difficult to know how or what to research (Grove *et al* 2013). While researchers may appreciate that if they do not know what questions they are asking, there is little hope of finding any answers, this may only add stress to an already tense situation.

Common problems encountered when developing a research question include: deciding which area to research from a range of issues that are of interest; knowing which area to focus on with no precise topic; and knowing the area and topic but finding it problematic to clearly communicate the question.

The authors suggest that once the research topic has been identified, the researcher should begin framing the research question by listing all the questions that could be researched. This may be a large or small list with questions interconnected or very different. Forming the right questions should be seen as an iterative process that occurs through continually engaging in reading and refining ones ideas at all stages (Maxwell 2013).

Hulley *et al*'s (2007) criteria for quantitative research are:

- Feasible.
- Interesting.
- Novel .
- Ethical.
- Relevant.

These criteria are collectively known as FINER.

Cooke *et al*'s (2012) characteristics of qualitative research are:

- Sample
- Phenomena of interest
- Design
- Evaluation
- Research type

These characteristics are collectively known as SPIDER. Researchers informed by these criteria and characteristics are likely to identify important aspects of research questions and develop successful research projects (Farrugia *et al* 2010). However, while researchers can refer to these criteria in

Table 2 Types of qualitative research question

Type	What it does
Contextual	Describes the form or nature of what exists
Explanatory	Examines the reasons for or associations between what exists
Evaluative	Appraises the effectiveness of what exists
Generative	Provides new ideas, aiding the development of theories, strategies or actions
Ideological	Advances the ideology of a position
Exploratory	Investigates a phenomenon where little is understood
Explanatory	Explains a phenomenon
Descriptive	Tries to describe a phenomenon
Emancipatory	Engages in social action related to a phenomenon

(Ritchie *et al* 2014, Marshall and Rossman 2011)

identifying important general aspects of a research question, a more specific format should be considered when developing specific research questions (Farrugia *et al* 2010).

Richardson *et al*'s (1995) format for quantitative research is:

- Population, patient, problem.
- Intervention.
- Comparison.
- Outcome.

This format is known as PICO.

Kahn *et al*'s (2003) format for qualitative research is:

- Population, problem.
- Exposure.
- Outcome or theme.

This format is known as PEO.

These methods have been used in evidence-based practice to frame and answer clinical questions (Huang *et al* 2006), develop strategies for searching literature (Schardt *et al* 2007), and frame research questions (Aslam and Emmanuel 2010). Once a research question has been identified, the research aim and objectives are developed. Developing a research question can be difficult, as can stating the research aim and objectives (Blaikie 2009).

Aims The researcher must be cognisant that the research's question, aim and objectives are inextricably linked. To determine the study's aim, the researcher must decide what its overall,

long-term aim is and describe the overall purpose (what should be accomplished) in general terms. This ultimately enables the reader to judge whether the researcher has achieved that purpose (Tully 2014). In this context, the Newman *et al* (2003) framework can be used to identify nine types of aim. These are to:

- Predict.
- Add to the knowledge base.
- Have a personal, social, institutional, and/or organisational impact.
- Measure change.
- Understand complex phenomena.
- Test new ideas.
- Generate new ideas.
- Inform constituencies.
- Examine the past.

Objectives Determining the research's aim leads naturally to determining its objectives. Research objectives are more specific than the aim and relate directly to the research question (Grove *et al* 2014, Parahoo 2014). They may be divided into 'primary' (bound to be achieved) and 'secondary' (incidental) objectives (Newell and Burnard 2011), and should be: closely related to the research question, cover all aspects of the problem, specific, ordered in a logical sequence, achievable, take into consideration the available resources, including time, and mutually exclusive of each other. They should also be stated using action verbs that can be evaluated,

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such as 'to describe', 'to identify', 'to measure' or 'to compare'.

Johnson and Christensen (2014) offer five typical research objectives: 'exploration', 'description', 'explanation', 'prediction' and 'influence'. Qualitative and quantitative research studies can be linked to one or more of these five research objectives (Onwuegbuzie and Leech 2006):

- 'Exploration' involves using mainly inductive methods to discover a concept, construct, phenomenon or situation and advance understanding, hypotheses or generalisations.
 - 'Description' involves identifying and describing the antecedents, nature and aetiology of a phenomenon.
 - 'Explanation' involves developing theory for the purpose of explaining the relationships among concepts or phenomena and determining reasons for the existence of events.
 - 'Prediction' refers to using pre-existing knowledge or theory to predict what will occur at a later point in time.
 - 'Influence' relates to manipulation of the setting or variable to produce an anticipated outcome.
- Doran's (1981) criteria can be applied when setting research objectives. These are:
- Specific: be exact about what you are going to accomplish.
 - Measurable: quantify the objectives.
 - Appropriate: align with the needs of the target audience.

- Realistic: do you have the resources to make the objective happen?
- Time specific: state when you will achieve the objective.

These criteria are collectively known as SMART.

It should be acknowledged that there is some debate as to whether research aims and objectives are required for all studies, and whether either or both are necessary. This is evident in published studies where only the research aim or the objectives are identified. This may be because only one was set or only one was identified in the published work.

Conclusion

Developing a good research question, aim and objective is a vital aspect of research. This can be challenging. It is essential that you take the time to do this systematically as the eventual success of the study relies on asking a suitable and clear question. The question must be grounded in research, coherent, clear and ask precisely what you want to find out. Knowing this will help you plan the project and identify the methodology, sample, data collection and data analysis necessary.

Developing the research question, aim and objective is an interactive and inductive process that takes place over time, through consultations with a supervisor, knowledgeable people, published literature, and use of the appropriate criteria (FINER and SPIDER) and format (PICO and PEO).

Online archive

For related information, visit our online archive and search using the keywords

Conflict of interest
None declared

▶ **Kloda LA, Bartlett JC** (2013) Formulating answerable questions: question negotiation in evidence-based practice. *Journal of the Canadian Health Libraries Association*. 34, 2, 55-60.

▶ **Lipowski EE** (2008) Developing great research questions. *American Journal of Health-System Pharmacy*. 65, 17, 1667-1670.

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Difficulties Faced by Jordanian Undergraduate EFL Students in Writing Research Problem Statement

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Abstract: The current study aims at investigating the difficulties encountering Jordanian EFL students in the writing problem statement section. The study grouped those difficulties into two main categories, namely academic skills difficulties and language skills difficulties. It has been noticed that undergraduate as well as postgraduate students, lack the required skills enabling them to conduct a well-constructed research article, which sparkles the idea of this study. In addition, no previous study has examined the challenges encountered by Jordanian EFL students when conducting a research paper in general and the problem statement section in particular. For the purpose of collecting the required data of the present study, twenty Jordanian EFL students had a teaching program about research writing skills. They were introduced thoroughly to all sections of the research article, with special emphasis on the research problem, section which is the main concern of this study following that the participants were given a month period to accomplish their research papers before submission. The results of the study reveal that the participants lack the academic skills enabling them to conduct a well-constructed problem statement section. More specifically, the vast majority of the participants fail to provide an overview of their topics and to identify their research gap within the current literature. The study also shows that the participants face serious grammatical errors when writing research problem section. It could be concluded that conducting a scholarly research article in general and problem statement, in particular, is a demanding issue and need to be further highlighted by researchers and academicians to identify the challenges facing learners.

Keywords: *EFL students, problem statement, research skills, language skills*

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INTRODUCTION

Research plays a substantial role in human's life as all advancements made by man during the past period of time are built primarily on research, whether theoretical or empirical. However, the process of conducting a scholarly research article is always challenging and demanding for postgraduate students and even for their professors because it requires implementing several processes simultaneously such as, preparing research instruments, finding the appropriate sample, data collection, data analysis and finally, the actual process of writing (Cuevas et al., 2016; Klein & Boscolo, 2016). In addition to referring to suitable resources, identifying a research problem, finding the research gap and fitting it into literature, analyzing the results of the study, explaining and justifying the research results. Implementing a research paper is the toughest experience the students may have to get a university degree as it consumes lots of time and requires great effort. This is in line with Alsied and Ibrahim (2017) who state that conducting a research is not a simple task because it needs hard work, insight into the subject under investigation, desire to carry out research and motivation.

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The difficulties encountering researchers in conducting a scholarly research paper are explained by [Phothongsunan \(2016\)](#), who subcategorized them into three main types, namely 1- discursive difficulties, 2- non-discursive difficulties and 3- other difficulties. Discursive difficulties are those that have to do with the language, including a poor command of the target language, misuse of vocabulary, violating rules of grammar, bad referencing and citation. This problem is reported by [Okamura \(2006\)](#) and [Fahy \(2008\)](#), who contend that language challenges in research writing is manifested by poor vocabulary, mother tongue interference, the problem of writing coherent paragraph and establishing well-constructed introduction, body and conclusion. According to [Al Fadda \(2012\)](#), the vast majority of challenges that non-native speakers of English encounter, especially Chinese, when writing in English are procedural challenges related to rhetorical patterning and genres, textual organization and writing the literature review.

Non-discursive difficulties refer to elusive factors comprising plagiarism, motivation, emotional and psychological factors. [Canagarajah \(2002\)](#) and [Salager \(2008\)](#) report some more non-discursive factors such as lack of resources, tight budget, weak internet connection, undependable mail service and lack of support from colleagues and supervisors. This is consistent with [Alsied and Ibrahim \(2017\)](#), who report that non-discursive challenges for Libyan researchers encompass lack of resources in the library, lack of training in research, the need for support from supervisors, and poor internet connection. Other obstacles of research writing are related to publication and the peer-review process. Despite the aforementioned challenges, carrying out a research article is indispensable and considered a privilege for the author ([Lonka et al., 2014](#)). In addition, implementing research papers open the door for the exchange of ideas and sharing the results and conclusion of research based on foundational knowledge in specific field.

Many attempts have been made to reach a working definition of research. According to [Pandey and Mishra \(2015\)](#), research is a two-word combination of "re" and "search" which indicates searching more and more. [Krause \(2020\)](#) provides a working definition of research writing as the process of utilizing evidence from journals, books, magazines, the internet and specialists to convince the readers about a specific subject. Research is also defined as "a process of enquiry and investigation; it is systematic, methodical and ethical; research can help solve practical problems and increase knowledge." ([Neville, 2007](#)). Research is reported as a manner of thinking and exploring several fields of knowledge in order to understand the specific issue and develop and test theories with the aim of making progress and advancement in the profession ([Kumar, 2016](#)). As stated by [Bocar \(2013\)](#), research is the process of data collection, analysis and interpretation and testing the applicability of the findings.

The process of writing a research paper is supposed to be systematic, well-structured and conform with the steps of research writing. According to Hourglass Model, as cited by [Swales \(1993\)](#), a research article is comprised of several components, including the title, abstract, introduction, body, discussion and references. Each section of the research paper is further divided into sub-sections. The introduction starts out with a broad discussion of the topic and narrows down to the specific topic of research. The introduction is composed of three sections.

1. Establish a territory: This section aims at providing the reader with a general background of the topic under investigation and view current studies on the topic.
2. Establish a niche: This section aims at showing the research problem and fitting the research gap within the current literature. The section also reveals research questions and assumptions.
3. Occupy the niche: This section presents the thesis statement of the research and outlines the structure of the paper ([Swales, 1993](#)).

The body section of a research paper describes the methods that have been utilized to answer the research questions and specify the problem mentioned in the introduction. This section includes parts such as the population of the study, the sample of the study and research instruments. The body helps answer questions such as how was the research questions addressed? And what was found ([Peat, Elliott, Baur, & Keena, 2002](#))? On the other hand, the discussion section generally presents background information about the topic, recaps the results of the study with special emphasis on explaining and justifying the results. This section also compares the results of the current study with previous literature and ends with a conclusion drawn from the results of the study.

In writing research, one of the most important parts that constitute a great troublesome for EFL in general is defining the research problem. More specifically, some researchers find it difficult to determine the exact problem to be addressed in the research, making him/her fail to fulfil the objectives of the study. [Ifeoma \(2019\)](#) argues that both undergraduate and postgraduate students encounter difficulty in identifying the problem of research. The problem

of the research is defined as the subject that the researcher intends to examine and explore, whether descriptively or experimentally. It is the focal and core point of the research, or in other words, it is the issue or the challenge that the researcher struggles with within the research article (Boudah, 2011).

It is crucial that the researcher starts out with his personal experience when determining the research problem to be addressed. Strictly speaking, the researcher is supposed to research topics that he has a genuine interest in because if he is not interested, it will be hard to complete the research. Another important point that needs to be considered when writing a research problem is narrowing down the emphasis of the research. A broad problem is featured as unclear and vague so that narrowing down the problem help identifying and addressing that problem and, in turn, designing good research questions (Boudah, 2011). When writing a research problem, three main elements need to be taken into consideration: First, the general overview of the focus topic to be investigated in the study. Second, explanation and justification of the problem: presenting logical reasons for the significance of addressing the problem, which can be attained through providing evidence from literature, recommending further research to be implemented or drawn from the researcher personal experience. Third, the research gap shows how the current study is different from previous studies within the research field. The research gap reveals that the problem has not been examined by past literature, and the current study attempts to bridge that gap. As it has been mentioned previously that a research problem illustrates the main topic or phenomenon to be explored in the study. A working definition of the research problem is advanced by Bryman (2007), who states that it is “a statement about an area of concern, a condition to be improved upon, a difficulty to be eliminated, or a troubling question that exists in theory or in practice that points to the need for meaningful understanding and deliberate investigation”. It is of paramount importance for the research problem to be direct, specific and clear and well-defined, otherwise the researcher will find himself lost, addressing a problem that does not exist. Problem statement is usually located in the end of introduction section, literature review or as a separate section following the introduction.

It has been noticed that researchers, postgraduate and undergraduate students find difficulty defining their research problem, in that the problem either too broadly or too narrowly defined (Ifeoma, 2019). This is consistent with Kombo and Tromp (2011), who argue that one of the basic challenges facing researchers is stating the problem unclearly, making them unable to choose the right methodology and fail to fulfil the objectives of the study. One more challenge with the research problem is that researchers struggle with determining the research gap and locating it into the existing literature.

Within the context of the present study, a considerable body of research has examined the challenges that confront researchers when writing scholarly articles (Alsied & Ibrahim, 2017; Cheung, 2013; Ifeoma, 2019; Pardede, 2018; Phothongsunan, 2016; Qasem & Zayid, 2019). The vast majority of these research place emphasis on general challenges and obstacles of research, excluding Pardede (2018), who explored the problem statement section. Lack of studies in this regard indicates that there is an urgent need to highlight the hindrances facing researchers and language learners when implementing a research study, especially the research problem section, which is the main concern of this study. Based on the aforesaid background, the present study aims at investigating the challenges encountering English foreign language learners EFL in the writing "research problem" section. Thus, the present study aims at answering the following questions:

1. Do Jordanian EFL students confront difficulties related to the lack of academic skills in the writing research problem section?
2. Do Jordanian EFL students confront difficulties related to the lack of language skills in the writing research problem section?

LITERATURE REVIEW

A considerable body of research has investigated the challenges of encountering learners as well as researchers in writing research. Those studies come up with a general consensus that there are lots of persistent difficulties pertinent to research writing, in the sense that some of the difficulties related to the researcher himself; lack of knowledge on research writing, while other difficulties related to lack of resources that help conduct research (Al-Khairi, 2013).

As it has been mentioned previously, previous literature elucidated the sources of difficulties in conducting a research paper, grouped them into two types; first, discursive difficulties which are pertinent to having poor skills of language such as misuse of vocabulary, violating rules of grammar, bad referencing and citation. Second, non-discursive

difficulties, which are related to plagiarism, motivation, emotional and psychological factors. Below are some of the studies that discussed the difficulties encountered by researchers (Negari, 2011).

Cheung (2013) examines the difficulties encountering first-year undergraduates when writing the discussion of the results section. The study also looks into the extent to which the students perceive the difficulties in writing research. The data for the purpose of the study is collected from a 12-week compulsory course entitled ALS 101 Academic Discourse Skills, offered in the first year of the undergraduate program. The researcher interviews eighteen undergraduate students in order to gain insight into the difficulties they experience in writing the discussion section. The results of the study indicate that the English language, including grammar and vocabulary, constitutes a challenge for the participants. The study also found that the students lack the required knowledge for writing the academic discussion section, where some unnecessary details were mentioned, and important findings ignored.

Sharing the same idea of examining lecturers' challenges when writing a research article, Phothongsunan (2016) interviews eighteen academics from two universities in Thailand. The interviews are made primarily to gain information on the issues that face the participants when writing academic papers for publication and how they tackle them. The findings of the study reveal that the participants in the study confront hindrances related to English language use comprising grammar, choice of vocabulary, organization of sentences and tone. The study also reveals that the participants face difficulties related to time constraints and lack of funding.

Komba (2016) explores the obstacles that postgraduate Tanzanian students struggle with when writing theses and dissertations. In addition, the study seeks to find a remedial solution in order to overcome those challenges. For the purpose of the study, the researcher reviews 39 theses and 64 dissertations from three universities in Tanzania. The findings of the study exhibit that half of the postgraduate students, whose theses and dissertations are reviewed, face various difficulties in writing research, including unsuitable presentation of chapters and having poor academic writing skills. The researcher concludes that research method courses taught at universities have to be revised with respect to their content. In line with Claudius, study, Alsied and Ibrahim (2017) examine the problems that Libyan EFL students face in research writing. In addition, the study investigates the attitudes of instructors regarding their students' skills in research. The sample of the study consists of 42 students and 4 instructors at Sebha University. The results of the study display that the participants face difficulties in writing research exhibited in the field of interest, choice of subject, developing a research problem, review of related literature, and reporting the results of the study. The results also reveal that the instructors look at the students' work negatively due to many reasons, including inadequate motivation and library resources, lack of background information and the need to offer more courses related to research writing.

In a newly issued study, Qasem and Zayid (2019) attempt at understanding the problems that encounter students in writing research projects in English. A total of 60 undergraduate students take part in the study at the University of Bisha. With the aim of collecting the needed data, a questionnaire is utilized, and interviews with teachers and students are done. The results of the study indicate that approximately 70% of the participants face English language-related issues. The study uncovers difficulties relevant to determining the research topic, insufficient background information about the topic, poor methodology and lack of references.

The aforementioned studies have clearly indicated various challenges that researchers encounter when carrying out a research study. Those studies reveal that the difficulties are either internal related to the researcher himself, such as lack of academic skills and having a poor command of the English language or external factors resulting from the unavailability of resources needed for research such as inadequacy of funding and resource books and bad internet connection. The vast majority of studies covering the internal challenges of research writing place emphases on obstacles in writing all sections of the research articles, while an important part, which is the problem statement section, is either totally neglected or swiftly covered. The research problem is the chief topic that a researcher seeks to investigate or the key issue a researcher is concerned to tackle. Thus, this study is implemented to bridge a gap in the literature by highlighting the problem statement section in order to understand the internal difficulties related to academic skills and language use.

METHODOLOGY

This section presents the methods used in collecting the required data to provide satisfactory answers to the research questions. The section includes two sub-sections, namely the sample of the study and instruments of the study.

Design of the Study

Research design is known as “how the researchers plan their studies so that they can obtain answers to the stated research questions using systematic approaches” (Rezai, 2016). The present study adopts a qualitative method for the purpose of answering the research questions, which is best exhibited by analyzing some research papers written by the participants in the study, with special emphasis on the problem statement section.

Sample of the Study

The sample of the study is comprised of twenty Jordanian EFL students pursuing their bachelor degree in English language and translation. The participants were third and fourth-year students enrolled in a research writing course in the summer semester in the academic year 2019/2020. All the participants are non-native speakers of English, and Arabic is their native language so that they are expected to have similar exposure to English, and the variation between them is not sharp.

Instruments of the Study

In collecting the data required for the present study, the participants are instructed to write a research paper about any topic related to language teaching, language learning or translation. Firstly, the participants subjected to a teaching program extended for two months, in that they were introduced to all details relevant to research writing. Strictly speaking, the students are presented with the steps of writing all research sections including title, abstract, introduction, problem statement, research questions, literature review, data collection, data analysis, results and conclusions. The process of writing each section was explained in details in order to get insight into writing a scholarly paper. Secondly, having introduced a particular section of research, the students were instructed to write a sample section within a specified period of time in order to verify the progress they have made. More specifically, when introducing, for instance, the research questions section, all details about this section are presented, such as the characteristics of a good research question and a bad research questions with illustrative examples and then the students were asked to apply what they have learned. Thirdly, after the end of the teaching program, the students were asked to choose a topic related to their major and start writing a research paper. They were given a period of no less than a month to write the paper. The vast majority of the students submitted their research papers on time, and then the papers were read and analyzed carefully by the researcher in order to explore their progress in research writing, especially the research problem section.

RESULTS

This section seeks to reveal the results of the study with reference to the two research questions that the study raised. To begin with, the section reveals the results pertinent to research question one, then displays the results of research question two. The question is formulated as:

Q1- Do Jordanian EFL students confront difficulties related to the lack of academic skills in the writing research problem section?

To answer this research question, the responses of students regarding writing a problem statement section is divided into three categories, according to the three elements of the research problem.

Table 1 *Academic Skills Pertinent to the Research Problem Section*

	Overview of the Problem	Justification of the Problem	Research Gap
Percentage of response	35%	60%	5%
Frequency of response	7	12	1

As it has been mentioned previously that the section of the research problem is comprised of three main components so that when a researcher misses one of those components, the research problem will not be identified, determined and addressed well, which results in making the problem too broad, failure to justify the importance of the problem and inability to locate the study within the current literature. Strictly speaking, the lack of academic skills needed to write the research problem leads to serious deficiencies in the research study as a whole and in turn, keeps the

research questions partially or totally unanswered. Table 1 above includes the responses of the participants with respect to providing the three main components of a research problem when writing the research problem section, namely, Overview of the problem, Justification of the problem and Research gap. The results indicate that 35% of the participants overviewed and provided background information of the topic to be investigated, with a frequency of 7 participants out of 20. Table 1 also shows that 60% of the participants in the study provided satisfactory justifications for the topic under investigation, with a frequency of 12 out of 20. In addition, Table 1 displays that only 5% of the participants attempted to locate their study within the current studies or to mention the research gap that their study is supposed to bridge. To be more specific, only 1 participant out of 20 tried to fit his study within the present literature. To gain further insight into the results of question one, it is important to consider pie Figure 1 below.

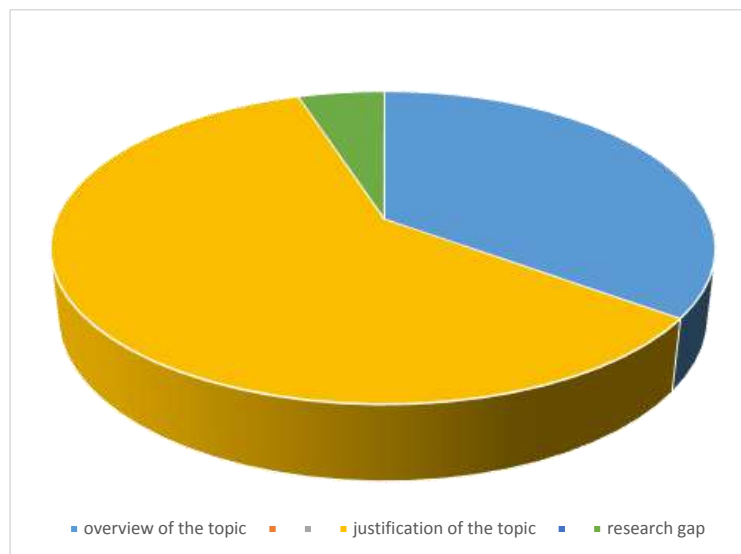


Figure 1 The Proportion of Academic Skills Pertinent to the Research Problem Section

It is obvious from Figure 1 that the biggest proportion of the chart is devoted to justification of the topic component, followed by an overview of the topic component, while the smallest proportion of the chart is for the research gap component. Those results indicate that the participants confront serious problems in identifying the research gap when writing the research problem section. The difficulties seem still existing when writing an overview of the topic, while little obstacles encountered by the participants in writing justification of the problem section. To sum up, it is evident that Jordanian EFL students lack the required knowledge of identifying the research problem in writing a research paper. The results of the research question two are depicted in Table 2 below. Question two is formulated as follows; Q2: Do Jordanian EFL students confront difficulties related to the lack of language skills in the writing research problem section?

To answer this research question, the errors related to language use committed by the participants are divided into two categories, namely grammatical errors and semantic errors. Grammatical errors include error relevant to the use of tense, order of words, agreement and determiners system. Semantic errors comprise errors pertinent to the choice and meaning of words and bigger units like phrases and clauses. Table 2 below depicts the results of the research question two.

Table 2 Grammatical and Semantic Errors Pertinent to the Research Problem Section

	Percentage of Errors	Frequency of Errors
Grammatical errors	94%	74
Semantic errors	6%	6

A close look at Table 2 above indicates that grammatical errors account for 94% of the total number of errors committed by the participants in the writing research problem section. The results also show that the frequency of grammatical errors is 74 errors. It is also apparent from Table 2 that semantic errors account only for 6% of total errors with a frequency of 6 errors. For an in-depth analysis of participant's errors related to language use, it is fitting to consider pie Figure 2 below.

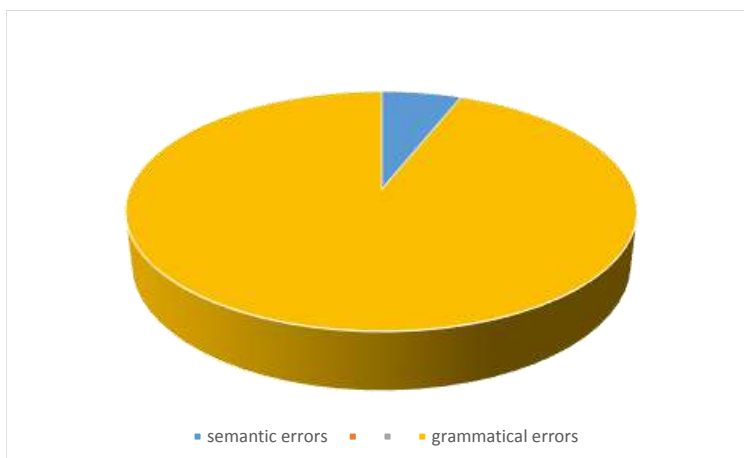


Figure 2 The Proportion of Grammatical and Semantic Errors Pertinent to the Research Problem Section

It is evident from Figure 2 that the biggest portion of errors related to language use in the writing research problem section is ascribed to grammatical errors. In contrast, a small portion of errors can be attributed to semantic errors. Those results clearly denote that the difficulties that the participants confront are grammar-related rather than semantic-related difficulties.

DISCUSSION OF RESULTS

The results of the study clearly indicate that Jordanian EFL students encounter serious difficulties in conducting the research problem section. Strictly speaking, only 35% of the participants in the study managed to present an overview or background information of the topic under investigation. 60% of the participants justified their research problem clearly and correctly, while only 5% of the participants were able to determine their research gap. This result is consistent with the results of some previous studies, indicating that EFL students, in general, face serious challenges in research writing (Cheung, 2013; Kombu, 2016; Phothongsunan, 2016; Qasem & Zayid, 2019). The results also depict that the great majority of language errors made by the participants are grammatical in nature, while very scant errors are semantic. This result goes with the findings of some studies, which reveal that English is a challenge for the participants when producing any piece of writing (Almahameed, 2017; Qasem & Zayid, 2019). To gain a thorough insight into the nature of difficulties and errors made by Jordanian EFL researchers in the writing research problem section, it is of paramount importance to analyze and discuss those difficulties with illustrative examples.

Difficulties Encountering Jordanian EFL students in Writing Research Problem Section

Having analyzed the research written by the participants, difficulties encountered by them can be grouped into two main categories.

Academic skills difficulties: The following is a sample of the problem statement section written by one of the participants in the study.

“Statement of the problem: In Jordan learning English is still challenging (Abdul Haq, 1982). Wahba, (1998) stated that problems of learning English in Jordanian educational courses (Zughoul and Taminian, 1984). Mostly in making errors in writing, lexis, syntax, and pronunciation. EFL students have many problems in all the language skills such as speaking, reading, listening, and writing. Rababah, (2003) stated that there have been a lot of weakness in English among school students who join the universities as English language majors. In fact, a lot of first-year university courses spent on English grammar vocabulary, oral skill, language use, and writing. Students are accepted in university programs without taking any international standardized tests such as TOEFL, IELTS but they had to take locally designed tests of English which lack of reliability.”

A close look at the research problem above indicates that the writer clearly and abundantly justifies this research problem in that Jordanian EFL students face serious challenges in learning English as a second language. The researcher supports his justification and explanation of the problem with the results of previous research. However, it is evident that the researcher did not provide a general overview of the nature of the topic to be investigated or the challenges confronted by learners. In addition, the researcher has not stated the position of his study within the current literature. More specifically, the researcher did not indicate whether his study contributes actively to the field of investigation, and nothing is said about the gap that the study is supposed to bridge in literature. One more sample of the research problem section written by the participants is below.

“Statement of the problem: Translating the news generally from English into Arabic can be a problematic matter for Arab translators due to the cultural differences between the two languages, Arabic and English, also Psychological problems and lack of self-confidence have a major role in the success or failure of the translation process. Simultaneous translation difficulties from English into Arabic representing by many aspects, including listening, understanding, analysis and reporting the message at a very specific time. Also, the interpreter faces the listener who receives the message from him.”

Analyzing the problem statement section above shows that the researcher describes the challenges that translators face when translating news, and he elaborates on the reasons beyond such difficulty. However, in-depth analysis and overview of the problem is needed, in the sense that the researcher is required to state more details with examples about the nature of problems in translation in order to enhance the understanding of the reader. The researcher is also needed to cite references that support his viewpoint. Furthermore, nothing is mentioned about the gap that the researcher's study attempts to bridge or to fit the study within the present literature.

Language use difficulties: Those difficulties can be either grammatical or semantic. Below are a few examples on the grammatical errors committed by the participants in the writing research problem section.

“Tense study include studying the structure, parts of speech also ”

Two errors are committed in the above sentence. The first one is a subject-verb agreement, in which no final "s" is added to the verb following the singular subject. In addition, the conjunction "and" is missing and should be added to connect the parts of the sentence above. The following is another example of grammatical errors taken from student's research.

“Grammar necessary in order to understand languages linguistic system”

In the above sentence, the verb is missing in that the verb to be "is" is required to make the sentence grammatical. The above-mentioned sentences are examples of grammatical difficulties that the participants encounter in the writing research problem section. That is to say, writing a problem statement section is demanding and needs professional skills to be accomplished well.

LIMITATIONS AND RECOMMENDATIONS

The sample of the current study is limited to Jordanian EFL students at Amman Arab University. The study is also limited to investigating the problem statement section, while other research sections beyond the scope of this study. It is recommended that future studies expand the scope of the investigation to other sections of a scholarly research paper, such as the literature review section and the results section. It is also recommended that future studies broaden the sample of the study to include postgraduate students.

CONCLUSION

The present study examined the challenges that Jordanian EFL students confront when implementing the research problem section. A total of twenty English majored students participated in the study. They were instructed to conduct a research article on a language-related topic after being presented to the main components and sections of the research article. Having analyzed the research papers written by the participants, it was found that the participants were unable to overview their research problem and to determine their research gap correctly. In addition, the study revealed that the participants committed a considerable amount of grammatical errors. It is recommended that more emphasis and teaching programs are allocated for teaching research writing skills in general and the problem statement section in particular.

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A FOUR STAGE FRAMEWORK FOR THE DEVELOPMENT OF A RESEARCH PROBLEM STATEMENT IN DOCTORAL DISSERTATIONS

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ABSTRACT

Aim/Purpose	Provide methodology suggesting steps to doctoral mentors to work with students in constructing their research problem statement in their dissertation.
Background	Writing a doctoral dissertation is a long journey, and it typically starts with writing the research problem statement. Students face challenges in articulating the research problem statement. Clearly articulating the research problems statement influences the success of the entire dissertation.
Methodology	This paper uses a widely used framework to describe student adjustment to graduate studies in general and to doctoral programs in particular.
Contribution	This study provides a framework for mentors and advisors to assist them in guiding students in writing their research problem statement.
Findings	Writing a research problem statement is difficult by itself. Following the methodological approach suggested in this study will help students with the task of writing their own.
Recommendations for Practitioners	A methodological approach to writing a research problem statement is helpful in mitigating the difficulties of writing the dissertation. This study tackles the difficulties with writing the research problem statement.
Recommendations for Researchers	More research needs to be done to expand the use of a methodological approach to writing in other sections of the dissertation.
Impact on Society	The findings of this research will help doctoral mentors/advisors as they guide students in completing the writing of their research problem statement

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Future Research	Future research should follow a similar methodological approach in guiding students in writing the other sections of the dissertation
Keywords	research problem statement, research dissertation, problem statement

INTRODUCTION

The dissertation is a labor of love requiring much work, sweat, and tears, as well as organization skills and extensive resources from others who are involved with the process. The final product is a document that one can recognize as a once-in-a-life-time achievement. We liken this experience to the task of building your own home. (Grant & Osanloo, 2016, p. 12).

Working on dissertations for doctoral programs is often lengthy, intricate, and subject to various factors that could influence the result of the dissertation (Black, 2017; Brill et al., 2014). Like the saying “The journey of a thousand miles starts with a first step,” the journey for completing doctoral dissertations often starts with defining the research problem. The problem statement becomes the centerpiece that other sections in the dissertation link to as the thesis builds (Burkard, 2014). Writing a problem statement is a complex step that involves multiple decisions that are influenced by numerous factors. However, once established, a good problem statement lays the foundation and paves the way to a complete dissertation.

Writing a doctoral research problem statement has been the subject of numerous studies (Blum & Preiss, 2005, Creswell, 2005, Ellis & Levy, 2009, Faryadi, 2018, Jacobs, 2013, Kerlinger & Lee, 2000). The point of emphasis from these studies is that writing a research problem statement is difficult and it is subject to the multiple factors that often influence its creation (Jacobs, 2013).

The complexity of writing a problem statement is increased further by the fact that the task of writing it (the research problem statement) is usually tackled at the beginning of the journey – a time when most students are struggling to adjust to the new environment of graduate studies or to the phase of dissertation writing (Gardner, 2009). A multi-faceted difficulty will be facing the students as they embark on writing (constructing, developing, or forming) their problem statements (Black, 2017). However, students are expected to write the problem statement at a time when they are still unable to fully comprehend the journey ahead.

In this paper, steps are suggested for doctoral faculty to help students construct their research problem statement. It takes into consideration the phases of adjusting to the new environment of graduate studies (or writing the dissertation). The remainder of this study is divided into the following sections:

- First, the theoretical framework is introduced. It forms the basis of this study and on which the recommendations will be based.
- Second, forming the research problem statement is explained, such as what is it and how it is typically written.
- In the third section, factors that make writing problem statements challenging are described.
- Factors that help with writing the research problem statement are elaborated on next.
- Next, suggestions are discussed on how to guide students into writing a problem statement.
- The last section provides the recommendations of this study.

THEORETICAL FRAMEWORK – THE BEELER’S MODEL

A theoretical foundation called “Beeler’s Framework” (Beeler, 1991) is used in this paper to suggest steps for academicians to mentor students through the stages of writing the problem statement. Beeler’s framework has been widely used among researchers in describing the adjustment of students to graduate studies. For example, Ali and Kohun (2007) used Beeler’s work to suggest a framework

for dealing with the feeling of social isolation in doctoral programs. Bloomberg and Volpe (2018) used it to create a road map for students and to provide suggestions for the completion of their doctoral dissertations. Lewis et al. (2003) used the model to explain the experience of African-American Ph.D. students at a predominantly white research institution.

Beeler's framework is about graduate students and their mental adjustment (academically and socially) and awareness as they embark on graduate studies. Beeler's framework suggests that students go through four phases in their mental adjustment to the requirements and the work in graduate studies:

- Phase 1: Unconscious incompetence
- Phase 2: Conscious incompetence
- Phase 3: Unconscious competence
- Phase 4: Conscious competence

The first stage of graduate studies starts upon entering the program. Many students have only a piece of limited knowledge about what is involved in the program, either academically or socially (Gardner, 2009); essentially, they do not know what they do not know. During this stage, students may question their decision to pursue the graduate degree, and doubts may linger about their performance. The second stage of adjustment takes place after the students gain knowledge about the academic requirements and what is ahead of them. Students at this stage become aware of their deficits when compared to the requirements of the program. Thus, their lack of knowledge becomes clear. By the third stage, students have gained some competence in their field of study but are largely unaware of their competence; thus, competency awareness takes place unconsciously. The fourth and last stage is the result of accumulating sufficient knowledge, and students become aware that they 'know what they know'. This conscious awareness of competence is cultivated by completing the degree requirements that they attempted. Figure 1 depicts the framework introduced by Beeler.



Figure1. Beeler's framework for adjustment to graduate studies (Beeler, 1991)

WHAT IS A RESEARCH PROBLEM STATEMENT?

At first glance, the question: "What is a Research Problem Statement?" may sound like the "duh" game that is played among some children when asked a question that has an obvious answer. But the answer to the question "What is a Research Problem Statement?" is not an obvious one, and this question may need to be answered before writing further about the topic. What is meant by a research problem statement will be clarified in this section.

First, what is a problem statement is presented in brief. Then the cause-and-effect relationship in problem statement is clarified. This step is necessary to complete before tackling what is meant by the research problem statement. Lastly, the research problem statement and how it is different to other problem statements are explained.

A PROBLEM STATEMENT

An old expression goes, "If it looks like a duck, swims like a duck, and quacks like a duck, then it probably is a duck." The same can be applied to understanding problem statements. A problem statement needs to sound like a problem when reading it or when talking about it. So, from the writing perspective, a problem statement needs to be written such that it sounds like one. If a statement –

which is ideally one sentence but could be more, although typically not more than two or three sentences at the most – sounds like a problem, and if it reads like a problem, then it is fair to say that one has a problem statement.

A reasonable question that could be asked at this point is how to make a statement sounds like a problem statement and what are the specifics in the statement that make it read like a problem. This question leads to the discussion of the “cause and effect relationship”, a concept that is important to shed light on in order to understand the research problem statement.

CAUSE AND EFFECT RELATIONSHIP

As the name indicates, a problem statement is a statement that is composed of a sentence or more, phrased to sound or to read like a researchable problem. Typically, it must have two parts, a cause and an effect. For instance, the statement could describe a loss or a missed opportunity and indicate a cause for the loss and the missed opportunity (Jacobs, 2013). In other words, there is a cause to the problem being stated, and there is an effect because of the cause (Williams, 2020).

A statement about the increase of air pollution in industrial cities, for example, might be a statement, but at that point, it does not have sufficient information to be considered a problem statement because it does not state an effect. However, say it was supplemented by another statement about the increasing cases of asthma in industrial cities that are caused by air pollution. Then, this statement pair presents the reader with a cause “air pollution in industrial cities” and an effect “increasing asthma in industrial cities.” Thus, it is be considered a problem statement: “The increase of air pollution in industrial cities results in increased instances of asthma.”

A relevant point to note here is that the effect could have other causes as well. Generally, however, the focus of a good problem statement is one cause and the relationship between that cause and the effect. Take the example of the air pollution statement again and the increase of asthma among residents of industrial cities. There could be other factors that made the effect (increasing asthma), such as increased pollen in the air that causes asthma. But the focus of the research problem statement is typically on one cause (Faradyi, 2018). In this case, it is air pollution. This relationship is depicted in Figure 2.

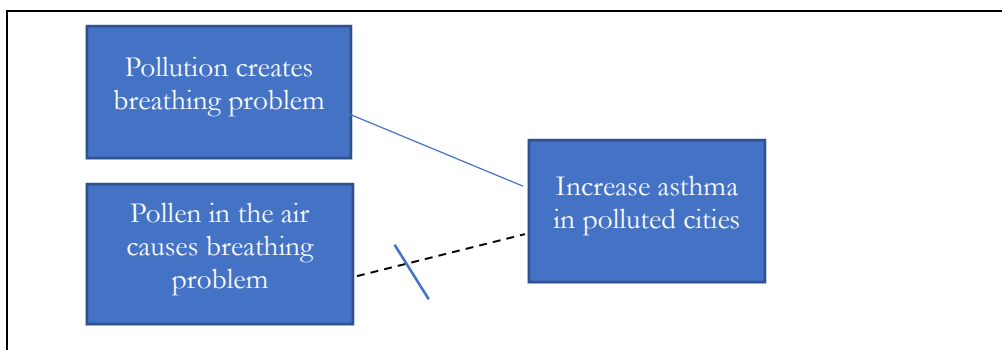


Figure 2. Cause and effect relationship illustrated

A RESEARCH WORTHY PROBLEM STATEMENT

By research worthy, it is meant that the problem being described (stated) must be supported by research data and different reviews. Also, the resolution sought from stating this problem must be achieved through research (Jacobs, 2013). In a research problem statement, there is no room for intuition and opinion. These must be put aside. The problem statement must be supported with research, and the anticipated resolution must be achieved through research as well.

Doctoral students are typically advised to tackle a viable research problem (Ellis & Levy, 2009). To understand what is meant by viable, we searched the dictionary to understand the meaning of the word “viable” and thesaurus to find alternative words that describe it. Webster dictionary defined viable as “capable of working, functioning, or developing adequately” (Merriam-Webster, n.d.). Thesaurus.com (n.d.) suggested the following alternatives: possible, usable, workable, doable.

We determined from this that, for a problem statement to be accepted as a research problem, it must meet two conditions: First, it must be workable or doable within the resources available to the researcher. Second, it must be usable. That is, the anticipated resolution of this problem is helpful, and the community will benefit from the resolutions sought in the research. In both cases, the element of research is present, and it is a must. That is, the problem, as stated, is supported by research, and the resolution to it must be achieved through research (Ellis & Levy, 2012).

CHALLENGES FACING THE WRITING OF PROBLEM STATEMENTS

Research that discusses the difficulty of writing a research problem statement is abundant. Kerlinger and Lee (2000), for example, noted that forming the problem statement in doctoral research is the most difficult and important task in writing a doctoral dissertation. Jacobs (2013) noted that research problems are not lying around waiting to be selected. Instead, the problem needs to be researched and selected based on the interest of the researcher. Jacobs further noted that writing a research problem statement demands skills far behind proficient writing skills.

A reasonable question to ask at this point is, “Why is writing a problem statement in a doctoral dissertation such a challenging task?” This section attempts to answer this question. Research writing and how it is different from other writing mechanisms is first explained. Then issues surrounding buzz words used to describe research problem statements are covered. Lastly, there is a discussion on the various sections in a dissertation that the problem statement is connected to and how this makes it more challenging to write the problem statement.

RESEARCH WRITING

There is consensus among researchers that writing a dissertation is accompanied by stress, anxiety, and anguish. For example, Pappa et al. (2017) discussed the distress that accompanies the writing of dissertations. Russell-Pinson and Harris (2019) wrote about the anguish and anxiety that accompany the writing of dissertations. Stadtlander et al. (2020) noted that writing dissertations (and hence the start by writing a research problem statement) is often accompanied by stress, anxiety, and anguish. The question that could be asked is, “What is in the doctoral dissertation writing that causes this distress, anxiety, and anguish?”

The doctoral dissertation introduces students to a different form of writing, one that is called “research writing” or “research for publication.” Research writing is different from other forms of writing (such as letter writing or journalistic writing) because there are explicit rules and conventions that guide research writing. Moreover, there are stated goals in research writing that the author must focus on. Aside from this, research writing must be based on research, facts, assertion, findings, resolutions, and suggestions, and all of these need to be based on research. The research must follow specific writing standards, such as American Psychological Association (APA) or Modern Language Association (MLA) (Burkard, 2014).

On the topic of writing standards (such as APA and MLA), it is worth noting that following a formal writing standard is helpful in many ways: it adds rigor; it provides standard and consistent ways to present the information; and it assists other researchers to understand the facts and find the resources that led to the facts (Gardner, 2009). For this reason, following a guideline represents the gold standard on which research writing is based, and information is presented. However, this

complicates the writing of the problem statement. Writing a good problem statement while adhering to strict writing requirements can be a challenging task.

THE BUZZ WORDS

In this research, we define buzz words as words that are not clearly defined in ordinary language. At the same time, we use the term “buzz words” to mean words that could give different meanings and that could be used in a variety of ways. The issue of buzz words has been discussed in the literature. Sacco (2014), for example, cautioned against using buzz words in writing because readers could come from different backgrounds. Sacco explained that unless words and phrases are understood in normal conversation across cultures and generations, it could be misleading and, therefore, counterproductive to include them in research writing.

The reason that “buzz words” are discussed in this research is that the literature that describes research problem statements is filled with buzz words that often make the literature difficult to comprehend. Repeated use of buzz words in describing what is meant by a research problem statement and the criteria that govern the articulation of it create added confusion for the students.

To explain this repeated use of buzz words, we searched literature that describes the writing of a research problem statement. In the list below, we present the buzz words we found from the literature review that describes what to include in articulating a research problem statement. We italicize the buzz words to distinguish them.

- Must be phrased so that it could be *explained in a sentence or few sentences* (Blum & Preiss, 2005)
- Must be concise, *well-conceived, and precise* (Jacobs, 2013)
- Research problem statements state a *specific condition* that needs *urgent attention* (Faraydi, 2018)
- Need to be based on *scholarly literature* (Jacobs, 2013)
- Makes a contribution to the *body of knowledge* (Ellis & Levy, 2012)
- It must be between *100 and 250 words in length* (Blum & Preiss, 2005)
- Must be significant enough to warrant dissertation research (Blum & Preiss, 2005)
- Based on literature and must describe a *gap in research* (Jacob, 2013)
- Premeditated investigation using *scientific methodology* (Faraydi, 2018)
- *Originality and significance* are requirements (Lovitts and Wert, 2009)
- “*original piece of empirical research*” (Stadtlander et al., 2020, p. 79)
- A problem statement that is *aligned with their passion* (Blum & Preiss, 2005)

The listing of that many words while explaining the concept could be inundating to the students, especially when words have multiple meanings. This is likely to confuse the students. In addition, the suggested brevity for defining a research problem statement adds to the difficulty. Blum and Preiss (2005), for example, suggested that problem statements need to be succinct. They further suggested that it must be worded so that the researcher can explain it easily and simplistically.

All these conditions lead to confusion. The novice researcher is overloaded with words that are not readily clear to them. At the same time, the research problem statement needs to be short sentences that do not exceed 250 words and could be explained by talking to people in an elevator traveling between two floors (Blum & Preiss, 2005). This confusion is illustrated in Figure 3; the center of the figure is the researcher that is bombarded by all these words.

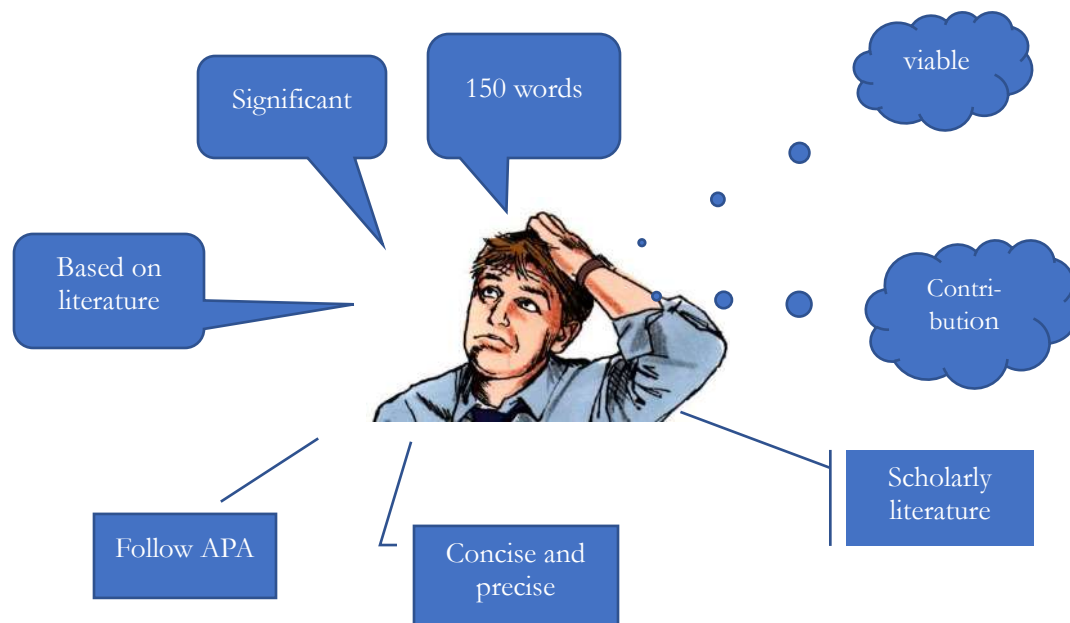


Figure 3. Confusion about buzz words in articulating the research problem statement

THE MULTI INTERLINKS TO OTHER SECTIONS

Jacobs (2013) noted that writing a problem statement represents a critical role for all remaining sections in the research. The critical role is played because other sections in the dissertation depend on defining the problem statement. The issue that makes it hard to write a research problem is its interconnection to several other sections in the dissertation. All sections need to be thought of or considered in advance when writing the problem statement. These sections could include the research purpose, research objectives, a theoretical framework, the method, identification of research, data collection, data analysis, data interpretation, and conclusions, all of which are normal elements of the dissertation process. As the student writes sections of the dissertation, each section touches on the problem statement. Thus, there is a strong interconnection between stating the problem statement and all these interconnected sections and refraining from deviating from it. Figure 4 suggests a drawing of this intricate relation of the problem statement with other sections in the dissertation.

This problem represents what is termed as a “look ahead” or a “peek” to other sections as the research problem statement was written. By look ahead or peek, we mean here that while the students are writing the research problem statement, they need to think about how it will affect other sections in the dissertation. For instance, defining a problem statement such that a student would not have resources for data collection would obviously complicate matters at the time of data collection. Similarly, developing a problem statement that leads to a research question that could not be answered, or worse, cannot be turned into a research question, creating problems later.

It is prudent to think about other sections while writing the problem statement. This undoubtedly adds another layer of difficulty to articulating the problem statement. Defining a research problem statement that is consistent with the other sections in the dissertation increases the challenge. Considering that the problem statement is typically written at the beginning of the dissertation, it means the student has little knowledge of what is ahead, so looking ahead is going to be difficult.

It is also important to note that the novice researcher may unintentionally or unknowingly deviate from the proposed research problem and discuss areas unrelated to the proposed research. A strong research problem statement is intended to not only ensure that the student touches on the statement

as all elements of the dissertation are written, but simultaneously refrains from meandering into elements not relevant to the research problem. Typically, the onus is on the students to maintain this focus as they write their research proposal and later, the thesis. However, the advisor also shoulders the responsibility of ensuring they reject work that is not focused, and that their student does not meander.



Figure 4. The Interconnections Between the Problem Statement and Other Sections in a Dissertation

HELPFUL FACTORS FOR WRITING THE RESEARCH PROBLEM STATEMENT

Hawley (2003) wrote a book titled “Being Bright is not Enough.” The book describes that writing a dissertation takes more than one set of skills. Jacob (2013) supports the same contention and additionally notes that writing a dissertation takes more than good writing skills. It is a mix of individual and institutional characteristics that must be present to help students write a dissertation. So, identifying helpful factors to writing the problem statement could assist students in their writing of the research problem statement.

Some of the factors that are considered helpful to students when writing their research problem statement are explained in this section. We begin by explaining self-efficacy and writing-efficacy and their role in writing dissertations in general and the research problem statement. Next, we discuss the importance of conducting some form of a literature review and how it helps students gain knowledge of what is written about their dissertation topic. Lastly, social support – a critical factor that is not often thought of when considering the dissertation – is discussed in this section.

WRITING EFFICACY AND SELF-EFFICACY

Self-efficacy has been the subject of discussions in many areas of research and is considered a key contributor to the success of individuals who persist and complete tasks (Bandura, 1993, Bruning et al., 2013). Bandura (2006) explained that self-efficacy is often associated with the capability of the individual to complete tasks. Bruning et al. (2013) described self-efficacy as the personal confidence that one has that they can complete their tasks successfully in the intended domain. In other words,

students with higher self-efficacy can tolerate more challenges and persist longer than those who have lower levels of self-efficacy.

Bandura (2006) identified four skills that contribute to an individual's self-efficacy: cognitive skills, motivational skills, emotional skills, and behavioral skills. In terms of the completion of academic tasks, Bandura (1993) noted that self-efficacy operates at three different levels when it comes to contributing to students' academic success. These include student's belief, the teacher's belief, and the faculty's belief in the self-efficacy of the students. McBrayer et al. (2018) conducted a study about the relationship between self-efficacy and time to degree completion of doctoral study. The study concluded that students with higher self-efficacy end up completing the writing of their dissertation in a shorter time.

A subsection of the concepts discussed regarding the writing of doctoral dissertation is the writing-efficacy of the student. Writing efficacy is connected to self-efficacy and explained by Stadtlander et al. (2020). It is described as how well a researcher can accomplish writing tasks based on various skills, such as composition, grammar, and other mechanical skills. Bruning et al. (2013) explained that writing efficacy is a function of three factors: writing ideation, writing convention, and writing self-regulation. In writing ideation, the researcher starts with writing ideas as they emerge in the mind, though how well the researcher expresses the ideas is also important. The writing convention is about the different conventions and rules that need to be followed when writing. Writing self-regulations are related to managing writing tasks and observing the writing habits that researchers develop.

Self-efficacy and writing-efficacy are necessary components in successfully completing the dissertation and writing the research problem statement as a first step to writing the dissertation. It is also a known fact that self-efficacy influences writing efficacy – that is, students with higher self-efficacy will make better progress. The question that could be asked regarding this study is how can mentors influence or contribute to enhancing both efficacies to successful writing the research problem statement.

Stadtlander et al. (2020) noted that a structured experience that is supervised by the mentor is helpful in improving self-efficacy of the students involved. Breitenbach (2019) suggested that designing a structured program with sufficient feedback could help more students complete their dissertation. Ewing et al. (2012) echoed similar contentions, and designing a structured dissertation writing process helps with the acquisition of research skills.

Structured experience in the dissertation writing (and writing the research problem statement in this study) can be helped by using a rubric type of feedback. Sparrow (2004) suggested that using rubrics improves teaching by explaining grading criteria. We argue that rubrics also help with writing efficacy and self-efficacy. Rubrics provide a clearer picture of what is expected. Therefore, students can approach the task with greater confidence. One other way for improving self-efficacy is through conducting a literature review.

IMPORTANCE OF CONDUCTING A LITERATURE REVIEW

The importance of conducting a literature review in writing a dissertation and, in particular, with writing a research problem statement cannot be underestimated. Randolph (2009), for example, noted the following about the purpose of conducting a literature review:

Conducting a literature review is a means of demonstrating an author's knowledge about a particular field of study, including vocabulary, theories, key variables and phenomena, and its methods and history. Conducting a literature review also informs the student of the influential researchers and research groups in the field.
(p. 2)

Jacobs (2013) noted the importance of conducting a literature review to narrow down the writing of the research problem statement. Wee and Banister (2016) suggested that the literature review

represents an added value for the researcher because they (the researcher) get up-to-date information on the issue being discussed in the problem statement. Torracco (2005), on the other hand, considers conducting a literature review very important because it gives the researcher exposure to the past writings on the topic and provides a framework on which to explore the future.

From the statements above, we can detect that conducting a literature review could often help the students clarify many points as they write the research problem statement (Columbaro, 2009). The literature review gives the students ideas on what problems were discussed in the past regarding the problem identified in their research. It gives them examples of how problems statements were written in the past, and they can model what they learn and use it to write their own problem statement (Ebrahim, 2012, Levy & Ellis, 2006).

While the literature review is essential in writing the research problem statement, it raises the question about the positioning of the literature review. Should the literature review be written in Chapter 2, or should it be moved to Chapter 1, before the problem statement? In our opinion, this is unnecessary. In order to understand what is written in the field about the problem statement, some literature review is essential and, therefore, expected in Chapter 1. However, a full scale literature review should be conducted in Chapter 2 and within the normal sequence of the dissertation chapters.

It must be noted that the student will, almost by definition, unearth new information when conducting a literature review. This is true in Chapter 2 as well as over the rest of the dissertation journey. Accordingly, if necessary, the research problem statement must be revisited and suitably modified or amended as needed to account for the new information. In the worst-case scenario, the student may find their specific problem suitably addressed in another publication. Rare as that occurrence may be, it would be naïve of any researcher to assume that will never happen to them.

THE ELEPHANT IN THE ROOM: SOCIAL SUPPORT

Social support, a major problem for students working on their dissertation, is the metaphorical elephant in the room. While widely accepted, the lack of social support is not addressed in sufficient depth in doctoral programs. Ali and Kohun (2007) consider the lack of social support at doctoral programs to be a major contributor to the high proportion of student attrition from doctoral programs. Devos et al. (2017) stress that student integration, socialization, and support from doctoral peers is important and will enhance student completion of their dissertations. Crauth (2017) wrote about socialization in doctoral programs, considering it a requisite for writing the doctoral dissertation and further suggesting that doctoral student attrition has social consequences. Ames et al. (2018) consider socialization as a prime factor in increasing retention in online doctoral programs.

Brill et al. (2014) conducted a study to suggest best practices in doctoral retention. The study addressed doctoral attrition from different points of view and provided suggestions on how to best work on a program that strives to keep the students in the program to complete their dissertations. Their suggestions focused in large parts on mentoring, and these come in the form of faculty mentoring and peer mentoring. This is echoed by Black (2017), who emphasized e-mentoring for doctoral students online. To summarize, the point discussed here is that social support and mentoring are prime support factors for dissertation writing success and for writing the research problem statement.

A particular point of emphasis is peer mentoring. This could come in the form of exchanging peer feedback, as suggested by Philippakos (2017). This kind of feedback helps with the social integration of the students into the program (Ali & Kohun, 2007). While writing about empowering students with peer mentoring, Krusemark & White (2020) reported on a 2017 Lehigh University study geared towards reducing the loss of prospective students and how to increase a sense of belonging once students were enrolled. Their study was on undergraduate students, but lessons learned were worth noting. For instance, one big takeaway from their study was the need for large-scale peer mentorship. An outcome of the 2017 study was Lehigh University leadership adopting, as one solution, a formalized

peer mentorship program, with clear goals, tracking, and assessments. A year later, the program has shown positive results (Krusemark & White, 2020).

At the doctoral level, it might initially appear that peer mentoring is not much more than fellow students reviewing the work of their peers for the purpose of giving feedback. However, there is more to the phenomenon since peer mentoring helps in several other ways. First, it helps give exposure to what the problem statements of their peers look like. By comprehending and analyzing the problem statements, peer groups can help refine each other's problem statements. Second, it helps with social integration into the program. Friendships could be developed, and feelings of belonging could be increased (Kumar & Johnson, 2017). Additionally, when a student receives feedback on their work, peer mentoring can provide an avenue for the airing of grievances. Work rejection and revision requests are an inevitable part of the doctoral journey. Conversing between peers helps the dejected student realize that they are not alone, that rejection is par for the course, that others have moved past this, and so will they (Columbaro, 2009).

CONSTRUCTING A RESEARCH PROBLEM STATEMENT

Blum and Preiss (2005) suggested a six-step approach for doctoral advisors to take when they coach students to write their problem statement. While the suggested steps do need to be specifically six steps, we agree that working with the students on writing their research problem statement could be best explained in many steps rather than a single step.

In this section, we list some suggestions on how to teach students to write their research problem statement. It will be divided into four stages according to Beeler's framework identified at the beginning of this study. In each stage, we identify the stage, suggest steps to follow at the identified stage, and recommend some questions to think about to stimulate thinking at that stage. This type of question is a substitute to formally using a grading rubric. While grading rubrics can be used to grade the research problem statement, we do not deem this necessary at every stage. We propose a set of questions at each stage aimed at evaluating whether the stage is complete.

STAGE 1: CRUDE STATEMENT OF THE PROBLEM

Beeler (1991) calls the first stage "Unconsciously incompetent." We describe this stage as "The students do not know that they do not know." In their early adjustment to doctoral study, students often do not know the details of what needs to be included in the problem statement, and they are often in a "fog" when it comes to these details. They often do not know about the writing conventions or the literature review. So, inundating them with long regulations about writing conventions and supporting literature is counterproductive, in our opinion. The approach we suggest is to work with students to list the problem in a crude statement. The focus of this is on three points at this stage. They can be explained as questions, as follows:

- Is it phrased as a problem statement (with cause and effect)?
- Is the problem statement researchable (research problem statement)?
- Is the research problem statement viable (time and resource constraints)?

The above could easily be part of a set of rubrics, as needed. We believe that these, or other questions in a similar vein, could stimulate thinking about the problem statement. This would move the student correctly to Stage 2.

STAGE 2: FINDING KEYWORDS

Beeler's model calls this stage "consciously incompetent." We call this stage "They know that they do not know." Awareness brought from Stage 1 is that the students are sensitive to what they do not know. They know that they need help phrasing the research problem statement. More constraints at this stage can be explained in terms of articulating the research problem statement. For example, a

look at the crude statement that was phrased in Stage 1 will be helpful. Look for “buzz words” or keywords to elaborate further. Also, connecting to a literature review is helpful at this stage. Typical helpful questions that could be asked at this stage include:

- Do you have keywords in the problem statement?
- Are the keywords explained or elaborated enough in the statement?
- Do you have supporting literature to support this as a problem that can be researched?

We believe that cogitating over and finding responses to questions like these will be helpful at this stage and help guide the student to the third stage.

STAGE 3: USING THE PEEL APPROACH

We call this step “They do not know that they know” as a clarification of Beeler’s model, which refers to it as “Unconsciously competent.” Competency was gained during the second stage. The student now knows more and can thus move to the next stage in writing. At this stage, we suggest using the Point Explain, Evaluate and Link (PEEL) method to gauge progress (Humphrey et al., 2015). This approach gives students the opportunity to look at their problem statement, begin the literature review, connect the problem statement with the different keywords, and make it sound like a research problem statement. Figure 5 below show a display of the PEEL approach.

P Point, Make the point	E Evidence Support opinion with evidence and data	E Explain how evidence support problem state- ment	L Link problem statement with other sections in dissertation
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Figure 5. The PEEL Approach

At this stage, the student may write what doctoral faculty sometimes refer to as “Naked Paragraphs.” These are paragraphs without citations or do not have references in them. In research studies, having too many “naked paragraphs” is a sign of insufficient research depth. Additionally, writing conventions and support can be enforced at this stage. Typical helpful questions at this stage can be:

- Do you have naked paragraphs in your problem statements?
- Are you connecting the problem statement with the research?
- Are you following writing conventions (such as APA) in your statement of the research problem?

We believe that questions like these will be helpful at this stage and will take them to the final stage.

STAGE 4: THE FINALIZATION STAGE

Beeler’s model calls this stage “Consciously Competent.” We call this step “They know that they know.” By now, students have become comfortable, their self-efficacy increases, and their writing efficacy increases as well. They also learn more about the other sections in the dissertation – the “lookahead ideas.” In other words, the students can think about what is ahead of them in writing the other sections of their dissertation. They can connect the problem statement to what is ahead of them and make adjustments if necessary.

The fourth and last step is to finalize the problem statement through supporting references or data (stats, figures, charts) to support that it is a problem worthy of research. Typical questions that can be asked at this stage:

- Do you have supporting data that this is a research worthy problem?
- Is the problem statement connected to other sections in the dissertation?
- Is the problem statement finalized in terms of writing conventions, viability, and research worthiness?

Table 1 summarizes our suggestions in four distinct stages that guide the students to write their research problem statement. The second column provides a name to that stage, based on our suggestions. The third column lists the questions that we deem helpful to ask at this stage. These questions are based on our findings in the previous section. They have been modified slightly to fit in the table.

Table 1 – Framework for the Completion of Writing Research Problem Statement

Adjustment stage	Suggestions for writing	Questions to ask
Stage 1	Phrasing the problem	Is it phrased as a problem statement (with cause and effect)? Is the problem statement researchable (research problem statement)? Is the research problem statement viable (time and resource constrain)?
Stage 2	Finding keywords to elaborate	Do you have keywords in the problem statement? Are the keywords explained or elaborated enough on it in the statement? Do you have supporting literature to back that this is a problem that can be researched?
Stage 3	The PEEL approach	Do you have naked paragraphs in your problem statement? Are you connecting the problem statement with research? Are you following writing conventions and APA in your statement of research problem?
Stage 4	Putting it all together. Supporting the problem statement with references and data.	Do you have supporting data that this is a research worthy problem? Is the problem statement connected to other sections in the dissertation? Is the problem statement finalized in terms of writing conventions, viability, and research worthiness?

THE ELEPHANT IN THE ROOM – AGAIN!

We must discuss social support once again at this juncture. This is because we consider social support to be essential to working on the dissertation. Different stages require different sets of steps to support the researcher. Community of Practice (CoP) and Community of Inquiry (CoI) are both recommended (Kumar et al., 2011). It is a good idea to include a support structure where students get feedback from their peers and other faculty in the program as they go through the four stages of completion. It must also be noted that social support is not limited to any specific stage of our framework. On the contrary, it could be applied to all the stages of the adjustment and the writing of the

research problem statement. It is also applicable to the entire dissertation journey. This will be further illustrated in the research recommendations section, below.

CONCLUSION AND RECOMMENDATIONS

After conducting this study, we present our recommendations as one figure that includes the previous table we finalized and suggests the importance of social support through all the stages we have reached. Because social support is needed at all stages and all levels, we included it as a shape that spans all levels. Figure 6 presents the completed table, including suggestions for social support.

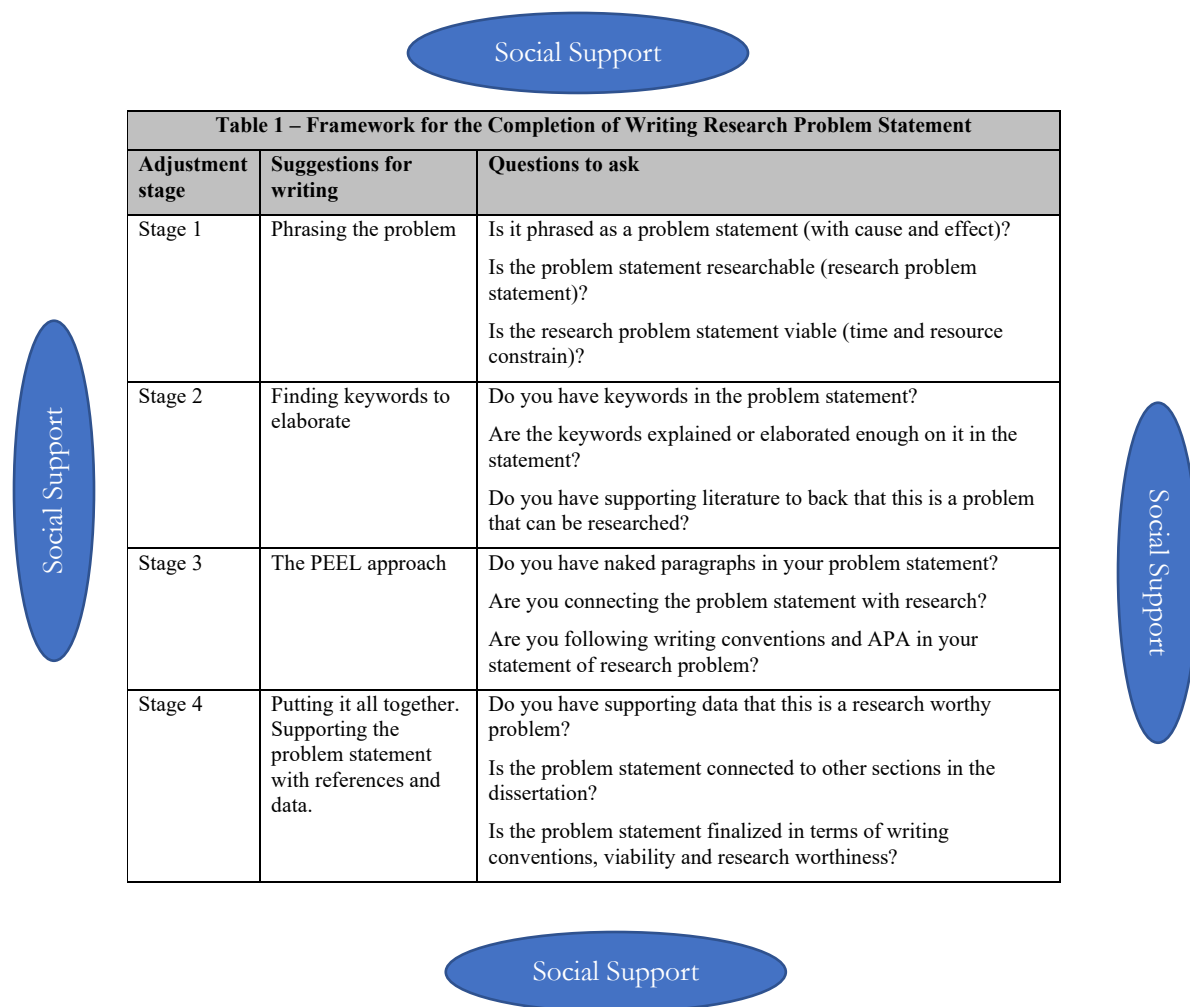


Figure 6. Suggested Framework for Writing Research Problem Statement

This study contributes to the field of doctoral studies by presenting a structured approach to assist students with the arduous task of writing their doctoral dissertation. The structured approach suggested in this study is consistent with the adjustment process that students go through at the beginning of their enrollment in the doctoral program or their beginning of writing the dissertation.

While the suggested structured approach is helpful, its utility could be further increased if illustrated with examples that inform actual student progress through each of the four stages. Thus, we intend to conduct a follow-up study to fill this gap: to provide typical examples of how students work their way through their construction of the research problem statement and the pitfalls they face. In addition, the follow-up study will introduce a rubric sheet that will provide instructions for completing each step, the objectives that are sought from completing them, and a grading sheet that can provide

different levels of evaluation feedback. This follow-up study is necessary to provide doctoral advisors with a complete model, a set of instructions, and evaluative feedback.

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How to Develop: Problem Statement, Topic, Objectives and Research Questions, Hypothesis

AN

A Seminar presentation

BY

Dr. Ambrose Azeta

Department of Software Engineering

Namibia University of Science and Technology

27 February, 2023

PRESENTATION OUTLINE

2

1. Introduction
2. Areas of Research in ICT
- 3. Computer Science Research**
 - 3.1 Formulation of Problem statement
 - 3.2 Research title
 - 3.3 Research Objectives
 - 3.4 Research Sub Objectives
 - 3.5 Research Questions and Hypothesis
 - 3.6 Research Methodology
 - 3.7 Computer Science Dissertation/thesis (sample outline)
 - 3.8 Computer Science publication/papers (sample outline)
- 4. Information Systems Research**
 - 4.1 Research Aim
 - 4.2 Research Objectives
 - 4.3 Mapping of Research Questions, Hypothesis and Objectives
 - 4.4 Confirmation of Research questions and Hypothesis
 - 4.5 Conclusion

2. AREAS OF RESEARCH IN ICT

3

- Computer Science
- Information Systems
- **Software Engineering**
- Cyber Security,
ETC
- My main research area is Software Engineering/
Software Development
- SE is applied in all domains

3. SOFTWARE ENGINEERING RESEARCH

4

- Model driven Engineering (MDE)
- Design using Unified Modelling Language (UML)
- System testing/Automated
- System Evaluation
- System maintenance and monitoring
- Automated software architecture and debugging
- Software defect prediction using machine learning

Definition of Concepts

5

What is Software Engineering (SE) ?

- Software engineering is the process of applying engineering principles to analyze the needs of a specific user and then design, build and test software applications which will meet those needs.
- Software engineering goes beyond the fundamentals of programming to create more complex software systems that can serve as critical systems for businesses and organizations.
- Software engineers need to understand how to write software and how software and hardware interact.

What is Computer-aided software engineering (CASE)

- *Computer-aided software engineering (CASE)* uses programs to assist in the creation and maintenance of other computer programs.
- Programming has historically been performed by (human) programmers or teams; CASE adds software to the programming team.

Three (3) ways of finding a Problem Statement

- 1) Existing manual/traditional system is not working well
- (2) Existing automated/computerized system exist but giving problem.
- (3) A New system. No existing system, rather the project has not been done before in the community.

Sample Problem Statement

7

SN	Project title	Sample problem statement
1	Development of Fake News and Advertisement Detection System	Individuals and groups have obligated themselves the task of spreading falsified information to other users with the aim of gaining some money, change the public's opinion on the integrity of reputable persons and organizations, potentially alter the results of an election or some other sinister reason.
2	Development of A Web-Based Medical Appointment System For Cove University	With the high number of appointments that are scheduled in clinics and hospitals every day, it is almost impossible for the clinic to achieve its goals. Definitely, there will be some dissatisfaction on both clinic and students. The clinic has to deal with urgent or emergency cases, absentees or no-shows among scheduled cases which may complicate the work of the receptionist who is booking these appointments manually through walk-ins.

3.1 FORMULATION OF PROBLEM (GAPS)

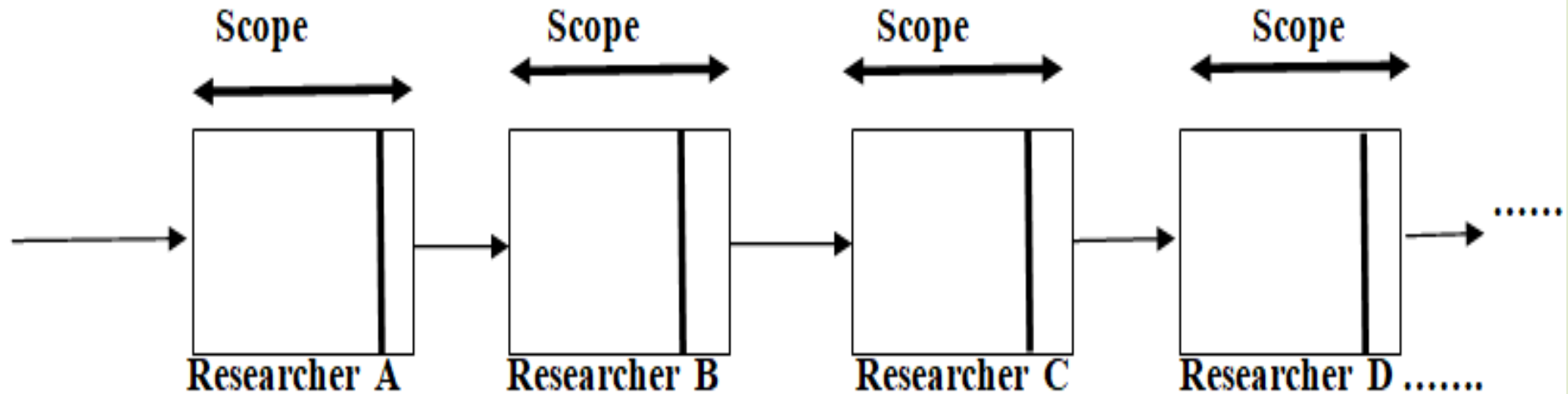
8

- ➡ State the problem you want to solve
- ➡ Cite the publication/papers that contains the limitations of a study or a problem that you want to solve, and could not be solved in the current research
- ➡ Gaps are normally discovered through systematic review

1. INTRODUCTION

9

RESEARCH CHAIN



Research is continuous (Research Advancements)

Some questions to address:

Which work are you extending?

What problem are you trying to solve?

What is the societal benefit of your research?

Etc

3.2 RESEARCH TOPIC

10

- Must contain: a technique/methodology (**Using** or **a**), domain (in), system or platform(**of** or **for**),

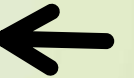
Examples

- Development of a model **for** result management system **in** E-Education **using** DevOps OR
- Development of **a** DevOps-based model **for** result management system **in** E-Education OR ←
- Development of **a** result management system **in** E-Education using DevOps

3.3 RESEARCH OBJECTIVE

11

- The objective of this study is to develop a devops-based model for result management system in e-education



3.4 RESEARCH SUB OBJECTIVES

12

The Sub **objectives** of this study are **To**:

- **To** carry out requirement elicitation of result management system
- **To** design a DevOps model for result management system
- **To** implement a result management system based on DevOps as proof of concept
- **To** evaluate the implemented system and benchmark with existing system

3.5 RESEARCH QUESTIONS AND HYPOTHESIS

13

Research Questions

- How can a DevOps model be developed for result management system?
- How can a result management system based on DevOps be evaluated.

Hypothesis

- **H1:** The development of a DevOps model for result management system is realistic
- **H1:** A result management system based on DevOps has a high usability evaluation compared to other models.

3.6 RESEARCH METHODOLOGY

14

- Tools/techniques such as software tools
- Research design/instruments such as questionnaire

3.7 CHAPTER OUTLINE OF DISSERTATION/THESIS

15

A Sample Thesis outline:

- ➡ Ch1-Introduction
- ➡ Ch2-Literature review
- ➡ Ch3-Research methodology
- ➡ Ch4-Results and Discussion
- ➡ Ch5-Conclusion

3.8 SECTION OUTLINE OF PUBLICATION/PAPERS

16

A Sample paper outline:

- ➡ Section1-Introduction
- ➡ Section2-Literature review
- ➡ Section3-Research methodology (System Design)
- ➡ Section4-Result and Discussion (System Implementation and Evaluation)
- ➡ Section5-Conclusion

4 INFORMATION SYSTEM RESEARCH

17

Research Title: *IMPACT OF ICT ON STUDENT ACADEMIC ACHIEVEMENT*

Formulation of Research Questions and Hypothesis

- **Research Question 1:** *To what extent does teaching with the aid of ICTs impact on student academic achievement?*
- **Hypothesis H1:** *Teaching with the aid of ICTs has a positive impact on student academic achievement.*
- **Research Question 2:** *To what extent does course student engagement impact on their academic achievement?*
- **Hypothesis H2:** *Student engagement has a positive impact on their academic achievement.*
- **Research Question 3:** *To what extent does student use of ICTs impact on academic achievement?*
- **Hypothesis H3:** *Student use of ICTs has a positive impact on academic achievement.*

4.1 RESEARCH AIM

18

The aim of this research is to explore the impact of ICT on student academic achievements

4.2 RESEARCH OBJECTIVES

19

The objectives are as follows:

- To determine if teaching with the aid of ICTs impact on student academic achievement
- To determine the extent to which course student engagement impact on student academic achievement
- To determine the extent to which student use of ICTs impact on student academic achievement

4.3 MAPPING OF RESEARCH QUESTIONS, HYPOTHESIS AND RESEARCH OBJECTIVES

20

SN	Research Questions (RQ)	Hypothesis (H)	Research Objectives (RO)
1	<i>To what extent does teaching with the aid of ICTs impact on student academic achievement</i>	<i>Teaching with the aid of ICTs has a positive impact on student academic achievement.</i>	<i>To determine if teaching with the aid of ICTs impact on student academic achievement</i>
2	<i>To what extent does course student engagement impact on their academic achievement?</i>	<i>Student engagement as a positive impact on their academic achievement</i>	<i>To determine the extent to which course student engagement impact on student academic achievement</i>
3	<i>To what extent does student use of ICTs impact on academic achievement?</i>	<i>Student use of ICTs has a positive impact on academic achievement</i>	<i>To determine the extent to which student use of ICTs impact on student academic achievement</i>

4.4 CONFIRMATION OF RESEARCH QUESTIONS AND HYPOTHESIS

21

RQ1: *To what extent does teaching with the aid of ICTs impact on student academic achievement?*

Construct B: Teaching with the aid of ICTs returned a strong positive correlation and prediction (2nd in weight(β) out of 5 predictors) of *student academic achievement*.

H1: *Teaching with the aid of ICTs has a positive impact on student academic achievement.*

H1 was supported since a significant positive relationship was established with model 1 & 2. (Unstd β = 5.954 & 5.848, $p < 0.01$). The finding is in agreement with the results of Bai, et al. (2016); Rugutt and Chemosit (2005.); Ilomaki (2008); Abuet al., (2011); Oye et al., (2012) and Angeli&Tsaggari (2016).

4.4 CONFIRMATION OF RESEARCH QUESTIONS AND HYPOTHESIS (CON'T)

22

RQ2: *To what extent does course student engagement impact on student academic achievement?*

Construct C: Student engagement, *had* a weak positive correlation with, and prediction (3rd in weight (β) out of 4 predictors) of Class Marks.

H2: *Student engagement has a positive impact on student academic achievement.*

H2 however, was supported since a significant positive relationship was established for model 1 & 2 (Unstd β = 1.086 & 1.140, $p < 0.05$). The finding is in line with Gunuc's (2014) conclusions.

5. PUBLICATION OF RESEARCH OUTPUT

23

Publishers of good Journal Outlets

- Nature
- Springer Nature
- Elsevier
- Sage
- Emerald
- Wiley
- University-based Journals
- Etc

Conference outlets

- ACM
- IEEE
- Springer
- Etc

Index Outlets/Databases

- Scopus
- Web of Science/ISI
- University databases/
repositories
- Etc

5. CONCLUSION

24

This presentations has no doubt achieved its objectives of highlighting the research methodology in Computer Science and related areas

THANK YOU

Effective Medical Writing

Pointers to getting your article published

Peh W C G, Ng K H

Writing the Introduction

ABSTRACT

The introduction section of a scientific paper aims to introduce a specific topic and to stimulate the reader's interest. It provides background information about what has already been done by others, supported by a limited number of relevant references. The reader should be informed about the purpose of the paper, what it will address, and how it relates to previous work.

Keywords: introduction, medical writing, research objectives, scientific paper structure

Singapore Med J 2008; 49(10): 756-758

INTRODUCTION

The introduction section of a scientific paper is normally placed at the beginning of the text proper, and follows the abstract. It is the first component of the conventional IMRAD (Introduction, Materials and methods, Results And Discussion) structure of an original article. Almost all papers have an introduction section, except perhaps for some types of short papers found in certain journals, such as case reports, commentaries and other paper categories. Authors should always check the individual journal's "Instructions to Authors" or "Author Guidelines" for the in-house style requirements.

Besides introducing a specific topic, the purpose of the introduction section is to stimulate the reader's interest, with the aim of leaving a positive initial impression of what is to come, so that he or she will be motivated to read the rest of the paper.⁽¹⁾ Inexperienced authors often have problems with this section. A very common problem is being unable to clearly state the objectives for writing the paper. The introduction can be divided into two components, namely: background information and purpose.

BACKGROUND INFORMATION

The first component of the introduction section aims to provide the reader with sufficient background information about the topic to be investigated or discussed, i.e. introduces the topic to the reader. This background information should include a brief summary of what has already been done in the particular area of interest.

The nature of a problem and its potential significance should be highlighted. References quoting relevant work by others should be provided, with particular care taken not to miss out important previous work. Editors and reviewers like to see that authors have made a systematic and comprehensive review of the literature. However, listing all the references found from an extensive literature search is not appropriate in the introduction section. Authors should also resist the temptation to provide an extensive critique of others' work in this section. Authors should instead confine themselves to providing a limited number of relevant references.

The length of the introduction section and depth of background information contained depend on the target audience and type of paper. The introduction section is usually longest for original articles, particularly those dealing with a complex subject, and shortest for case reports. If a certain topic is aimed at a general medical audience, the introduction may need to be more descriptive and have greater explanatory details in order to better introduce the topic; while for the same topic written for a subspecialty audience, the authors can assume that the reader would be familiar with the topic and all that is required is a brief background to show why the study was conducted and why the topic is important. The background information usually starts with more general and broad-based facts, and then moves on to specifics. As a rule, the introduction should be as succinct and focused as possible.

PURPOSE

The second component of the introduction section aims to inform the reader about the purpose of the article and how it relates to previous work. The author should be able to provide information about how the hypothesis for a study, clinical or investigative observation, or idea for a procedure or its modification came about, either from his personal thoughts and experience, or from referenced work by others. Authors should highlight any new, different or special aspects of what he or she aims to undertake in the study. Editors will not be interested in publishing, and the reader will not be interested in reading, studies that are merely a repetition of what had already been done by others.

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The reader should be able to understand why the authors chose to undertake the study, and the rationale for the authors' research. The reviewers, and later on the reader, need to be convinced of the importance of the research question, and feel stimulated enough to read the rest of the paper. The identification and definition of a particular clinical or research problem is therefore vital. The reader should be informed about how the problem will be addressed. The purpose for writing the paper should be precisely stated at the end of the introduction section. The main and secondary objectives should be clear, and ideally comprise no more than two sentences. The author should not include data or conclusions from the work being reported.

Box 1. Components of the Introduction:

1. Background information
 - a. What have others done?
 - b. Provide evidence: supported by limited number of relevant references.
2. Purpose of study
 - a. Why undertake this research?
 - b. How does it relate to what has already been written?
 - c. What is so different or special about your research?

Box 2. Common errors:

- Overlong and rambling introduction section
- Extensive listing of references
- Extensive critique of others' work
- Important previous work missing
- Objectives not clearly stated
- Inclusion of data or conclusions from the work being reported

SUMMARY

The introduction section of a scientific paper should be concise and yet informative. The reader should be able to understand why the author's work is important in the context of what is already known, and be stimulated enough to read the rest of the article.

Box 3. Take home points:

1. The introduction should be succinct and focused.
2. Provide sufficient background information supported by relevant references.
3. Clearly state the purpose for writing the paper and how it relates to previous work.

REFERENCE

1. Day RA, Gastel B. How to Write and Publish a Scientific Paper. 6th ed. Burnham: Greenwood Press, 2006.

SINGAPORE MEDICAL COUNCIL CATEGORY 3B CME PROGRAMME
Multiple Choice Questions (Code SMJ 200810A)

	True	False
Question 1. The introduction section:		
(a) Should be made as long as possible.	☐	☐
(b) Should contain at least 100 references.	☐	☐
(c) Should always be structured.	☐	☐
(d) Aims to stimulate the reader's interest.	☐	☐
Question 2. The following are components of the introduction section:		
(a) Background information.	☐	☐
(b) Detailed statistical analysis.	☐	☐
(c) Results.	☐	☐
(d) Purpose of study.	☐	☐
Question 3. The following statements regarding the introduction section are true:		
(a) It provides sufficient background for understanding the rest of the paper.	☐	☐
(b) Extensive critique of others' work is required.	☐	☐
(c) If the authors disagree with certain pieces of work, reference to these may be omitted.	☐	☐
(d) Its length and depth should be tailored to the target audience.	☐	☐
Question 4. In the introduction section, the purpose should address the following:		
(a) Why undertake this research?	☐	☐
(b) How does it relate to what has already been written?	☐	☐
(c) What is so different or special about the undertaken research?	☐	☐
(d) What new data was found?	☐	☐
Question 5. When writing the introduction section, the following rules apply:		
(a) Always read the particular journal's author guidelines.	☐	☐
(b) The purpose for writing the paper should be concise.	☐	☐
(c) The purpose for writing the paper is almost always contained within the opening two sentences of the introduction section.	☐	☐
(d) A particular clinical or research problem should be identified.	☐	☐

Doctor's particulars:

Name in full: _____

MCR number: _____ Specialty: _____

Email address: _____

SUBMISSION INSTRUCTIONS:

(1) Log on at the SMJ website: <http://www.sma.org.sg/cme/smj> and select the appropriate set of questions. (2) Select your answers and provide your name, email address and MCR number. Click on "Submit answers" to submit.

RESULTS:

(1) Answers will be published in the SMJ December 2008 issue. (2) The MCR numbers of successful candidates will be posted online at www.sma.org.sg/cme/smj by 15 December 2008. (3) All online submissions will receive an automatic email acknowledgment. (4) Passing mark is 60%. No mark will be deducted for incorrect answers. (5) The SMJ editorial office will submit the list of successful candidates to the Singapore Medical Council.

Deadline for submission: (October 2008 SMJ 3B CME programme): 12 noon, 25 November 2008.

Data Science Cluster

Ruusa Ipinge

Areas of Specialisation

area of specialization in Data Science

- Specific Machine learning, especially its application on data for good
- I am also a certified General Data Protection (GDPR) whereby I am concerned about setting up policies
- I am also working on mobilizing digitalising Namibia

Digitalising Namibia

- Some of the projects I am working on is detecting water leakage through machine learning
- Using machine learning to identify those that are in extreme poverty
- Developing a recommendation system to help students choose their major
- Digitalizing data to identify vulnerable people for grants assistance for different ministries
- Digitalizing a Cumulative Card for the ministry of education
- Digitalizing a medical card for ministry of health

Data Science Workshops and Conferences

- We have a NUST star show your talent end of the year hackathon
- We have Namibia Indaba X hackathon coming soon in April , MAy
- We will be soon hosting a GDPR workshop
- And there will be indaba hackathon in Ghana, contact me on how to apply
- So Many Conferences around the world in Data Science
-



Digital Forensics and Information Security Research Cluster

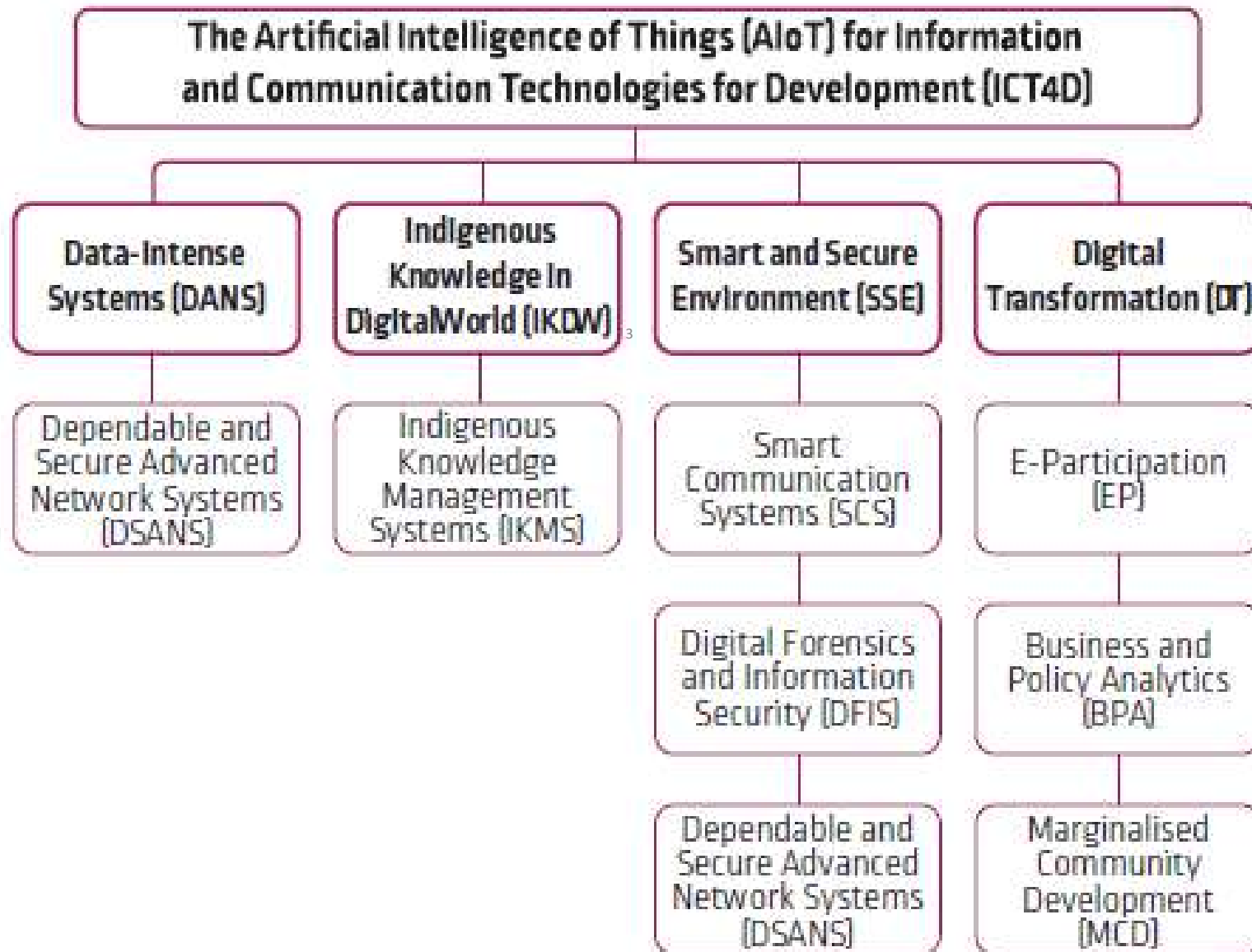
Digital Forensics and Information Security Research Cluster

Smart and Secure Environment





- FCI Research Focus Areas & Clusters
- Background
- Focus and Activities
- Team & Projects
- Graduates
- Students
- Publications



Responsible Journalism (RJ)

Journalism and Media Technology (JMT)



FCI Research Areas & Clusters

FCI Research Areas:

1. **Data-intensive Systems (DANS)**
2. Indigenous Knowledge in Digital World (IKDW)
3. **Smart & Secure Environment (SSE)**
4. Digital Transformation (DT)

FCI Research Clusters:

1. Dependable and Secure Advanced Network Systems – **DANS & SSE**
2. Indigenous Knowledge Management Systems (IKMS)- **IKDW**
3. Smart Communication Systems - **SSE**
4. **Smart Communication Systems (SCS) - SSE**
5. Digital Forensics & Information Security (DFIS) – **SSE**
6. E- Participation (EP) – **DT**
7. Business & Policy Analytics (BPA) - **DT**
8. Marginalised Community Development (MCD)- **DT**



About Us

Smart and Secure Environment

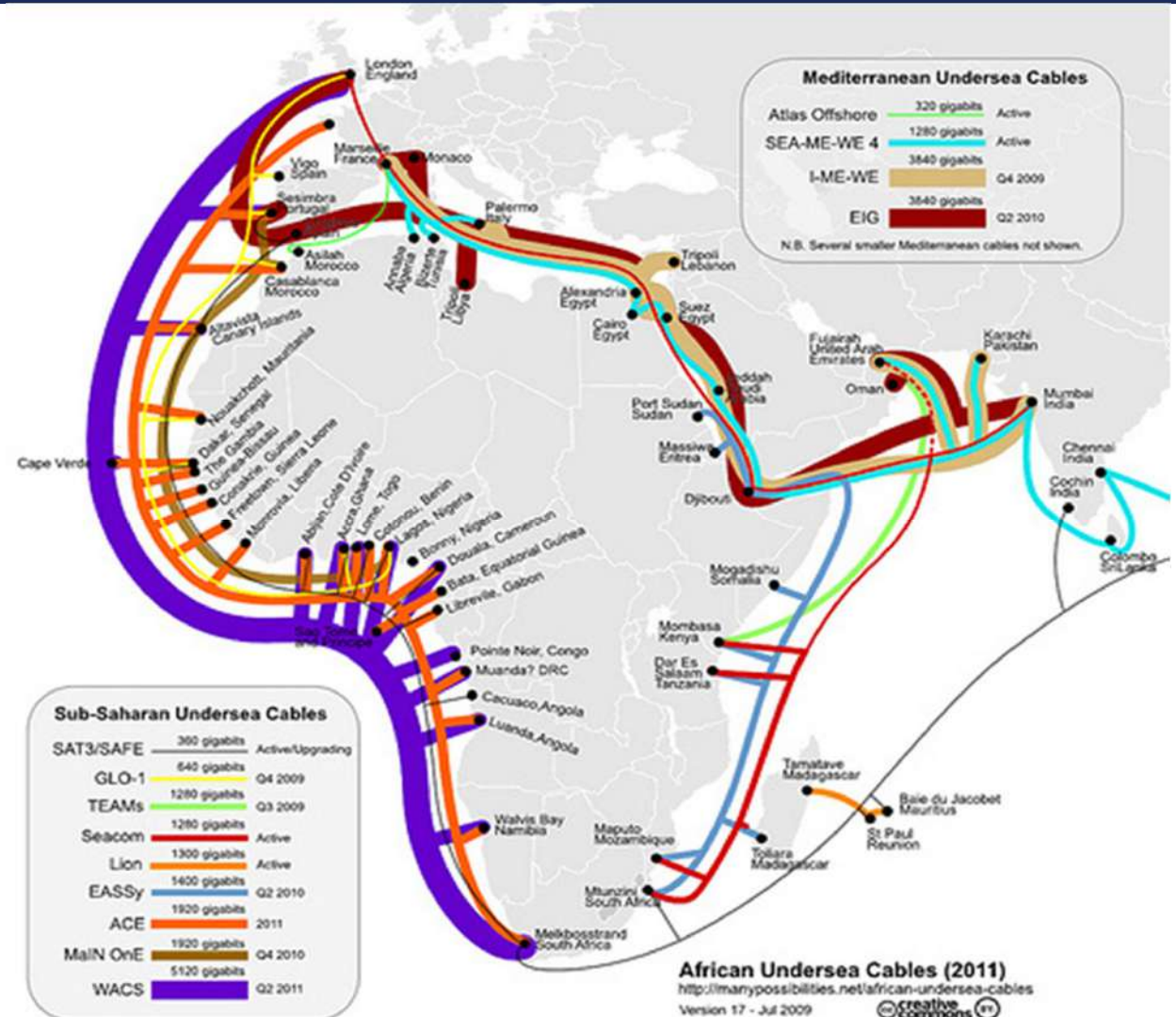
Digital Forensics and Information Security Research Cluster

Focus Areas

- The Internet of Things, AI and Emerging technologies security.
- Cyber Security
- Child Online Protection research.
- Security of organizational and industry operations research at various levels of complexity.
- Digital, Mobile and Cloud Forensics.
- Human Computer Interaction security
- Cloud Computing Security
- AIoT for ICT4D



Background





Established in 2012, the research group focuses on making the internet more secure and a safe place for civil society and government.

A team of researchers who are constantly seeking partnership with all stakeholders who share the same aspiration of ensuring online activities are secured.

This group among other things promote publications, post graduate research projects as well as enable cooperation with industry and international partners.

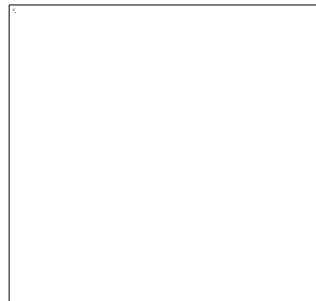
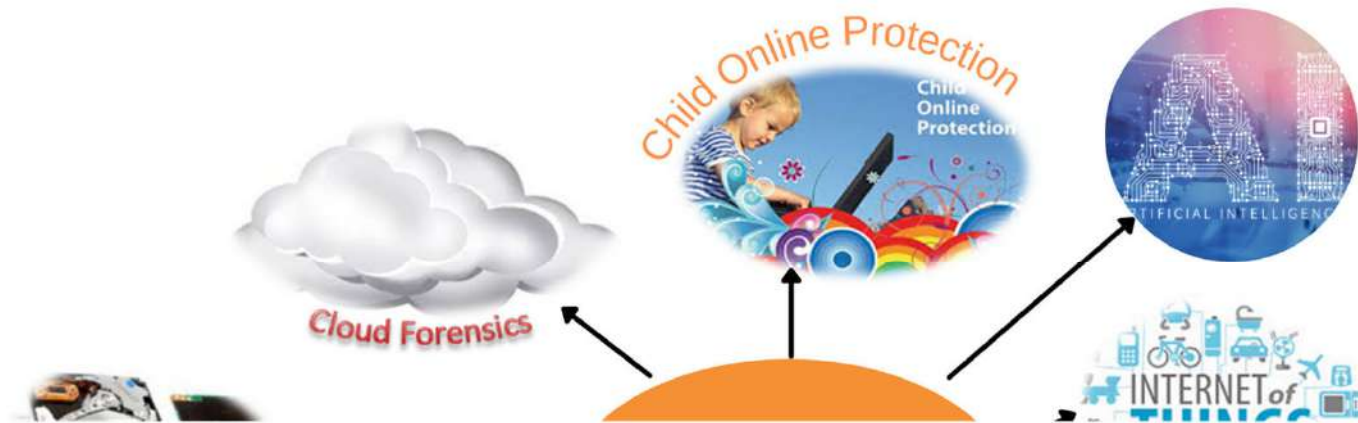
We always believe in engaging with the community around to propose viable solutions based on understanding their needs.





Digital Forensics and Information Security Research Cluster

Focus areas





AIoT for ICT4D





Cluster Focus Areas





Among other things, our research group is working on the following research areas:-

- Child Online Protection research.
- Security of organizational and industry operations research at various levels of complexity.
- The Internet of Things, AI and Emerging technologies security.
- Digital, Mobile and Cloud Forensics.
- Human Computer Interaction security
- Cloud Computing Security



Digital Forensics and Information Security Research Cluster

Members

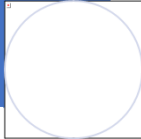
- PhD IT (User Experience/Information Security), MSc Computer Science: Cluster leader

Fungai Bhunu Shava
(FBS)



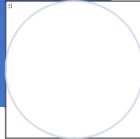
- PhD Computer Science (Critical Infrastructure security), MIT

Mercy Bere Chitauro (MC)



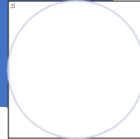
- PhD Informatics (HCISec) student, MSc Business Informatics, BIT: Syst. Admin and Netw, ND: Information Technology

Gabriel Nhinda (GN)



- PhD Computer Science (Cloud Forensics) student, MSc Computer Science

Isaac Nhamu
(IN)



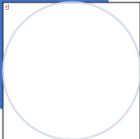
- PhD Computer Science student (AI and Security)
- Master of IT, PGCHE, CCNI
- Digital Forensics,

Julius Silaa
(JS)



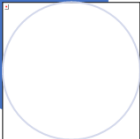
- PhD student, Master of Computer Science, BHons Communication Networks, Cloud computing, E governance in education, Systems security

Shadreck Chitauro (SC)



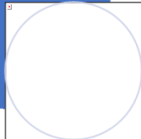
- MSc (Network Security), BHons Communication Networks: Network Security, InfoSec

Edward Nepolo (EN)



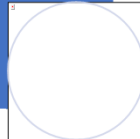
- PhD: Cybersecurity – student Master of Computer Science (Network Security), B Compt Sc Honours Information Security, B Compt Sc Systems Administration, CCNA1

Uakomba Mbasuva (UM)



- PhD Computer Science, Postdoctoral Fellow (Internet of Things (IoT) & AI) Security, MSc Computer Science

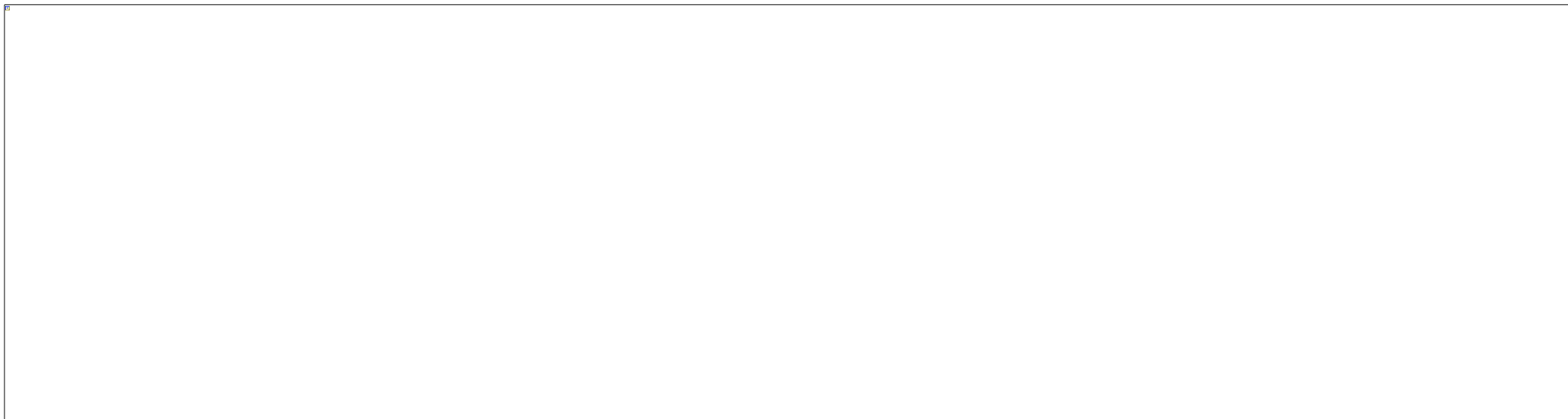
Attlee M. Gamundani
(AMG)

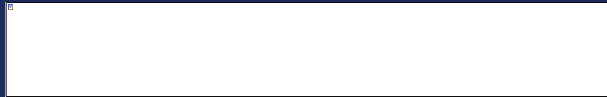




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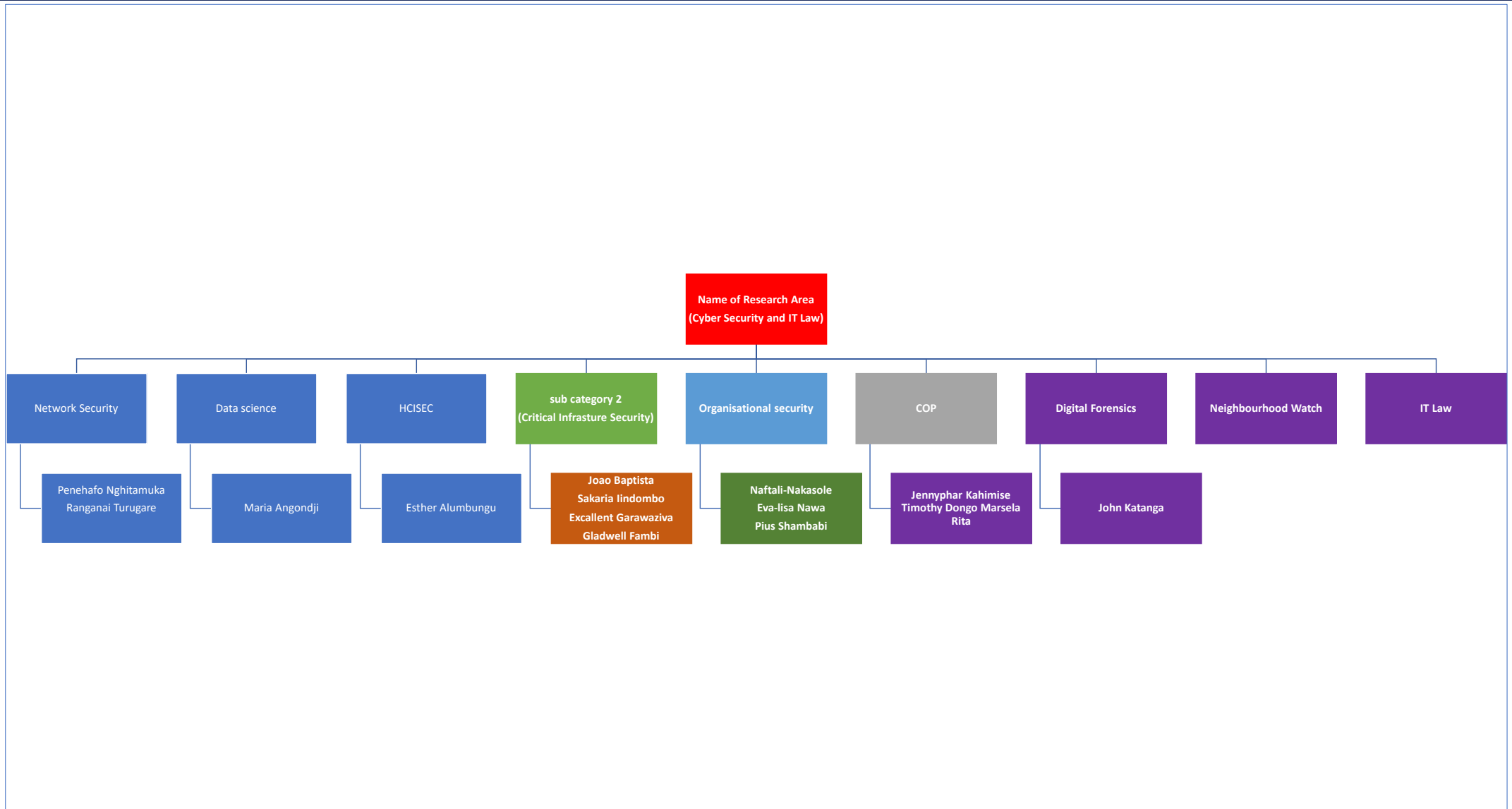
Research interests FBS





Research Interests MC

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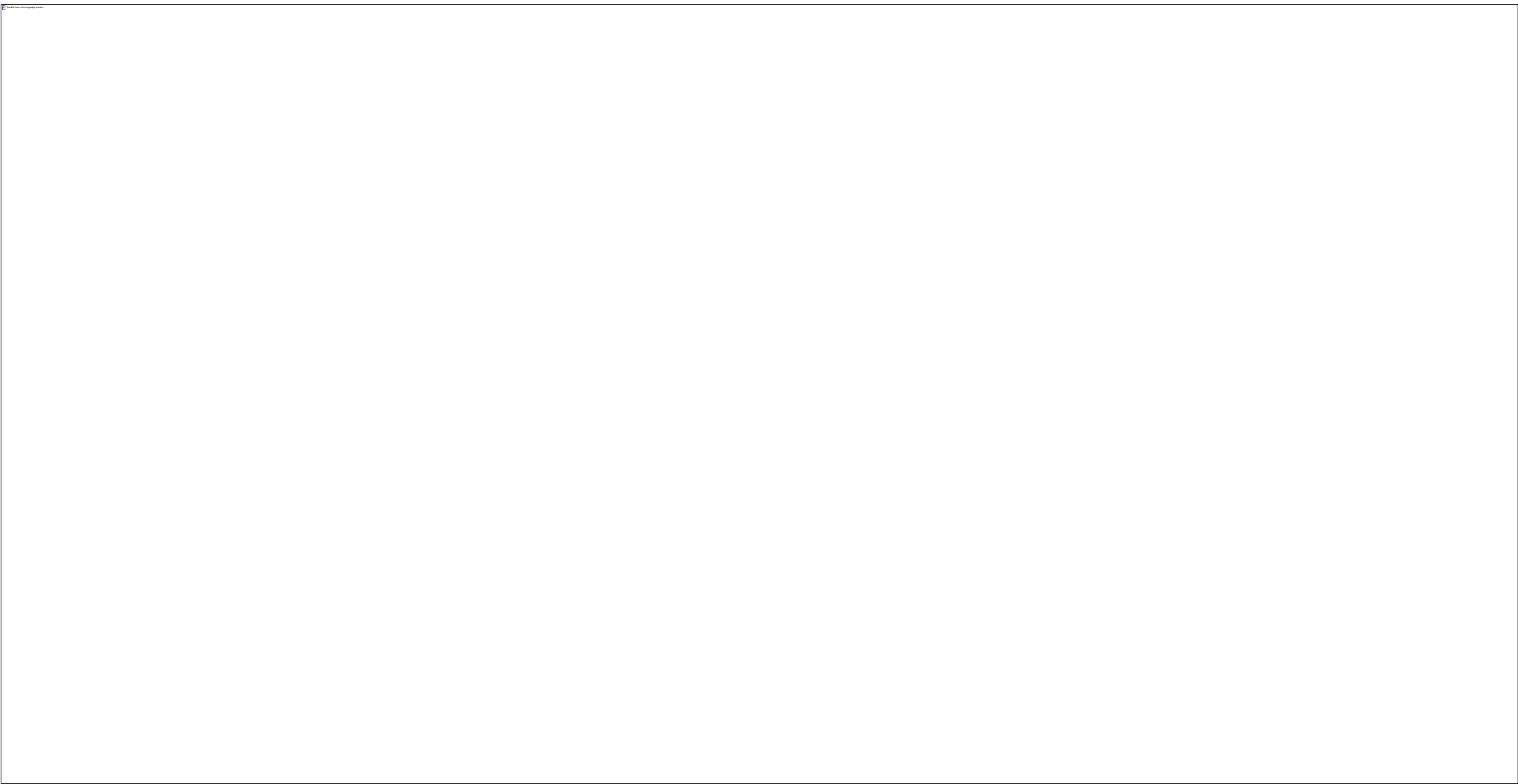




Digital Forensics and Information Security Research Cluster

Research Interests

Julius Silaa

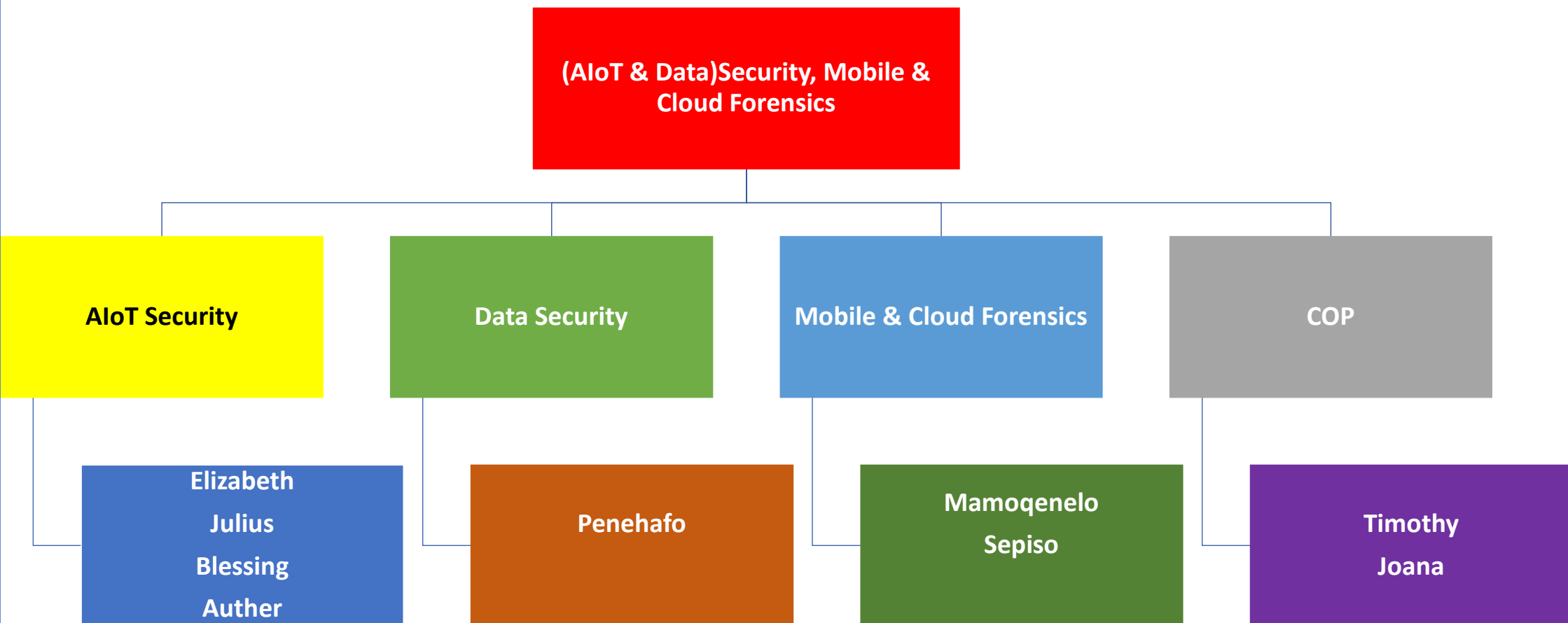




Digital Forensics and Information Security Research Cluster

Research Interests

Attlee M. Gamundani





Digital Forensics and Information Security Research Cluster

Research Interests

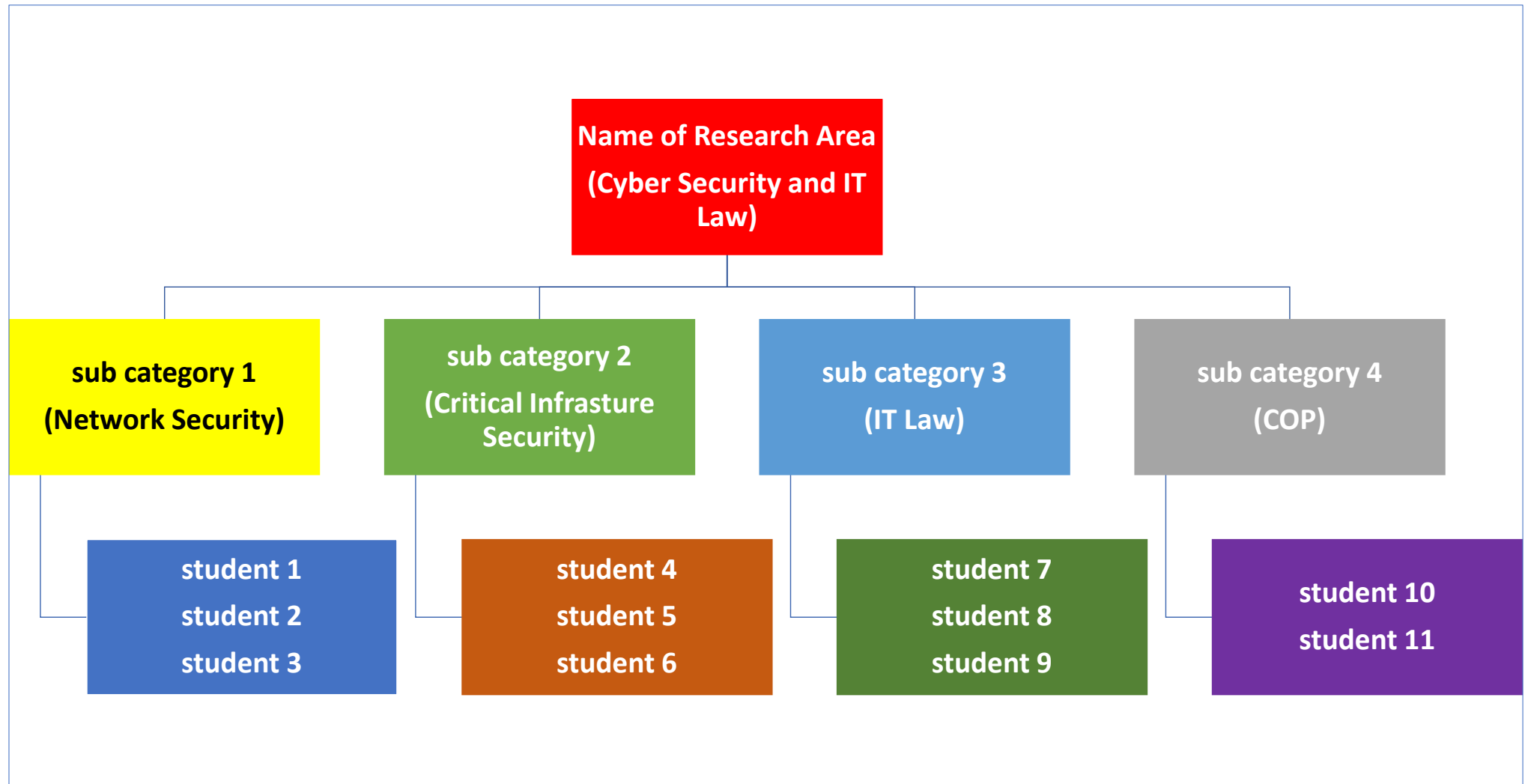
Isaac Nhamu

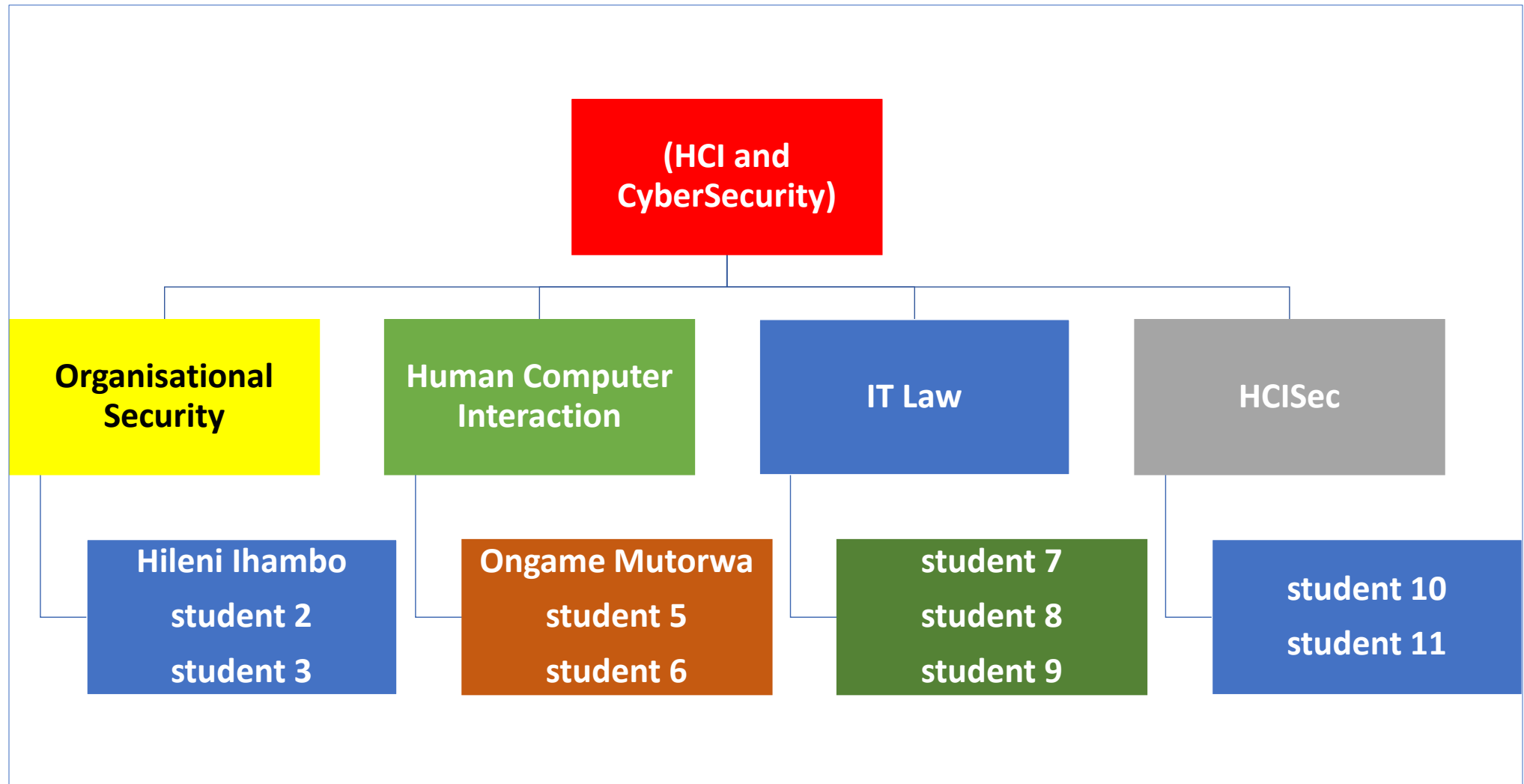


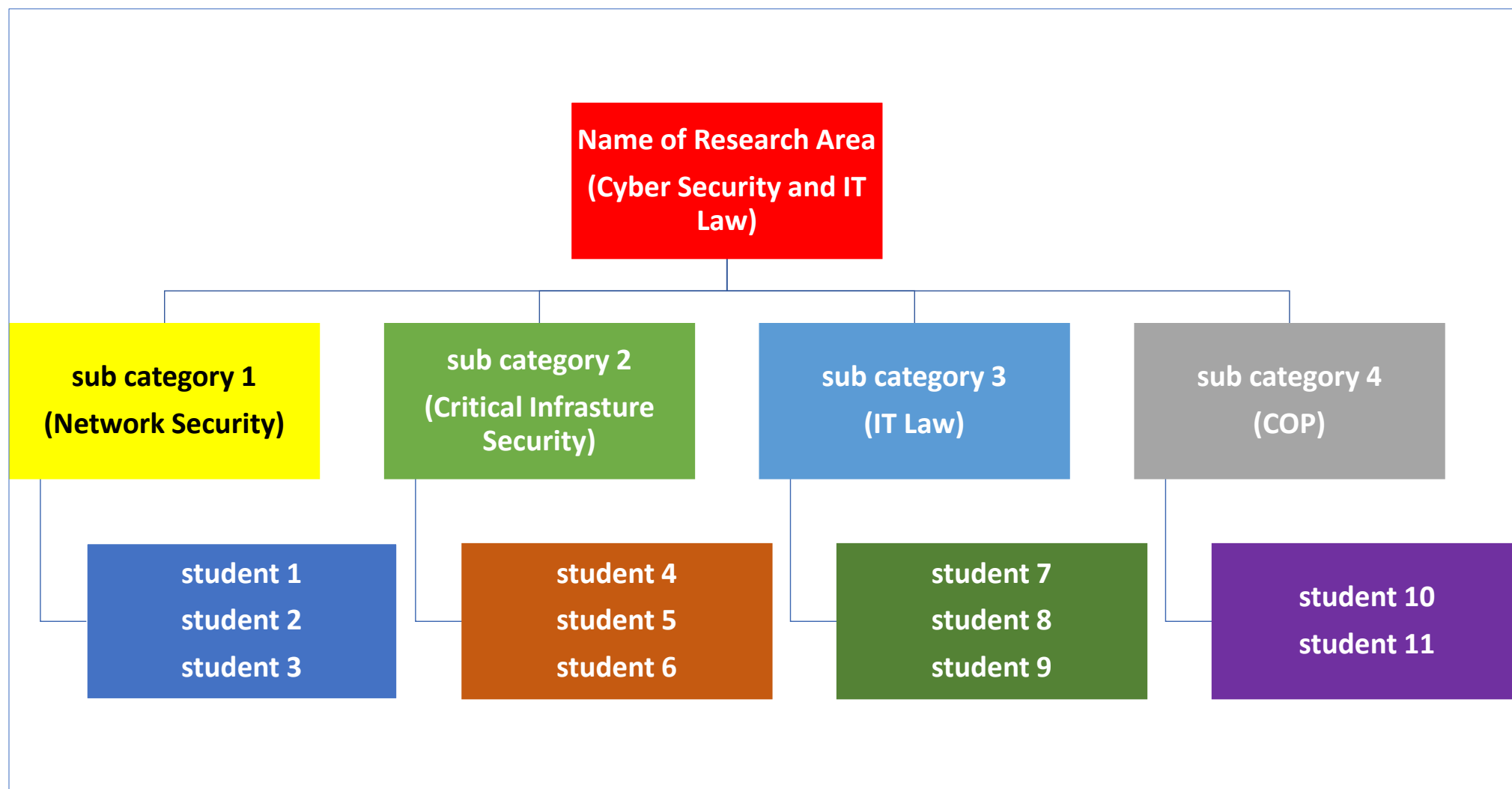


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Research Interests Shadreck Chitauro





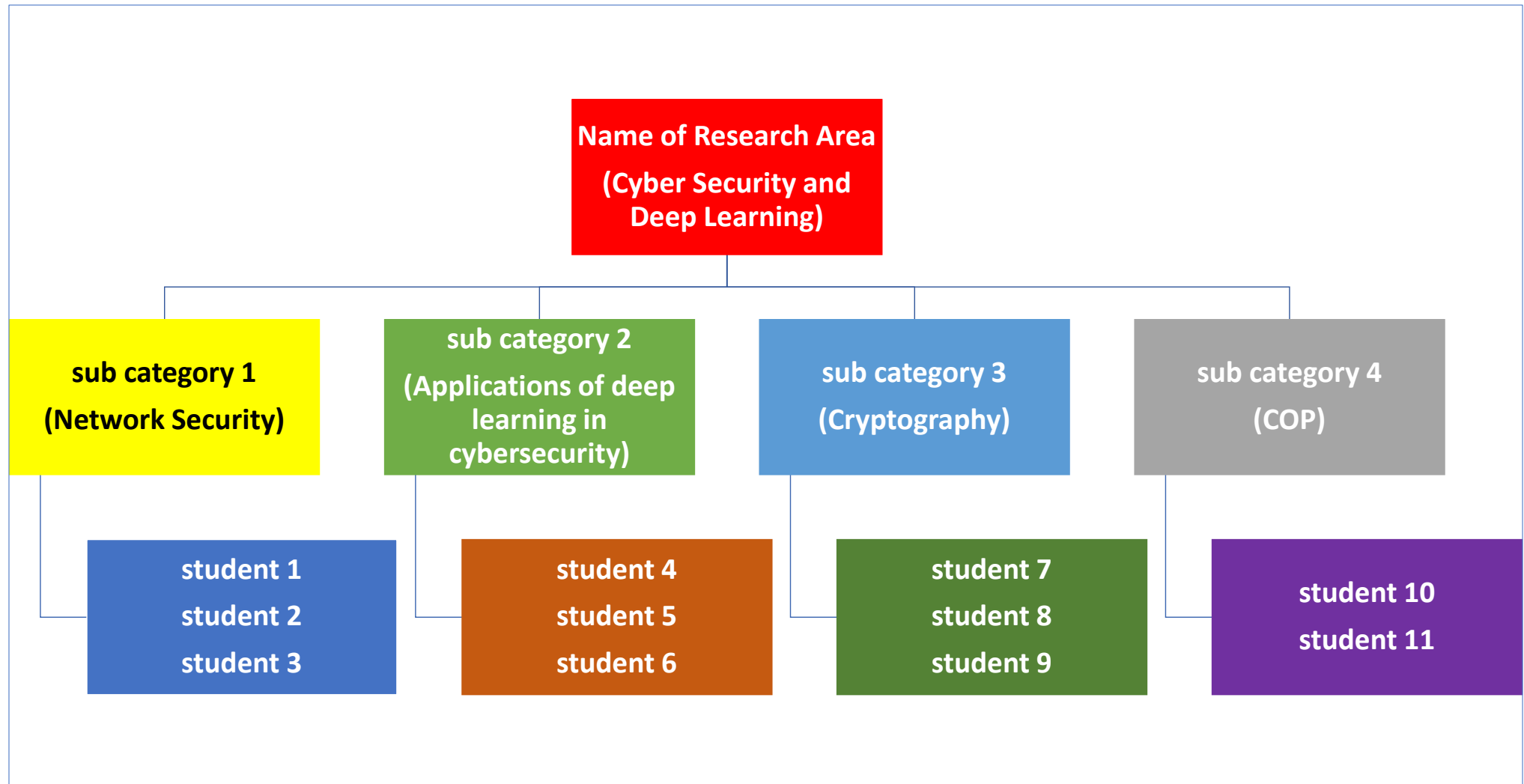




Digital Forensics and Information Security Research Cluster

Research Interests

Uakomba E. Mbasuva





Honours Graduates so far

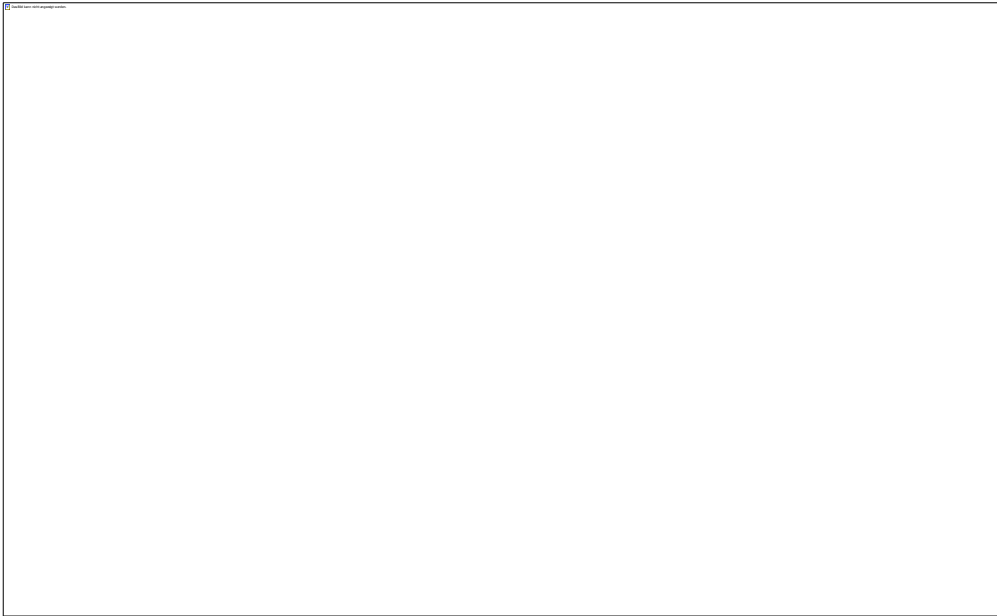
Digital Forensics and Information Security Research Cluster

Joel Eelu (2011)	Liina Tshilunga (2011)	Martha Shipanga (2012)	Paulus Shigwedha (2013)	Diana (2014)	Dea-da Ondeja kandjii (2014)	Moses N. Ndiikongela(2014)	Martin Kanyangela(2014)
Julia Nambili (2014)	Samuel Olusegun Jegede (2014)	Martin Hamukwaya (2015)	Barminas Steve Zeraua (2015)	Klaudia Uuzombala (2015)	Rodney Koujo(2015)	Konjeni Elago(2015)	Andrisia Aluteni (2015)
Christian Mingeli (2016)	Lucia Munango (2016)	Joseph Andreas(2016)	Ruben lipinge Stefanus (2016)	Rosetha Kays (2016)	Aliyu Dan Sharif Rabi (2016)	Adolf Samakupa (2017)	Aruni Kanyimba (2017)
Nathaniel Kanyinga (2017)	Joshua Kamwi (2017)	Ivy Mbala (2017)	Selestina Amandus (2018)	Victoria Ndafapawa Haidula (2018)	Lucas Mbambo Nekare (2018)	Mathew M. Mbwalala (2018)	Eunike T. Ndemwiimba (2019)
Pius Shambabi (2019)	Margaret Ndapewa Paulus (2019)	Petrus Kahare Sikongo (2019)	Marsela Rita (2020)	Joana Nkoma (2020)	Anène Hoff (2020)	Venessa Tjizumaue (2014)	Juso-Ilyod Ekandjo (2014)
Kafita Kalenikumwe (2015)	Stanford Musarurwa (2015)	Ben Mwashekuna (2016)	Elizabeth Shipena (2016)	Esther Shigwedha (2016)	Ndeshihafela Haita (2016)	Salmon Msati (2017)	Moses Masaku (2018)
lina-Marie Shiweva (2018)	Thomas Shomwele (2019)	Raimbert Namukuwa (2019)	Abisai Kamati (2019)	Diina Alfeus (2020)	Kleopas Kapalanga (2020)	Daymien Harris (2020)	Petrus Kambinda (2020)



..Honours Graduates so far

Digital Forensics and Information Security Research Cluster





Digital Forensics and Information Security Research Cluster

MSc Graduates so far

Apr. 2016

- Licky Erastus

Oct. 2017

- Cephas Pahla

Apr. 2018

- Stanford Musarurwa

Apr. 2019

- Diana Tjirare

Apr. 2019

- Eunike Nghihalwa

Apr. 2019

- Victoria Shakela

Apr. 2019

- Joris Ikany

2020

- Mamoqenelo Morolong

2021

- Jennyphar Kahimise

2021

- Paulus Iita Shigwedha

2022

- Lonnie Matsa

2022

- Eva-Lisa Nawa

2022

- Gabriel Musa

2022

- Naemi Gerson



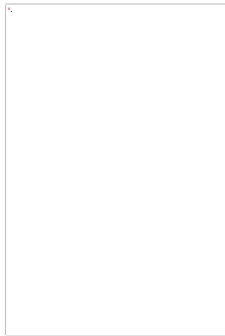
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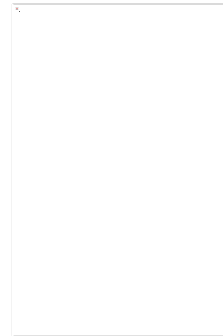


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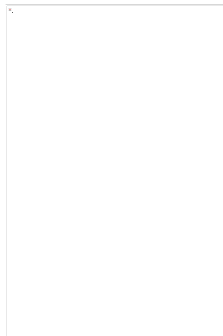
PhD Graduates this far



Dr Licky R. Erastus
(2019)



Dr Attlee M.
Gamundani (2019)



Dr Mercy
Chitauro(2020)



Digital Forensics and Information Security Research Cluster

Current MSc Students



Naemi
Gerson

Ester
Alumbungu

Eva Lisa
Nawa

Vicentius
Caley

Gabriel
Musa

Lonnie
Matsa

Juso Lloyd
Ekandjo

Raimbert
Namukuwa

Joshua
Kamwi

Hloaele
Komete

Hileni
Ihambo

Elizabeth
Shipena

Penehafo
Nghitamuka

Blessing
Sibanda

Auther
Makuvaza

Pius
Shambabi

Joao
Baptista

Naftali
Nakasole





Digital Forensics and Information Security Research Cluster

Current PhD students

Maria Kanyama

Ranganai Turugare

Timothy Dongo

Julius Silaa

Tendai Marengereke

Gabriel Nhinda

Mamoqenelo Morolong

Jennyphar Kahimise



Publications 2022

[Shihepo, E., Bhunu Shava, F., & Chitauro, M. \(2022\). Review of BYOD security awareness within Namibian enterprises. Presented at the 1st Zimbabwe Conference on Information Communication & Technology \(ZCICT\) 2022 \(9 to 10 November\).](#)

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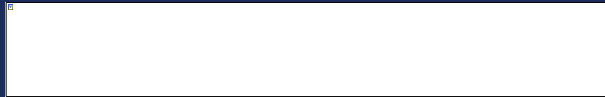
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City Resort, 19 – 21 October 2022.

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Rusere, K., Musarurwa, S., **Bhunu Shava, F.**, Maravanyika, M. (2022). Security Concerns on Social Media Customer-to-Customer Online Transactions. A Case of Namibia Facebook Users. Presented at IST-Africa 2022. May 2022., www.IST-Africa.org/Conference2022



Nhinda, G., & **Bhunu Shava, F.** (2021). Towards the use of Participatory Methods in Cybersecurity research in rural Africa: a grassroots approach. 3rd International Multidisciplinary Information Technology and Engineering Conference (IMITEC), 2021, pp. 1-7, doi: 10.1109/IMITEC52926.2021.9714649. <https://ieeexplore.ieee.org/document/9714649>

Shigwedha, P., & **Bhunu Shava, F.** (2021). Designing a Secure Integration Model for E-Governance Platforms. 3rd International Multidisciplinary Information Technology and Engineering Conference (IMITEC), 2021, pp. 1-6, doi: 10.1109/IMITEC52926.2021.9714556. <https://ieeexplore.ieee.org/document/9714556>

[Nawa, E., Chitauro, M. & **Bhunu Shava, F.** \(2021\). Assessing Patterns of Cybercrimes Associated with Online Transactions in Namibia Banking Institutions' Cyberspace," 2021 3rd International Multidisciplinary Information Technology and Engineering Conference \(IMITEC\), 2021, pp. 1-6, doi: 10.1109/IMITEC52926.2021.9714697. <https://ieeexplore.ieee.org/document/9714697>](https://ieeexplore.ieee.org/document/9714697)

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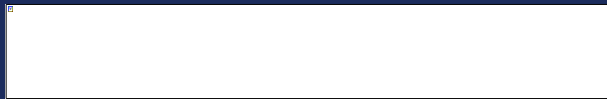
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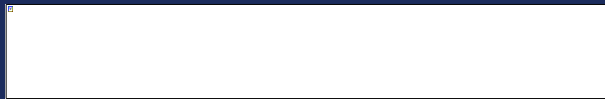
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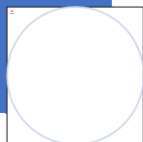


Digital Forensics and Information Security Research Cluster

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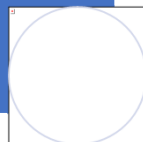
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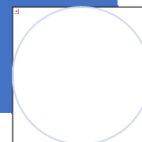
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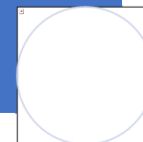
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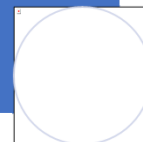
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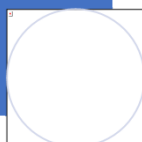
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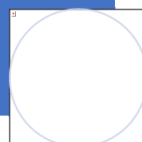
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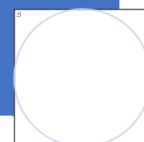
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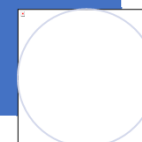
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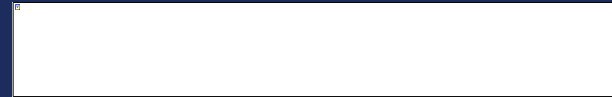
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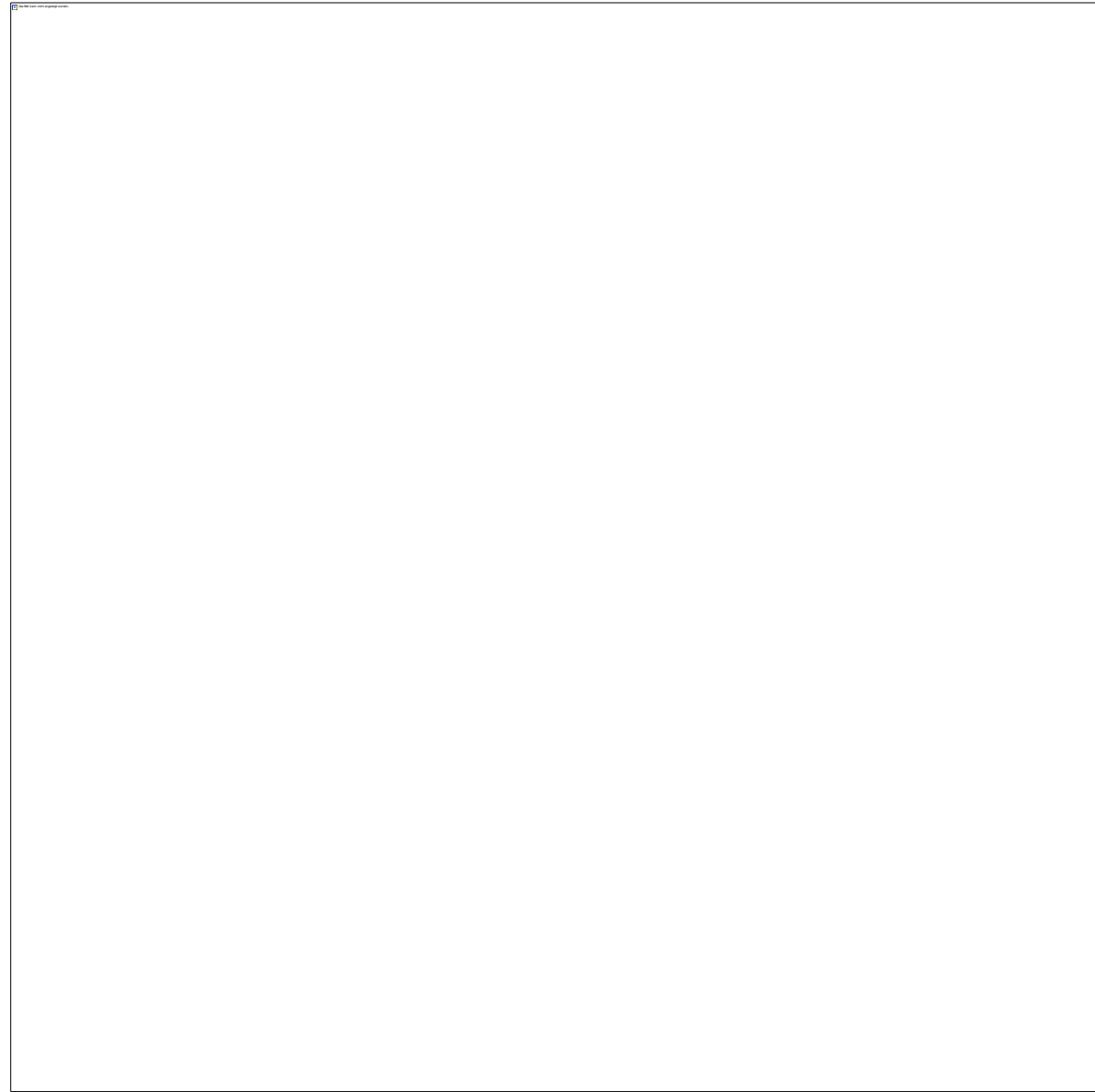


Digital Forensics and Information Security Research Cluster

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Learning objectives:

At the end of this session, participants will be able to:

- Recognize benefits of engaging social / scientific research.
define what a 'Research' is.
- List the major characteristics of research/ research proposal;.
- Describe steps of conducting a social / scientific research.
prepare a brief summary of a research proposal.
- Describe the main components of a research report.

Introduction

- Scientific / social research plays a very important role in our efforts to advance technological / knowledge envelope
- Research helps us create **new knowledge** and develop proper tools for the use of existing knowledge.
- Not only does it enable researchers/practitioners to diagnose problems, research also provides evidence for policies and decisions on socio-economic development

What is research?

Research is the systematic collection, analysis and interpretation of data to answer a certain question or solve a problem.

Characteristics of research:

- It demands a clear **statement of the problem**.
- It requires clear objectives and a plan (it is not aimlessly looking for something in order to come across a solution).
- It builds on existing data, using both positive and negative findings.
- New data should be *systematically* collected and analyzed to answer the original research objectives.

Research Proposal – Definitions

“... the meaning of *proposal* ... suggests looking forward, to what the researcher *plans to do in the future*. ...

The proposal lays out the **problem** for research, describes exactly **how** the research will be conducted, and outlines in precise detail the **resources** – both factual and instrumental – the researcher will use to achieve the **desired results**.”

Leedy & Omrod, Practical Research. 8th ed, p. 117, 115.

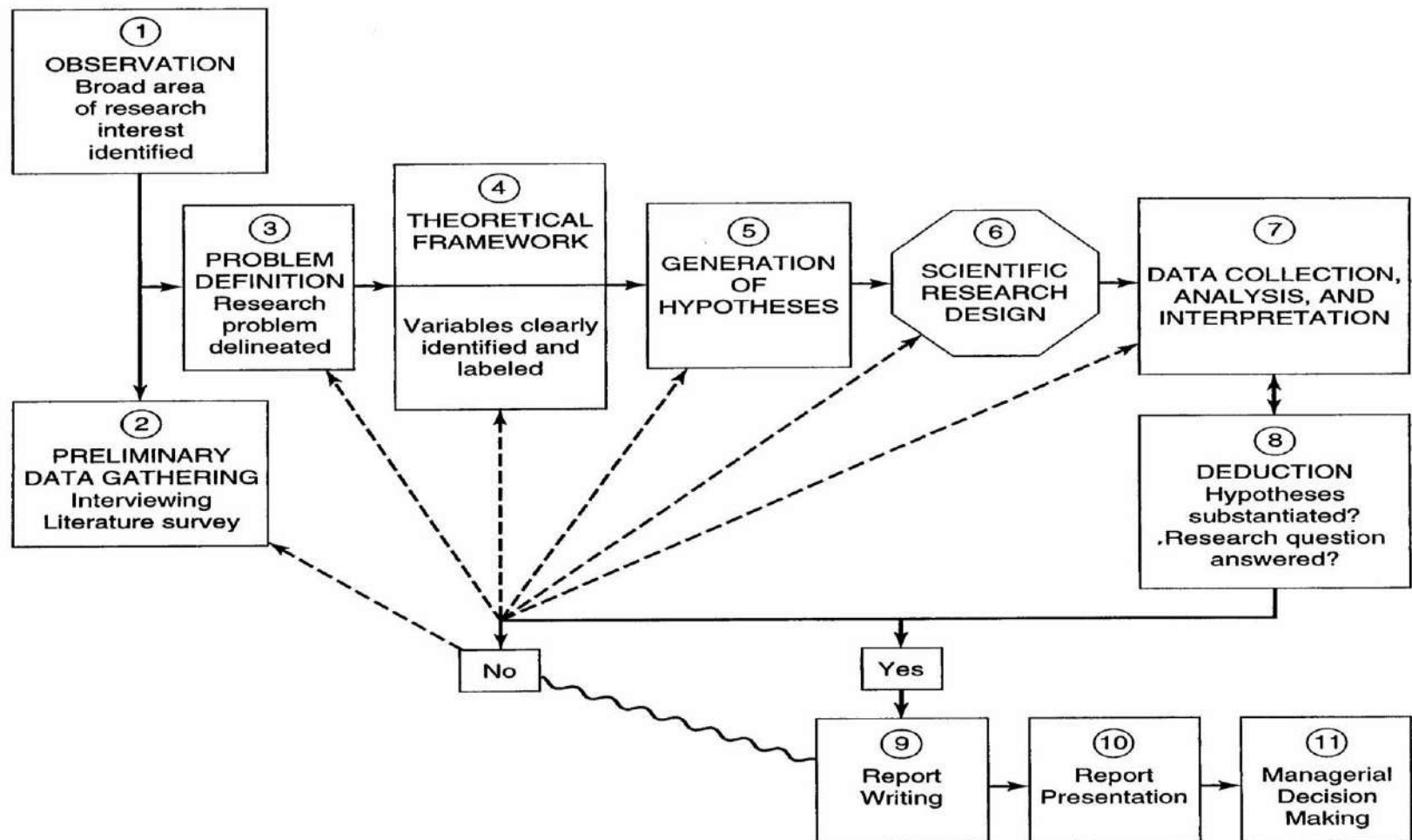
Essentially, to make the reader to understand :-

- What you are going to do
- Rational of the research
- Objectives of the research
- Methodology
- Expected output

What is a proposal ?

- An outline of the research **problem**
 - A statement of the significance of the problem
 - A review of related studies
 - A discussion of the procedures
 - Methods for data collection and analysis
 - A note about the study's limitations
- A good proposal should consist of the first three chapters of the thesis
 - It should :-
 - **Begin with an Introduction - statement of the problem/background information (Chapter 1)**
 - **A review of the Literature (Chapter 2)**
 - **Defining of the Research Methodology (Chapter 3)**

The Research Process For Applied and Basic Research



Steps of conducting a research:

A. Prioritizing and selecting a research topic

B. Review of literature and other existing information

C. Development of a Research Proposal

D. Implementation of study:

- i. Data collection
- ii. Data processing and analysis
- iii. Interpretation of results
- iv. Final report writing
- v. Presenting the results: Scientific, presentation at meetings, seminars, workshops or conferences, and presentation for administrators and policy-makers

Steps of conducting a research:

A. Prioritizing and selecting a research topic

Criteria for selecting a research topic:

1. **Relevance**: The topic you choose should be a priority problem.

Questions to be asked include:

- How large or widespread/depth is the problem?
- Who is affected?
- How severe is the problem?

2. Avoidance of **duplication**:

- Before you decide to carry out a study, it is important that you find out whether the suggested topic has been investigated before, either within the proposed study area or in another area with similar conditions.
- If the topic has been researched, the results should be reviewed to explore whether major questions that deserve further investigation remain unanswered.
- If not, another topic should be chosen

Steps of conducting a research:

B. Literature review

Why is it important to review already available information when preparing for a research?

- It prevents you from **duplicating** work that has been done before.
- It helps you to find out **what others have learned** and reported on the problem you want to study. This may assist you in refining your **statement of the problem**.
- It helps you to become **more familiar** with the various research methods that might be used in your study.
- It should provide you with convincing arguments for **why** your particular research project is needed.
- Learning what is known and unknown
- Learning how field of knowledge was developed (history)
- Showing you understand your field
- Confirming your own research is worthwhile
- Explaining how knowledge is developed over time
- Justifying your research
- Identifying how you will make a contribution

See Example,,,pg 31

Steps of conducting a research:

C. Development of a research proposal

Contents

- a. Title of the research
- b. Introduction: Background information and Statement of the research problem (Scientific justification for the study)
- c. Research objectives
- d. Research hypothesis
- e. Methodology
- f. Work plan
- g. Plan for utilization and dissemination of research results
- h. References
- i. Annexes

C. Development of a research proposal

a. Title of the research proposal

- A good title should be short, accurate, and concise
- It should make the central objectives of the study clear to the reader
- It is important to specify what population will be investigated, and where it will be conducted.
- **E.g. Factors Influencing e-Bidding Adoption in the Public Sector**

C. Development of a research proposal

b. Introduction (Background information and Statement of the research problem)

This section should convince the reader of the relevance of the study

It should provide enough background data for an outsider to understand the different aspects of the problem, or the different factors influencing the problem and the context in which it occurs.

Your review of available literature and reports should further illustrate why the problem is important, not only in your own working area but probably also beyond

Note: Be selective, remembering that this section serves to justify your study, not to display your ability to read literature.

See example..pg 1

C. Development of a research proposal

c. Research objectives

Research objectives are the goal to be achieved by a research.

Why should research objectives be developed?

The formulation of objectives will help you to:

- Focus the study (narrowing it down to essentials)
- Avoid the collection of data which are not strictly necessary for understanding and solving the problem you have identified
- Organize the study in clearly defined parts or phases
 - Properly formulated, specific objectives will facilitate the development of your research methodology and will help to orient the collection, analysis, interpretation and utilization of data.
 - How should you state your objectives?
- The general and specific objectives should be included as stated in the proposal

Example Objective To examine the User Factors and System Factors that Influence eBidding Utilization among Government Sourcing Officials in Malaysian Public Sector

Example Specific Objective:

1. To identify the variables that may influence the adoption of eBidding by government users
2. To examine the effects of the variables on the adoption of eBidding by government users
3. To propose a framework to analyze the adoption of eBidding by government users
4. To examine if some of the variables have moderating or mediating effects on the relationships established as stated in objective 2

The Use of a Theoretical Framework in a Research Proposal

The **theoretical framework** plays an important role in guiding the entire process of the research study

- **Theories** are constructed in order to explain, predict and master phenomena (e.g. relationships, events, or the behavior). In many instances we are constructing models of reality.
- A theory makes generalizations about observations and consists of an interrelated, coherent set of ideas and models.

Having examined the different kinds of variables that could operate in a situation and how the relationships among these can be established, it is **now possible** to see how we can develop the conceptual model or the theoretical framework for our research.

If the framework is logically sound and substantiated by previous research studies, there is a strong possibility that the predictions or hypotheses evolving from that framework will be supported

The Use of a Theoretical Framework in a Research Proposal

- The **variables** considered relevant to the study should be clearly defined.
- A **conceptual model** that describes the relationships between the variables in the model should be given.
- A **clear explanation** of why we expect these relationships to exist.
- A **conceptual framework** is used in research to outline possible courses of action or to present a preferred approach to an idea or thought.
- The **theoretical framework** elaborates the relationships among the variables, explains the theory underlying these relations, and describes the nature and direction of the relationships.
- The theoretical framework provides the logical base for developing **testable hypotheses**.
- see example pg 64

Example of Theories Employed in Theoretical Framework

E.g. The eBidding adoption study and the model proposed will be based on:

- ❖ **Unified Theory of Acceptance and Use of Technology (UTAUT)**
(Venkatesh et al., 2003)

An Information System (IS) framework for assessing an individual's intention to use an IS technology

- ❖ **Information System Success Model (Delone and Mclean, 2003)**

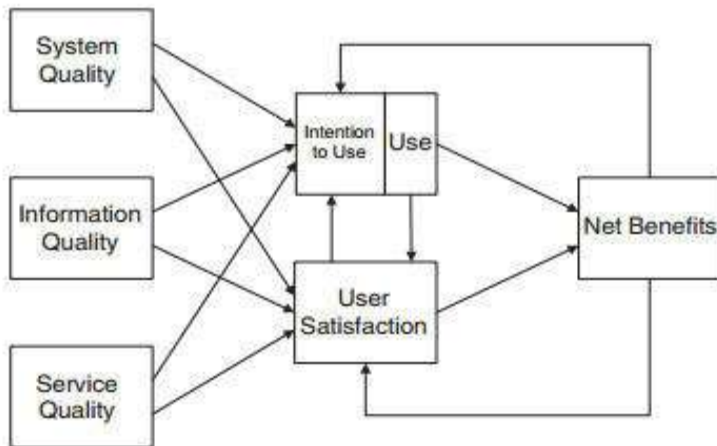
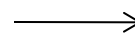
A system success can be evaluated in terms of information, system, and service quality; these characteristics affect the subsequent use or intention to use and user satisfaction

- ❖ **Personal Innovativeness in Information Technology (Agarwal and Prasad, 1998)**

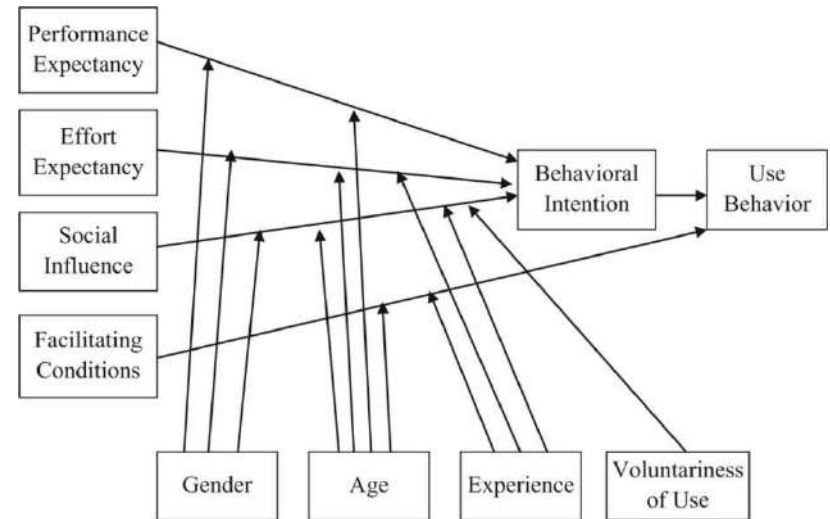
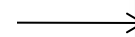
Domain-specific individual trait which reflects the willingness of a person to try out a new information technology

Example : Models related to Technology/Innovation adoption/acceptance/use

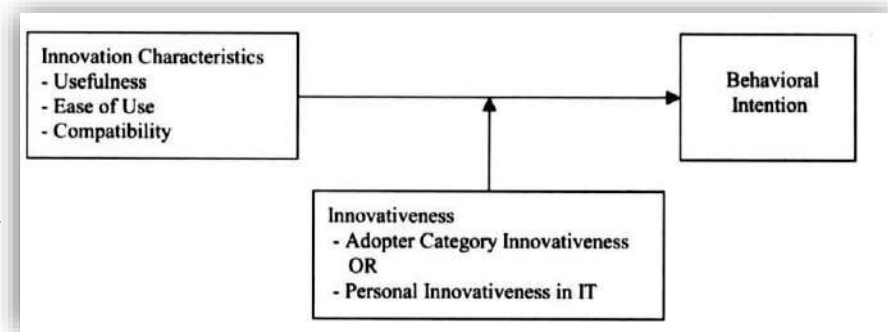
Unified Theory of Acceptance and Use of Technology (UTAUT) (Venkatesh et al., 2003)



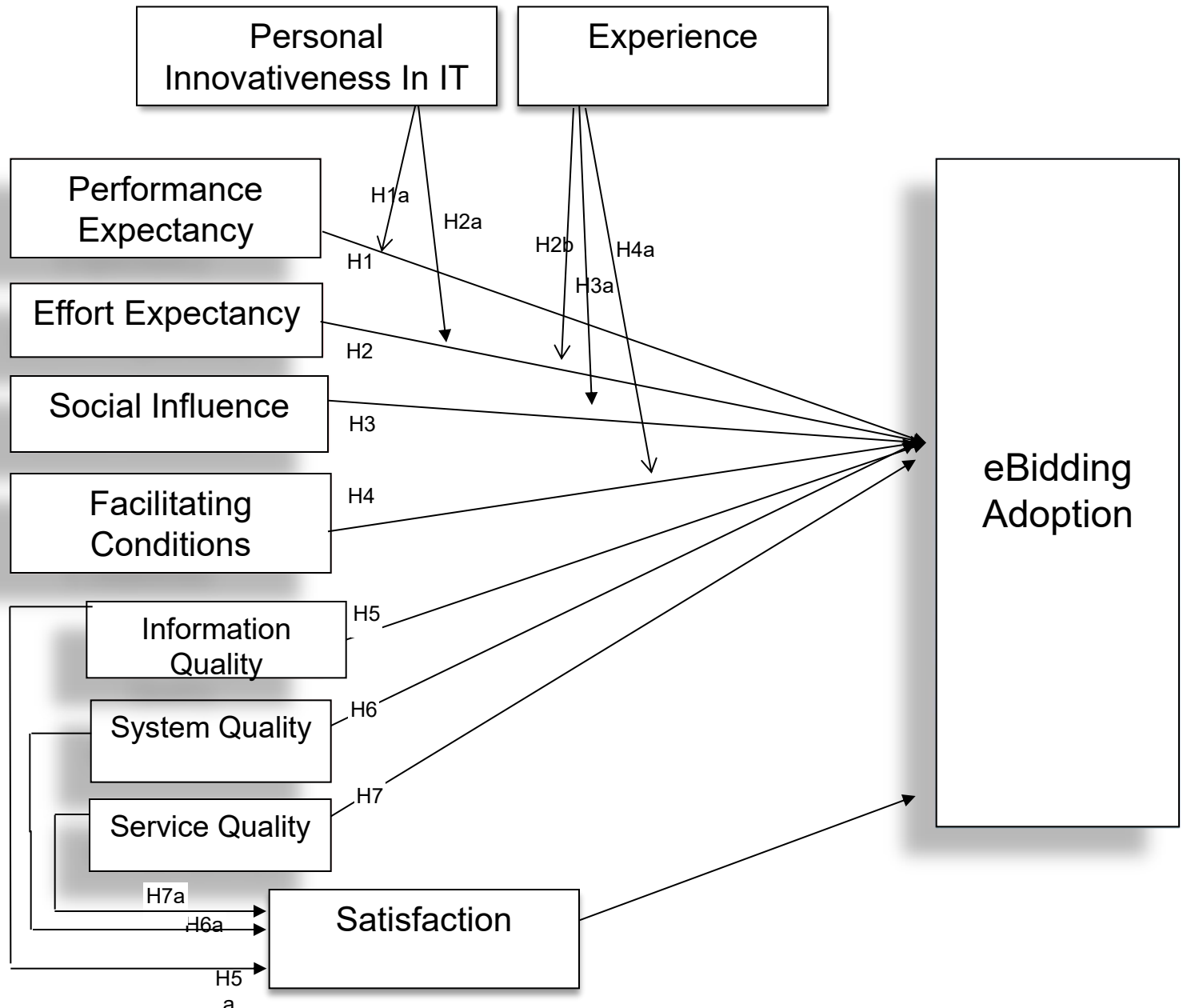
Moderator Model of Personal Innovativeness in Information Technology (PIIT) (Agarwal & Prasad, 1998)



IS Updated Success Model (Delone and Mclean, 2003)



Example :Proposed Research Model



C. Development of a research proposal

d. Research hypothesis

A hypothesis can be defined as a prediction or explanation of the relationship between : one or more independent variables (PREDISPOSING/RISK FACTORS) and one dependent variable (OUTCOME/CONDITION/DISEASE)).

- a logical relationship between two or more variables expressed in the form of a testable statement. A hypothesis, in other words, translates the problem statement into a precise, clear prediction of expected outcomes.
- It must be emphasized that hypotheses are not meant to be haphazard guesses, but should reflect the depth of knowledge, imagination and experience of the investigator must be based on established theory/model
- Therefore, in the process of formulating hypotheses, all variables relevant to the study should be identified.
- See example..pg 76 & 77

C. Development of a research proposal

e. Methodology

This section summarizes the most important points of the research design including:

Variables: It is necessary to identify the variables that will be involved in the research project being designed.

Independent (predisposing/risk factors/antecedents),

Dependent (outcome), **see example**

It is also necessary to specify whether these variables are Numerical (continuous/discrete), or Categorical (ordinal/nominal).

Operational definition of different study variables should be clearly phrased.

The methodology you followed for the collection of your data should be described in detail.

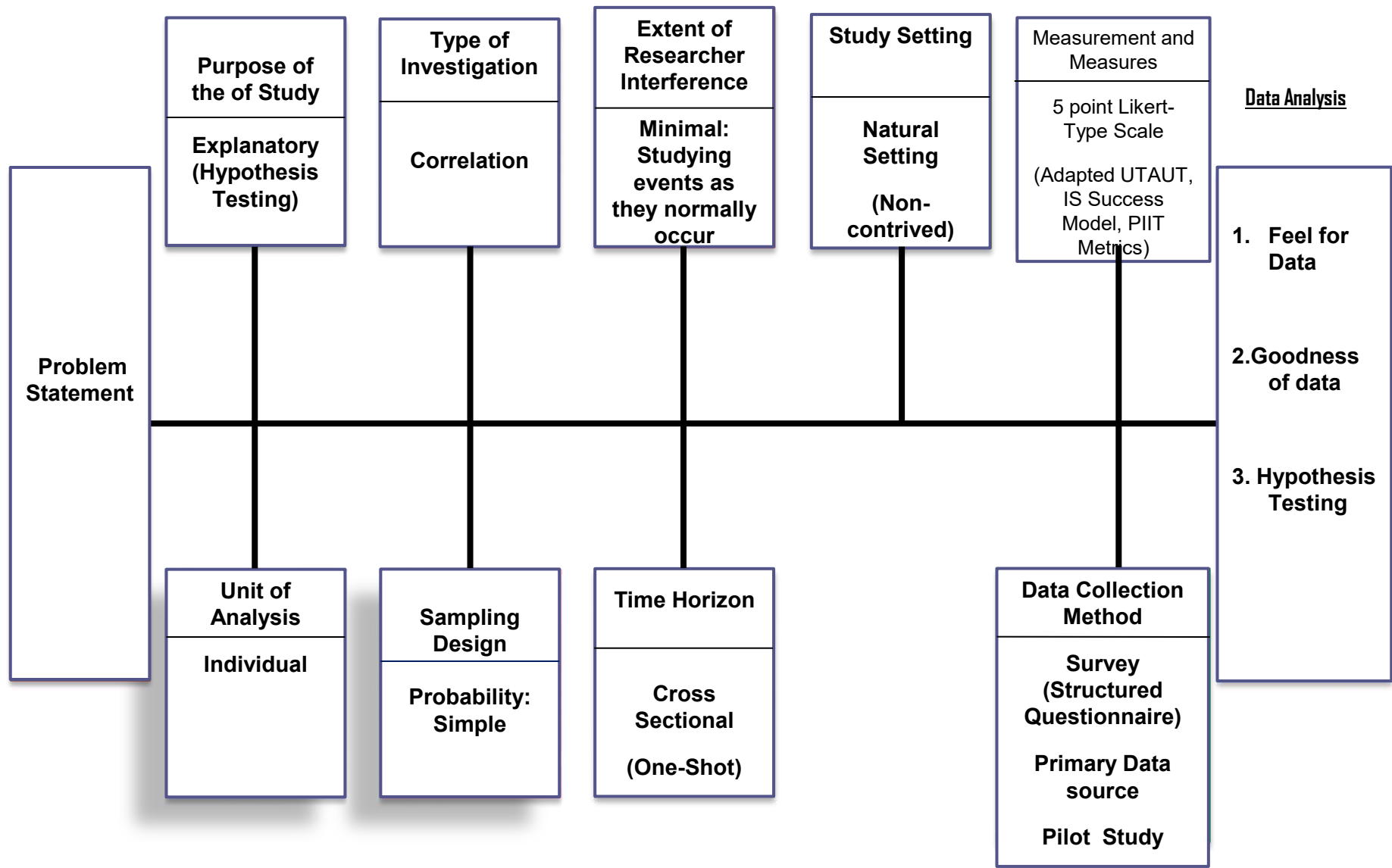
The methodology section should include a description of:

- the study type
- major study variables on which data was collected
- the study population, sampling method and the size of the sample
- data-collection techniques used
- how the data was collected and by whom
- procedures used for data analysis, including statistical tests **(see example)**

Example : Operationalizing Research Instrument

No	Section	Variables	No. of Statements	Scale of Measurement	References
1	A	Performance expectancy	7	Interval	Venkatesh et al., (2003)
2	B	Effort Expectancy	7	Interval	Venkatesh et al., (2003)
3	C	Social influence	7	Interval	Venkatesh et al., (2003)
4	D	Facilitating Conditions	7	Interval	Venkatesh et al., (2003)
5	E	System Quality	7	Interval	Delone and Mclean, (2003)
6	F	Information Quality	7	Interval	Delone and Mclean, (2003)
7	G	Service Quality	7	Interval	Delone and Mclean, (2003)
8	H	Satisfaction	7	Interval	Wixom and Todd, (2005)
9	I	Experience	5	Interval	Venkatesh et al., (2003)
10	J	Personal Innovativeness in IT	5	Interval	Agarwal and Karahanna, (2000)
11	K	Actual Use	4	Interval Nominal	Delone and Mclean, (2003)
12		Respondent's profile			

An Example of Research Design



Significance of the Study

- Why this study is significant?
- Why do it now?
- What will it contribute to the existing research literature?
- What implications your findings may have?
- Who will benefit from it?

[Scholarly acceptability] **See Example**

...References

The references in your text can be numbered in the sequence in which they appear in the report and then listed in this order in the list of references

Harvard Citation or author/date system references made in the text are listed in alphabetical order by the surname of the first author in the bibliography at the end of the text.

e.g.

Reference within essay: In his survey of the mating habits of frogs, Bloggs (1998) refuted that ...

Entry in bibliography:

Bloggs, B. 1998. Frogs and their mating habits . London: Tadpole Press

The Vancouver System or footnotes/endnotes or numeric system; references are numbered consecutively in the order in which they are first cited in the text.

e.g.

Reference within essay:

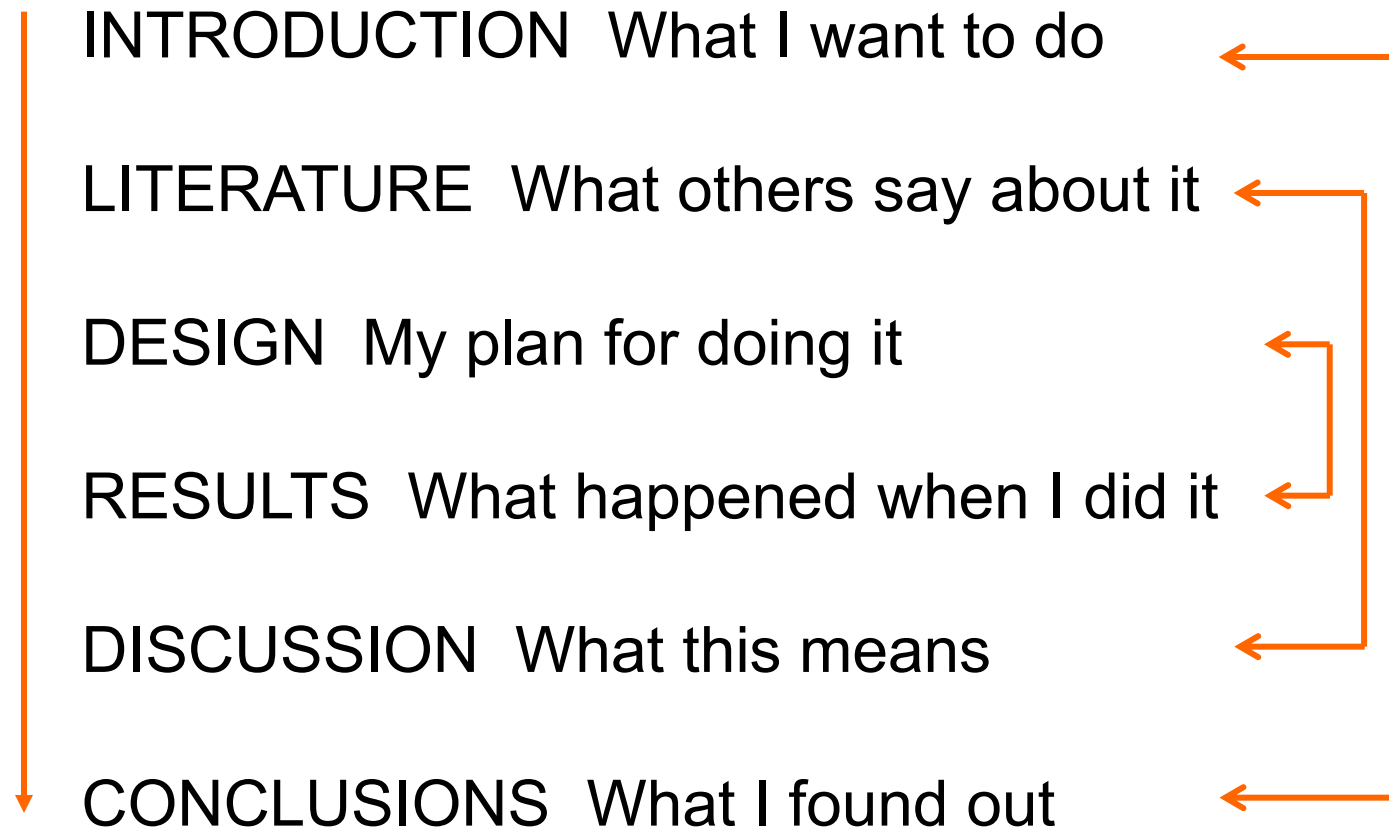
In his survey of the mating habits of frogs, Bloggs (1) refuted that ...

Entry in bibliography:

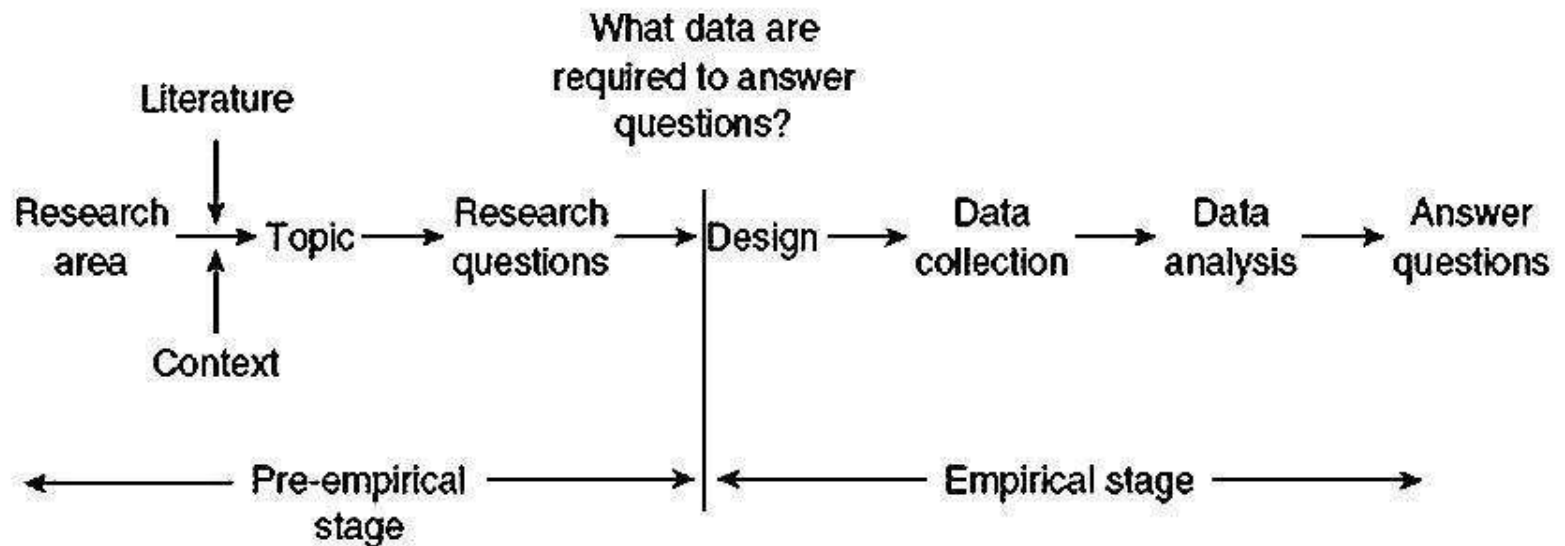
1. Bloggs B. Frogs and their mating habits. London : Tadpole Press; 1998

Thank you

The classical thesis structure



Simplified model of research



Punch, K. F. (2007). *Developing Effective Research Proposals* (2 ed.). Los Angeles, CA: SAGE Publications.

Writing a Thesis

.....Overall Main Components of a Thesis :

- I. Title or cover page
- II. Abstract
- III. Introduction
- IV. Objectives
- V. Methodology
- VI. Research results (findings)
- VII. Discussion
- VIII. Conclusions
- IX. Recommendations
- X. References
- XI. Annexes or appendices (data collection tools, tables)

Main Components of a Thesis

- **Chapter 1 : Introduction**

- Background of the study
- Overview of research subject
- Broad to specific
- Problem statements
- Objectives of study
- Practical implications
- Theoretical contributions
- Study Limitations

- **Chapter 2: Literature Review**

- Past research related to study
 - Global & local publications – journal and books
- Overview of theories related to research subjects – at least 5 or 3 key theories
- Review and critique

Main Components of a Thesis

- **Chapter 3: Theoretical Framework**

- Introduction
- Conceptual development
 - Models/theories comparisons
 - Which models as base of proposed model
 - Proposed research model
- Hypotheses development
 - Independent variables
 - Dependent variables
 - Moderating / mediating variables

- **Chapter 4: Methodology**

- Introduction
- Population and sampling
- Data Collection and Sampling procedure (target ; sampling population ; size)
- Questionnaire design
 - Measurement of variables
 - Research design
 - SEM- Measurement model

Main Components of a Thesis

- **Chapter 5 : Data Analysis & Discussion**

- Introduction
- Feel for data
- Frequency distribution of demographics
 - Positions ; locations ; years of working ; gender
- Testing Goodness of Data
 - Reliability and validity
 - Results from SEM analysis
 - Significant or not significant related to hypotheses

- **Chapter 6 : Conclusions**

- Introduction
- Discussions on results – conform to findings in literature ; if not why..?
- Implications of Research Findings
 - Theoretical Implications
 - Practical Implications
 - Policy Implications
- Recommendations for Future Research
- Limitations

...CONCLUSIONS

- The conclusions should follow logically from the discussion of the findings.
- As the discussion will follow the sequence in which the findings have been presented (which in turn depends on your objectives) the conclusions should logically follow the same order.

....RECOMMENDATIONS

- The recommendations should follow logically from the discussion of the findings.
- Recommendations may be summarized according to the groups towards which they are directed, for example:
 - ↳ policy-makers / stakeholders
 - ↳ program managers/implementing officials at different levels
 - ↳ potential clients
 - ↳ the community at large

....Plan for utilization and dissemination of research results

- The proposal should indicate what reports or other means of disseminating research findings are planned.

- Any or all of the following are appropriate for disseminating the results of the study:

Progress reports

Final report

Publications

Seminars, workshops, and conferences

Discussion with policymakers and program managers

....ANNEXES OR APPENDICES

The annexes should contain any additional information needed to enable professionals to follow your research procedures and data analysis.

Examples of information that can be presented in annexes are:

- Tables referred to in the text but not included in order to keep the report short.
- Interview schedule/ questionnaires (and/or other data collection tools).
- Informed consent form
- Institutional/Ethical approval for the study

Further Readings

1. Creswell, J. W. (2009). *Research Design: Qualitative, Quantitative, and Mixed Methods Approaches*. Los Angeles, CA: SAGE, p. 97–127.
2. Hernon, P. (2007). Editorial: what is problem statement? *Library & Information Science Research*, 29, 307–309.
3. Hernon, P., & Metoyer-Duran, C. (1993). Problem statements: An exploratory study of their function, significance, and form. *Library & Information Science Research*, 15, 71–92.
4. Metoyer-Duran, C., & Hernon, P. (1994). Problem statements in research proposals and published research: A case study of researchers' viewpoints. *Library & Information Science Research*, 16, 105–118.
5. Nitecki, D. A. (2010). Guest editorial: Finalizing a research paper—findings through conclusion. *Library & Information Science Research*, 32, 1–3.
6. Sproull, N. L. (1988). *Handbook of research methods: A guide for practitioners and students in the social sciences*. London: Scarecrow Press, p. 140.
7. Teddlie, C., & Tashakkori, A. (2009). *Foundations of Mixed Methods Research : Integrating Quantitative and Qualitative Approaches in the Social and Behavioral Sciences*. Los Angeles, CA: SAGE, p. 120–124.

BREAK FOR JUMAAT PRAYER

UNESCO Chair on Secure High-performance Computing for Higher Education and Research

2022

Smart and Secure Environment (SSE)

AT A GLANCE

Prof Dharm Singh Jat

UNESCO Chairholder

E:unescochair-shpcher@nust.na

<http://unescochair-shpcher.nust.na/>



The UNESCO Chair on Secure High-performance Computing for Higher Education and Research established at NUST is part of the UNESCO-NUST Partnership within the UNESCO/UNITWIN Science Chairs Programme.

UNESCO Chair on Secure High-performance
Computing for Higher Education and Research





Goal of the UNESCO Chair

The purpose on the UNESCO Chair on Secure High-performance Computing for Higher Education and Research shall be to

- Promote an integrated system of research, training, information and documentation on digital technologies.**
- It will facilitate collaboration between high-level, internationally-recognised researchers and teaching staff from the University and other institutions in Namibia, as well as elsewhere in Africa, Europe and in other regions of the world.**

Specific objectives

- 1. Establish the high-level Super Computing facility in Namibia, which is relevant for the development of Namibian society and the wider region;**
- 2. Establish programmes leading to a Ph.D. degree in the related field of Smart and Secure Environment (SSE), while assuring gender equality and offering a high-level training programme for graduate and postgraduate students;**
- 3. Promote research in the field of SSE in Namibia, the Southern African Development Community (SADC) region and in other parts of the world and exchange of lecturer across borders;**



Specific objectives

- 4. Create a platform for national and international exchanges of ideas through international and regional conferences to be held on a regular basis, as well as workshops involving leading computer scientists;**
- 5. Collaborate with UNESCO, Category 2 centres under the auspices of UNESCO and other UNESCO Chairs to develop relevant programmes and activities.**

The significance of the Chair and expected outcomes to national development

The UNESCO Chair on "SHPCR" will contribute to accelerate the development of the Namibian society and building the foundations for technology by creating a critical mass of high-level expertise in the related field of the Chair.

The UNESCO Research Chair for SHPCR will develop

- (i) supercomputing facilities for researchers**
- (ii) big data analytical techniques**
- (iii) smart communication and**
- (iv) cyber security solution
for society.**



Industry 4.0: Introduction

**The fourth Industrial Revolution
emerges from AI and the Internet of
Things**





4th Industrial Revolution -Introduction

The key essence of Industry 4.0 Concept is the enhancement of efficiency in the process of industry by way the utilization of connectivity, computing, data, IoT and Artificial Intelligence.

The five characteristics of Industry 4.0

- 1. Industry 4.0 is interconnection**
- 2. Industry 4.0 is data**
- 3. Industry 4.0 is Integration**
- 4. Industry 4.0 is Innovation**
- 5. Industry 4.0 is transition**

Segment of Enterprise in Industry 4.0

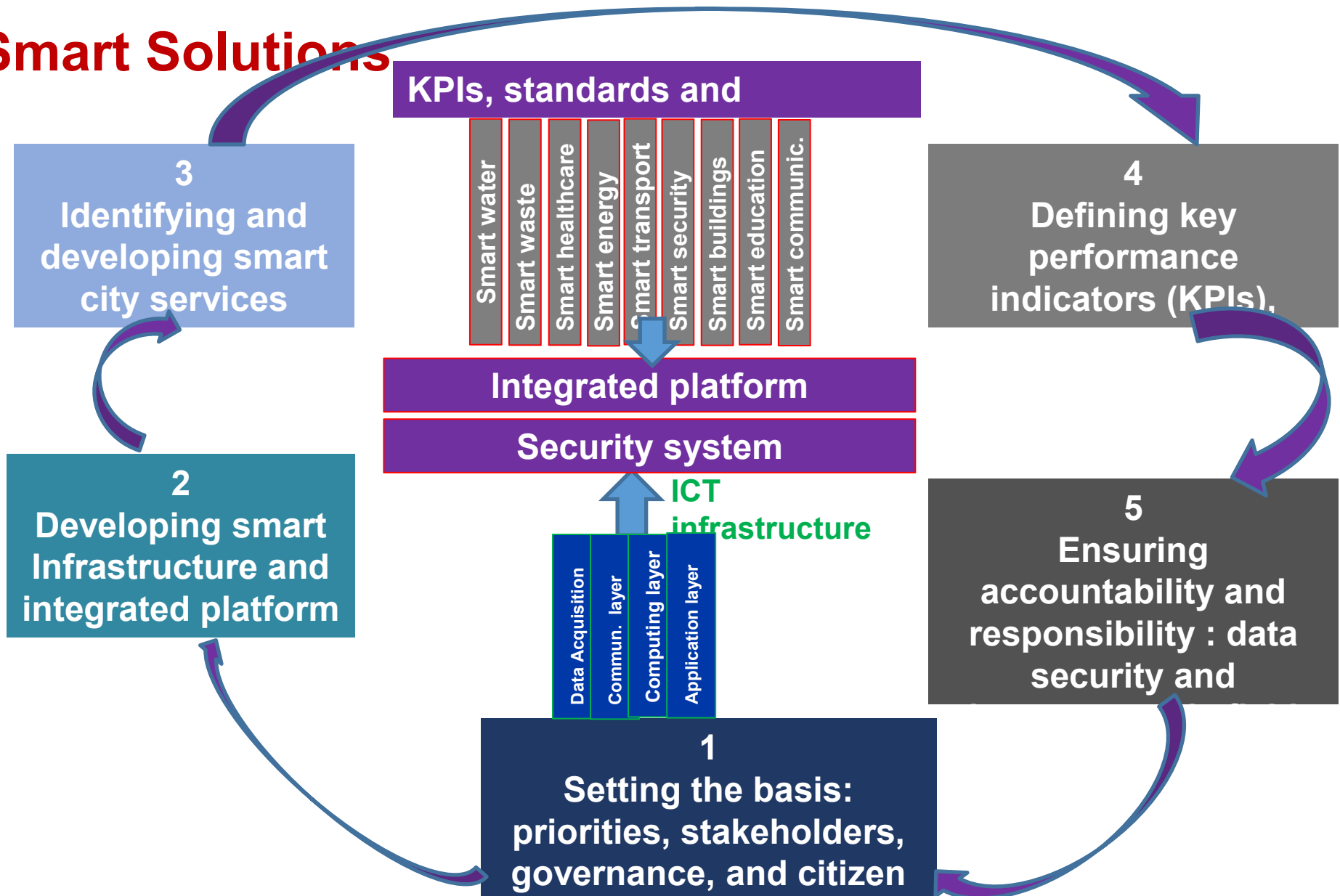
Technology Provider

- Artificial Intelligence (AI)
- Internet of Things
- Edge Computing
- Blockchain
- Virtual & Augmented Reality
- 3D Printing
- Robotics / Automation

Infrastructure Provider

- Internet Service Provider (ISP)
 - HPC System
 - Cloud Computing
 - Telecommunication
-

Smart Solutions



Smart Solutions

E-Governance and Citizen Services

- Public Information and Grievance Redressal
- Electronic Service Delivery
- Citizen Engagement
- Citizens-City's Eyes and Ears
- Video Crime Monitoring

SMART Health and Education

- Tele-Medicine
- Remote Teaching and Learning

Waste Management

- Waste to Energy and Fuel
- Waste to Compost
- Waste Water to be Treated
- Recycling and Reduction of Construction and Demolition (C&D) Waste

Water Management

- Smart Meters and Management
- Leakage Identification, Preventive maintenance
- Water Quality Monitoring

Energy Management

- Smart Meters and Management
- Renewable Sources of Energy
- Energy Efficient and Green Buildings

Urban Mobility

- Smart Parking
- Intelligent Traffic Management
- Intelligent Multi-Model Transport

HPC-AI infrastructure...

- The HPC and AI system will strengthen application development of packages such as
 - ✓ **Bioinformatics, Climatology, Astrophysics, Computational Fluid Dynamics, Computational Chemistry, Molecular Dynamics, Finite Element Analysis, Agriculture etc.,**

The HPC facility has significantly contributed to the fight against the COVID-19 pandemic since mid-March 2020 through faster simulations, medical imaging, and forecasting.



HPC: Applications **PARAM!Arub System**

Machine Learning / Deep Learning Application Development

Most of the popular python based machine learning/deep learning libraries are installed on PARAM!Arub system.

Weather Research and Forecasting (WRF)

The Weather Research and Forecasting (WRF) Model is a next-generation mesoscale numerical weather prediction system designed to serve both operational forecasting and atmospheric research needs.

Code Aster

Code_Aster is a free and open source software package for finite element analysis, and numerical simulation in structural mechanics.

Ferret

Ferret is an interactive computer visualization and analysis environment designed to meet the needs of oceanographers and meteorologists analyzing large and complex gridded data sets.

HPC: Applications

Abinit 9.0.2

ABINIT is a package whose main program allows one to find the total energy, charge density and electronic structure of systems made of electrons and nuclei (molecules and periodic solids) within Density Functional Theory (DFT), using pseudopotentials and a plane wave basis.

Openmx

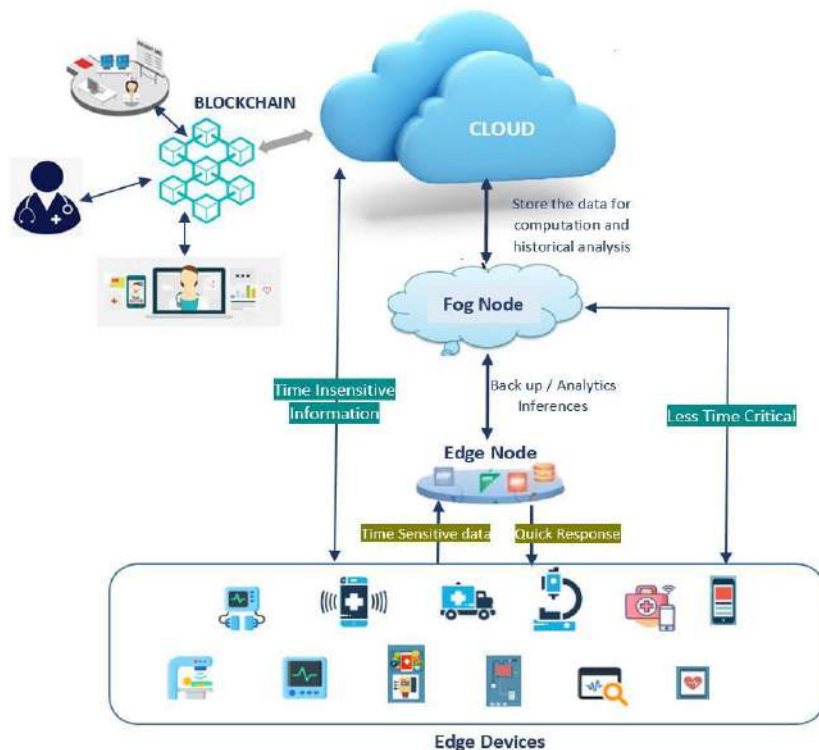
OpenMx is free and open source software for use with R that allows estimation of a wide variety of advanced multivariate statistical models. OpenMx consists of a library of functions and optimizers that allow you to quickly and flexibly define an SEM model and estimate parameters given observed data.

OOFEM

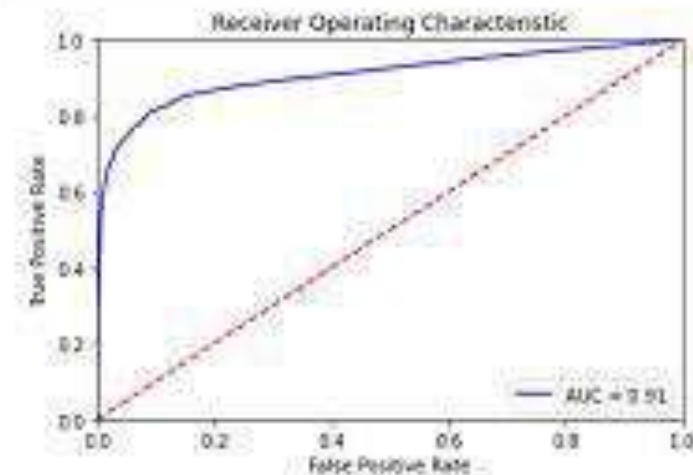
OOFEM is parallel, object-oriented finite element code for solving mechanical, transport and fluid mechanics problems.

Ongoing PhD Research

Design and implementation of blockchain-based secure edge computing paradigm for time constraint applications

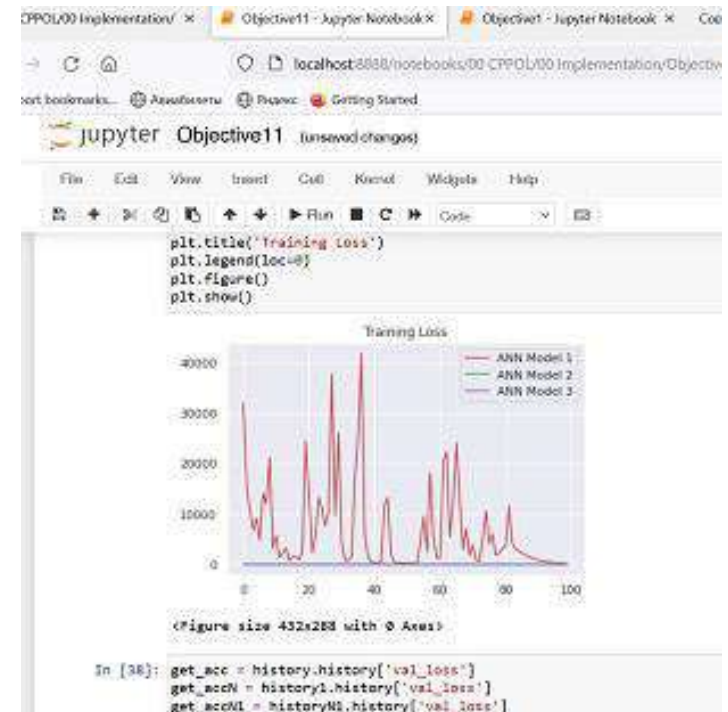
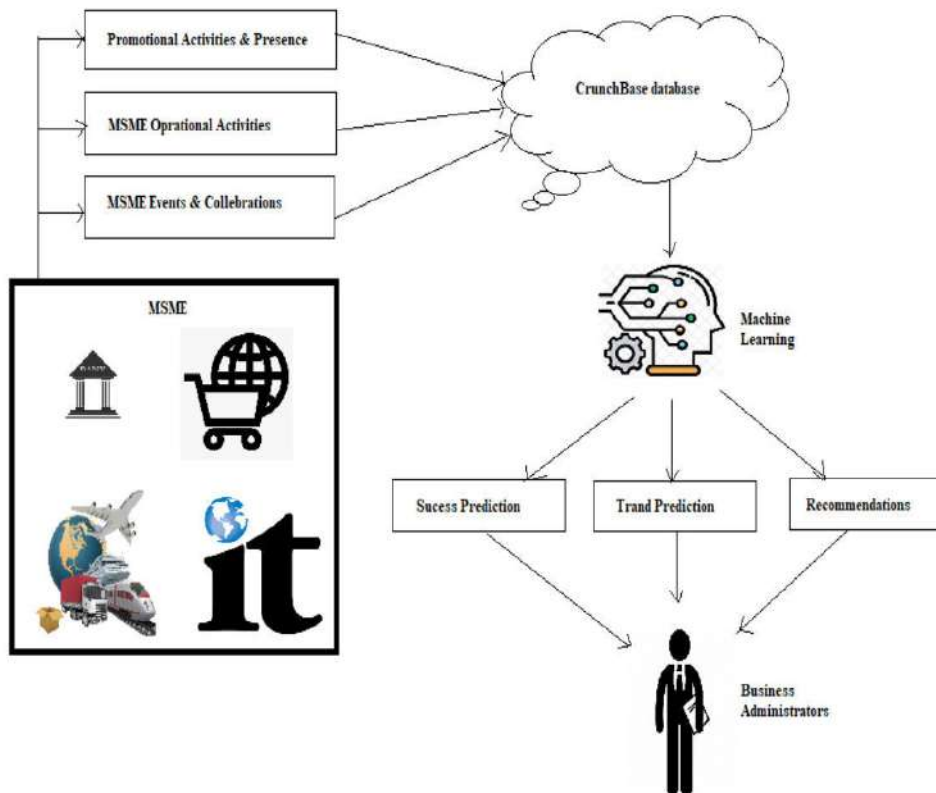


```
plt.title('Receiver Operating Characteristic')
plt.plot(fpr, tpr, 'b', label = 'AUC = %0.2f' % roc_auc)
plt.legend(loc = 'lower right')
plt.plot([0, 1], [0, 1], 'r--')
plt.xlim([0, 1])
plt.ylim([0, 1])
plt.ylabel('True Positive Rate')
plt.xlabel('False Positive Rate')
plt.show()
```



Ongoing PhD Research

Design and Develop a Prediction Model for Improving Start-Up Success Using Machine Learning



Ongoing Master Research

- 1. Design and Development of System for Post Infection Behaviour Analysis**
- 2. Design of a Security Threats Identification Model for Internet of Things (IoT) Smart Office Devices**
- 3. Enhancing the Effectiveness of an Email Phishing Awareness Tool for Organisations in Namibia**
- 4. A Chain of Custody Method for Preserving Volatile Digital Evidence for Admissibility in Namibian Courts**
- 5. Designing an Email Forensic Readiness Architecture for the Banking Sector**
- 6. Designing an improved secure user authentication system for the public cloud infrastructure as a service in a hybrid cloud**



Work Package 5: Development and application of open-source H2-Nexus-System planning tool

Objectives

- Development of an open-source software tool that allows users to simulate and optimize the design of H2-Nexus-Systems (back-end).
- Customize the sizing tool for the purpose of decentral GH2 feasibility studies (front-end) to allow easy application and usage of tool for a broad stakeholder group.
- Show economic and ecological benefits of H2-Nexus-Systems by simulation of case studies with planning tool.

Enhancing energy system modelling: <https://oemof.org/>



Project: Green Hydrogen Energy-Water-Food Nexus Model for Economic Prosperity

Work Package 6: Development and Application of Digital Technologies for H2-Nexus System

Objectives

1. Design and Develop of an efficient IoT-enabled Real-Time Monitoring System for Green Hydrogen Sensors.
2. Design and develop a prediction model for improving key performance indicators of hydrogen production plant using Artificial Intelligence.
3. To develop HPC application for electronic-structure calculations and materials modelling at the nanoscale and processes for hydrogen generation and storage, and for future generations using HPC applications.
4. Use high performance computing facilities (HPC) to develop high strength and corrosion-resistant materials.
5. To develop a reference model as a guideline to execute the digital twin for the H2-Nexus System To develop AI-Enabled Digital Twins Technology for the H2-Nexus System.
6. To demonstrate the impact of the proposed digital twin Technology through a real-world Green Hydrogen manufacturing plant.

UNESCO SHPCHER Team



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Dr Attlee M. Gamundani



Mr Arpit Jain





Conclusion

-
- The intelligent Edge is a foundation for future digital business and its ecosystems and by bringing powerful computing capabilities closer to the edge where data originates and processed with machine learning helping to accelerate the Fourth Industrial Revolution.
 - Edge computing is still in its early stages and has potential to discover the way for more efficient computing technology for enabling future Internet of Things .
 - Edge Computing using Blockchain technology paradigm may provide data filtering at the edge, security, ensure the data integrity, and privacy with the help of decentralised and distributed ledger of trust.
 - The connectivity of smart city and Industry 4.0 may create huge opportunities to join the physical and the digital world and it can develop the public and private sector's effectiveness, expand financial benefits and improve the standard of living.



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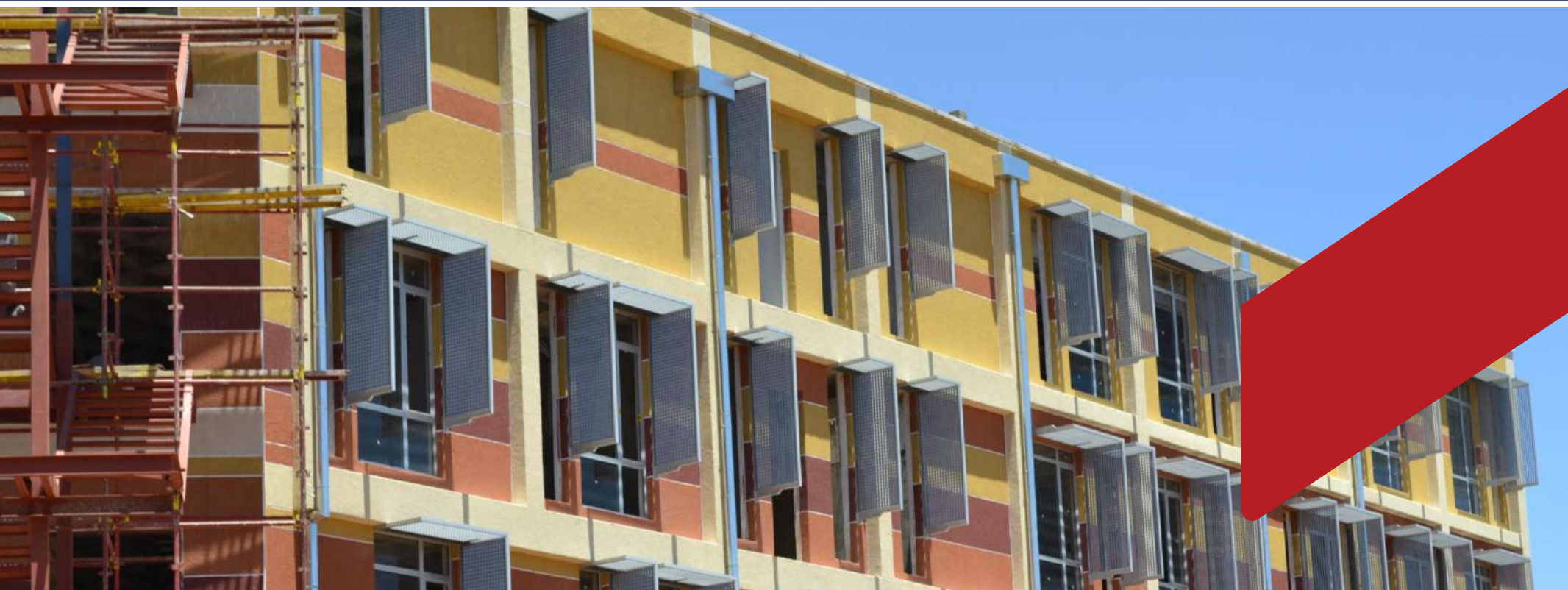
Faculty of Computing and Informatics

Defining a Research Problem

Dr Irja Shaanika

Dept. of Informatics

Date: February 2023





- Constructing the research problem
- Defining your research objectives
- Defining your research questions
- Defining your Research Topic



What is a research Problem

A paragraph that conveys the issues and context that gave rise to the problem.

According to Bryman(2007)

” it’s a statement about an area of concern, a condition to be improved upon, a difficulty to be eliminated, or a troubling question that exists in theory or in practice that points to the need for meaningful understanding and deliberate investigation ”

A research Problem Must be :

- Direct
- Specific
- Clear and Well defined



Challenges in defining the problem statement

- The problem is either too broad
- Either too narrowly
- Determining the research gap and locating it into the existing literature



Research Problem Formulation

1. Identify your interest area.
2. What is the problem?
3. Where does it exist?
4. Who does it affect?
5. How is it a problem?
6. What are the implications



Defining the Research Problem

The problem statement must state what the problem is –AND NOT ONLY PROVIDE a narrative of challenges and solutions.

“the application of the implemented NAMCIS has presented challenges when it comes to the amalgamation of reports and statistics of court data as well as the administration and maintenance of the systems. The challenges are encountered when analysing and reporting data from different locations which is time-consuming and cumbersome, and often leads to errors and inaccurate information (Shi et al., 2019).”

A research problem statement must be stated as the absence of, the lack of, the inadequate of, the insufficient of, the insufficient of, the deficiency of, etc

“The lack of integration of Namibian court information system across all lower courts in Namibia has led to inconsistency in decision-making as a result of data inaccuracy and data unreliability.”



Research Problem = Background + Problem + Implications

- **Background Introductory to the subject**

Organisations from various industries have become dependent on information systems to automate and enable business processes' efficiency and effectiveness (Yahaya et al., 2015). Thus, in Namibia, the lower courts have also implemented an information system known as the Namibian court information system (NAMCIS).

- **State the problem**

However, the lack of integration of the Namibian court information system across the lower courts has led to inconsistency in decision-making as a result of data inaccuracy and data unreliability. Moreover, the lack of integration contributes to the cumbersome systems administration of the court information systems due to repetitive technical maintenance and systems administration tasks.

- **Consequences of the problem**

The application of the implemented NAMCIS has presented challenges when it comes to the amalgamation of reports and statistics of court data as well as the administration and maintenance of the systems. The challenges are encountered when analysing and reporting data from different locations which is time-consuming and cumbersome, and often leads to errors and inaccurate information (Shi et al., 2019). From the perspective of system administration, the systems are administered and maintained individually, an exercise which is time-consuming and costly as activities such as fixing bugs, upgrades, running backups and user-accounts administration are carried out separately and repetitively for each system.



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The proposed solution

Thus, this study proposed a model that can be used to guide and govern the integration of the Namibian court information system. Otherwise, the problem and challenges articulated above will persist.



Citations in Problem Statement

Very Important – Support your problem statement with relevant citations.

e.g Few SME's receive funding from Development of Bank of Namibia. The 2022-2023 BON financial report indicates that of the 50 applications received, only 4 were successfully funded(BON Report,2023).



Defining Research Aim

- The comprehensive goal/solution to the problem identified
- The researcher must determine what is the overall long term by describing the overall purpose (what should be accomplished) in general terms.

“to design a model that will guide and govern the integration of the Namibian lower courts' information systems”



Defining Research Objectives

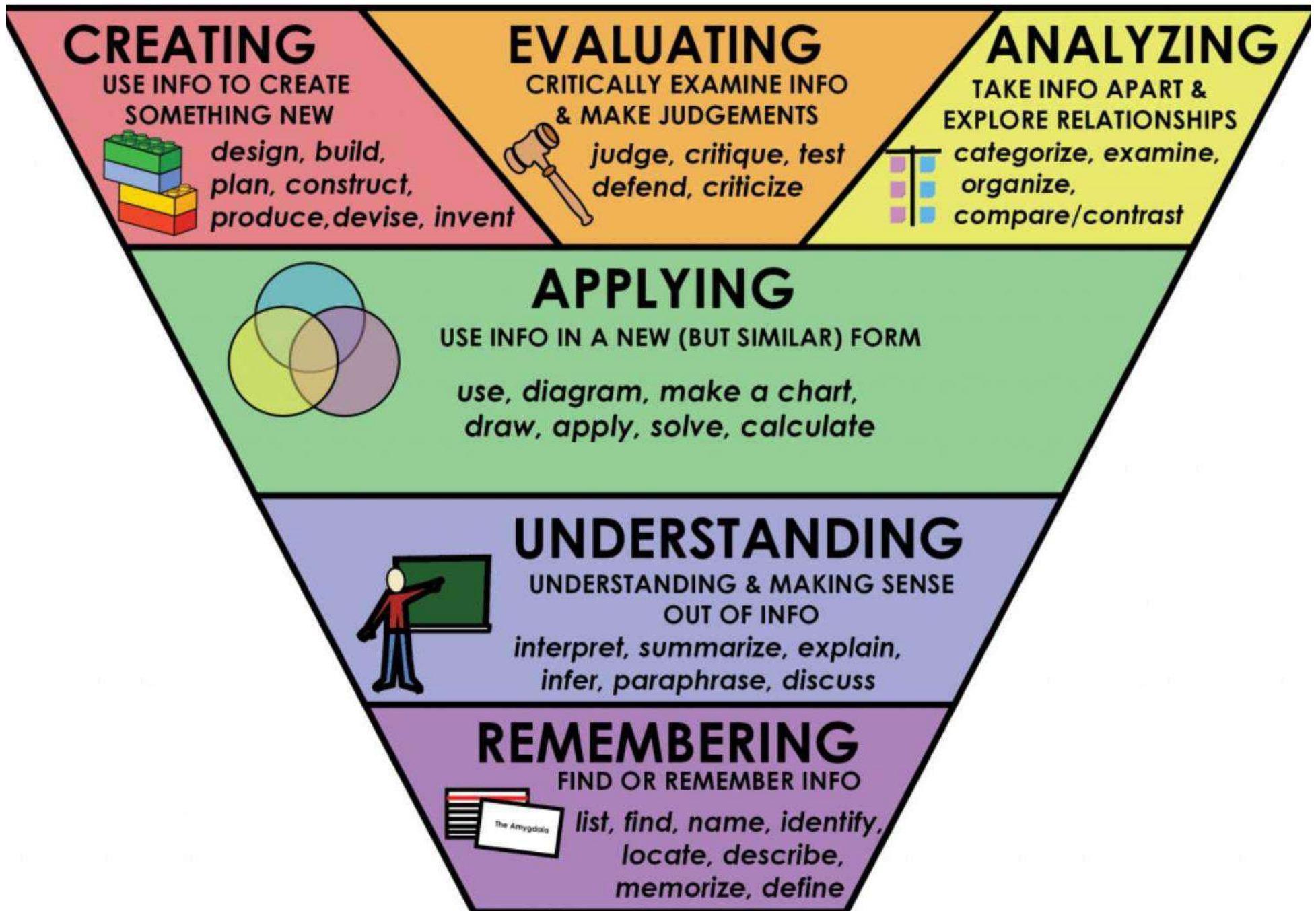
How will the defined aim be achieved ?

Milestones that need to be set towards resolving the problem.

Characteristics:

- Cover all aspects of the problem,
- Specific,
- Ordered in a logical sequence,
- Achievable, take into consideration the available resources, including time,
- Mutually exclusive of each other.
- They should also be stated using action verbs
 - Use Bloom Taxonomy

BLOOM'S TAXONOMY





Example of Objectives

- i. To examine the state of court information systems integration in Namibian lower courts.
- i. To examine factors that could impact the integration of the Namibian lower courts information system.
- ii. To investigate how the integration of Namibian Lower Courts' information systems will improve data reporting and systems administration.



Defining Research Questions

- Directly related to the Research Objectives
 - Matching to the research Objectives
 - Phrased in the form of a question
-
- What is the state of court information systems integration in Namibian lower courts?
 - What are the factors that could impact the integration of the Namibian lower courts information system?
 - How will the integration of Namibian Lower Courts' information systems improve data reporting and systems administration?



Objectives to Research Questions

Aim:	Main Question:
To develop an ICT framework that will support patient-centric healthcare services in Malawi.	What are the components of an ICT framework that will support a patient-centric approach in Malawi?
1. To identify the challenges underlying the healthcare ICT development in the public sector in Malawi.	1. What are the challenges underlying the healthcare ICT development in the public sector in Malawi?
2. To identify ICT-related factors that can support the patient-centric healthcare in Malawi.	2. Which are the ICT-related factors that can contribute to patient-centric healthcare in Malawi?
3. To identify the requirements of an ICT framework that can support a patientcentric approach in Malawi.	3. What are the requirements for an ICT framework that can support a patientcentric approach in Malawi?
4. To identify the benefits of an ICT framework that can support a patient centric approach in Malawi	4. What are the benefits of an ICT framework that can support a patient-centric approach in Malawi?



Components Relationship





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Faculty of Computing and Informatics

Thank You.



Writing Your Proposal Introduction

BY : IRJA SHAANIKA

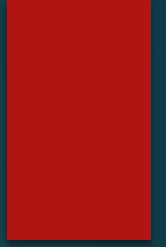
Why “Introduction” ?

- ▶ Getting your readers on-board
- ▶ Giving readers an overview what your study entails
- ▶ Will awaken readers interest

Introduction entails

- ▶ A brief introduction to the study
 - ▶ Concise yet Informative
- ▶ Adequate background information of the study
 - ▶ Discuss general background, Move on to the specifics, stressing the magnitude of the problem
 - ▶ Supported with limited literature.
- ▶ Quote key papers and findings
 - ▶ highlight the knowledge gaps and how your study provides further information on the area.
 - ▶ Describes how and why the research questions have arisen.
- ▶ Explicitly state the importance of your research
 - ▶ reader may not necessarily make the leap in logic that is obvious to you.
- ▶ The introduction should end with the aim being clearly stated.

How “Introduction”



Be guided by objectives

- ▶ Objectives - Your map(guide).
- ▶ Ensure you are not diverting from the focus of your study

Have themes /keywords

- ▶ Main topics/headings.
- ▶ Based on problem statement, objectives, topic.
- ▶ May be a single word or several words(may include phrases)
- ▶ Will help readers locate/search your study in databases.

Identify keywords sequence

- ▶ Order of keywords : which one should come first, second....., last? Why
- ▶ Which order will help my readers follow and understand my study very well?
- ▶ Most cases : study domain > what is being studied
- ▶ Example: Securing ATMs cash withdrawals
- ▶ ATM> Cash withdrawals> Security
- ▶ Example: Implementing **Electronic Healthcare Records** in **Public Hospitals**
- ▶ ?????

Maintain connectivity/flow among themes

- ▶ The conclusion of one term should lead into the introduction of the following term.
- ▶ (Information Systems : IS Integration)
- ▶ **Information Systems:** It is no doubt the implementation of ICT offers unlimited benefits to organisations. However, for benefits such as collaboration and ease of information flow, **information systems integration** is mandatory.
- ▶ **IS integration:** **Information Systems integration** refers to the interconnectivity between the implemented systems. In his study David (2019) discussed types of IS integration which are $()()$. According to Shaanika (2019), the type of IS integration an organisation select is influenced by various factors such as underlying infrastructures and technical skills.

Advice!!!

- ▶ Always consider the structure of the (Ws)...H... when writing
 - ▶ What
 - ▶ Where
 - ▶ When, How
 - ▶ Why
- ▶ Do not leave statements hanging.....



Thank you:

Any questions??

Writing an Introduction

Dr Munyaradzi Maravanyika

Key Goals of an Introduction

1. Present your topic and get the reader interested
2. Provide background or summarize existing research
3. Position your own approach
4. Detail your specific research problem and problem statement
5. Give an overview of the paper's structure

Definition of Research Introduction

- A research introduction is the first section of a research paper or thesis that provides an overview of the study's background, purpose, scope, and significance.
- It serves as the foundation upon which the rest of the study is built.
- The research introduction should be well-crafted to capture the reader's attention and motivate them to read further.

Step 1: Introduce your topic

- The first job of the introduction is to tell the reader what your topic is and why it's interesting or important.
- This is generally accomplished with a strong opening hook.
- The hook is a striking opening sentence that clearly conveys the relevance of your topic.
- Think of an interesting fact or statistic, a strong statement, a question, or a brief anecdote that will get the reader wondering about your topic.
- Don't feel that your hook necessarily has to be deeply impressive or creative.
- Clarity and relevance are still more important than catchiness.
- The key thing is to guide the reader into your topic and situate your ideas.

Make others care

- When you do research, you do it because you hope it will make a positive difference in the real world, which means you need to write for other people.
- If other people don't read your writing, your research will make no difference.
- This means you have to get other people, besides yourself, to care about the 'itch' you identified.
- If you can't do this, you will be given the biggest insult a researcher can get, which is, "So what?"
- The best way to make others care is to create a problem for them.

Step 2: Describe the background

- Argumentative background:
 - You'll explore some general background here
 - After you've caught your reader's attention, specify a bit more, providing context and narrowing down your topic.
- Empirical Background
 - In a more empirical paper, this is the place to review previous research and establish how yours fits in.
 - Begin by establishing the kinds of research that have been done, and end with limitations or gaps in the research that you intend to respond to.

Provide Background Information

- After the hook, provide some background information on the topic
- This can include historical context, current trends, or any other relevant information
- The purpose of this is to help the reader understand the importance of the research and why it is relevant

Identify the gap

- Explain the gap by showing that there is big difference between what we want and what we have.
- Find academic sources and credible data that you can use to support the claims that there is a gap.
- Without credible sources to back you up, it will be difficult for you to convince your chosen audience that the problem is serious enough to warrant their attention.
- Bear in mind that your audience probably has many researchers presenting problems to them, all of which compete for their attention, so you need to do all you can to convince them that they should care about your problem.

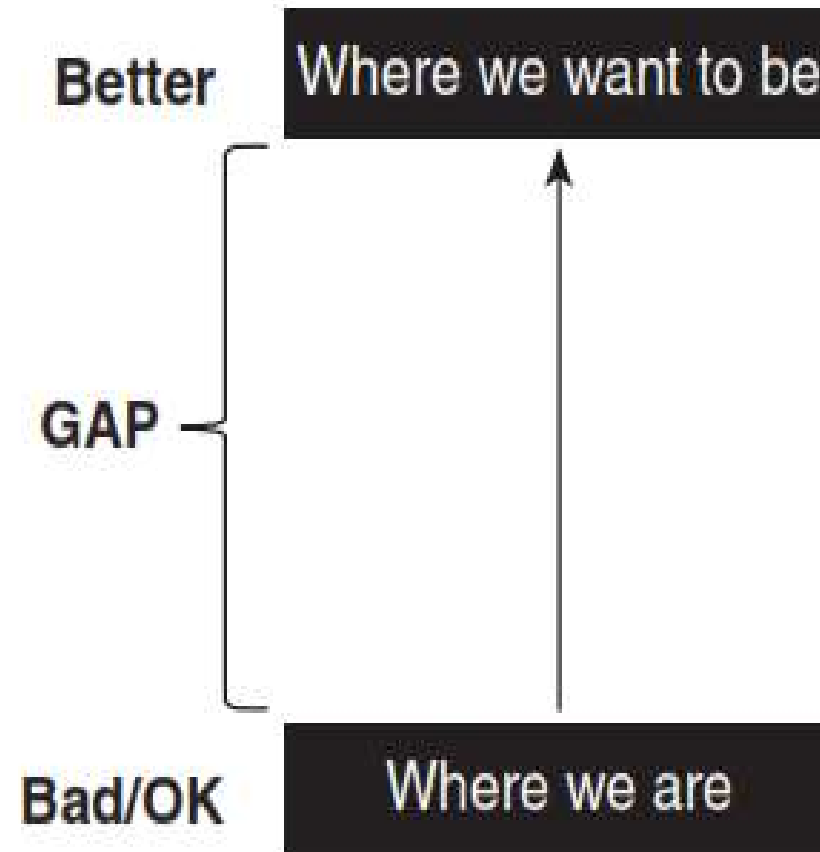
Identify the Gap

To close this gap, you have to take action, and if this action is difficult, then you have a problem.

What makes this gap a problem?

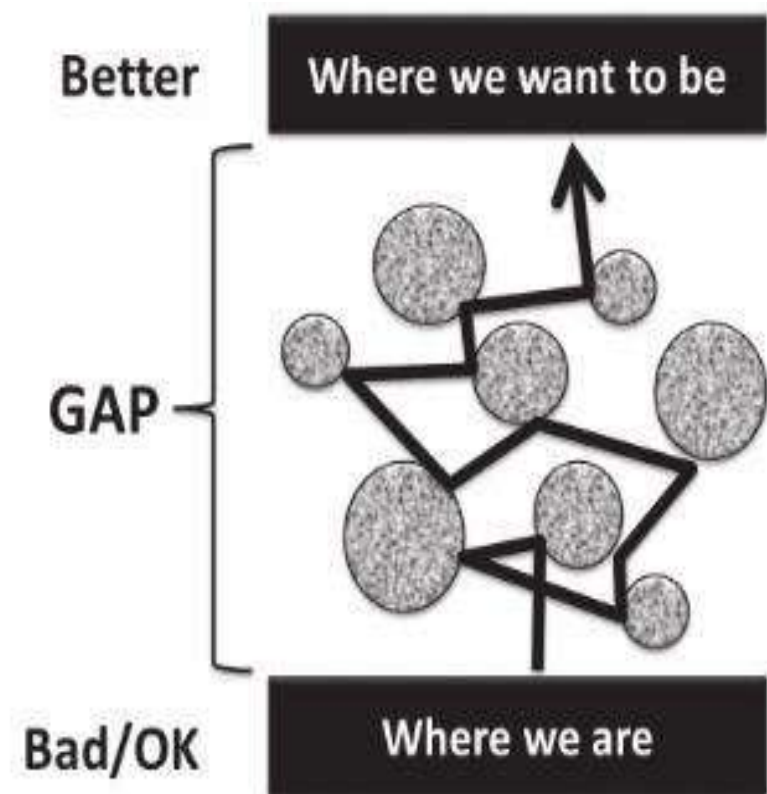
Looking at the origin of the word gives us a clue.

It comes from the Greek word ***problema***, which means 'obstacle to action'.



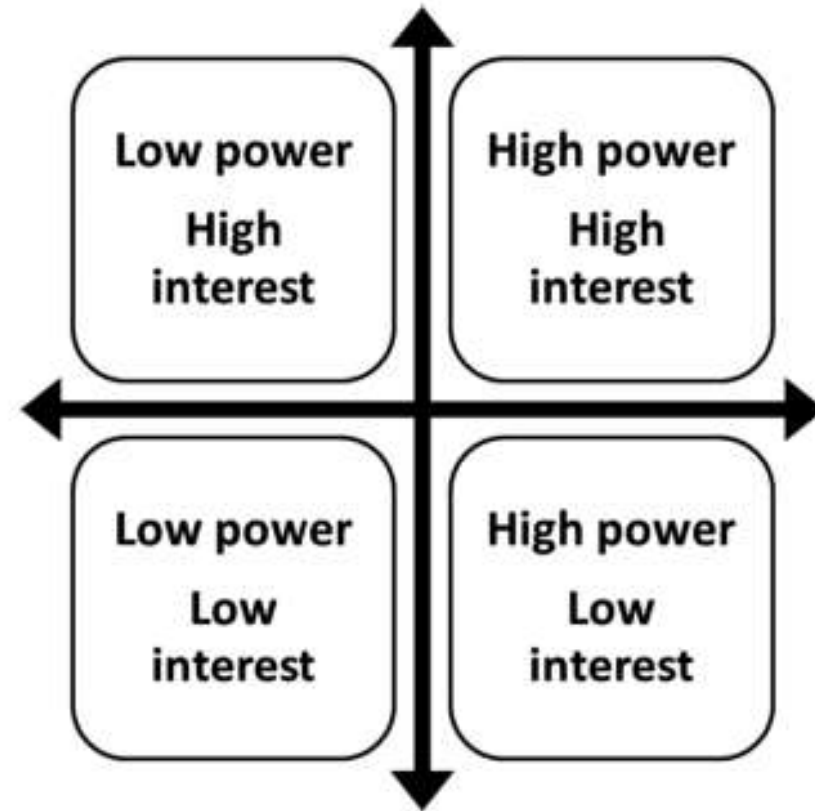
Identify the obstacles

- Look inside the gap and identify the main obstacles to solving the problem and how they make it difficult to take action to solve the problem.



Identify your audience

- A problem is always a problem for someone and, often, a group of people or multiple groups of people.
- So, you need to have a person, entity or some group of people in mind when you try to discuss the problem.
- This also gives your writing further focus to ensure that you get to the point quickly.
- We always write messages, reports and e-mails with someone in mind, and research should be the same.



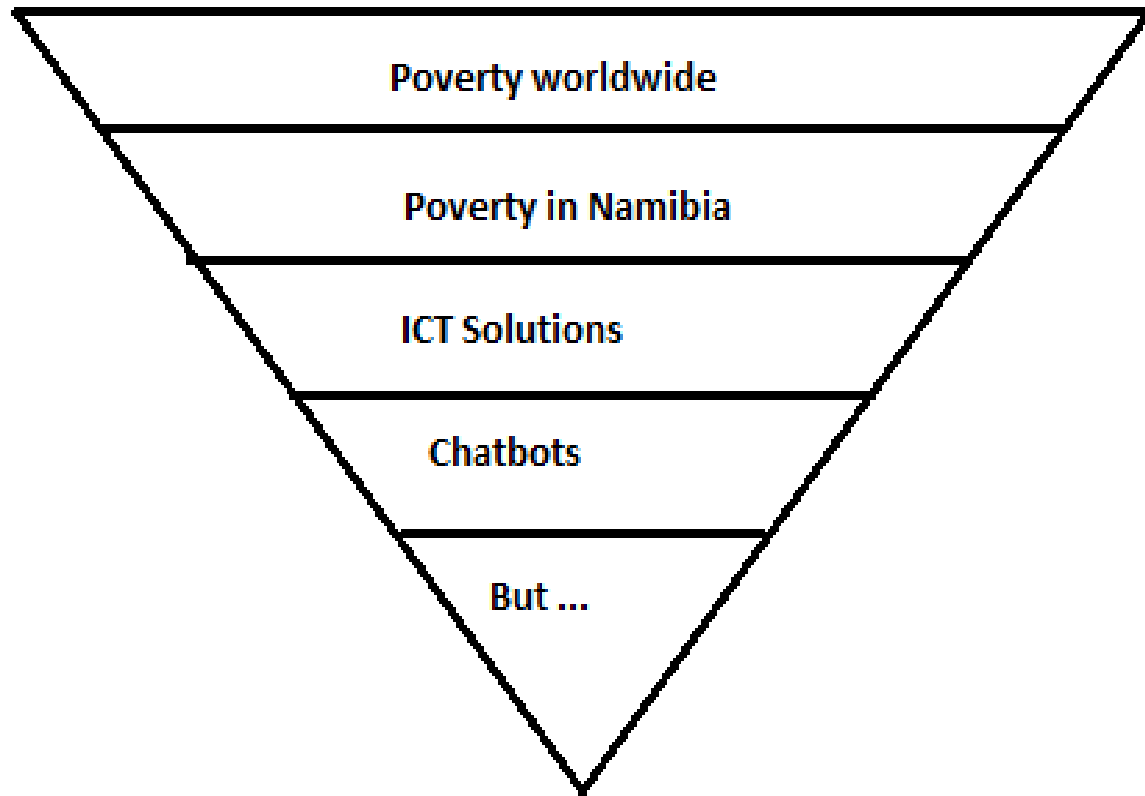
Identify an appropriate context

- When communicating the problem, you have to be careful because not everyone will agree with you that your problem is that important, and may even refute its existence entirely.
- And if a reader disagrees with you early on, it is a lot more difficult to convince them at a later stage.
- Context gives us the broad background to the problem or tells the story that led up to the problem.
- In the context, there is usually little that readers can disagree with because you don't explain the problem yet.
- It creates some common ground between you and the reader, and it is from here that you can lead them to the problem.
- There are two main ways by which you can create context, and you should be able to do it in one or two paragraphs at most.

Narrative

- Relate the chain of events from a time just before the problem emerged to the actual emergence of the problem.
- Brevity is of the essence here – don't lose the reader's interest by giving them a long history lesson.
- The narrative, or chain of events that led to the problem's existence, is usually the easiest and most persuasive way of contextualising the problem.
- This is because it draws on historical facts that are indisputable and because most people are less critical when they are told a story.
- However, the narrative only works if there was a time before and after the problem.
- If the problem has always existed, you need to contextualise the problem using the next approach.

Context as Inverted Pyramid



- This is why I call it an ‘inverted (upside-down) pyramid’—because you go from the very general to the very specific.
- Again, being brief is of the essence, because you will bore your readers and they will lose concentration if you spend too much time on vague and general discussions.

Contextualising

- Familiarising the audience with the place, time and setting – also helps the reader orient themselves and see where you are coming from.
- We do this in most stories: we don't start with what went wrong, but with 'once upon a time'.
- We also do this in movies: we don't start with the action, but first with a view of the place and gradually zoom in to where the action is taking place.

Writing: 'background to the research' section

- First, write down the context
- The context leads to the real-world problem, so then you write down the explanation of the problem
- This should not take more than one page, and of this one page you should spend more time on explaining the actual problem in the real world, because it is this that convinces people that they should care.
- The background starts with a general discussion of an issue in the first paragraph.
- It then moves to a discussion of the issue in the second paragraph, it led to a particular law. In the third paragraph, it explains the problem with that law.
- Facts are stated and academic references back up all the important claims.

Writing: 'background to the research' section

- In the background section, you focused on the real-world obstacles that make it difficult to solve the real-world problem.
- There is another kind of obstacle that needs to exist to create the need for research.
- This one does not lie in the real world, but in the world of knowledge.
- We think about that in the next step

Determine why current studies are not good enough

- So far, you have shown that there is a problem in the real world and you identified real-world obstacles that make it difficult to do something about it – like lack of funding, wrong policies, corruption etc.
- Now you have to show that research plays a role in the existence of, and solution to this real-world problem.
- Real-world problems, however, cannot be solved by research, because research is not a person who can walk around and change the world.
- So, let us look at what role research plays in the existence of a problem.
- It could be that the research, up until that point, was not good enough to help people solve the real-world problem, so that further research would give people the knowledge they need to address the real-world problem.
- This is where you can step in.

The most common kinds of gaps are:

- **Time gaps:** You find previous studies about the real-world problem, but notice that the studies are outdated and are therefore no longer relevant. New updated research is therefore needed based on the latest data.
- **Place gaps:** You find previous studies, but notice that the studies are only applied to some places. New research is therefore needed that applies the relevant ideas to places – cities, industries, countries etc. – neglected by researchers in the past.
- **People gaps:** You find previous studies, but notice that the studies are only applied to some people or groups. New research is therefore needed that applies the relevant ideas to the groups neglected by researchers in the past.
- **Method gaps:** You find previous studies, but notice that the studies use methods that are wrong, inappropriate or outdated. New research is therefore needed using the best available or new methods.

- The challenge is to convince your audience that, given all the research out there about your real-world problem, you have found something inadequate.
- Here are some ways to do that:

If you found that there is confusion about an issue in the form of a disagreement, then:

1. You mention the studies (by referencing them) that take one view on the real-world problem;
2. Mention the studies (by referencing them) that take the opposing view;
3. Explain what it is they disagree on; and
4. Explain what research is still needed to settle the disagreement (and this is the research you will do).

If you found that there is confusion in the form of an unsolved puzzle, then:

1. You explain the puzzle with reference to studies that show that there is a puzzle;
2. Mention the studies (by referencing them) that have attempted, and failed, to solve the puzzle; and
3. Explain what research is still needed that might help to solve the puzzle, and state that this is the research you will do.

Step 3: Establish your research problem

- Try to lead into the problem on the basis of your discussion of the literature.
- Think in terms of these questions:
 - What research gap is your work intended to fill?
 - What limitations in previous work does it address?
 - What contribution to knowledge does it make?

State the Problem

- The next step is to state the problem or issue that the research addresses
- This should be a clear and concise statement that identifies the specific problem that the research aims to solve
- It should be written in a way that is easy for the reader to understand

Make the connection between your problem and the existing research using phrases

Phrase	Connection
Although x has been studied in detail, insufficient attention has been paid to y .	You will address a previously overlooked aspect of your topic.
The implications of x study deserve to be explored further.	You will build on something suggested by a previous study, exploring it in greater depth.
It is generally assumed that x . However, this paper suggests that y ...	You will depart from the consensus on your topic, establishing a new position.

Contrasting the real-world problem and the knowledge problem

	Real-world problem	Knowledge problem
Where is the problem?	In the everyday reality of people and things	In the studies on the real-world problem
Who experiences it?	Ordinary people, practitioners and those with the power to address the problem	Academics with an interest in the research
How do we prove it?	Provide facts in support of the seriousness of the problem	Show that the studies are inadequate

Step 4: Specify your research question

- The research question is the question you want to answer in empirical research.
- Present your research question clearly and directly, with a minimum of discussion at this point.
- The rest of the paper will be taken up with discussing and investigating this question; here you just need to express it.
- If your research involved testing hypotheses, these should be stated along with your research question.

Outline the Research Questions

- After stating the problem, outline the research questions that the study will address
- These questions should be specific and focused, and they should relate directly to the problem statement
- This will give the reader a clear understanding of what the study aims to achieve

Highlight the Significance of the Study

- Explain why the study is important and what its potential impact could be
- This can include practical applications, theoretical contributions, or any other relevant information
- The purpose of this is to show the reader why the study is worth their attention and why it matters

Mention Limitations

- It is important to mention any limitations of the study
- This can include things like sample size, data collection methods, or any other potential issues that could impact the validity of the study
- This will help the reader understand the context of the study and how it should be interpreted

End of Presentation



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OF SCIENCE AND TECHNOLOGY

Faculty of Computing and Informatics
Department of Computer Sciences / Informatics

Title:

Name:

Supervisor:

Date: 2020





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OF SCIENCE AND TECHNOLOGY

Faculty of Computing and Informatics
Department of Computer Sciences

Outline

1. Introduction/Background
2. Problem Statement
3. Preliminary Literature review
4. Research Objectives and Questions
5. Research Methodology
6. Limitations, Delineation, Assumptions
7. Timeline
8. Conclusion
9. References



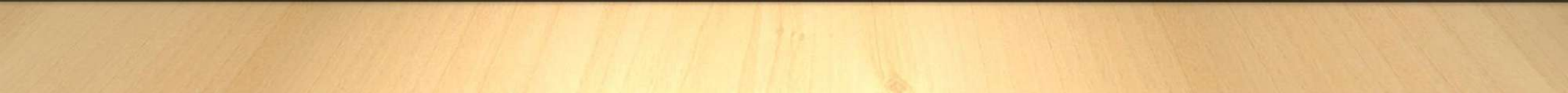
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Research Methodology

BY DR. SHAANIKA

13 APRIL 2023





Lesson Objectives

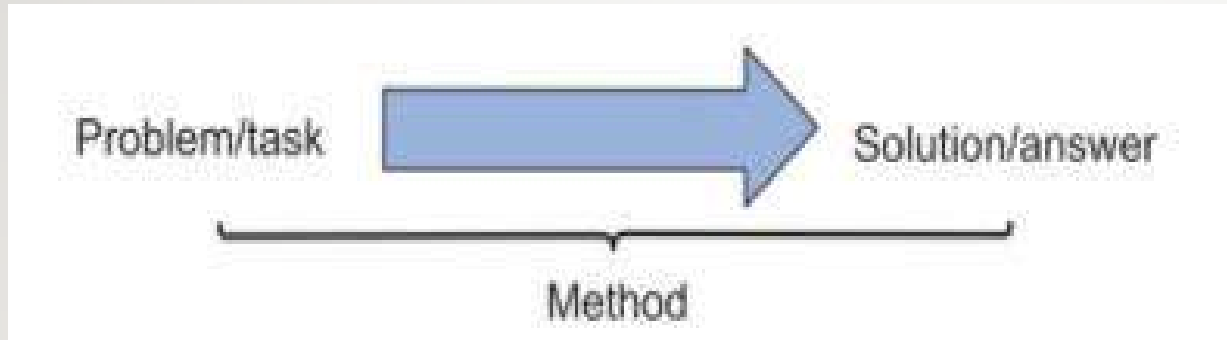
- Understand the need for methodology in research
- Explain the differences between quantitative, qualitative and mixed methods research
- Be able to select a methodology for your study

What is methodology

- A way in which we approach problems and seek answers.
 - Methodology :covers data collection techniques, data collection process and data analysis.
- A carrier to the desired destination.
- Selected in alignment with research aims, objectives and questions.

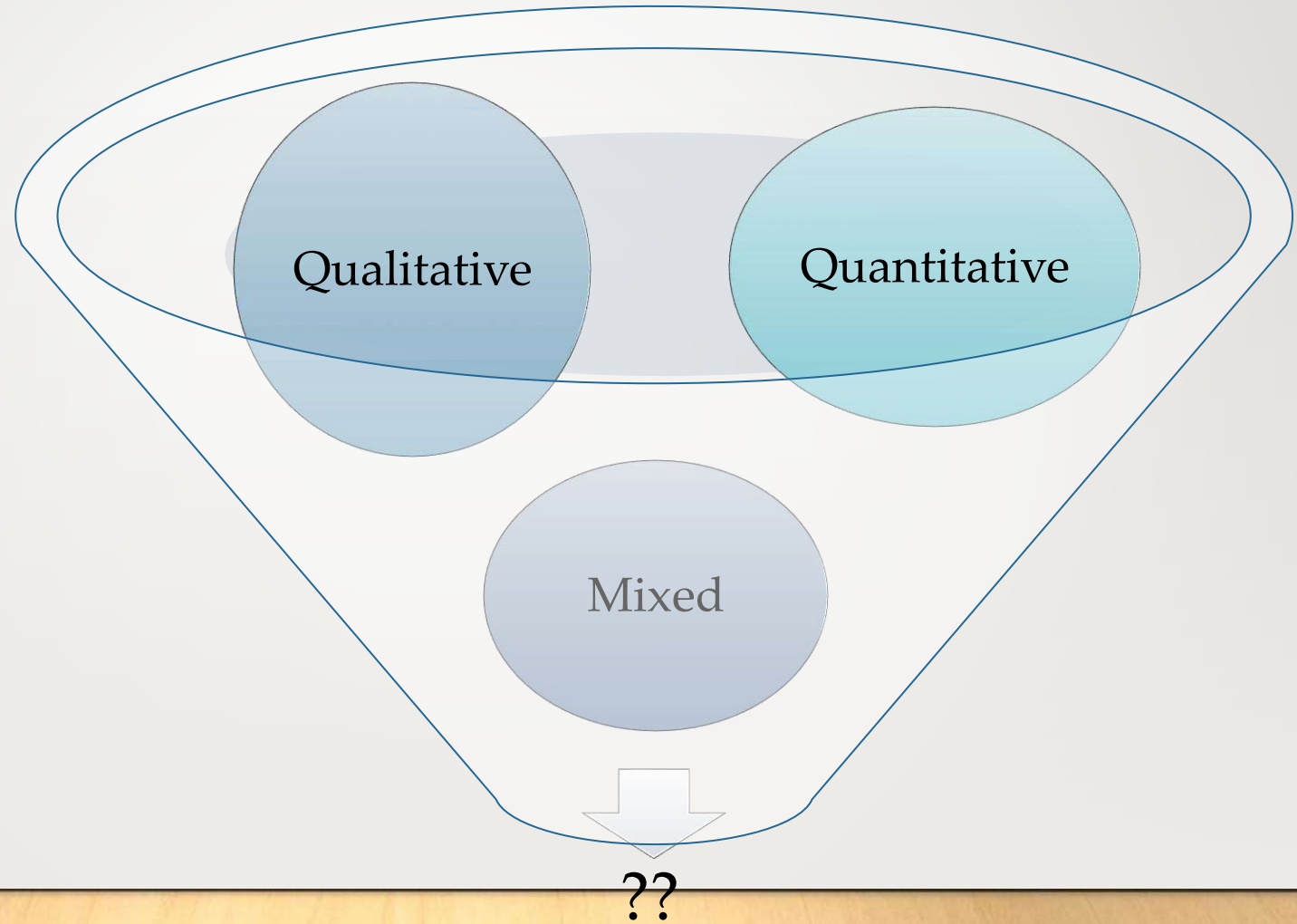
What is research methodology ?

- A vehicle undertaken to achieve study objectives.



- Logic or ways of reasoning to arrive at solutions.
- Rules of communication, i.e. to explain how the findings have been achieved (Ghauri, Grønhaug & Strange, 2020).

Methodological choice



Qualitative research method

- A form of social inquiry -explore and uncover explanations, rather than deduce them from measurements.
- Study participants' meanings and the relationships between them.
- Natural- study participants in their natural environment
- Subjective-Interpretive -each individual experience the world from their own point of view

Quantitative research method

- Main focus is counts and measurements of things
- Tends to be confirmatory
- Often begins with hypothesis and variables that needs to be tested and the results obtained are used for generalisation
- Reality should be studied objectively by the researchers who should put a distance between themselves.

Mixed research method

- Social phenomena are so complex, different kinds of methods are needed to best understand these complexities.
- Combines the use of quantitative and qualitative.
- Integration of subjectivity and objectivity.
- An objective need to be achieved by applying both methods.

Mixed Methods Designs

Research Design	Process	Purpose	Level of Interaction	Priority	Example from Literature
Convergent parallel (concurrent)	Qual Quant	To obtain different but complementary data to answer a single research question	Data collected & analysed independently	Equal	Peters and Cotton (2013) collected both mailed surveys and undertook unstructured interviews with women with physical disability in order to gain a broad understanding of the barriers and facilitating factors associated with in accessing and experiencing breast and cervical cancer screening services.
Sequential explanatory	QUANT → qual	Qualitative data are collected to explain the quantitative findings	Quant data frames qualitative data collection	Quantitative dominant	Pfaff et al. (2014) used a mailed survey to measure perceived confidence in interprofessional collaboration amongst new graduate nurses. Following analysis of the survey data they conducted interviews with 16 new graduate nurses to explain and expand upon the quantitative findings.
Sequential exploratory	QUAL → quant	Quant data builds on qualitative findings to provide generalizability	Qualitative data frames quant data collection	Qualitative dominant	Hamshire, Willgoss and Wibberley (2013) conducted a series of interviews with nursing students to explore their experiences and expectations of their nursing course. These interview findings informed the development of an online survey that was administered to 1080 students across nine UK universities.
Embedded / Nested	Qual (quant) Or Quant (qual)	To obtain different data to answer a complementary research question	Embedded dataset provides answers to a complementary research question.	Can be either Qualitative or Quantitative dominant	Kinser et al. (2013) conducted a randomised controlled trial of an 8 week yoga intervention for women with major depression. Outcomes included depression severity, stress, anxiety and rumination. Qualitative interviews were embedded in the trial to explore the feasibility and acceptability of the intervention.

Methodologic choice

- A closer look at your objectives.
- What is the best way to resolve your problem: explanations or measurements , experiments?.
- Look at the potency/weakness of each method in relation to your research problem.
- The methods used will influence the type of data collection techniques.

Exercise : Identify type of research

- Performance evaluations of two algorithms (memory usage, processing time, etc)
- Users perceptions about a particular movie. (Visual quality, acceptance, interesting, etc)
- To understand the perceptions of using new mobile applications to disseminate information related to COVID-19
- Study the performance of two systems producing a certain product e.g Bottle milk production



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Faculty of Computing and Informatics

RIT812s: Research Methodology

Dr Irja Shaanika
Dept. of Informatics

Date: February 2023





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What is Research?





*“Research is an **original contribution** to the **existing stock of knowledge** making for its advancement. It is the **pursuit of truth** with the help of study, observation, comparison and experiment. In short, the search for knowledge through objective and **systematic** method of finding solution to a problem is research.”*



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What is Scientific Research

*“If a research contributes
to a **body of science**, and follows the **scientific method**.”*



And what is Methodology ?

*“Methodology is a research strategy that translates **ontological** and **epistemological** principles into guidelines that show **how** research is to be conducted and principles, procedures, and practices that **govern** research.”*



The Functions of Research

The main function of research is to improve research procedures through the refinement and extension of knowledge.

The improvement is associated with various aspects:

1. Research encourages scientific and inductive thinking, besides promoting the development of logical habits of thinking and organisation.
2. The function of research is to make a decision concerning the refinement or extension of knowledge in this particular area.
3. Research assumes significant role in the formation of policies, for the government and business. Research also tries to solve various operational and planning problems.



The Functions of Research.....

4. Another function of research is to aid administrators to improve the Social systems. Research is equally important to social scientists for analysing social relationships and seeking explanations to various social problems.

5. Research helps in carrier development.



Classification of Research

1. Basic level(Basic research is designed to add an organized body of scientific knowledge and does not necessarily produce results of immediate practical value).
2. Applied level (to solve an immediate practical problem and the goal of adding to scientific knowledge is secondary).



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Faculty of Computing and Informatics

Thank You.

Ethics in Research

By Dr Shaanika





Table of Content

- Ethics: the concept
- Stakeholders in research
- Ethical issue to consider concerning
 - Research Participants
 - Collecting Information
 - Seeking informed consent
 - Providing incentives



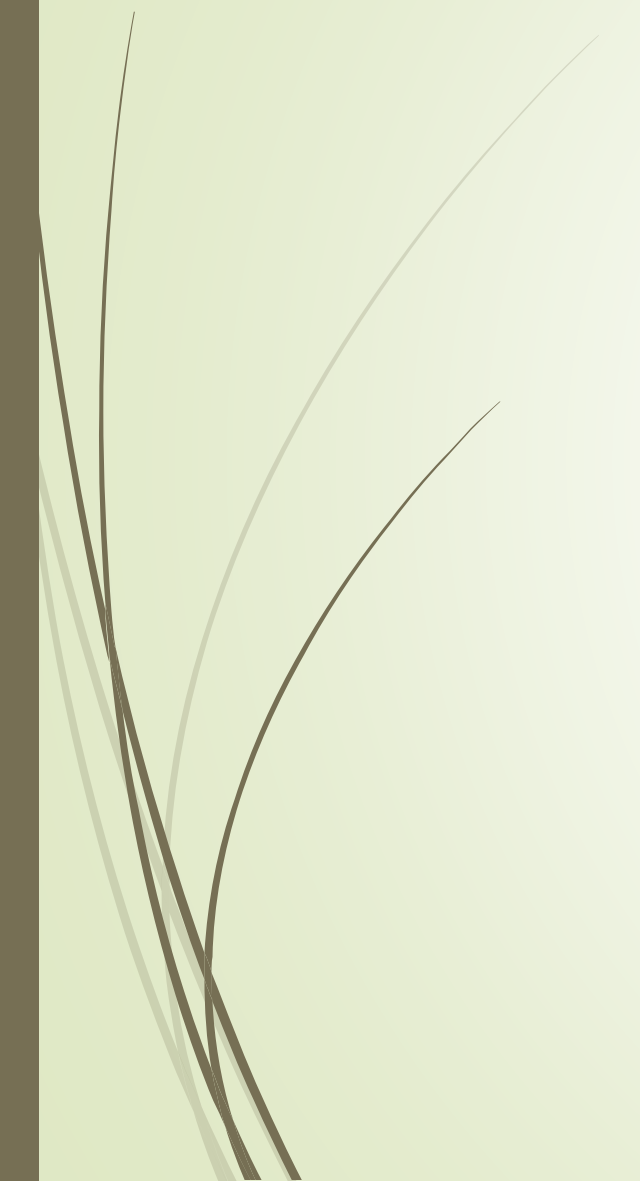
Ethics: the concept

- Babbie (2007) describes ethics as conforming to standards of conducts in accordance with professionalism or environmental culture.
- Ethical considerations in research reflect significant *moral concerns about practice and responsibilities* of researchers in the environments that they intend to conduct their studies.
- Ethical Standard (factors to consider)
 - Participant's consent, freewill to enroll in the interview process
 - Confidentiality and anonymity(how will it be maintained)
 - Data retention -protection





Why Do ethics Matter

- Crucial for maintaining research integrity.
 - Research misconduct can have dire consequences.
 - Following ethics protects both the researcher and participants.
- 



University Research Policy

- See document attached on ELearning



Stakeholders in research

- NUST
- Researcher
- Supervisors
- Organisations(Cases)
- Participants



Research Participants

- From study population –
 - From whom the information required to find answers to your research question is obtained.
 - Decide specifically and clearly who constitutes your study population, in order to select the appropriate respondents.



Collecting Information

- Data collection tools(interviews, questionnaires, observations)
- Informed consent
 - Organisation
 - Individual participants



Seeking informed consent

- A copy of the Approved proposal signed by both supervisor and student.
- A copy of the questionnaire/interview(if any)/ data collection protocol/tools
- Filled in Ethical Clearance Application Form
- Completed Informed Consent form
- Letter Requesting permission from target partner/institution.

Plagiarism

Common Types of Plagiarism

- ✓ **Complete plagiarism:** Submit another authors work in your name
- ✓ **Source-based plagiarism:** Reference an incorrect or non-existent source
- ✓ **Direct plagiarism:** Copy text from another document word-to-word
- ✓ **Self or auto plagiarism:** Reuse a major part of your own work without attribution
- ✓ **Paraphrasing plagiarism:** Make minor changes and use others' writing
- ✓ **Mosaic plagiarism:** Interlay someone else's phrases or text within your work.
- ✓ **Accidental plagiarism:** Unintentional paraphrasing or copying due to neglect
- ✓ **Inaccurate authorship:** Authorship instead of acknowledgment to contributors and vice versa

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Thank You

➡ Any Questions





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Department of Computer Science

Research Methodology

ASSUMPTIONS,

LIMITATIONS AND,

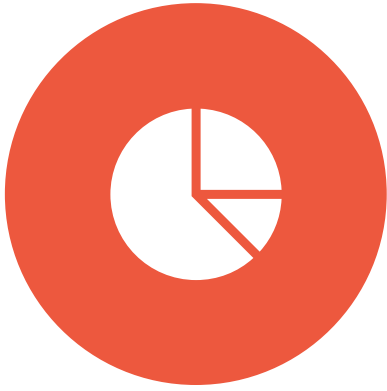
DELINIATIONS IN THE RESEARCH PROCESS

ETHICAL CONSIDERATIONS
CHAPTER OUTLINES

Why are we here?

During the process of your thesis writing, you might suddenly realize that your research has innate flaws, of which some you have control of, and others you don't. Being able to recognize and accurately describe these problems is what makes you a distinctive researcher.

What are they?



LIMITATIONS.



**DELINEATION
S.**



ASSUMPTIONS.

Limitations?

- ❖ These are potential **weaknesses** of the study that are **out** of the researcher's control.
- ❖ They are difficult to contain.
- ❖ In most cases, they are closely associated with the chosen research design, funding constraints, or other factors.
- ❖ Do not worry about limitations because limitations affect almost all research projects, as well as most things in life.
- ❖ However, they should be **clearly acknowledged** as they may affect the study design, results and ultimately, conclusions.

Common examples of limitations

❖ Access

If your study depends having access to organisations, people, or documents, and for some reasons, this access is denied, or limited in one way or the other, the reason for this need to be stated and described.

❖ Time

Another common limitation is time. If your study was limited to a certain amount of time, your results are affected and are depended on conditions occurring during that time and by the operations of society during that time period (e.g., economy, social trends).

❖ Lack of prior research studies on the topic

Citing prior research forms the basis of your literature review, and helps lay a foundation for understanding your problem. A limitation like this one can serve as an opportunity to describe the need for further research.

❖ Resources (e.g. finance)

❖ Participant dropout,

❖ Limitations of instrument used, etc.

Why should I state my limitations?

- ❖ **Always admit and identify the study's limitations**

Not mentioning them means you are giving your supervisor an opportunity to point them out on your behalf, and this can lead to failure as it will be assumed that you have willingly ignored them

- ❖ **They are an opportunity for you, as a researcher to make suggestions for future research.**

- ❖ **All studies have limitations**

- ❖ **The discussion of Your limitations should therefore be related and restricted to the problem under investigation.**

Delineations?

- ❖ This are the boundaries that you, the researcher, sets for your own study.
- ❖ They define the scope of your study, and unlike the limitations, they are in your control.
- ❖ *Delimitations are set so that the study's aims and objectives do not become impossible to achieve.*

Listed below are examples of delimitations:

- ❖ Research objectives,
- ❖ Research questions,
- ❖ Methodology,
- ❖ Variables and populations chosen as targets to study.

Things to consider on Delineations

- ❖ When you are stating your delimitations, clearly inform the reader why you chose a certain aspect of this particular subject/problem and not the other.
- ❖ The answer might simply be that you were curious about the topic and/or wanted to improve it.
- ❖ In any case, you should clearly list the other options available and the reasons why you did not choose these options immediately after you list your delimitations.
- ❖ Limit your delimitations to the things that the reader might reasonably expect you to do, but you, for clearly explained reasons, have decided not to do/cover
- ❖ Remember that delineations are not weaknesses, and you don't have to apologize for them. Good, strong research projects have clear boundaries.

Assumptions

- ❖ Assumptions are things that are accepted as true, or at least plausible, by researchers and anyone who will read your thesis/paper.
- ❖ In other words, any scholar reading your paper will assume that certain aspects of your study is true given your population, statistical test, research design, or other delimitations.
- ❖ They can't be controlled, but they need to exist for your study to be credible, or essential.

For Example:

If you are conducting a survey, you need to assume that people will answer truthfully. Or, if you are choosing a certain sample, you need to assume that this sample is a true representative of the population that you wish to make inferences to.

Summary on the DLA

	Delimitations	Limitations	Assumptions
<i>Purpose</i>	To determine the parameters/scope of the study	To determine the weakness associated to decisions made in the study	To make a study relevant
<i>Control</i>	There is a level of control over where to draw the boundaries	Difficult to control but some can be addressed in the study	Can't be totally controlled but they should exist to make the study relevant

Ethical considerations

Research Ethics Clearance is an instrument to measure the ethical fulfilment of a research process. Approval from the Ethics Committee must be obtained before research begins. In other words, Research Clearance Ethics is a reference for researchers in upholding the values of integrity, honesty, and fairness in conducting research.

The main purpose of obtaining ethical clearance is to:

- protect research subjects/respondents from physical harm (threats),
- psychological harm (stress, regret),
- social harm (stigma, hatred from society) and most importantly,
- legal consequences (sued) as result of participating in a research.

Who needs ethical clearance?

Researchers can apply for an ethical clearance if their research meets the following criteria:

- All research involving humans must go through a process of ethical clearance. Excluded from this is research that uses secondary data, literature reviews or published data.
- Research that uses a review of confidential materials (medical records from hospital/health clinic) must go through an ethical clearance process (although it still requires permission from the issuing institution).
- Likewise, research that uses information from closed/non-public media, such as statistical reviews from an institution (employees, clients, patients, users, service providers, service records, financial records, company records) must go through ethical clearance.
- Studies that use additional methods, which deals directly with humans such as interviews, Focus Group Discussions (FGD) and others, still require an ethical clearance process, although the main method used is a review of publicly available material.

What is needed?

Supervisor + student has to validate that the following documents have been fully completed before sharing documents with the Ethical Clearance committee.

Checklist:

1. A copy of the Approved Proposal signed by both supervisor and student.
2. A copy of the Questionnaires/ Interview (if any)/ data collection protocol/tool.
3. A filled-in Ethical Clearance Application Form
4. A written/completed Informed Consent Form (ICF) - **Information Sheet, Certificate of Consent**
5. A copy of Draft letter requesting Permission from target partner/institution (alternatively, a permission letter from them)

Thesis Outline

The final thesis will have the following chapters

Chapter One: The Introduction and background information

Chapter Two: Literature review

Chapter Three: Methodology - Design of proposed framework/system

Chapter Four: The controller design and its classifiers

Chapter Five: Findings, discussion and contributions

Chapter Six: Conclusion and Future work

References

Marwat, K. (2015). *Limitations, delimitations, assumptions and keywords of research*. [PowerPoint slide]. Comsats Wah Campus. Retrieved from <https://www.slideshare.net/kamalshah3975/limitationsdelimitations-assumptions-and-keywords>

Simon, M. K. (2011). *Dissertation and scholarly research: Recipes for success* (2011 Ed.). Seattle, WA, Dissertation Success, LLC. <http://dissertationrecipes.com/wp-content/uploads/2011/04/AssumptionslimitationsdelimitationsX.pdf>

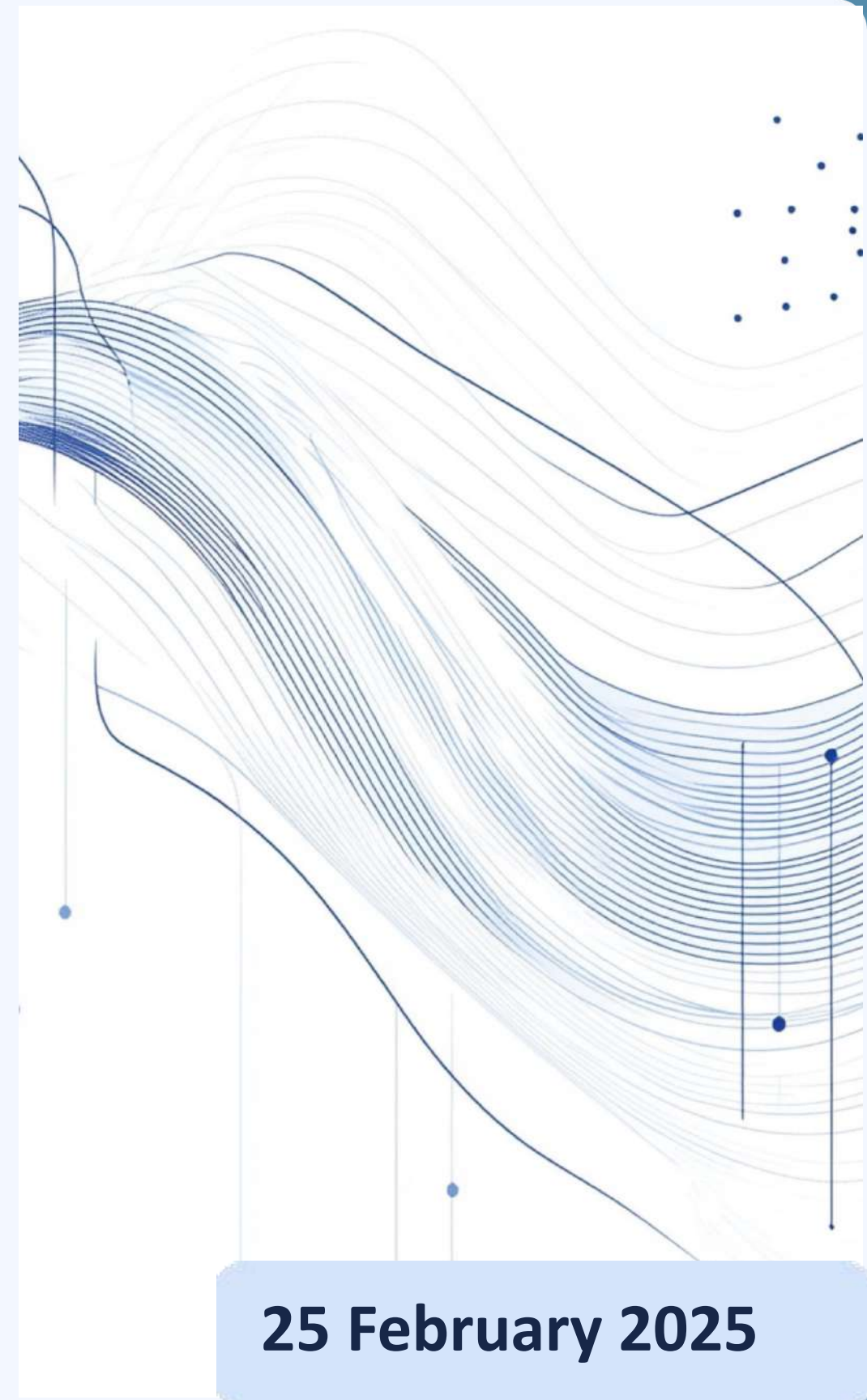
Thank you

Any Questions?

Creating an Effective Problem Statement

Prof Jude Osakwe

25 February 2025



CONTENTS

1. Understanding the Purpose

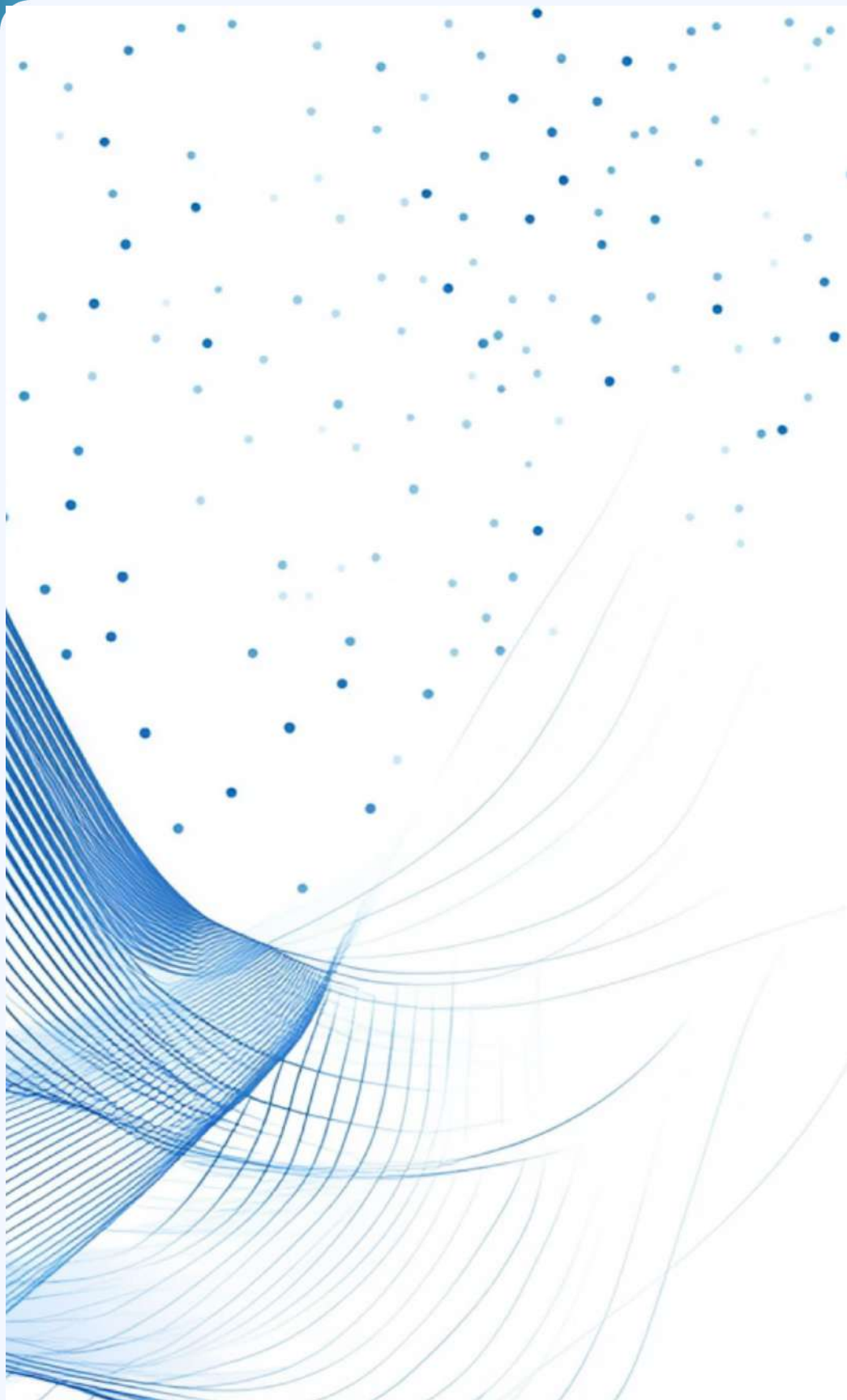
3. Putting It All Together: A Template

2. Steps to Writing a Problem Statement

4. Tips for Writing a Strong Problem Statement Statement



1. Understanding the Purpose



Importance of a Problem Statement

1

Clear Definition of the Issue

- Clearly define the issue at hand.
- Ensure immediate understanding of the problem.

2

Justification for a Solution

- Explain the significance of the problem.
- Highlight the importance of solving it.

3

Focus for Efforts

- Provide a clear target for efforts.
- Ensure alignment on the goal.

4

Support Statements with References

2. Steps to Writing a Problem Statement Statement



Describe the Ideal Situation (The "What Should Be")

Vision of the Desired State

- Paint a picture of the optimal scenario.
- Use specific and concrete language.

Establishing a Baseline

- Quantify the ideal situation with metrics or benchmarks.
- Help measure progress later.

Describe the Current Reality (The "What Is")



Articulation of the Problem

- Clearly describe the current situation.
- Focus on observable effects or symptoms.

Objective and Factual Description

- Base description on data and evidence.
- Avoid opinions and use statistics.



Providing Context

- Explain relevant background or circumstances.
- Include information about the organization or industry.





Explain the Consequences (The "So What")

Articulating the Impact

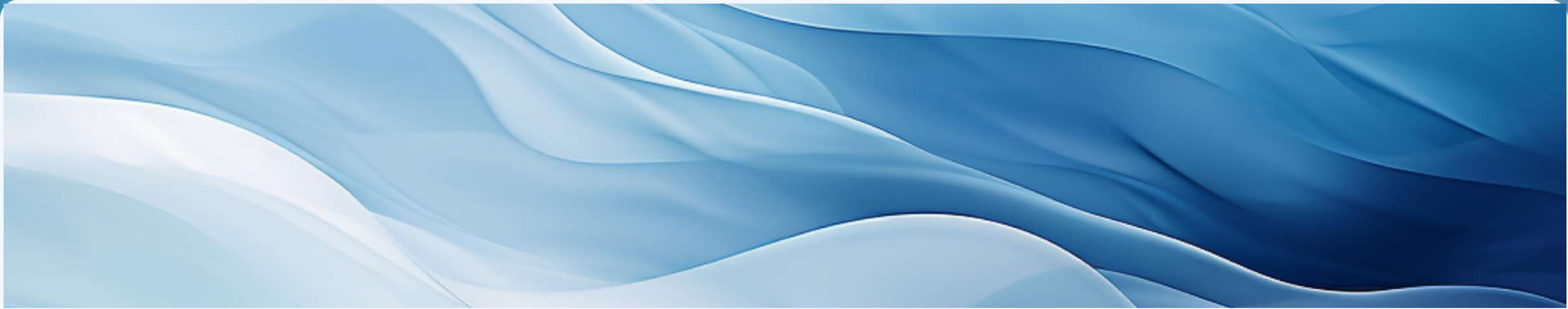
- Describe the negative consequences consequences of the problem.
- Identify who is affected and how.

Quantifying the Impact

- Use numbers to illustrate the impact. impact.
- Provide specific examples.

Long-term Effects

- Consider immediate and future implications.
- Relate consequences to organizational goals.



Suggest a Solution (Optional, but often helpful - The "What Could Be")

Outlining a Potential Direction

- Briefly state what needs to be done.
- Focus on the desired outcome.

Keeping it Broad

- Avoid specificity about the solution.
- Encourage brainstorming and exploration.

Reiterating Benefits

- Summarize the positive impact of solving the problem.

3. Putting It All Together: A Template



3. Putting It All Together: A Template

1

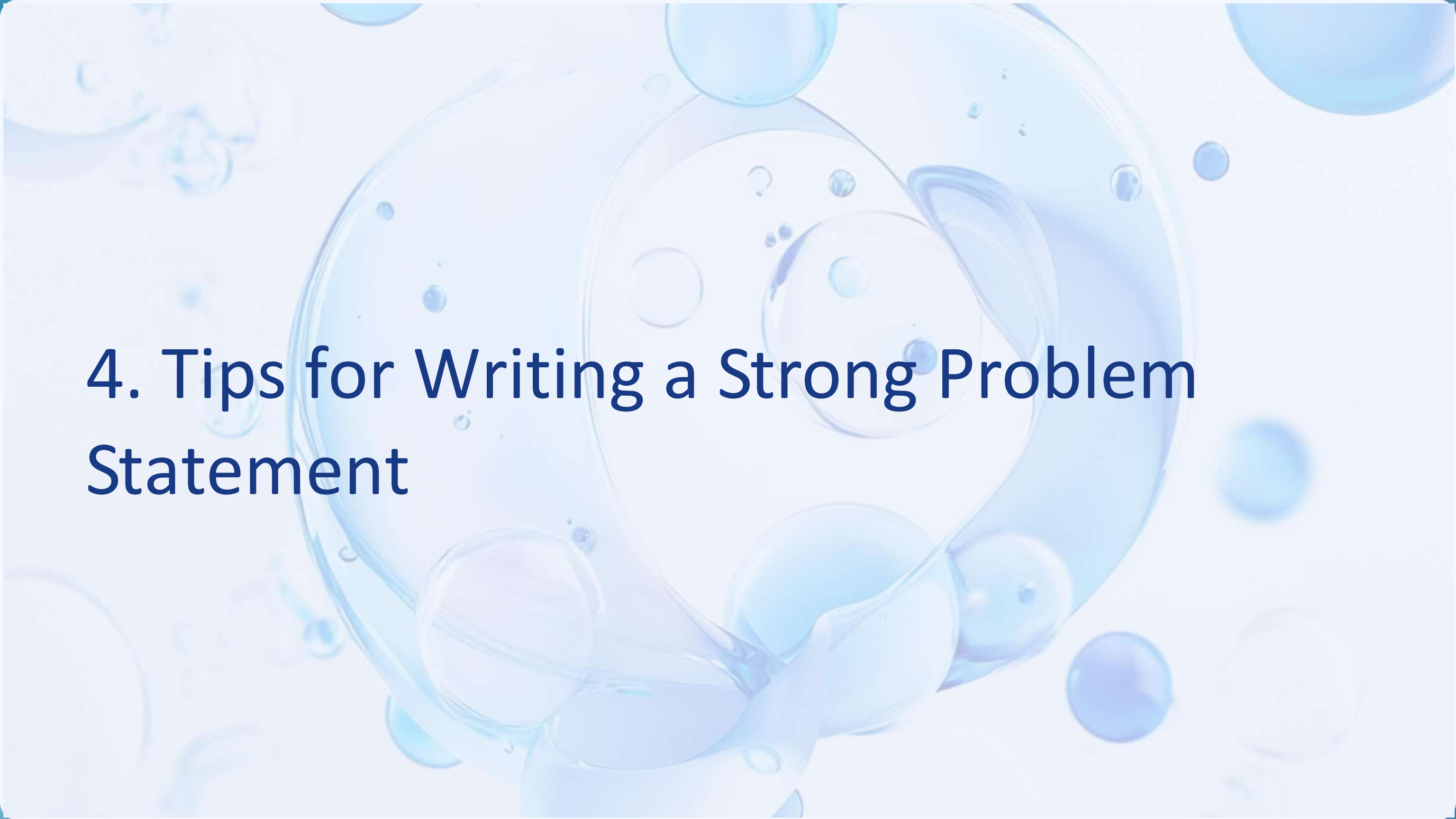
Structure of the Problem Statement

- Ideal: "Ideally, [desired state/situation]..."
- Reality: "However, currently [problem/issue]..."
- Consequences: "As a result, [negative consequences/impact]..."
- Suggestion (Optional): "To address this, we need to [general solution direction] in order to [desired outcome/benefit]..."

2

Example of a Problem Statement

- Ideal: "Ideally, our customers should be able to easily resolve their inquiries through our online chat system within 5 minutes..."
- Reality: "However, currently, the average wait time for customers using our online chat system is 15 minutes..."
- Consequences: "As a result, customer satisfaction scores related to online chat have decreased by 25%..."
- Suggestion: "To address this, we need to improve the efficiency and effectiveness of our online chat system..."



4. Tips for Writing a Strong Problem Statement

4. Tips for Writing a Strong Problem Statement(1)

1

Conciseness and Clarity

- Use precise language and avoid jargon.
- Keep it short while conveying necessary information.

2

Focus on One Problem

- Break down multiple issues into separate statements.

3

Specific and Measurable

- Use data and metrics to support claims.



4. Tips for Writing a Strong Problem Statement(2)

1

Avoid Blame

- Focus on the problem, not assigning fault.

2

Get Feedback

- Share with others for input and understanding.

3

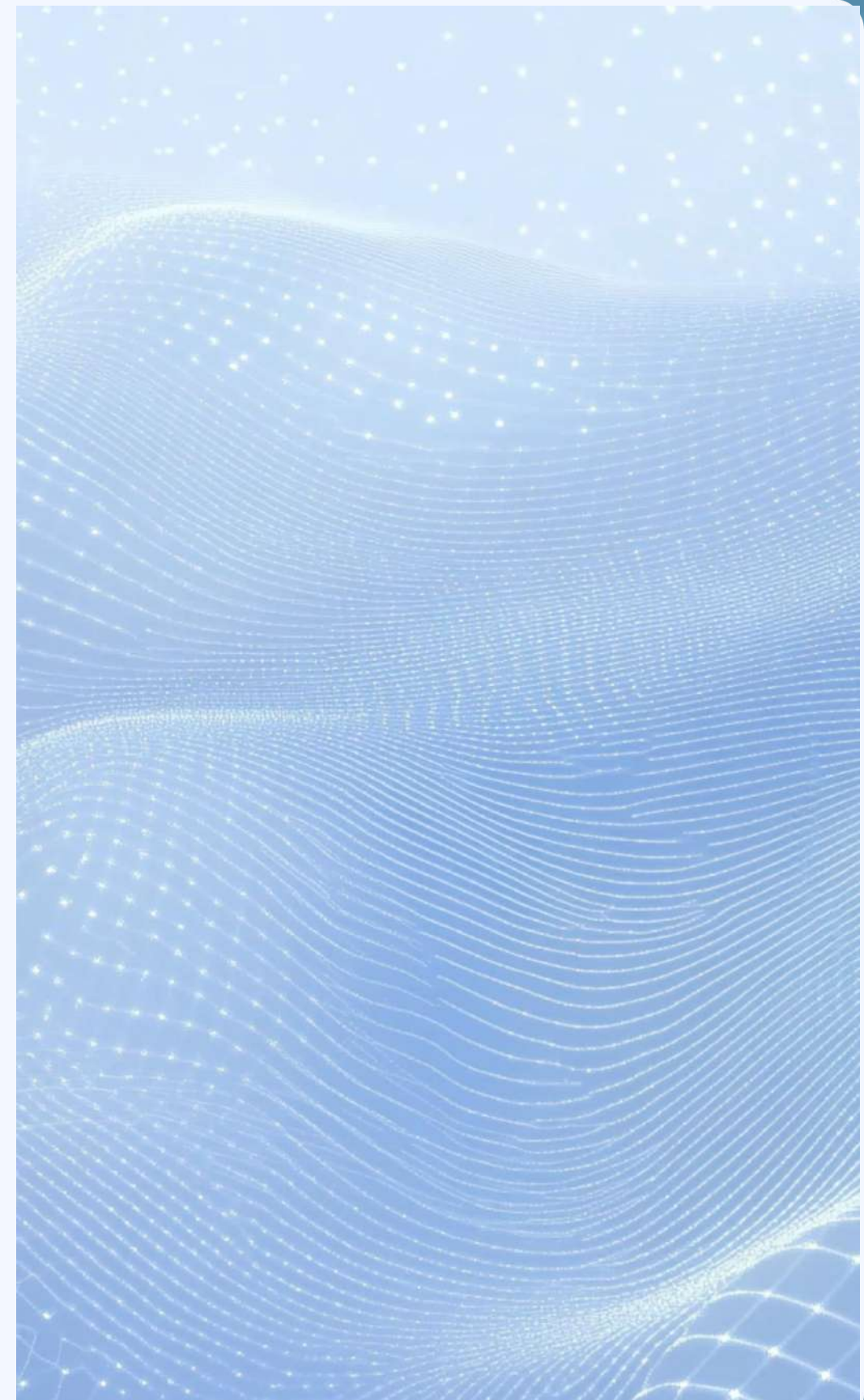
Revise as Needed

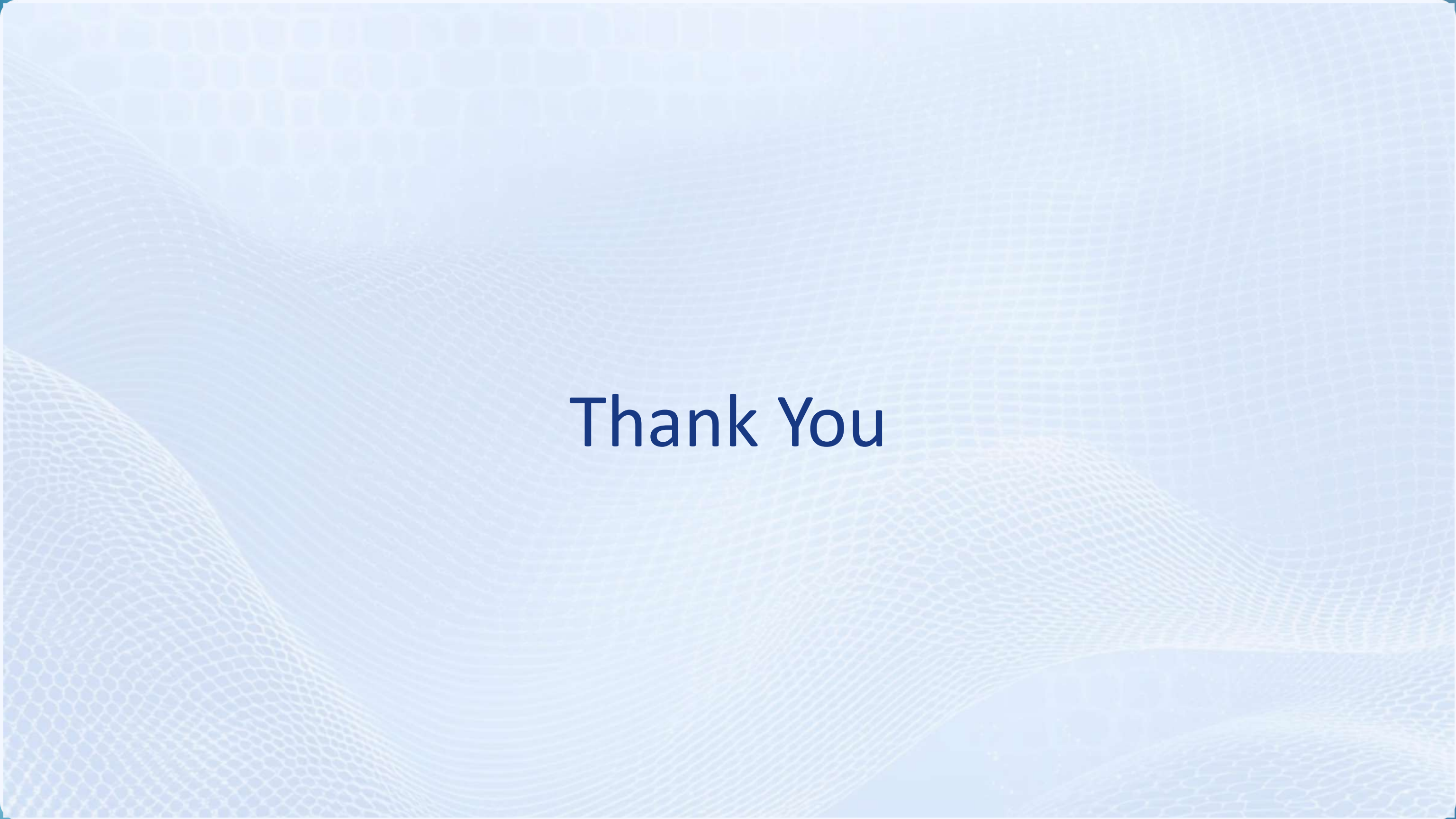
- Be prepared to refine and update the statement.

4

Consider the Audience

- Tailor language and framing for the intended readers.





Thank You



NAMIBIA UNIVERSITY
OF SCIENCE AND TECHNOLOGY

Faculty of Computing and Informatics

2024

Basics of research?



Meaning of Research

Research in common parlance refers to a search for knowledge. Research is an art of scientific investigation. Dictionary definition of research is a careful investigation or inquiry specially through search for new facts in any branch of knowledge. Some people consider research as a movement from the known to the unknown.



What is research?

The systematic investigation into and study of materials, sources, etc, in order to establish facts and reach new conclusions. b. an endeavour to discover new or collate old facts etc by the scientific study of a subject or by a course of critical investigation



What is research?

Research is what we do when we have a question or a problem we want to resolve

We may already think we know the answer to our question already

We may think the answer is obvious, common sense even

But until we have subjected our problem to rigorous scientific scrutiny, our 'knowledge' remains little more than guesswork or at best, intuition.



What is research?

First priority is to formulate your question

Then figure out how you are going to answer it

- How have others answered it?
- How does your proposal fit in with what others have done?
- How will you know when you have answered it?

Then you can present your answer



OBJECTIVES OF RESEARCH

The purpose of research is to discover answers to questions through the application of scientific procedures. The main aim of research is to find out the truth which is hidden and which has not been discovered as yet. Each research study has its own specific purpose.



Characteristics of Research

Research is directed towards the solution of a problem.

Based upon observable experience or empirical evidence.

Demands accurate observation and description.

Involves gathering new data from primary sources or using existing data for a new purpose.

Requires expertise

Objective and logical

Involves the quest for answers to unsolved problems.

Requires courage.

Should be carefully recorded and reported.



Research Categories

Basic Research

- Conducted to verify the acceptability of a given theory or to know more about certain concepts.
- Also known as fundamental research, „theoretical“ research or „pure“ research.

Applied Research

- conducted when a decision must be made about a specific real-life problem.
 - It is of two types:
- ▮ Applied Research can further be divided into two groups:
 - ▮ Problem Solving Research
 - ▮ Problem Oriented Research



Research Process

The research process involves several key stages, including problem identification, literature review, data collection, analysis and reporting. Each stage is essential for producing valid and reliable research outcomes



Quantitative Research

Utilises statistical analysis to quantify observations and generalise results. Emphasizes objectivity and control in data collection and analysis.



Qualitative Research

Focuses on understanding human behavior and experiences through in-depth observations and interviews. Values subjective interpretation and context.



Mixed Methods Research

Combines quantitative and qualitative approaches to provide a comprehensive understanding of a research problem. Offers diverse perspectives and insights.



Experimental Research

Employs controlled experiments to establish cause-and-effect relationships. Requires strict adherence to predefined protocols and conditions.





Survey Research

Utilises Questionnaires and interviews to gather data from a sample population. Provides insights into attitudes, behaviors, and opinions.



Ethnographic Research

Involves immersion in a culture to understand social phenomena.
Requires prolonged fieldwork and participant observation.

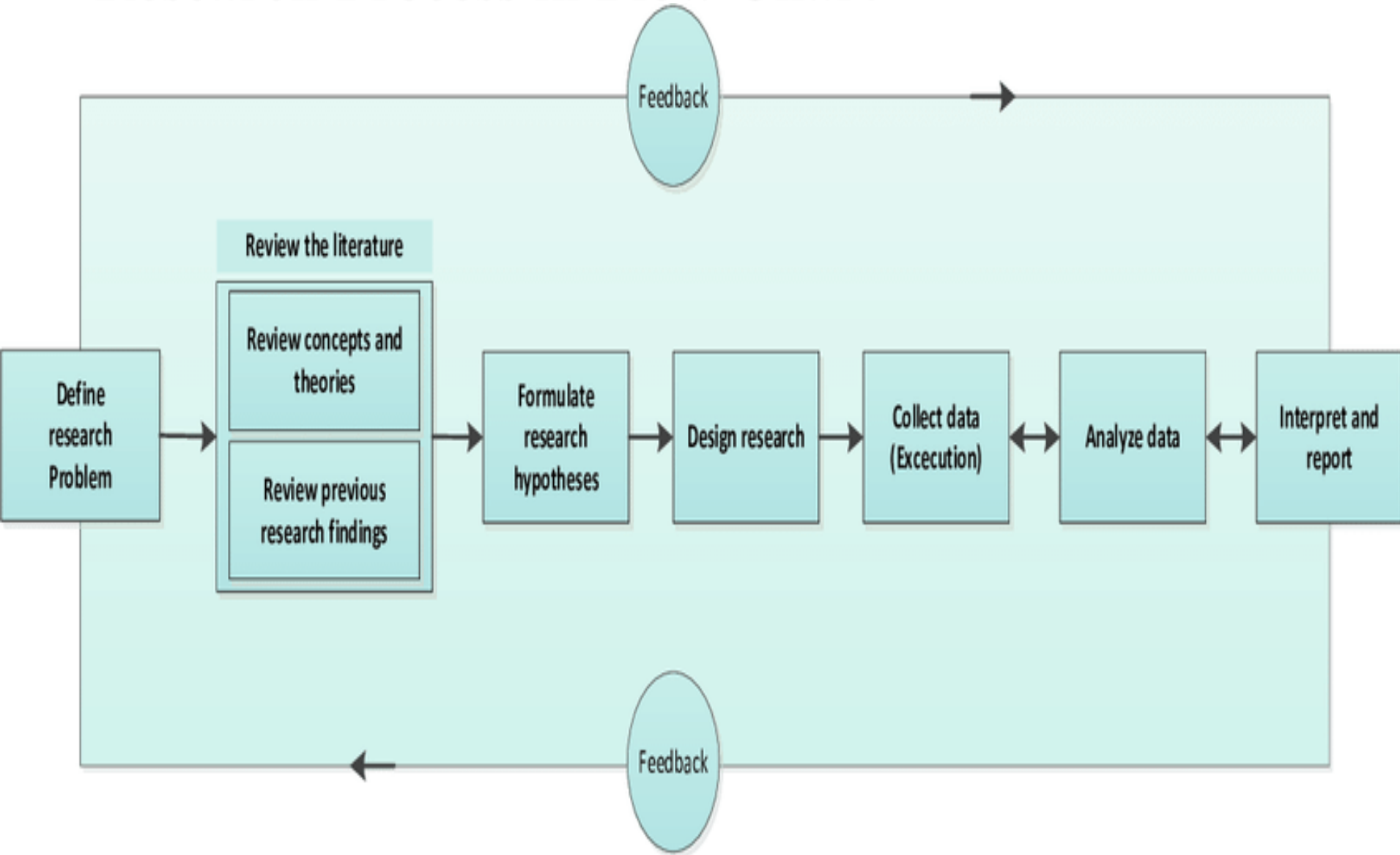


Meta-Analysis

Synthesizes findings from multiple studies to draw overarching conclusions. Provides a comprehensive overview of existing research.



Research Process in Flow Chart





Steps involved in Research Process in Research Methodology

Formulating the research problem: There are two types of research problems, viz., those which relate to states of nature and those which relate to relationships between variables. At the very outset the researcher must single out the problem he wants to study, i.e., he must decide the general area of interest or aspect of a subject-matter that he would like to inquire into.

Extensive literature survey: Once the problem is formulated, a brief summary of it should be written down. It is compulsory for a research writing a thesis to write synopsis of the topic and submit to his/her supervisor. At this juncture the researcher should undertake extensive literature survey connected with the problem.



Research Design

A **research design** is a strategy for answering your research question using empirical data. Creating a research design means making decisions about:

- Your overall research objectives and approach
- Whether you'll rely on Primary or secondary research
- Your sampling methods or criteria for selecting subjects
- Your data collection methods
- The procedures you will follow to collect data
- Your data analysis methods

A well-planned research design helps ensure that your methods match your research objectives and that you use the right kind of analysis for your data.



Determining sample design: All the items under consideration in any field of inquiry constitute „universe“ or „population“. A complete enumeration of all the items in the „population“ is known as a census inquiry. It can be presumed that in such an inquiry when all the items are covered no element of chance is left and highest accuracy is obtained. But in practice this may not be true.

Collecting the data: In dealing with any real life problem it is often found that data at hand are inadequate, and hence, it becomes necessary to collect data that are appropriate. There are several ways of collecting the appropriate data which differ considerably in context of money costs, time and other resources at the disposal of the researcher.



Execution of the project: Execution of the project is a very important step in the research process. If the execution of the project proceeds on correct lines, the data to be collected would be adequate and dependable. The researcher should see that the project is executed in a systematic manner and in time.

Analysis of data: After the data have been collected, the researcher turns to the task of analyzing them. The analysis of data requires a number of closely related operations such as establishment of categories, the application of these categories to raw data through coding, tabulation and then drawing statistical inferences. The unwieldy data should necessarily be condensed into a few manageable groups and tables for further analysis. Thus, researcher should classify the raw data into some purposeful and usable categories.



The main text of the report should have the following parts:

Introduction: It should contain a clear statement of the objective of the research and an explanation of the methodology adopted in accomplishing the research. The scope of the study along with various limitations should as well be stated in this part.

Summary of findings: After introduction there would appear a statement of finding sand recommendations in non-technical language. If the findings are extensive, they should be summarized.

Main report: The main body of the report should be presented in logical sequence and broken-down into readily identifiable sections.

Conclusion: Towards the end of the main text, researcher should again put down the results of his research clearly and precisely. In fact, it is the final summing up.

At the end of the report, appendices should be enlisted in respect of all technical data. Bibliography, i.e., list of books, journals, reports, etc., consulted, should also be given in the end.

Index should also be given specially in a published research report.



Ethical Considerations

Ethical considerations are paramount in research to ensure the protection of participants and the integrity of the study. Adhering to ethical guidelines and obtaining informed consent are fundamental principles.

THANK YOU



**NAMIBIA UNIVERSITY
OF SCIENCE AND TECHNOLOGY**

-RESEARCH PROPOSAL TEMPLATE-

PROPOSED TITLE

(Type the proposed title in capital letters, the title should be in the middle of the page)

by

Name(s) & Surname

Student Number

Submitted in fulfillment of the requirements for the degree

XXXXXXXXXXXXXXXXXXXXXX

in the

(Department of XXXXXXXXX)

at

NAMIBIA UNIVERSITY OF SCIENCE AND TECHNOLOGY

Supervisor:

(Title, initials and surname of supervisor)

Co-supervisor:

(Title, initials and surname of supervisor)

Date of submission:

dd/mm/yyyy

METADATA

TITLE: (your title)

STUDENT NAME:

SUPERVISOR:

CO-SUPERVISOR:

DEPARTMENT:

QUALIFICATION:

SPECIALISATION:

CLUSTER:

KEYWORDS:

TYPE OF RESEARCH: Applied Research

METHODOLOGY:

STATUS: Proposal

DOCUMENT DATE: (date of submission)

SPONSOR (Cluster/ Research lab):

1. Topic:
2. Problem Statement
3. Aim
4. Objectives
5. Research Questions

Supervisor Signature:

NUST & FCI LOGO (here)

-RESEARCH PROPOSAL TEMPLATE-

PROPOSED TITLE

(Type the proposed title in capital letters, the title should be in the middle of the page)

by

Name(s) & Surname

Student Number

Submitted in fulfillment of the requirements for the degree

XXXXXXXXXXXXXXXXXXXXX

in the

(Department of XXXXXXXXX)

at

NAMIBIA UNIVERSITY OF SCIENCE AND TECHNOLOGY

Supervisor:

(Title, initials and surname of supervisor)

Co-supervisor:

(Title, initials and surname of supervisor)

Date of submission:

dd/mm/yyyy

METADATA

TITLE: (your title)

STUDENT NAME:

SUPERVISOR:

CO-SUPERVISOR:

DEPARTMENT:

QUALIFICATION:

SPECIALISATION:

CLUSTER:

KEYWORDS:

TYPE OF RESEARCH: Applied Research

METHODOLOGY:

STATUS: Proposal

DOCUMENT DATE: (date of submission)

SPONSOR (Cluster/ Research lab):

DECLARATION

I, ...(name (s))... , hereby declare that the work contained in the proposal for the degree.....project, entitled:

“.....(title of the project).....”,

is my own original work and that I have not previously in its entirety or in part submitted it at any university or other higher education institution for the award of a degree.

I further declare that I will fully acknowledge any sources of information I will use for the research in accordance with the Institution rules.

Signature: _____ Date: _____

SIGNATURE OF THE SUPERVISOR

I, _____, herewith declare that I accept this proposal for my supervision

Signature: _____ Date: _____

INSTRUCTIONS:

- ✓ The body of your proposal must address the sections listed below; however you are free to adapt to the type and nature of your investigation.
- ✓ Use Calibri font 12; line spacing 1.5; justify-alignment (except if intended)
- ✓ Page numbering: use “Page X of Y”, or “x/y” at bottom of page
- ✓ Avoid use of “first person singular” like “I”, “me”, etc.
- ✓ The length of the proposal write-up should be between 10 to 15 pages (up to 20 for PhD), including everything.
- ✓ Remove all instructions given in this template in your final write-up.

PROPOSED TITLE

Type the proposed title in capital letters, the title should be in the middle of the page; an unambiguous title that conveys the focus of the research

ABSTRACT

(short summary of the study in about 200 -250 words; the content should include: context, problem, methods/solutions, significance, (future..)); (This is the last section of your proposal to be written)

(Also see Appendix A for more insight)

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ABSTRACT	V
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4 IMPORTANCE / SIGNIFICANCE/BENEFITS OF THE STUDY/VALUE	2
5 RESEARCH METHODOLOGY (APPROACH TO THE SOLUTION)	2
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1 INTRODUCTION/BACKGROUND

Note that a research proposal is always future directed. It is an “architects plan” which explains what the researcher intends to do. It is generally written in the future tense. Here you give the context or background and introduction to the research.

2 PROBLEM STATEMENT

(Main statement; may be followed by related questions) (and/ or) Hypothesis (assumptions) or / and the type of envisaged solution)

Divide this part into three:

- Motivate the reason / need for the research by, for example, explaining the dilemma and problem. What will you be investigating in order to come to a conclusion about the problem that you have identified.
- Immediately after this one should state any HYPOTHESIS, IF ANY, if it refers to basic statement related to the problem; otherwise if the hypothesis is related to the type of suggested solution then it will be stated after “Research Objectives or Research Questions

3 RESEARCH OBJECTIVE(S)

(Main objective and sub-objectives as major tasks to be done which will bring you to, or help you reach a given milestone)

- Here you also have to mention possible research objectives (what do you want to achieve with this study). Usually these are 3-5 steps or milestones or sub-objectives you will follow to reach the purpose. Remember when readers get to your conclusion of your study they will go back to this part to see whether your stated purpose or objectives have been achieved.
- (alternatively) **RESEARCH QUESTIONS**: Main and sub-questions. Each (sub) Research question should be answered when at least one Research (sub) objective has been met.

4 PRELIMINARY LITERATURE REVIEW and or Theoretical Underpinnings/ Framework

(Review of existing state of knowledge (understanding, methodologies, solutions, projections...) on the main questions/ issues of the research that your Keywords and or practices (refer to the main variables of the problem) encapsulates.

- **More than 60% should deal with the problem focus** (one or some of this: problem domain, the process, methodology, solution, significance); and **less than 40%** on the remaining aspect.
- Advised to use sub-sections corresponding to knowledge/ issues encapsulated in the main questions/ or objectives of the study
- The section should end with a short preliminary conclusion, and link to your initially indicated problem statement).

Normally your full literature review (in the final Thesis) must adequately cover the following issues:

- Conceptual and operational definitions of the key constructs mentioned in the title and hypotheses
- A discussion of relevant previous research or conceptual work on the constructs
- A discussion of possible relationships between key constructs or expected differences between groups on the key constructs included in the study
- A discussion of possible measurement approaches and scales for each of the key constructs included in the study
- Demonstrating your ability to critically review and logically present the materials in a coherent manner.

It must provide enough theoretical support for the hypotheses formulated, if any, for testing.

Use relevant, current and comprehensive sources from authority in the field

5 IMPORTANCE / SIGNIFICANCE/BENEFITS OF THE STUDY/VALUE

State why this study is important, and to whom it may have any benefits/values. Why is your work worth doing? There is usually a theoretical (oftentimes academic) and a practical significance (to the social, human, industry context, etc.) which needs attention/explanation.

6 RESEARCH METHODOLOGY (Approach to the solution)

This section should consist of the following:

- Your Methodology (blueprint view of the inquiry)
- The research Design: Name and discuss the overall approach you will use to test your thesis statement or research questions. Qualitative or quantitative or triangulation, back this with references. Here you can also highlight the strength and weaknesses of the design.
- Your research Methods: Each design has certain methods, indicate the methodology in detail, how you apply it and justify this to the reader. Sub-section here should be research/data collection instruments, participants and data analysis.

6.1 Research/data collection instruments

Describe the instruments you will use for data collection, what you expect them to deliver and why you believe they will get you reliable data. Also indicate the reliability and validity of the instruments. How do you know the instruments will get the data you want and that this data will measure what you say it should. Will the data be sufficient for your purposes?

6.2 Participants/population

Which sampling technique did you apply for selecting your participants? Why only these and what will their feedback provide/mean to assist you to answer the research questions? Provide a detailed table of how many participants from which levels in the company/business you will include in the study.

6.3 Data analysis

Once you have data of whatever nature you need to do something with it in order to turn it into evidence. To turn it into information you need to analyse it. You may apply statistical analysis techniques or some form of textual analysis or a combination of the two. Whatever you choose you need to explain it to your readers so that they understand how you will arrive at your conclusions.

(see [Appendix B](#) for a more hierarchical presentation of the same.)

- Limitations of the method: All methods have limitations. To identify what limitations your method have, brainstorm how else you might have tackled your problem.

7 DELINEATIONS AND LIMITATIONS

7.1 Delineation

covers you back and explains exactly what you are responsible for and by detailing what you are not responsible for and why. This is the scope of your work.

7.2 Limitations

are those things which you will not cover because of money, time and also has to do with the method you have chosen and its limitations (scope, context, resources/time/finance/skills, etc.)

[and (optional) **Risks** (impact of constraints, and ethics issues)] ; (It is advised at proposal level you make use of bulleting items)

8 ASSUMPTIONS

Assumptions are things that you take to be true without checking whether they are true. You also expect your reader to believe them without offering further evidence. Try to limit these, and provide evidence but if you cannot get rid of it, rather name it than leaving it for examiners to criticize it later on. If there are no assumptions then do mention it here.

Assumptions are part of the premises when reasoning; they can be used while presenting your design.

9 RISK/ FEASIBILITY ANALYSIS AND ETHICAL CONSIDERATIONS

List here if there will be any ethical consent to be gathered from participants or your population and other ethical issues. Indicate that documents will be kept by you/at NUST for a period of two years. That participant information will be kept confidential and that participants can withdraw from the study at any time, that their participation is voluntarily and if your project has been provided with any ethical Faculty number.

For Risk/ Feasibility Analysis do refer to section 6 also because they are linked to Constraints/Limitations; (make use of bulleting items)

10 WORK PLAN

(Matching methods/tasks with sub-objectives over time; Gantt-chart can be used; landscape paper format too)

Task number	Task (sub-objectives/ milestones)	(Resources)	Deliverables (interim achievements)	Start and Finish dates	Duration	Dependency Tasks number

(Detailed Resources Needed) (including **Budget**, if any)

Number	Resources & Quantity	Specification	Purpose	Costs	Availability

11 CHAPTER OUTLINE/ OVERVIEW

Here you should indicate what each chapter of the planned research write-up will focus on:

Chapter One: Introduction

Chapter Two: Literature review (Note this can span over more than one chapter depending on the scope of investigation; e.g. covering state-of-the-art “theory”, state-of-the-art- “practices”, state-of-the-art- “techniques”)

Next Chapter: Research methodology

Next Chapter: Findings/results and discussion

Next Chapter: Significance of the findings (=> contribution)

Next Chapter: Conclusions, recommendations and future works

REFERENCES / BIBLIOGRAPHY

The list of references at the end of your report contains details on all materials consulted (books, proceedings, journal articles, e-mail messages and web pages) and that have been cited in the text. Please make sure that you use the correct referencing method in the text as well as in the list of references at the end of your report.

Bibliography may be chosen if the list of sources should include materials used but could not be cited in the text (check with your supervisor)

Please use the **APA method** of referencing as explained in the CITATION guide as provided by NUST Library and from the FCI research Coordinators.

APPENDIX A: ABSTRACT CONTENT

Item in abstract	Example
Problem specification	The lack of <i>[state what is missing that causes the problem you have identified]</i> remains an open problem.
Main ways of attacking the problem(<i>show knowledge of area</i>)	The most common approaches to <i>[your problem area]</i> are based on <i>[state broad underlying assumption/theory]</i> . These approaches <i>[list main features]</i> . While such approaches have the advantage of <i>[say what they are good at]</i> , they cannot <i>[identify limitations]</i> .
Recent approaches (<i>show problem hasn't been solved recently</i>)	In recent years, <i>[alternative 'leading edge' type of approach/system/model/theory]</i> has gained increasing support, because <i>[say why it is getting popular]</i> .
Specific recent approach that you are going to develop further, and its limitations (<i>establish small research gap</i>)	<i>[Specify one form if the new approach]</i> is one form of approach/system/model/theory that <i>[say what this form is particularly good at]</i> . However, attempts at implementing/trialing the <i>[specific approach]</i> have exposed limitations. <i>[describe some different attempts and their respective limitations]</i> .
Aim of the thesis. Main features of proposed solution.	This thesis is to design/develop and test a new <i>[approach/system/model/theory]</i> , which <i>[summarise its main features]</i> . It will do this by <i>[indicate methodology/methods]</i> .
Expected benefits/contribution. How these will be evaluated.	It is expected that this approach/system/model/theory will overcome the limitations of <i>[say what you expect it will better than existing approaches]</i> . The new approach/system/model/theory will be evaluated by <i>[say how the benefits will be demonstrated]</i> .

(Source: Muyingi et al. (unpublished). "The complete research guidebook full of things your supervisor should have told you")

APPENDIX B: METHODOLOGY

Methodology can be restructured in Hierarchy:

1-The research Paradigm (philosophical view (ontology, epistemology, axiology, and rhetoric) of knowledge and its relationship with the researcher; e.g. Positivism, Interpretivism, Pragmatism

2-The Main Approach / Methodology : High-level map/ Set of Principles and Focal Approach (Descriptive; Explanatory; Exploratory, Theory testing/building/ constructive; Comparative; Longitudinal; Action Research; Participatory; Blended methodologies)

3- The Main Design or Argumentation Flow or Problem Solving Approach or Strategy (Deductive/ Inductive/ etc.)

4-The Methods (How to apply the approach: (Data type: qualitative or Quantitative or Mixed); and (Procedures)

5-Tools/Techniques/Instruments: (how to materialise the methods):

a)-The instruments/ tools to Collect Data

b)-The instruments/ tools to Analyse Data

c)-The instruments/ tools to Evaluate/ test Findings)

APPENDIX C: TECHNICAL CARE CHECKLIST FOR FINAL RESEARCH PROPOSAL

Please check that your final research proposal complies with the specific requirements set out below. Students often lose unnecessary marks because of sloppy technical care. Ask yourself the following questions ...

TECHNICAL CARE	✓ or x
1. PRELIMINARIES (Before the main body text starts on page 1)	
Is the layout of and information supplied on the title page correct?	
Have I included an Abstract (on a separate page) before the Table of Contents?	
Does the Abstract provide a <u>complete</u> summary of the whole research proposal?	
Is the Table of Contents complete and correct?	
Is the List of Figures complete and correct?	
Is the List of Tables complete and correct?	
Do the wordings of headings correspond meticulously with the wording in the Table of Contents?	
Do the headings numbers in the text correspond meticulously with the numbers in the Table of Contents?	
Do the figure / table captions listed in the List of Tables and List of Figures correspond a 100% with the captions used in the text?	
Have all tables and figures been supplied with correct captions (situated <u>above</u> the table or figure)?	
Have all tables and figures been supplied with correct and complete source references where appropriate (situated <u>below</u> the table or figure)?	
Have all tables and figures been numbered correctly? (Figures and tables are numbered independently starting from 1. Do not include section numbers in the numbering of tables / figures)	
Are the preliminary pages (i.e. Executive Summary, Table of Contents, List of Figures and List of Tables) numbered in Roman numerals: (i), (ii), (iii), (iv)?	
2. TEXT	
Are all pages in the body of the text numbered correctly with Arabic figures (1, 2, 3, etc. out of total page)?	
Have I used only one method of emphasizing (<i>italics</i> , or bold or <u>underlining</u>) throughout the report?	

Is there a line open between paragraphs?	
Are the paragraphs in my report not perhaps too long? Can't I split long paragraphs to improve readability?	
Have I made any "sweeping" or unsubstantiated statements, such as "there is no literature available on this topic" or "this research will contribute to the body of knowledge"?	
Have I used an impersonal, objective writing style (One symptom of a subjective writing style is that the researcher refers to him-/herself directly. References to "I", "we" and even "the researcher" should be avoided.)	
3. REFERENCING TECHNIQUES	
Are all sources (books, journals, e-mail messages, web pages) that have been cited in the text, also been included in the List of References at the end of the report and vice versa?	
Is the list of references strictly in the required order of the referencing style you are using ?	
Have I used the correct revised (abridged) Harvard referencing method in the text, as well as in the list of references?	
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Are all the appendices numbered up to the last page?	
Have all the appendices been tagged / "flagged" for easy cross-referencing?	
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5. GENERAL TECHNICAL CARE	
<i>Have I checked my proposal for spelling and grammatical errors?</i>	
<i>Have I used capital letters and abbreviations correctly?</i>	
Have I consistently rounded off all numeric values to three (3) decimals?	
Has text of the whole document been "justified" (i.e. aligned evenly along both the left and right margins to form a square box)?	
Are all headings numbered correctly?	
Headings in capital letters (first order headings) should not be underlined.	
There should be no full stops at the end of headings.	
Have I used an A4 paper setting?	

Have all pages printed correctly?	
Have I consistently used one and a half (1½) line spacing in the body of the text?	
Is the font size of the body text correct (12 pt)?	
Have I consistently used the same font type (i.e. Ariel or Times New Roman) and font size (12 pt) for the body text?	
Is the left margin set to at least 2 cm?	

Postgraduate Research Progress Report Template (one page)

Faculty of Computing and Informatics (FCI)

Year under Review: (tick-off)	One	Two	Three	Four	Five	Six	Seven	Eight
Semester End: (tick-off)								
Student Name, Surname								
Student Number								
Degree								
Research Title								
1. Research Problem in brief (Problem Statement)								
2. Research Objectives (use bullet points)		•						
3. Evaluation	Activities Completed/ Achievements? What outputs have been Delivered? (use bullet points)							
	Status (If No, please provide details: How far from schedule, budget or objectives? What actions are you taking to resolve the delays? What support is required?)	On schedule?						
		On budget?						
		On target?						
	Activities in Progress (use bullet points)							
	Activities Outstanding (use bullet points)							
4. Adjustments Were there any adjustments to the scope? (use bullet points, If yes, please provide details: What was changed? How will it impact your schedule, budget or objectives?)								
5. Challenges to Date (use bullet points)								

6. Risks (use bullet points)	
7. Students Comments	
8. Supervisor Comments	
Student's Signature	
Supervisor's Name and Signature	

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Student's Signature	
Supervisor's Name and Signature	



NAMIBIA UNIVERSITY
OF SCIENCE AND TECHNOLOGY

-RESEARCH PROPOSAL TEMPLATE-

PROPOSED TITLE

(Type the proposed title in capital letters, the title should be in the middle of the page)

by

Name(s) & Surname

Student Number

Submitted in fulfillment of the requirements for the degree

XXXXXXXXXXXXXXXXXXXXXX

in the

(Department of XXXXXXXXX)

at

NAMIBIA UNIVERSITY OF SCIENCE AND TECHNOLOGY

Supervisor:

(Title, initials and surname of supervisor)

Co-supervisor:

(Title, initials and surname of supervisor)

Date of submission:

dd/mm/yyyy

METADATA

TITLE: (your title)

STUDENT NAME:

SUPERVISOR:

CO-SUPERVISOR:

DEPARTMENT:

QUALIFICATION:

SPECIALISATION:

CLUSTER:

KEYWORDS:

TYPE OF RESEARCH: Applied Research

METHODOLOGY:

STATUS: Proposal

DOCUMENT DATE: (date of submission)

SPONSOR (Cluster/ Research lab):

1. Topic:
2. Problem Statement
3. Aim
4. Objectives
5. Research Questions

Supervisor Signature:



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- 5. Research Questions**
- 6. Preliminary Literature Review**
- 7. Importance / significance/benefits of the study/value**
- 8. Reference List**



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5. Research Questions
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8. Research Methodology
 - a. Research data collection Instruments
 - b. Participants Population
 - c. Data Analysis
9. Delineations and Limitations.
 - a. Delineations
 - b. Limitations
10. Assumptions,
11. Risk/ Feasibility Analysis And Ethical Considerations
12. Work Plan
13. Chapter Outline/ Overview

14. Reference List

NUST & FCI LOGO (here)

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METHODOLOGY:

STATUS: Proposal

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SPONSOR (Cluster/ Research lab):

DECLARATION

I, ...(name (s))... , hereby declare that the work contained in the proposal for the degree.....project, entitled:

“.....(title of the project).....”,

is my own original work and that I have not previously in its entirety or in part submitted it at any university or other higher education institution for the award of a degree.

I further declare that I will fully acknowledge any sources of information I will use for the research in accordance with the Institution rules.

Signature: _____ Date: _____

SIGNATURE OF THE SUPERVISOR

I, _____, herewith declare that I accept this proposal for my supervision

Signature: _____ Date: _____

INSTRUCTIONS:

- ✓ The body of your proposal must address the sections listed below; however you are free to adapt to the type and nature of your investigation.
- ✓ Use Calibri font 12; line spacing 1.5; justify-alignment (except if intended)
- ✓ Page numbering: use “Page X of Y”, or “x/y” at bottom of page
- ✓ Avoid use of “first person singular” like “I”, “me”, etc.
- ✓ The length of the proposal write-up should be between 10 to 15 pages (up to 20 for PhD), including everything.
- ✓ Remove all instructions given in this template in your final write-up.

PROPOSED TITLE

Type the proposed title in capital letters, the title should be in the middle of the page; an unambiguous title that conveys the focus of the research

ABSTRACT

(short summary of the study in about 200 -250 words; the content should include: context, problem, methods/solutions, significance, (future..)); (This is the last section of your proposal to be written)

(Also see Appendix A for more insight)

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3 PRELIMINARY LITERATURE REVIEW AND OR THEORETICAL UNDERPINNINGS/ FRAMEWORK	2
4 IMPORTANCE / SIGNIFICANCE/BENEFITS OF THE STUDY/VALUE	2
5 RESEARCH METHODOLOGY (APPROACH TO THE SOLUTION)	2
5.1 RESEARCH/DATA COLLECTION INSTRUMENTS	3
5.2 PARTICIPANTS/POPULATION	3
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6 DELINEATIONS AND LIMITATIONS	3
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1 INTRODUCTION/BACKGROUND

Note that a research proposal is always future directed. It is an “architects plan” which explains what the researcher intends to do. It is generally written in the future tense. Here you give the context or background and introduction to the research.

2 PROBLEM STATEMENT

(Main statement; may be followed by related questions) (and/ or) Hypothesis (assumptions) or / and the type of envisaged solution)

Divide this part into three:

- Motivate the reason / need for the research by, for example, explaining the dilemma and problem. What will you be investigating in order to come to a conclusion about the problem that you have identified.
- Immediately after this one should state any HYPOTHESIS, IF ANY, if it refers to basic statement related to the problem; otherwise if the hypothesis is related to the type of suggested solution then it will be stated after “Research Objectives or Research Questions

3 RESEARCH OBJECTIVE(S)

(Main objective and sub-objectives as major tasks to be done which will bring you to, or help you reach a given milestone)

- Here you also have to mention possible research objectives (what do you want to achieve with this study). Usually these are 3-5 steps or milestones or sub-objectives you will follow to reach the purpose. Remember when readers get to your conclusion of your study they will go back to this part to see whether your stated purpose or objectives have been achieved.
- (alternatively) **RESEARCH QUESTIONS**: Main and sub-questions. Each (sub) Research question should be answered when at least one Research (sub) objective has been met.

4 PRELIMINARY LITERATURE REVIEW and or Theoretical Underpinnings/ Framework

(Review of existing state of knowledge (understanding, methodologies, solutions, projections...) on the main questions/ issues of the research that your Keywords and or practices (refer to the main variables of the problem) encapsulates.

- **More than 60% should deal with the problem focus** (one or some of this: problem domain, the process, methodology, solution, significance); and **less than 40%** on the remaining aspect.
- Advised to use sub-sections corresponding to knowledge/ issues encapsulated in the main questions/ or objectives of the study
- The section should end with a short preliminary conclusion, and link to your initially indicated problem statement).

Normally your full literature review (in the final Thesis) must adequately cover the following issues:

- Conceptual and operational definitions of the key constructs mentioned in the title and hypotheses
- A discussion of relevant previous research or conceptual work on the constructs
- A discussion of possible relationships between key constructs or expected differences between groups on the key constructs included in the study
- A discussion of possible measurement approaches and scales for each of the key constructs included in the study
- Demonstrating your ability to critically review and logically present the materials in a coherent manner.

It must provide enough theoretical support for the hypotheses formulated, if any, for testing.

Use relevant, current and comprehensive sources from authority in the field

5 IMPORTANCE / SIGNIFICANCE/BENEFITS OF THE STUDY/VALUE

State why this study is important, and to whom it may have any benefits/values. Why is your work worth doing? There is usually a theoretical (oftentimes academic) and a practical significance (to the social, human, industry context, etc.) which needs attention/explanation.

6 RESEARCH METHODOLOGY (Approach to the solution)

This section should consist of the following:

- Your Methodology (blueprint view of the inquiry)
- The research Design: Name and discuss the overall approach you will use to test your thesis statement or research questions. Qualitative or quantitative or triangulation, back this with references. Here you can also highlight the strength and weaknesses of the design.
- Your research Methods: Each design has certain methods, indicate the methodology in detail, how you apply it and justify this to the reader. Sub-section here should be research/data collection instruments, participants and data analysis.

6.1 Research/data collection instruments

Describe the instruments you will use for data collection, what you expect them to deliver and why you believe they will get you reliable data. Also indicate the reliability and validity of the instruments. How do you know the instruments will get the data you want and that this data will measure what you say it should. Will the data be sufficient for your purposes?

6.2 Participants/population

Which sampling technique did you apply for selecting your participants? Why only these and what will their feedback provide/mean to assist you to answer the research questions? Provide a detailed table of how many participants from which levels in the company/business you will include in the study.

6.3 Data analysis

Once you have data of whatever nature you need to do something with it in order to turn it into evidence. To turn it into information you need to analyse it. You may apply statistical analysis techniques or some form of textual analysis or a combination of the two. Whatever you choose you need to explain it to your readers so that they understand how you will arrive at your conclusions.

(see [Appendix B](#) for a more hierarchical presentation of the same.)

- Limitations of the method: All methods have limitations. To identify what limitations your method have, brainstorm how else you might have tackled your problem.

7 DELINEATIONS AND LIMITATIONS

7.1 Delineation

covers you back and explains exactly what you are responsible for and by detailing what you are not responsible for and why. This is the scope of your work.

7.2 Limitations

are those things which you will not cover because of money, time and also has to do with the method you have chosen and its limitations (scope, context, resources/time/finance/skills, etc.)

[and (optional) **Risks** (impact of constraints, and ethics issues)] ; (It is advised at proposal level you make use of bulleting items)

8 ASSUMPTIONS

Assumptions are things that you take to be true without checking whether they are true. You also expect your reader to believe them without offering further evidence. Try to limit these, and provide evidence but if you cannot get rid of it, rather name it than leaving it for examiners to criticize it later on. If there are no assumptions then do mention it here.

Assumptions are part of the premises when reasoning; they can be used while presenting your design.

9 RISK/ FEASIBILITY ANALYSIS AND ETHICAL CONSIDERATIONS

List here if there will be any ethical consent to be gathered from participants or your population and other ethical issues. Indicate that documents will be kept by you/at NUST for a period of two years. That participant information will be kept confidential and that participants can withdraw from the study at any time, that their participation is voluntarily and if your project has been provided with any ethical Faculty number.

For Risk/ Feasibility Analysis do refer to section 6 also because they are linked to Constraints/Limitations; (make use of bulleting items)

10 WORK PLAN

(Matching methods/tasks with sub-objectives over time; Gantt-chart can be used; landscape paper format too)

Task number	Task (sub-objectives/ milestones)	(Resources)	Deliverables (interim achievements)	Start and Finish dates	Duration	Dependency Tasks number

(Detailed Resources Needed) (including **Budget**, if any)

Number	Resources & Quantity	Specification	Purpose	Costs	Availability

11 CHAPTER OUTLINE/ OVERVIEW

Here you should indicate what each chapter of the planned research write-up will focus on:

Chapter One: Introduction

Chapter Two: Literature review (Note this can span over more than one chapter depending on the scope of investigation; e.g. covering state-of-the-art “theory”, state-of-the-art- “practices”, state-of-the-art- “techniques”)

Next Chapter: Research methodology

Next Chapter: Findings/results and discussion

Next Chapter: Significance of the findings (=> contribution)

Next Chapter: Conclusions, recommendations and future works

REFERENCES / BIBLIOGRAPHY

The list of references at the end of your report contains details on all materials consulted (books, proceedings, journal articles, e-mail messages and web pages) and that have been cited in the text. Please make sure that you use the correct referencing method in the text as well as in the list of references at the end of your report.

Bibliography may be chosen if the list of sources should include materials used but could not be cited in the text (check with your supervisor)

Please use the **APA method** of referencing as explained in the CITATION guide as provided by NUST Library and from the FCI research Coordinators.

APPENDIX A: ABSTRACT CONTENT

Item in abstract	Example
Problem specification	The lack of <i>[state what is missing that causes the problem you have identified]</i> remains an open problem.
Main ways of attacking the problem(<i>show knowledge of area</i>)	The most common approaches to <i>[your problem area]</i> are based on <i>[state broad underlying assumption/theory]</i> . These approaches <i>[list main features]</i> . While such approaches have the advantage of <i>[say what they are good at]</i> , they cannot <i>[identify limitations]</i> .
Recent approaches (<i>show problem hasn't been solved recently</i>)	In recent years, <i>[alternative 'leading edge' type of approach/system/model/theory]</i> has gained increasing support, because <i>[say why it is getting popular]</i> .
Specific recent approach that you are going to develop further, and its limitations (<i>establish small research gap</i>)	<i>[Specify one form if the new approach]</i> is one form of approach/system/model/theory that <i>[say what this form is particularly good at]</i> . However, attempts at implementing/trialing the <i>[specific approach]</i> have exposed limitations. <i>[describe some different attempts and their respective limitations]</i> .
Aim of the thesis. Main features of proposed solution.	This thesis is to design/develop and test a new <i>[approach/system/model/theory]</i> , which <i>[summarise its main features]</i> . It will do this by <i>[indicate methodology/methods]</i> .
Expected benefits/contribution. How these will be evaluated.	It is expected that this approach/system/model/theory will overcome the limitations of <i>[say what you expect it will better than existing approaches]</i> . The new approach/system/model/theory will be evaluated by <i>[say how the benefits will be demonstrated]</i> .

(Source: Muyingi et al. (unpublished). "The complete research guidebook full of things your supervisor should have told you")

APPENDIX B: METHODOLOGY

Methodology can be restructured in Hierarchy:

1-The research Paradigm (philosophical view (ontology, epistemology, axiology, and rhetoric) of knowledge and its relationship with the researcher; e.g. Positivism, Interpretivism, Pragmatism

2-The Main Approach / Methodology : High-level map/ Set of Principles and Focal Approach (Descriptive; Explanatory; Exploratory, Theory testing/building/ constructive; Comparative; Longitudinal; Action Research; Participatory; Blended methodologies)

3- The Main Design or Argumentation Flow or Problem Solving Approach or Strategy (Deductive/ Inductive/ etc.)

4-The Methods (How to apply the approach: (Data type: qualitative or Quantitative or Mixed); and (Procedures)

5-Tools/Techniques/Instruments: (how to materialise the methods):

a)-The instruments/ tools to Collect Data

b)-The instruments/ tools to Analyse Data

c)-The instruments/ tools to Evaluate/ test Findings)

APPENDIX C: TECHNICAL CARE CHECKLIST FOR FINAL RESEARCH PROPOSAL

Please check that your final research proposal complies with the specific requirements set out below. Students often lose unnecessary marks because of sloppy technical care. Ask yourself the following questions ...

TECHNICAL CARE	✓ or x
1. PRELIMINARIES (Before the main body text starts on page 1)	
Is the layout of and information supplied on the title page correct?	
Have I included an Abstract (on a separate page) before the Table of Contents?	
Does the Abstract provide a <u>complete</u> summary of the whole research proposal?	
Is the Table of Contents complete and correct?	
Is the List of Figures complete and correct?	
Is the List of Tables complete and correct?	
Do the wordings of headings correspond meticulously with the wording in the Table of Contents?	
Do the headings numbers in the text correspond meticulously with the numbers in the Table of Contents?	
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SPONSOR (Cluster/ Research lab):

DECLARATION

I, ...(name (s))... , hereby declare that the work contained in the proposal for the degree.....project, entitled:

“.....(title of the project).....”,

is my own original work and that I have not previously in its entirety or in part submitted it at any university or other higher education institution for the award of a degree.

I further declare that I will fully acknowledge any sources of information I will use for the research in accordance with the Institution rules.

Signature: _____ Date: _____

SIGNATURE OF THE SUPERVISOR

I, _____, herewith declare that I accept this proposal for my supervision

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INSTRUCTIONS:

- ✓ The body of your proposal must address the sections listed below; however you are free to adapt to the type and nature of your investigation.
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(short summary of the study in about 200 -250 words; the content should include: context, problem, methods/solutions, significance, (future..)); (This is the last section of your proposal to be written)

(Also see Appendix A for more insight)

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(Main statement; may be followed by related questions) (and/ or) Hypothesis (assumptions) or / and the type of envisaged solution)

Divide this part into three:

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6.1 Research/data collection instruments

Describe the instruments you will use for data collection, what you expect them to deliver and why you believe they will get you reliable data. Also indicate the reliability and validity of the instruments. How do you know the instruments will get the data you want and that this data will measure what you say it should. Will the data be sufficient for your purposes?

6.2 Participants/population

Which sampling technique did you apply for selecting your participants? Why only these and what will their feedback provide/mean to assist you to answer the research questions? Provide a detailed table of how many participants from which levels in the company/business you will include in the study.

6.3 Data analysis

Once you have data of whatever nature you need to do something with it in order to turn it into evidence. To turn it into information you need to analyse it. You may apply statistical analysis techniques or some form of textual analysis or a combination of the two. Whatever you choose you need to explain it to your readers so that they understand how you will arrive at your conclusions.

(see [Appendix B](#) for a more hierarchical presentation of the same.)

- Limitations of the method: All methods have limitations. To identify what limitations your method have, brainstorm how else you might have tackled your problem.

7 DELINEATIONS AND LIMITATIONS

7.1 Delineation

covers you back and explains exactly what you are responsible for and by detailing what you are not responsible for and why. This is the scope of your work.

7.2 Limitations

are those things which you will not cover because of money, time and also has to do with the method you have chosen and its limitations (scope, context, resources/time/finance/skills, etc.)

[and (optional) **Risks** (impact of constraints, and ethics issues)] ; (It is advised at proposal level you make use of bulleting items)

8 ASSUMPTIONS

Assumptions are things that you take to be true without checking whether they are true. You also expect your reader to believe them without offering further evidence. Try to limit these, and provide evidence but if you cannot get rid of it, rather name it than leaving it for examiners to criticize it later on. If there are no assumptions then do mention it here.

Assumptions are part of the premises when reasoning; they can be used while presenting your design.

9 RISK/ FEASIBILITY ANALYSIS AND ETHICAL CONSIDERATIONS

List here if there will be any ethical consent to be gathered from participants or your population and other ethical issues. Indicate that documents will be kept by you/at NUST for a period of two years. That participant information will be kept confidential and that participants can withdraw from the study at any time, that their participation is voluntarily and if your project has been provided with any ethical Faculty number.

For Risk/ Feasibility Analysis do refer to section 6 also because they are linked to Constraints/Limitations; (make use of bulleting items)

10 WORK PLAN

(Matching methods/tasks with sub-objectives over time; Gantt-chart can be used; landscape paper format too)

Task number	Task (sub-objectives/ milestones)	(Resources)	Deliverables (interim achievements)	Start and Finish dates	Duration	Dependency Tasks number

(Detailed Resources Needed) (including **Budget**, if any)

Number	Resources & Quantity	Specification	Purpose	Costs	Availability

11 CHAPTER OUTLINE/ OVERVIEW

Here you should indicate what each chapter of the planned research write-up will focus on:

Chapter One: Introduction

Chapter Two: Literature review (Note this can span over more than one chapter depending on the scope of investigation; e.g. covering state-of-the-art “theory”, state-of-the-art- “practices”, state-of-the-art- “techniques”)

Next Chapter: Research methodology

Next Chapter: Findings/results and discussion

Next Chapter: Significance of the findings (=> contribution)

Next Chapter: Conclusions, recommendations and future works

REFERENCES / BIBLIOGRAPHY

The list of references at the end of your report contains details on all materials consulted (books, proceedings, journal articles, e-mail messages and web pages) and that have been cited in the text. Please make sure that you use the correct referencing method in the text as well as in the list of references at the end of your report.

Bibliography may be chosen if the list of sources should include materials used but could not be cited in the text (check with your supervisor)

Please use the **APA method** of referencing as explained in the CITATION guide as provided by NUST Library and from the FCI research Coordinators.

APPENDIX A: ABSTRACT CONTENT

Item in abstract	Example
Problem specification	The lack of <i>[state what is missing that causes the problem you have identified]</i> remains an open problem.
Main ways of attacking the problem(<i>show knowledge of area</i>)	The most common approaches to <i>[your problem area]</i> are based on <i>[state broad underlying assumption/theory]</i> . These approaches <i>[list main features]</i> . While such approaches have the advantage of <i>[say what they are good at]</i> , they cannot <i>[identify limitations]</i> .
Recent approaches (<i>show problem hasn't been solved recently</i>)	In recent years, <i>[alternative 'leading edge' type of approach/system/model/theory]</i> has gained increasing support, because <i>[say why it is getting popular]</i> .
Specific recent approach that you are going to develop further, and its limitations (<i>establish small research gap</i>)	<i>[Specify one form if the new approach]</i> is one form of approach/system/model/theory that <i>[say what this form is particularly good at]</i> . However, attempts at implementing/trialing the <i>[specific approach]</i> have exposed limitations. <i>[describe some different attempts and their respective limitations]</i> .
Aim of the thesis. Main features of proposed solution.	This thesis is to design/develop and test a new <i>[approach/system/model/theory]</i> , which <i>[summarise its main features]</i> . It will do this by <i>[indicate methodology/methods]</i> .
Expected benefits/contribution. How these will be evaluated.	It is expected that this approach/system/model/theory will overcome the limitations of <i>[say what you expect it will better than existing approaches]</i> . The new approach/system/model/theory will be evaluated by <i>[say how the benefits will be demonstrated]</i> .

(Source: Muyingi et al. (unpublished). "The complete research guidebook full of things your supervisor should have told you")

APPENDIX B: METHODOLOGY

Methodology can be restructured in Hierarchy:

1-The research Paradigm (philosophical view (ontology, epistemology, axiology, and rhetoric) of knowledge and its relationship with the researcher; e.g. Positivism, Interpretivism, Pragmatism

2-The Main Approach / Methodology : High-level map/ Set of Principles and Focal Approach (Descriptive; Explanatory; Exploratory, Theory testing/building/ constructive; Comparative; Longitudinal; Action Research; Participatory; Blended methodologies)

3- The Main Design or Argumentation Flow or Problem Solving Approach or Strategy (Deductive/ Inductive/ etc.)

4-The Methods (How to apply the approach: (Data type: qualitative or Quantitative or Mixed); and (Procedures)

5-Tools/Techniques/Instruments: (how to materialise the methods):

a)-The instruments/ tools to Collect Data

b)-The instruments/ tools to Analyse Data

c)-The instruments/ tools to Evaluate/ test Findings)

APPENDIX C: TECHNICAL CARE CHECKLIST FOR FINAL RESEARCH PROPOSAL

Please check that your final research proposal complies with the specific requirements set out below. Students often lose unnecessary marks because of sloppy technical care. Ask yourself the following questions ...

TECHNICAL CARE	✓ or x
1. PRELIMINARIES (Before the main body text starts on page 1)	
Is the layout of and information supplied on the title page correct?	
Have I included an Abstract (on a separate page) before the Table of Contents?	
Does the Abstract provide a <u>complete</u> summary of the whole research proposal?	
Is the Table of Contents complete and correct?	
Is the List of Figures complete and correct?	
Is the List of Tables complete and correct?	
Do the wordings of headings correspond meticulously with the wording in the Table of Contents?	
Do the headings numbers in the text correspond meticulously with the numbers in the Table of Contents?	
Do the figure / table captions listed in the List of Tables and List of Figures correspond a 100% with the captions used in the text?	
Have all tables and figures been supplied with correct captions (situated <u>above</u> the table or figure)?	
Have all tables and figures been supplied with correct and complete source references where appropriate (situated <u>below</u> the table or figure)?	
Have all tables and figures been numbered correctly? (Figures and tables are numbered independently starting from 1. Do not include section numbers in the numbering of tables / figures)	
Are the preliminary pages (i.e. Executive Summary, Table of Contents, List of Figures and List of Tables) numbered in Roman numerals: (i), (ii), (iii), (iv)?	
2. TEXT	
Are all pages in the body of the text numbered correctly with Arabic figures (1, 2, 3, etc. out of total page)?	
Have I used only one method of emphasizing (<i>italics</i> , or bold or <u>underlining</u>) throughout the report?	

Is there a line open between paragraphs?	
Are the paragraphs in my report not perhaps too long? Can't I split long paragraphs to improve readability?	
Have I made any "sweeping" or unsubstantiated statements, such as "there is no literature available on this topic" or "this research will contribute to the body of knowledge"?	
Have I used an impersonal, objective writing style (One symptom of a subjective writing style is that the researcher refers to him-/herself directly. References to "I", "we" and even "the researcher" should be avoided.)	
3. REFERENCING TECHNIQUES	
Are all sources (books, journals, e-mail messages, web pages) that have been cited in the text, also been included in the List of References at the end of the report and vice versa?	
Is the list of references strictly in the required order of the referencing style you are using ?	
Have I used the correct revised (abridged) Harvard referencing method in the text, as well as in the list of references?	
Have I correctly indicated the sources of all tables / figures taken or adapted from the literature? (Such references are provided <u>below</u> the table or figure)	
4. APPENDICES	
Do all the appendices have appropriate descriptive titles?	
Are all the appendices numbered up to the last page?	
Have all the appendices been tagged / "flagged" for easy cross-referencing?	
Are all the appendices included in the Table of Contents with their correct page numbers?	
5. GENERAL TECHNICAL CARE	
<i>Have I checked my proposal for spelling and grammatical errors?</i>	
<i>Have I used capital letters and abbreviations correctly?</i>	
Have I consistently rounded off all numeric values to three (3) decimals?	
Has text of the whole document been "justified" (i.e. aligned evenly along both the left and right margins to form a square box)?	
Are all headings numbered correctly?	
Headings in capital letters (first order headings) should not be underlined.	
There should be no full stops at the end of headings.	
Have I used an A4 paper setting?	

Have all pages printed correctly?	
Have I consistently used one and a half (1½) line spacing in the body of the text?	
Is the font size of the body text correct (12 pt)?	
Have I consistently used the same font type (i.e. Ariel or Times New Roman) and font size (12 pt) for the body text?	
Is the left margin set to at least 2 cm?	

Postgraduate Research Progress Report Template (one page)

Faculty of Computing and Informatics (FCI)

Year under Review: (tick-off)	One	Two	Three	Four	Five	Six	Seven	Eight
Semester End: (tick-off)								
Student Name, Surname								
Student Number								
Degree								
Research Title								
1. Research Problem in brief (Problem Statement)								
2. Research Objectives (use bullet points)		•						
3. Evaluation	Activities Completed/ Achievements? What outputs have been Delivered? (use bullet points)							
	Status (If No, please provide details: How far from schedule, budget or objectives? What actions are you taking to resolve the delays? What support is required?)	On schedule?						
		On budget?						
		On target?						
	Activities in Progress (use bullet points)							
	Activities Outstanding (use bullet points)							
4. Adjustments Were there any adjustments to the scope? (use bullet points, If yes, please provide details: What was changed? How will it impact your schedule, budget or objectives?)								
5. Challenges to Date (use bullet points)								

6. Risks (use bullet points)	
7. Students Comments	
8. Supervisor Comments	
Student's Signature	
Supervisor's Name and Signature	



**NAMIBIA UNIVERSITY
OF SCIENCE AND TECHNOLOGY**

-RESEARCH PROPOSAL TEMPLATE-

PROPOSED TITLE

(Type the proposed title in capital letters, the title should be in the middle of the page)

by

Name(s) & Surname

Student Number

Submitted in fulfillment of the requirements for the degree

XXXXXXXXXXXXXXXXXXXXXX

in the

(Department of XXXXXXXXX)

at

NAMIBIA UNIVERSITY OF SCIENCE AND TECHNOLOGY

Supervisor:

(Title, initials and surname of supervisor)

Co-supervisor:

(Title, initials and surname of supervisor)

Date of submission:

dd/mm/yyyy

METADATA

TITLE: (your title)

STUDENT NAME:

SUPERVISOR:

CO-SUPERVISOR:

DEPARTMENT:

QUALIFICATION:

SPECIALISATION:

CLUSTER:

KEYWORDS:

TYPE OF RESEARCH: Applied Research

METHODOLOGY:

STATUS: Proposal

DOCUMENT DATE: (date of submission)

SPONSOR (Cluster/ Research lab):

1. Topic:
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4. Objectives
5. Research Questions

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 - b. Limitations
10. Assumptions,
11. Risk/ Feasibility Analysis And Ethical Considerations
12. Work Plan
13. Chapter Outline/ Overview

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NUST & FCI LOGO (here)

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Describe the instruments you will use for data collection, what you expect them to deliver and why you believe they will get you reliable data. Also indicate the reliability and validity of the instruments. How do you know the instruments will get the data you want and that this data will measure what you say it should. Will the data be sufficient for your purposes?

6.2 Participants/population

Which sampling technique did you apply for selecting your participants? Why only these and what will their feedback provide/mean to assist you to answer the research questions? Provide a detailed table of how many participants from which levels in the company/business you will include in the study.

6.3 Data analysis

Once you have data of whatever nature you need to do something with it in order to turn it into evidence. To turn it into information you need to analyse it. You may apply statistical analysis techniques or some form of textual analysis or a combination of the two. Whatever you choose you need to explain it to your readers so that they understand how you will arrive at your conclusions.

(see [Appendix B](#) for a more hierarchical presentation of the same.)

- Limitations of the method: All methods have limitations. To identify what limitations your method have, brainstorm how else you might have tackled your problem.

7 DELINEATIONS AND LIMITATIONS

7.1 Delineation

covers you back and explains exactly what you are responsible for and by detailing what you are not responsible for and why. This is the scope of your work.

7.2 Limitations

are those things which you will not cover because of money, time and also has to do with the method you have chosen and its limitations (scope, context, resources/time/finance/skills, etc.)

[and (optional) **Risks** (impact of constraints, and ethics issues)] ; (It is advised at proposal level you make use of bulleting items)

8 ASSUMPTIONS

Assumptions are things that you take to be true without checking whether they are true. You also expect your reader to believe them without offering further evidence. Try to limit these, and provide evidence but if you cannot get rid of it, rather name it than leaving it for examiners to criticize it later on. If there are no assumptions then do mention it here.

Assumptions are part of the premises when reasoning; they can be used while presenting your design.

9 RISK/ FEASIBILITY ANALYSIS AND ETHICAL CONSIDERATIONS

List here if there will be any ethical consent to be gathered from participants or your population and other ethical issues. Indicate that documents will be kept by you/at NUST for a period of two years. That participant information will be kept confidential and that participants can withdraw from the study at any time, that their participation is voluntarily and if your project has been provided with any ethical Faculty number.

For Risk/ Feasibility Analysis do refer to section 6 also because they are linked to Constraints/Limitations; (make use of bulleting items)

10 WORK PLAN

(Matching methods/tasks with sub-objectives over time; Gantt-chart can be used; landscape paper format too)

Task number	Task (sub-objectives/ milestones)	(Resources)	Deliverables (interim achievements)	Start and Finish dates	Duration	Dependency Tasks number

(Detailed Resources Needed) (including **Budget**, if any)

Number	Resources & Quantity	Specification	Purpose	Costs	Availability

11 CHAPTER OUTLINE/ OVERVIEW

Here you should indicate what each chapter of the planned research write-up will focus on:

Chapter One: Introduction

Chapter Two: Literature review (Note this can span over more than one chapter depending on the scope of investigation; e.g. covering state-of-the-art “theory”, state-of-the-art- “practices”, state-of-the-art- “techniques”)

Next Chapter: Research methodology

Next Chapter: Findings/results and discussion

Next Chapter: Significance of the findings (=> contribution)

Next Chapter: Conclusions, recommendations and future works

REFERENCES / BIBLIOGRAPHY

The list of references at the end of your report contains details on all materials consulted (books, proceedings, journal articles, e-mail messages and web pages) and that have been cited in the text. Please make sure that you use the correct referencing method in the text as well as in the list of references at the end of your report.

Bibliography may be chosen if the list of sources should include materials used but could not be cited in the text (check with your supervisor)

Please use the **APA method** of referencing as explained in the CITATION guide as provided by NUST Library and from the FCI research Coordinators.

APPENDIX A: ABSTRACT CONTENT

Item in abstract	Example
Problem specification	The lack of <i>[state what is missing that causes the problem you have identified]</i> remains an open problem.
Main ways of attacking the problem(<i>show knowledge of area</i>)	The most common approaches to <i>[your problem area]</i> are based on <i>[state broad underlying assumption/theory]</i> . These approaches <i>[list main features]</i> . While such approaches have the advantage of <i>[say what they are good at]</i> , they cannot <i>[identify limitations]</i> .
Recent approaches (<i>show problem hasn't been solved recently</i>)	In recent years, <i>[alternative 'leading edge' type of approach/system/model/theory]</i> has gained increasing support, because <i>[say why it is getting popular]</i> .
Specific recent approach that you are going to develop further, and its limitations (<i>establish small research gap</i>)	<i>[Specify one form if the new approach]</i> is one form of approach/system/model/theory that <i>[say what this form is particularly good at]</i> . However, attempts at implementing/trialing the <i>[specific approach]</i> have exposed limitations. <i>[describe some different attempts and their respective limitations]</i> .
Aim of the thesis. Main features of proposed solution.	This thesis is to design/develop and test a new <i>[approach/system/model/theory]</i> , which <i>[summarise its main features]</i> . It will do this by <i>[indicate methodology/methods]</i> .
Expected benefits/contribution. How these will be evaluated.	It is expected that this approach/system/model/theory will overcome the limitations of <i>[say what you expect it will better than existing approaches]</i> . The new approach/system/model/theory will be evaluated by <i>[say how the benefits will be demonstrated]</i> .

(Source: Muyingi et al. (unpublished). "The complete research guidebook full of things your supervisor should have told you")

APPENDIX B: METHODOLOGY

Methodology can be restructured in Hierarchy:

1-The research Paradigm (philosophical view (ontology, epistemology, axiology, and rhetoric) of knowledge and its relationship with the researcher; e.g. Positivism, Interpretivism, Pragmatism

2-The Main Approach / Methodology : High-level map/ Set of Principles and Focal Approach (Descriptive; Explanatory; Exploratory, Theory testing/building/ constructive; Comparative; Longitudinal; Action Research; Participatory; Blended methodologies)

3- The Main Design or Argumentation Flow or Problem Solving Approach or Strategy (Deductive/ Inductive/ etc.)

4-The Methods (How to apply the approach: (Data type: qualitative or Quantitative or Mixed); and (Procedures)

5-Tools/Techniques/Instruments: (how to materialise the methods):

a)-The instruments/ tools to Collect Data

b)-The instruments/ tools to Analyse Data

c)-The instruments/ tools to Evaluate/ test Findings)

APPENDIX C: TECHNICAL CARE CHECKLIST FOR FINAL RESEARCH PROPOSAL

Please check that your final research proposal complies with the specific requirements set out below. Students often lose unnecessary marks because of sloppy technical care. Ask yourself the following questions ...

TECHNICAL CARE	✓ or x
1. PRELIMINARIES (Before the main body text starts on page 1)	
Is the layout of and information supplied on the title page correct?	
Have I included an Abstract (on a separate page) before the Table of Contents?	
Does the Abstract provide a <u>complete</u> summary of the whole research proposal?	
Is the Table of Contents complete and correct?	
Is the List of Figures complete and correct?	
Is the List of Tables complete and correct?	
Do the wordings of headings correspond meticulously with the wording in the Table of Contents?	
Do the headings numbers in the text correspond meticulously with the numbers in the Table of Contents?	
Do the figure / table captions listed in the List of Tables and List of Figures correspond a 100% with the captions used in the text?	
Have all tables and figures been supplied with correct captions (situated <u>above</u> the table or figure)?	
Have all tables and figures been supplied with correct and complete source references where appropriate (situated <u>below</u> the table or figure)?	
Have all tables and figures been numbered correctly? (Figures and tables are numbered independently starting from 1. Do not include section numbers in the numbering of tables / figures)	
Are the preliminary pages (i.e. Executive Summary, Table of Contents, List of Figures and List of Tables) numbered in Roman numerals: (i), (ii), (iii), (iv)?	
2. TEXT	
Are all pages in the body of the text numbered correctly with Arabic figures (1, 2, 3, etc. out of total page)?	
Have I used only one method of emphasizing (<i>italics</i> , or bold or <u>underlining</u>) throughout the report?	

Is there a line open between paragraphs?	
Are the paragraphs in my report not perhaps too long? Can't I split long paragraphs to improve readability?	
Have I made any "sweeping" or unsubstantiated statements, such as "there is no literature available on this topic" or "this research will contribute to the body of knowledge"?	
Have I used an impersonal, objective writing style (One symptom of a subjective writing style is that the researcher refers to him-/herself directly. References to "I", "we" and even "the researcher" should be avoided.)	
3. REFERENCING TECHNIQUES	
Are all sources (books, journals, e-mail messages, web pages) that have been cited in the text, also been included in the List of References at the end of the report and vice versa?	
Is the list of references strictly in the required order of the referencing style you are using ?	
Have I used the correct revised (abridged) Harvard referencing method in the text, as well as in the list of references?	
Have I correctly indicated the sources of all tables / figures taken or adapted from the literature? (Such references are provided <u>below</u> the table or figure)	
4. APPENDICES	
Do all the appendices have appropriate descriptive titles?	
Are all the appendices numbered up to the last page?	
Have all the appendices been tagged / "flagged" for easy cross-referencing?	
Are all the appendices included in the Table of Contents with their correct page numbers?	
5. GENERAL TECHNICAL CARE	
<i>Have I checked my proposal for spelling and grammatical errors?</i>	
<i>Have I used capital letters and abbreviations correctly?</i>	
Have I consistently rounded off all numeric values to three (3) decimals?	
Has text of the whole document been "justified" (i.e. aligned evenly along both the left and right margins to form a square box)?	
Are all headings numbered correctly?	
Headings in capital letters (first order headings) should not be underlined.	
There should be no full stops at the end of headings.	
Have I used an A4 paper setting?	

Have all pages printed correctly?	
Have I consistently used one and a half (1½) line spacing in the body of the text?	
Is the font size of the body text correct (12 pt)?	
Have I consistently used the same font type (i.e. Ariel or Times New Roman) and font size (12 pt) for the body text?	
Is the left margin set to at least 2 cm?	

Systematic Literature Review

Introduction

1. Who is who?
2. What is systematic literature review?
3. How do I do a systematic literature review?
4. How to present a systematic literature review?
5. Conclusion.

Introduction

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What is Systematic Literature Review?

Systematic Literature Review:

Is a review of a clearly formulated question(s) that uses systematic and explicit methods to **identify, select and critically appraise relevant research** and to **collect and analyse** data from the studies that are included in the review.

(Cochrane Collaboration, 2014).

Why Systematic Literature Review?

SLR aims to address the problem being researched by **identifying**, **critically evaluating** and **integrating** the findings of all relevant **high-quality individual studies**.

Systematic Literature Review?

A good SR should achieve the following;

1. Establish to what extent existing research has progressed towards clarifying a problem,
2. Identify relations, contradictions, gaps and inconsistencies in the literature, and explore reasons.
3. Comment on, evaluate, extend, or develop theory.
4. Provide implications for practice and policy.
5. Describe directions for future research.

(Baumeister & Leary, 1997; Bem, 1995; Cooper, 2003)

Systematic Literature Review?

A good SLR is characterised by being;

- **Objective:** *Explains criteria for inclusion & exclusion of findings.*
- **Systematic:** *Methodical & organised.*
- **Transparent:** *Clear judgement on inclusion & exclusion.*
- **Replicable:** *Allows for update and further review at a later stage.*

SLR should adopt a JUDGE AND JURY approach, rather than JUDGE AND LAWYER approach.

How do I do a SLR?

Key stages in conducting a SLR.

1. Scoping,
2. Planning,
3. Identification/ searching,
4. Screening,
5. Eligibility.

Scoping

Formulate your research questions

Example:

1. What can be done to improve the currently used workload framework?
2. How best can we implement a real-time system to detect and prevent fraud?

This should help you to be focused and narrow your literature search.

Planning

Break your research questions down into individual concepts to create search terms.

- Identify key terms/ ideas from the RQ, *(eg: workload frameworks)*

Formulate preliminary inclusion & exclusion criteria

- Based on your RQ *(eg: only real-time systems are considered)*

Make a record of the details of your searches and results obtained.

Identification /Searching

- Select databases that are relevant to your topic.
- Consider using Boolean search operators (NOT, AND, OR).
- Consider searching by publication year.
- Consider additional searches on alternative sources to ensure diversity.
- Consider published works.

Screening

- Read titles and abstracts of identified articles.
- Export references of selected articles to a citation manager (EndNote, RefWorks, Ms Word, etc.)

Eligibility

- Sift the full-text version of potentially eligible articles and extract relevant information.
- Add the included material onto your rough SLR document.
- Paraphrase and reference.
- Use turnitin to check your plagiarism level.

How to Present a Systematic Review

- Provide a theoretical and empirical background to the literature, explaining key terms, definitions and concepts;
- Describe the characteristics of included studies.
- Critically evaluate and integrate the results in an **unbiased and systematic way**, including stating whether the results are heterogeneous and discussing possible reasons.
- Balance conflict resolution:

(by identifying inconsistencies in study results & points of contention in theories, conceptualizations and methods in the literature)

Cont...

- Summarize and discuss the findings and conclusions of the review in a balanced and impartial way.
- Explicitly and intuitively link your conclusions to the evidence reviewed.
- Discuss the strengths and limitations of the literature articles.
- Establish to what extent existing research has progressed towards clarifying a particular problem.
- Comment on, evaluate, extend, or propose a new theory.
- Draw conclusions and make recommendations for practice

Common Errors with literature reviews

Common Errors

- **List-like writing that lacks synthesis**
- Relying on material that is likely to be out-of-date
- Exclusion of landmark studies
- Lack of a clear organisational structure
- Not discriminating between relevant and irrelevant materials
- **Not being sufficiently critical**

List-like Writing that Lacks Synthesis

Resulting from failing to integrate arguments into the review.



DO NOT SUMMARISE YOUR READINGS.

Example 1:

During the past decade, many researchers have shown interest in computer security. Their findings are reported here. Smith (2004) developed a model investigation on *(the next paragraph describes Smith's model)*... Jones (2005) showed that... *(the following paragraph says what Jones asserted)*... Harry (2008) makes a clear distinction between... *(the following paragraph states what Harry says)*.

Example 2:

This study brings together work in two areas of computer security. According to Robbins et al (1992) the early theories of computer security were concerned with unauthorised access.

Current thinking however does not discount these theories, but simply builds upon them to include contemporary approaches. Scholl (2007), argue that there are three basic approaches to security. He proposed that these approaches can be seen as an elaboration of Peter (2014)'s original ideas...

Not being sufficiently critical

Note that, the purpose of a literature review:

1. Is not just to summarise what is currently known about a topic.
2. Is to provide a detailed justification for your research.

That is, ultimately it is in the form of an argument and this argument is developed by pointing out "holes" or "Gaps" in the reviewed sources.

The End.

Thank you for attending, hope you
learnt something.

How to Develop: Problem Statement, Topic, Objectives and Research Questions, Hypothesis

AN

A Seminar presentation

BY

Dr. Ambrose Azeta

Department of Software Engineering

Namibia University of Science and Technology

27 February, 2023

PRESENTATION OUTLINE

2

1. Introduction
2. Areas of Research in ICT
- 3. Computer Science Research**
 - 3.1 Formulation of Problem statement
 - 3.2 Research title
 - 3.3 Research Objectives
 - 3.4 Research Sub Objectives
 - 3.5 Research Questions and Hypothesis
 - 3.6 Research Methodology
 - 3.7 Computer Science Dissertation/thesis (sample outline)
 - 3.8 Computer Science publication/papers (sample outline)
- 4. Information Systems Research**
 - 4.1 Research Aim
 - 4.2 Research Objectives
 - 4.3 Mapping of Research Questions, Hypothesis and Objectives
 - 4.4 Confirmation of Research questions and Hypothesis
 - 4.5 Conclusion

2. AREAS OF RESEARCH IN ICT

3

- Computer Science
- Information Systems
- **Software Engineering**
- Cyber Security,
ETC
- My main research area is Software Engineering/
Software Development
- SE is applied in all domains

3. SOFTWARE ENGINEERING RESEARCH

4

- Model driven Engineering (MDE)
- Design using Unified Modelling Language (UML)
- System testing/Automated
- System Evaluation
- System maintenance and monitoring
- Automated software architecture and debugging
- Software defect prediction using machine learning

Definition of Concepts

5

What is Software Engineering (SE) ?

- Software engineering is the process of applying engineering principles to analyze the needs of a specific user and then design, build and test software applications which will meet those needs.
- Software engineering goes beyond the fundamentals of programming to create more complex software systems that can serve as critical systems for businesses and organizations.
- Software engineers need to understand how to write software and how software and hardware interact.

What is Computer-aided software engineering (CASE)

- *Computer-aided software engineering* (CASE) uses programs to assist in the creation and maintenance of other computer programs.
- Programming has historically been performed by (human) programmers or teams; CASE adds software to the programming team.

Three (3) ways of finding a Problem Statement

- 1) Existing manual/traditional system is not working well
- (2) Existing automated/computerized system exist but giving problem.
- (3) A New system. No existing system, rather the project has not been done before in the community.

Sample Problem Statement

7

SN	Project title	Sample problem statement
1	Development of Fake News and Advertisement Detection System	Individuals and groups have obligated themselves the task of spreading falsified information to other users with the aim of gaining some money, change the public's opinion on the integrity of reputable persons and organizations, potentially alter the results of an election or some other sinister reason.
2	Development of A Web-Based Medical Appointment System For Cove University	With the high number of appointments that are scheduled in clinics and hospitals every day, it is almost impossible for the clinic to achieve its goals. Definitely, there will be some dissatisfaction on both clinic and students. The clinic has to deal with urgent or emergency cases, absentees or no-shows among scheduled cases which may complicate the work of the receptionist who is booking these appointments manually through walk-ins.

3.1 FORMULATION OF PROBLEM (GAPS)

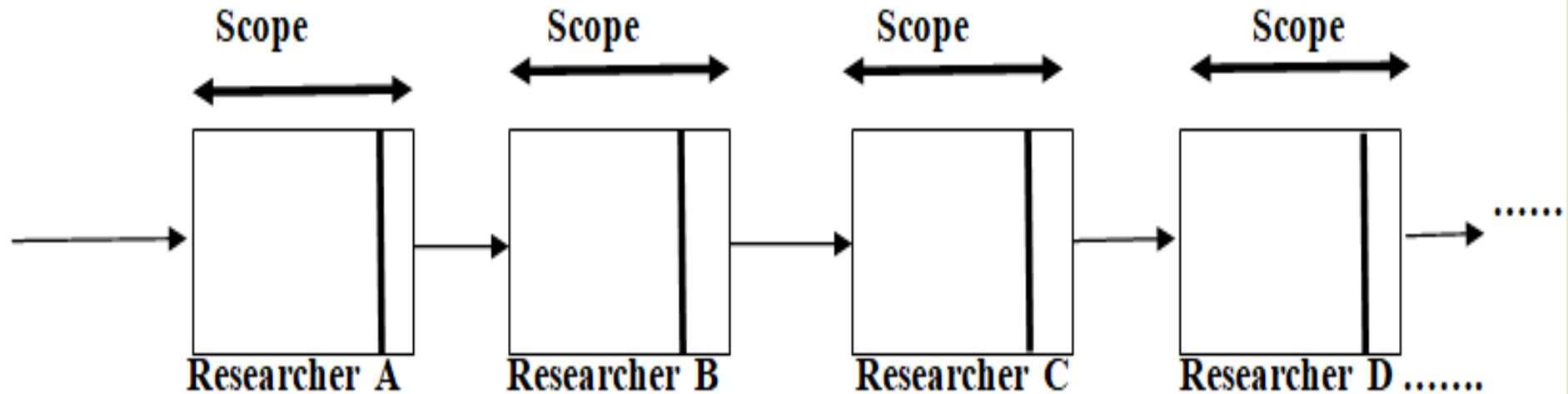
8

- ➡ State the problem you want to solve
- ➡ Cite the publication/papers that contains the limitations of a study or a problem that you want to solve, and could not be solved in the current research
- ➡ Gaps are normally discovered through systematic review

1. INTRODUCTION

9

RESEARCH CHAIN



Research is continuous (Research Advancements)

Some questions to address:

Which work are you extending?

What problem are you trying to solve?

What is the societal benefit of your research?

Etc

3.2 RESEARCH TOPIC

10

- Must contain: a technique/methodology (**Using** or **a**), domain (in), system or platform(**of** or **for**),

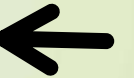
Examples

- Development of a model **for** result management system **in** E-Education **using** DevOps OR
- Development of **a** DevOps-based model **for** result management system **in** E-Education OR ←
- Development of **a** result management system **in** E-Education using DevOps

3.3 RESEARCH OBJECTIVE

11

- The objective of this study is to develop a devops-based model for result management system in e-education



3.4 RESEARCH SUB OBJECTIVES

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The Sub **objectives** of this study are **To**:

- **To** carry out requirement elicitation of result management system
- **To** design a DevOps model for result management system
- **To** implement a result management system based on DevOps as proof of concept
- **To** evaluate the implemented system and benchmark with existing system

3.5 RESEARCH QUESTIONS AND HYPOTHESIS

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Research Questions

- How can a DevOps model be developed for result management system?
- How can a result management system based on DevOps be evaluated.

Hypothesis

- **H1:** The development of a DevOps model for result management system is realistic
- **H1:** A result management system based on DevOps has a high usability evaluation compared to other models.

3.6 RESEARCH METHODOLOGY

14

- Tools/techniques such as software tools
- Research design/instruments such as questionnaire

3.7 CHAPTER OUTLINE OF DISSERTATION/THESIS

15

A Sample Thesis outline:

- ➡ Ch1-Introduction
- ➡ Ch2-Literature review
- ➡ Ch3-Research methodology
- ➡ Ch4-Results and Discussion
- ➡ Ch5-Conclusion

3.8 SECTION OUTLINE OF PUBLICATION/PAPERS

16

A Sample paper outline:

- ➡ Section1-Introduction
- ➡ Section2-Literature review
- ➡ Section3-Research methodology (System Design)
- ➡ Section4-Result and Discussion (System Implementation and Evaluation)
- ➡ Section5-Conclusion

4 INFORMATION SYSTEM RESEARCH

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Research Title: *IMPACT OF ICT ON STUDENT ACADEMIC ACHIEVEMENT*

Formulation of Research Questions and Hypothesis

- **Research Question 1:** *To what extent does teaching with the aid of ICTs impact on student academic achievement?*
- **Hypothesis H1:** *Teaching with the aid of ICTs has a positive impact on student academic achievement.*
- **Research Question 2:** *To what extent does course student engagement impact on their academic achievement?*
- **Hypothesis H2:** *Student engagement has a positive impact on their academic achievement.*
- **Research Question 3:** *To what extent does student use of ICTs impact on academic achievement?*
- **Hypothesis H3:** *Student use of ICTs has a positive impact on academic achievement.*

4.1 RESEARCH AIM

18

The aim of this research is to explore the impact of ICT on student academic achievements

4.2 RESEARCH OBJECTIVES

19

The objectives are as follows:

- To determine if teaching with the aid of ICTs impact on student academic achievement
- To determine the extent to which course student engagement impact on student academic achievement
- To determine the extent to which student use of ICTs impact on student academic achievement

4.3 MAPPING OF RESEARCH QUESTIONS, HYPOTHESIS AND RESEARCH OBJECTIVES

20

SN	Research Questions (RQ)	Hypothesis (H)	Research Objectives (RO)
1	<i>To what extent does teaching with the aid of ICTs impact on student academic achievement</i>	<i>Teaching with the aid of ICTs has a positive impact on student academic achievement.</i>	<i>To determine if teaching with the aid of ICTs impact on student academic achievement</i>
2	<i>To what extent does course student engagement impact on their academic achievement?</i>	<i>Student engagement as a positive impact on their academic achievement</i>	<i>To determine the extent to which course student engagement impact on student academic achievement</i>
3	<i>To what extent does student use of ICTs impact on academic achievement?</i>	<i>Student use of ICTs has a positive impact on academic achievement</i>	<i>To determine the extent to which student use of ICTs impact on student academic achievement</i>

4.4 CONFIRMATION OF RESEARCH QUESTIONS AND HYPOTHESIS

21

RQ1: *To what extent does teaching with the aid of ICTs impact on student academic achievement?*

Construct B: Teaching with the aid of ICTs returned a strong positive correlation and prediction (2nd in weight(β) out of 5 predictors) of *student academic achievement*.

H1: *Teaching with the aid of ICTs has a positive impact on student academic achievement.*

H1 was supported since a significant positive relationship was established with model 1 & 2. (Unstd β = 5.954 & 5.848, $p < 0.01$). The finding is in agreement with the results of Bai, et al. (2016); Rugutt and Chemosit (2005.); Ilomaki (2008); Abuet al., (2011); Oye et al., (2012) and Angeli&Tsaggari (2016).

4.4 CONFIRMATION OF RESEARCH QUESTIONS AND HYPOTHESIS (CON'T)

22

RQ2: *To what extent does course student engagement impact on student academic achievement?*

Construct C: Student engagement, *had* a weak positive correlation with, and prediction (3rd in weight (β) out of 4 predictors) of Class Marks.

H2: *Student engagement has a positive impact on student academic achievement.*

H2 however, was supported since a significant positive relationship was established for model 1 & 2 (Unstd β = 1.086 & 1.140, $p < 0.05$). The finding is in line with Gunuc's (2014) conclusions.

5. PUBLICATION OF RESEARCH OUTPUT

23

Publishers of good Journal Outlets

- Nature
- Springer Nature
- Elsevier
- Sage
- Emerald
- Wiley
- University-based Journals
- Etc

Conference outlets

- ACM
- IEEE
- Springer
- Etc

Index Outlets/Databases

- Scopus
- Web of Science/ISI
- University databases/
repositories
- Etc

5. CONCLUSION

24

This presentations has no doubt achieved its objectives of highlighting the research methodology in Computer Science and related areas

THANK YOU

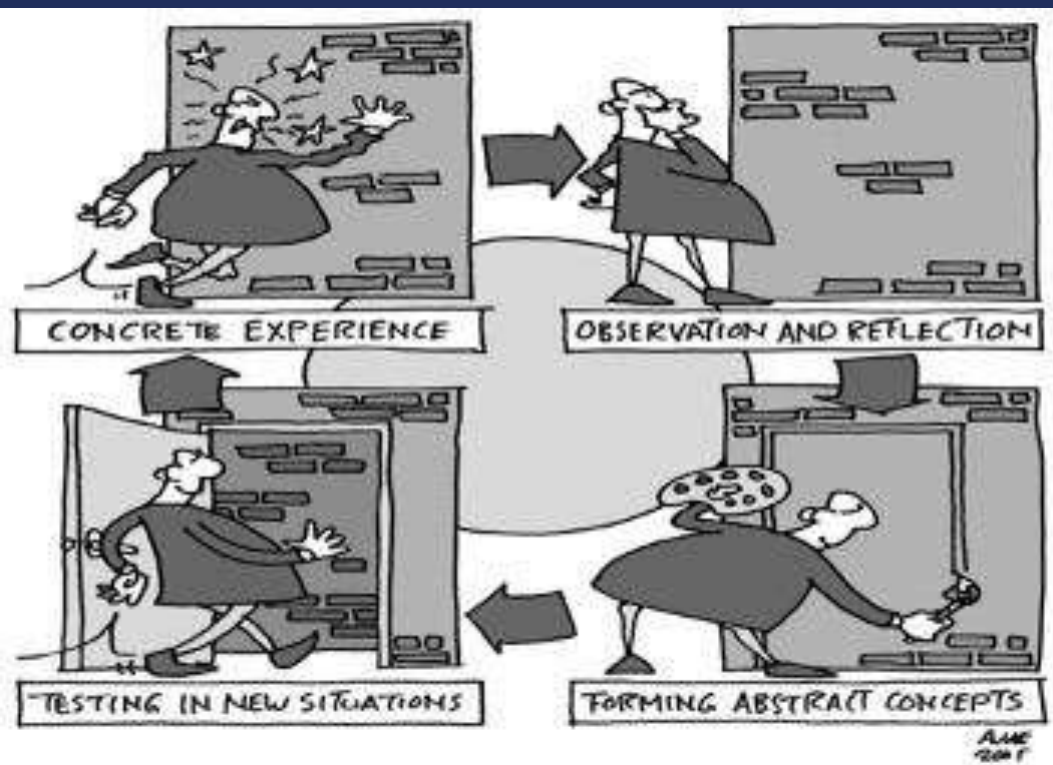


NAMIBIA UNIVERSITY
OF SCIENCE AND TECHNOLOGY

Faculty of Computing & Informatics

Action Research (AR) & Design-based Research

Towards relevant SW development research



Heike
Winschiers-Theophilus

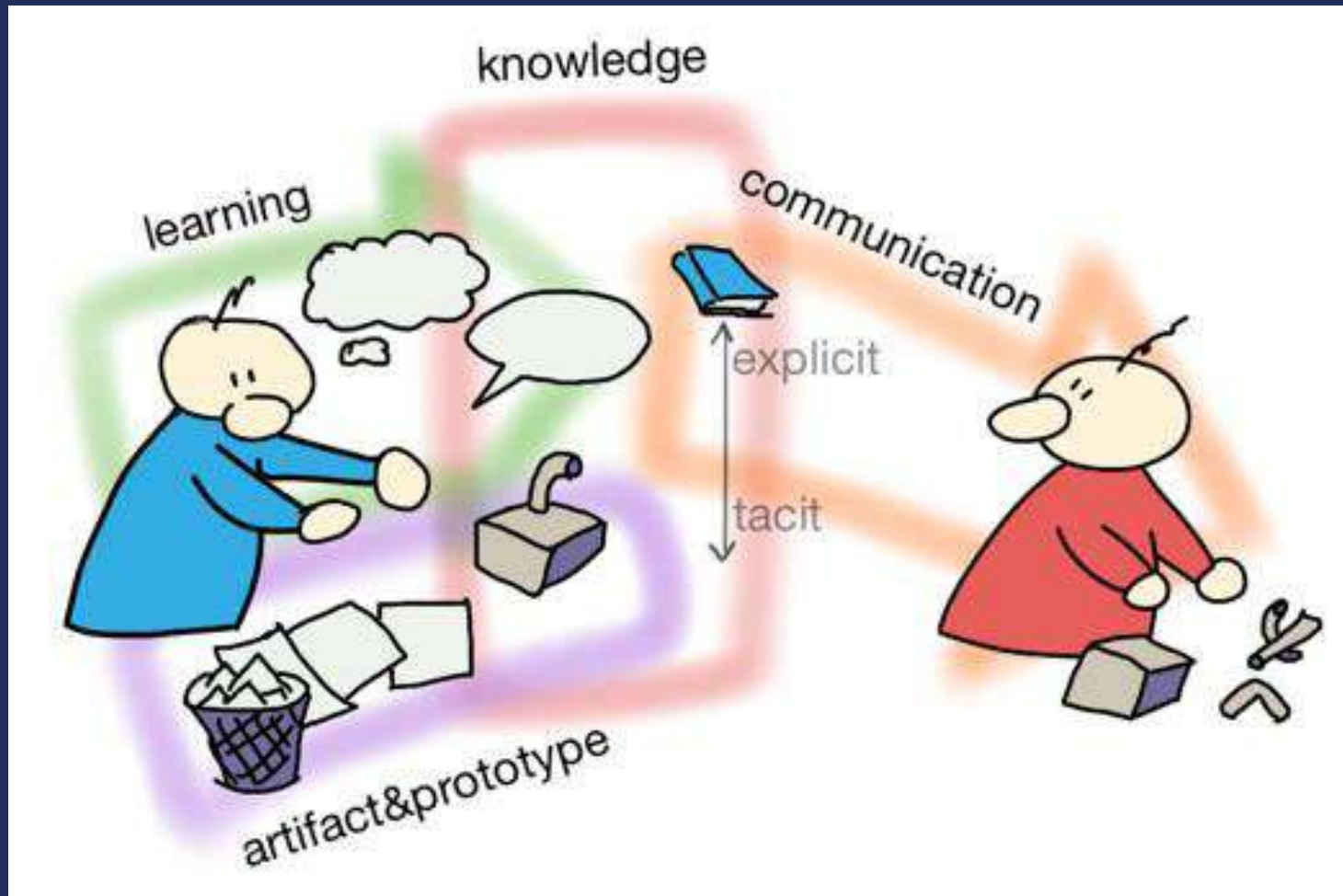
Terminology

	Working Definition	Key Associated Terms
Doing Research	Work done with the intention to produce knowledge for use by others	Question, hypothesis, theory, investigation, interpretation, generalization, validation, discovery
Doing Design	Work done with the intention to produce a feasible solution to improve a given situation	Idea and concept generation, synthesis, development, integration, discovery, prototyping, invention, implementation, realization
Artifact	Object (often material) created during a design process	Sketch, blueprint, brief, specifications, vision, proposal, recommendation, business plan
Prototype	Artifact used in research that can realize the (inter)action that is studied	Implementation, realization, test , exploration, solution, proof of concept, construction
Knowledge	Understanding about the world that can be communicated to others	Theory, book, publication, expertise
Design Practice	The ways in which design professionals conduct their work	Brief, contract, client, stakeholder, studio
Experiment	Specific: a piece of controlled, hypothesis-testing research; General: an explorative confrontation with real-world situations	Specific: hypothesis, statistics, (in)dependent variables; General: trial, intervention, exploration



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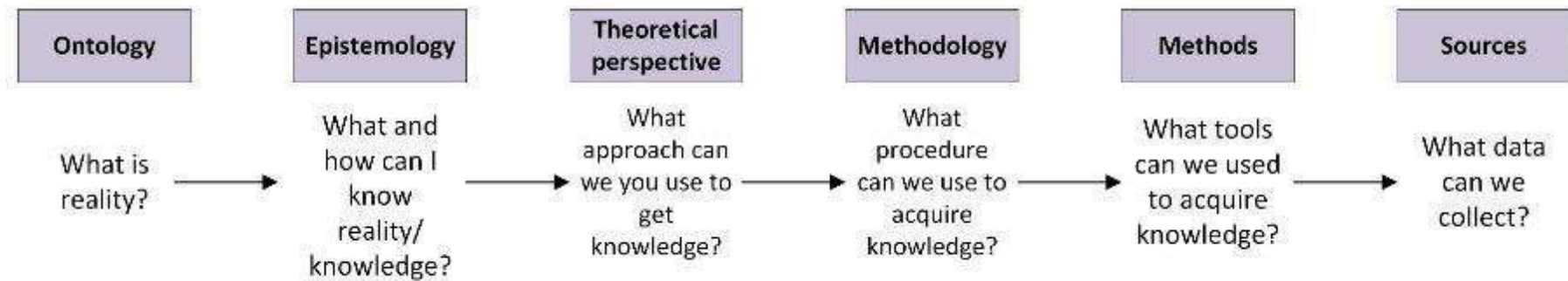
Faculty of Computing & Informatics





Paradigm	Ontology <i>What is reality?</i>	Epistemology <i>How can I know reality?</i>	Theoretical Perspective <i>Which approach do you use to know something?</i>	Methodology <i>How do you go about finding out?</i>	Method <i>What techniques do you use to find out?</i>
Positivism	There is a single reality or truth (more realist).	Reality can be measured and hence the focus is on reliable and valid tools to obtain that.	Positivism Post-positivism	Experimental research Survey research	Usually quantitative, could include: Sampling Measurement and scaling Statistical analysis Questionnaire Focus group Interview
Constructivist / Interpretive	There is no single reality or truth. Reality is created by individuals in groups (less realist).	Therefore, reality needs to be interpreted. It is used to discover the underlying meaning of events and activities.	Interpretivism (reality needs to be interpreted) • Phenomenology • Symbolic interactionism • Hermeneutics Critical Inquiry Feminism	Ethnography Grounded Theory Phenomenological research Heuristic inquiry Action Research Discourse Analysis Feminist Standpoint research etc	Usually qualitative, could include: Qualitative interview Observation Participant Non participant Case study Life history Narrative Theme identification etc
Pragmatism	Reality is constantly renegotiated, debated, interpreted in light of its usefulness in new unpredictable situations.	The best method is one that solves problems. Finding out is the means, change is the underlying aim.	Deweyan pragmatism <i>Research through design</i>	Mixed methods Design-based research Action research	Combination of any of the above and more, such as data mining expert review, usability testing, physical prototype
Subjectivism	Reality is what we perceive to be real	All knowledge is purely a matter of perspective.	Postmodernism Structuralism Post-structuralism	Discourse theory Archaeology Genealogy Deconstruction etc.	Autoethnography Semiotics Literary analysis Pastiche Intertextuality etc.
Critical	Realities are socially constructed entities that are under constant internal influence.	Reality and knowledge is both socially constructed and influenced by power relations from within society	Marxism Queer theory feminism	critical discourse analysis, critical ethnography action research ideology critique	Ideological review Civil actions open-ended interviews, focus groups, open-ended questionnaires, open-ended observations, and journals.

AR and Design-based research a Pragmatic Approach



Adapted from Hay (2002) pg. 64 and Crotty (1998)

Pragmatism	Reality is constantly renegotiated, debated, interpreted in light of its usefulness in new unpredictable situations.	The best method is one that solves problems. Finding out is the means, change is the underlying aim.	Deweyan pragmatism <i>Research through design</i>	Mixed methods Design-based research Action research	Combination of any of the above and more, such as data mining expert review, usability testing, physical prototype
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Value-laden/
subjective

Dependability
co-constructed

Learning
through
change

Open-ended/
iterative

Triangulation

Action Research

Motivations

- To make academic research relevant, researchers should try out their theories with practitioners in real situations and real organizations
- The emphasis is more on what practitioners do than on what they say they do

Key Assumptions

- Social settings cannot be reduced for study, and
- Actions brings understanding

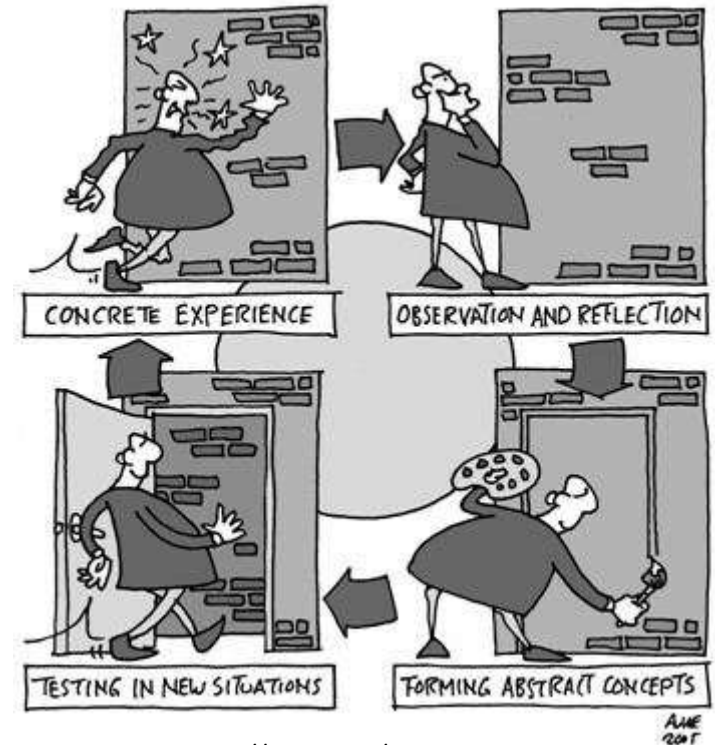
AR characteristics

- Based on **real life problems**
- Taking actions, aiming for change
- Knowledge creation (Learning) come from evaluation and reflection of the actions
- **Collaboration** with participants, democratic, interdisciplinary
- Researcher is immersed
- Situational, subjective
- Transferable rather than generalizable
- Interpretative, theory building

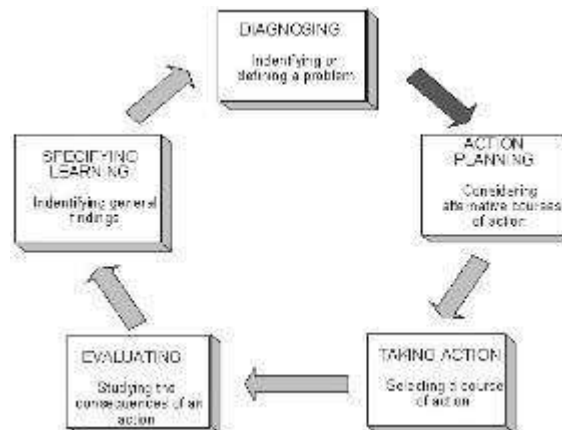
A cyclical approach



<https://edresearch.nmsu.edu/cro-research-publications-2/action-research-initiatives/ar/>

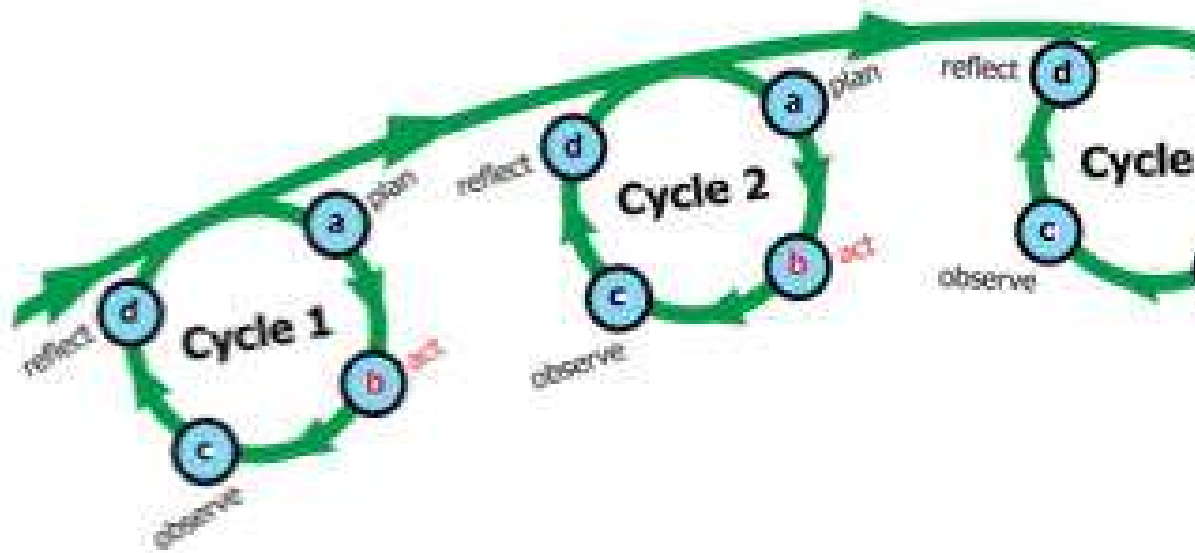


http://loop2.org/?page_id=175

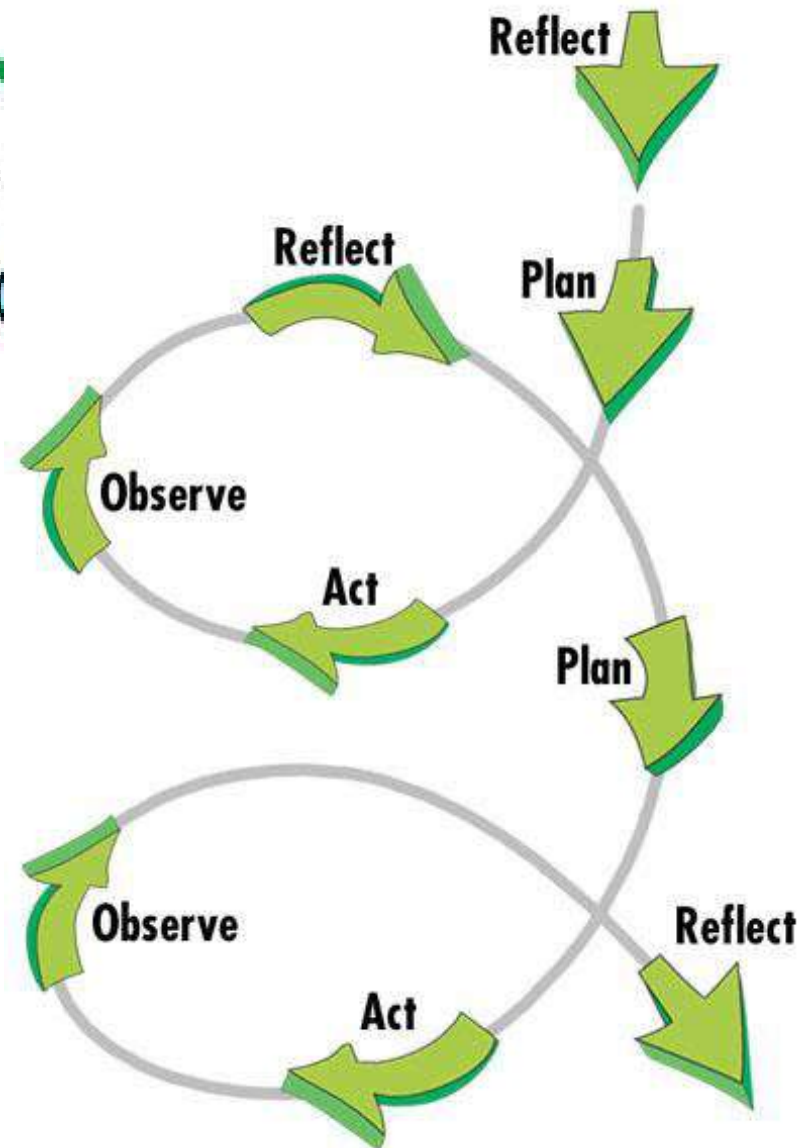


<http://www.vkmaheshwari.com/WP/?p=402>

An iterative approach



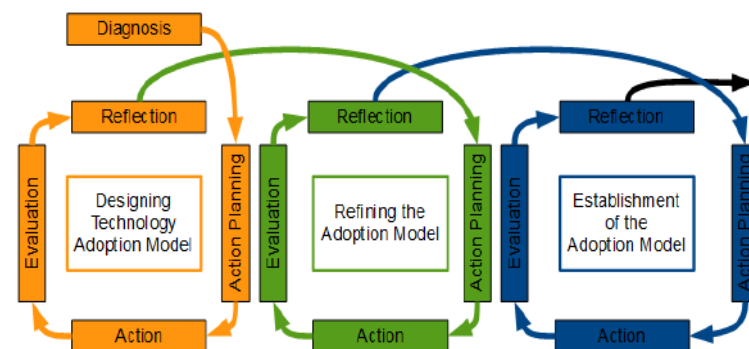
<http://robertsonandchang.com/services/learning-labs-action-research-for-innovation-and-excellence>



Developing a Technology adoption model for youth services in an informal settlement

Master Thesis by Asnath Kambunga

Cycle	Phase	Question/objective	Method	Finding/results
Cycle 1 Design Technology Adoption strategy	Diagnosis 1 (D1)	Which are the current technologies known to the Havana Youth? How are the youth currently adopting those technologies?	1. Questionnaire 2. Focus groups 3. Digital services expo workshop	Few technologies known, Yet no adoption, extracted factors affecting adoption in Havana
	Action Planning 1 (P1)	How is the rest of the world in similar setting as Havana adopting technologies?	1. Systematic literature review	Extracted factors affecting adoption Created strategy Created adoption time schedule
	Action 1 (A1)	How should new technologies be introduced in Havana to ensure better adoption?	1. Introduction to the adoption strategy workshop session 2. Training on the technology usage 3. Joint design of adoption campaign;	The campaign materials for deployment were not clear, youth were not committed
	Evaluation 1 (E1)	What were the action taken and their outcomes?	1. Synthesise results from AP1 and A1	Technology not adopted as the youth seems not to understand their roles and responsibilities, nor the system.
	Reflection 1 (R1)	What are the lesson learned to be considered in AP2	Reflect upon the action and its outcome. Summarise the lesson learned	Lesson learned to be carried over to AP2
Cycle 2 Refine Technology Adoption strategy	Action Planning 2 (AP2)	What are the elements that contribute to the ambassadors' understanding of their role in the technology adoption campaign?	Dedicated commitment session, Campaign material refinement Content training	Commitment session Campaigns materials redesign workshop, training plan
	Action 2 (A2)	How should the ambassadors understand their role in the technology adoption campaign to lead to better adoption of the technology?	1. Dedicated commitment session 2. Campaign materials refinement workshop 3. Content training	Youth set adoption/ deployment goals, better understanding of the system, partial execution of the adoption plan
	Evaluation 2 (E2)	What do the youth think of the whole adoption process?	Questionnaires	Youth not motivated
	Reflection 2 (R2)	What are the lesson learned to be considered in AP3	1. Summarise the lesson learned	Lesson learned to be carried over to AP3
Cycle 3 Establishme nt of the Technology Adoption strategy	Action Planning 3 (AP3)	what are the elements that will improve the technology adoption strategy	1. Scrutinize results from E2	Developed a refined adoption strategy, Developed a deployment schedule
	Action 3 (A3)	What are the best methods to implement the proposed adoption strategy in AP3?	1. Workshops 2. Meetings 3. Training	System deployed, motivated youth
	Evaluation 3 (E3)	How effective was the adoption strategy?	1. Questionnaires 2. Discussion among researchers	Committed ambassadors, Successful deployment
	Reflection 3 (R3)	What are the lesson learned to be considered in the development of a technology adoption model	1. Summarise the lesson learned	Social influence and capability plays a role in adoption



Kambunga, A. P., Winschiers-Theophilus, H., & Goagoses, N. (2018, December). Re-conceptualizing technology adoption in informal settlements based on a Namibian application. In *Proceedings of the Second African Conference for Human Computer Interaction: Thriving Communities* (pp. 1-10)

Data Collection Methods

- Semi-structured interviews
- Participant observations
- Analysis of documents
- Use of checklists: data registers, analysis tools, and health workers
- Software prototyping process and logs
- Group discussions and Training workshops
- Video/still pictures
- Analysis of press media reports

Data Analysis and Presentation

Interviews, Observations, Questionnaires, and site documents work together to support the research claims.

Empirical materials are presented in

- Descriptive statistics (quantitative data)
 - Model and measurement instrument for evaluating user satisfaction (7-point scale)
- Qualitative excerpts of encoded user reactions
- Software evaluation via criteria such as reliability and usability
- Logs and Screen shots of programs

Secondary Sources of Data

- Documents from the field
- Photos and videos

Validity of AR

Values

- Credibility
- Transferability
- Dependability
- Confirmability

Means

- Prolonged engagement, verbatim principle, multiple perspectives, workability
- Share context knowledge
- Transparency in description
- Present enough evidences

Hayes, G.R. 2011. The relationship of action research to human-computer interaction. ACM Trans. Comput. Hum. Interact. 18, 3, Article 15 (July 2011), 20 pages.
DOI=10.1145/1993060.1993065 <http://doi.acm.org/10.1145/1993060.1993065>

Ethics in AR

- Conventional ones +
- Participants: from subject to collaborator
- Relationship
- Sustainability of change
- Choice of action

Design-based research

“The designing act of creating prototypes is in itself a potential generator of knowledge (if only its insights do not disappear into the prototype, but are fed back into the disciplinary and cross-disciplinary platforms that can fit these insights into the growth of theory).”

—(Stappers 2007)

<https://www.interaction-design.org/literature/book/the-encyclopedia-of-human-computer-interaction-2nd-ed/research-through-design>

“we intend the term design research (sic) to mean an intention to produce knowledge and not the work to more immediately inform the development of a commercial product”. Zimmerman, Forlizzi, and Evenson (2007, p. 494)

RtD is defined as “a research approach that employs methods and processes from design practice [sic] as a legitimate method of inquiry” Zimmerman, Stolterman, and Forlizzi (2010)

Zimmerman, J., Forlizzi, J., & Evenson, S. (2007, April). Research through design as a method for interaction design research in HCI. In *Proceedings of the SIGCHI conference on Human factors in computing systems* (pp. 493-502)

Zimmerman, J., Stolterman, E., & Forlizzi, J. (2010, August). An analysis and critique of Research through Design: towards a formalization of a research approach. In *proceedings of the 8th ACM conference on designing interactive systems* (pp. 310-319)

Research through Design

“ the investigation of knowledge through purposeful design. ... the study of design and the process of knowledge production that occurs through the act of design”

Faste, T., & Faste, H. (2012, August). Demystifying “design research”: Design is not research, research is design. In *IDSA education symposium* (Vol. 2012, p. 15)

USER INTERFACE ACTIONS FOR THE RESTORATION OF EYE-TRACKED ATTENTION LOSS

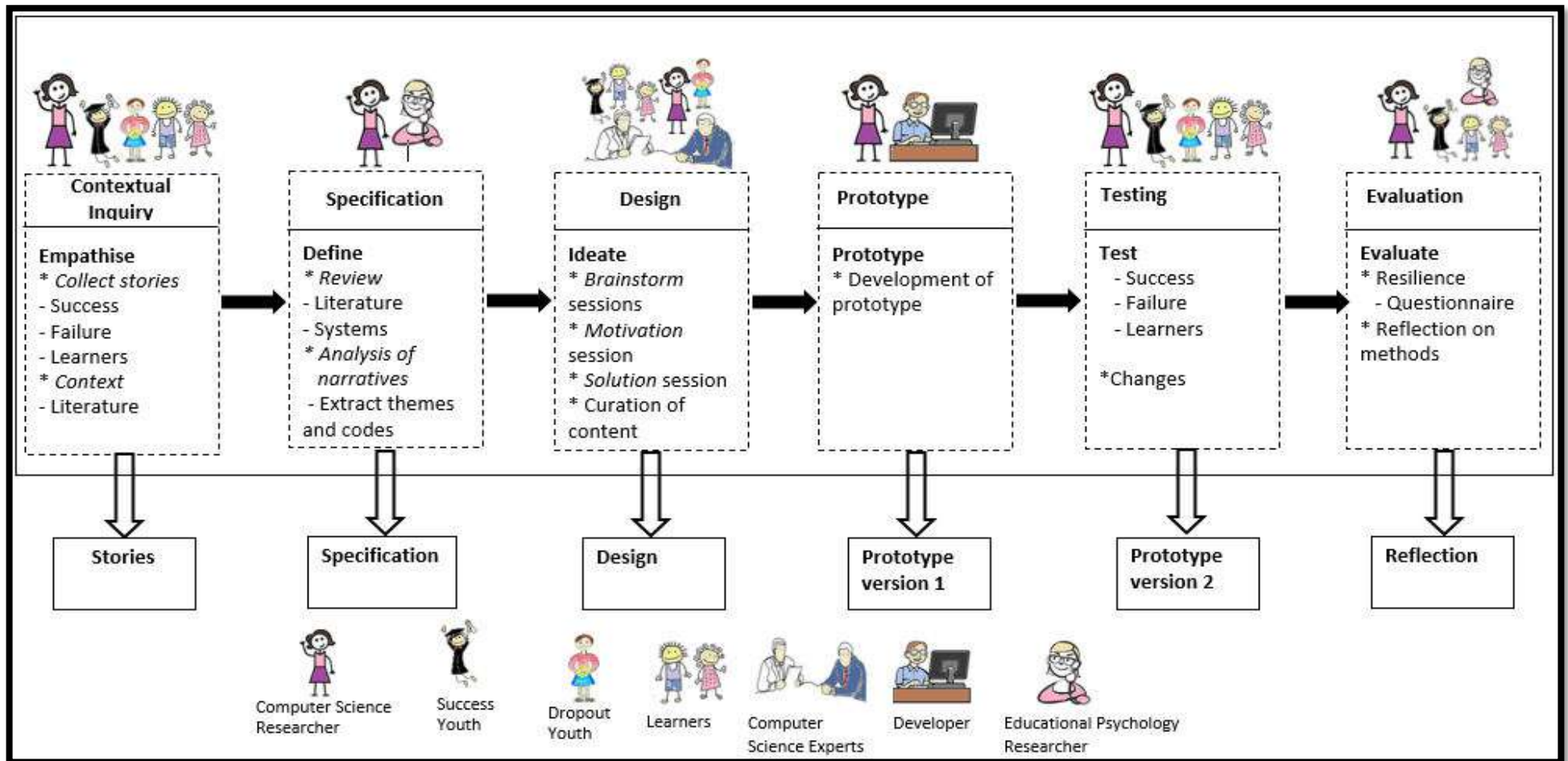
Honors thesis by Baptista Jacinto Joao

A Designed Experiment



RtD process

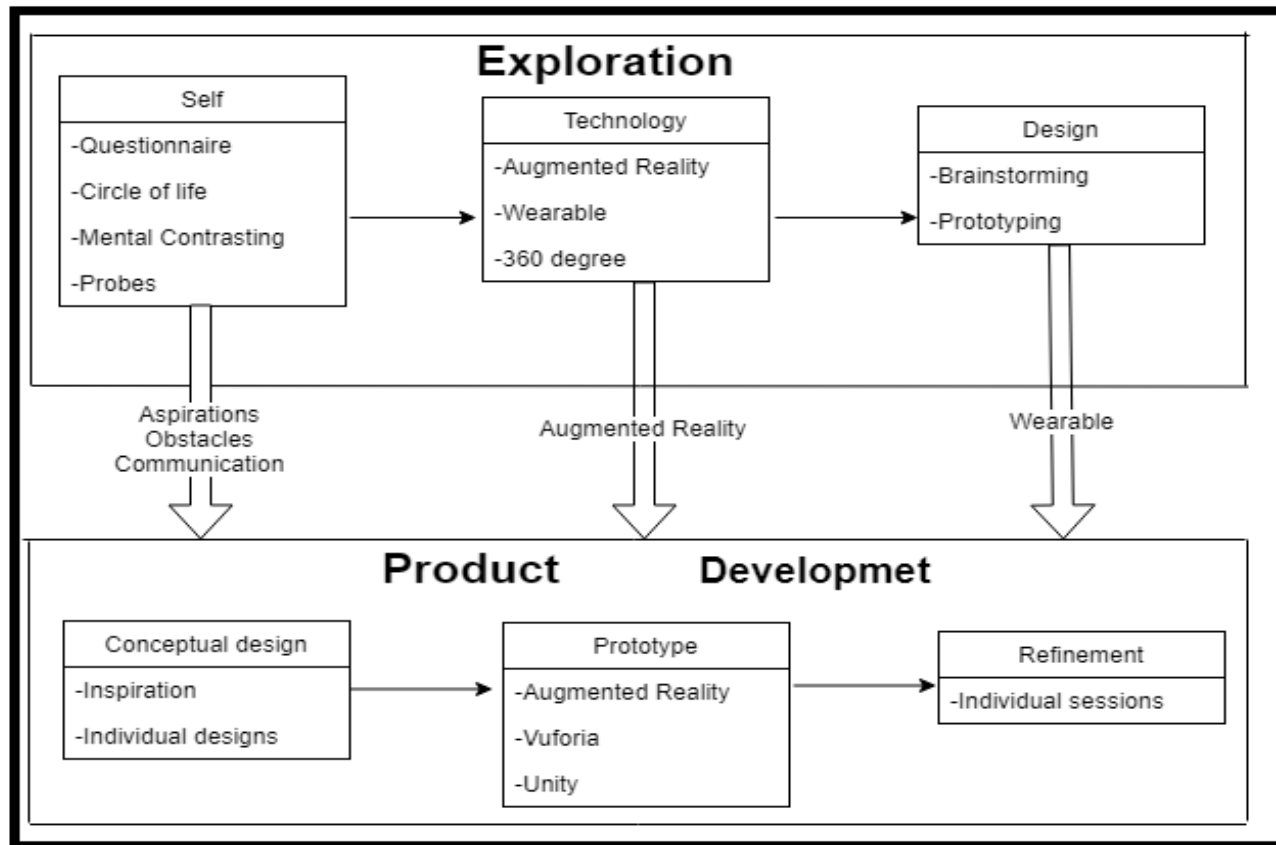
Technology to increase resilience to decrease school drop out



Kays, R. (2020) DEVELOPING TECHNOLOGY TO STRENGTHEN RESILIENCE IN SAN CHILDREN TO REDUCE SCHOOL DROPOUTS, Master Thesis

RtD process

Co-design of personal AR wearables to enhance social well-being





PAMIBIA UNIVERSITY
OF SCIENCE AND TECHNOLOGY

Faculty of Computing & Informatics





Problem Statement: A Comprehensive Guide

Prof. Jude Osakwe

CONTENTS

01-

**Understanding the
Problem Statement**

02-

**Defining the "Ideal
Situation"**

03-

**Assessing the "Current
Reality"**

04-

**Analyzing the
"Consequences"**



05-

**Suggesting a Direction
("What Could Be")**

06-

Template & Examples



01 Understanding the Problem Statement

Purpose & Importance

Clearly Define Problem

The problem statement should immediately identify the core issue being addressed. A well-defined problem statement leaves no ambiguity about the specific challenge at hand.

Justify Need for Solution

It should strongly argue why solving the problem is crucial and significant. Emphasize the importance of finding a resolution and highlight the potential benefits of doing so.

Focus Project Efforts

The problem statement should provide a sharp, concise target for all project activities. By setting a clear objective, it helps streamline efforts and avoid scope creep.

Reference Point for Alignment

It should act as a constant point of reference throughout the project lifecycle to ensure the team remains focused on the primary goal. This helps maintain alignment and direction.

Key Elements Overview

Ideal Situation

1

Describe the desired state or how things should ideally function absent the identified problem. This sets the context for understanding the gap.

2

Current Reality

Clearly articulate the problem itself by detailing the current undesirable situation. Describe the issue as it exists and provide a contrast to the ideal.

Consequences

3

Explain the negative impacts and implications of the problem if left unresolved. Emphasize the repercussions and their effects on stakeholders.

4

Suggestion (Optional)

Optionally, offer a high-level suggestion of a potential direction for solving the problem. This provides a preliminary idea of what might be done.



02 Defining the "Ideal Situation"

Vision of the Desired State

Paint a Picture of Success

Create a vivid depiction of how things will operate when the problem is resolved. Describe the scenario where objectives are met and improvements are realized.

Remove the Problem

Imagine a scenario where the identified issue no longer exists and highlight the resulting improvements. Focus on operational excellence and smooth processes.

Future Vision

Illustrate a future state that underscores the benefits of having addressed and corrected the problem. Emphasize the value of long-term positive outcomes.

Specificity is Key



01

Concrete Language

Use precise, unambiguous language to clearly articulate the desired outcome. Avoid broad or abstract terms that can be misinterpreted.

02

Avoid Vague Generalities

Steer clear of general statements; instead, provide specific details about what success looks like. Focus on quantifiable and observable improvements.

03

Customer Example

"Customers should complete online purchases within 3 minutes with no errors." This makes it easy to understand and measure the desired improvement in customer experience

Establishing a Baseline

01

Quantify Ideal Situation

Use quantifiable metrics to define the ideal state. This helps establish a clear, measurable target for improvement.

02

Metrics and Benchmarks

Define specific benchmarks and key performance indicators (KPIs) that will be used to track progress. Include metrics such as time, cost, and satisfaction levels.

03

Measuring Future Progress

Explain how these metrics will be used to assess progress and demonstrate the impact of implemented solutions. Ensure there is a method for ongoing tracking.



03 Assessing the "Current Reality"

Articulating the Problem

Deviating from the Ideal

Explain how the current situation deviates from the envisioned ideal state. Clearly show the gap that needs to be bridged.

Heart of Problem Statement

Highlight that this description is at the core of the problem statement. This is where the actual problem is defined and brought to light.

Current Obstacles

Describe the specific obstacles or challenges that are preventing the achievement of the ideal state. Understand why the ideal state is not currently attained.

Competitive Advantage

Data-Driven Approach

Base the description of the current reality on tangible data and evidence, not personal opinions. Use factual information to underpin the problem statement.

Avoid Opinions, Use Facts

Refrain from including subjective viewpoints or interpretations in the description of the problem. Maintain a neutral, fact-based perspective.

Statistics and Examples

Support the problem description with specific statistics, examples, and relevant data points. Include tangible information that provides a clear view of the issue.

Focusing on Symptoms



01

Observable Effects

Concentrate on the observable effects or symptoms of the underlying issue. Describe the tangible outcomes rather than speculating on root causes.

02

Postpone Root Cause Analysis

Defer any attempts to identify the root cause of the problem at this stage. Focus solely on describing the symptoms that are being observed.

03

Current Performance Problems

Highlight the performance issues or deficiencies that are directly related to the identified problem. Describe the current state of poor performance.

Providing Context



Background Information

Include relevant background information that provides context for the problem. This can include details about the organization, industry, or market.

Organizational Factors

Explain organizational factors or internal dynamics that are contributing to the problem. Understand how internal processes or structures impact the situation.

Target Audience Details

Provide details about the target audience or stakeholders who are affected by the problem. Describe who is impacted and how they are influenced by the issue.



04 Analyzing the "Consequences"

Articulating the Impact



Negative Consequences

Clearly describe the negative consequences resulting from the problem. Highlight the adverse effects and detrimental outcomes.



Who is Affected? How?

Identify the specific individuals, groups, or stakeholders that are affected, and explain how they are impacted. Understand the breadth and depth of the problem's reach.



Pain points

Showcase specific problems that are painful or difficult to endure. Understand and describe problems from key stakeholders' perspectives

Quantifying the Impact

01 Numbers Make it Compelling

Emphasize the importance of quantifying the impact of the problem whenever possible. Use numerical data to make the consequences more compelling and understandable.

02 Examples of Quantification

-"Reduced online sales conversion by 15%" or "Customer service costs increased by \$20,000 per month" are tangible examples of quantification.

03 Demonstrate Actual Costs

Quantify the financial, operational, or reputational costs associated with the problem. Show the direct financial burden and indirect expenses incurred.



Long-Term Implications

“ Immediate vs. Future

Consider both the immediate consequences and the potential long-term implications of the problem. Understand the short-term and long-term effects.

“ Effects of Inaction

Describe the effects of not addressing the problem, including potential risks and missed opportunities. Highlight the dangers of inaction and complacency.

“ Negative Brand Impact

Emphasize how and how long unchecked problems will impact customer loyalty and brand image



Organizational Goals

Hindered Progress

Explain how the problem is impeding the organization's ability to achieve its targets.
Demonstrate the impact on performance and productivity.

01

Strategic Objectives

Connect the consequences of the problem to the organization's overall strategic objectives. Show how the problem hinders progress toward key goals.

02

03

Goal Achievements Impact

Showcase specific impacts to goals and their achievements based on the effects of problem





05 Suggesting a Direction ("What Could Be")

Outlining Potential Solutions



High Level Solutions

Summarize potential high level solutions to give context to possible ways of addressing the problem



Focus and Motivation

Use the suggestion to provide focus and motivation for the problem-solving effort. Inspire the team and provide a sense of direction.



General Statement of Action

Provide a general statement about the actions needed to address the problem. Outline the broad steps that should be taken.



Desired Outcome



01

Result of Solving Problem

Clearly state the desired result of solving the problem. Describe the ideal outcome that should be achieved.

02

Focus on Benefits

Emphasize the benefits that will be realized once the problem is resolved. Highlight the positive impacts and improvements.

03

Measurable Outcomes

Use numeric and measurable outcomes to demonstrate the effects of addressing a specific problem

Keeping it Broad



01

Avoid Specificity

Avoid being too specific or prescriptive about the solution at this stage. Maintain flexibility and encourage innovative thinking.

02

Open to Brainstorming

Encourage brainstorming and exploration of different approaches to solving the problem. Foster a creative and collaborative environment.

03

Solution Agnostic

Remain solution-agnostic and avoid predetermining the specific solution that should be implemented. Encourage open-mindedness and flexibility.

Reiterate Benefits

01

Positive Impact Summary

Summarize the positive impact that will result from successful implementation of a solution. Reinforce the value of resolving the problem.

02

Implementing a Solution

Showcase the opportunities provided by the problem that should be addressed when implementing a solution

03

Highlight Improvements

Describe the improvements that can be anticipated once the solution is in place. Focus on tangible benefits and positive changes.



06 Template & Examples

Putting It All Together

“ Ideal, Reality, Consequences

Utilize the problem statement structure template: "Ideally, [desired state]; However, [current problem]; As a result, [negative consequences]." Ensure all sections flow together logically.

“ Problem Statement Structure

Follow the logical structure of Ideal state, Current Reality, and defined Consequences to make a clear and compelling case for change and to provide a clear understanding.

“ Filling In-Depth Template

Review and refine the comprehensive problem statement template to ensure all dimensions of the problem are captured in detail, setting the stage for effective solutions.



Customer Service Example



01

Online Retailer Scenario

Imagine an online retail business aiming to improve its poor customer service experience. Customer satisfaction is down, returns are up, and response times are slow.

02

Specific Examples of Template use

Ideal: "Customers can easily resolve inquiries in under 3 minutes."
Reality: "Average resolution time is 10 minutes." Consequence: "Customer churn increased by 15%."

03

Customer Satisfaction Metrics

Monitor metrics such as Net Promoter Score (NPS), customer service resolution times, and customer feedback scores to measure the impact of improvements.

•Thank you for



Research Methodology

Simulations

Peter Gallert

Department of Computer Science
School of Information Technology
Faculty of Computing and Informatics
Namibia University of Science and Technology

20 April 2023



Outline

- 1 Introduction
- 2 Example 1: Packet Tracer
- 3 Example 2: ns-3



Brain in a jar



Figure: “Brain in a vat”. Picture by Alexander Wivel, CC-BY-SA

- If you were just a brain in a jar
- and had no sensory organs whatsoever
- with some computer to give input
- but no way to experience reality
- There would be no way to figure that out! (contrary to *The Matrix* Trilogy)
- A.k.a a **simulation**



What is a simulation?

- A **simulation** is a simplified imitation of a real-world process over time
- Simulations use a **model** of perceived reality
- Some characteristics are deemed important, others are not
- By using the simulation (the model), we **abstract from** all unimportant details
- Example: flight simulator
 - **Important:** pilot input, weather, airport, airplane type, mechanics, emergencies, ...
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Why simulations?

1 Money:

- Cisco "pod" (3 routers, 3 switches, and accessories): 100,000 N\$
- Cisco Packet Tracer: 0 N\$
- And that's still reasonably cheap! There are simulations for power grids, satellite launches, and the global financial system

2 Safety: How do you learn to

- Land a Boeing 747?
- Transplant a human heart?
- React to a pandemic?
- Reanimate an accident victim?
- Extinguish a fire in a residential building?

3 Convenience:

- Book lab 5, travel to campus, do a networking lab,
- Or: Fill in a set of forms, get HoD approval, take a Cisco pod home,
- Or: Install Packet Tracer



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Simulations in science

- Empirical scientific method: **experiment**
- Experiment is only valid if it is **repeatable**
- Many factors hinder repeatability:
 - The person / team running the experiment
 - Environmental factors: ambient temperature, time of day / year
 - Equipment manufacturing faults and deviations
 - Detail of the documentation
- A **simulation** helps suppress outside factors (abstraction)
- Also simplifies documentation
- Some simulations are **better than the real thing!**



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Packet Tracer



- **Packet Tracer** is a network simulator for teaching
- Developed by Cisco Systems, a leading networking hardware manufacturer
- **Real-time simulator** for routers, switches, firewalls, cell phones, access points, et cetera
- GUI, drop devices on screen, connect, configure
- Simulation mode: "follow" a packet on its way through the network
- Simple download, all operating systems, free of charge (account creation at www.netacad.com required)



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Main screen

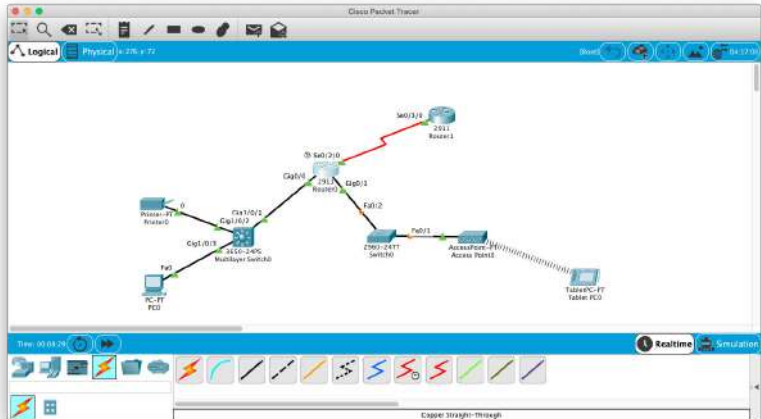


Figure: Packet Tracer screenshot with some connected devices. Fair use. 

Packet Tracer properties

- **Modelled:** Device configuration (hardware and software), protocol behaviour and performance
- **Not modelled:** Space considerations, reduced feature set
- Only selected commands and hardware options—those taught in, and sold to, the Cisco Academies



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- **Modelled:** Device configuration (hardware and software), protocol behaviour and performance
- **Not modelled:** Space considerations, reduced feature set
- Only selected commands and hardware options—those taught in, and sold to, the Cisco Academies



Packet Tracer hardware configuration



Figure: Packet Tracer configuration window: hardware. Fair use.



Packet Tracer software configuration



Figure: Packet Tracer configuration window: Cisco IOS command line interface. Fair use.

Advantages, disadvantages

- **Good for teaching computer networking:**

- 1 Point-and-click, easy to learn
- 2 Realistic simulation of hardware installation and configuration mistakes
- 3 Runs on all major operating systems, even on Android
- 4 Accompanied by an online course (CCNA) to teach network device configuration

- **Not suitable for performance estimates:**

- 1 Timers independent of host performance
- 2 Illogical result, for instance for Quality of Service impact

- **Proprietary and vendor-dependent:**

- 1 Non-Cisco devices cannot be simulated
- 2 Many "fantasy devices" that do not exist in this form



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Outline

- 1 Introduction
- 2 Example 1: Packet Tracer
- 3 Example 2: ns-3



ns-3



- **Network simulator (ns)** is a network simulator for research
- Currently in version 3, **ns-3**
- Developed by its user community worldwide
- **Discrete event simulator**, calculating **events**, their **effects**, **duration**, and **output**
- **Input:** C++ source code
- **Output:** (all optional)
 - `.pcap` packet capture file for display with Wireshark
 - `.tr` bare text trace file of chronological events
 - `.xml` graphic for display with offline animator NetAnim
 - Command line messages



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Command line

```

peter@debian: ~/work/ns-3-allinone/ns-3-dev
File Edit View Search Terminal Help
peter@debian:~/work/ns-3-allinone/ns-3-dev$ ./waf --run stest
Waf: Entering directory `/home/peter/work/ns-3-allinone/ns-3-dev/build'
[2809/2857] Compiling scratch/stest.cc
[2818/2857] Linking build/scratch/stest
Waf: Leaving directory `/home/peter/work/ns-3-allinone/ns-3-dev/build'
Build commands will be stored in build/compile_commands.json
'build' finished successfully (5.047s)
Total Bytes Received: 680448
peter@debian:~/work/ns-3-allinone/ns-3-dev$ ./waf --run stest
Waf: Entering directory `/home/peter/work/ns-3-allinone/ns-3-dev/build'
[2810/2857] Compiling scratch/stest.cc
[2818/2857] Linking build/scratch/stest
Waf: Leaving directory `/home/peter/work/ns-3-allinone/ns-3-dev/build'
Build commands will be stored in build/compile_commands.json
'build' finished successfully (2.570s)
At time +2s client sent 1024 bytes to 10.1.2.4 port 55
At time +2.01799s server received 1024 bytes from 10.1.3.3 port 49153
At time +2.01799s server sent 1024 bytes to 10.1.3.3 port 49153
At time +2.03371s client received 1024 bytes from 10.1.2.4 port 55
At time +3s client sent 1024 bytes to 10.1.2.4 port 55
At time +3.00529s server received 1024 bytes from 10.1.3.3 port 49153
At time +3.00529s server sent 1024 bytes to 10.1.3.3 port 49153
At time +3.01058s client received 1024 bytes from 10.1.2.4 port 55
peter@debian:~/work/ns-3-allinone/ns-3-dev$

```

Figure: Command line screenshot of ns-3, with simulation output. Own work



ns-3 properties

- **Modelled:** Protocol behaviour and performance
- **Not modelled:** most device configuration steps, hardware restrictions
- With ns-3, all devices **behave exactly as they should:**
 - Implementing all aspects of the simulated protocols
 - Delivering exactly the performance as determined by standards and configuration
 - Only runtime differs
- But many default values make little sense, for instance Ethernet with 4.2 GBps and no delay



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ns-3 C++ script

```

// Step 4: Define global declarations
GlobalValue::Bind ("ChecksumEnabled", BooleanValue (true));
// Calculate FCSs

// Step 5: Create nodes and node containers
NodeContainer routers;
routers.Create (2);

NodeContainer whkAP;
whkAP.Add (routers.Get (0)); // attach AP to router Windhoek
NodeContainer whkSTA;
whkSTA.Create (12);

NodeContainer rehoLAN;
rehoLAN.Add (routers.Get (1)); // attach switch to router Rehoboth
rehoLAN.Create (8);

// Step 6: Create connections
PointToPointHelper whkreho; // Define parameters of the Windhoek-Rehoboth WAN
whkreho.SetDeviceAttribute ("DataRate", StringValue ("5Mbps"));
whkreho.SetChannelAttribute ("Delay", StringValue ("2ns"));

CsmaHelper rehoswitch; // Define parameters of the Rehoboth LAN
rehoswitch.SetChannelAttribute ("DataRate", StringValue ("100Mbps"));
rehoswitch.SetChannelAttribute ("Delay", TimeValue (NanoSeconds (6560)));

Ssid whkssid = Ssid ("Windhoek WiFi");

```

Figure: C++ source code to produce a simulation. Own work

Advantages, disadvantages

- **Ideal for scientific experiments:**

- 1 The same C++ script will produce the same output, on any computer. Only runtime can differ
- 2 Control variables do not change
- 3 Easy control of the dependent variable, easy re-run with different values. For instance, same data transfer with 802.11a, 11n, 11ax
- 4 Manufacturing faults or marketing tricks irrelevant

- **Steep learning curve:**

- 1 Requires Linux, and only Linux
- 2 Tricky installation, lots of prerequisites, must be compiled on target machine
- 3 Lots of networking knowledge needed: Any logical mistake in the C++ script breaks compilation
- 4 Some understanding of C++ required—classes, inheritance, screen output, and general syntax



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NetAnim

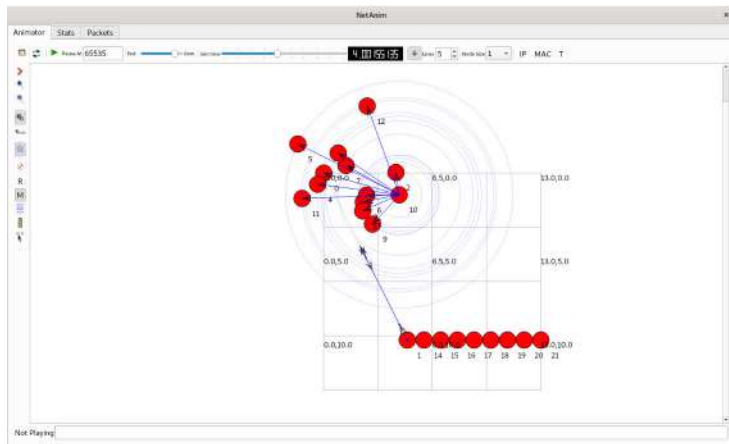


Figure: Graphical display of an ns-3 simulation with NetAnim. Own work



End of Presentation

Any questions?





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OF SCIENCE AND TECHNOLOGY

Faculty of Computing and Informatics

School of Computer Science

Literature Review

Presented by: Dr A. M. Gamundani

Senior Lecturer : Cyber Security



**You are what you do,
not what you say you'll do**



Agenda

- Getting you to think about the purpose of a literature review
- Getting you to think about what your own literature review will look like
- Critical thinking, reading and writing
- Planning and structuring options



**Review the
Available
Literature**

**Publish
Findings**

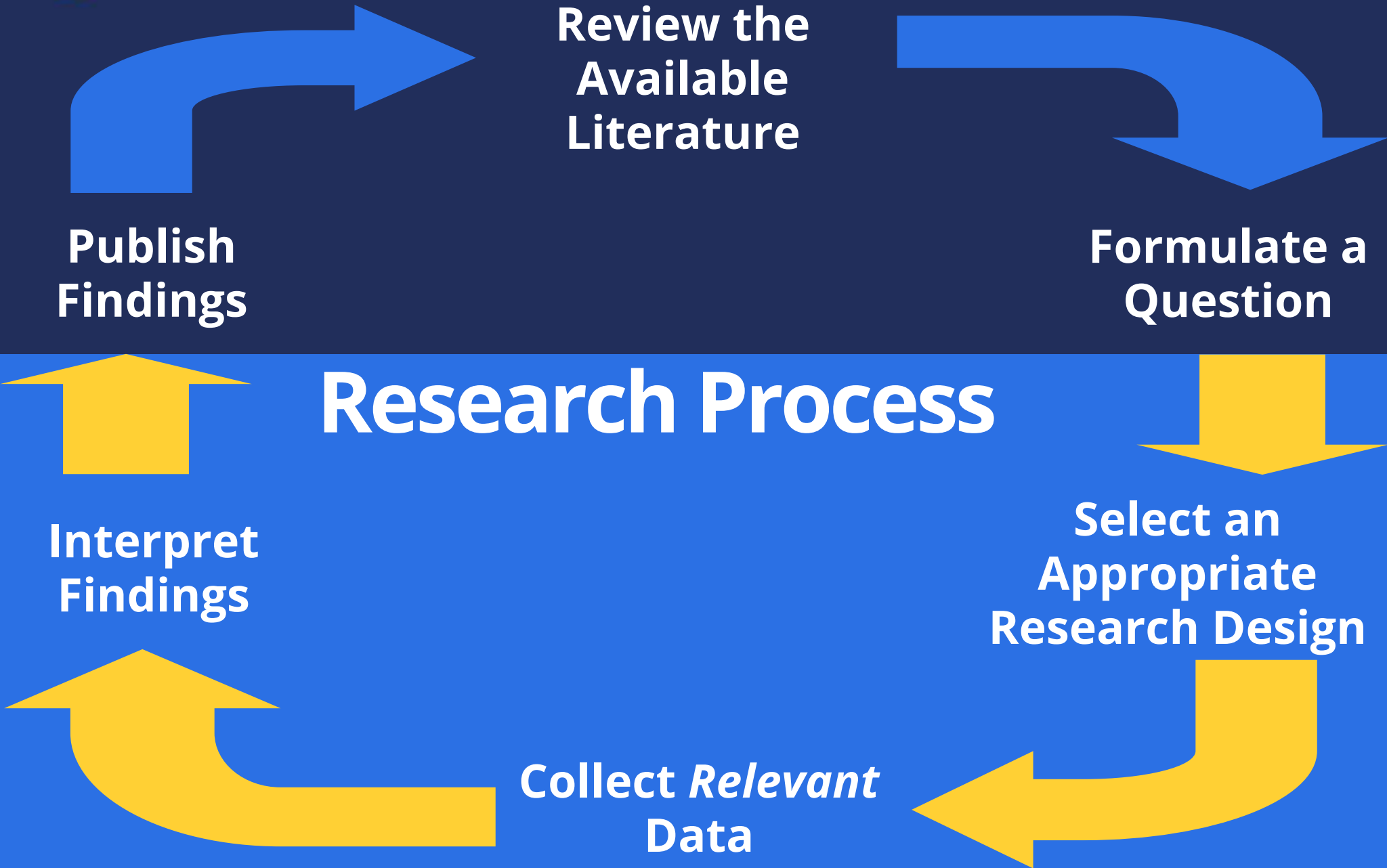
**Formulate a
Question**

Research Process

**Interpret
Findings**

**Select an
Appropriate
Research Design**

**Collect *Relevant*
Data**





What is a literature review (for)?

- When thinking about what a literature review *is*, we need to think about what it is *for*.
- What is the purpose of a literature review?
- *Activity 1:- What is literature review in your own understanding?*



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What is a literature review (for)?

Establishes the terms and context.

- *How else will you define exactly what you're looking at and where its limits are?*

Presents a survey of preceding literature on the topic.

- *How else will you know what's been done already?*

Explores ways that others have solved similar questions/problems.

- *How else will you select an appropriate methodology and approach?*

Outlines the relationship of these texts to each other.

- *How else will you know what the different perspectives and debates are, and where you are coming from?*

Evaluates the quality and relevance of the literature.

- *How else will you be able to build on or reject it?*

Establishes the gaps or inadequacies.

- *How else will you justify your own contribution?*

Demonstrates your scholarly rigor.

- *How else can I have faith in your conclusions?*

Literature Review

A critical analysis of existing research in your field - highlights both the strengths and weaknesses of existing research

Allows you to gain a critical understanding of your field

Opportunity to think about what has been done in your field; opportunity to think about the similarities, patterns, trends and also differences across the existing research

By identifying strengths and weakness, you will be able to think about what has not/needs to be done in your field

The gap in the literature is your justification for your research

MORE THAN JUST A CHAPTER...



- A literature review is a process as well as an outcome!
- **Literature review as an outcome:** appears in the final draft of your thesis as part of your introduction or as a separate chapter.
- **Literature review as a process:** critical engagement (thinking, reading and writing) with relevant research on your topic. It is a crucial and formative stage of your thesis journey.





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CRITICAL THINKING LEVELS



Undergraduate level:

backing your points up with appropriate, authoritative sources



Masters Level:

demonstrating that you understand how knowledge is created



PhD Level:

demonstrating that you can create new, original knowledge



Professional academic:

peer review, building a coherent body of work of your own and assuring others' work





READING CRITICALLY

THINK ABOUT:

What were the research aims of the paper/book?

1

Is the research aim achieved?
If so, how did they do it?

2

Are there any problems with
their methodology?

3

Was it a strong or a weak
research model?

4

How will this research help with
your own research?

5

What can you take from it?
What needs to be avoided?
What are you doing differently?



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Critical thinking

(a three-year-old's guide)

Why?

How does that
work?

What's that
made of?

But X says...

How do you
know?

What's that
for?

What does
that mean?

Says who?

What happens
if...

So what?



What critical thinking can mean in terms of the literature review

- Understanding research on its own terms – testing its viability
- Understanding research in relation to other arguments
- Critiquing research in relation to what you want to do



Starting to think about your own literature review

Who are the key players in my field? This could be anything from academics, governing bodies, schools of thought etc. (Sources!)

What are the main ideas/debates in my field?

How have these ideas changed over time?

What are some of the problems with these ideas/debates?
Is there a problem with the methodology?

What are you going to do differently?



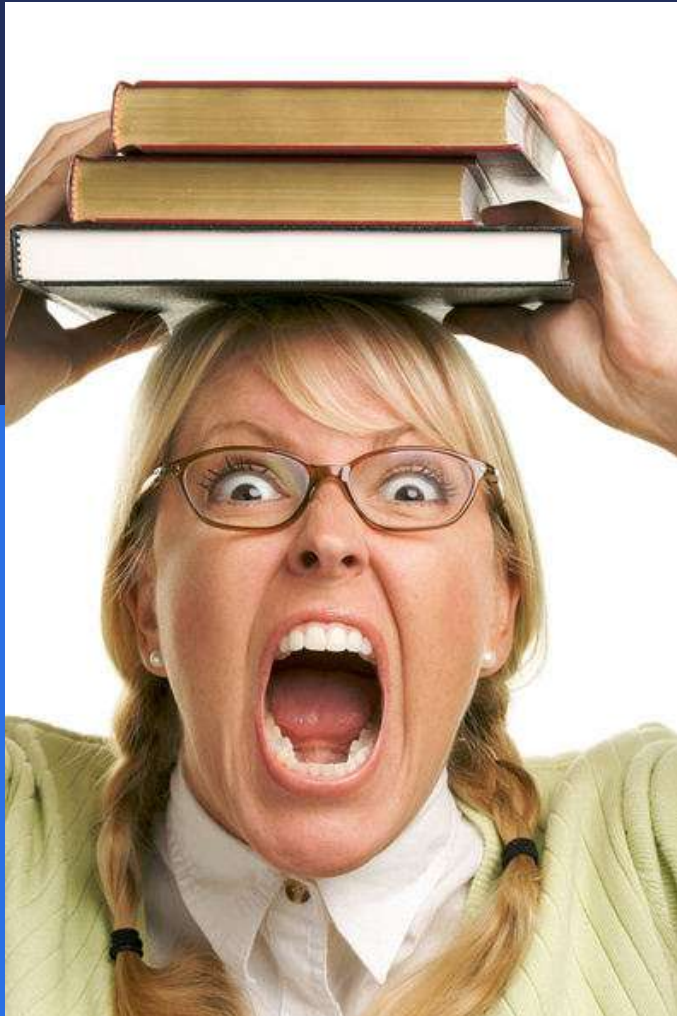
Key players and sources

First stage of the literature review is to identify the key people in your field and collate all relevant sources about your topic.

Ask yourself:

- What research and theory is there on my topic?
- What are the key sources (books, articles) on my topic?
- Who are the main theorists and researchers in this area?
- How has the topic/problem been investigated over time?

Where to start?!



YOU DON'T NEED TO READ
EVERYTHING – you can't!

- You don't need to read every text
- You don't need to read every word



(" Artificial Intelligence" OR "AI"
OR " Machine Learning" OR
"ML") AND (" Ethics" OR "
Governance")



Main ideas/debates

Once you have the relevant sources you can begin to think about what the key ideas, debates, methodologies etc. are in your field.

You can also think about how these ideas have changed over time.

Ask yourself:

- How has the topic or problem been defined?
- Are there any trends and patterns across the literature?
- What methodological assumptions and approaches have been used?
- What are the agreements and disagreements between theorists on my topic?



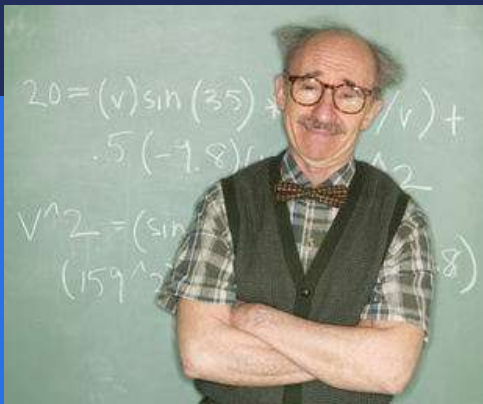
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Organising your material: Identifying a debate

Scholar X



Scholar Y



- disagrees with
- agrees with (*school of thought?*)
- builds on the conclusions of
- confirms the findings of
- has reservations about



Thinking critically

When identifying the key ideas, themes and methodologies in your field, it is important to think critically about them

This will allow you to identify a 'gap' in the literature

Ask yourself:

- What are the strengths and weaknesses of these debates?
- What evidence is lacking, inconclusive or limited?
- What will you add to the topic? What will you do differently?



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- What's your story?



Map your story (literature review):

- What is your topic?
- Who are the key people in your field?
- What are the key resources?
- What are the key ideas in your field?
- What methodologies have been used?
- What are some of the strengths and weaknesses of existing research?
- What will your contribution be?
- How will it be different?

(NB: If you can't answer some of these questions, make a note of this. It will come in handy later!)



Planning is about organising the structure of your literature review (your story will help with this!)

How will you organise the information?

- Chronologically?
- Thematically?
- By trends/approaches/techniques?
- Major debates/controversies?
- Probably a combination of these



You can:

- Jot down ideas
- Move them around
- Create links
- Put text on nodes
- Swap between 'Diagram' and 'Outline'
- Transfer to Word



Writing the review

Start with an overview

Decide on organising principles (themes, trends, methodology, chronology, controversies – usually a combination of some of these)

Use headings for the different sections of the review

Provide summative signposts of where your argument is leading

Summarise your review/highlight 'gap' in research



Tips for clear writing:

- Clear introduction: overview of topic, aim of review and structure
- Clear paragraph structure
- Make sure the subject of your sentence is clear
- Don't assume knowledge
- Make sure key terminology and difficult ideas are always explained thoroughly (ask your yourself: does it make sense?)
- Be objective and balanced
- Use signposts to orientate the reader



Where appropriate, begin sections and paragraphs with a statement which synthesises or analyses, rather than just describes

Use signposting words to demonstrate how texts relate to each other and also what you think of them

- Eg. However, yet, moreover, indeed, similarly etc



Paragraphs and flow

Paragraph:

- Topic sentence
- Discussion of topic
- Closing sentence

Thematic and grammatical links

- Logical progression from one paragraph to the next
- Demonstrate links in your language



Use of citations in the literature review

Two types of citations:

Integral: The author's name appears in the sentence.

- Example (author-date system): Lillis (2001) argues that both tutors and students often lack explicit knowledge of the conventions governing the construction of academic texts.

Non-integral: The author's name appears outside sentence.

- Example: Both tutors and students often lack explicit knowledge of the conventions governing the construction of academic texts (Lillis, 2001).



- **Whose voice is dominant - the writer's or the original author's?**
- The moon is made of cheese (Brie 1999).
- Brie (1999) argues that the moon is made of cheese.
- As Brie (1999) points out, the moon is made of cheese.
- According to Brie (1999), the moon is made of cheese.
However,
- Brie (1999) argues out that the moon is made of cheese.
However,



Learning from models

Look at samples of good theses in your field

Read reviews in prestigious journals in your field



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The End of Session

All the best in your research!!



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Faculty of Computing and Informatics

2022

RESEARCH REPORT

Proud hosts for the The International Conference on Information Systems
and Emerging Technologies (ICSET) and International Conference on
Data Science, Machine Learning and Artificial Intelligence (DSMLAI); 23 -
25 November 2022



**Dr Suama L Hamunyela
(Acting Associate Dean:
Research and Innovation
(2022))**

I am thrilled to present to you the 2022 Faculty of Computing and Informatics (FCI) research outputs report. 2022 was a year of re-envisioning and reconstructing research and innovation steps after a hard-hitting pandemic. Despite the aftermath feelings experienced this year, the FCI research outputs continues to rise. We witnessed graduations of thirty (32) Masters, and a total of sixty-five (65) peer reviewed publications. The statistics are a confirmation that FCI is set towards achieving its vision of being the leading enabler of societal digital transformation.

At FCI, research outputs translate to tangible products and research publications. The research publications are a mirror reflection of work done on the ground and at times an output of scientific investigations geared towards shaping new knowledge and understanding. This year, FCI rigorously ensured comprehensive knowledge creation within the auspices of enabling technologies for the 4IR national mandates. The concepts of Artificial Intelligence, Human Computer Interaction (HCI), Analytics, Blockchain, Cybersecurity and others were investigated. Peer review being the mechanism for ensuring quality, is deemed fundamental towards benchmarking the quality of the research outputs. FCI research outputs emergences from local, regional, and international research platforms.

Still the Faculty has a lot to do to accelerate in innovations and partnerships spheres. With the current level of enthusiasm and passion, I am confident that FCI will in the coming year thrive in all aspects of research, innovations, and partnerships.

Here are a few 2023 events to consider:

- careedconference@uws.ac.uk
- <https://chi2023.acm.org/>
- <https://indore.hosting.acm.org/acmwir2023/>
- ICISSET2023 and
- AfriChi 2023

A handwritten signature in black ink, appearing to read 'Suama L Hamunyela'.

Dr Suama L Hamunyela

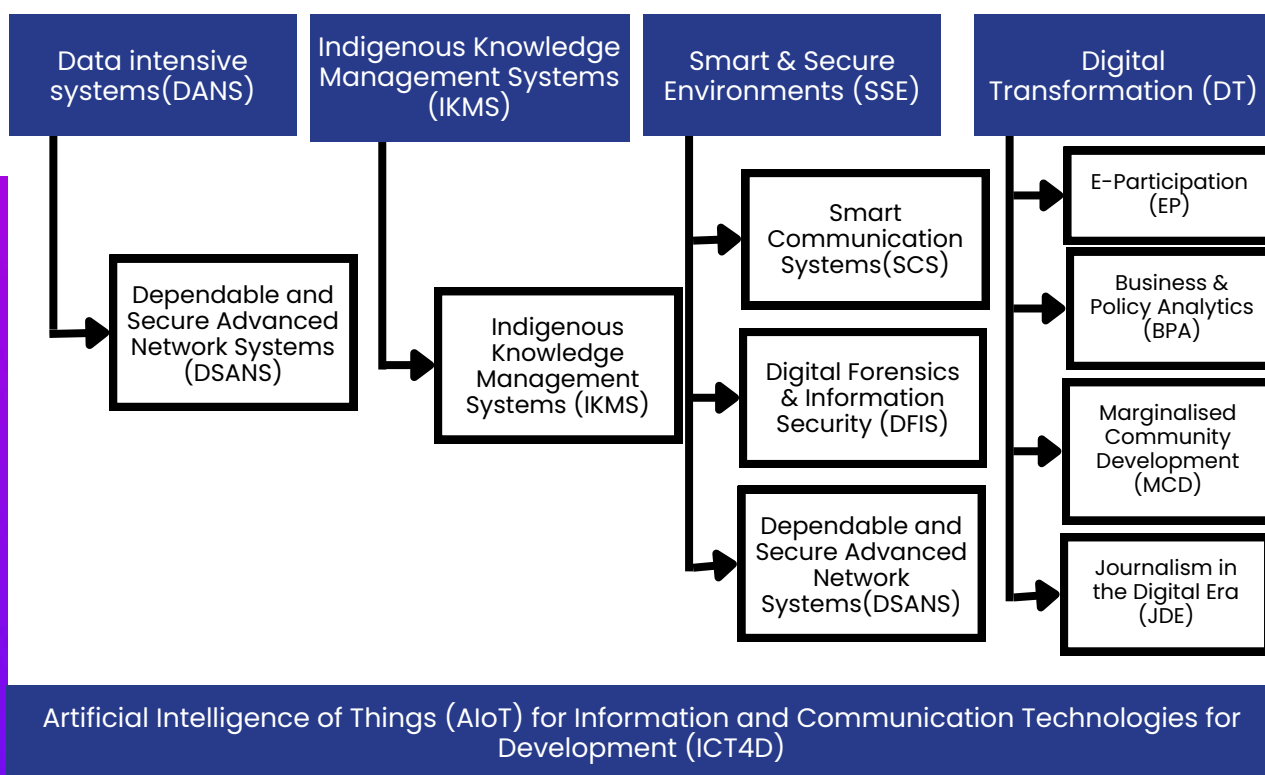
Acting Associate Dean: Research and Innovation (2022)

02 Overview of FCI Research and Innovations

The Faculty of Computing and Informatics continues to excel in research, innovations and partnership at the Namibia University of Science and Technology (NUST). The Faculty has nine (9) active research clusters with several projects across the breadth of the Faculty's four (4) research niche areas: Digital Transformation, Smart and Secure Environments, Indigenous Knowledge in Digital World, and Big Data: Infrastructure & Analytics.

The faculty holds three prestigious awards of UNESCO Research Chairs, one in Digital Technologies for Gender and the other in Smart and Secure Environment covering areas of Smart Communication, Big Data: Infrastructure and Analytics and Cyber Security and a third prestigious award of UNESCO Research Chairs, in Technology Co-design with Indigenous and Marginalized Communities was secured. Staff accountable for research activities including supervision and publications are three (2) Full professors, three (5) Associate Professors and eleven (15) Senior Lecturers.

In the 2022 academic year, the FCI graduated, eight (8) Master of Computer Science and one (9) Master of Informatics students and fifteen (15) Master of Journalism and Media Technology. FCI recorded a total of 7 Journal publications and 37 conference papers, a total of six (6) peer reviewed journal publications, and 4 book chapters. The Faculty's publication output soared with support of sponsors such as the Google DARA who provided financial support, NCRST, SASSCAL and other international partners such as EU. The Computing discipline uses a conference-publishing model more compared to the journal model, and thus the high impact outlets are peer-reviewed conferences. Faculty members served as reviewers of prestigious journals and conferences such as: 2022 DSMLAI, ICISSET, CHI 2022, IEEE Transactions on Industrial Informatics, Journal: African Journal of Science, Technology, Innovation and Development and Global Humanitarian Technology Conference (GHTC) 2021.



FCI Research Clusters

03 Research activities

The three flag ship research activities under the Faculty of Computing and Informatics are in: -

- Indigenous Knowledge in Digital World;
- Cyber Security and Digital Forensics;
- Emerging Technologies (Data Science and AI).

04 Brief profiles of key pacesetters



Prof. Fungai Bhunu Shava: is an Associate Professor in Cybersecurity within the Faculty of Computing and Informatics, Acting Executive Dean, and the leader of the Digital Forensics and Information Security Research Cluster. She is the founding and former chairperson of the Namibia National Cyber Security Competition and the coordinator of Africa Region to International Cybersecurity Competition (AFRICC). She supervises postgraduates and manages a few research projects in the areas of Information Security and Digital Forensics. She has published more than 50 papers in reputable journals and conference proceedings in her fields of specialisation namely Human Computer Interaction, User Experience, Network Security, Information Security in under-serviced communities.

Her work focuses on the secure use of ICTs as enabling tools for socio-economic development in developing economies and societies. She is also an International Visitors Leadership Program (IVLP) alumni and an Application Security Specialist with IBM Security AppScan v8.7. She is also responsible for other projects such as the Child Online Protection, Cisco Academy, and Instructor Training Centre at NUST. She has reviewed conference and journal papers in her areas of specialisation. She serves on the National cybersecurity stakeholder committees including the SADC Cyber-Infrastructure Technical Experts meetings, Africa to International Cybersecurity Competition and the International Cybersecurity Challenge. She has co-organised and spearheaded: Child Online Protection hackathons; WiSci STEAM Camp 2019 in collaboration with US Embassy and GirlUp; TechCamp Namibia: Cybersecurity for Everyone in collaboration with the USA government and Tech and Innovation Bazaars in collaboration with MTC. She participated in the Federal Department of Foreign Affairs, Switzerland (Geneva Dialogue on Responsible Behaviour in Cyberspace) as well as United Nations Institute for Disarmament Research (UNIDR) Cyber Stability Conferences. She actively contributes to Safety and Security Expo; Women in Computing Conferences, Google Developer's Circle, Namibia Internet Governance Forum (NamiIGF), Child online Protection Taskforce, Namibia; IEEE and ACM chapters in Namibia.



Dr Attlee M. Gamundani is a Senior Lecturer in the Cybersecurity Department. He is the Research lead for the Artificial Intelligence of Things (AIoT) for Information and Communication Technologies for Development (ICT4D) cluster. He is an alumni of the United Nations University (UNU) Institute in Macau where he completed his PostDoc in 2021 focusing on AI Ethics and Governance. Dr. Gamundani is serving as the country (Namibia) representative for Responsible AI Network-Africa (RAIN-Africa).

He was appointed as an Advisory Board Member and Operations - Fourth Industrial Revolution (4IR) committee for the Pan African Information Communication Technology Association (PAICTA) in 2020. Dr Gamundani's long CV consists among other publications, over 17 Journal publications, as well as over 50 Conference publications covering IoT Security, Cloud Computing Security, ICT4D, AI Security and Data Security. He delivered keynote addresses on security issues at national and international forums as well as co-authored junior level ICT Teacher's Textbooks (Grade 4, 5, 6 and 7) in Namibia. He was elevated to the Senior Member IEEE (SMIEEE) status as testimony of his wide contributions to the computer science domain. He is also a Check Point Certified Security Administrator (CCSA).

Among his main community engagement activities is the annual Namibia National Cyber Security Competition (NNCSC) which he currently chairs. He serves on the Africa Region to International Cybersecurity Competition (AFRICC) and the International Cybersecurity Competition committees.

Among other funded projects Dr Gamundani is locally executing as Co-PI or PI are the Lacuna Project (US\$48 000); Smart Parking project (N\$200 000- Funded by MTC); Making Windhoek Safer project (N\$56 000 - Funded by MTC). In 2016 he worked with the DFIS team on the Child Online Protection Research Project funded by UNICEF (US\$30 000).

Quarter 1 - 2022

- IoT Smart Gardening project (GIPE, Indonesia);
- IoT Smart Parking Project - CoW(MTC Funded);
- Visiting Faculty - Ashesi University, Ghana (Data Structures & Algorithms);
- Postgraduate Research Seminar Facilitator (Universität Leipzig, Germany).

Quarter 2 - 2022

- Personal Health Info System (GIPE, Peru);
- Cybersecurity Awareness Radio Shows (5 Shows at PAICTA & 1 Energy 100 FM -8th NNCSC Competition Overview);
- GIPE Spring School (Germany);
- Jury Member (ICC, Greece)
- 4IR AI Expo;
- 4IR RID Stream Report.

Quarter 4 - 2022

- Smart City Hunt (IoT Design);
- FinTech Digital Security Forum;
- TN Innovation Week (Hackathon);
- International Cybersecurity Webinar Hosting;
- Cybersecurity Capacity Building :The Global Constellation Annual Conference.

Quarter 3 - 2022

- Deep Learning Indaba X (Hackathon);
- IoT Innovation Challenge (Hackathon);
- NAMDEB AR/VR Challenge(Hackathon);
- SARFAI Ministerial round table Program Planning;
- SARFAI Forum Invited Speaker.
- Postgraduate Research Seminar - UNAM & Botho University , Lesotho.

DR GAMUNDANI'S 2022 RESEARCH PROFILE MILESTONES

Prof Jude Osakwe is an Associate Professor in the Department of Informatics. He is a member The Nigeria Computer Society and a member of the Digital transformation in the Department of Informatics. His research areas of interests are ICT in Education, Business Analytics and ICT for Development. Professor Osakwe is involved in various activities including being a Local Chair of the International Conference on Information Systems and Emerging Technology 2022 (ICISSET). Together with the FCI team, he sourced conference material worth 2500USD, and 40 000ND sponsorship for the ICISSET conference which will publish in the SSRN Elsevier proceedings. Graduated 3 Masters in Informatics, two (2) Journal articles Presentation ten (10) conference papers in international conference. Professor Osakwe is also a reviewer in Scientific Africa (Elsevier)



Dr Lameck Amugongo is a passionate Data Scientist. He is the DARA Big Data project lead at FCI. The DARA Big Data Project provides intensive data science skills training to students from the 8 African countries who partnered to build the mid-frequency instrument of the Square Kilometre Array (SKA) telescope in Southern Africa.

Dr Amugongo holds a PhD in Cancer Science from Manchester University. His research involves investigating image-based data mining methods to analyse how lung tumors change during treatment. For the year 2022, Dr Amugongo presented at the following conferences.

- Deep Learning Indaba 2022, Tunis (Tunisia).
- UNESCO Southern Africa sub-regional forum on Artificial intelligence (SARFAI) 2022 (Windhoek).
- Black in AI workshop co-located at Neural Information Processing Systems, New Orleans (Louisiana, US).

Together with other data scientists in the Faculty, Dr Amugongo was awarded a grant (£ 8000) through DBD/OAD Infrastructure call and a DARA Big Data grant (£ 3500).

The energetic Dr Amugongo organised the following workshops at NUST, [HTTPS:](https://)

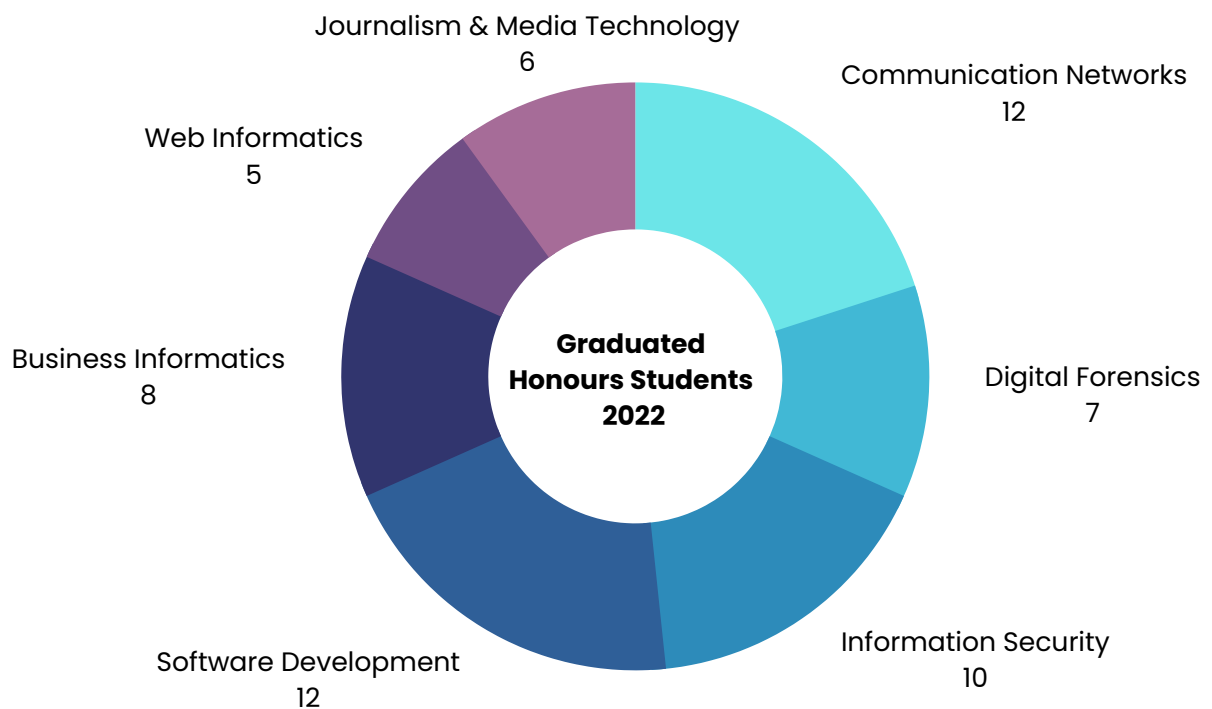
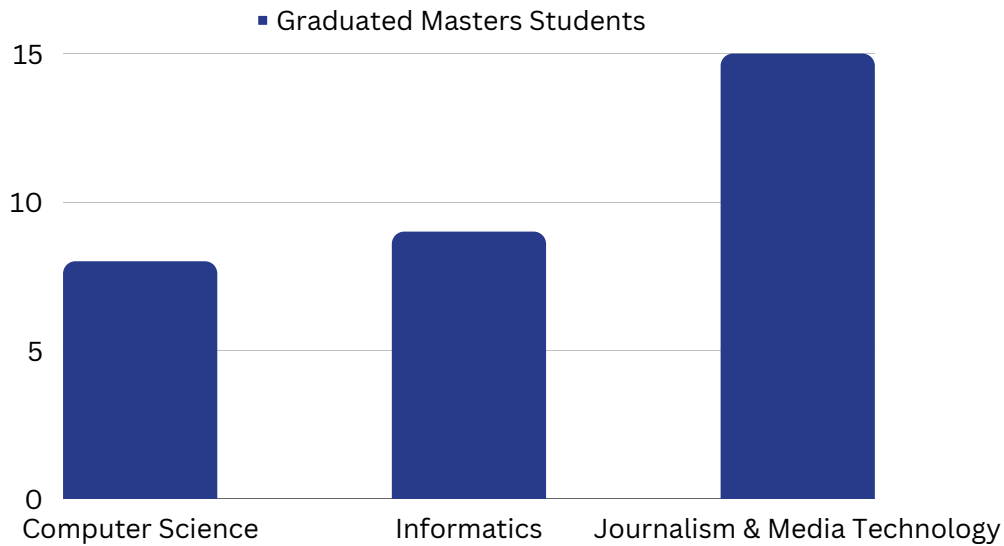
- NUST-DARA Data science school 2022.
- NAMDEB VR/AR Hackathon 2022 and
- Indaba X Namibia 2022.

He is currently working on EU Horizon funded projects with collaborators at the Institute for Ethics in Artificial Intelligence at Technical University of Munich.

05 FCI Research Output (Publications) Jan 2022 – Nov 2022



06 FCI Research Output (Graduations) Jan 2022 – Nov 2022



6 graduated with Postgraduate Certificates

07 Journal Articles

1. Ahmed Farouk, Shahid Mumtaz, Dharm Singh Jat, Heena Rathore, Zhe Li (Eds) (2022). Special Issue Computation and Modelling of Economic and Financial Complex Systems Discrete Dynamics in Nature and Society, vol. 2022. <https://www.hindawi.com/journals/ddns/si/705941/page/2/>
2. Pietari Keskinen, Heike Winschiers-Theophilus, Shilumbe Chivuno-Kuria, Anna Müller, Marko Nieminen: Digital microwork as a livelihood strategy in a Namibian informal settlement. *Electron. J. Inf. Syst. Dev. Ctries.* 88(1) (2022)
3. Heike Winschiers-Theophilus, Naska Goagoses, Erkki Rötönen, Tariq Zaman: Pushing political, cultural, and geographical boundaries: Distributed co-design with children from Namibia, Malaysia and Finland. *Int. J. Child Comput. Interact.* 31: 100439 (2022)
4. Heike Winschiers-Theophilus: Bridging worlds: indigenous knowledge in the digital world. *JCDL* 2022: 1
5. Blessing Guembe, Ambrose Azeta, Sanjay Misra, Victor Osamor, Luis Fernandez-Sanz and Vera Pospelova (2022), "The emerging Threat of AI-driven Attacks: A Review". *An International Journal of Applied Artificial Intelligence*. Vol 36 Issue 1. Published by Taylor & Francis Ltd. DOI10.1080/08839514.2022.2037254
6. Sitto, K., Lubinga, E., Chiumbu, S., Sobane, K. & Mpofu N. (2022). Evaluating government use of digital media during Covid-19 in Southern Africa. *World Medical & Health Policy*: 1–18
7. Amugongo, L. M., Green, A., Cobben, D., van Herk, M., McWilliam, A., & Osorio, E. V. (2022). Identification of modes of tumor regression in non-small cell lung cancer patients during radiotherapy. *Medical Physics*, 49(1), 370–381.
8. Khan, A., Milne-Ives, M., Meinert, E., Iyawa, G. E., Jones, R. B., & Josephraj, A. N. (2022). A scoping review of digital twins in the context of the Covid-19 pandemic. *Biomedical Engineering and Computational Biology*, 13, 11795972221102115.

08 Book Chapters

1. Jain, A., Jat, D.S. (2022). A Review on Consensus Protocol of Blockchain Technology. In: Nagar, A.K., Jat, D.S., Marín-Raventós, G., Mishra, D.K. (eds) *Intelligent Sustainable Systems. Lecture Notes in Networks and Systems*, vol 334. Springer, Singapore. https://doi.org/10.1007/978-981-16-6369-7_72. https://link.springer.com/chapter/10.1007/978-981-16-6369-7_72#citeas.
2. Meyer, M., Gamundani, A. M., Becker, K., Malpartida, D., Nugroho, A., Ochoa-Luna, J., ... & Winschiers-Theophilus, H. GLOBAL INTERCULTURAL PROJECT EXPERIENCE (GIPE): A Distributed Interdisciplinary Project-Based Learning. Matalda Carmo (Ed.) *Education Applications & Development VII*, published by inScience Press, 2022
3. Idemudia, E., Iyamu, T., Ndayizigamiye, P., Shaanika, I. (Eds.). (2021) *Using Information Technology Advancements to Adapt to Global Pandemics* (pp. 66–113). IGI Global
4. Chiumbu, S., Mpofu N. & Sobane, K. (2022). Framing COVID 19: Fear appeal messaging in the print media in Namibia & South Africa. In Dralega, C. A. & Napakol, A. (Eds.). (2022). *Covid-19 and the media in Sub-Saharan Africa: Media viability, framing and health communication*. Emerald Publishing

1. Jat, D.S., Jain, A. (2022). Implementation of Blockchain-Enabled Healthcare System using Hyperledger Fabric, DSMLAI '21': Proceedings of the International Conference on Data Science, Machine Learning and Artificial Intelligence August 2021 Pages 37–47 <https://doi.org/10.1145/3484824.3484914>
2. Jat, D.S., Kamati, T. (2022). Honeypot Behaviour Patterns Learning for Mass Email Marketing Detection, DSMLAI '21': Proceedings of the International Conference on Data Science, Machine Learning and Artificial Intelligence August 2021 Pages 48–52 <https://doi.org/10.1145/3484824.3484891>
3. Misra, A.K., Jat, D.S., Mishra D.K. (2022). Machine Intelligence for Predicting New Start-ups Success: A Survey, DSMLAI '21': Proceedings of the International Conference on Data Science, Machine Learning and Artificial Intelligence August 2021 Pages 99–105 <https://doi.org/10.1145/3484824.3484919>
4. Singh, O., Yadav, M., Yadav, P., Rishiwal, V., Thakur, P., Jat, D.S. (2021). QoS-Attentive Learning-based Routing for Scalable WSNs, DSMLAI '21': Proceedings of the International Conference on Data Science, Machine Learning and Artificial Intelligence August 2021 Pages 305–310 <https://doi.org/10.1145/3484824.3484906>
5. Jain, A., Jat, D.S. (2023). An Analysis of Supervised Machine Learning Algorithms for COVID-19 Diagnosis. In: Yang, X.S., Sherratt, S., Dey, N., Joshi, A. (eds) Proceedings of Seventh International Congress on Information and Communication Technology. Lecture Notes in Networks and Systems, vol 465. Springer, Singapore. https://doi.org/10.1007/978-981-19-2397-5_71.
6. Rusere, K., Musarurwa, S., Bhunu Shava, F., Maravanyika, M. (2022). Security Concerns on Social Media Customer-to-Customer Online Transactions. A Case of Namibia Facebook Users. Presented at IST-Africa 2022. May 2022., www.IST-Africa.org/Conference2022
7. Morolong M.P., Bhunu Shava F., Shilongo V.G. (2022) Designing an Email Security Awareness Program for State-Owned Enterprises in Namibia. In: Senjyu T., Mahalle P., Perumal T., Joshi A. (eds) IOT with Smart Systems. Smart Innovation, Systems and Technologies, vol 251. Springer, Singapore. https://doi.org/10.1007/978-981-16-3945-6_67 pp 679–688
8. Nelago E, Osakwe J, Iyawa G., Ujakpa M. (2022). Leveraging Technology to Enhance Effective Electronic Policing in Developing Countries. Presented at IST-Africa 2022. May 2022., www.IST-Africa.org/Conference2022
9. Osakwe J., Iyaloo N. W., Tarzoor T., Iyawa G., Ujakpa M. (2022). Determinants of Information Systems Resources for Business Organisations' Competitive Advantage: A Resource-Based View Approach. Presented at IST-Africa 2022. May 2022., www.IST-Africa.org/Conference2022.
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11. Haixula P. H. Osakwe J. (2022). Effectiveness of Outsourced Information Systems in Public Sector. Presented at IST-Africa 2022. May 2022., www.IST-Africa.org/Conference2022.
12. Osakwe J., Iyawa G., Ankome T. (2022). Learning Analytics Tools for Enhancing Students' Performance: A Global Perspective. Presented at IST-Africa 2022. May 2022., www.IST-Africa.org/Conference2022.
13. Shilamba M., Osakwe J., Iyawa G., Ujakpa M. (2022). The Use of Electronic Notification Systems (e-Notification system) for Registering Births and Deaths: Findings from a Developing Country. Presented at IST-Africa 2022. May 2022., www.IST-Africa.org/Conference2022.
14. Ujakpa M., Osakwe J., Iyawa G., Iyaloo W., Hashiyana V., Simasiku G. (2022). Development of the Greenwell Matongo Community Library Management Information System (LMIS). Presented at IST-Africa 2022. May 2022., www.IST-Africa.org/Conference2022.
15. Ujakpa M., Iyawa G. Osakwe J. Mutasa S., Suresh N., Sakeus S. (2022). Maternity Management Information System for Katutura Hospital Maternity Department, Windhoek, Namibia. Presented at IST-Africa 2022. May 2022., www.IST-Africa.org/Conference2022.
16. Ujakpa M., Mutayala M., Osakwe J., Iyawa G., Shililifa E. (2022). School Management Information System (SMIS) for the Elite College. Presented at IST-Africa 2022. May 2022., www.IST-Africa.org/Conference2022.
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19. Marie Louise Juul Søndergaard, Gopinaath Kannabiran, Simran Chopra, Nadia Campo Woytuk, Dilrukshi Gamage, Ebtisam Alabdulqader, Heather McKinnon, Heike Winschiers-Theophilus, Shaowen Bardzell: Feminist Voices about Ecological Issues in HCI. CHI Extended Abstracts 2022: 90:1-90:7
20. Emilie Maria Nybo Arendtthorp, Kasper Rodil, Heike Winschiers-Theophilus, Christof Magoath: Overcoming Legacy Bias: Re-Designing Gesture Interactions in Virtual Reality With a San Community in Namibia. CHI 2022: 555:1-555:18
21. Tariq Zaman, David Lim, Isaac Makosa, Naska Goagoses, Erkki Rötönen, Helvi Itenge, Daniel Yong Wen Tan, Sydney Mutelo, Heike Winschiers-Theophilus: A Children-Created Virtual Learning Space Station. IMX 2022: 241-244
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25. Muchinenyika, S.H., Gamundani, A.M. & Stanley C (2022). Conceptual Design of a Sensorless Smart Parking Payment Solution. 2022 IEEE 9th International Conference on Sciences of Electronics, Technologies of Information and Telecommunications (SETIT). <http://www.setit.rnu.tn/Final-Version2022/41.pdf>
26. Shaanika, I. The Use of Mixed-Methods as a Research Strategy in Information Systems Studies. Proceedings of the 13th International Conference on Society and Information Technologies (ICSIT 2022), Iyamu, T., & Shaanika, I. (2022). Assessing Business Architecture Readiness in Organisations. In ICEIS (2) (pp. 506-514).
27. Nakasole, N., Chitauo, M., & Hamunyela, S. Analysing ICT initiatives towards Smart Policing to Assist African Law Enforcement in Combating Cybercrime. SETIT, Genoa-Italy & Sfax-Tunisia 28-30 May 2022. <http://www.setit.rnu.tn/Final-Version2022/33.pdf>
28. Simon Hlekisana Muchinenyika, Hippolyte N. Muyingi, "An Analysis of Screen Resolution Effects on Battery Endurance: A Case of Smartphones," in 2022 IEEE 9th International Conference on Sciences of Electronics, Technologies of Information and Telecommunications (SETIT), 2022, pp 356-360
29. Ajayi, L.K., Azeta, A., Odun-Ayo, I., Aniemeka, E.T. (2022). Acoustic Nudging-Based Model for Vocabulary Reformulation in Continuous Yorùbá Speech Recognition. In: Gervasi, O., Murgante, B., Hendrix, E.M.T., Tanar, D., Apduhan, B.O. (eds). International Conference on Computational Science and Its Applications – ICCSA 2022. pp 494-508. Lecture Notes in Computer Science, vol 13375. Springer. Malaga, Spain, July 4-7, 2022. https://doi.org/10.1007/978-3-031-10522-7_34.
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32. Shihepo, E., Bhunu Shava, F., & Chitauo, M. (2022). Review of BYOD security awareness within Namibian enterprises. Presented at the 1st Zimbabwe Conference on Information Communication & Technology (ZCICT) 2022 9th – 10th November 2022, Harare, Zimbabwe (Virtual)

33. Musa, G., Bhunu Shava, F. (2022). Feature selection for Evaluating Deep Reinforcement Learning DDoS attack Detection Mechanisms using CICIDS2017 Dataset 74, International Conference on Information Systems and Emerging Technologies, (ICISSET 2022), 23- 25 November, Windhoek, Namibia.
34. Nawa, E.T., Bhunu Shava, F., & Chitauro, M. (2022). Developing a Cybersecurity Framework for the Namibian Banking Institutions 69, International Conference on Information Systems and Emerging Technologies, (ICISSET 2022), 23- 25 November, Windhoek, Namibia.
35. Gerson, N., & Bhunu Shava, F. (2022). Designing A Security Model for Ehealth System in Namibia 70, International Conference on Information Systems and Emerging Technologies, (ICISSET 2022), 23- 25 November, Windhoek, Namibia.
36. Rita, M., Bhunu Shava, F., & Chitauro, M. (2022). Tech4Good, an analysis on Artificial Intelligent Powered Chatbots with Child Online Protection in Mind 3, International Conference on Information Systems and Emerging Technologies, (ICISSET 2022), 23- 25 November, Windhoek, Namibia.
37. Kanyama, M. N., Bhunu Shava, F., Gamundani, A.M., & Hartmann, A. (2022). Machine Learning Anomaly Detection Techniques Applicable to Smart Water Metering Networks – A Systematic Review. 23WNSYMP (23rd WaterNet/WARFSA/GWPSA Symposium).
38. Kavikairua J., Bhunu Shava, F. (2022). Algorithm to Impact Children’s Online Behaviour and Raise their Cyber Security Awareness. International Conference on Artificial Intelligence, Big Data, Computing and Data Communication Systems (icABCD), 2022, pp. 1-7, doi: 10.1109/icABCD54961.2022.9856017.
<https://ieeexplore.ieee.org/document/9856017>
39. Rusere, K., Musarurwa, S., Bhunu Shava, F., Maravanyika, M. (2022). Security Concerns on Social Media Customer-to-Customer Online Transactions. A Case of Namibia Facebook Users. Presented at IST-Africa 2022. May 2022., www.IST-Africa.org/Conference2022

10 Books Authored/Co-Authored

1. Luhach A.K., Jat D.S., Bin Ghazali K.H., Gao XZ., Lingras P. (eds) (2022). Advanced Informatics for Computing Research. ICAICR 2021. Communications in Computer and Information Science, CCIS, volume 1575). Springer Nature Switzerland AG 2022. <https://doi.org/10.1007/978-3-031-09469-9>
2. Poonia, K. R., Singh, V., Jat, D.S., Diván, M. S., Khan, S. M. (2022). Proceedings of Third International Conference on Sustainable Computing, SUSCOM 2021 Book Title Proceedings of Third International Conference on Sustainable Computing: SUSCOM 2021, Series ISSN 2194-5357. <https://link.springer.com/book/9789811645372>.
3. Nagar, A.K, Jat, D.S., Marín-Raventós, G., Mishra, D.K. (Eds.2022), Intelligent Sustainable Systems, Selected Papers of Worlds4 2021, Volume 1, Springer. <https://link.springer.com/book/10.1007/978-981-16-6309-3>
4. Nagar, A.K, Jat, D.S., Marín-Raventós, G., Mishra, D.K. (Eds.2022), Intelligent Sustainable Systems, Selected Papers of Worlds4 2021, Volume 2, Springer. <https://doi.org/10.1007/978-981-16-6369-7>

11 Conference Presentations

1. Ajai Mishra, Dharm Singh Jat, and Durgesh Kumar Mishra; *An Experimental Study of Machine Learning Algorithms for Predicting Startup Success* 6th World Conference on Smart Trends in Systems Security and Sustainability organised by Global knowledge research foundation 24-27 August 2022
2. Arpit Jain, Dharm Singh Jat, *Edge Enabled Prognosis using Cognitive Ensemble Machine Learning Model for Time Constraint Applications*. 6th World Conference on Smart Trends in Systems Security and Sustainability organised by Global knowledge research foundation 24-27 August 2022.
3. Attlee M. Gamundani; *Building Responsible AI for Africa by defining the Socio-technical Constructs*. Southern Africa sub-Regional Forum on Artificial Intelligence, 7-9 September 2022, Safari Court Hotel and Conference Centre, Windhoek, Namibia
4. Lameck Amugongo; *AI Ethics from an African Context – A Case Study in the Health Sector*. Southern Africa sub-Regional Forum on Artificial Intelligence, 7-9 September 2022, Safari Court Hotel and Conference Centre, Windhoek, Namibia
5. Jose Quenum; *Session 1 Moderator – Status of AI development and utilisation in Southern Africa*. Southern Africa sub-Regional Forum on Artificial Intelligence, 7-9 September 2022, Safari Court Hotel and Conference Centre, Windhoek, Namibia
6. Attlee M. Gamundani and Shakela Viktoria; *Distilling the encounters towards building a Namibian cybersecurity awareness platform*. Global Constellation Annual Conference: Linking Cybersecurity Capacity Research to development in Africa, 2-3 November 2022, Park Inn by Radisson, Newlands, Cape Town.

7. Gabriel T. Nhinda and Fungai Bhunu Shava; Using co-design to craft cybersecurity secure practices for rural communities in Africa: A case study from Northern Namibia. Global Constellation Annual Conference: Linking Cybersecurity Capacity Research to development in Africa, 2-3 November 2022, Park Inn by Radisson, Newlands, Cape Town.

8. Nobert R. Jere , Sonwabo Mdwaba and Attlee M. Gamundani ; Co-designing a cybersecurity citizen centric framework for South Africa. Global Constellation Annual Conference: Linking Cybersecurity Capacity Research to development in Africa, 2-3 November 2022, Park Inn by Radisson, Newlands, Cape Town.

9. Nobert R. Jere, Attlee M. Gamundani and Sonwabo Mdwaba; Cybersecurity a national pandemic: Who is safer from cyber-attacks? Global Constellation Annual Conference: Linking Cybersecurity Capacity Research to development in Africa, 2-3 November 2022, Park Inn by Radisson, Newlands, Cape Town.

10. Kanyama Maria N, Fungai Bhunu Shava, Attlee M. Gamundani and Andreas Hartman; Machine learning anomaly detection techniques applicable to smart water metering networks- A systematic review. Integrated Water Resource Management for Sustainable Development in Eastern and Southern Africa, 23rd WaterNet/WARFSA/GWP-SA Symposium, Rustenburg, South Africa, 19th -21st October 2022.

11. Manfred Meyer, Katja Becker, Attlee M. Gamundani and Colin Stanley; A Global Intercultural Project Experience (GIPE): Reflections on Combining Online and On-site Project-Based Learning across four Continents. International Conference on Transformative Teaching: Focus on Pedagogy 2022, 15-17 November 2022. A virtual conference: Florida State University, University of Dundee, Zayed University.



Smart City Hunt - November 2022 (NUST, MTC, NCRST, City of Windhoek and UNDP)

12 Others (Keynotes, Talks, Magazines, e.t.c)

1. Prof Dharm Singh Jat, Session Chair and speaker, 7th International Conference on Information & Communication Technology (ICICT2022), 21-24 February 2022, London, UK
2. Prof Dharm Singh Jat invited talk on Smart Computing and Communication: Enabling future IoTs, International Conference on "Empowering Smart Future through Scientific Development and Technology organized by The Research Committee & Department of Computer Applications (MCA), Patna Women's College (Autonomous), Patna University, 5TH MAY 2022.
3. Prof Dharm Singh Jat contributed as Resource Person in AICTE Training and Learning (ATAL) Academy Online Elementary FDP on "Security in Blockchain and Cyber-Physical Systems Design" from 17/01/2022 to 21/01/2022 at Government Engineering College, Rajkot.
4. Prof Dharm Singh Jat sharing knowledge as a Speaker for the Region 10 IEEE Computer Society Mega DVP Program on the topic "Smart Computing and Communication: Enabling Future Internet of Things" on May 14th, 2022
5. Prof Dharm Singh Jat, Inaugural Speaker, 6th World Conference on Smart Trends in Systems Security and Sustainability organised by Global knowledge research foundation 24-27 August 2022.
6. Prof Fungai Bhunu Shava - Empowering Women and Girls in Science - promoting gender equity in science for women and girls PAICTA. (https://www.youtube.com/watch?v=G64-Cpkh_o&t=29s)
7. Dr Attlee M. Gamundani -Tech Power Talk – Cybersecurity – February 15, 2022 (radio 78.5 Pan-African Voice) Radio Show (<https://www.youtube.com/watch?v=pIOfnlZgKEs>)
8. Dr Attlee M. Gamundani -Tech Power Talk – Cybersecurity – March 8, 2022 (radio 78.5 Pan-African Voice) Radio Show (<https://www.youtube.com/watch?v=hbF5JqsUOUo>)
9. Dr Attlee M. Gamundani -Tech Power Talk – Cybersecurity – April 5, 2022 (radio 78.5 Pan-African Voice) Radio Show (<https://www.youtube.com/watch?v=y-eNLz3pdQw>)
10. Dr Attlee M. Gamundani -Tech Power Talk – Cybersecurity – May 17, 2022 (radio 78.5 Pan-African Voice) Radio Show (<https://www.youtube.com/watch?v=FcuQ2tFBwGU>)
11. Hickok, M., Dorsey, C., O'Brien, T., Baur, D., Ingram, K., Chauhan, C., & Gamundani, A. (2022). Case Study: The Distilling of a Biased Algorithmic Decision System through a Business Lens. Available at SSRN. (<https://ssrn.com/abstract=4019672>)



13 Postgraduate Thesis Outputs 2022

Master of Computer Science (NQF-9) – [8]

Graduate Name	Thesis Title
MUSA Gabriel P	Efficient Hybrid Deep Reinforcement Learning Mechanism for Distributed Denial of Service Attack Detection in Software Defined Networks
GERSON Naemi	Designing a Security Model for Ehealth System in Namibia
JONAS Josua	Multi-Cloud Serverless Function Composition using a Hierarchical Task Network Planner
KORUHAMA Kahuhu A	A Genealogy Tool to Account for the Dual Descent Kinship among Ovaherero Descents
MATSA Lonnie S	Design of an Intrusion Detection System for Distributed Denial of Service Attacks in Software Defined Networks Using Hybrid Deep Learning
SHEHU Yakubu	An Optimized Neural-Network Based Algorithm for Semi-Automated Identification and Quantification of Wildlife on High Resolution Drone
SHINEDIMA Gabriel N	Development of an Interactive World Map Game Co-Designed by Primary School Learners in Namibia
SIBANDA Blessing K	Mobile Application for Plant Leaf Disease Classification in a Resource-Constrained Environment



TN Mobile Innovation Week



UNESCO, Southern Africa sub-Regional Forum on AI

Master of Informatics (NQF-9) - [9]

Graduate Name	Thesis Title
GABRIEL Emeritha N	A Conceptual model for the adoption of an online program recommender system for the Namibian Higher Educational Institutions.
MARTHIN Alfriede N	A Model for Teaching Physical Science in Namibian High Schools through the Use of ICT Platforms
NAHOLE Lukas	A Strategy for Consumer Adoption and Usage of Point-of-Sale E-Payment System in Namibia Payment System in Namibia.
KUBERSKY Brandon R	Machine Learning Model to Analyse Cattle Movement with GPS-Enabled Device Tracking in Namibia-Hardap Communal Lands
SHIPULWA Fillipus S	An Automated Adaptive Risk Management System for Public Financial Information Systems: A Case Study of the Ministry of Finance in Namibia
NAKASOLE Naftali	Developing a Smart Policing Model to Assist Namibian Law Enforcement Agencies in Combating Cybercrimes
NABOTH Rosalia N N	Developing an Enterprise Resource Planning System Conversion Model: A Case Study of the Namibia Power Corporation (PTY)
NAWA Eva-Lisa T	Developing A Cybersecurity Framework for the Banking Sector of Namibia
SHALUMBU Iyaloo S	Towards the Development of an Adaptive Strategic Framework for Digital Transformation. A Case Study of Namibia Statistics Agency

Master of Journalism and Media Technology (NQF-9) - [15]

Graduate Name	Thesis Title
NTEMTWA Angela K	An exploration into the perceptions of the public relations profession among selected corporate organisations in Windhoek, Namibia. NUST
MUSHAYAVANHU Kudakwashe	Framing of the COVID-19 pandemic in Namibian National Newspapers: An analysis of The Namibian, Namibian Sun and New Era. NUST
MATSI Fransina A	Investigating the effectiveness of NAMCOL's corporate communication strategy in changing brand perceptions among selected stakeholders in Windhoek, Namibia. NUST
KOOPER Immanuel	Is Language a Barrier to Public's Participation in the Law-making Process?
MATUNDU Marina	A Reception Analysis of Omurari FM's Okaruveze Kovaremane Radio Programme for People with Disabilities
TJITUKA Floyd P	The Role of Oral Media and Radio in Promoting and Preserving Indigenous Knowledge Systems: A Case Study of Otjiherero-Speaking Communities in Kunene Region
MATTY Fidel	Examining stakeholders' views on the implications of delayed enactments of the access to information bill on media reporting in Namibia
MUNDODZI Faith	A Reception Analysis of Selected Cartoon Messages on COVID-19 by Upper Primary School Learners at Highlands Christian School in Windhoek
SHIHEPO Timoteus N	An investigation into how the Namibian print media industry is coping with contemporary economic conditions
ZVIYITA Itai	An examination of ethical issues confronting the operations of selected hybrid media organisations in the digital age in Namibia

MARE Hilary	Investigating the Implications of COVID-19 on the Sustainability of the Print Media in Namibia: A Case Study of the Namibian, Confidente and New Era
KAPENDA Jairus J J	Assessing the use of social media in the Public Relations Practices of the Government Institution Pension Fund (GIPF) of Namibia
HAMBUREE Belinda	An Evaluation of Print Media Campaigns for Promoting Road Safety in Namibia from January to December 2019
IITA Ndamonako A	Investigating Stakeholder Engagement in Public Relations Practices of two Namibian Universities. (The International University of Management and Namibia University of Science and Technology
ADETONA David A	An investigation into how newspapers adhere to copyright regulations in Namibia



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**NAMIBIA UNIVERSITY
OF SCIENCE AND TECHNOLOGY**

Faculty Name: *Computing and Informatics*

Name of Department: *Computer Science*

COURSE OUTLINE FOR RESEARCH METHODOLOGY(RIT812S)

STATEMENT ABOUT ACADEMIC HONESTY AND INTEGRITY

All staff and students of the Namibia University of Science and Technology (NUST), upon signing their employment contracts and registration forms, commit themselves to abide by the policies and rules of the institution. The core activity of NUST is learning and in this respect, academic honesty and integrity are very important to ensure that learning is valid, reliable, and credible.

NUST, therefore, does not condone any form of academic dishonesty, including plagiarism and cheating on tests and assessments, amongst other such practices. NUST requires students to always do their own assignments and to produce their own academic work, unless given a group assignment.

Academic Dishonesty includes, but is not limited to:

- Using the ideas, words, works or inventions of someone else as if it is your own work.
- Using the direct words of someone else without quotation marks, even if it is referenced.
- Copying from writings (books, articles, webpages, other students' assignments, etc.), published or unpublished, without referencing.
- Syndication of a piece of work, all or part of an assignment, by a group of students, unless the assignment was a legitimate group assignment.
- The borrowing and use of another person's assignment, with or without their knowledge or permission.
- Infringing copyright, including documents, copied or cut and pasted from the internet.
- Asking someone else to prepare an assignment for you or to write or sit an assessment for you, whether this is against payment or not.
- Re-submitting work done already for another course or programme as new work, so-called self-plagiarism.
- Bringing notes into an examination or test venue, regardless of whether the notes were used to copy or not.
- Receiving any outside assistance in any form or shape during an examination or test.

All forms of academic dishonesty are viewed as misconduct under NUST Student Rules and Regulations. Students who make themselves guilty of academic dishonesty will be brought before a Disciplinary Committee and may be suspended from studying for a certain time or may be expelled. All students who are found guilty of academic dishonesty shall have an appropriate endorsement on their academic record, which will never be erased.

Campus activities that involve physical contact, whether in a meeting, laboratory, assessment, tutorial/lecture will be held under strict COVID-19 National Health and Safety Protocols. Students not adhering to such National Regulations, i.e. wearing masks correctly to cover both the mouth and nose, social distancing of at least 1.5m, hand sanitising and refraining from campus activities when experiencing COVID-19 symptoms, will not be allowed into a venue.

Course Information

Course Code and Title:	RIT812S, Research Methodology
Department(s):	Computer Science, Informatics, Cyber security and Software Engineering
Programmes:	Bachelor of Computer Science Honours (Communications Networks) Bachelor of Computer Science Honours (Software Development) Bachelor of Computer Science Honours (Information Security) Bachelor of Computer Science Honours (Digital Forensics) Bachelor of Informatics Honours (Web Informatics) Bachelor of Informatics Honours (Business Informatics)
Notational Hours:	150 hours (Contact: 45 hours; Directed self-learning and Self-Directed learning: 65 hours; Assessment: 40 hours)
NQF Level and Credit:	NQF level 8, 15 credits
Semester Offered:	1
Pre-Requisites:	None
Course Equivalencies:	Research Methodology (RIT811S)
Course Objectives:	The course aims to equip students with essential research skills in order to plan and execute a scientific or technical research project, focusing on applied research in Computing and Informatics areas of her/his own interest. Students will get the needed conceptual, writing, and design skills to develop and present a comprehensive research project proposal.

Course Delivery Methods:

The course will be facilitated through the following learning activities:

- One and a half hours, twice per week, of ONLINE contact with students.
- Introduction of course topics and progress reports assignments via Zoom or Microsoft Teams.
- Online lectures to be delivered through Microsoft Teams
- Lecturers and students to interact through online platforms: Emails, Microsoft Teams and WhatsApp.

Lectures: The theoretical foundations will be presented through online platform(Ms Teams). Students are encouraged to embrace online learning and partake in online teaching that will be taking place on **Tuesday**

and Thursday at the time as allocated on the courses timetable. The students are expected to deepen their knowledge and understanding of the covered topics through guided self-study.

Course format:

Contact: 45 hours (Online discussions between students and lecturer(s))

Directed self-learning and Self-Directed learning: 65 hours (out-of-class study time, reading prescribed material, organisational overhead, organising and conducting team meetings, online consultations with a lecturer and fellow students, collecting prescribed material, etc)

Assessment: 40 hours (All online: Proposal Presentation, Final Thesis Submission)

Course Resources:

There are several facilities that students can use as a course resource at NUST and at home.

- a) **NUST level:** Prospectus; The Library; student services especially the Writing Centre; Social Worker; Centre for Teaching and Learning; Internet is available the whole day for research.
- b) **Course Level:** Course Outline; Electronic learning resources on E-Learning (Moodle) materials will be developed.

Quality Assurance: Moderation of assessments will be done in accordance with NUST's general rules and guidelines on moderation.

Student Support and Learning Resources:

Institutional support is provided through the NUST Library, the Department of the Students Services, and the Student Representative Council. Faculty support is provided through new-student orientation activities and the FCI student representatives. Departmental and course-based support is provided through supervised practicals, customized course material and through interactions with lecturers, Head of Department, and the Deputy Head of Department.

Effective Start Date: Monday, 12 February 2025

Course Coordinating Team Information:

Full Name: Prof Jude Osakwe
Email: josakwe@nust.na
Office phone: 0612072952
Office location: Polyheights 406

Office hours:

08H00 am – 16H30 pm (consultations via emails)

Consultation hours: All lecturers (involved in the teachings) are available for consultation with students.

Consultation times can be found on the doors of lecturers' offices. You are welcome to make an appointment with the lecturer via email.

Lecturers Information:

(if a specific topic was not clear, please contact the respective lecturer for the topic via email)

Table 1: List of Lecturers who shall facilitate sessions for this course via online- Semester 1, 2024.

#	Full Name(s)	E-mail Address	Office Phone #	Office Location
1.	Prof Dharm SINGH	dsingh@nust.na	207 2991	PolyHeights, 2 nd Floor, Office 202
2.	Prof Heike WINSCHIERS-THEOPHILUS	hwinschiers@nust.na	207 2168	FCI House
3.	Prof Guy Zodi LUSILAO	gzodi@nust.na	207 2774	FCI House
4.	Dr Suama HAMUNYELA	slhamunyela@nust.na	207 2922	FCI House
5.	Prof Jose QUENUM	jqenum@nust.na	207 2235	Science & Tech Building
6.	Prof Fungai BHUNU SHAVA	fbshava@nust.na	207 2510	FCI House
7.	Prof Samuel AKINSOLA	sakinsola@nust.na	207 2704	PolyHeights, 4 th Floor, Office 401
8.	Prof Jude OSAKWE	josakwe@nust.na	207 2952	PolyHeights, 4 th Floor, Office 401
9.	Prof Gloria E. IYAWA	giyawa@nust.na	207 2760	FCI House
10.	Dr Irja SHAANIKA	ishaanika@nust.na	207 2620	FCI House
11.	Dr Colin STANLEY	cstanley@nust.na	207 2752	FCI House
12.	Dr Maravanyika MUNYARADZI	mmaravanyika@nust.na	207 2263	FCI House
13.	Dr Mercy CHITAURO	mchitauro@nust.na	207 2039	FCI House
14.	Dr Ambrose AZETA	aazeta@nust.na	207 2673	FCI House
15.	Dr Richard MALIWATU	rmaliwatu@nust.na	207 2206	FCI Housse

Possible Supervisors:

Possible supervisors and their areas of interests are listed below:

NOTE: Each supervisor on average can supervise only 4 students, on a first come first serve basis, based on their area of interests and specialisation.

Table 2 List of supervisors and their areas of interests

Full Name(s)	E-Mail/TEL	Area of Specialization/Research Interest
Prof Dharm SINGH JAT	dsingh@nust.na +264 61 207 2991	Multimedia communications, wireless technologies, mobile communication systems, and video transmission over wired/wireless networks, Network security, Internet of things and Smart and Secure Environment
Prof Heike WINSCHIERS-THEOPHILUS	hwinschiers@nust.na +264 61 207 2168	Participatory and Interaction Design, HCI, intangible Digital Cultural Heritage, and Indigenous Knowledge Preservation, Youth and rural development (Community-based co-design with indigenous and marginalised communities Cluster)
Prof Guy-Alain ZODI LUSILAO	gzodi@nust.na +264 61 207 2774	Multimedia Wireless Networks, Cloud Networking, Performance Analysis in wired and wireless Networks, Traffic Engineering, Wireless Sensor Networks, Network Security, Fog Networking and Internet of Things
Prof Jose QUENUM	jquenum@nust.na +264 61 207 2235	Big Data Infrastructure, Process Coordination/Messaging, Programming Languages and Concurrency Control, Virtual Machine Monitor for Big Data Infrastructure, Data Structures for Video Indexing
Prof Fungai BHUNU SHAVA	fbshava@nust.na +264 61 207 2510	Data communication/ computer networking security/ User experience in Security/ Human Computer Interaction Security and Cyber Security
Prof Suama L. HAMUNYELA	slhamunyela@nust.na +264 61 207 2922	E-governance/ICT4D, E-health and digital health, process management
Prof Jude OSAKWE	josakwe@nust.na +264 61 207 2952	ICT in Education, Business Information Systems, EGovernment, Data Governance/Data Analytics.
Prof Samuel AKINSOLA	sakinsola@nust.na +264 61 207 2704	Business Information Systems, Knowledge Support/Knowledge Management, Cluster: Health Informatics

Prof Gloria IYAWA	giyawa@nust.na +264 61 207 2760	Digital Health, Health Informatics and E-Government.
Dr Irja SHAANIKA	ishaanika@nust.na +264 61 207 2620	Area of Healthcare, Enterprise systems, Business Processes, Enterprise Architecture
Dr Munyaradzi MARAVANYIKA	mmaravanyika@nust.na +264 61 207 2263	ICT in education, environmental and wildlife management, intelligent systems, AI systems
Dr Mercy CHITAURO	mchitauro@nust.na +264 61 207 2039	Critical Infrastructure security, Network Security, IT law
Dr Atlee GAMUNDANI	agamundani@nust.na +264 61 207 2099	Internet of Things (IoT) Security, Cloud Computing Security, Smart and Secure environments, Data Security.
Dr Ambrose AZETA	aazeta@nust.na +264 61 207 2673	Software Engineering
Dr Richard MALIWATU	rmaliwatu@nust.na +264 61 207 2206	ICT service delivery to rural and underserved communities, Data communication/computer networking, data processing and analytics, Data mining, Machine Learning.
Mr Sebastian Mukumbira	smukumbira@nust.na +264 61 207 2895	Participatory and Interaction Design, Indigenous Knowledge Preservation, Rangeland Monitoring (Community-based codesign with pastoralists), Spatial Data Infrastructures
Mr Isaac Nyamu	inhamu@nust.na +264 61 207 2074	Digital Forensics, Database Security and Data Protection, Cloud Computing (Security), Cyber Security, Software Development with an emphasis on security, Systems Administration
Mr Admire KACHEPA	akachepea@nust.na +264 61 207 2476	Business processes their analysis and management; Implementation of ERPs in organisations; Implementation of Information Systems
Ms Albertina SHILONGO	ashilongo@nust.na +264 61 207 2691	Information Systems, Big data, Computer networks
Mr Eliazer MBAEVA	embaeva@nust.na +264 61 207 2700	Web Applications Engineering (WAE), Semantic Web (Ontologies), Software/Requirements Engineering, Business Process (Re)-Engineering, Indigenous Knowledge Preservation

Ms Teresa CHIKOHORA	tchikohora@nust.na +264 61 207 2888	Data Mining, Technology Adoption by Small to Medium Enterprises, Business Intelligence ,Educational Technologies
Mr Gabriel NHINDA	gnhinda@nust.na +264 61 207 2724	Human Computer Interaction, Human Computer Interaction Security, Education
Mr Gereon KAPUIRE	gkapuire@nust.na +264 61 207 2334	HCI, Digital Cultural heritage and Indigenous Knowledge Preservation, Community-Based Co-Design, Ethics

Mr Simon MUCHINENYIKA	smuchinenyika@nust.na +264 61 207 2054	Optimal Energy Savings in Mobile Communication / Software Development, Multiculturalism Crowdsourcing
Mr Julius SILAA	jsilaa@nust.na +264 61 207 2037	IT Security, Bitcoins
Mr Johnson BILLAWER	jbillawer@nust.na +264 61 207 2705	Informatics (Enterprise Systems/ERP, E-Business, E-government)
Ms Jovita MATEUS	jmateus@nust.na +264 61 207 2771	Multimedia Applications, Wireless Sensor Networks, Cloud/Fog Computing, IoT, Network Security
Ms Loini IYAMBO	liiyambo@nust.na +264 61 207 2446	Communication networks, Internet of Things
Mr Peter GALLERT	pgallert@nust.na +264 61 207 2268	Wikipedia, computer networks
Ms Sinte MUTELO	smutelo@nust.na +264 61 207 2648	Healthcare, Health Informatics, Information and Communication Technologies for Development (ICT4D), and Business Information Systems.
Mr Steven TJIRASO	stjiraso@nust.na +264 61 207 2097	Indigenous Knowledge Research , Knowledge representation and reasoning, logic.
Mr Edward NEPOLO	enepolo@nust.na +264 61 207 2520	Data Communication Networks and Cyber Security.
Ms Teresia ANKOME	tankome@nust.na +264 61 207 2420	Communication networks, network security, Internet Of Things.
Ms Ndinelago NASHADI	nnashandi@nust.na +264 61 207 2911	Information systems, Persuasive technologies, Big Data Analysis, Culture and Computer Science, Tracking technologies
Ms Ruusa IIPINGE	ripinge@nust.na	Big data, Data for Good, health informatics, Data Science, artificial Intelligence

Mr Nkululeko MTHEMBO	nmthembo@nust.na +264 61 207 2747	ICT in Education/ICT for Development E-Business/ E commerce /E-government Digital Economy
Ms Rosseta KAYS	rkays@nust.na +264 61 207 2405	Marginalised Community Development/ Co-designing technologies / Software Engineering and Human-Computer Interaction
Ms Vicky SHAKELA	vshakela@nust.na +264 61 207 2856	Cyber Security and Communication Networks
Ms Shilumbe CHIVUNO-KURIA	schivuno@nust.na +264 61 207 2057	Human Computer Interactions, Co-designing technological services with unemployed youth.
Mr. Shadreck CHITAURO	schitauro@nust.na +264 61 207 2689	Cloud Computing, Network Security and E-Learning Systems Implementations.

It is the student responsibility to ensure his/her progress reports are signed by the supervisor and submitted ONTIME.. All submissions (progress reports, presentations, proposal documents) should be submitted online and not through the lecturer emails. LATE SUBMISSIONS WILL NOT BE ACCEPTED. Students are encouraged to be in touch with their supervisors, discuss with their supervisors content, topics and informations shared in class. Communicate to your supervisors your submission due dates on time and not the week the specific activity is due. Please note your supervisors are dealing with other matters in the Department hence its your responsibility to keep them up to date with your progress. ALL SUBMISSION DUE DATES ARE STATED IN THE COURSE OUTLINE

Student Readiness:

Technology & Equipment Readiness:

This course will only utilise already existing NUST equipment (on top of the Class Group WhatsApp).

Student Commitments and Contact Times:

- a) **Class Attendance:** Attend all online-classes, and be on time, and make appropriate notes. If a student fails to attend an online class, it is their responsibility to catch-up on what was covered. Students have access to the recommended books from the library. If any special presentations are used in class, it will be made available (text, audio, video) and do not need to be copied by the student.
- b) **Self-Directed Study:** After every class, and before the start of the next class, sort, complete, and annotate own lecture notes, and select further reading in case any particular item was not understood on the first attempt. Students must spend their notational hours revising, practicing or/and completing homework for the course. Students should take the reading list (prescribed and recommended reading seriously. Read all prescribed material to the extent that the student could write a one-page essay, summarising the content, without looking at the material again.

- c) **Communications:** Maintain active communication with your group lecturer and respective supervisors. Students are expected to use established channels of communication and platform (EMAILS, WHATSAPP) to solve course-related issues on time. The end of the semester is not the best time to fix issues. Use the available time to deal with matters at hand.

Recommended Reading:

Grønmo, S. (2019). *Social research methods: Qualitative, quantitative and mixed methods approaches*. Sage.

Gravetter, F. J., & Forzano, L. A. B. (2018). *Research methods for the behavioral sciences*. Cengage Learning.

Bell, E., Bryman, A., & Harley, B. (2018). *Business research methods*. Oxford university press.

Hansen, A., & Machin, D. (2018). *Media and communication research methods*. Macmillan International Higher Education.

Student Learning:

Specific Learning Outcomes:

After completing the course students will, through a written research proposal, show evidence of their ability to:

- Evaluate and apply the types, principles and procedures of research in Computing and Informatics disciplines.
- Select appropriate methods for system requirements, data gathering and analysis of sets of data in research;
- Synthesise, interpret and integrate the results and draw conclusions from data;
- Apply the basics of publishing and adhere to recommended referencing styles, e.g. APA guidelines;
- Communicate research process planning effectively.

Comprehensive Learning Outcome:

Produce and present a comprehensive research proposal.

Course Schedule:

Semester Week	Specific Date	Topic	Activity/Deliverables	
1	11-13 February	Setting Research Methodology Class Tone	• Course outline • Expectations • Process of supervision • Proposal template, • Process around proposal & • Deadlines Research Ethics	
2	18-20 February		• Ethical Considerations • University Research Policy • Paraphrasing /Plagiarism/Turnitin tool	
3	25 February 2025	Types of Scientific Research	• What is research • Types of Research	
4	27 February 2025	Cluster Presentations (I)	Data-intensive Systems (DANS): • Dependable (and secure) Advanced Network Systems • Infrastructure & Analytics (BDIA)(Big Data)	
		Cluster Presentations (II)	Smart & Secure Environment (SSE) • Smart Communication Systems • Digital Forensics and Information Security • Dependable and Secure Advanced Network Systems(DSANS)	

		Cluster Presentations (III)	Digital Transformation • E-Participation • Business and Policy Analytics(BPA) • Marginalised Community Development (MCD) • Journalism in the Digital Era(JDE)	
		Cluster Presentations (IV)	Indigenous Knowledge Management Systems(IKMS) • Indigenous Knowledge Management Systems- • Marginalised Community Development	
4	27 February 2024	Research Problem (I)	Formulation of: • Problem Statement • Topic • Aim • Research objectives • Research Questions & • Hypothesis	
5	04 March 2025	Research Problem (II)	Formulation of: • Problem Statement • Topic • Aim • Research objectives • Research Questions & Hypothesis	
6	6 March 2024	Data sources and APA Referencing (I)	APA Citations and referencing Searching for Information, credible information source, access to various NUST Library databases/portals (ACM, IEEE, etc.).	

7	11 March 2025	Data sources and APA Referencing (II)	Referencing tools • Medley Referencing manager • Zetero	
8	13 March 2025	Proposal Introduction Writing (I)	Introduction • What is proposal Introduction • How to write your proposal Introduction	
9	18 March 2025	Proposal Introduction Writing (II)	Introduction • What is proposal Introduction • How to write your proposal Introduction	

10	25 March 2025	Writing Literature review (I)	Literature Review • What is literature Review? • Why Literature Review? • How to write your study literature?	
11	27 March 2025	Writing Literature review (II)	• Writing Literature review /Organising your literature review • How to write a Literature Review • And questions on literature review	
	31 March – 04 April Institutional Holiday			
12	08 April 2025	Methodology (I)	• Qualitative Approach (Inductive) • Quantitative Approach (Deductive) • Mixed Approach (Abductive Approach)	
13	11 April 2025	Methodology (II)	Methodology SW Development (Action Research, Design Research, Waterfall, Agile, Prototyping, Spiral, XP, etc)	
	15 April 2025	Methodology (II)	Research Designs (Approach to planning) • Case study • Experimental • Theoretical	
14	17 April 2025	Methodology (IV)	Systems and Communication Networks • Modelling and Simulation Approach • Simulation Tools	

	Friday 18 April Until Monday 21 April 2024 Easter Weekend			
15	22 April 2025	Methodology (V)	Data Collection • Before data collection • Participants and stakeholders privacy • Data collection techniques’ and tools • Data Analysis	
16	24 April 2025	write up (I)	• Delineations and limitations • Assumptions • Risk/ feasibility analysis and ethical considerations • Chapter Outlines	
17	29 April 2024	Analysis and write up (II)	Developing your study models/frameworks	
18	1 st May 2024	Monday, Workers Day Observed, 1 st May		
19	04 May 2024	Cassinga Day , 4 th May		
20	06 May	• Evaluating your models/frameworks • Discussing & Recommendation Conclusions		
21	Graduation Ceremony 09 May			
22	13 May 2024	Consultation on Proposals Ethical	Questions/Answering Session Clearance Application Documents	
23	09 May 2024	Proposal Presentation Etiquettes	• Presentation outlines • Presentation checklist/Structure • How to Present your presentation	
23 May 2025 -- Deadline for submissions of Proposal Presentations (ppt)				
04- 06 June 2025 -- All RIT Proposal Presentations				
10 June 2025 - All RIT Final Proposal Submission				

Important Dates:

NOTE: The following dates are subject to change based on the needs of the students at the lecturer's prerogative. Students will be notified ahead of time of any changes.

Exact Date to be announced

Proposal Presentations Exact Date to be announced

Final Proposal Submission Assessment and Evaluation:

Continuous Assessment will apply, consisting of:

- Completion of milestones(Progress reports).
- Research proposal presentation.
- Final research proposal on a topic chosen by the student and approved by the supervisor or a senior academic.

Assessments	Weight
Proposal Online Presentations (Evaluated by: Supervisor & Assessor)	40
Final Written Proposal (Evaluated by : Supervisor & Moderators)	60
Total:	100%

Minimum pass required for this course: 50%

Course Policies:**General Academic Policies:**

It is the student's responsibility to be familiar with and adhere to the NUST's Policies. These Policies can be found in the NUST Prospectus or online at www.polytechnic.edu.na/prospectus.

Submission of reports, presentations and final proposal:

All Late Submissions will not be accepted after the due date and time and thus marked as 0%.

Failure to Pay Fees:

A student who fails to pay his/her fees may not be allowed to write the examination and if allowed, the results will be withheld until all outstanding fees are paid in full.

Important Student Services at NUST

There are a variety of services that you can use at NUST. These services are to your advantage – Use them!!! They include the following:

- Student Counselling and Career Development - Dean of Students Office.
- Writing centre and student academic problems – Centre for Teaching & Learning (CTL)
- Campus Health and Wellness Centre (CHWC) - Dean of Student's office/ NUST Clinic

Authorisation:

This course is authorised for use by:

Head of Department

Date

ACKNOWLEDGEMENT BY STUDENT

(To be completed by all students on the course, detached from the course outline and kept on record in the Department)

I, _____, student Number _____, hereby acknowledge that I have received the course outline for Research Methodology (RIT812S) in semester 1 (2024), and that I have familiarised myself with its content, in particular the statement about academic honesty and integrity. I agree to abide by the Policies and arrangements spelt out in this course outline.

Signature of student

Date

Setting a research question, aim and objective

Cite this article as: Doody O, Bailey ME (2016) Setting a research question, aim and objective. *Nurse Researcher*. 23, 4, 19-23.

Date of submission: November 4 2014. Date of acceptance: May 27 2015.

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This article has been subject to double-blind review and has been checked using antiplagiarism software

Author guidelines
journals.rcni.com/r/nr-author-guidelines

Abstract

Aim To describe the development of a research question, aim and objective.

Background The first steps of any study are developing the research question, aim and objective. Subsequent steps develop from these and they govern the researchers' choice of population, setting, data to be collected and time period for the study. Clear, succinctly posed research questions, aims and objectives are essential if studies are to be successful.

Discussion Researchers developing their research questions, aims and objectives generally experience difficulties. They are often overwhelmed trying to convert what they see as a relevant issue from practice

into research. This necessitates engaging with the relevant published literature and knowledgeable people.

Conclusion This paper identifies the issues to be considered when developing a research question, aim and objective. Understanding these considerations will enable researchers to effectively present their research question, aim and objective.

Implications for practice To conduct successful studies, researchers should develop clear research questions, aims and objectives.

Keywords research question, research aim, research objective, study development, nursing research, novice researchers

Introduction

DEFINING THE research question is a particularly significant step in research as it narrows the research aim and objective down to specific areas the study will address (Creswell 2014, Johnson and Christensen 2014). Research questions are vital as they guide the choice of methodology, methods, sample, sample size, data collection instrument and data analysis techniques (Lipowski 2008).

Selecting and constructing a good research question is one of any study's most difficult aspects (Blaikie 2007, 2009, Aslam and Emmanuel 2010). Despite this, there is little direction on creating good research questions (Lipowski 2008). They do not appear spontaneously and the ability to pose good questions is a skill that can be refined and developed with direction and collaboration from colleagues and supervisors (Thabane *et al* 2009). Importantly, research questions are not static: they change, adapt or are refined when

researchers work through the different steps of the design cycle, reviewing the literature, integrating theory and developing a conceptual framework (Hennink *et al* 2011).

Novice researchers are often overwhelmed and bewildered when trying to convert what they see as relevant issues from practice into a study. This difficulty arises as they progress their work and they feel they know what they need to answer.

Only by consulting a supervisor and/or other knowledgeable people about the research and critical evaluation of the relevant published literature can researchers develop research questions, aims and objectives. Developing a research question is an interactive and inductive endeavour that takes place over time.

A clear research question is vital and sets the scene for development of a clear research aim and objective that are inextricably linked (Martindale and Taylor 2014).

Table 1 Types of quantitative research question

Type	Question
Descriptive	- Seeks to quantify responses to one or more variables - Often begins with 'What is...?' or 'What are...?'
Comparative	- Seeks to compare two or more groups for some outcome variable - Often uses words such as 'compare' - When it involves two groups it can be written as: 'What is the difference in (dependent variable) between (group 1) and (group 2)?' - Can be extended to three or more groups by replacing the word 'between' with 'among' - Can be causal in nature, such as the effect of X on Y - Such causal questions are implicitly comparative in nature when a comparison is made between a group where X is involved and a group where X is not involved
Relationship	- Is concerned with trends between or among two or more variables - Often uses words such as 'relate', 'relationship', 'association' and 'trend' - Relationship questions involving two variables usually can be written using: 'What is the relationship between (independent variable) and (dependent variable) among (population)?' - Can be extended to three or more variables by replacing the word 'between' with 'among'

(Onwuegbuzie and Leech 2006)

Researchers should:

- Develop a research problem that matches your interests, background, training and publications.
- Base research on current evidence.
- The research question should logically present each step in a progression from what is known to important issues that are not.
- Do not try to be 'research trendy'. Do not follow a trend or issue with little evidence from the literature or little idea of how it will add to the knowledge base.
- Avoid topics. Look at clearly defined research problems instead.
- Stay focused. You will find many things of interest along the path, but you should ask yourself: 'Is this related to what I want to uncover or just a point of interest?'

Research questions

Research questions can be developed from theoretical knowledge, previous research or experience, or a practical need at work (Parahoo 2014). They have numerous roles, such as identifying the importance of the research, providing a framework, direction and cohesion,

and setting the limits by revealing the boundaries of the study and highlighting the type of data to be collected (White 2009).

Quantitative research questions These tend to be precise and can be categorised as 'descriptive', 'comparative' or 'relationship' (Table 1). Words such as 'do', 'does', 'is' or 'are' should be avoided as they invite 'yes'/'no' responses.

Appropriate questions should highlight the population, dependent variables and design (historical, descriptive, correlational, experimental, causal-comparative, quasi-experimental), and the link between the research question and the design (Onwuegbuzie and Leech 2006, Kloda and Bartlett 2013). In comparative/relationship research questions, independent variables should also be highlighted.

Qualitative research questions are flexible, adaptable and non-directional (Creswell 2013). They seek to determine or discover a process, or define experiences. They are normally used to try to understand particular educational, familial or social processes or experiences that happen in a specific location and/or context (Marshall and Rossman 2011). They usually describe and address 'what' and 'how' questions, avoiding words such as 'affect', 'influence', 'compare' and 'relate'.

Qualitative research questions These can represent broad or central areas of research or specific areas or sub-categories (Creswell 2013). Sub-questions can address important concerns and difficulties to be resolved or describe the context. Like quantitative questions, qualitative questions link to research design, including historical/narrative, case study, ethnography, phenomenology, grounded theory and autoethnography.

Generally, these questions are non-directional and use words that state that the study will: 'discover' (grounded theory), 'explain' or 'seek to understand' (ethnography), 'explore a process' (case study) or 'describe the experiences' (phenomenology).

Researchers can often find developing qualitative questions difficult (Agee 2009). To help researchers think about the different types and purposes of their studies, constructs for question types are presented in Table 2. Comparative questions are often overlooked in qualitative research. Onwuegbuzie and Leech (2007) described four measures that compare:

1. Participants pairwise (pairwise sampling designs).
2. Two or more subgroups (subgroup sampling designs).

3. Two or more members of the same subgroup, wherein one or more members represents a sub-sample of the full sample (nested sampling designs).
4. Two or more subgroups that are extracted from different levels of a study (multilevel sampling designs).

Mixed-methods studies Researchers conducting these studies can choose to develop separate quantitative questions and qualitative questions or develop a mixed methods question depending on the study (Creswell 2009). Separate quantitative and qualitative questions are appropriate when the approaches, not the mixed methods or integrative component of the study, are the focus (Tashakkori and Teddlie 2010). This identifies the significance of the qualitative and quantitative phases of the study and their collective power (Creswell and Plano Clarke 2011).

The development of a mixed methods question should reflect the procedures or the content and not include separate quantitative and qualitative questions (Creswell 2009, Tashakkori and Teddlie 2010). This suggests that there is some integration or linking between the study's quantitative and qualitative phases (Tashakkori and Teddlie 2010, Creswell and Plano Clark 2011).

Overall, good research questions are the basis of good research as they enable researchers to identify what they want to know and to work from knowing less towards knowing more (Payne and Payne 2004). Identifying what you want to know more about is essential. Researchers can be vague at the beginning of a study about what they want to know and unclear questions can lead to an unfocused project (White 2009). Researchers should aim to write clear, articulated questions that enable them to find the answers in a focused, clear way.

Criteria One difficulty in writing good research questions is that there are potentially an unlimited number of them. Without a clear, focused research question, it is difficult to know how or what to research (Grove *et al* 2013). While researchers may appreciate that if they do not know what questions they are asking, there is little hope of finding any answers, this may only add stress to an already tense situation.

Common problems encountered when developing a research question include: deciding which area to research from a range of issues that are of interest; knowing which area to focus on with no precise topic; and knowing the area and topic but finding it problematic to clearly communicate the question.

The authors suggest that once the research topic has been identified, the researcher should begin framing the research question by listing all the questions that could be researched. This may be a large or small list with questions interconnected or very different. Forming the right questions should be seen as an iterative process that occurs through continually engaging in reading and refining ones ideas at all stages (Maxwell 2013).

Hulley *et al*'s (2007) criteria for quantitative research are:

- Feasible.
- Interesting.
- Novel .
- Ethical.
- Relevant.

These criteria are collectively known as FINER.

Cooke *et al*'s (2012) characteristics of qualitative research are:

- Sample
- Phenomena of interest
- Design
- Evaluation
- Research type

These characteristics are collectively known as SPIDER. Researchers informed by these criteria and characteristics are likely to identify important aspects of research questions and develop successful research projects (Farrugia *et al* 2010). However, while researchers can refer to these criteria in

Table 2 Types of qualitative research question

Type	What it does
Contextual	Describes the form or nature of what exists
Explanatory	Examines the reasons for or associations between what exists
Evaluative	Appraises the effectiveness of what exists
Generative	Provides new ideas, aiding the development of theories, strategies or actions
Ideological	Advances the ideology of a position
Exploratory	Investigates a phenomenon where little is understood
Explanatory	Explains a phenomenon
Descriptive	Tries to describe a phenomenon
Emancipatory	Engages in social action related to a phenomenon

(Ritchie *et al* 2014, Marshall and Rossman 2011)

identifying important general aspects of a research question, a more specific format should be considered when developing specific research questions (Farrugia *et al* 2010).

Richardson *et al*'s (1995) format for quantitative research is:

- Population, patient, problem.
- Intervention.
- Comparison.
- Outcome.

This format is known as PICO.

Kahn *et al*'s (2003) format for qualitative research is:

- Population, problem.
- Exposure.
- Outcome or theme.

This format is known as PEO.

These methods have been used in evidence-based practice to frame and answer clinical questions (Huang *et al* 2006), develop strategies for searching literature (Schardt *et al* 2007), and frame research questions (Aslam and Emmanuel 2010). Once a research question has been identified, the research aim and objectives are developed. Developing a research question can be difficult, as can stating the research aim and objectives (Blaikie 2009).

Aims The researcher must be cognisant that the research's question, aim and objectives are inextricably linked. To determine the study's aim, the researcher must decide what its overall,

long-term aim is and describe the overall purpose (what should be accomplished) in general terms. This ultimately enables the reader to judge whether the researcher has achieved that purpose (Tully 2014). In this context, the Newman *et al* (2003) framework can be used to identify nine types of aim. These are to:

- Predict.
- Add to the knowledge base.
- Have a personal, social, institutional, and/or organisational impact.
- Measure change.
- Understand complex phenomena.
- Test new ideas.
- Generate new ideas.
- Inform constituencies.
- Examine the past.

Objectives Determining the research's aim leads naturally to determining its objectives. Research objectives are more specific than the aim and relate directly to the research question (Grove *et al* 2014, Parahoo 2014). They may be divided into 'primary' (bound to be achieved) and 'secondary' (incidental) objectives (Newell and Burnard 2011), and should be: closely related to the research question, cover all aspects of the problem, specific, ordered in a logical sequence, achievable, take into consideration the available resources, including time, and mutually exclusive of each other. They should also be stated using action verbs that can be evaluated,

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such as 'to describe', 'to identify', 'to measure' or 'to compare'.

Johnson and Christensen (2014) offer five typical research objectives: 'exploration', 'description', 'explanation', 'prediction' and 'influence'. Qualitative and quantitative research studies can be linked to one or more of these five research objectives (Onwuegbuzie and Leech 2006):

- 'Exploration' involves using mainly inductive methods to discover a concept, construct, phenomenon or situation and advance understanding, hypotheses or generalisations.
 - 'Description' involves identifying and describing the antecedents, nature and aetiology of a phenomenon.
 - 'Explanation' involves developing theory for the purpose of explaining the relationships among concepts or phenomena and determining reasons for the existence of events.
 - 'Prediction' refers to using pre-existing knowledge or theory to predict what will occur at a later point in time.
 - 'Influence' relates to manipulation of the setting or variable to produce an anticipated outcome.
- Doran's (1981) criteria can be applied when setting research objectives. These are:
- Specific: be exact about what you are going to accomplish.
 - Measurable: quantify the objectives.
 - Appropriate: align with the needs of the target audience.

- Realistic: do you have the resources to make the objective happen?
- Time specific: state when you will achieve the objective.

These criteria are collectively known as SMART.

It should be acknowledged that there is some debate as to whether research aims and objectives are required for all studies, and whether either or both are necessary. This is evident in published studies where only the research aim or the objectives are identified. This may be because only one was set or only one was identified in the published work.

Conclusion

Developing a good research question, aim and objective is a vital aspect of research. This can be challenging. It is essential that you take the time to do this systematically as the eventual success of the study relies on asking a suitable and clear question. The question must be grounded in research, coherent, clear and ask precisely what you want to find out. Knowing this will help you plan the project and identify the methodology, sample, data collection and data analysis necessary.

Developing the research question, aim and objective is an interactive and inductive process that takes place over time, through consultations with a supervisor, knowledgeable people, published literature, and use of the appropriate criteria (FINER and SPIDER) and format (PICO and PEO).

Online archive

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Conflict of interest
None declared

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Difficulties Faced by Jordanian Undergraduate EFL Students in Writing Research Problem Statement

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Abstract: The current study aims at investigating the difficulties encountering Jordanian EFL students in the writing problem statement section. The study grouped those difficulties into two main categories, namely academic skills difficulties and language skills difficulties. It has been noticed that undergraduate as well as postgraduate students, lack the required skills enabling them to conduct a well-constructed research article, which sparkles the idea of this study. In addition, no previous study has examined the challenges encountered by Jordanian EFL students when conducting a research paper in general and the problem statement section in particular. For the purpose of collecting the required data of the present study, twenty Jordanian EFL students had a teaching program about research writing skills. They were introduced thoroughly to all sections of the research article, with special emphasis on the research problem, section which is the main concern of this study following that the participants were given a month period to accomplish their research papers before submission. The results of the study reveal that the participants lack the academic skills enabling them to conduct a well-constructed problem statement section. More specifically, the vast majority of the participants fail to provide an overview of their topics and to identify their research gap within the current literature. The study also shows that the participants face serious grammatical errors when writing research problem section. It could be concluded that conducting a scholarly research article in general and problem statement, in particular, is a demanding issue and need to be further highlighted by researchers and academicians to identify the challenges facing learners.

Keywords: *EFL students, problem statement, research skills, language skills*

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INTRODUCTION

Research plays a substantial role in human's life as all advancements made by man during the past period of time are built primarily on research, whether theoretical or empirical. However, the process of conducting a scholarly research article is always challenging and demanding for postgraduate students and even for their professors because it requires implementing several processes simultaneously such as, preparing research instruments, finding the appropriate sample, data collection, data analysis and finally, the actual process of writing (Cuevas et al., 2016; Klein & Boscolo, 2016). In addition to referring to suitable resources, identifying a research problem, finding the research gap and fitting it into literature, analyzing the results of the study, explaining and justifying the research results. Implementing a research paper is the toughest experience the students may have to get a university degree as it consumes lots of time and requires great effort. This is in line with Alsied and Ibrahim (2017) who state that conducting a research is not a simple task because it needs hard work, insight into the subject under investigation, desire to carry out research and motivation.

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The difficulties encountering researchers in conducting a scholarly research paper are explained by [Phothongsunan \(2016\)](#), who subcategorized them into three main types, namely 1- discursive difficulties, 2- non-discursive difficulties and 3- other difficulties. Discursive difficulties are those that have to do with the language, including a poor command of the target language, misuse of vocabulary, violating rules of grammar, bad referencing and citation. This problem is reported by [Okamura \(2006\)](#) and [Fahy \(2008\)](#), who contend that language challenges in research writing is manifested by poor vocabulary, mother tongue interference, the problem of writing coherent paragraph and establishing well-constructed introduction, body and conclusion. According to [Al Fadda \(2012\)](#), the vast majority of challenges that non-native speakers of English encounter, especially Chinese, when writing in English are procedural challenges related to rhetorical patterning and genres, textual organization and writing the literature review.

Non-discursive difficulties refer to elusive factors comprising plagiarism, motivation, emotional and psychological factors. [Canagarajah \(2002\)](#) and [Salager \(2008\)](#) report some more non-discursive factors such as lack of resources, tight budget, weak internet connection, undependable mail service and lack of support from colleagues and supervisors. This is consistent with [Alsied and Ibrahim \(2017\)](#), who report that non-discursive challenges for Libyan researchers encompass lack of resources in the library, lack of training in research, the need for support from supervisors, and poor internet connection. Other obstacles of research writing are related to publication and the peer-review process. Despite the aforementioned challenges, carrying out a research article is indispensable and considered a privilege for the author ([Lonka et al., 2014](#)). In addition, implementing research papers open the door for the exchange of ideas and sharing the results and conclusion of research based on foundational knowledge in specific field.

Many attempts have been made to reach a working definition of research. According to [Pandey and Mishra \(2015\)](#), research is a two-word combination of "re" and "search" which indicates searching more and more. [Krause \(2020\)](#) provides a working definition of research writing as the process of utilizing evidence from journals, books, magazines, the internet and specialists to convince the readers about a specific subject. Research is also defined as "a process of enquiry and investigation; it is systematic, methodical and ethical; research can help solve practical problems and increase knowledge." ([Neville, 2007](#)). Research is reported as a manner of thinking and exploring several fields of knowledge in order to understand the specific issue and develop and test theories with the aim of making progress and advancement in the profession ([Kumar, 2016](#)). As stated by [Bocar \(2013\)](#), research is the process of data collection, analysis and interpretation and testing the applicability of the findings.

The process of writing a research paper is supposed to be systematic, well-structured and conform with the steps of research writing. According to Hourglass Model, as cited by [Swales \(1993\)](#), a research article is comprised of several components, including the title, abstract, introduction, body, discussion and references. Each section of the research paper is further divided into sub-sections. The introduction starts out with a broad discussion of the topic and narrows down to the specific topic of research. The introduction is composed of three sections.

1. Establish a territory: This section aims at providing the reader with a general background of the topic under investigation and view current studies on the topic.
2. Establish a niche: This section aims at showing the research problem and fitting the research gap within the current literature. The section also reveals research questions and assumptions.
3. Occupy the niche: This section presents the thesis statement of the research and outlines the structure of the paper ([Swales, 1993](#)).

The body section of a research paper describes the methods that have been utilized to answer the research questions and specify the problem mentioned in the introduction. This section includes parts such as the population of the study, the sample of the study and research instruments. The body helps answer questions such as how was the research questions addressed? And what was found ([Peat, Elliott, Baur, & Keena, 2002](#))? On the other hand, the discussion section generally presents background information about the topic, recaps the results of the study with special emphasis on explaining and justifying the results. This section also compares the results of the current study with previous literature and ends with a conclusion drawn from the results of the study.

In writing research, one of the most important parts that constitute a great troublesome for EFL in general is defining the research problem. More specifically, some researchers find it difficult to determine the exact problem to be addressed in the research, making him/her fail to fulfil the objectives of the study. [Ifeoma \(2019\)](#) argues that both undergraduate and postgraduate students encounter difficulty in identifying the problem of research. The problem

of the research is defined as the subject that the researcher intends to examine and explore, whether descriptively or experimentally. It is the focal and core point of the research, or in other words, it is the issue or the challenge that the researcher struggles with within the research article (Boudah, 2011).

It is crucial that the researcher starts out with his personal experience when determining the research problem to be addressed. Strictly speaking, the researcher is supposed to research topics that he has a genuine interest in because if he is not interested, it will be hard to complete the research. Another important point that needs to be considered when writing a research problem is narrowing down the emphasis of the research. A broad problem is featured as unclear and vague so that narrowing down the problem help identifying and addressing that problem and, in turn, designing good research questions (Boudah, 2011). When writing a research problem, three main elements need to be taken into consideration: First, the general overview of the focus topic to be investigated in the study. Second, explanation and justification of the problem: presenting logical reasons for the significance of addressing the problem, which can be attained through providing evidence from literature, recommending further research to be implemented or drawn from the researcher personal experience. Third, the research gap shows how the current study is different from previous studies within the research field. The research gap reveals that the problem has not been examined by past literature, and the current study attempts to bridge that gap. As it has been mentioned previously that a research problem illustrates the main topic or phenomenon to be explored in the study. A working definition of the research problem is advanced by Bryman (2007), who states that it is “a statement about an area of concern, a condition to be improved upon, a difficulty to be eliminated, or a troubling question that exists in theory or in practice that points to the need for meaningful understanding and deliberate investigation”. It is of paramount importance for the research problem to be direct, specific and clear and well-defined, otherwise the researcher will find himself lost, addressing a problem that does not exist. Problem statement is usually located in the end of introduction section, literature review or as a separate section following the introduction.

It has been noticed that researchers, postgraduate and undergraduate students find difficulty defining their research problem, in that the problem either too broadly or too narrowly defined (Ifeoma, 2019). This is consistent with Kombo and Tromp (2011), who argue that one of the basic challenges facing researchers is stating the problem unclearly, making them unable to choose the right methodology and fail to fulfil the objectives of the study. One more challenge with the research problem is that researchers struggle with determining the research gap and locating it into the existing literature.

Within the context of the present study, a considerable body of research has examined the challenges that confront researchers when writing scholarly articles (Alsied & Ibrahim, 2017; Cheung, 2013; Ifeoma, 2019; Pardede, 2018; Phothongsunan, 2016; Qasem & Zayid, 2019). The vast majority of these research place emphasis on general challenges and obstacles of research, excluding Pardede (2018), who explored the problem statement section. Lack of studies in this regard indicates that there is an urgent need to highlight the hindrances facing researchers and language learners when implementing a research study, especially the research problem section, which is the main concern of this study. Based on the aforesaid background, the present study aims at investigating the challenges encountering English foreign language learners EFL in the writing "research problem" section. Thus, the present study aims at answering the following questions:

1. Do Jordanian EFL students confront difficulties related to the lack of academic skills in the writing research problem section?
2. Do Jordanian EFL students confront difficulties related to the lack of language skills in the writing research problem section?

LITERATURE REVIEW

A considerable body of research has investigated the challenges of encountering learners as well as researchers in writing research. Those studies come up with a general consensus that there are lots of persistent difficulties pertinent to research writing, in the sense that some of the difficulties related to the researcher himself; lack of knowledge on research writing, while other difficulties related to lack of resources that help conduct research (Al-Khairiy, 2013).

As it has been mentioned previously, previous literature elucidated the sources of difficulties in conducting a research paper, grouped them into two types; first, discursive difficulties which are pertinent to having poor skills of language such as misuse of vocabulary, violating rules of grammar, bad referencing and citation. Second, non-discursive

difficulties, which are related to plagiarism, motivation, emotional and psychological factors. Below are some of the studies that discussed the difficulties encountered by researchers (Negari, 2011).

Cheung (2013) examines the difficulties encountering first-year undergraduates when writing the discussion of the results section. The study also looks into the extent to which the students perceive the difficulties in writing research. The data for the purpose of the study is collected from a 12-week compulsory course entitled ALS 101 Academic Discourse Skills, offered in the first year of the undergraduate program. The researcher interviews eighteen undergraduate students in order to gain insight into the difficulties they experience in writing the discussion section. The results of the study indicate that the English language, including grammar and vocabulary, constitutes a challenge for the participants. The study also found that the students lack the required knowledge for writing the academic discussion section, where some unnecessary details were mentioned, and important findings ignored.

Sharing the same idea of examining lecturers' challenges when writing a research article, Phothongsunan (2016) interviews eighteen academics from two universities in Thailand. The interviews are made primarily to gain information on the issues that face the participants when writing academic papers for publication and how they tackle them. The findings of the study reveal that the participants in the study confront hindrances related to English language use comprising grammar, choice of vocabulary, organization of sentences and tone. The study also reveals that the participants face difficulties related to time constraints and lack of funding.

Komba (2016) explores the obstacles that postgraduate Tanzanian students struggle with when writing theses and dissertations. In addition, the study seeks to find a remedial solution in order to overcome those challenges. For the purpose of the study, the researcher reviews 39 theses and 64 dissertations from three universities in Tanzania. The findings of the study exhibit that half of the postgraduate students, whose theses and dissertations are reviewed, face various difficulties in writing research, including unsuitable presentation of chapters and having poor academic writing skills. The researcher concludes that research method courses taught at universities have to be revised with respect to their content. In line with Claudius, study, Alsied and Ibrahim (2017) examine the problems that Libyan EFL students face in research writing. In addition, the study investigates the attitudes of instructors regarding their students' skills in research. The sample of the study consists of 42 students and 4 instructors at Sebha University. The results of the study display that the participants face difficulties in writing research exhibited in the field of interest, choice of subject, developing a research problem, review of related literature, and reporting the results of the study. The results also reveal that the instructors look at the students' work negatively due to many reasons, including inadequate motivation and library resources, lack of background information and the need to offer more courses related to research writing.

In a newly issued study, Qasem and Zayid (2019) attempt at understanding the problems that encounter students in writing research projects in English. A total of 60 undergraduate students take part in the study at the University of Bisha. With the aim of collecting the needed data, a questionnaire is utilized, and interviews with teachers and students are done. The results of the study indicate that approximately 70% of the participants face English language-related issues. The study uncovers difficulties relevant to determining the research topic, insufficient background information about the topic, poor methodology and lack of references.

The aforementioned studies have clearly indicated various challenges that researchers encounter when carrying out a research study. Those studies reveal that the difficulties are either internal related to the researcher himself, such as lack of academic skills and having a poor command of the English language or external factors resulting from the unavailability of resources needed for research such as inadequacy of funding and resource books and bad internet connection. The vast majority of studies covering the internal challenges of research writing place emphases on obstacles in writing all sections of the research articles, while an important part, which is the problem statement section, is either totally neglected or swiftly covered. The research problem is the chief topic that a researcher seeks to investigate or the key issue a researcher is concerned to tackle. Thus, this study is implemented to bridge a gap in the literature by highlighting the problem statement section in order to understand the internal difficulties related to academic skills and language use.

METHODOLOGY

This section presents the methods used in collecting the required data to provide satisfactory answers to the research questions. The section includes two sub-sections, namely the sample of the study and instruments of the study.

Design of the Study

Research design is known as “how the researchers plan their studies so that they can obtain answers to the stated research questions using systematic approaches” (Rezai, 2016). The present study adopts a qualitative method for the purpose of answering the research questions, which is best exhibited by analyzing some research papers written by the participants in the study, with special emphasis on the problem statement section.

Sample of the Study

The sample of the study is comprised of twenty Jordanian EFL students pursuing their bachelor degree in English language and translation. The participants were third and fourth-year students enrolled in a research writing course in the summer semester in the academic year 2019/2020. All the participants are non-native speakers of English, and Arabic is their native language so that they are expected to have similar exposure to English, and the variation between them is not sharp.

Instruments of the Study

In collecting the data required for the present study, the participants are instructed to write a research paper about any topic related to language teaching, language learning or translation. Firstly, the participants subjected to a teaching program extended for two months, in that they were introduced to all details relevant to research writing. Strictly speaking, the students are presented with the steps of writing all research sections including title, abstract, introduction, problem statement, research questions, literature review, data collection, data analysis, results and conclusions. The process of writing each section was explained in details in order to get insight into writing a scholarly paper. Secondly, having introduced a particular section of research, the students were instructed to write a sample section within a specified period of time in order to verify the progress they have made. More specifically, when introducing, for instance, the research questions section, all details about this section are presented, such as the characteristics of a good research question and a bad research questions with illustrative examples and then the students were asked to apply what they have learned. Thirdly, after the end of the teaching program, the students were asked to choose a topic related to their major and start writing a research paper. They were given a period of no less than a month to write the paper. The vast majority of the students submitted their research papers on time, and then the papers were read and analyzed carefully by the researcher in order to explore their progress in research writing, especially the research problem section.

RESULTS

This section seeks to reveal the results of the study with reference to the two research questions that the study raised. To begin with, the section reveals the results pertinent to research question one, then displays the results of research question two. The question is formulated as:

Q1- Do Jordanian EFL students confront difficulties related to the lack of academic skills in the writing research problem section?

To answer this research question, the responses of students regarding writing a problem statement section is divided into three categories, according to the three elements of the research problem.

Table 1 *Academic Skills Pertinent to the Research Problem Section*

	Overview of the Problem	Justification of the Problem	Research Gap
Percentage of response	35%	60%	5%
Frequency of response	7	12	1

As it has been mentioned previously that the section of the research problem is comprised of three main components so that when a researcher misses one of those components, the research problem will not be identified, determined and addressed well, which results in making the problem too broad, failure to justify the importance of the problem and inability to locate the study within the current literature. Strictly speaking, the lack of academic skills needed to write the research problem leads to serious deficiencies in the research study as a whole and in turn, keeps the

research questions partially or totally unanswered. Table 1 above includes the responses of the participants with respect to providing the three main components of a research problem when writing the research problem section, namely, Overview of the problem, Justification of the problem and Research gap. The results indicate that 35% of the participants overviewed and provided background information of the topic to be investigated, with a frequency of 7 participants out of 20. Table 1 also shows that 60% of the participants in the study provided satisfactory justifications for the topic under investigation, with a frequency of 12 out of 20. In addition, Table 1 displays that only 5% of the participants attempted to locate their study within the current studies or to mention the research gap that their study is supposed to bridge. To be more specific, only 1 participant out of 20 tried to fit his study within the present literature. To gain further insight into the results of question one, it is important to consider pie Figure 1 below.

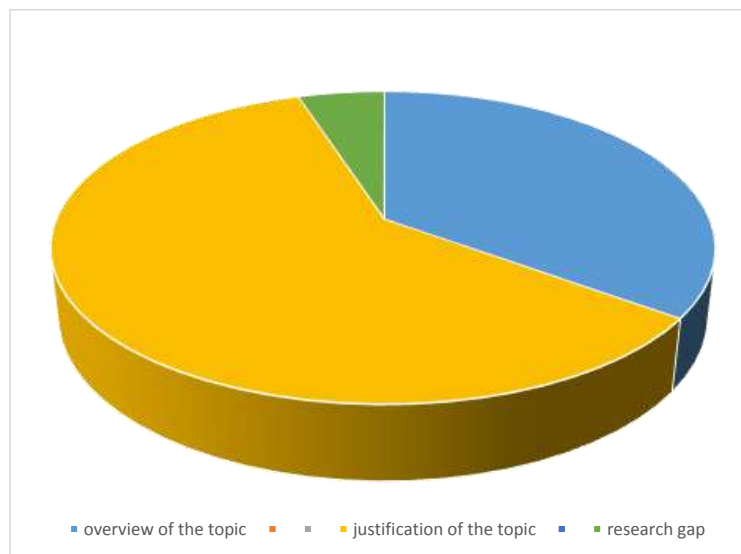


Figure 1 *The Proportion of Academic Skills Pertinent to the Research Problem Section*

It is obvious from Figure 1 that the biggest proportion of the chart is devoted to justification of the topic component, followed by an overview of the topic component, while the smallest proportion of the chart is for the research gap component. Those results indicate that the participants confront serious problems in identifying the research gap when writing the research problem section. The difficulties seem still existing when writing an overview of the topic, while little obstacles encountered by the participants in writing justification of the problem section. To sum up, it is evident that Jordanian EFL students lack the required knowledge of identifying the research problem in writing a research paper. The results of the research question two are depicted in Table 2 below. Question two is formulated as follows; Q2: Do Jordanian EFL students confront difficulties related to the lack of language skills in the writing research problem section?

To answer this research question, the errors related to language use committed by the participants are divided into two categories, namely grammatical errors and semantic errors. Grammatical errors include error relevant to the use of tense, order of words, agreement and determiners system. Semantic errors comprise errors pertinent to the choice and meaning of words and bigger units like phrases and clauses. Table 2 below depicts the results of the research question two.

Table 2 *Grammatical and Semantic Errors Pertinent to the Research Problem Section*

	Percentage of Errors	Frequency of Errors
Grammatical errors	94%	74
Semantic errors	6%	6

A close look at Table 2 above indicates that grammatical errors account for 94% of the total number of errors committed by the participants in the writing research problem section. The results also show that the frequency of grammatical errors is 74 errors. It is also apparent from Table 2 that semantic errors account only for 6% of total errors with a frequency of 6 errors. For an in-depth analysis of participant's errors related to language use, it is fitting to consider pie Figure 2 below.

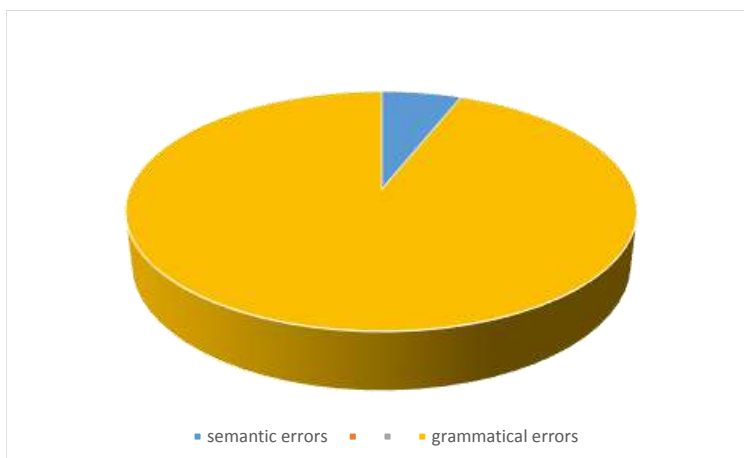


Figure 2 The Proportion of Grammatical and Semantic Errors Pertinent to the Research Problem Section

It is evident from Figure 2 that the biggest portion of errors related to language use in the writing research problem section is ascribed to grammatical errors. In contrast, a small portion of errors can be attributed to semantic errors. Those results clearly denote that the difficulties that the participants confront are grammar-related rather than semantic-related difficulties.

DISCUSSION OF RESULTS

The results of the study clearly indicate that Jordanian EFL students encounter serious difficulties in conducting the research problem section. Strictly speaking, only 35% of the participants in the study managed to present an overview or background information of the topic under investigation. 60% of the participants justified their research problem clearly and correctly, while only 5% of the participants were able to determine their research gap. This result is consistent with the results of some previous studies, indicating that EFL students, in general, face serious challenges in research writing (Cheung, 2013; Komba, 2016; Phothongsunan, 2016; Qasem & Zayid, 2019). The results also depict that the great majority of language errors made by the participants are grammatical in nature, while very scant errors are semantic. This result goes with the findings of some studies, which reveal that English is a challenge for the participants when producing any piece of writing (Almahameed, 2017; Qasem & Zayid, 2019). To gain a thorough insight into the nature of difficulties and errors made by Jordanian EFL researchers in the writing research problem section, it is of paramount importance to analyze and discuss those difficulties with illustrative examples.

Difficulties Encountering Jordanian EFL students in Writing Research Problem Section

Having analyzed the research written by the participants, difficulties encountered by them can be grouped into two main categories.

Academic skills difficulties: The following is a sample of the problem statement section written by one of the participants in the study.

“Statement of the problem: In Jordan learning English is still challenging (Abdul Haq, 1982). Wahba, (1998) stated that problems of learning English in Jordanian educational courses (Zughoul and Taminian, 1984). Mostly in making errors in writing, lexis, syntax, and pronunciation. EFL students have many problems in all the language skills such as speaking, reading, listening, and writing. Rababah, (2003) stated that there have been a lot of weakness in English among school students who join the universities as English language majors. In fact, a lot of first-year university courses spent on English grammar vocabulary, oral skill, language use, and writing. Students are accepted in university programs without taking any international standardized tests such as TOEFL, IELTS but they had to take locally designed tests of English which lack of reliability.”

A close look at the research problem above indicates that the writer clearly and abundantly justifies this research problem in that Jordanian EFL students face serious challenges in learning English as a second language. The researcher supports his justification and explanation of the problem with the results of previous research. However, it is evident that the researcher did not provide a general overview of the nature of the topic to be investigated or the challenges confronted by learners. In addition, the researcher has not stated the position of his study within the current literature. More specifically, the researcher did not indicate whether his study contributes actively to the field of investigation, and nothing is said about the gap that the study is supposed to bridge in literature. One more sample of the research problem section written by the participants is below.

“Statement of the problem: Translating the news generally from English into Arabic can be a problematic matter for Arab translators due to the cultural differences between the two languages, Arabic and English, also Psychological problems and lack of self-confidence have a major role in the success or failure of the translation process. Simultaneous translation difficulties from English into Arabic representing by many aspects, including listening, understanding, analysis and reporting the message at a very specific time. Also, the interpreter faces the listener who receives the message from him.”

Analyzing the problem statement section above shows that the researcher describes the challenges that translators face when translating news, and he elaborates on the reasons beyond such difficulty. However, in-depth analysis and overview of the problem is needed, in the sense that the researcher is required to state more details with examples about the nature of problems in translation in order to enhance the understanding of the reader. The researcher is also needed to cite references that support his viewpoint. Furthermore, nothing is mentioned about the gap that the researcher's study attempts to bridge or to fit the study within the present literature.

Language use difficulties: Those difficulties can be either grammatical or semantic. Below are a few examples on the grammatical errors committed by the participants in the writing research problem section.

“Tense study include studying the structure, parts of speech also ”

Two errors are committed in the above sentence. The first one is a subject-verb agreement, in which no final "s" is added to the verb following the singular subject. In addition, the conjunction "and" is missing and should be added to connect the parts of the sentence above. The following is another example of grammatical errors taken from student's research.

“Grammar necessary in order to understand languages linguistic system”

In the above sentence, the verb is missing in that the verb to be "is" is required to make the sentence grammatical. The above-mentioned sentences are examples of grammatical difficulties that the participants encounter in the writing research problem section. That is to say, writing a problem statement section is demanding and needs professional skills to be accomplished well.

LIMITATIONS AND RECOMMENDATIONS

The sample of the current study is limited to Jordanian EFL students at Amman Arab University. The study is also limited to investigating the problem statement section, while other research sections beyond the scope of this study. It is recommended that future studies expand the scope of the investigation to other sections of a scholarly research paper, such as the literature review section and the results section. It is also recommended that future studies broaden the sample of the study to include postgraduate students.

CONCLUSION

The present study examined the challenges that Jordanian EFL students confront when implementing the research problem section. A total of twenty English majored students participated in the study. They were instructed to conduct a research article on a language-related topic after being presented to the main components and sections of the research article. Having analyzed the research papers written by the participants, it was found that the participants were unable to overview their research problem and to determine their research gap correctly. In addition, the study revealed that the participants committed a considerable amount of grammatical errors. It is recommended that more emphasis and teaching programs are allocated for teaching research writing skills in general and the problem statement section in particular.

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A FOUR STAGE FRAMEWORK FOR THE DEVELOPMENT OF A RESEARCH PROBLEM STATEMENT IN DOCTORAL DISSERTATIONS

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ABSTRACT

Aim/Purpose	Provide methodology suggesting steps to doctoral mentors to work with students in constructing their research problem statement in their dissertation.
Background	Writing a doctoral dissertation is a long journey, and it typically starts with writing the research problem statement. Students face challenges in articulating the research problem statement. Clearly articulating the research problems statement influences the success of the entire dissertation.
Methodology	This paper uses a widely used framework to describe student adjustment to graduate studies in general and to doctoral programs in particular.
Contribution	This study provides a framework for mentors and advisors to assist them in guiding students in writing their research problem statement.
Findings	Writing a research problem statement is difficult by itself. Following the methodological approach suggested in this study will help students with the task of writing their own.
Recommendations for Practitioners	A methodological approach to writing a research problem statement is helpful in mitigating the difficulties of writing the dissertation. This study tackles the difficulties with writing the research problem statement.
Recommendations for Researchers	More research needs to be done to expand the use of a methodological approach to writing in other sections of the dissertation.
Impact on Society	The findings of this research will help doctoral mentors/advisors as they guide students in completing the writing of their research problem statement

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Future Research	Future research should follow a similar methodological approach in guiding students in writing the other sections of the dissertation
Keywords	research problem statement, research dissertation, problem statement

INTRODUCTION

The dissertation is a labor of love requiring much work, sweat, and tears, as well as organization skills and extensive resources from others who are involved with the process. The final product is a document that one can recognize as a once-in-a-life-time achievement. We liken this experience to the task of building your own home. (Grant & Osanloo, 2016, p. 12).

Working on dissertations for doctoral programs is often lengthy, intricate, and subject to various factors that could influence the result of the dissertation (Black, 2017; Brill et al., 2014). Like the saying “The journey of a thousand miles starts with a first step,” the journey for completing doctoral dissertations often starts with defining the research problem. The problem statement becomes the centerpiece that other sections in the dissertation link to as the thesis builds (Burkard, 2014). Writing a problem statement is a complex step that involves multiple decisions that are influenced by numerous factors. However, once established, a good problem statement lays the foundation and paves the way to a complete dissertation.

Writing a doctoral research problem statement has been the subject of numerous studies (Blum & Preiss, 2005, Creswell, 2005, Ellis & Levy, 2009, Faryadi, 2018, Jacobs, 2013, Kerlinger & Lee, 2000). The point of emphasis from these studies is that writing a research problem statement is difficult and it is subject to the multiple factors that often influence its creation (Jacobs, 2013).

The complexity of writing a problem statement is increased further by the fact that the task of writing it (the research problem statement) is usually tackled at the beginning of the journey – a time when most students are struggling to adjust to the new environment of graduate studies or to the phase of dissertation writing (Gardner, 2009). A multi-faceted difficulty will be facing the students as they embark on writing (constructing, developing, or forming) their problem statements (Black, 2017). However, students are expected to write the problem statement at a time when they are still unable to fully comprehend the journey ahead.

In this paper, steps are suggested for doctoral faculty to help students construct their research problem statement. It takes into consideration the phases of adjusting to the new environment of graduate studies (or writing the dissertation). The remainder of this study is divided into the following sections:

- First, the theoretical framework is introduced. It forms the basis of this study and on which the recommendations will be based.
- Second, forming the research problem statement is explained, such as what is it and how it is typically written.
- In the third section, factors that make writing problem statements challenging are described.
- Factors that help with writing the research problem statement are elaborated on next.
- Next, suggestions are discussed on how to guide students into writing a problem statement.
- The last section provides the recommendations of this study.

THEORETICAL FRAMEWORK – THE BEELER’S MODEL

A theoretical foundation called “Beeler’s Framework” (Beeler, 1991) is used in this paper to suggest steps for academicians to mentor students through the stages of writing the problem statement. Beeler’s framework has been widely used among researchers in describing the adjustment of students to graduate studies. For example, Ali and Kohun (2007) used Beeler’s work to suggest a framework

for dealing with the feeling of social isolation in doctoral programs. Bloomberg and Volpe (2018) used it to create a road map for students and to provide suggestions for the completion of their doctoral dissertations. Lewis et al. (2003) used the model to explain the experience of African-American Ph.D. students at a predominantly white research institution.

Beeler's framework is about graduate students and their mental adjustment (academically and socially) and awareness as they embark on graduate studies. Beeler's framework suggests that students go through four phases in their mental adjustment to the requirements and the work in graduate studies:

- Phase 1: Unconscious incompetence
- Phase 2: Conscious incompetence
- Phase 3: Unconscious competence
- Phase 4: Conscious competence

The first stage of graduate studies starts upon entering the program. Many students have only a piece of limited knowledge about what is involved in the program, either academically or socially (Gardner, 2009); essentially, they do not know what they do not know. During this stage, students may question their decision to pursue the graduate degree, and doubts may linger about their performance. The second stage of adjustment takes place after the students gain knowledge about the academic requirements and what is ahead of them. Students at this stage become aware of their deficits when compared to the requirements of the program. Thus, their lack of knowledge becomes clear. By the third stage, students have gained some competence in their field of study but are largely unaware of their competence; thus, competency awareness takes place unconsciously. The fourth and last stage is the result of accumulating sufficient knowledge, and students become aware that they 'know what they know'. This conscious awareness of competence is cultivated by completing the degree requirements that they attempted. Figure 1 depicts the framework introduced by Beeler.



Figure1. Beeler's framework for adjustment to graduate studies (Beeler, 1991)

WHAT IS A RESEARCH PROBLEM STATEMENT?

At first glance, the question: "What is a Research Problem Statement?" may sound like the "duh" game that is played among some children when asked a question that has an obvious answer. But the answer to the question "What is a Research Problem Statement?" is not an obvious one, and this question may need to be answered before writing further about the topic. What is meant by a research problem statement will be clarified in this section.

First, what is a problem statement is presented in brief. Then the cause-and-effect relationship in problem statement is clarified. This step is necessary to complete before tackling what is meant by the research problem statement. Lastly, the research problem statement and how it is different to other problem statements are explained.

A PROBLEM STATEMENT

An old expression goes, "If it looks like a duck, swims like a duck, and quacks like a duck, then it probably is a duck." The same can be applied to understanding problem statements. A problem statement needs to sound like a problem when reading it or when talking about it. So, from the writing perspective, a problem statement needs to be written such that it sounds like one. If a statement –

which is ideally one sentence but could be more, although typically not more than two or three sentences at the most – sounds like a problem, and if it reads like a problem, then it is fair to say that one has a problem statement.

A reasonable question that could be asked at this point is how to make a statement sounds like a problem statement and what are the specifics in the statement that make it read like a problem. This question leads to the discussion of the “cause and effect relationship”, a concept that is important to shed light on in order to understand the research problem statement.

CAUSE AND EFFECT RELATIONSHIP

As the name indicates, a problem statement is a statement that is composed of a sentence or more, phrased to sound or to read like a researchable problem. Typically, it must have two parts, a cause and an effect. For instance, the statement could describe a loss or a missed opportunity and indicate a cause for the loss and the missed opportunity (Jacobs, 2013). In other words, there is a cause to the problem being stated, and there is an effect because of the cause (Williams, 2020).

A statement about the increase of air pollution in industrial cities, for example, might be a statement, but at that point, it does not have sufficient information to be considered a problem statement because it does not state an effect. However, say it was supplemented by another statement about the increasing cases of asthma in industrial cities that are caused by air pollution. Then, this statement pair presents the reader with a cause “air pollution in industrial cities” and an effect “increasing asthma in industrial cities.” Thus, it is be considered a problem statement: “The increase of air pollution in industrial cities results in increased instances of asthma.”

A relevant point to note here is that the effect could have other causes as well. Generally, however, the focus of a good problem statement is one cause and the relationship between that cause and the effect. Take the example of the air pollution statement again and the increase of asthma among residents of industrial cities. There could be other factors that made the effect (increasing asthma), such as increased pollen in the air that causes asthma. But the focus of the research problem statement is typically on one cause (Faradyi, 2018). In this case, it is air pollution. This relationship is depicted in Figure 2.

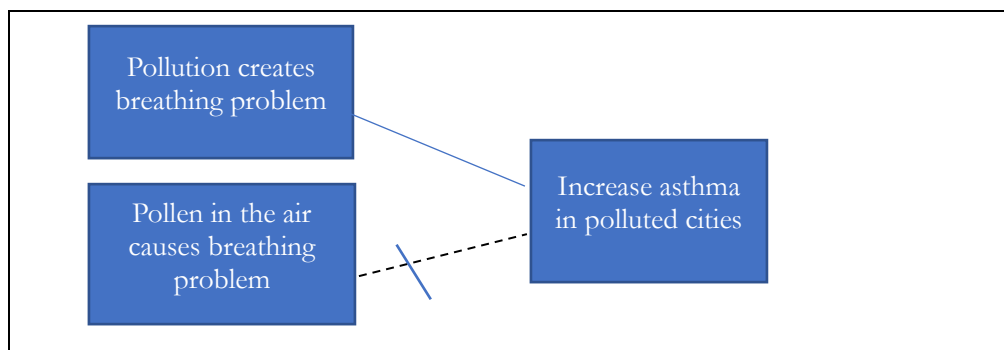


Figure 2. Cause and effect relationship illustrated

A RESEARCH WORTHY PROBLEM STATEMENT

By research worthy, it is meant that the problem being described (stated) must be supported by research data and different reviews. Also, the resolution sought from stating this problem must be achieved through research (Jacobs, 2013). In a research problem statement, there is no room for intuition and opinion. These must be put aside. The problem statement must be supported with research, and the anticipated resolution must be achieved through research as well.

Doctoral students are typically advised to tackle a viable research problem (Ellis & Levy, 2009). To understand what is meant by viable, we searched the dictionary to understand the meaning of the word “viable” and thesaurus to find alternative words that describe it. Webster dictionary defined viable as “capable of working, functioning, or developing adequately” (Merriam-Webster, n.d.). Thesaurus.com (n.d.) suggested the following alternatives: possible, usable, workable, doable.

We determined from this that, for a problem statement to be accepted as a research problem, it must meet two conditions: First, it must be workable or doable within the resources available to the researcher. Second, it must be usable. That is, the anticipated resolution of this problem is helpful, and the community will benefit from the resolutions sought in the research. In both cases, the element of research is present, and it is a must. That is, the problem, as stated, is supported by research, and the resolution to it must be achieved through research (Ellis & Levy, 2012).

CHALLENGES FACING THE WRITING OF PROBLEM STATEMENTS

Research that discusses the difficulty of writing a research problem statement is abundant. Kerlinger and Lee (2000), for example, noted that forming the problem statement in doctoral research is the most difficult and important task in writing a doctoral dissertation. Jacobs (2013) noted that research problems are not lying around waiting to be selected. Instead, the problem needs to be researched and selected based on the interest of the researcher. Jacobs further noted that writing a research problem statement demands skills far behind proficient writing skills.

A reasonable question to ask at this point is, “Why is writing a problem statement in a doctoral dissertation such a challenging task?” This section attempts to answer this question. Research writing and how it is different from other writing mechanisms is first explained. Then issues surrounding buzz words used to describe research problem statements are covered. Lastly, there is a discussion on the various sections in a dissertation that the problem statement is connected to and how this makes it more challenging to write the problem statement.

RESEARCH WRITING

There is consensus among researchers that writing a dissertation is accompanied by stress, anxiety, and anguish. For example, Pappa et al. (2017) discussed the distress that accompanies the writing of dissertations. Russell-Pinson and Harris (2019) wrote about the anguish and anxiety that accompany the writing of dissertations. Stadtlander et al. (2020) noted that writing dissertations (and hence the start by writing a research problem statement) is often accompanied by stress, anxiety, and anguish. The question that could be asked is, “What is in the doctoral dissertation writing that causes this distress, anxiety, and anguish?”

The doctoral dissertation introduces students to a different form of writing, one that is called “research writing” or “research for publication.” Research writing is different from other forms of writing (such as letter writing or journalistic writing) because there are explicit rules and conventions that guide research writing. Moreover, there are stated goals in research writing that the author must focus on. Aside from this, research writing must be based on research, facts, assertion, findings, resolutions, and suggestions, and all of these need to be based on research. The research must follow specific writing standards, such as American Psychological Association (APA) or Modern Language Association (MLA) (Burkard, 2014).

On the topic of writing standards (such as APA and MLA), it is worth noting that following a formal writing standard is helpful in many ways: it adds rigor; it provides standard and consistent ways to present the information; and it assists other researchers to understand the facts and find the resources that led to the facts (Gardner, 2009). For this reason, following a guideline represents the gold standard on which research writing is based, and information is presented. However, this

complicates the writing of the problem statement. Writing a good problem statement while adhering to strict writing requirements can be a challenging task.

THE BUZZ WORDS

In this research, we define buzz words as words that are not clearly defined in ordinary language. At the same time, we use the term “buzz words” to mean words that could give different meanings and that could be used in a variety of ways. The issue of buzz words has been discussed in the literature. Sacco (2014), for example, cautioned against using buzz words in writing because readers could come from different backgrounds. Sacco explained that unless words and phrases are understood in normal conversation across cultures and generations, it could be misleading and, therefore, counterproductive to include them in research writing.

The reason that “buzz words” are discussed in this research is that the literature that describes research problem statements is filled with buzz words that often make the literature difficult to comprehend. Repeated use of buzz words in describing what is meant by a research problem statement and the criteria that govern the articulation of it create added confusion for the students.

To explain this repeated use of buzz words, we searched literature that describes the writing of a research problem statement. In the list below, we present the buzz words we found from the literature review that describes what to include in articulating a research problem statement. We italicize the buzz words to distinguish them.

- Must be phrased so that it could be *explained in a sentence or few sentences* (Blum & Preiss, 2005)
- Must be concise, *well-conceived, and precise* (Jacobs, 2013)
- Research problem statements state a *specific condition* that needs *urgent attention* (Faraydi, 2018)
- Need to be based on *scholarly literature* (Jacobs, 2013)
- Makes a contribution to the *body of knowledge* (Ellis & Levy, 2012)
- It must be between *100 and 250 words in length* (Blum & Preiss, 2005)
- Must be significant enough to warrant dissertation research (Blum & Preiss, 2005)
- Based on literature and must describe a *gap in research* (Jacob, 2013)
- Premeditated investigation using *scientific methodology* (Faraydi, 2018)
- *Originality and significance* are requirements (Lovitts and Wert, 2009)
- “*original piece of empirical research*” (Stadtlander et al., 2020, p. 79)
- A problem statement that is *aligned with their passion* (Blum & Preiss, 2005)

The listing of that many words while explaining the concept could be inundating to the students, especially when words have multiple meanings. This is likely to confuse the students. In addition, the suggested brevity for defining a research problem statement adds to the difficulty. Blum and Preiss (2005), for example, suggested that problem statements need to be succinct. They further suggested that it must be worded so that the researcher can explain it easily and simplistically.

All these conditions lead to confusion. The novice researcher is overloaded with words that are not readily clear to them. At the same time, the research problem statement needs to be short sentences that do not exceed 250 words and could be explained by talking to people in an elevator traveling between two floors (Blum & Preiss, 2005). This confusion is illustrated in Figure 3; the center of the figure is the researcher that is bombarded by all these words.

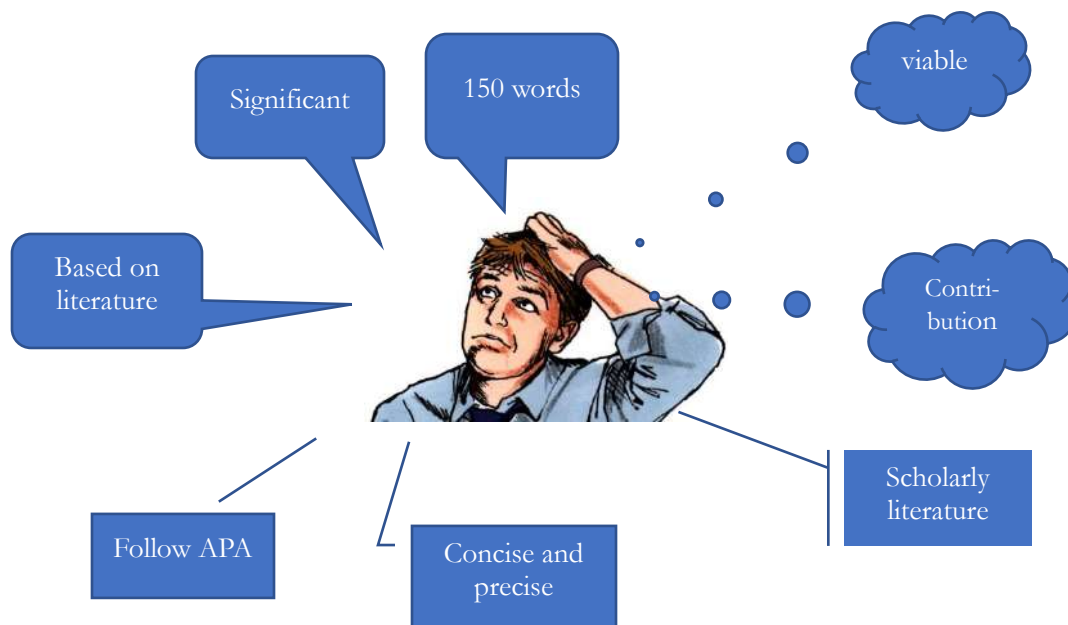


Figure 3. Confusion about buzz words in articulating the research problem statement

THE MULTI INTERLINKS TO OTHER SECTIONS

Jacobs (2013) noted that writing a problem statement represents a critical role for all remaining sections in the research. The critical role is played because other sections in the dissertation depend on defining the problem statement. The issue that makes it hard to write a research problem is its interconnection to several other sections in the dissertation. All sections need to be thought of or considered in advance when writing the problem statement. These sections could include the research purpose, research objectives, a theoretical framework, the method, identification of research, data collection, data analysis, data interpretation, and conclusions, all of which are normal elements of the dissertation process. As the student writes sections of the dissertation, each section touches on the problem statement. Thus, there is a strong interconnection between stating the problem statement and all these interconnected sections and refraining from deviating from it. Figure 4 suggests a drawing of this intricate relation of the problem statement with other sections in the dissertation.

This problem represents what is termed as a “look ahead” or a “peek” to other sections as the research problem statement was written. By look ahead or peek, we mean here that while the students are writing the research problem statement, they need to think about how it will affect other sections in the dissertation. For instance, defining a problem statement such that a student would not have resources for data collection would obviously complicate matters at the time of data collection. Similarly, developing a problem statement that leads to a research question that could not be answered, or worse, cannot be turned into a research question, creating problems later.

It is prudent to think about other sections while writing the problem statement. This undoubtedly adds another layer of difficulty to articulating the problem statement. Defining a research problem statement that is consistent with the other sections in the dissertation increases the challenge. Considering that the problem statement is typically written at the beginning of the dissertation, it means the student has little knowledge of what is ahead, so looking ahead is going to be difficult.

It is also important to note that the novice researcher may unintentionally or unknowingly deviate from the proposed research problem and discuss areas unrelated to the proposed research. A strong research problem statement is intended to not only ensure that the student touches on the statement

as all elements of the dissertation are written, but simultaneously refrains from meandering into elements not relevant to the research problem. Typically, the onus is on the students to maintain this focus as they write their research proposal and later, the thesis. However, the advisor also shoulders the responsibility of ensuring they reject work that is not focused, and that their student does not meander.



Figure 4. The Interconnections Between the Problem Statement and Other Sections in a Dissertation

HELPFUL FACTORS FOR WRITING THE RESEARCH PROBLEM STATEMENT

Hawley (2003) wrote a book titled “Being Bright is not Enough.” The book describes that writing a dissertation takes more than one set of skills. Jacob (2013) supports the same contention and additionally notes that writing a dissertation takes more than good writing skills. It is a mix of individual and institutional characteristics that must be present to help students write a dissertation. So, identifying helpful factors to writing the problem statement could assist students in their writing of the research problem statement.

Some of the factors that are considered helpful to students when writing their research problem statement are explained in this section. We begin by explaining self-efficacy and writing-efficacy and their role in writing dissertations in general and the research problem statement. Next, we discuss the importance of conducting some form of a literature review and how it helps students gain knowledge of what is written about their dissertation topic. Lastly, social support – a critical factor that is not often thought of when considering the dissertation – is discussed in this section.

WRITING EFFICACY AND SELF-EFFICACY

Self-efficacy has been the subject of discussions in many areas of research and is considered a key contributor to the success of individuals who persist and complete tasks (Bandura, 1993, Bruning et al., 2013). Bandura (2006) explained that self-efficacy is often associated with the capability of the individual to complete tasks. Bruning et al. (2013) described self-efficacy as the personal confidence that one has that they can complete their tasks successfully in the intended domain. In other words,

students with higher self-efficacy can tolerate more challenges and persist longer than those who have lower levels of self-efficacy.

Bandura (2006) identified four skills that contribute to an individual's self-efficacy: cognitive skills, motivational skills, emotional skills, and behavioral skills. In terms of the completion of academic tasks, Bandura (1993) noted that self-efficacy operates at three different levels when it comes to contributing to students' academic success. These include student's belief, the teacher's belief, and the faculty's belief in the self-efficacy of the students. McBrayer et al. (2018) conducted a study about the relationship between self-efficacy and time to degree completion of doctoral study. The study concluded that students with higher self-efficacy end up completing the writing of their dissertation in a shorter time.

A subsection of the concepts discussed regarding the writing of doctoral dissertation is the writing-efficacy of the student. Writing efficacy is connected to self-efficacy and explained by Stadtlander et al. (2020). It is described as how well a researcher can accomplish writing tasks based on various skills, such as composition, grammar, and other mechanical skills. Bruning et al. (2013) explained that writing efficacy is a function of three factors: writing ideation, writing convention, and writing self-regulation. In writing ideation, the researcher starts with writing ideas as they emerge in the mind, though how well the researcher expresses the ideas is also important. The writing convention is about the different conventions and rules that need to be followed when writing. Writing self-regulations are related to managing writing tasks and observing the writing habits that researchers develop.

Self-efficacy and writing-efficacy are necessary components in successfully completing the dissertation and writing the research problem statement as a first step to writing the dissertation. It is also a known fact that self-efficacy influences writing efficacy – that is, students with higher self-efficacy will make better progress. The question that could be asked regarding this study is how can mentors influence or contribute to enhancing both efficacies to successful writing the research problem statement.

Stadtlander et al. (2020) noted that a structured experience that is supervised by the mentor is helpful in improving self-efficacy of the students involved. Breitenbach (2019) suggested that designing a structured program with sufficient feedback could help more students complete their dissertation. Ewing et al. (2012) echoed similar contentions, and designing a structured dissertation writing process helps with the acquisition of research skills.

Structured experience in the dissertation writing (and writing the research problem statement in this study) can be helped by using a rubric type of feedback. Sparrow (2004) suggested that using rubrics improves teaching by explaining grading criteria. We argue that rubrics also help with writing efficacy and self-efficacy. Rubrics provide a clearer picture of what is expected. Therefore, students can approach the task with greater confidence. One other way for improving self-efficacy is through conducting a literature review.

IMPORTANCE OF CONDUCTING A LITERATURE REVIEW

The importance of conducting a literature review in writing a dissertation and, in particular, with writing a research problem statement cannot be underestimated. Randolph (2009), for example, noted the following about the purpose of conducting a literature review:

Conducting a literature review is a means of demonstrating an author's knowledge about a particular field of study, including vocabulary, theories, key variables and phenomena, and its methods and history. Conducting a literature review also informs the student of the influential researchers and research groups in the field.
(p. 2)

Jacobs (2013) noted the importance of conducting a literature review to narrow down the writing of the research problem statement. Wee and Banister (2016) suggested that the literature review

represents an added value for the researcher because they (the researcher) get up-to-date information on the issue being discussed in the problem statement. Torraco (2005), on the other hand, considers conducting a literature review very important because it gives the researcher exposure to the past writings on the topic and provides a framework on which to explore the future.

From the statements above, we can detect that conducting a literature review could often help the students clarify many points as they write the research problem statement (Columbaro, 2009). The literature review gives the students ideas on what problems were discussed in the past regarding the problem identified in their research. It gives them examples of how problems statements were written in the past, and they can model what they learn and use it to write their own problem statement (Ebrahim, 2012, Levy & Ellis, 2006).

While the literature review is essential in writing the research problem statement, it raises the question about the positioning of the literature review. Should the literature review be written in Chapter 2, or should it be moved to Chapter 1, before the problem statement? In our opinion, this is unnecessary. In order to understand what is written in the field about the problem statement, some literature review is essential and, therefore, expected in Chapter 1. However, a full scale literature review should be conducted in Chapter 2 and within the normal sequence of the dissertation chapters.

It must be noted that the student will, almost by definition, unearth new information when conducting a literature review. This is true in Chapter 2 as well as over the rest of the dissertation journey. Accordingly, if necessary, the research problem statement must be revisited and suitably modified or amended as needed to account for the new information. In the worst-case scenario, the student may find their specific problem suitably addressed in another publication. Rare as that occurrence may be, it would be naïve of any researcher to assume that will never happen to them.

THE ELEPHANT IN THE ROOM: SOCIAL SUPPORT

Social support, a major problem for students working on their dissertation, is the metaphorical elephant in the room. While widely accepted, the lack of social support is not addressed in sufficient depth in doctoral programs. Ali and Kohun (2007) consider the lack of social support at doctoral programs to be a major contributor to the high proportion of student attrition from doctoral programs. Devos et al. (2017) stress that student integration, socialization, and support from doctoral peers is important and will enhance student completion of their dissertations. Crauth (2017) wrote about socialization in doctoral programs, considering it a requisite for writing the doctoral dissertation and further suggesting that doctoral student attrition has social consequences. Ames et al. (2018) consider socialization as a prime factor in increasing retention in online doctoral programs.

Brill et al. (2014) conducted a study to suggest best practices in doctoral retention. The study addressed doctoral attrition from different points of view and provided suggestions on how to best work on a program that strives to keep the students in the program to complete their dissertations. Their suggestions focused in large parts on mentoring, and these come in the form of faculty mentoring and peer mentoring. This is echoed by Black (2017), who emphasized e-mentoring for doctoral students online. To summarize, the point discussed here is that social support and mentoring are prime support factors for dissertation writing success and for writing the research problem statement.

A particular point of emphasis is peer mentoring. This could come in the form of exchanging peer feedback, as suggested by Philippakos (2017). This kind of feedback helps with the social integration of the students into the program (Ali & Kohun, 2007). While writing about empowering students with peer mentoring, Krusemark & White (2020) reported on a 2017 Lehigh University study geared towards reducing the loss of prospective students and how to increase a sense of belonging once students were enrolled. Their study was on undergraduate students, but lessons learned were worth noting. For instance, one big takeaway from their study was the need for large-scale peer mentorship. An outcome of the 2017 study was Lehigh University leadership adopting, as one solution, a formalized

peer mentorship program, with clear goals, tracking, and assessments. A year later, the program has shown positive results (Krusemark & White, 2020).

At the doctoral level, it might initially appear that peer mentoring is not much more than fellow students reviewing the work of their peers for the purpose of giving feedback. However, there is more to the phenomenon since peer mentoring helps in several other ways. First, it helps give exposure to what the problem statements of their peers look like. By comprehending and analyzing the problem statements, peer groups can help refine each other's problem statements. Second, it helps with social integration into the program. Friendships could be developed, and feelings of belonging could be increased (Kumar & Johnson, 2017). Additionally, when a student receives feedback on their work, peer mentoring can provide an avenue for the airing of grievances. Work rejection and revision requests are an inevitable part of the doctoral journey. Conversing between peers helps the dejected student realize that they are not alone, that rejection is par for the course, that others have moved past this, and so will they (Columbaro, 2009).

CONSTRUCTING A RESEARCH PROBLEM STATEMENT

Blum and Preiss (2005) suggested a six-step approach for doctoral advisors to take when they coach students to write their problem statement. While the suggested steps do need to be specifically six steps, we agree that working with the students on writing their research problem statement could be best explained in many steps rather than a single step.

In this section, we list some suggestions on how to teach students to write their research problem statement. It will be divided into four stages according to Beeler's framework identified at the beginning of this study. In each stage, we identify the stage, suggest steps to follow at the identified stage, and recommend some questions to think about to stimulate thinking at that stage. This type of question is a substitute to formally using a grading rubric. While grading rubrics can be used to grade the research problem statement, we do not deem this necessary at every stage. We propose a set of questions at each stage aimed at evaluating whether the stage is complete.

STAGE 1: CRUDE STATEMENT OF THE PROBLEM

Beeler (1991) calls the first stage "Unconsciously incompetent." We describe this stage as "The students do not know that they do not know." In their early adjustment to doctoral study, students often do not know the details of what needs to be included in the problem statement, and they are often in a "fog" when it comes to these details. They often do not know about the writing conventions or the literature review. So, inundating them with long regulations about writing conventions and supporting literature is counterproductive, in our opinion. The approach we suggest is to work with students to list the problem in a crude statement. The focus of this is on three points at this stage. They can be explained as questions, as follows:

- Is it phrased as a problem statement (with cause and effect)?
- Is the problem statement researchable (research problem statement)?
- Is the research problem statement viable (time and resource constraints)?

The above could easily be part of a set of rubrics, as needed. We believe that these, or other questions in a similar vein, could stimulate thinking about the problem statement. This would move the student correctly to Stage 2.

STAGE 2: FINDING KEYWORDS

Beeler's model calls this stage "consciously incompetent." We call this stage "They know that they do not know." Awareness brought from Stage 1 is that the students are sensitive to what they do not know. They know that they need help phrasing the research problem statement. More constraints at this stage can be explained in terms of articulating the research problem statement. For example, a

look at the crude statement that was phrased in Stage 1 will be helpful. Look for “buzz words” or keywords to elaborate further. Also, connecting to a literature review is helpful at this stage. Typical helpful questions that could be asked at this stage include:

- Do you have keywords in the problem statement?
- Are the keywords explained or elaborated enough in the statement?
- Do you have supporting literature to support this as a problem that can be researched?

We believe that cogitating over and finding responses to questions like these will be helpful at this stage and help guide the student to the third stage.

STAGE 3: USING THE PEEL APPROACH

We call this step “They do not know that they know” as a clarification of Beeler’s model, which refers to it as “Unconsciously competent.” Competency was gained during the second stage. The student now knows more and can thus move to the next stage in writing. At this stage, we suggest using the Point Explain, Evaluate and Link (PEEL) method to gauge progress (Humphrey et al., 2015). This approach gives students the opportunity to look at their problem statement, begin the literature review, connect the problem statement with the different keywords, and make it sound like a research problem statement. Figure 5 below show a display of the PEEL approach.

P Point, Make the point	E Evidence Support opinion with evidence and data	E Explain how evidence support problem state- ment	L Link problem statement with other sections in dissertation
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Figure 5. The PEEL Approach

At this stage, the student may write what doctoral faculty sometimes refer to as “Naked Paragraphs.” These are paragraphs without citations or do not have references in them. In research studies, having too many “naked paragraphs” is a sign of insufficient research depth. Additionally, writing conventions and support can be enforced at this stage. Typical helpful questions at this stage can be:

- Do you have naked paragraphs in your problem statements?
- Are you connecting the problem statement with the research?
- Are you following writing conventions (such as APA) in your statement of the research problem?

We believe that questions like these will be helpful at this stage and will take them to the final stage.

STAGE 4: THE FINALIZATION STAGE

Beeler’s model calls this stage “Consciously Competent.” We call this step “They know that they know.” By now, students have become comfortable, their self-efficacy increases, and their writing efficacy increases as well. They also learn more about the other sections in the dissertation – the “lookahead ideas.” In other words, the students can think about what is ahead of them in writing the other sections of their dissertation. They can connect the problem statement to what is ahead of them and make adjustments if necessary.

The fourth and last step is to finalize the problem statement through supporting references or data (stats, figures, charts) to support that it is a problem worthy of research. Typical questions that can be asked at this stage:

- Do you have supporting data that this is a research worthy problem?
- Is the problem statement connected to other sections in the dissertation?
- Is the problem statement finalized in terms of writing conventions, viability, and research worthiness?

Table 1 summarizes our suggestions in four distinct stages that guide the students to write their research problem statement. The second column provides a name to that stage, based on our suggestions. The third column lists the questions that we deem helpful to ask at this stage. These questions are based on our findings in the previous section. They have been modified slightly to fit in the table.

Table 1 – Framework for the Completion of Writing Research Problem Statement

Adjustment stage	Suggestions for writing	Questions to ask
Stage 1	Phrasing the problem	Is it phrased as a problem statement (with cause and effect)? Is the problem statement researchable (research problem statement)? Is the research problem statement viable (time and resource constrain)?
Stage 2	Finding keywords to elaborate	Do you have keywords in the problem statement? Are the keywords explained or elaborated enough on it in the statement? Do you have supporting literature to back that this is a problem that can be researched?
Stage 3	The PEEL approach	Do you have naked paragraphs in your problem statement? Are you connecting the problem statement with research? Are you following writing conventions and APA in your statement of research problem?
Stage 4	Putting it all together. Supporting the problem statement with references and data.	Do you have supporting data that this is a research worthy problem? Is the problem statement connected to other sections in the dissertation? Is the problem statement finalized in terms of writing conventions, viability, and research worthiness?

THE ELEPHANT IN THE ROOM – AGAIN!

We must discuss social support once again at this juncture. This is because we consider social support to be essential to working on the dissertation. Different stages require different sets of steps to support the researcher. Community of Practice (CoP) and Community of Inquiry (CoI) are both recommended (Kumar et al., 2011). It is a good idea to include a support structure where students get feedback from their peers and other faculty in the program as they go through the four stages of completion. It must also be noted that social support is not limited to any specific stage of our framework. On the contrary, it could be applied to all the stages of the adjustment and the writing of the

research problem statement. It is also applicable to the entire dissertation journey. This will be further illustrated in the research recommendations section, below.

CONCLUSION AND RECOMMENDATIONS

After conducting this study, we present our recommendations as one figure that includes the previous table we finalized and suggests the importance of social support through all the stages we have reached. Because social support is needed at all stages and all levels, we included it as a shape that spans all levels. Figure 6 presents the completed table, including suggestions for social support.

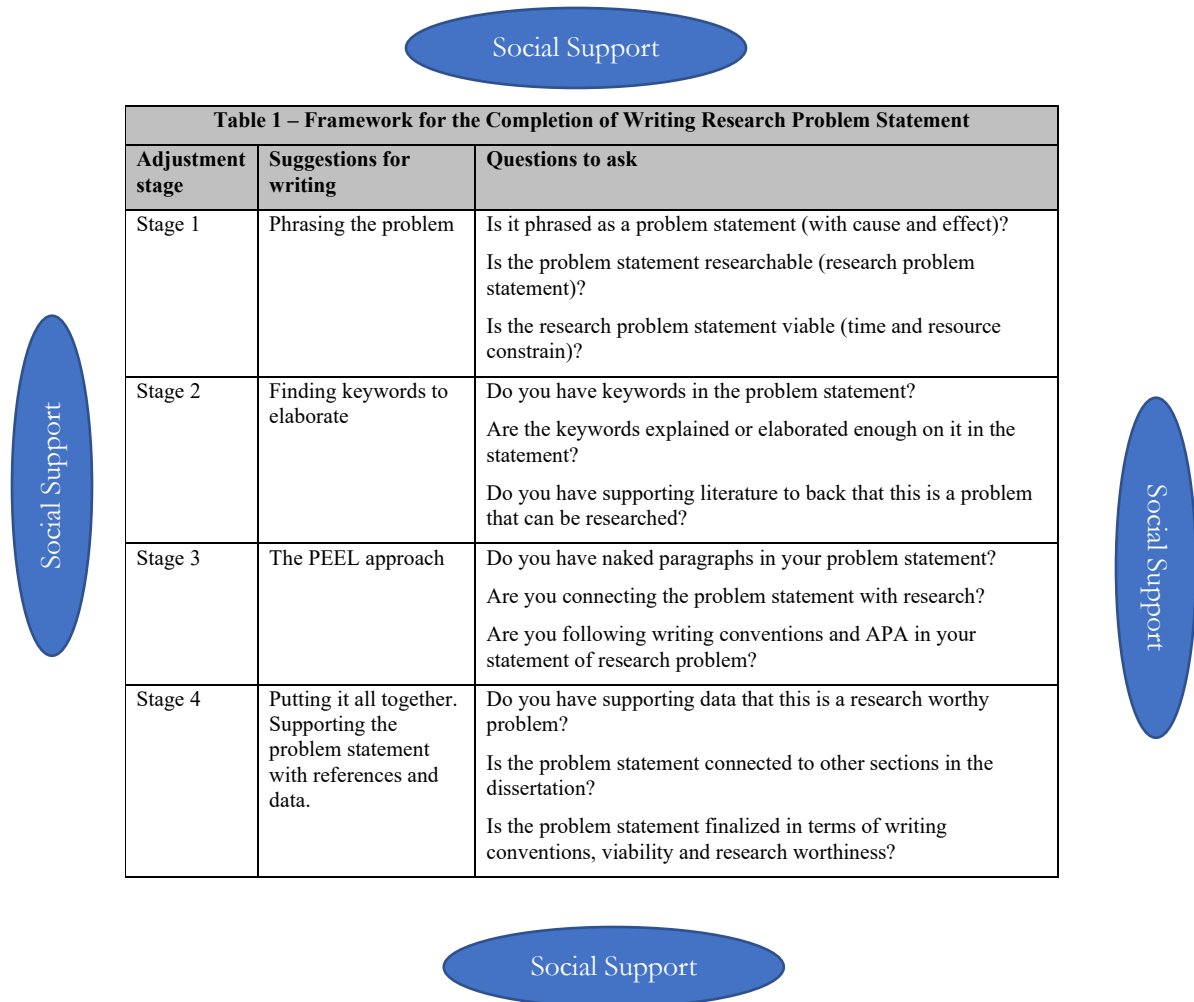


Figure 6. Suggested Framework for Writing Research Problem Statement

This study contributes to the field of doctoral studies by presenting a structured approach to assist students with the arduous task of writing their doctoral dissertation. The structured approach suggested in this study is consistent with the adjustment process that students go through at the beginning of their enrollment in the doctoral program or their beginning of writing the dissertation.

While the suggested structured approach is helpful, its utility could be further increased if illustrated with examples that inform actual student progress through each of the four stages. Thus, we intend to conduct a follow-up study to fill this gap: to provide typical examples of how students work their way through their construction of the research problem statement and the pitfalls they face. In addition, the follow-up study will introduce a rubric sheet that will provide instructions for completing each step, the objectives that are sought from completing them, and a grading sheet that can provide

different levels of evaluation feedback. This follow-up study is necessary to provide doctoral advisors with a complete model, a set of instructions, and evaluative feedback.

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Shardul Y. Pandya, Ph.D. Dr. Pandya currently serves as Core Faculty and Sr. Doctoral Dissertation Advisor with the School of Business, Technology and Healthcare Administration at Capella University. He has a Bachelor's degree in Industrial and Production Engineering (dual major) from the BMS College of Engineering at Bangalore University, a Master of Science degree in Mechanical Engineering from the College of Engineering at the Colorado State University at Fort Collins, CO, and a Ph.D. in Engineering Management from the College of Engineering at Old Dominion University in Norfolk, VA. Over the years, he has worked and consulted at various manufacturing organizations, as well as in the areas of Data Mining and Database Marketing. He has taught university level courses at all levels, and approximately 20 doctoral students have successfully defended their dissertations under his supervision. His public LinkedIn profile is here: <https://www.linkedin.com/in/shardulpandya/>

How to Develop: Problem Statement, Topic, Objectives and Research Questions, Hypothesis

AN

A Seminar presentation

BY

Dr. Ambrose Azeta

Department of Software Engineering

Namibia University of Science and Technology

27 February, 2023

PRESENTATION OUTLINE

2

1. Introduction
2. Areas of Research in ICT
- 3. Computer Science Research**
 - 3.1 Formulation of Problem statement
 - 3.2 Research title
 - 3.3 Research Objectives
 - 3.4 Research Sub Objectives
 - 3.5 Research Questions and Hypothesis
 - 3.6 Research Methodology
 - 3.7 Computer Science Dissertation/thesis (sample outline)
 - 3.8 Computer Science publication/papers (sample outline)
- 4. Information Systems Research**
 - 4.1 Research Aim
 - 4.2 Research Objectives
 - 4.3 Mapping of Research Questions, Hypothesis and Objectives
 - 4.4 Confirmation of Research questions and Hypothesis
 - 4.5 Conclusion

2. AREAS OF RESEARCH IN ICT

3

- Computer Science
- Information Systems
- **Software Engineering**
- Cyber Security,
ETC
- My main research area is Software Engineering/
Software Development
- SE is applied in all domains

3. SOFTWARE ENGINEERING RESEARCH

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- Model driven Engineering (MDE)
- Design using Unified Modelling Language (UML)
- System testing/Automated
- System Evaluation
- System maintenance and monitoring
- Automated software architecture and debugging
- Software defect prediction using machine learning

Definition of Concepts

5

What is Software Engineering (SE) ?

- Software engineering is the process of applying engineering principles to analyze the needs of a specific user and then design, build and test software applications which will meet those needs.
- Software engineering goes beyond the fundamentals of programming to create more complex software systems that can serve as critical systems for businesses and organizations.
- Software engineers need to understand how to write software and how software and hardware interact.

What is Computer-aided software engineering (CASE)

- *Computer-aided software engineering (CASE)* uses programs to assist in the creation and maintenance of other computer programs.
- Programming has historically been performed by (human) programmers or teams; CASE adds software to the programming team.

Three (3) ways of finding a Problem Statement

- 1) Existing manual/traditional system is not working well
- (2) Existing automated/computerized system exist but giving problem.
- (3) A New system. No existing system, rather the project has not been done before in the community.

Sample Problem Statement

7

SN	Project title	Sample problem statement
1	Development of Fake News and Advertisement Detection System	Individuals and groups have obligated themselves the task of spreading falsified information to other users with the aim of gaining some money, change the public's opinion on the integrity of reputable persons and organizations, potentially alter the results of an election or some other sinister reason.
2	Development of A Web-Based Medical Appointment System For Cove University	With the high number of appointments that are scheduled in clinics and hospitals every day, it is almost impossible for the clinic to achieve its goals. Definitely, there will be some dissatisfaction on both clinic and students. The clinic has to deal with urgent or emergency cases, absentees or no-shows among scheduled cases which may complicate the work of the receptionist who is booking these appointments manually through walk-ins.

3.1 FORMULATION OF PROBLEM (GAPS)

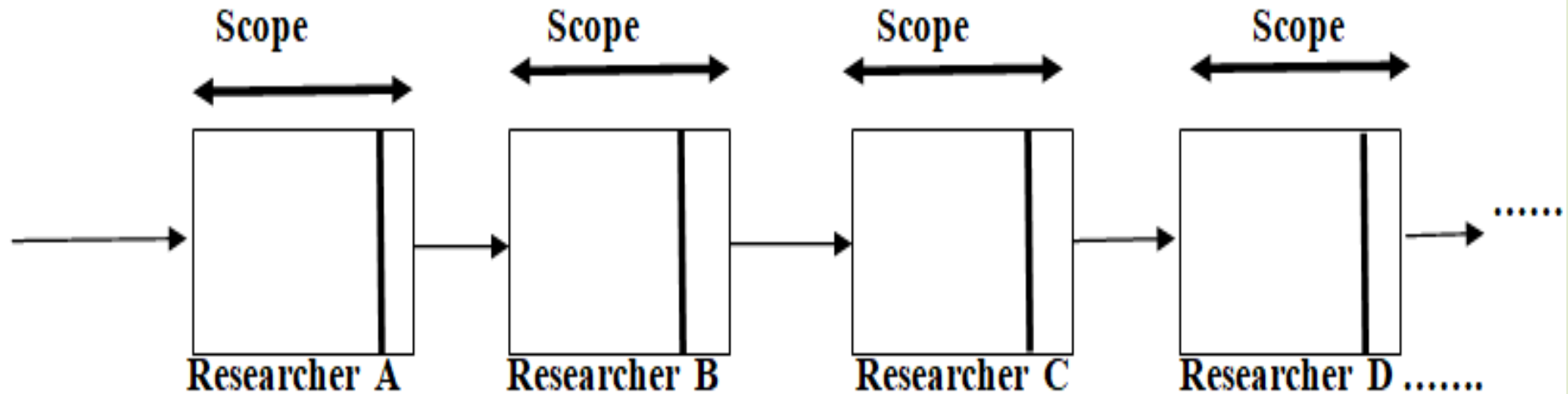
8

- ➡ State the problem you want to solve
- ➡ Cite the publication/papers that contains the limitations of a study or a problem that you want to solve, and could not be solved in the current research
- ➡ Gaps are normally discovered through systematic review

1. INTRODUCTION

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RESEARCH CHAIN



Research is continuous (Research Advancements)

Some questions to address:

Which work are you extending?

What problem are you trying to solve?

What is the societal benefit of your research?

Etc

3.2 RESEARCH TOPIC

10

- Must contain: a technique/methodology (**Using** or **a**), domain (in), system or platform(**of** or **for**),

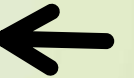
Examples

- Development of a model **for** result management system **in** E-Education **using** DevOps OR
- Development of **a** DevOps-based model **for** result management system **in** E-Education using DevOps ←
- Development of **a** result management system **in** E-Education using DevOps

3.3 RESEARCH OBJECTIVE

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- The objective of this study is to develop a devops-based model for result management system in e-education



3.4 RESEARCH SUB OBJECTIVES

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The Sub **objectives** of this study are **To**:

- **To** carry out requirement elicitation of result management system
- **To** design a DevOps model for result management system
- **To** implement a result management system based on DevOps as proof of concept
- **To** evaluate the implemented system and benchmark with existing system

3.5 RESEARCH QUESTIONS AND HYPOTHESIS

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Research Questions

- How can a DevOps model be developed for result management system?
- How can a result management system based on DevOps be evaluated.

Hypothesis

- **H1:** The development of a DevOps model for result management system is realistic
- **H1:** A result management system based on DevOps has a high usability evaluation compared to other models.

3.6 RESEARCH METHODOLOGY

14

- Tools/techniques such as software tools
- Research design/instruments such as questionnaire

3.7 CHAPTER OUTLINE OF DISSERTATION/THESIS

15

A Sample Thesis outline:

- ➡ Ch1-Introduction
- ➡ Ch2-Literature review
- ➡ Ch3-Research methodology
- ➡ Ch4-Results and Discussion
- ➡ Ch5-Conclusion

3.8 SECTION OUTLINE OF PUBLICATION/PAPERS

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A Sample paper outline:

- ➡ Section1-Introduction
- ➡ Section2-Literature review
- ➡ Section3-Research methodology (System Design)
- ➡ Section4-Result and Discussion (System Implementation and Evaluation)
- ➡ Section5-Conclusion

4 INFORMATION SYSTEM RESEARCH

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Research Title: *IMPACT OF ICT ON STUDENT ACADEMIC ACHIEVEMENT*

Formulation of Research Questions and Hypothesis

- **Research Question 1:** *To what extent does teaching with the aid of ICTs impact on student academic achievement?*
- **Hypothesis H1:** *Teaching with the aid of ICTs has a positive impact on student academic achievement.*
- **Research Question 2:** *To what extent does course student engagement impact on their academic achievement?*
- **Hypothesis H2:** *Student engagement has a positive impact on their academic achievement.*
- **Research Question 3:** *To what extent does student use of ICTs impact on academic achievement?*
- **Hypothesis H3:** *Student use of ICTs has a positive impact on academic achievement.*

4.1 RESEARCH AIM

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The aim of this research is to explore the impact of ICT on student academic achievements

4.2 RESEARCH OBJECTIVES

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The objectives are as follows:

- ➡ To determine if teaching with the aid of ICTs impact on student academic achievement
- ➡ To determine the extent to which course student engagement impact on student academic achievement
- ➡ To determine the extent to which student use of ICTs impact on student academic achievement

4.3 MAPPING OF RESEARCH QUESTIONS, HYPOTHESIS AND RESEARCH OBJECTIVES

20

SN	Research Questions (RQ)	Hypothesis (H)	Research Objectives (RO)
1	<i>To what extent does teaching with the aid of ICTs impact on student academic achievement</i>	<i>Teaching with the aid of ICTs has a positive impact on student academic achievement.</i>	<i>To determine if teaching with the aid of ICTs impact on student academic achievement</i>
2	<i>To what extent does course student engagement impact on their academic achievement?</i>	<i>Student engagement as a positive impact on their academic achievement</i>	<i>To determine the extent to which course student engagement impact on student academic achievement</i>
3	<i>To what extent does student use of ICTs impact on academic achievement?</i>	<i>Student use of ICTs has a positive impact on academic achievement</i>	<i>To determine the extent to which student use of ICTs impact on student academic achievement</i>

4.4 CONFIRMATION OF RESEARCH QUESTIONS AND HYPOTHESIS

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RQ1: *To what extent does teaching with the aid of ICTs impact on student academic achievement?*

Construct B: Teaching with the aid of ICTs returned a strong positive correlation and prediction (2nd in weight(β) out of 5 predictors) of *student academic achievement*.

H1: *Teaching with the aid of ICTs has a positive impact on student academic achievement.*

H1 was supported since a significant positive relationship was established with model 1 & 2. (Unstd β = 5.954 & 5.848, $p < 0.01$). The finding is in agreement with the results of Bai, et al. (2016); Rugutt and Chemosit (2005.); Ilomaki (2008); Abuet al., (2011); Oye et al., (2012) and Angeli&Tsaggari (2016).

4.4 CONFIRMATION OF RESEARCH QUESTIONS AND HYPOTHESIS (CON'T)

22

RQ2: *To what extent does course student engagement impact on student academic achievement?*

Construct C: Student engagement, *had* a weak positive correlation with, and prediction (3rd in weight (β) out of 4 predictors) of Class Marks.

H2: *Student engagement has a positive impact on student academic achievement.*

H2 however, was supported since a significant positive relationship was established for model 1 & 2 (Unstd β = 1.086 & 1.140, $p < 0.05$). The finding is in line with Gunuc's (2014) conclusions.

5. PUBLICATION OF RESEARCH OUTPUT

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Publishers of good Journal Outlets

- Nature
- Springer Nature
- Elsevier
- Sage
- Emerald
- Wiley
- University-based Journals
- Etc

Conference outlets

- ACM
- IEEE
- Springer
- Etc

Index Outlets/Databases

- Scopus
- Web of Science/ISI
- University databases/
repositories
- Etc

5. CONCLUSION

24

This presentations has no doubt achieved its objectives of highlighting the research methodology in Computer Science and related areas

THANK YOU

Effective Medical Writing

Pointers to getting your article published

Peh W C G, Ng K H

Writing the Introduction

ABSTRACT

The introduction section of a scientific paper aims to introduce a specific topic and to stimulate the reader's interest. It provides background information about what has already been done by others, supported by a limited number of relevant references. The reader should be informed about the purpose of the paper, what it will address, and how it relates to previous work.

Keywords: introduction, medical writing, research objectives, scientific paper structure

Singapore Med J 2008; 49(10): 756-758

INTRODUCTION

The introduction section of a scientific paper is normally placed at the beginning of the text proper, and follows the abstract. It is the first component of the conventional IMRAD (Introduction, Materials and methods, Results And Discussion) structure of an original article. Almost all papers have an introduction section, except perhaps for some types of short papers found in certain journals, such as case reports, commentaries and other paper categories. Authors should always check the individual journal's "Instructions to Authors" or "Author Guidelines" for the in-house style requirements.

Besides introducing a specific topic, the purpose of the introduction section is to stimulate the reader's interest, with the aim of leaving a positive initial impression of what is to come, so that he or she will be motivated to read the rest of the paper.⁽¹⁾ Inexperienced authors often have problems with this section. A very common problem is being unable to clearly state the objectives for writing the paper. The introduction can be divided into two components, namely: background information and purpose.

BACKGROUND INFORMATION

The first component of the introduction section aims to provide the reader with sufficient background information about the topic to be investigated or discussed, i.e. introduces the topic to the reader. This background information should include a brief summary of what has already been done in the particular area of interest.

The nature of a problem and its potential significance should be highlighted. References quoting relevant work by others should be provided, with particular care taken not to miss out important previous work. Editors and reviewers like to see that authors have made a systematic and comprehensive review of the literature. However, listing all the references found from an extensive literature search is not appropriate in the introduction section. Authors should also resist the temptation to provide an extensive critique of others' work in this section. Authors should instead confine themselves to providing a limited number of relevant references.

The length of the introduction section and depth of background information contained depend on the target audience and type of paper. The introduction section is usually longest for original articles, particularly those dealing with a complex subject, and shortest for case reports. If a certain topic is aimed at a general medical audience, the introduction may need to be more descriptive and have greater explanatory details in order to better introduce the topic; while for the same topic written for a subspecialty audience, the authors can assume that the reader would be familiar with the topic and all that is required is a brief background to show why the study was conducted and why the topic is important. The background information usually starts with more general and broad-based facts, and then moves on to specifics. As a rule, the introduction should be as succinct and focused as possible.

PURPOSE

The second component of the introduction section aims to inform the reader about the purpose of the article and how it relates to previous work. The author should be able to provide information about how the hypothesis for a study, clinical or investigative observation, or idea for a procedure or its modification came about, either from his personal thoughts and experience, or from referenced work by others. Authors should highlight any new, different or special aspects of what he or she aims to undertake in the study. Editors will not be interested in publishing, and the reader will not be interested in reading, studies that are merely a repetition of what had already been done by others.

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The reader should be able to understand why the authors chose to undertake the study, and the rationale for the authors' research. The reviewers, and later on the reader, need to be convinced of the importance of the research question, and feel stimulated enough to read the rest of the paper. The identification and definition of a particular clinical or research problem is therefore vital. The reader should be informed about how the problem will be addressed. The purpose for writing the paper should be precisely stated at the end of the introduction section. The main and secondary objectives should be clear, and ideally comprise no more than two sentences. The author should not include data or conclusions from the work being reported.

Box 1. Components of the Introduction:

1. Background information
 - a. What have others done?
 - b. Provide evidence: supported by limited number of relevant references.
2. Purpose of study
 - a. Why undertake this research?
 - b. How does it relate to what has already been written?
 - c. What is so different or special about your research?

Box 2. Common errors:

- Overlong and rambling introduction section
- Extensive listing of references
- Extensive critique of others' work
- Important previous work missing
- Objectives not clearly stated
- Inclusion of data or conclusions from the work being reported

SUMMARY

The introduction section of a scientific paper should be concise and yet informative. The reader should be able to understand why the author's work is important in the context of what is already known, and be stimulated enough to read the rest of the article.

Box 3. Take home points:

1. The introduction should be succinct and focused.
2. Provide sufficient background information supported by relevant references.
3. Clearly state the purpose for writing the paper and how it relates to previous work.

REFERENCE

1. Day RA, Gastel B. How to Write and Publish a Scientific Paper. 6th ed. Burnham: Greenwood Press, 2006.

SINGAPORE MEDICAL COUNCIL CATEGORY 3B CME PROGRAMME
Multiple Choice Questions (Code SMJ 200810A)

	True	False
Question 1. The introduction section:		
(a) Should be made as long as possible.	☐	☐
(b) Should contain at least 100 references.	☐	☐
(c) Should always be structured.	☐	☐
(d) Aims to stimulate the reader's interest.	☐	☐
Question 2. The following are components of the introduction section:		
(a) Background information.	☐	☐
(b) Detailed statistical analysis.	☐	☐
(c) Results.	☐	☐
(d) Purpose of study.	☐	☐
Question 3. The following statements regarding the introduction section are true:		
(a) It provides sufficient background for understanding the rest of the paper.	☐	☐
(b) Extensive critique of others' work is required.	☐	☐
(c) If the authors disagree with certain pieces of work, reference to these may be omitted.	☐	☐
(d) Its length and depth should be tailored to the target audience.	☐	☐
Question 4. In the introduction section, the purpose should address the following:		
(a) Why undertake this research?	☐	☐
(b) How does it relate to what has already been written?	☐	☐
(c) What is so different or special about the undertaken research?	☐	☐
(d) What new data was found?	☐	☐
Question 5. When writing the introduction section, the following rules apply:		
(a) Always read the particular journal's author guidelines.	☐	☐
(b) The purpose for writing the paper should be concise.	☐	☐
(c) The purpose for writing the paper is almost always contained within the opening two sentences of the introduction section.	☐	☐
(d) A particular clinical or research problem should be identified.	☐	☐

Doctor's particulars:

Name in full: _____

MCR number: _____ Specialty: _____

Email address: _____

SUBMISSION INSTRUCTIONS:

(1) Log on at the SMJ website: <http://www.sma.org.sg/cme/smj> and select the appropriate set of questions. (2) Select your answers and provide your name, email address and MCR number. Click on "Submit answers" to submit.

RESULTS:

(1) Answers will be published in the SMJ December 2008 issue. (2) The MCR numbers of successful candidates will be posted online at www.sma.org.sg/cme/smj by 15 December 2008. (3) All online submissions will receive an automatic email acknowledgment. (4) Passing mark is 60%. No mark will be deducted for incorrect answers. (5) The SMJ editorial office will submit the list of successful candidates to the Singapore Medical Council.

Deadline for submission: (October 2008 SMJ 3B CME programme): 12 noon, 25 November 2008.

Data Science Cluster

Ruusa Ipinge

Areas of Specialisation

area of specialization in Data Science

- Specific Machine learning, especially its application on data for good
- I am also a certified General Data Protection (GDPR) whereby I am concerned about setting up policies
- I am also working on mobilizing digitalising Namibia

Digitalising Namibia

- Some of the projects I am working on is detecting water leakage through machine learning
- Using machine learning to identify those that are in extreme poverty
- Developing a recommendation system to help students choose their major
- Digitalizing data to identify vulnerable people for grants assistance for different ministries
- Digitalizing a Cumulative Card for the ministry of education
- Digitalizing a medical card for ministry of health

Data Science Workshops and Conferences

- We have a NUST star show your talent end of the year hackathon
- We have Namibia Indaba X hackathon coming soon in April , MAy
- We will be soon hosting a GDPR workshop
- And there will be indaba hackathon in Ghana, contact me on how to apply
- So Many Conferences around the world in Data Science
-



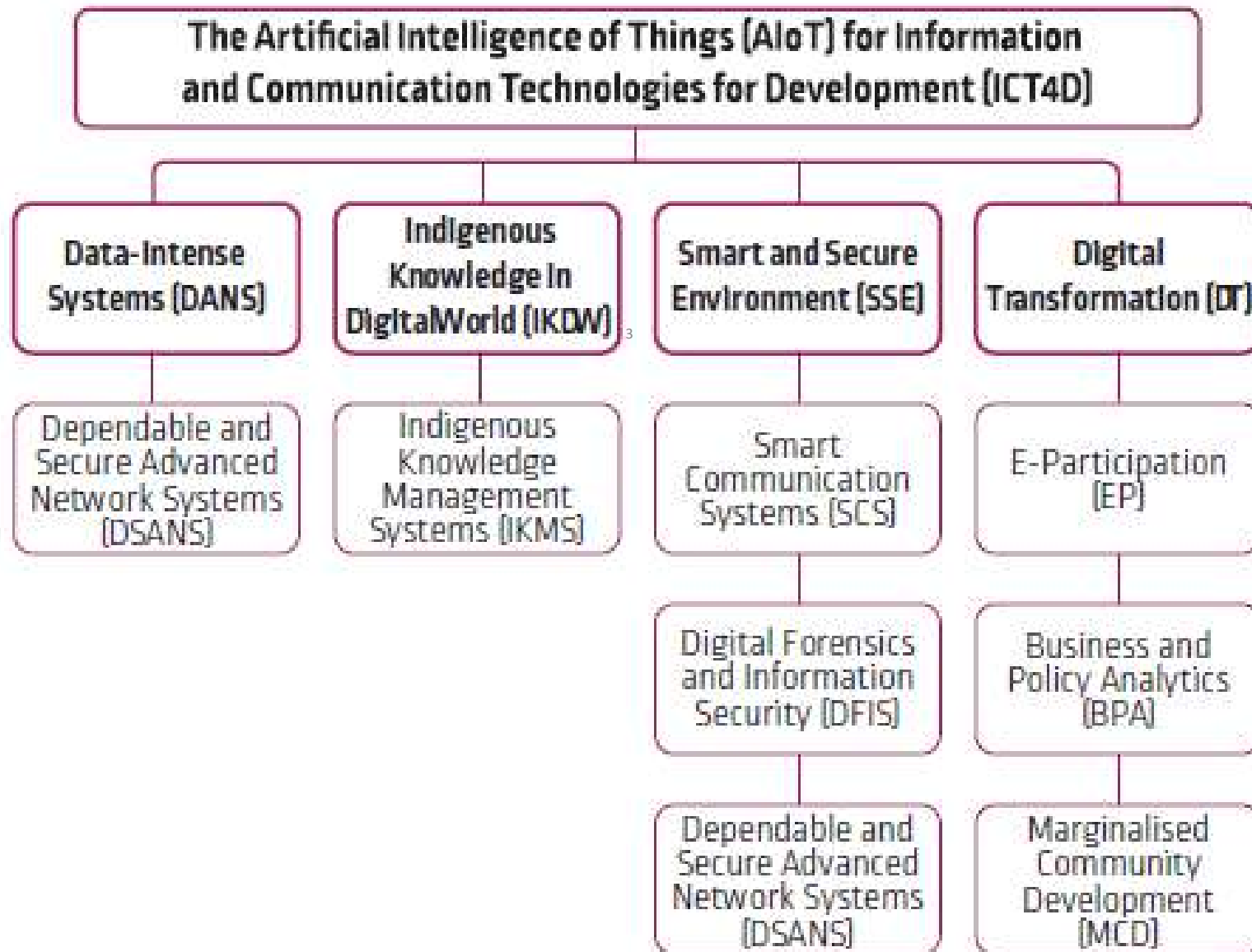
Digital Forensics and Information Security Research Cluster

Digital Forensics and Information Security Research Cluster Smart and Secure Environment





- FCI Research Focus Areas & Clusters
- Background
- Focus and Activities
- Team & Projects
- Graduates
- Students
- Publications



Responsible Journalism (RJ)

Journalism and Media Technology (JMT)



FCI Research Areas & Clusters

FCI Research Areas:

1. **Data-intensive Systems (DANS)**
2. Indigenous Knowledge in Digital World (IKDW)
3. **Smart & Secure Environment (SSE)**
4. Digital Transformation (DT)

FCI Research Clusters:

1. Dependable and Secure Advanced Network Systems – **DANS & SSE**
2. Indigenous Knowledge Management Systems (IKMS)- **IKDW**
3. Smart Communication Systems - **SSE**
4. **Smart Communication Systems (SCS) - SSE**
5. Digital Forensics & Information Security (DFIS) – **SSE**
6. E- Participation (EP) – **DT**
7. Business & Policy Analytics (BPA) - **DT**
8. Marginalised Community Development (MCD)- **DT**



About Us

Smart and Secure Environment

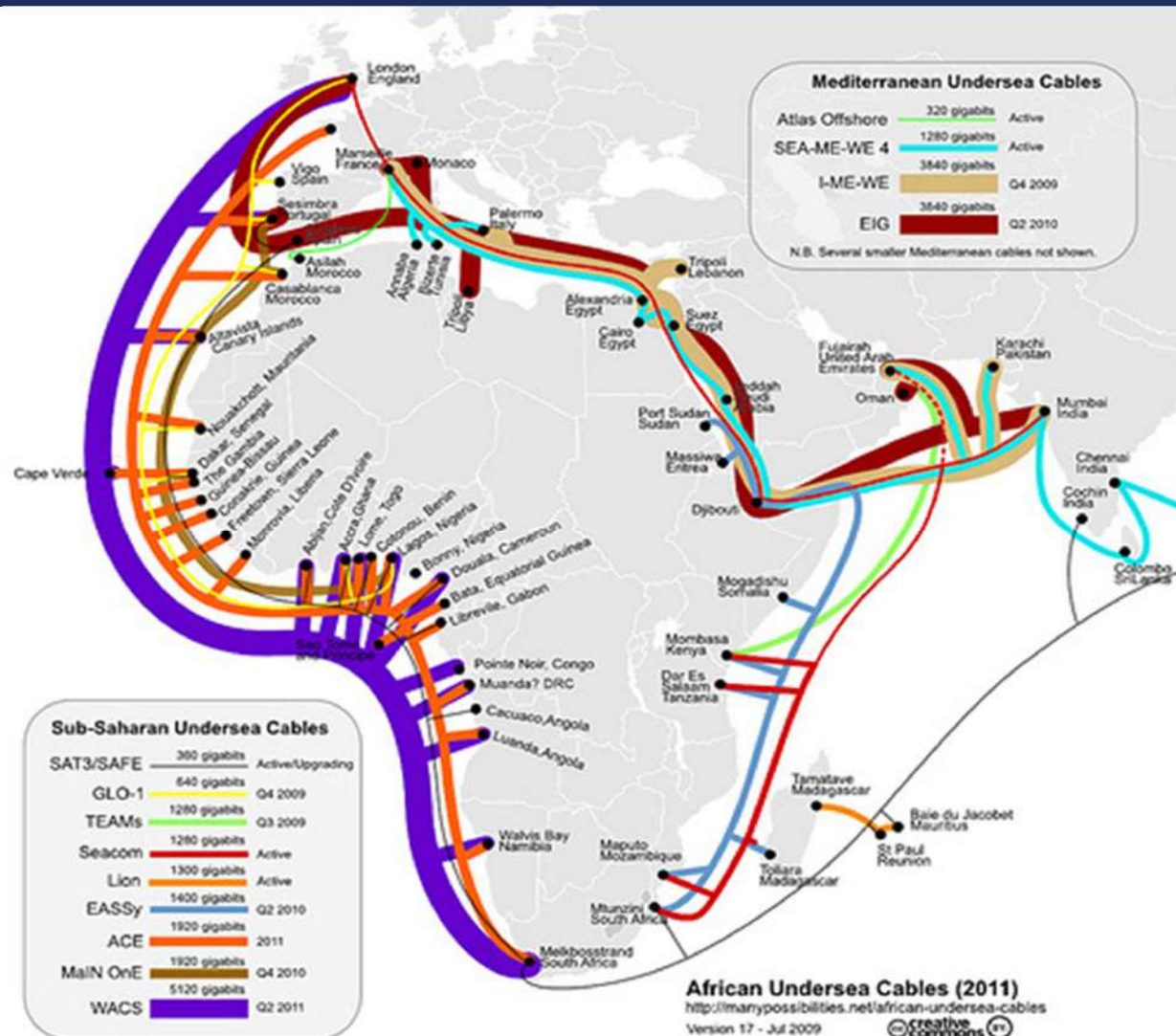
Digital Forensics and Information Security Research Cluster

Focus Areas

- The Internet of Things, AI and Emerging technologies security.
- Cyber Security
- Child Online Protection research.
- Security of organizational and industry operations research at various levels of complexity.
- Digital, Mobile and Cloud Forensics.
- Human Computer Interaction security
- Cloud Computing Security
- AIoT for ICT4D



Background





Established in 2012, the research group focuses on making the internet more secure and a safe place for civil society and government.

A team of researchers who are constantly seeking partnership with all stakeholders who share the same aspiration of ensuring online activities are secured.

This group among other things promote publications, post graduate research projects as well as enable cooperation with industry and international partners.

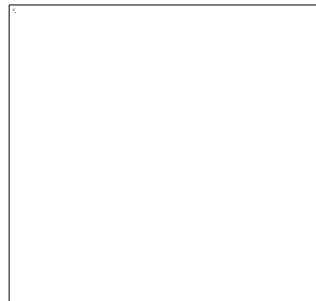
We always believe in engaging with the community around to propose viable solutions based on understanding their needs.





Digital Forensics and Information Security Research Cluster

Focus areas





AIoT for ICT4D





Cluster Focus Areas





Among other things, our research group is working on the following research areas:-

- Child Online Protection research.
- Security of organizational and industry operations research at various levels of complexity.
- The Internet of Things, AI and Emerging technologies security.
- Digital, Mobile and Cloud Forensics.
- Human Computer Interaction security
- Cloud Computing Security



Digital Forensics and Information Security Research Cluster

Members

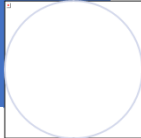
- PhD IT (User Experience/Information Security), MSc Computer Science: Cluster leader

Fungai Bhunu Shava
(FBS)



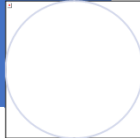
- PhD Computer Science (Critical Infrastructure security), MIT

Mercy Bere Chitauro (MC)



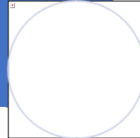
- PhD Informatics (HCISec) student, MSc Business Informatics, BIT: Syst. Admin and Netw, ND: Information Technology

Gabriel Nhinda (GN)



- PhD Computer Science (Cloud Forensics) student, MSc Computer Science

Isaac Nhamu
(IN)



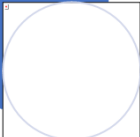
- PhD Computer Science student (AI and Security)
- Master of IT, PGCHE, CCNI
- Digital Forensics,

Julius Silaa
(JS)



- PhD student, Master of Computer Science, BHons Communication Networks, Cloud computing, E governance in education, Systems security

Shadreck Chitauro (SC)



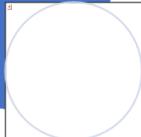
- MSc (Network Security), BHons Communication Networks: Network Security, InfoSec

Edward Nepolo (EN)



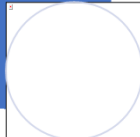
- PhD: Cybersecurity – student Master of Computer Science(Network Security) , B Compt Sc Honours Information Security, B Compt Sc Systems Administration, CCNA1

Uakomba Mbasuva (UM)



- PhD Computer Science, Postdoctoral Fellow (Internet of Things (IoT) & AI) Security, MSc Computer Science

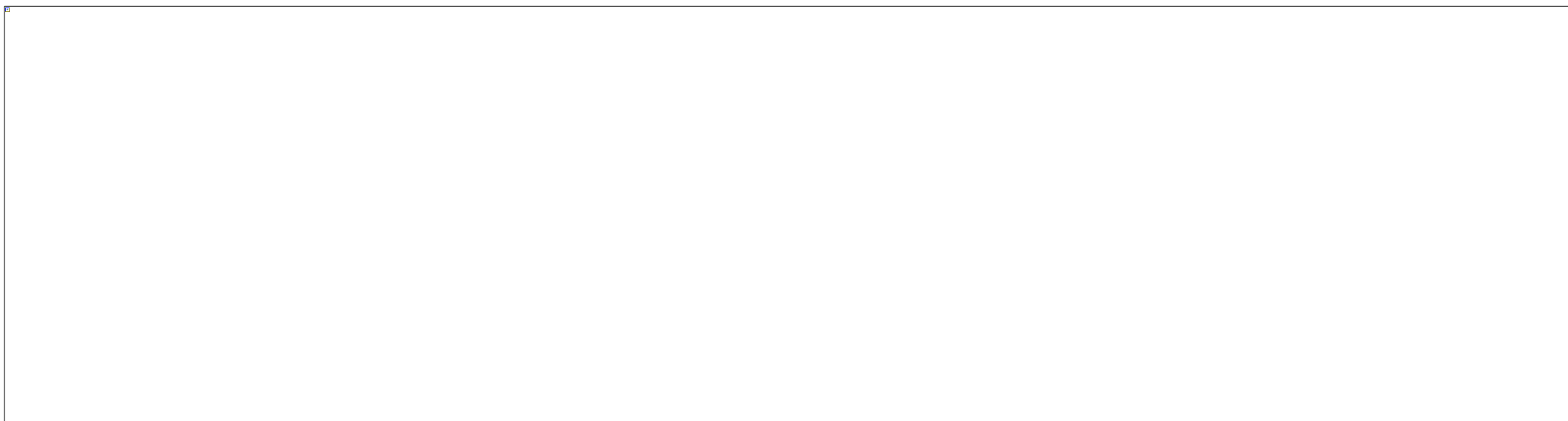
Attlee M. Gamundani
(AMG)

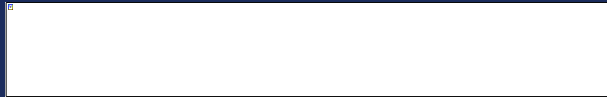




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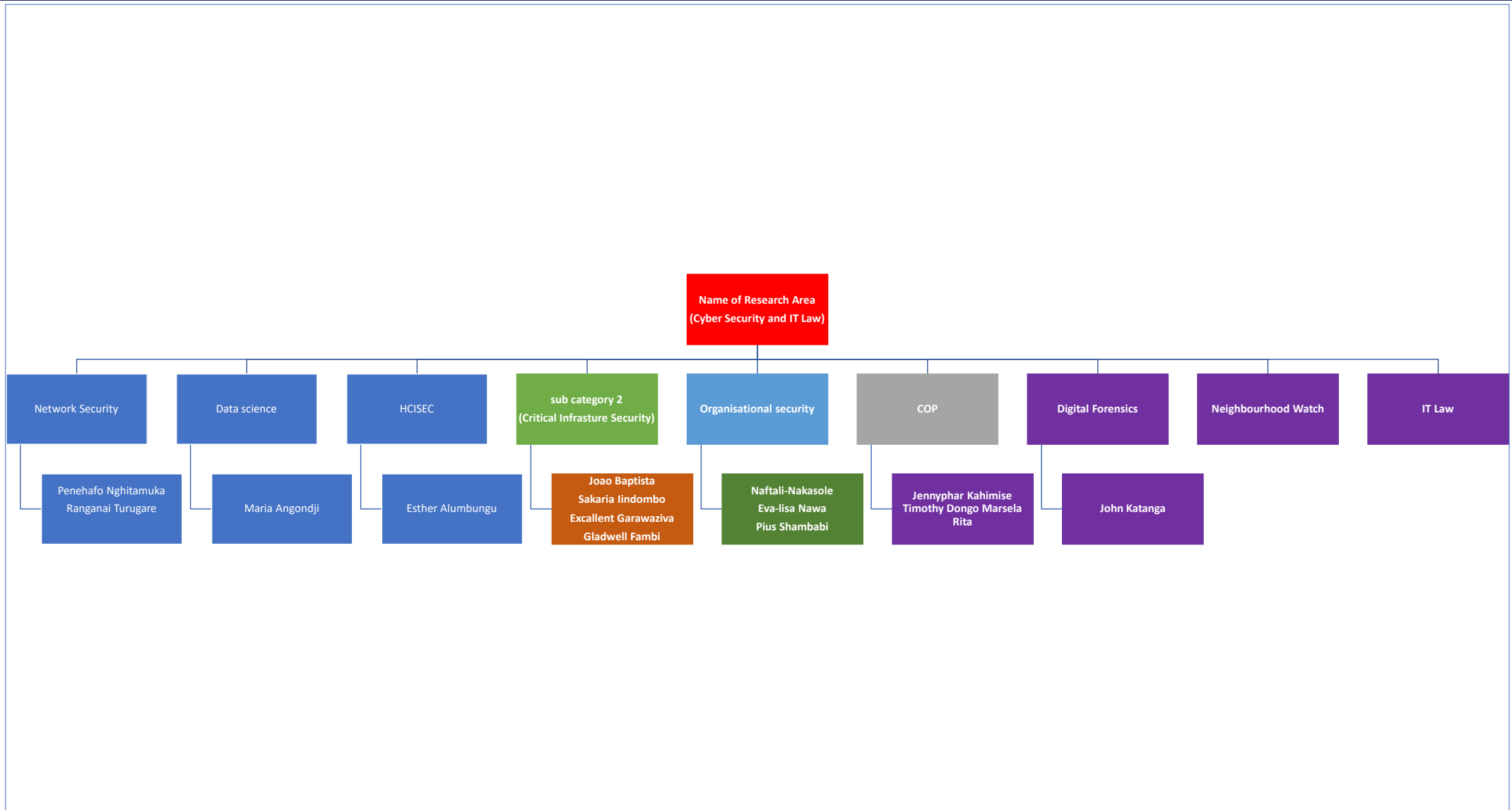
Research interests FBS





Research Interests MC

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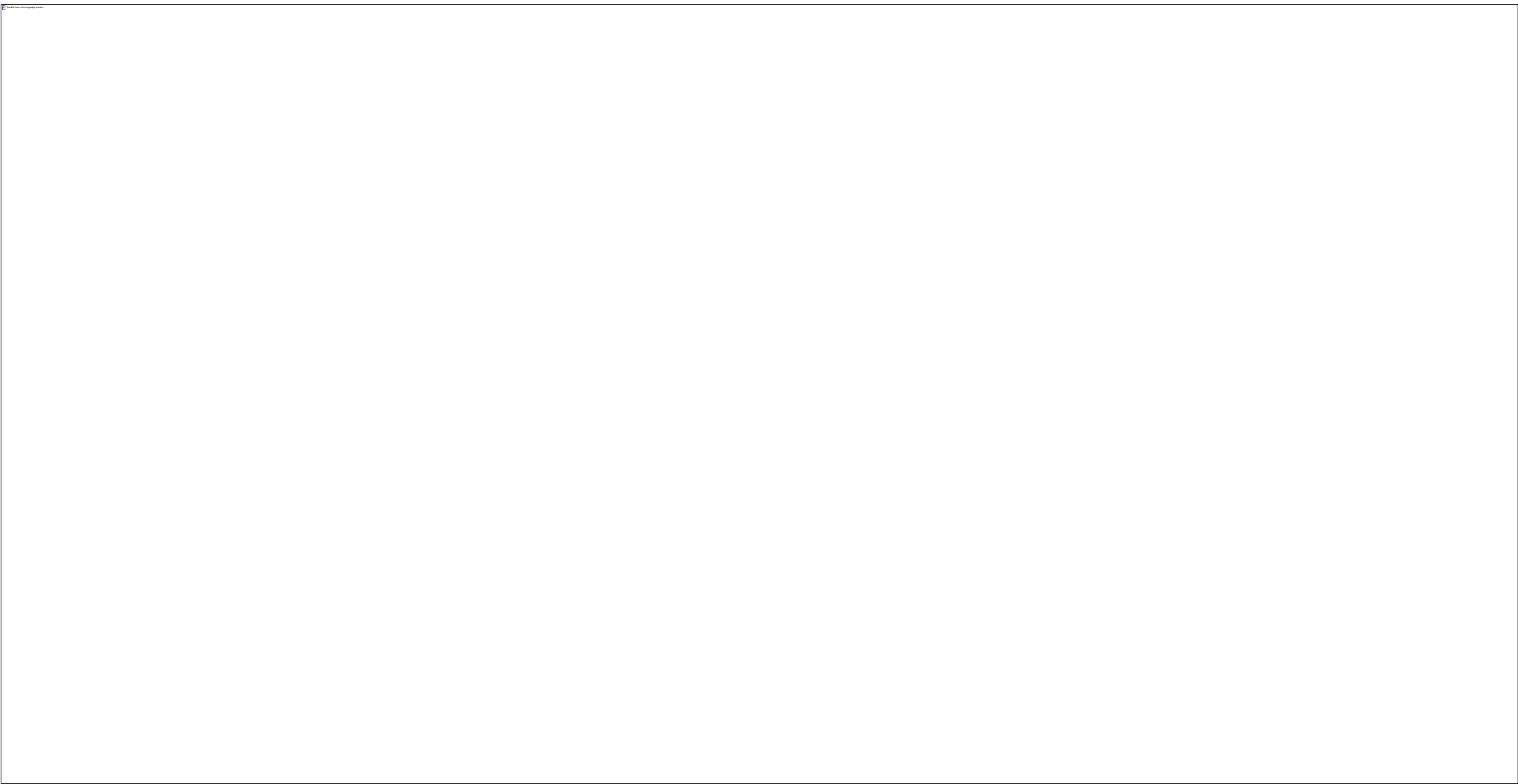




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Research Interests

Julius Silaa

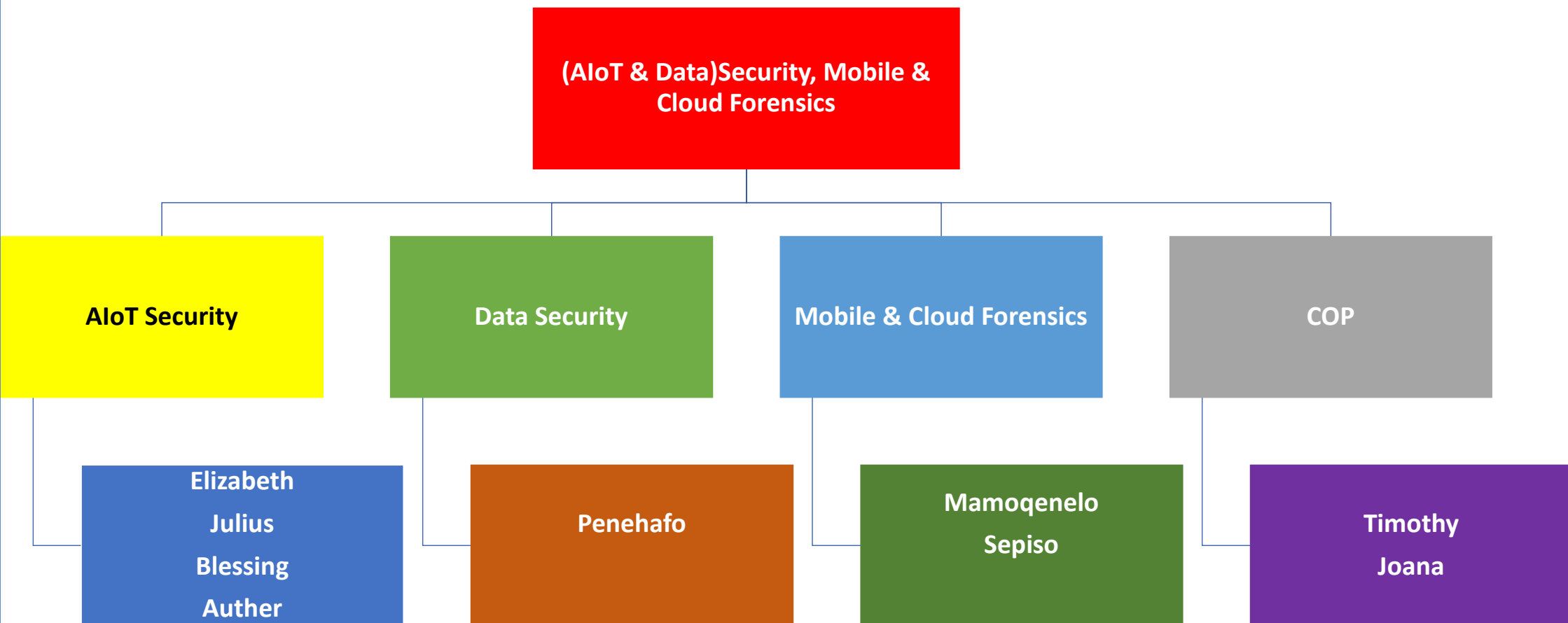




Digital Forensics and Information Security Research Cluster

Research Interests

Attlee M. Gamundani





Digital Forensics and Information Security Research Cluster

Research Interests

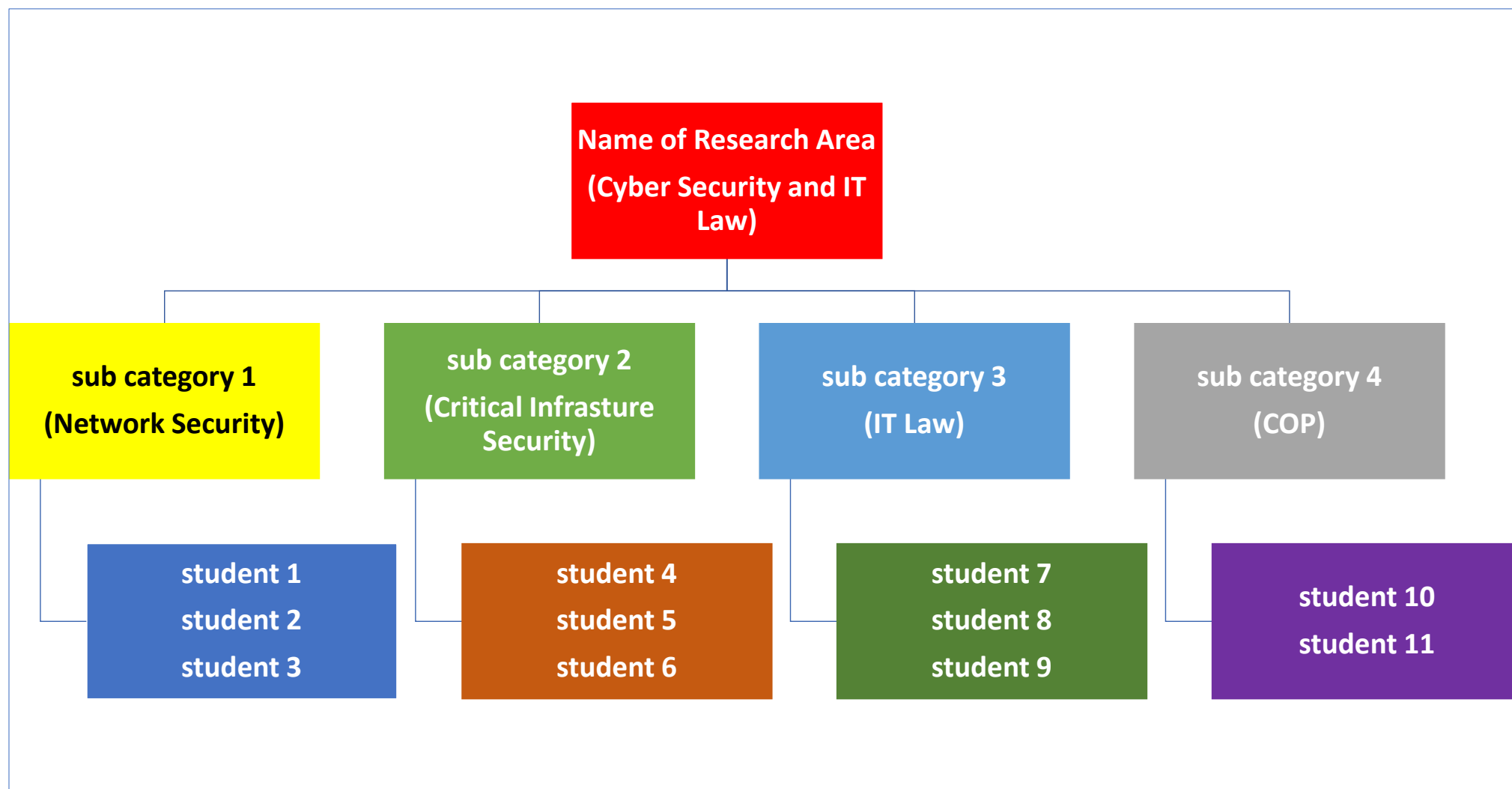
Isaac Nhamu

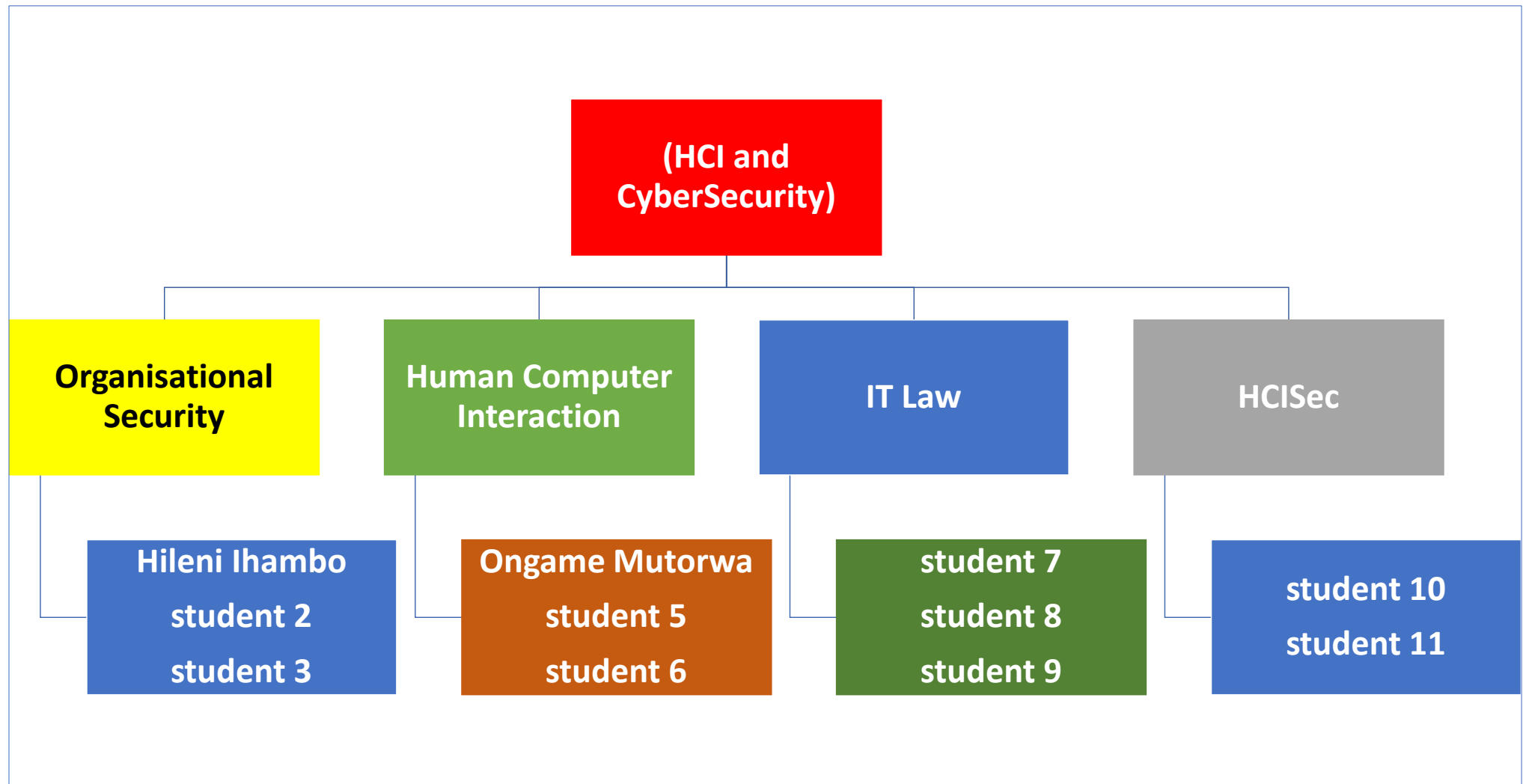


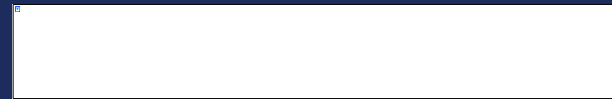


Digital Forensics and Information Security Research Cluster

Research Interests Shadreck Chitauro

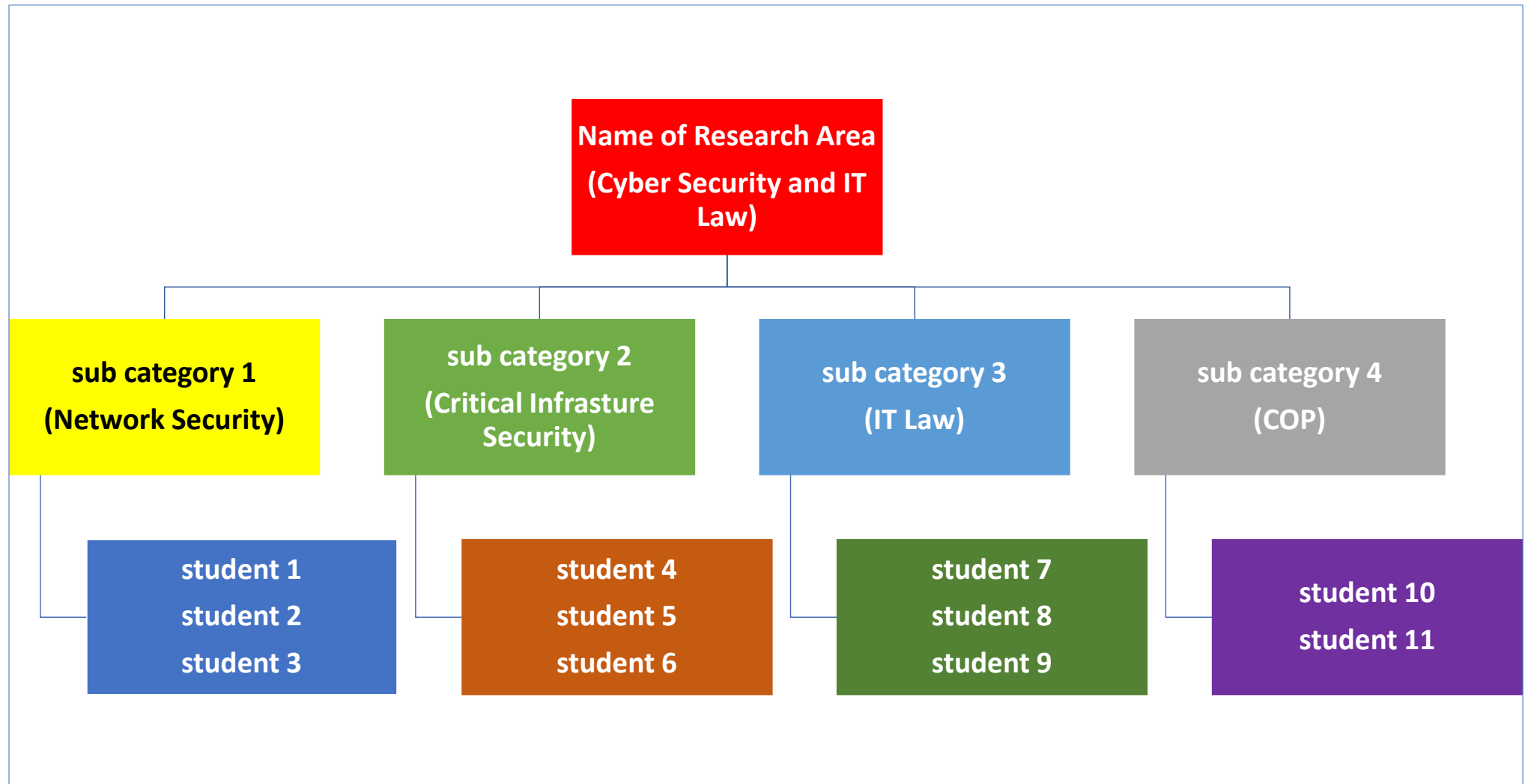






Digital Forensics and Information Security Research Cluster

Research Interests Edward Nepolo

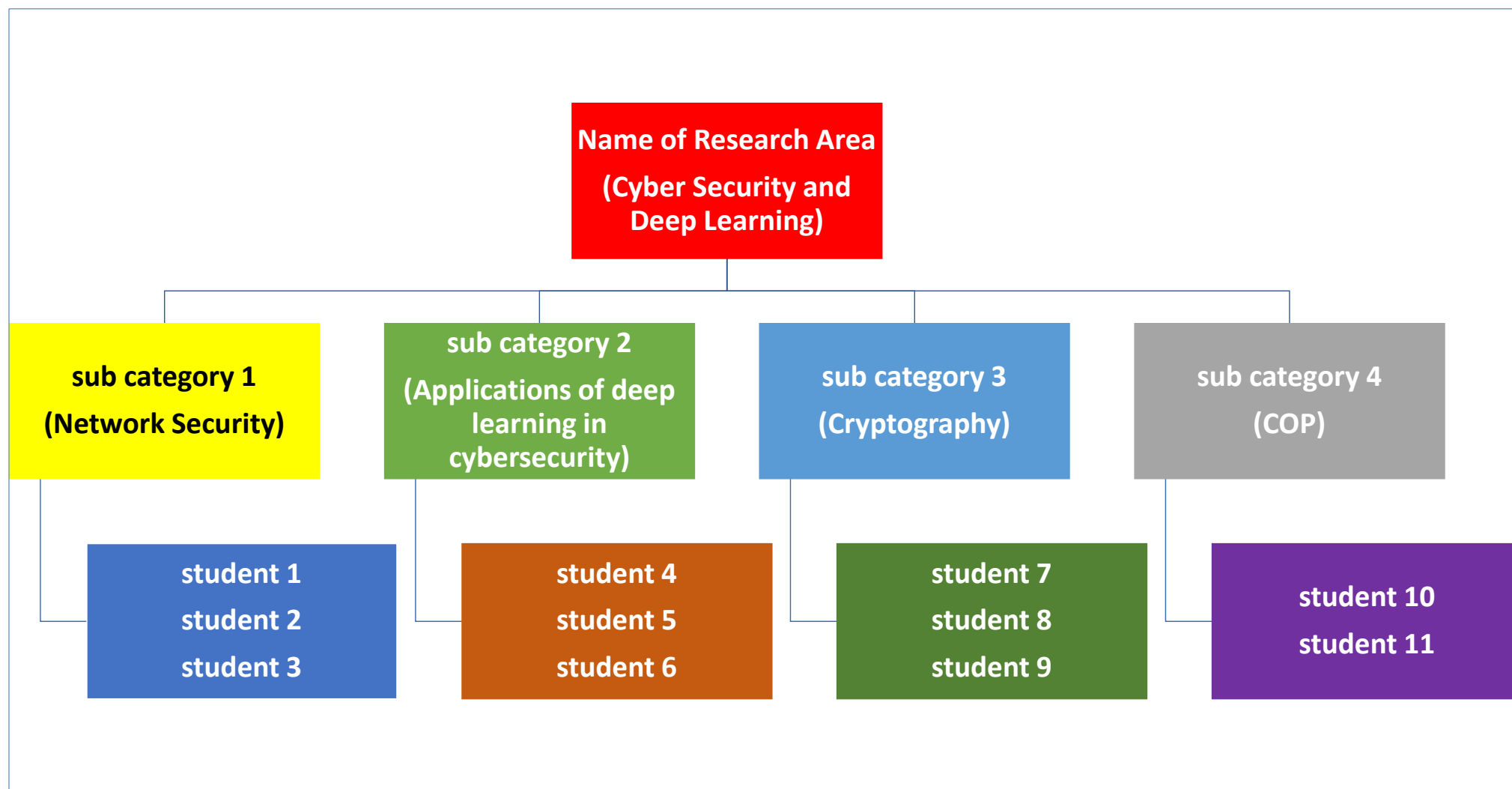




Digital Forensics and Information Security Research Cluster

Research Interests

Uakomba E. Mbasuva





Honours Graduates so far

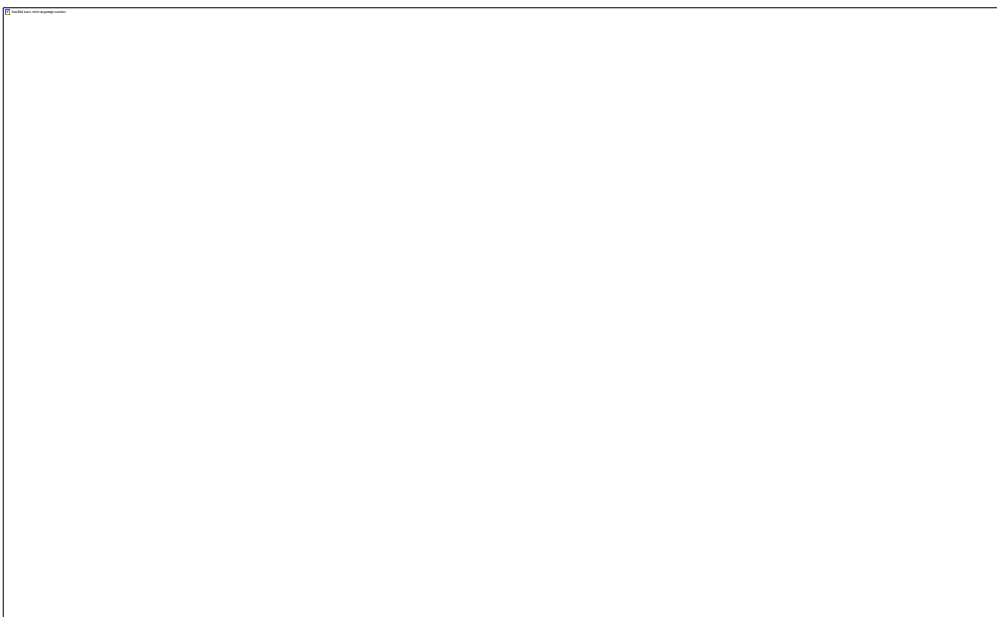
Digital Forensics and Information Security Research Cluster

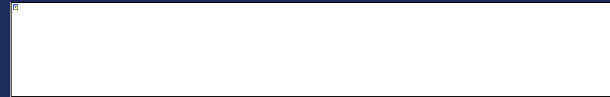
Joel Eelu (2011)	Liina Tshilunga (2011)	Martha Shipanga (2012)	Paulus Shigwedha (2013)	Diana (2014)	Dea-da Ondeja kandjii (2014)	Moses N. Ndiikongela(2014)	Martin Kanyangela(2014)
Julia Nambili (2014)	Samuel Olusegun Jegede (2014)	Martin Hamukwaya (2015)	Barminas Steve Zeraua (2015)	Klaudia Uuzombala (2015)	Rodney Koujo(2015)	Konjeni Elago(2015)	Andrisia Aluteni (2015)
Christian Mingeli (2016)	Lucia Munango (2016)	Joseph Andreas(2016)	Ruben lipinge Stefanus (2016)	Rosetha Kays (2016)	Aliyu Dan Sharif Rabi (2016)	Adolf Samakupa (2017)	Aruni Kanyimba (2017)
Nathaniel Kanyinga (2017)	Joshua Kamwi (2017)	Ivy Mbala (2017)	Selestina Amandus (2018)	Victoria Ndafapawa Haidula (2018)	Lucas Mbambo Nekare (2018)	Mathew M. Mbwalala (2018)	Eunike T. Ndemwiimba (2019)
Pius Shambabi (2019)	Margaret Ndapewa Paulus (2019)	Petrus Kahare Sikongo (2019)	Marsela Rita (2020)	Joana Nkoma (2020)	Anène Hoff (2020)	Venessa Tjizumaue (2014)	Juso-Ilyod Ekandjo (2014)
Kafita Kalenikumwe (2015)	Stanford Musarurwa (2015)	Ben Mwashekuna (2016)	Elizabeth Shipena (2016)	Esther Shigwedha (2016)	Ndeshihafela Haita (2016)	Salmon Msati (2017)	Moses Masaku (2018)
Iina-Marie Shiweva (2018)	Thomas Shomwele (2019)	Raimbert Namukuwa (2019)	Abisai Kamati (2019)	Diina Alfeus (2020)	Kleopas Kapalanga (2020)	Daymien Harris (2020)	Petrus Kambinda (2020)



..Honours Graduates so far

Digital Forensics and Information Security Research Cluster





Digital Forensics and Information Security Research Cluster

MSc Graduates so far

Apr. 2016

- Licky Erastus

Oct. 2017

- Cephas Pahla

Apr. 2018

- Stanford Musarurwa

Apr. 2019

- Diana Tjirare

Apr. 2019

- Eunike Nghihalwa

Apr. 2019

- Victoria Shakela

Apr. 2019

- Joris Ikany

2020

- Mamoqenelo Morolong

2021

- Jennyphar Kahimise

2021

- Paulus Iita Shigwedha

2022

- Lonnie Matsa

2022

- Eva-Lisa Nawa

2022

- Gabriel Musa

2022

- Naemi Gerson



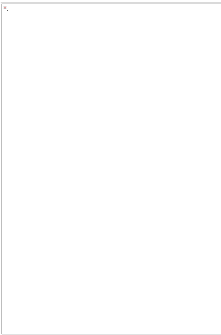
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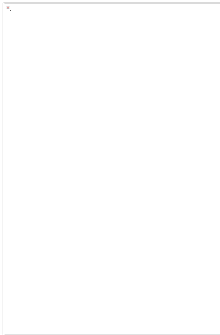


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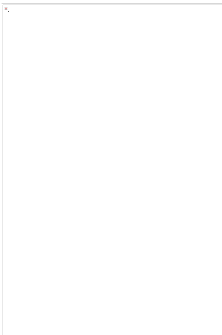
PhD Graduates this far



Dr Licky R. Erastus
(2019)



Dr Attlee M.
Gamundani (2019)



Dr Mercy
Chitauro(2020)



Digital Forensics and Information Security Research Cluster

Current MSc Students



Naemi
Gerson

Ester
Alumbungu

Eva Lisa
Nawa

Vicentius
Caley

Gabriel
Musa

Lonnie
Matsa

Juso Lloyd
Ekandjo

Raimbert
Namukuwa

Joshua
Kamwi

Hloaele
Komete

Hileni
Ihambo

Elizabeth
Shipena

Penehafo
Nghitamuka

Blessing
Sibanda

Auther
Makuvaza

Pius
Shambabi

Joao
Baptista

Naftali
Nakasole





Digital Forensics and Information Security Research Cluster

Current PhD students

Maria Kanyama

Ranganai Turugare

Timothy Dongo

Julius Silaa

Tendai Marengereke

Gabriel Nhinda

Mamoqenelo Morolong

Jennyphar Kahimise



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[Shihepo, E., Bhunu Shava, F., & Chitauro, M. \(2022\). Review of BYOD security awareness within Namibian enterprises. Presented at the 1st Zimbabwe Conference on Information Communication & Technology \(ZCICT\) 2022 \(9 to 10 November\).](#)

Musa, G., Bhunu Shava, F. (2023). Feature Selection for Evaluating Deep Reinforcement Learning DDoS attack Detection Mechanisms using the CICIDS2017 Dataset (November 23, 2022). Available at SSRN: <https://ssrn.com/abstract=4331696> or <http://dx.doi.org/10.2139/ssrn.4331696> .

Nawa, E.T., Bhunu Shava, F., Chitauro, M. (2023). Developing a Cybersecurity Framework for the Namibian Banking Institutions (November 23, 2022). Available at SSRN: <https://ssrn.com/abstract=4332733> or <http://dx.doi.org/10.2139/ssrn.4332733>

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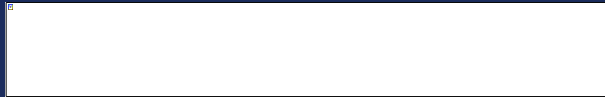
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City Resort, 19 – 21 October 2022.

Kavikairua J., **Bhunu Shava, F.** (2022). Algorithm to Impact Children's Online Behaviour and Raise their Cyber Security Awareness. International Conference on Artificial Intelligence, Big Data, Computing and Data Communication Systems (icABCD), 2022, pp. 1-7, doi: 10.1109/icABCD54961.2022.9856017. <https://ieeexplore.ieee.org/document/9856017>

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Shigwedha, P., & **Bhunu Shava, F.** (2021). Designing a Secure Integration Model for E-Governance Platforms. 3rd International Multidisciplinary Information Technology and Engineering Conference (IMITEC), 2021, pp. 1-6, doi: 10.1109/IMITEC52926.2021.9714556. <https://ieeexplore.ieee.org/document/9714556>

[Nawa, E., Chitauro, M. & **Bhunu Shava, F.** \(2021\). Assessing Patterns of Cybercrimes Associated with Online Transactions in Namibia Banking Institutions' Cyberspace," 2021 3rd International Multidisciplinary Information Technology and Engineering Conference \(IMITEC\), 2021, pp. 1-6, doi: 10.1109/IMITEC52926.2021.9714697. <https://ieeexplore.ieee.org/document/9714697>](https://ieeexplore.ieee.org/document/9714697)

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1. Shilongo, V.G., Morolong, M., & Bhunu Shava, F. (2021). Designing an Email Security Awareness Program for State-Owned Enterprises in Namibia. To be presented at the 5th International Conference on ICT for Intelligent Systems (ICTIS – 2021) in Ahmedabad, India: April 23-24, 2021.
2. Shambabi, P.T., Musarurwa, S., & Bhunu Shava, F. (2021). Assessing Organisational Information Security Culture Among Workforce in Universities: A Case of Namibia. To be presented at IST-Africa 2021. Online: 10-14 May 2021. ISBN: 978-1-905824-66-3, www.IST-Africa.org/Conference2021
3. Silaa, J., Morolong, M., Gamundani, A. M., & Bhunu Shava, F. (2021). End User Online Security & Privacy Protection. Presented at IST-Africa 2021. Online: 10-14 May 2021. ISBN: 978-1-905824-66-3, www.IST-Africa.org/Conference2021
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5. Namukuwa, R., Chitauro, M. & Bhunu Shava, F. (2021). Existing Optimal Water Network Designs for Addressing Demand and Supply Needs for NamWater. Presented at IST-Africa 2021. Online: 10-14 May 2021. ISBN: 978-1-905824-66-3



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1. Erastus, L. R., Jere, N. R., & Bhunu Shava, F. (2020). Secure Information Infrastructure Framework Components for a Smart City: A Case Study of Windhoek. In P. Cunningham, & M. Cunningham (Ed.), IST-Africa 2020. Kampala, Uganda: IST-Africa 2020 Conference Proceedings, IMC International Information Management Corporation. 18-22 May 2020. ISBN: 978-1-905824-64-9, INSPEC Accession Number: 19810155. <https://ieeexplore.ieee.org/document/9144030> , www.IST-Africa.org/Conference2020
2. Morolong, M., Gamundani, A. M., & Bhunu Shava, F. (2020). Bring Your Own Device (BYOD) Information Security Risks: Case of Lesotho. 15th International Conference on Cyber Warfare and Security (ICCWS 2020). Norfolk, Virginia. 12 - 13 March 2020. DOI:10.34190/ICCWS.20.101, <https://www.proquest.com/openview/9ee8f7fcf05bacdc74d1dc9ef2c17b1f/1?pq-origsite=gscholar&cbl=396500>
3. Kahimise, J., & Bhunu Shava, F. (2020). An Analysis on Social Networking Threats. 15th International Conference on Cyber Warfare and Security (ICCWS 2020). Norfolk, Virginia. 12 - 13 March 2020. DOI:10.34190/ICCWS.20.127, <https://www.proquest.com/conference-papers-proceedings/analysis-social-networking-threats/docview/2455894388/se-2?accountid=201395>
5. Gerson, N., & Bhunu Shava, F. (2020). A Review of Security System Assessment Tools Suitable for eHealth in Namibia. 15th International Conference on Cyber Warfare and Security (ICCWS 2020). Norfolk, Virginia. 12 - 13 March 2020. DOI:10.34190/ICCWS.20.115, <https://www.proquest.com/docview/2390159170?pq-origsite=gscholar&fromopenview=true>
6. Erastus, L. R., Jere, N. R., & Bhunu Shava, F. (2020). A Secure Smart City Infrastructure framework for e-Service Delivery within a Developing Country: A case of Windhoek in Namibia. Fifth International Congress on Information and Communication Technology (ICICT 2020). London, UK. February 20-21, 2020.



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1. Chitauro, M., Bhunu Shava, F., Gamundani, A.M., Nhamu, I., Chitauro, S., & Phillips, A. (2019). Overcoming Barriers: Enhancing Cyber Security Education and Awareness Through Cloud Based Cyber Security Competitions. In Proceedings of UbuntuNet-Connect. Zanzibar, Tanzania, UbuntuNet – ConnectUbuntu <https://repository.ubuntunet.net/handle/10.20374/290>
2. Chitauro, M., Muyingi, H., John, S. & Chitauro, S (2019). A Bio-Immunology Inspired Industrial Control System Security Model. In the proceedings of First International Conference on Sustainable Technologies for Computational Intelligence, 29 -30 March, Jaipur, Rajasthan, India. In Press
3. Chitauro, M., Muyingi, H., John, S. & Chitauro, S (2019). A Survey of APT Defence Techniques. In Proceedings of 14th International Conference on Cyber Warfare and Security, 28 February – 1 March, Stellenbosch, South Africa. In press.
4. Gamundani, A., Nyakabau, K., Nepolo, E. Congestion Control in Wireless LAN using the Predictive Control Model, 14th International Conference on Electronics & Communications, Renewable Energy and IoTs: Vision 2040, 7-8 September 2019 at Chandigarh, India.
5. Kahimise, J., & Bhunu Shava, F. (2019). An analysis of children's online activities and behaviours that expose them to cybercrimes. 27th Telecommunications Forum (TELFOR) 2019. pp. 1-4. Serbia, Belgrade, IEEE-Xplore. <https://ieeexplore-ieee-org.eresources.nust.na/stamp/stamp.jsp?tp=&arnumber=8971089>
6. Morolong, M., Bhunu Shava, F., & Gamundani, A. M. (2019). Sensitive Data Leakage Through Android Applications in a Bring Your Own Device (BYOD) Workplace. In P. Cunningham, & M. Cunningham (Ed.), IST-Africa 2019. Nairobi: IST-Africa 2019 Conference Proceedings, IMC International Information Management Corporation. DOI: 10.23919/ISTAFRICA.2019.8764833 . INSPEC Accession Number: 18849472. ISBN: 978-1-905824-62-5 www.IST-



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1. Chikohora, E., Gamundani, A., & Chikohora, T. (2018). Adaptive Algorithm for Parameterization of Feature Extraction Techniques in Remote Sensing Image Processing. In IST-Africa 2018 Conference Proceedings (pp. 1–10). Gaborone: IIMC International Information Management Corporation. . ISBN: 978-1-905824-59-5.
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3. Gamundani, A. M., & Nekare, L. M. (2018). A Review of New Trends in Cyber Attacks: A Zoom into Distributed Database Systems. In IST-Africa 2018 Conference Proceedings (pp. 1–9). Gaborone: IIMC International Information Management Corporation, 2018.
4. Gamundani, A. M., Phillips, A., & Muyingi, H. N., (2018). An Overview of potential authentication threats and attacks on Internet of Things (IoT): A focus on Smart Home Applications. In Proceedings of the 11th IEEE International Conference on Internet of Things (iThings-2018) Halifax, Canada. <http://cse.stfx.ca/~iThings2018/acceptedlist.htm>
5. Gamundani, A. M., Phillips, A., & Muyingi, H.N., (2018). A Review and Costing of Lightweight Authentication Schemes for Internet of Things (IoT): Towards design of an authentication architecture for smart home applications. In Proceedings of the 6th International Workshop on Applications and Techniques in Cyber Security 2018 (ATCS 2018), in Conjunction with SecureComm 2018, Singapore.
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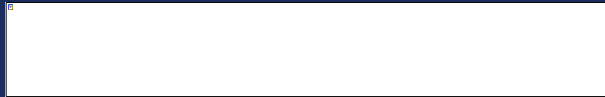
Publications 2018

1. Gamundani, A.M., Phillips, A., & MUYINGI, H.N. (2018). Privacy Presevation and Security Dilemma: Relationship proposition for IoT authentication. In Proceedings of the International Conference on Recent Innovations in Electrical, Electronics & Commununication Engineering (ICRIEECE) (pp. 45). 27-28 July 2018, KIIT University, Bhubaneswar, India
2. Maoneke, P.B., Bhunu Shava, F., Gamundani, A.M., Bere-Chitauro, M & Nhamu, I (forthcoming 2018). ICTs Use and Cyberspace Risks Faced By Adolescents in Namibia. In the proceedings of AfriCHI: The 2nd African Conference for Human Computer Interaction, Windhoek, Namibia, 3-7 December 2018. ACM DL
3. Nghihalwa, E., & Bhunu Shava, F. (2018). A Secure Cloud Adoption Framework (SCAF) for the Namibian government information technology departments. In Proceeding of World Conference on Smart Trends in Systems, Security and Sustainability (WS4 2018): London, UK
4. Nghihalwa, E., & Bhunu Shava, F. (2018). An assessment of cloud computing readiness in the Namibian government Information Technology departments. (forthcoming) at the 19th IEEE Mediterranean Electrotechnical Conference MELECON '18 (pp.92-97). Marrakesh: IEEE.
5. Nhamu, I. & Gamundani, A.M (forthcoming 2018). Using virtualised laboratory environments to enhance hands-on learning experiences in cyber security courses. In Proceedings of UbuntuNet-Connect. 22-23 November, Hotel verde, Zanzibar, Tanzania.



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1. Erastus, L. R., Jere, N. R., & Bhunu Shava, F. (2017). A Security Model for Namibian Government Services. In P. Cunningham, & M. Cunningham (Ed.), *IST-Africa 2017*. Windhoek: IST-Africa 2017 Conference Proceedings, IIMC International Information Management Corporation. <http://ieeexplore.ieee.org/document7892633/>, DOI: 10.1109/ICTBIG.2016.7892633,
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3. Pahla, C., Bhunu Shava, F., & Musingi, H. N.-N. (2017). Analysis of group mobility for opportunistic campus network. *IEEE Tensymp*. Cochin, Kerala: IEEE. doi:978-1-5090-6255-3/17/
4. Tjirare, D., & Bhunu Shava, F. (2017). Gap Analysis of the ISO/IEC 27000 Standard Implementation in Namibia. *IST-Africa, 2017*. Windhoek: IST-Africa 2017 Conference Proceedings, IIMC International Information Management Corporation.
5. Bhunu Shava, F., Gamundani, A.M., Chitau, M., Nhamu, I., & Mikka, J. (2016). A frame-work for addressing child online protection issues in Namibia: Societal Governance. 2016 Sanord Conference, Uppsala Universitet, Uppsala Sweden, 7-9 September 2016. <http://akkonferens.slu.se/sanord2016/wpcontent/uploads/sites/24/2015/11/SANORD->



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1. Bhunu Shava, F., Gamundani, A.M., Chitauro, M., Nhamu, I., & Mikka, J. (2016). A frame-work for addressing child online protection issues in Namibia: Societal Governance. 2016 Sanord Conference, Uppsala Universitet, Uppsala Sweden, 7-9 September 2016.
<http://akkonferens.slu.se/sanord2016/wpcontent/uploads/sites/24/2015/11/SANORD->
2. Bhunu Shava, F., Bere, M., Nhamu, I. & Gamundani, A. M. (2015). Desinging A Knowledge Based Economy Model For Developing Countries With A Focus On Security. (SANORD 2015, Windhoek, 1-3 December, 2015)
3. Gamundani, A.M., Nhamu, I., Bhunu Shava, F., & Bere, M. (2015). Social networking as both a cause for concern and a window for hope - A focus on child online protection in Namibia (ACCC2015), Port Elizabeth, 3-5 November 2015.
4. Erastus, L., Jere, N.R., & Bhunu Shava, F. (2015). Multimodal Biometrics for better security: A case of Namibian Government Departments. Natitonal Research Symposium, Hilton Hotel: Windhoek.
5. Erastus, L. R., Jere, N. R., & Bhunu Shava, F. (2015). Exploring Challenges of Biometric Technology Adoption: A Namibian Review (ETNCC2015). *International Conference on Emerging Trends in Network and Computer Communication* (pp 91- 94). Windhoek: IEEE Xplore.
6. Bhunu Shava, F. and van Greunen, D. *Designing user security metrics for a security awareness at Higher and Tertiary institution*. Proceedings of the 8th International Development Informatics Association conference (IDIA 2014). Port Elizabeth. ISBN: 978-0-620-63498-4, 280-296
7. Bhunu Shava, F. and van Greunen, D. *Factors affecting user experience with security features: A case study of an academic institution in Namibia*, IEEE, doi 978- 1-4799-0808-0/13. Presented at ISSA 2013 (International Information Security South Africa Conference)



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3. Bere, M., Bhunu Shava, F., Gamundani , A. M., & Nhamu, I. (2015, November). How Advanced Persistent Threat Exploit Humans. *IJCSI*, 12(6), 170-174.
4. Gamundani, A. M., Bhunu Shava, F., Bere, M., & Kandjimi, H. (2015, April 2015 28). A preliminary survey on childonline protection initiatives: A focus on Namibia. *International Research Journal of Computer Science*, 2(4).

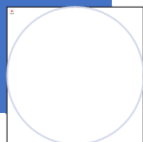


Digital Forensics and Information Security Research Cluster

Contacts

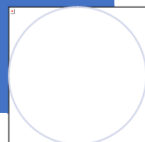
- fbshava@nust.na

Fungai
Bhunu Shava
(FBS)



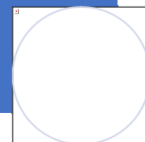
- mchitauro@nust.na

Mercy Bere
Chitauro (MC)



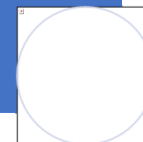
- gnhinda@nust.na

Gabriel
Nhinda (GN)



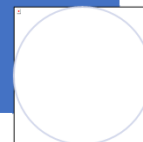
- inhamu@nust.na

Isaac Nhamu
(IN)



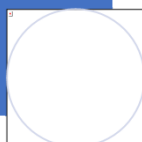
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- Tel:061-20172037

Julius Silaa
(JS)



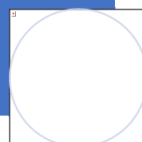
- schitauro@nust.na

Shadreck
Chitauro (SC)



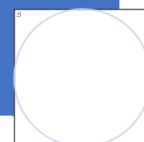
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Edward
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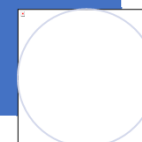
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Digital Forensics and Information Security Research Cluster

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Bhunu Shava

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Learning objectives:

At the end of this session, participants will be able to:

- Recognize benefits of engaging social / scientific research.
define what a 'Research' is.
- List the major characteristics of research/ research proposal;.
- Describe steps of conducting a social / scientific research.
prepare a brief summary of a research proposal.
- Describe the main components of a research report.

Introduction

- Scientific / social research plays a very important role in our efforts to advance technological / knowledge envelope
- Research helps us create **new knowledge** and develop proper tools for the use of existing knowledge.
- Not only does it enable researchers/practitioners to diagnose problems, research also provides evidence for policies and decisions on socio-economic development

What is research?

Research is the systematic collection, analysis and interpretation of data to answer a certain question or solve a problem.

Characteristics of research:

- It demands a clear **statement of the problem**.
- It requires clear objectives and a plan (it is not aimlessly looking for something in order to come across a solution).
- It builds on existing data, using both positive and negative findings.
- New data should be *systematically* collected and analyzed to answer the original research objectives.

Research Proposal – Definitions

“... the meaning of *proposal* ... suggests looking forward, to what the researcher *plans to do in the future*. ...

The proposal lays out the **problem** for research, describes exactly **how** the research will be conducted, and outlines in precise detail the **resources** – both factual and instrumental – the researcher will use to achieve the **desired results**.”

Leedy & Omrod, Practical Research. 8th ed, p. 117, 115.

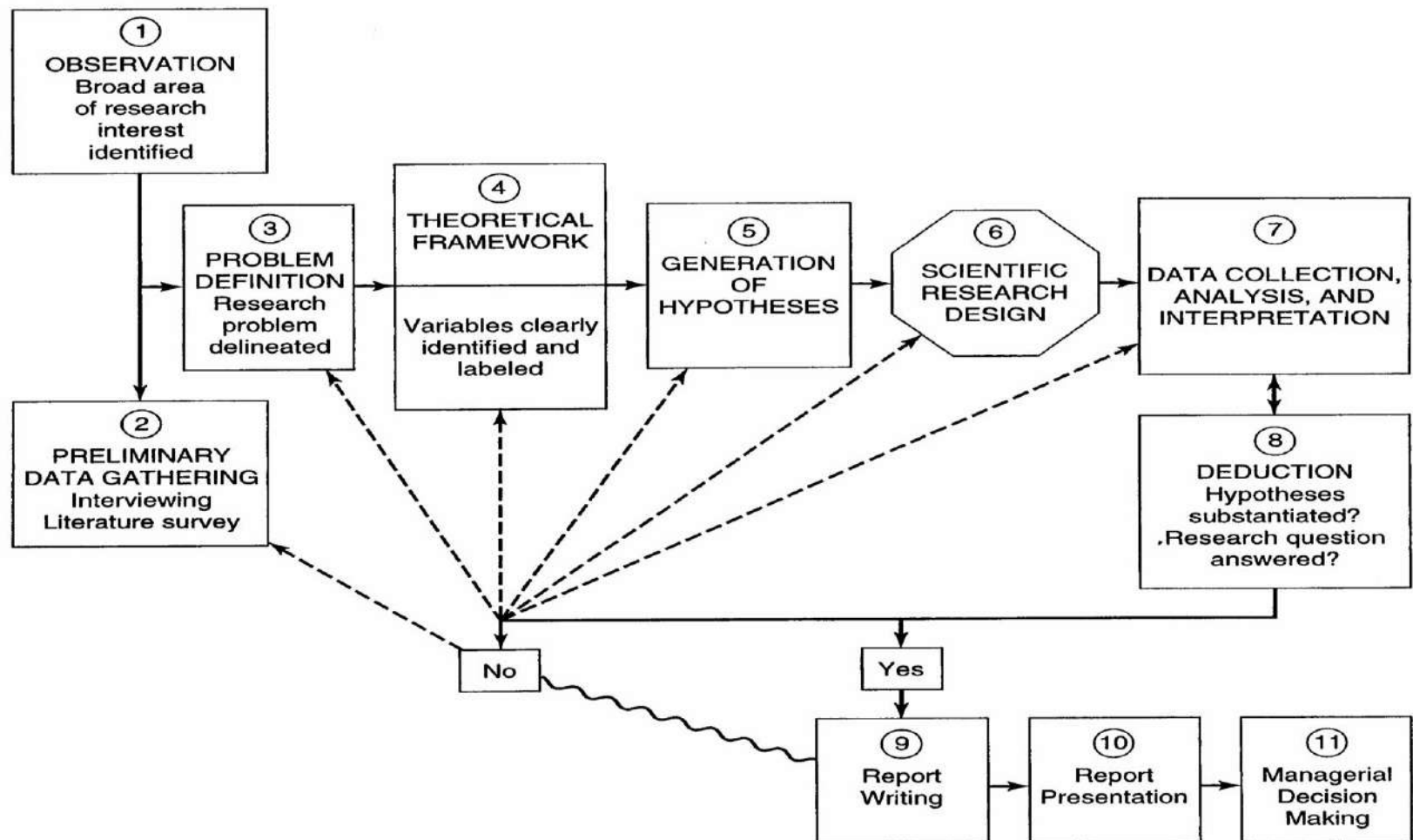
Essentially, to make the reader to understand :-

- What you are going to do
- Rational of the research
- Objectives of the research
- Methodology
- Expected output

What is a proposal ?

- An outline of the research **problem**
 - A statement of the significance of the problem
 - A review of related studies
 - A discussion of the procedures
 - Methods for data collection and analysis
 - A note about the study's limitations
- A good proposal should consist of the first three chapters of the thesis
 - It should :-
 - **Begin with an Introduction - statement of the problem/background information (Chapter 1)**
 - **A review of the Literature (Chapter 2)**
 - **Defining of the Research Methodology (Chapter 3)**

The Research Process For Applied and Basic Research



Steps of conducting a research:

A. Prioritizing and selecting a research topic

B. Review of literature and other existing information

C. Development of a Research Proposal

D. Implementation of study:

- i. Data collection
- ii. Data processing and analysis
- iii. Interpretation of results
- iv. Final report writing
- v. Presenting the results: Scientific, presentation at meetings, seminars, workshops or conferences, and presentation for administrators and policy-makers

Steps of conducting a research:

A. Prioritizing and selecting a research topic

Criteria for selecting a research topic:

1. **Relevance**: The topic you choose should be a priority problem.

Questions to be asked include:

- How large or widespread/depth is the problem?
- Who is affected?
- How severe is the problem?

2. Avoidance of **duplication**:

- Before you decide to carry out a study, it is important that you find out whether the suggested topic has been investigated before, either within the proposed study area or in another area with similar conditions.
- If the topic has been researched, the results should be reviewed to explore whether major questions that deserve further investigation remain unanswered.
- If not, another topic should be chosen

Steps of conducting a research:

B. Literature review

Why is it important to review already available information when preparing for a research?

- It prevents you from **duplicating** work that has been done before.
- It helps you to find out **what others have learned** and reported on the problem you want to study. This may assist you in refining your **statement of the problem**.
- It helps you to become **more familiar** with the various research methods that might be used in your study.
- It should provide you with convincing arguments for **why** your particular research project is needed.
- Learning what is known and unknown
- Learning how field of knowledge was developed (history)
- Showing you understand your field
- Confirming your own research is worthwhile
- Explaining how knowledge is developed over time
- Justifying your research
- Identifying how you will make a contribution

See Example,,,pg 31

Steps of conducting a research:

C. Development of a research proposal

Contents

- a. Title of the research
- b. Introduction: Background information and Statement of the research problem (Scientific justification for the study)
- c. Research objectives
- d. Research hypothesis
- e. Methodology
- f. Work plan
- g. Plan for utilization and dissemination of research results
- h. References
- i. Annexes

C. Development of a research proposal

a. Title of the research proposal

- A good title should be short, accurate, and concise
- It should make the central objectives of the study clear to the reader
- It is important to specify what population will be investigated, and where it will be conducted.
- **E.g. Factors Influencing e-Bidding Adoption in the Public Sector**

C. Development of a research proposal

b. Introduction (Background information and Statement of the research problem)

This section should convince the reader of the relevance of the study

It should provide enough background data for an outsider to understand the different aspects of the problem, or the different factors influencing the problem and the context in which it occurs.

Your review of available literature and reports should further illustrate why the problem is important, not only in your own working area but probably also beyond

Note: Be selective, remembering that this section serves to justify your study, not to display your ability to read literature.

[See example..pg 1](#)

C. Development of a research proposal

c. Research objectives

Research objectives are the goal to be achieved by a research.

Why should research objectives be developed?

The formulation of objectives will help you to:

- Focus the study (narrowing it down to essentials)
- Avoid the collection of data which are not strictly necessary for understanding and solving the problem you have identified
- Organize the study in clearly defined parts or phases
 - Properly formulated, specific objectives will facilitate the development of your research methodology and will help to orient the collection, analysis, interpretation and utilization of data.
 - How should you state your objectives?
- The general and specific objectives should be included as stated in the proposal

Example Objective To examine the User Factors and System Factors that Influence eBidding Utilization among Government Sourcing Officials in Malaysian Public Sector

Example Specific Objective:

1. To identify the variables that may influence the adoption of eBidding by government users
2. To examine the effects of the variables on the adoption of eBidding by government users
3. To propose a framework to analyze the adoption of eBidding by government users
4. To examine if some of the variables have moderating or mediating effects on the relationships established as stated in objective 2

The Use of a Theoretical Framework in a Research Proposal

The **theoretical framework** plays an important role in guiding the entire process of the research study

- **Theories** are constructed in order to explain, predict and master phenomena (e.g. relationships, events, or the behavior). In many instances we are constructing models of reality.
- A theory makes generalizations about observations and consists of an interrelated, coherent set of ideas and models.

Having examined the different kinds of variables that could operate in a situation and how the relationships among these can be established, it is **now possible** to see how we can develop the conceptual model or the theoretical framework for our research.

If the framework is logically sound and substantiated by previous research studies, there is a strong possibility that the predictions or hypotheses evolving from that framework will be supported

The Use of a Theoretical Framework in a Research Proposal

- The **variables** considered relevant to the study should be clearly defined.
- A **conceptual model** that describes the relationships between the variables in the model should be given.
- A **clear explanation** of why we expect these relationships to exist.
- A **conceptual framework** is used in research to outline possible courses of action or to present a preferred approach to an idea or thought.
- The **theoretical framework** elaborates the relationships among the variables, explains the theory underlying these relations, and describes the nature and direction of the relationships.
- The theoretical framework provides the logical base for developing **testable hypotheses**.
- see example pg 64

Example of Theories Employed in Theoretical Framework

E.g. The eBidding adoption study and the model proposed will be based on:

- ❖ **Unified Theory of Acceptance and Use of Technology (UTAUT)**
(Venkatesh et al., 2003)

An Information System (IS) framework for assessing an individual's intention to use an IS technology

- ❖ **Information System Success Model (Delone and Mclean, 2003)**

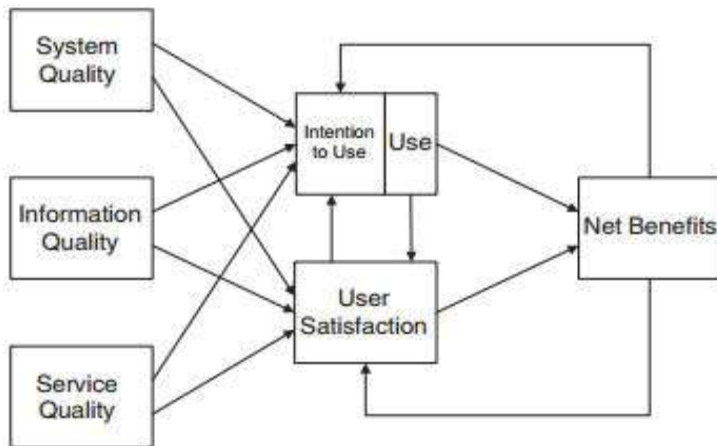
A system success can be evaluated in terms of information, system, and service quality; these characteristics affect the subsequent use or intention to use and user satisfaction

- ❖ **Personal Innovativeness in Information Technology (Agarwal and Prasad, 1998)**

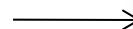
Domain-specific individual trait which reflects the willingness of a person to try out a new information technology

Example : Models related to Technology/Innovation adoption/acceptance/use

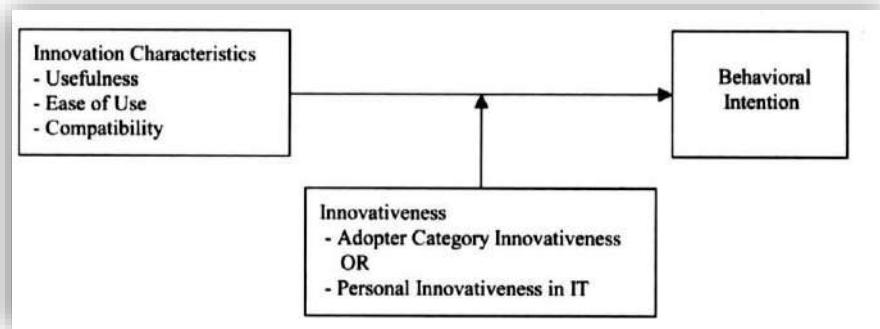
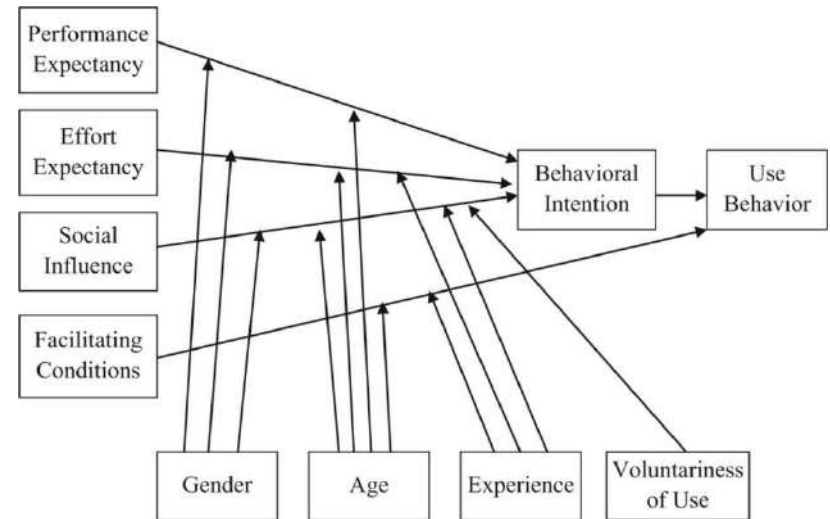
Unified Theory of Acceptance and Use of Technology (UTAUT) (Venkatesh et al., 2003)



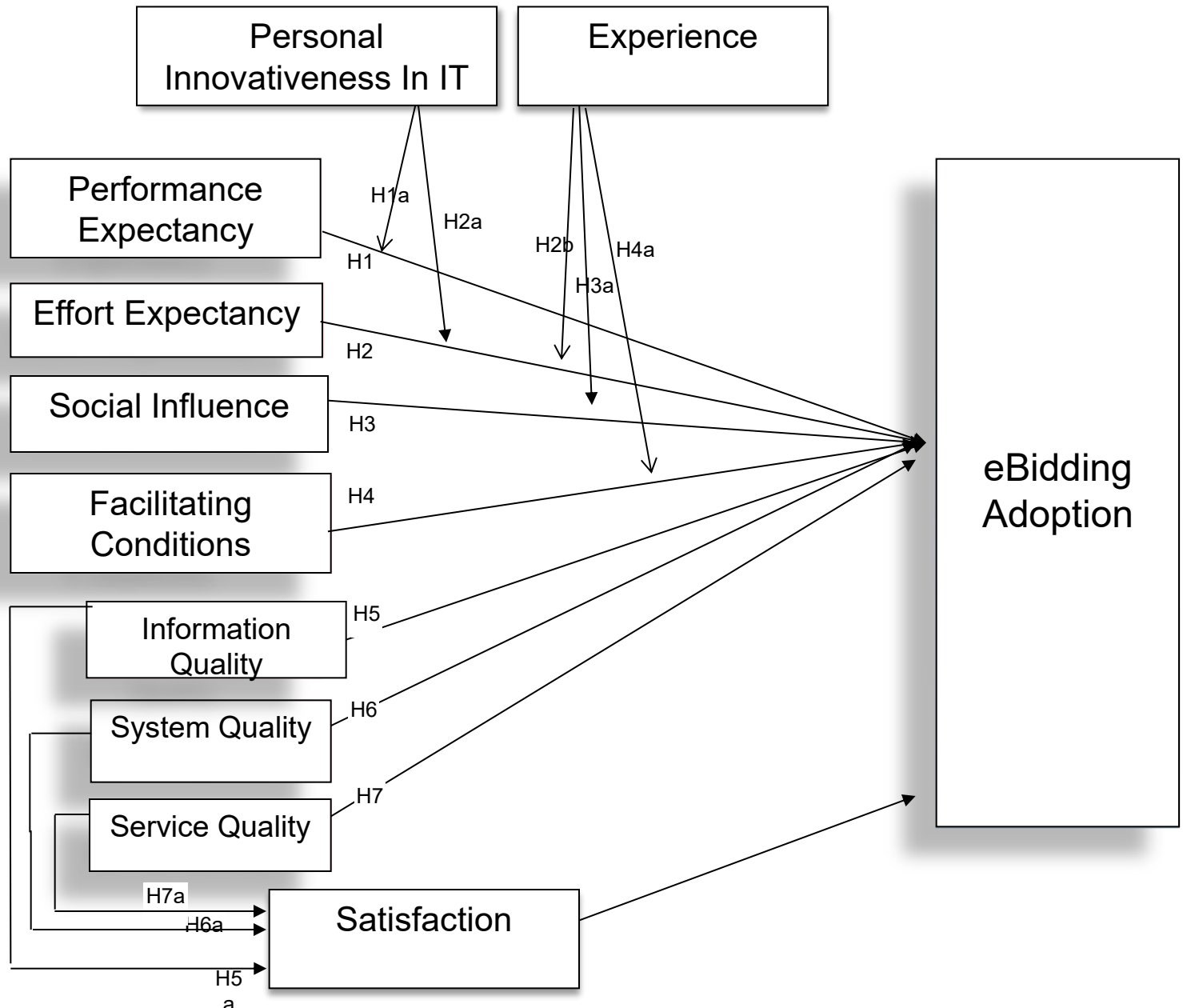
Moderator Model of Personal Innovativeness in Information Technology (PIIT) (Agarwal & Prasad, 1998)



IS Updated Success Model (Delone and Mclean, 2003)



Example :Proposed Research Model



C. Development of a research proposal

d. Research hypothesis

A hypothesis can be defined as a prediction or explanation of the relationship between : one or more independent variables (PREDISPOSING/RISK FACTORS) and one dependent variable (OUTCOME/CONDITION/DISEASE)).

- a logical relationship between two or more variables expressed in the form of a testable statement. A hypothesis, in other words, translates the problem statement into a precise, clear prediction of expected outcomes.
- It must be emphasized that hypotheses are not meant to be haphazard guesses, but should reflect the depth of knowledge, imagination and experience of the investigator must be based on established theory/model
- Therefore, in the process of formulating hypotheses, all variables relevant to the study should be identified.
- See example..pg 76 & 77

C. Development of a research proposal

e. Methodology

This section summarizes the most important points of the research design including:

Variables: It is necessary to identify the variables that will be involved in the research project being designed.

Independent (predisposing/risk factors/antecedents),

Dependent (outcome), **see example**

It is also necessary to specify whether these variables are Numerical (continuous/discrete), or Categorical (ordinal/nominal).

Operational definition of different study variables should be clearly phrased.

The methodology you followed for the collection of your data should be described in detail.

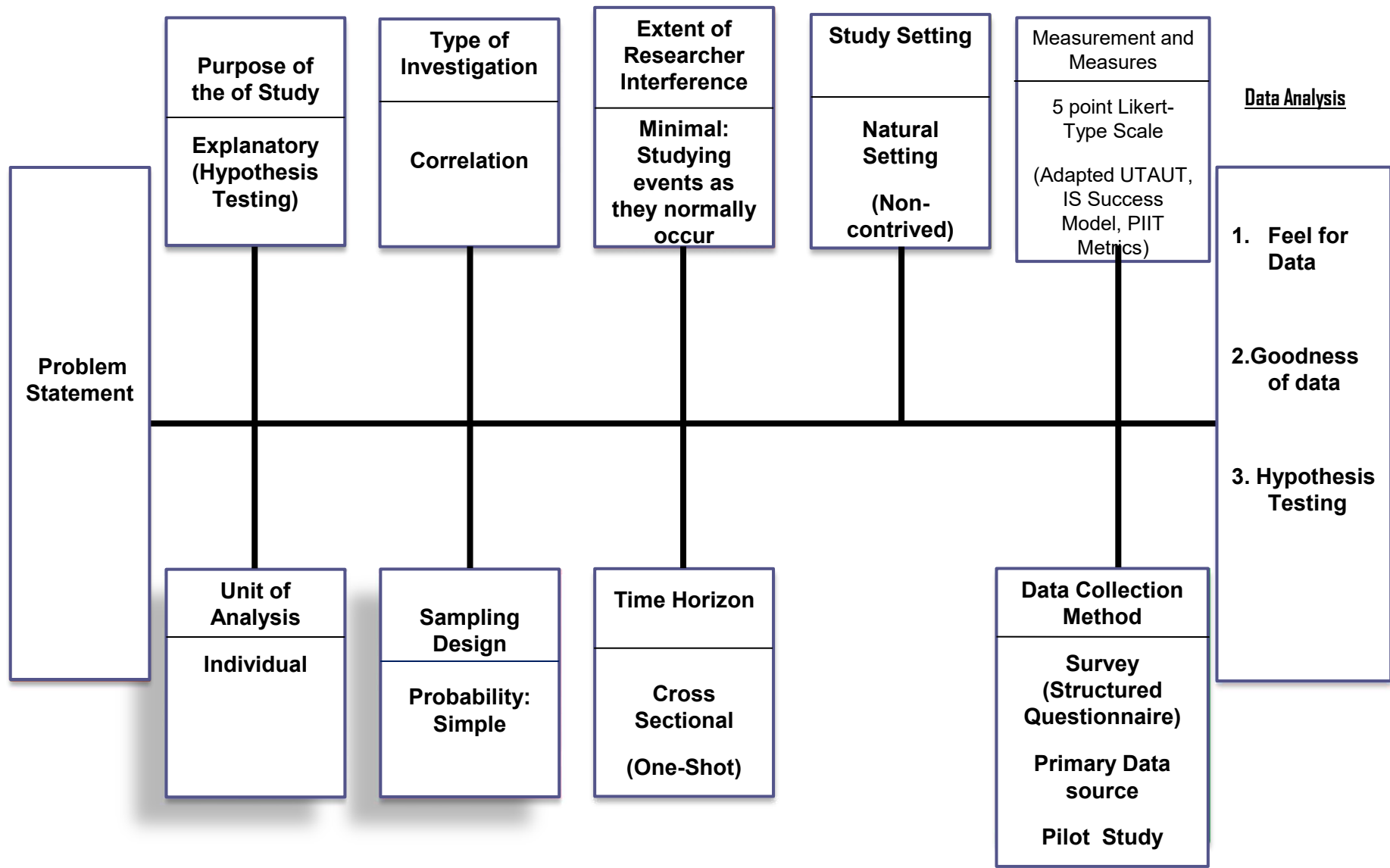
The methodology section should include a description of:

- the study type
- major study variables on which data was collected
- the study population, sampling method and the size of the sample
- data-collection techniques used
- how the data was collected and by whom
- procedures used for data analysis, including statistical tests **(see example)**

Example : Operationalizing Research Instrument

No	Section	Variables	No. of Statements	Scale of Measurement	References
1	A	Performance expectancy	7	Interval	Venkatesh et al., (2003)
2	B	Effort Expectancy	7	Interval	Venkatesh et al., (2003)
3	C	Social influence	7	Interval	Venkatesh et al., (2003)
4	D	Facilitating Conditions	7	Interval	Venkatesh et al., (2003)
5	E	System Quality	7	Interval	Delone and Mclean, (2003)
6	F	Information Quality	7	Interval	Delone and Mclean, (2003)
7	G	Service Quality	7	Interval	Delone and Mclean, (2003)
8	H	Satisfaction	7	Interval	Wixom and Todd, (2005)
9	I	Experience	5	Interval	Venkatesh et al., (2003)
10	J	Personal Innovativeness in IT	5	Interval	Agarwal and Karahanna, (2000)
11	K	Actual Use	4	Interval Nominal	Delone and Mclean, (2003)
12		Respondent's profile			

An Example of Research Design



Significance of the Study

- Why this study is significant?
- Why do it now?
- What will it contribute to the existing research literature?
- What implications your findings may have?
- Who will benefit from it?

[Scholarly acceptability] **See Example**

...References

The references in your text can be numbered in the sequence in which they appear in the report and then listed in this order in the list of references

Harvard Citation or author/date system references made in the text are listed in alphabetical order by the surname of the first author in the bibliography at the end of the text.

e.g.

Reference within essay: In his survey of the mating habits of frogs, Bloggs (1998) refuted that ...

Entry in bibliography:

Bloggs, B. 1998. Frogs and their mating habits . London: Tadpole Press

The Vancouver System or footnotes/endnotes or numeric system; references are numbered consecutively in the order in which they are first cited in the text.

e.g.

Reference within essay:

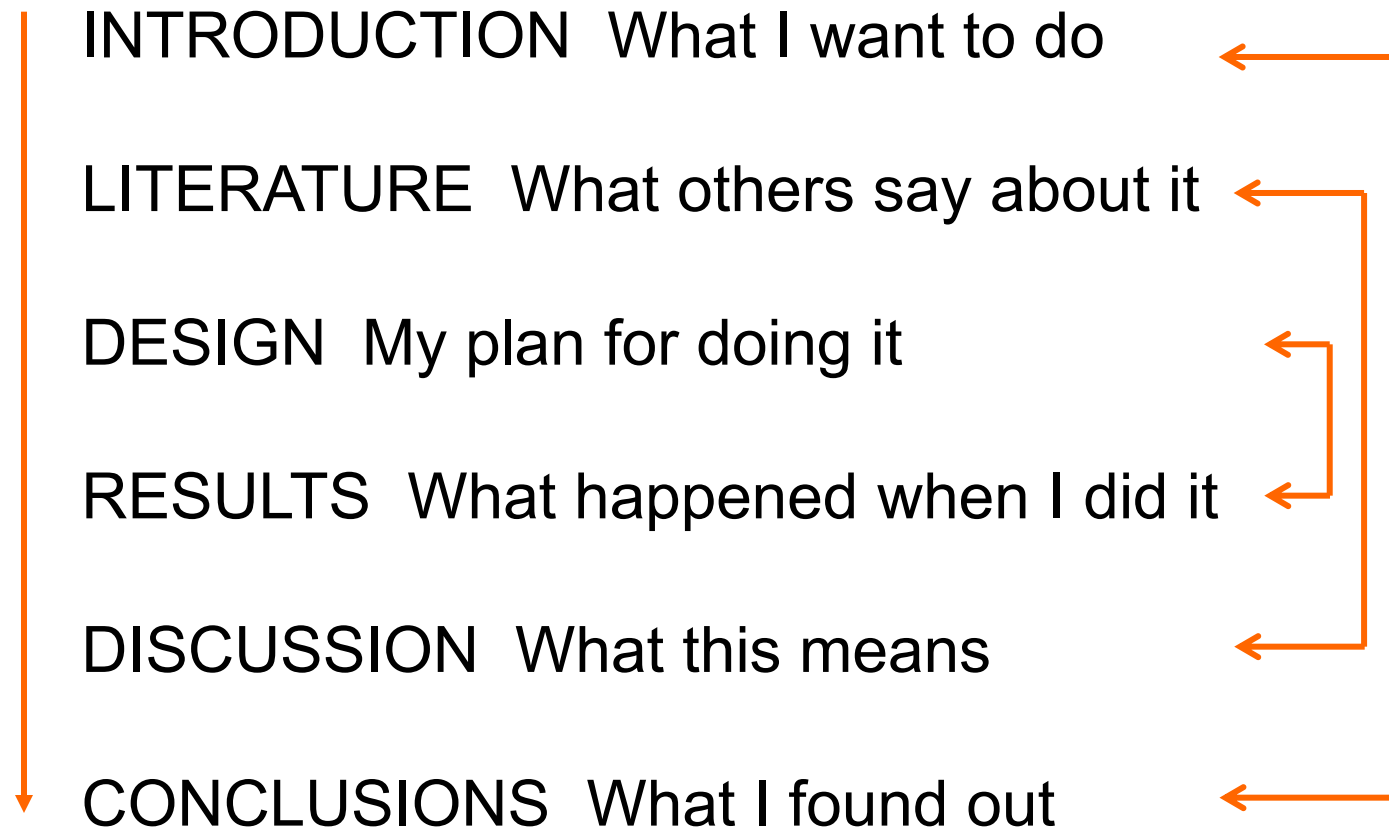
In his survey of the mating habits of frogs, Bloggs (1) refuted that ...

Entry in bibliography:

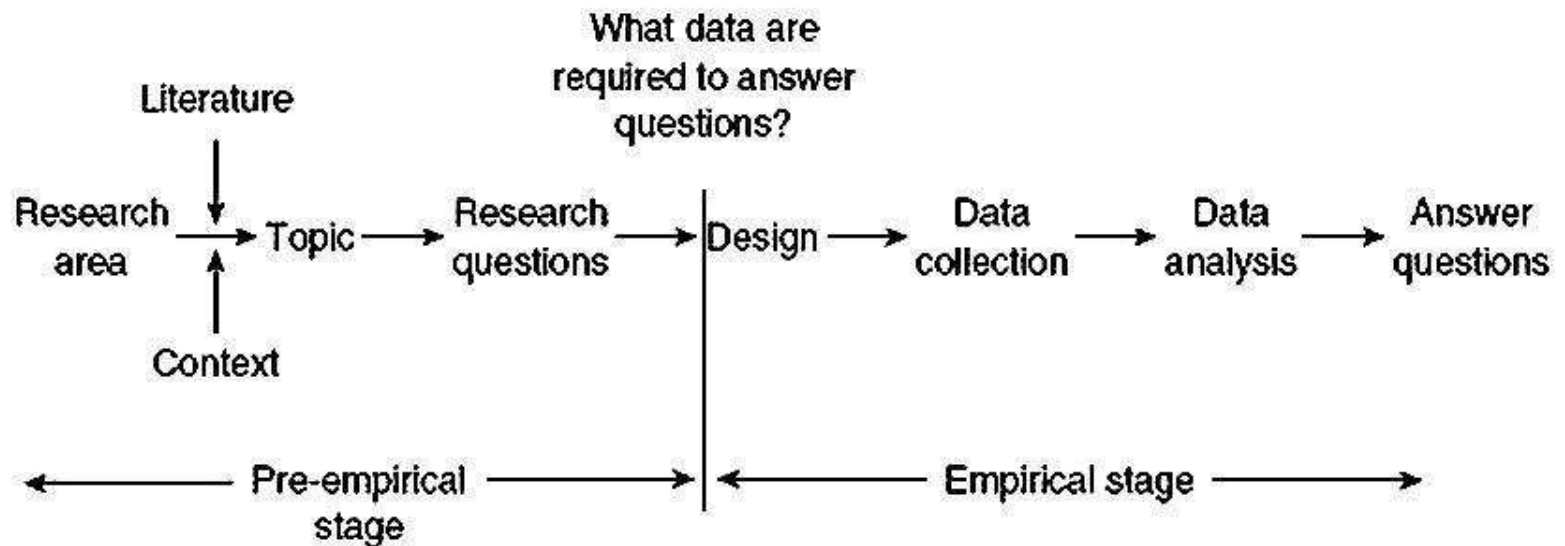
1. Bloggs B. Frogs and their mating habits. London : Tadpole Press; 1998

Thank you

The classical thesis structure



Simplified model of research



Punch, K. F. (2007). *Developing Effective Research Proposals* (2 ed.). Los Angeles, CA: SAGE Publications.

Writing a Thesis

.....Overall Main Components of a Thesis :

- I. Title or cover page
- II. Abstract
- III. Introduction
- IV. Objectives
- V. Methodology
- VI. Research results (findings)
- VII. Discussion
- VIII. Conclusions
- IX. Recommendations
- X. References
- XI. Annexes or appendices (data collection tools, tables)

Main Components of a Thesis

- **Chapter 1 : Introduction**

- Background of the study
- Overview of research subject
- Broad to specific
- Problem statements
- Objectives of study
- Practical implications
- Theoretical contributions
- Study Limitations

- **Chapter 2: Literature Review**

- Past research related to study
 - Global & local publications – journal and books
- Overview of theories related to research subjects – at least 5 or 3 key theories
- Review and critique

Main Components of a Thesis

- **Chapter 3: Theoretical Framework**

- Introduction
- Conceptual development
 - Models/theories comparisons
 - Which models as base of proposed model
 - Proposed research model
- Hypotheses development
 - Independent variables
 - Dependent variables
 - Moderating / mediating variables

- **Chapter 4: Methodology**

- Introduction
- Population and sampling
- Data Collection and Sampling procedure (target ; sampling population ; size)
- Questionnaire design
 - Measurement of variables
 - Research design
 - SEM- Measurement model

Main Components of a Thesis

- **Chapter 5 : Data Analysis & Discussion**

- Introduction
- Feel for data
- Frequency distribution of demographics
 - Positions ; locations ; years of working ; gender
- Testing Goodness of Data
 - Reliability and validity
 - Results from SEM analysis
 - Significant or not significant related to hypotheses

- **Chapter 6 : Conclusions**

- Introduction
- Discussions on results – conform to findings in literature ; if not why..?
- Implications of Research Findings
 - Theoretical Implications
 - Practical Implications
 - Policy Implications
- Recommendations for Future Research
- Limitations

...CONCLUSIONS

- The conclusions should follow logically from the discussion of the findings.
- As the discussion will follow the sequence in which the findings have been presented (which in turn depends on your objectives) the conclusions should logically follow the same order.

....RECOMMENDATIONS

- The recommendations should follow logically from the discussion of the findings.
- Recommendations may be summarized according to the groups towards which they are directed, for example:
 - ↳ policy-makers / stakeholders
 - ↳ program managers/implementing officials at different levels
 - ↳ potential clients
 - ↳ the community at large

....Plan for utilization and dissemination of research results

- The proposal should indicate what reports or other means of disseminating research findings are planned.
- Any or all of the following are appropriate for disseminating the results of the study:

Progress reports

Final report

Publications

Seminars, workshops, and conferences

Discussion with policymakers and program managers

....ANNEXES OR APPENDICES

The annexes should contain any additional information needed to enable professionals to follow your research procedures and data analysis.

Examples of information that can be presented in annexes are:

- Tables referred to in the text but not included in order to keep the report short.
- Interview schedule/ questionnaires (and/or other data collection tools).
- Informed consent form
- Institutional/Ethical approval for the study

Further Readings

1. Creswell, J. W. (2009). *Research Design: Qualitative, Quantitative, and Mixed Methods Approaches*. Los Angeles, CA: SAGE, p. 97–127.
2. Hernon, P. (2007). Editorial: what is problem statement? *Library & Information Science Research*, 29, 307–309.
3. Hernon, P., & Metoyer-Duran, C. (1993). Problem statements: An exploratory study of their function, significance, and form. *Library & Information Science Research*, 15, 71–92.
4. Metoyer-Duran, C., & Hernon, P. (1994). Problem statements in research proposals and published research: A case study of researchers' viewpoints. *Library & Information Science Research*, 16, 105–118.
5. Nitecki, D. A. (2010). Guest editorial: Finalizing a research paper—findings through conclusion. *Library & Information Science Research*, 32, 1–3.
6. Sproull, N. L. (1988). *Handbook of research methods: A guide for practitioners and students in the social sciences*. London: Scarecrow Press, p. 140.
7. Teddlie, C., & Tashakkori, A. (2009). *Foundations of Mixed Methods Research : Integrating Quantitative and Qualitative Approaches in the Social and Behavioral Sciences*. Los Angeles, CA: SAGE, p. 120–124.



BREAK FOR JUMAAT PRAYER

UNESCO Chair on Secure High-performance Computing for Higher Education and Research

2022

Smart and Secure Environment (SSE)

AT A GLANCE

Prof Dharm Singh Jat

UNESCO Chairholder

E:unescochair-shpcher@nust.na

<http://unescochair-shpcher.nust.na/>



The UNESCO Chair on Secure High-performance Computing for Higher Education and Research established at NUST is part of the UNESCO-NUST Partnership within the UNESCO/UNITWIN Science Chairs Programme.

UNESCO Chair on Secure High-performance
Computing for Higher Education and Research





Goal of the UNESCO Chair

The purpose on the UNESCO Chair on Secure High-performance Computing for Higher Education and Research shall be to

- Promote an integrated system of research, training, information and documentation on digital technologies.**
- It will facilitate collaboration between high-level, internationally-recognised researchers and teaching staff from the University and other institutions in Namibia, as well as elsewhere in Africa, Europe and in other regions of the world.**

Specific objectives

- 1. Establish the high-level Super Computing facility in Namibia, which is relevant for the development of Namibian society and the wider region;**
- 2. Establish programmes leading to a Ph.D. degree in the related field of Smart and Secure Environment (SSE), while assuring gender equality and offering a high-level training programme for graduate and postgraduate students;**
- 3. Promote research in the field of SSE in Namibia, the Southern African Development Community (SADC) region and in other parts of the world and exchange of lecturer across borders;**

Specific objectives

- 4. Create a platform for national and international exchanges of ideas through international and regional conferences to be held on a regular basis, as well as workshops involving leading computer scientists;**
- 5. Collaborate with UNESCO, Category 2 centres under the auspices of UNESCO and other UNESCO Chairs to develop relevant programmes and activities.**

The significance of the Chair and expected outcomes to national development

The UNESCO Chair on "SHPCR" will contribute to accelerate the development of the Namibian society and building the foundations for technology by creating a critical mass of high-level expertise in the related field of the Chair.

The UNESCO Research Chair for SHPCR will develop

- (i) supercomputing facilities for researchers**
- (ii) big data analytical techniques**
- (iii) smart communication and**
- (iv) cyber security solution**
for society.

Industry 4.0: Introduction

**The fourth Industrial Revolution
emerges from AI and the Internet of
Things**



4th Industrial Revolution -Introduction

The key essence of Industry 4.0 Concept is the enhancement of efficiency in the process of industry by way the utilization of connectivity, computing, data, IoT and Artificial Intelligence.

The five characteristics of Industry 4.0

- 1. Industry 4.0 is interconnection**
- 2. Industry 4.0 is data**
- 3. Industry 4.0 is Integration**
- 4. Industry 4.0 is Innovation**
- 5. Industry 4.0 is transition**

Segment of Enterprise in Industry 4.0

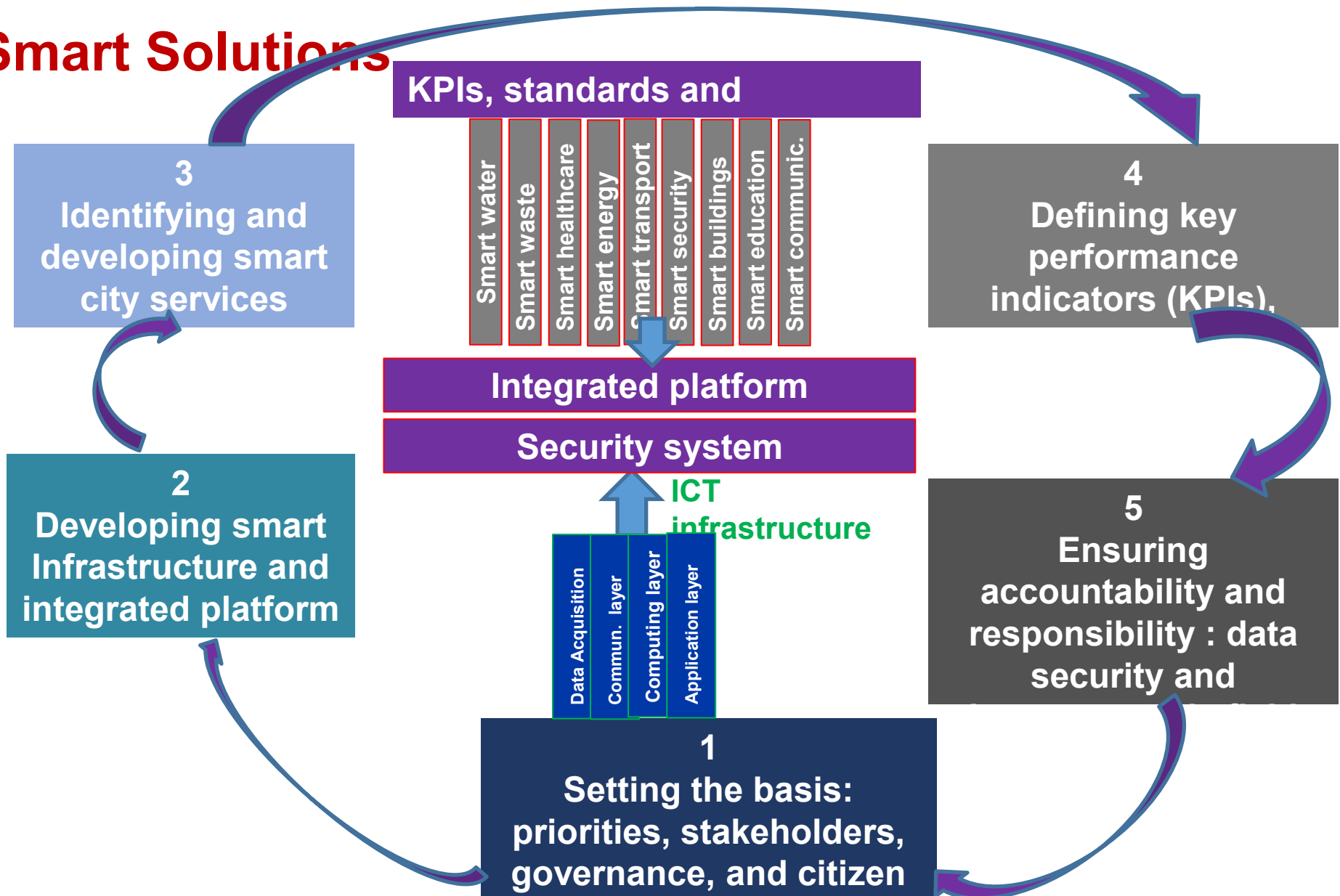
Technology Provider

- Artificial Intelligence (AI)
- Internet of Things
- Edge Computing
- Blockchain
- Virtual & Augmented Reality
- 3D Printing
- Robotics / Automation

Infrastructure Provider

- Internet Service Provider (ISP)
 - HPC System
 - Cloud Computing
 - Telecommunication
-

Smart Solutions



Smart Solutions

E-Governance and Citizen Services

- Public Information and Grievance Redressal
- Electronic Service Delivery
- Citizen Engagement
- Citizens-City's Eyes and Ears
- Video Crime Monitoring

SMART Health and Education

- Tele-Medicine
- Remote Teaching and Learning

Waste Management

- Waste to Energy and Fuel
- Waste to Compost
- Waste Water to be Treated
- Recycling and Reduction of Construction and Demolition (C&D) Waste

Water Management

- Smart Meters and Management
- Leakage Identification, Preventive maintenance
- Water Quality Monitoring

Energy Management

- Smart Meters and Management
- Renewable Sources of Energy
- Energy Efficient and Green Buildings

Urban Mobility

- Smart Parking
- Intelligent Traffic Management
- Intelligent Multi-Model Transport

HPC-AI infrastructure...

- The HPC and AI system will strengthen application development of packages such as
 - ✓ **Bioinformatics, Climatology, Astrophysics, Computational Fluid Dynamics, Computational Chemistry, Molecular Dynamics, Finite Element Analysis, Agriculture etc.,**

The HPC facility has significantly contributed to the fight against the COVID-19 pandemic since mid-March 2020 through faster simulations, medical imaging, and forecasting.



HPC: Applications **PARAM!Arub System**

Machine Learning / Deep Learning Application Development

Most of the popular python based machine learning/deep learning libraries are installed on PARAM!Arub system.

Weather Research and Forecasting (WRF)

The Weather Research and Forecasting (WRF) Model is a next-generation mesoscale numerical weather prediction system designed to serve both operational forecasting and atmospheric research needs.

Code Aster

Code_Aster is a free and open source software package for finite element analysis, and numerical simulation in structural mechanics.

Ferret

Ferret is an interactive computer visualization and analysis environment designed to meet the needs of oceanographers and meteorologists analyzing large and complex gridded data sets.

HPC: Applications

Abinit 9.0.2

ABINIT is a package whose main program allows one to find the total energy, charge density and electronic structure of systems made of electrons and nuclei (molecules and periodic solids) within Density Functional Theory (DFT), using pseudopotentials and a plane wave basis.

Openmx

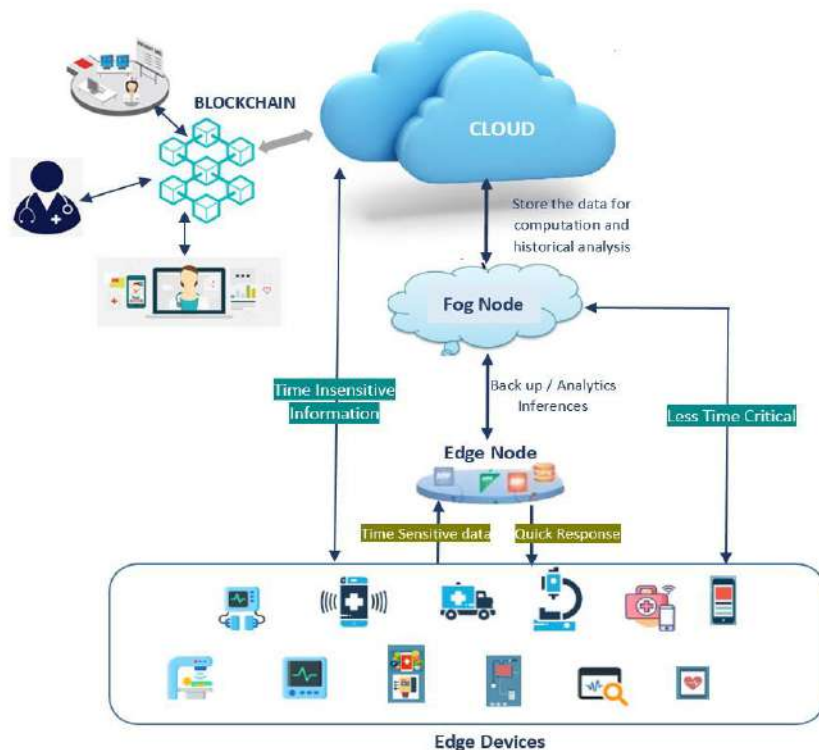
OpenMx is free and open source software for use with R that allows estimation of a wide variety of advanced multivariate statistical models. OpenMx consists of a library of functions and optimizers that allow you to quickly and flexibly define an SEM model and estimate parameters given observed data.

OOFEM

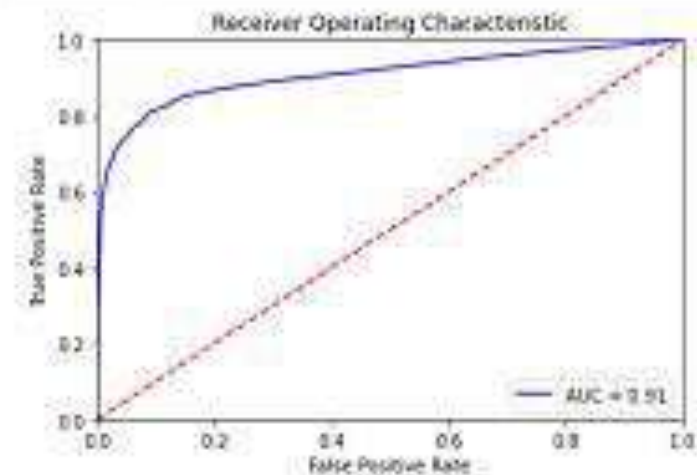
OOFEM is parallel, object-oriented finite element code for solving mechanical, transport and fluid mechanics problems.

Ongoing PhD Research

Design and implementation of blockchain-based secure edge computing paradigm for time constraint applications

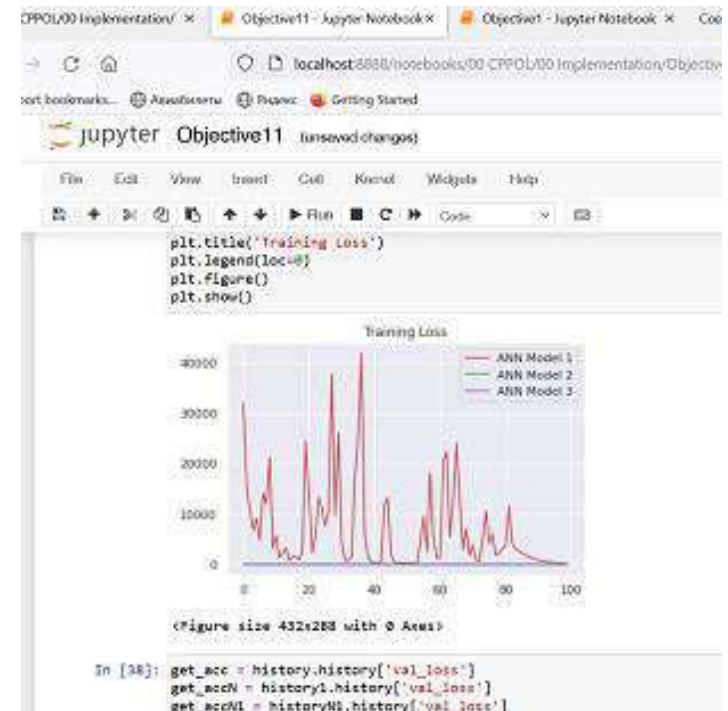
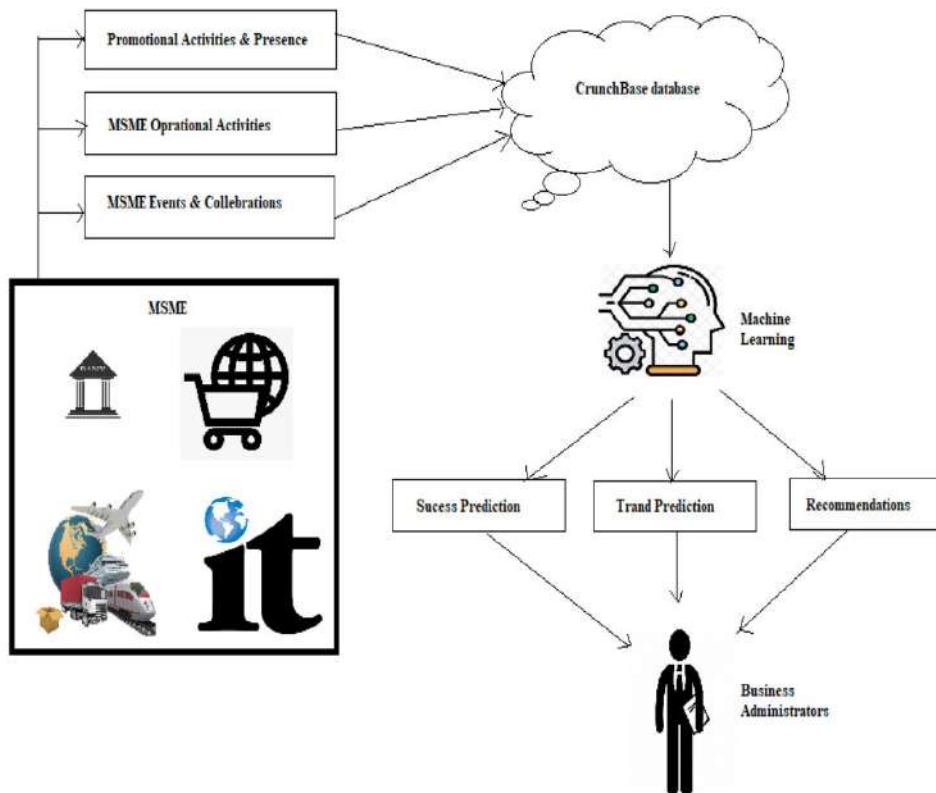


```
plt.title('Receiver Operating Characteristic')
plt.plot(fpr, tpr, 'b', label = 'AUC = %0.2f' % roc_auc)
plt.legend(loc = 'lower right')
plt.plot([0, 1], [0, 1], 'r--')
plt.xlim([0, 1])
plt.ylim([0, 1])
plt.ylabel('True Positive Rate')
plt.xlabel('False Positive Rate')
plt.show()
```



Ongoing PhD Research

Design and Develop a Prediction Model for Improving Start-Up Success Using Machine Learning



Ongoing Master Research

- 1. Design and Development of System for Post Infection Behaviour Analysis**
- 2. Design of a Security Threats Identification Model for Internet of Things (IoT) Smart Office Devices**
- 3. Enhancing the Effectiveness of an Email Phishing Awareness Tool for Organisations in Namibia**
- 4. A Chain of Custody Method for Preserving Volatile Digital Evidence for Admissibility in Namibian Courts**
- 5. Designing an Email Forensic Readiness Architecture for the Banking Sector**
- 6. Designing an improved secure user authentication system for the public cloud infrastructure as a service in a hybrid cloud**



Work Package 5: Development and application of open-source H2-Nexus-System planning tool

Objectives

- Development of an open-source software tool that allows users to simulate and optimize the design of H2-Nexus-Systems (back-end).
- Customize the sizing tool for the purpose of decentral GH2 feasibility studies (front-end) to allow easy application and usage of tool for a broad stakeholder group.
- Show economic and ecological benefits of H2-Nexus-Systems by simulation of case studies with planning tool.

Enhancing energy system modelling: <https://oemof.org/>



Project: Green Hydrogen Energy-Water-Food Nexus Model for Economic Prosperity

Work Package 6: Development and Application of Digital Technologies for H2-Nexus System

Objectives

1. Design and Develop of an efficient IoT-enabled Real-Time Monitoring System for Green Hydrogen Sensors.
2. Design and develop a prediction model for improving key performance indicators of hydrogen production plant using Artificial Intelligence.
3. To develop HPC application for electronic-structure calculations and materials modelling at the nanoscale and processes for hydrogen generation and storage, and for future generations using HPC applications.
4. Use high performance computing facilities (HPC) to develop high strength and corrosion-resistant materials.
5. To develop a reference model as a guideline to execute the digital twin for the H2-Nexus System To develop AI-Enabled Digital Twins Technology for the H2-Nexus System.
6. To demonstrate the impact of the proposed digital twin Technology through a real-world Green Hydrogen manufacturing plant.

UNESCO SHPCHER Team



Prof Dharm Singh Jat



Prof Fungai BhunuShava



Prof Guy Lusilao Zodi



Prof Jose Quenum



Dr Attlee M. Gamundani



Mr Arpit Jain





Conclusion

-
- The intelligent Edge is a foundation for future digital business and its ecosystems and by bringing powerful computing capabilities closer to the edge where data originates and processed with machine learning helping to accelerate the Fourth Industrial Revolution.
 - Edge computing is still in its early stages and has potential to discover the way for more efficient computing technology for enabling future Internet of Things .
 - Edge Computing using Blockchain technology paradigm may provide data filtering at the edge, security, ensure the data integrity, and privacy with the help of decentralised and distributed ledger of trust.
 - The connectivity of smart city and Industry 4.0 may create huge opportunities to join the physical and the digital world and it can develop the public and private sector's effectiveness, expand financial benefits and improve the standard of living.



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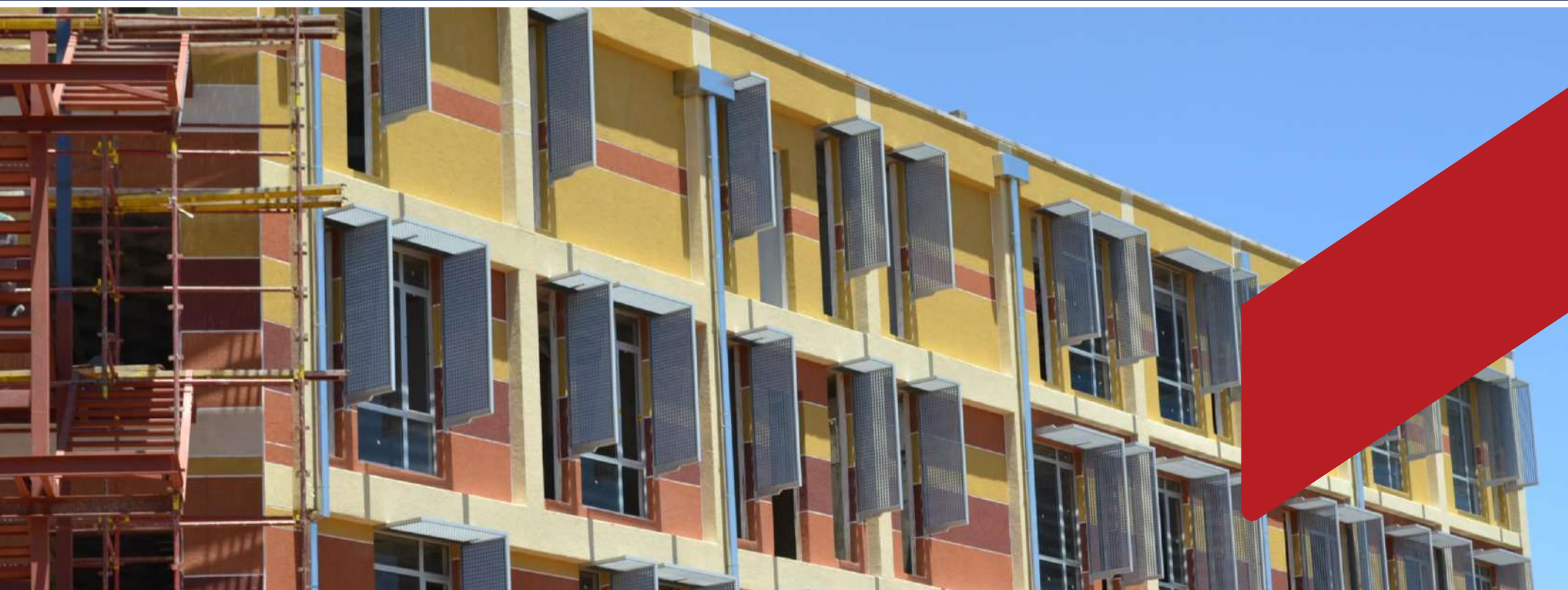
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Defining a Research Problem

Dr Irja Shaanika

Dept. of Informatics

Date: February 2023





- Constructing the research problem
- Defining your research objectives
- Defining your research questions
- Defining your Research Topic



What is a research Problem

A paragraph that conveys the issues and context that gave rise to the problem.

According to Bryman(2007)

” it’s a statement about an area of concern, a condition to be improved upon, a difficulty to be eliminated, or a troubling question that exists in theory or in practice that points to the need for meaningful understanding and deliberate investigation ”

A research Problem Must be :

- Direct
- Specific
- Clear and Well defined



Challenges in defining the problem statement

- The problem is either too broad
- Either too narrowly
- Determining the research gap and locating it into the existing literature



Research Problem Formulation

1. Identify your interest area.
2. What is the problem?
3. Where does it exist?
4. Who does it affect?
5. How is it a problem?
6. What are the implications



Defining the Research Problem

The problem statement must state what the problem is –AND NOT ONLY PROVIDE a narrative of challenges and solutions.

“the application of the implemented NAMCIS has presented challenges when it comes to the amalgamation of reports and statistics of court data as well as the administration and maintenance of the systems. The challenges are encountered when analysing and reporting data from different locations which is time-consuming and cumbersome, and often leads to errors and inaccurate information (Shi et al., 2019).”

A research problem statement must be stated as the absence of, the lack of, the inadequate of, the insufficient of, the insufficient of, the deficiency of, etc

“The lack of integration of Namibian court information system across all lower courts in Namibia has led to inconsistency in decision-making as a result of data inaccuracy and data unreliability.”



Research Problem = Background + Problem + Implications

- **Background Introductory to the subject**

Organisations from various industries have become dependent on information systems to automate and enable business processes' efficiency and effectiveness (Yahaya et al., 2015). Thus, in Namibia, the lower courts have also implemented an information system known as the Namibian court information system (NAMCIS).

- **State the problem**

However, the lack of integration of the Namibian court information system across the lower courts has led to inconsistency in decision-making as a result of data inaccuracy and data unreliability. Moreover, the lack of integration contributes to the cumbersome systems administration of the court information systems due to repetitive technical maintenance and systems administration tasks.

- **Consequences of the problem**

The application of the implemented NAMCIS has presented challenges when it comes to the amalgamation of reports and statistics of court data as well as the administration and maintenance of the systems. The challenges are encountered when analysing and reporting data from different locations which is time-consuming and cumbersome, and often leads to errors and inaccurate information (Shi et al., 2019). From the perspective of system administration, the systems are administered and maintained individually, an exercise which is time-consuming and costly as activities such as fixing bugs, upgrades, running backups and user-accounts administration are carried out separately and repetitively for each system.



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The proposed solution

Thus, this study proposed a model that can be used to guide and govern the integration of the Namibian court information system. Otherwise, the problem and challenges articulated above will persist.



Citations in Problem Statement

Very Important – Support your problem statement with relevant citations.

e.g Few SME's receive funding from Development of Bank of Namibia. The 2022-2023 BON financial report indicates that of the 50 applications received, only 4 were successfully funded(BON Report,2023).



Defining Research Aim

- The comprehensive goal/solution to the problem identified
- The researcher must determine what is the overall long term by describing the overall purpose (what should be accomplished) in general terms.

“to design a model that will guide and govern the integration of the Namibian lower courts' information systems”



Defining Research Objectives

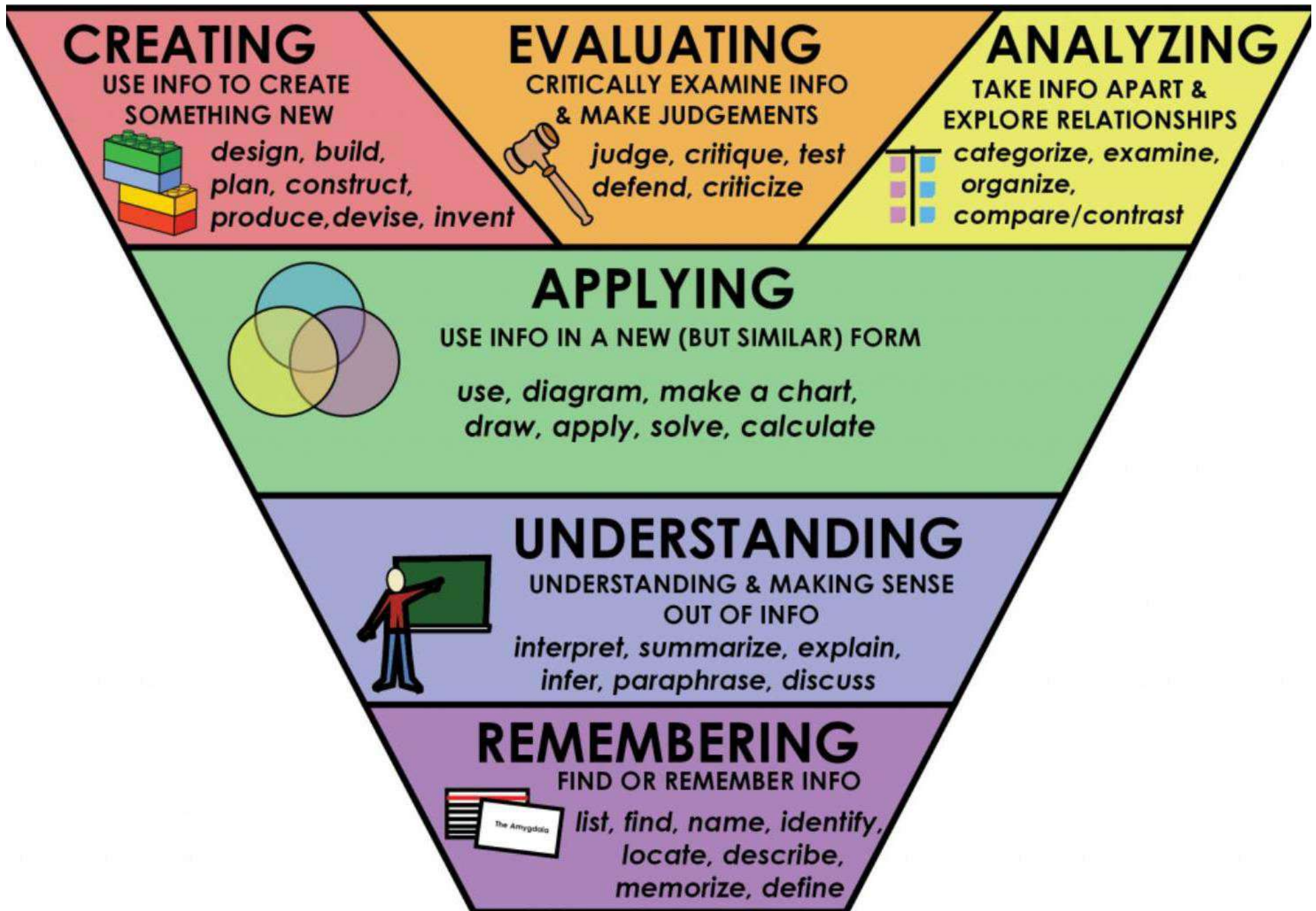
How will the defined aim be achieved ?

Milestones that need to be set towards resolving the problem.

Characteristics:

- Cover all aspects of the problem,
- Specific,
- Ordered in a logical sequence,
- Achievable, take into consideration the available resources, including time,
- Mutually exclusive of each other.
- They should also be stated using action verbs
 - Use Bloom Taxonomy

BLOOM'S TAXONOMY





Example of Objectives

- i. To examine the state of court information systems integration in Namibian lower courts.
- i. To examine factors that could impact the integration of the Namibian lower courts information system.
- ii. To investigate how the integration of Namibian Lower Courts' information systems will improve data reporting and systems administration.



Defining Research Questions

- Directly related to the Research Objectives
 - Matching to the research Objectives
 - Phrased in the form of a question
-
- What is the state of court information systems integration in Namibian lower courts?
 - What are the factors that could impact the integration of the Namibian lower courts information system?
 - How will the integration of Namibian Lower Courts' information systems improve data reporting and systems administration?



Objectives to Research Questions

Aim:	Main Question:
To develop an ICT framework that will support patient-centric healthcare services in Malawi.	What are the components of an ICT framework that will support a patient-centric approach in Malawi?
1. To identify the challenges underlying the healthcare ICT development in the public sector in Malawi.	1. What are the challenges underlying the healthcare ICT development in the public sector in Malawi?
2. To identify ICT-related factors that can support the patient-centric healthcare in Malawi.	2. Which are the ICT-related factors that can contribute to patient-centric healthcare in Malawi?
3. To identify the requirements of an ICT framework that can support a patientcentric approach in Malawi.	3. What are the requirements for an ICT framework that can support a patientcentric approach in Malawi?
4. To identify the benefits of an ICT framework that can support a patient centric approach in Malawi	4. What are the benefits of an ICT framework that can support a patient-centric approach in Malawi?



Components Relationship





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Thank You.



Writing Your Proposal Introduction

BY : IRJA SHAANIKA

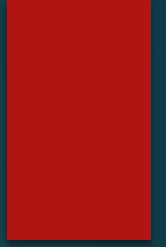
Why “Introduction” ?

- ▶ Getting your readers on-board
- ▶ Giving readers an overview what your study entails
- ▶ Will awaken readers interest

Introduction entails

- ▶ A brief introduction to the study
 - ▶ Concise yet Informative
- ▶ Adequate background information of the study
 - ▶ Discuss general background, Move on to the specifics, stressing the magnitude of the problem
 - ▶ Supported with limited literature.
- ▶ Quote key papers and findings
 - ▶ highlight the knowledge gaps and how your study provides further information on the area.
 - ▶ Describes how and why the research questions have arisen.
- ▶ Explicitly state the importance of your research
 - ▶ reader may not necessarily make the leap in logic that is obvious to you.
- ▶ The introduction should end with the aim being clearly stated.

How “Introduction”



Be guided by objectives

- ▶ Objectives - Your map(guide).
- ▶ Ensure you are not diverting from the focus of your study

Have themes /keywords

- ▶ Main topics/headings.
- ▶ Based on problem statement, objectives, topic.
- ▶ May be a single word or several words(may include phrases)
- ▶ Will help readers locate/search your study in databases.

Identify keywords sequence

- ▶ Order of keywords : which one should come first, second....., last? Why
- ▶ Which order will help my readers follow and understand my study very well?
- ▶ Most cases : study domain > what is being studied
- ▶ Example: Securing ATMs cash withdrawals
- ▶ ATM> Cash withdrawals> Security
- ▶ Example: Implementing **Electronic Healthcare Records** in **Public Hospitals**
- ▶ ?????

Maintain connectivity/flow among themes

- ▶ The conclusion of one term should lead into the introduction of the following term.
- ▶ (Information Systems : IS Integration)
- ▶ **Information Systems:** It is no doubt the implementation of ICT offers unlimited benefits to organisations. However, for benefits such as collaboration and ease of information flow, **information systems integration** is mandatory.
- ▶ **IS integration:** **Information Systems integration** refers to the interconnectivity between the implemented systems. In his study David (2019) discussed types of IS integration which are $()()$. According to Shaanika (2019), the type of IS integration an organisation select is influenced by various factors such as underlying infrastructures and technical skills.

Advice!!!

- ▶ Always consider the structure of the (Ws)...H... when writing
 - ▶ What
 - ▶ Where
 - ▶ When, How
 - ▶ Why
- ▶ Do not leave statements hanging.....



Thank you:

Any questions??

Writing an Introduction

Dr Munyaradzi Maravanyika

Key Goals of an Introduction

1. Present your topic and get the reader interested
2. Provide background or summarize existing research
3. Position your own approach
4. Detail your specific research problem and problem statement
5. Give an overview of the paper's structure

Definition of Research Introduction

- A research introduction is the first section of a research paper or thesis that provides an overview of the study's background, purpose, scope, and significance.
- It serves as the foundation upon which the rest of the study is built.
- The research introduction should be well-crafted to capture the reader's attention and motivate them to read further.

Step 1: Introduce your topic

- The first job of the introduction is to tell the reader what your topic is and why it's interesting or important.
- This is generally accomplished with a strong opening hook.
- The hook is a striking opening sentence that clearly conveys the relevance of your topic.
- Think of an interesting fact or statistic, a strong statement, a question, or a brief anecdote that will get the reader wondering about your topic.
- Don't feel that your hook necessarily has to be deeply impressive or creative.
- Clarity and relevance are still more important than catchiness.
- The key thing is to guide the reader into your topic and situate your ideas.

Make others care

- When you do research, you do it because you hope it will make a positive difference in the real world, which means you need to write for other people.
- If other people don't read your writing, your research will make no difference.
- This means you have to get other people, besides yourself, to care about the 'itch' you identified.
- If you can't do this, you will be given the biggest insult a researcher can get, which is, "So what?"
- The best way to make others care is to create a problem for them.

Step 2: Describe the background

- Argumentative background:
 - You'll explore some general background here
 - After you've caught your reader's attention, specify a bit more, providing context and narrowing down your topic.
- Empirical Background
 - In a more empirical paper, this is the place to review previous research and establish how yours fits in.
 - Begin by establishing the kinds of research that have been done, and end with limitations or gaps in the research that you intend to respond to.

Provide Background Information

- After the hook, provide some background information on the topic
- This can include historical context, current trends, or any other relevant information
- The purpose of this is to help the reader understand the importance of the research and why it is relevant

Identify the gap

- Explain the gap by showing that there is big difference between what we want and what we have.
- Find academic sources and credible data that you can use to support the claims that there is a gap.
- Without credible sources to back you up, it will be difficult for you to convince your chosen audience that the problem is serious enough to warrant their attention.
- Bear in mind that your audience probably has many researchers presenting problems to them, all of which compete for their attention, so you need to do all you can to convince them that they should care about your problem.

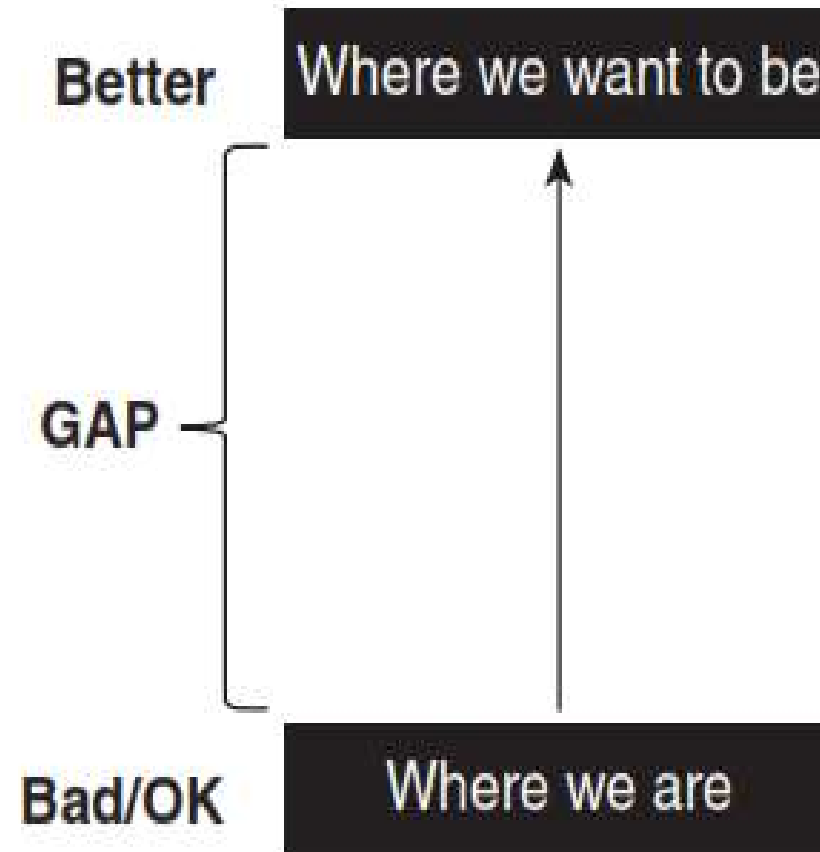
Identify the Gap

To close this gap, you have to take action, and if this action is difficult, then you have a problem.

What makes this gap a problem?

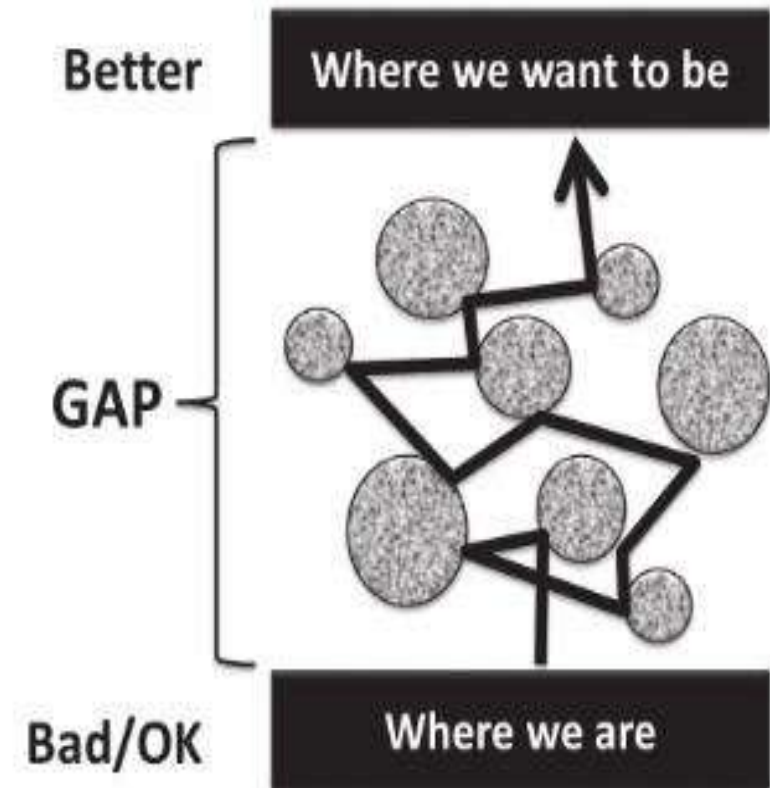
Looking at the origin of the word gives us a clue.

It comes from the Greek word ***problema***, which means 'obstacle to action'.



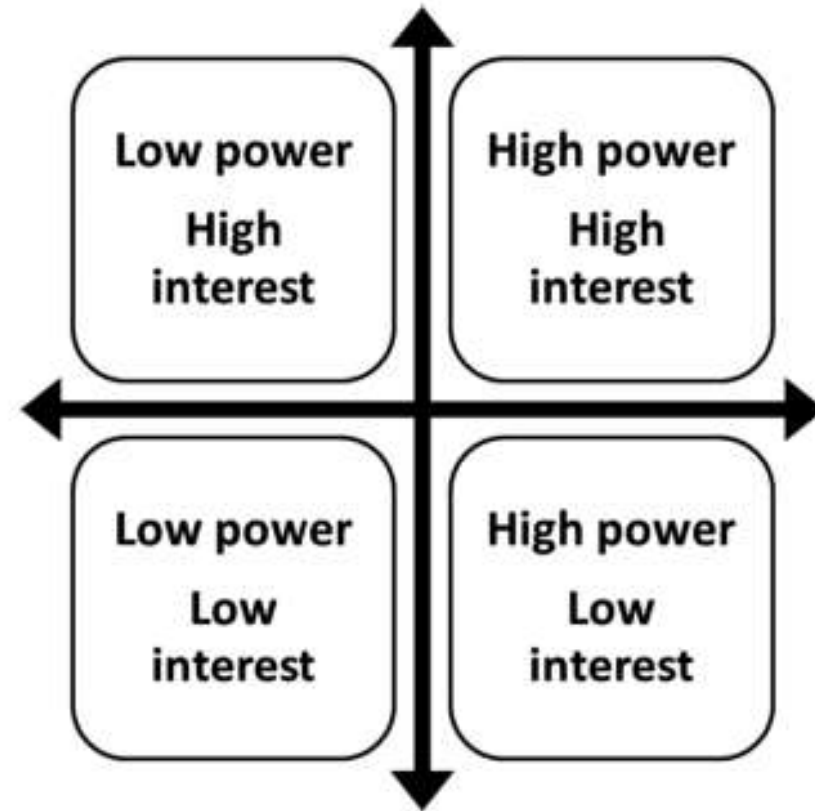
Identify the obstacles

- Look inside the gap and identify the main obstacles to solving the problem and how they make it difficult to take action to solve the problem.



Identify your audience

- A problem is always a problem for someone and, often, a group of people or multiple groups of people.
- So, you need to have a person, entity or some group of people in mind when you try to discuss the problem.
- This also gives your writing further focus to ensure that you get to the point quickly.
- We always write messages, reports and e-mails with someone in mind, and research should be the same.



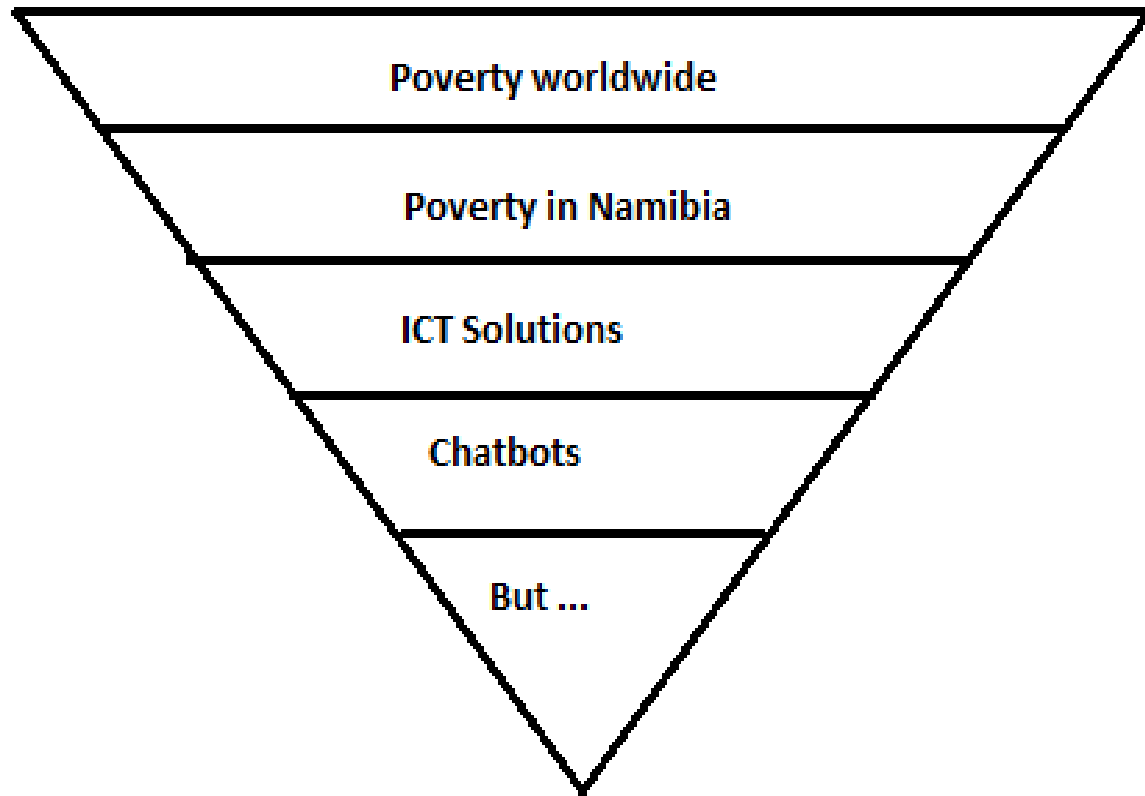
Identify an appropriate context

- When communicating the problem, you have to be careful because not everyone will agree with you that your problem is that important, and may even refute its existence entirely.
- And if a reader disagrees with you early on, it is a lot more difficult to convince them at a later stage.
- Context gives us the broad background to the problem or tells the story that led up to the problem.
- In the context, there is usually little that readers can disagree with because you don't explain the problem yet.
- It creates some common ground between you and the reader, and it is from here that you can lead them to the problem.
- There are two main ways by which you can create context, and you should be able to do it in one or two paragraphs at most.

Narrative

- Relate the chain of events from a time just before the problem emerged to the actual emergence of the problem.
- Brevity is of the essence here – don't lose the reader's interest by giving them a long history lesson.
- The narrative, or chain of events that led to the problem's existence, is usually the easiest and most persuasive way of contextualising the problem.
- This is because it draws on historical facts that are indisputable and because most people are less critical when they are told a story.
- However, the narrative only works if there was a time before and after the problem.
- If the problem has always existed, you need to contextualise the problem using the next approach.

Context as Inverted Pyramid



- This is why I call it an ‘inverted (upside-down) pyramid’—because you go from the very general to the very specific.
- Again, being brief is of the essence, because you will bore your readers and they will lose concentration if you spend too much time on vague and general discussions.

Contextualising

- Familiarising the audience with the place, time and setting – also helps the reader orient themselves and see where you are coming from.
- We do this in most stories: we don't start with what went wrong, but with 'once upon a time'.
- We also do this in movies: we don't start with the action, but first with a view of the place and gradually zoom in to where the action is taking place.

Writing: 'background to the research' section

- First, write down the context
- The context leads to the real-world problem, so then you write down the explanation of the problem
- This should not take more than one page, and of this one page you should spend more time on explaining the actual problem in the real world, because it is this that convinces people that they should care.
- The background starts with a general discussion of an issue in the first paragraph.
- It then moves to a discussion of the issue in the second paragraph, it led to a particular law. In the third paragraph, it explains the problem with that law.
- Facts are stated and academic references back up all the important claims.

Writing: 'background to the research' section

- In the background section, you focused on the real-world obstacles that make it difficult to solve the real-world problem.
- There is another kind of obstacle that needs to exist to create the need for research.
- This one does not lie in the real world, but in the world of knowledge.
- We think about that in the next step

Determine why current studies are not good enough

- So far, you have shown that there is a problem in the real world and you identified real-world obstacles that make it difficult to do something about it – like lack of funding, wrong policies, corruption etc.
- Now you have to show that research plays a role in the existence of, and solution to this real-world problem.
- Real-world problems, however, cannot be solved by research, because research is not a person who can walk around and change the world.
- So, let us look at what role research plays in the existence of a problem.
- It could be that the research, up until that point, was not good enough to help people solve the real-world problem, so that further research would give people the knowledge they need to address the real-world problem.
- This is where you can step in.

The most common kinds of gaps are:

- **Time gaps:** You find previous studies about the real-world problem, but notice that the studies are outdated and are therefore no longer relevant. New updated research is therefore needed based on the latest data.
- **Place gaps:** You find previous studies, but notice that the studies are only applied to some places. New research is therefore needed that applies the relevant ideas to places – cities, industries, countries etc. – neglected by researchers in the past.
- **People gaps:** You find previous studies, but notice that the studies are only applied to some people or groups. New research is therefore needed that applies the relevant ideas to the groups neglected by researchers in the past.
- **Method gaps:** You find previous studies, but notice that the studies use methods that are wrong, inappropriate or outdated. New research is therefore needed using the best available or new methods.

- The challenge is to convince your audience that, given all the research out there about your real-world problem, you have found something inadequate.
- Here are some ways to do that:

If you found that there is confusion about an issue in the form of a disagreement, then:

1. You mention the studies (by referencing them) that take one view on the real-world problem;
2. Mention the studies (by referencing them) that take the opposing view;
3. Explain what it is they disagree on; and
4. Explain what research is still needed to settle the disagreement (and this is the research you will do).

If you found that there is confusion in the form of an unsolved puzzle, then:

1. You explain the puzzle with reference to studies that show that there is a puzzle;
2. Mention the studies (by referencing them) that have attempted, and failed, to solve the puzzle; and
3. Explain what research is still needed that might help to solve the puzzle, and state that this is the research you will do.

Step 3: Establish your research problem

- Try to lead into the problem on the basis of your discussion of the literature.
- Think in terms of these questions:
 - What research gap is your work intended to fill?
 - What limitations in previous work does it address?
 - What contribution to knowledge does it make?

State the Problem

- The next step is to state the problem or issue that the research addresses
- This should be a clear and concise statement that identifies the specific problem that the research aims to solve
- It should be written in a way that is easy for the reader to understand

Make the connection between your problem and the existing research using phrases

Phrase	Connection
Although x has been studied in detail, insufficient attention has been paid to y .	You will address a previously overlooked aspect of your topic.
The implications of x study deserve to be explored further.	You will build on something suggested by a previous study, exploring it in greater depth.
It is generally assumed that x . However, this paper suggests that y ...	You will depart from the consensus on your topic, establishing a new position.

Contrasting the real-world problem and the knowledge problem

	Real-world problem	Knowledge problem
Where is the problem?	In the everyday reality of people and things	In the studies on the real-world problem
Who experiences it?	Ordinary people, practitioners and those with the power to address the problem	Academics with an interest in the research
How do we prove it?	Provide facts in support of the seriousness of the problem	Show that the studies are inadequate

Step 4: Specify your research question

- The research question is the question you want to answer in empirical research.
- Present your research question clearly and directly, with a minimum of discussion at this point.
- The rest of the paper will be taken up with discussing and investigating this question; here you just need to express it.
- If your research involved testing hypotheses, these should be stated along with your research question.

Outline the Research Questions

- After stating the problem, outline the research questions that the study will address
- These questions should be specific and focused, and they should relate directly to the problem statement
- This will give the reader a clear understanding of what the study aims to achieve

Highlight the Significance of the Study

- Explain why the study is important and what its potential impact could be
- This can include practical applications, theoretical contributions, or any other relevant information
- The purpose of this is to show the reader why the study is worth their attention and why it matters

Mention Limitations

- It is important to mention any limitations of the study
- This can include things like sample size, data collection methods, or any other potential issues that could impact the validity of the study
- This will help the reader understand the context of the study and how it should be interpreted

End of Presentation



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Faculty of Computing and Informatics
Department of Computer Sciences / Informatics

Title:

Name:

Supervisor:

Date: 2020





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Faculty of Computing and Informatics
Department of Computer Sciences

Outline

1. Introduction/Background
2. Problem Statement
3. Preliminary Literature review
4. Research Objectives and Questions
5. Research Methodology
6. Limitations, Delineation, Assumptions
7. Timeline
8. Conclusion
9. References



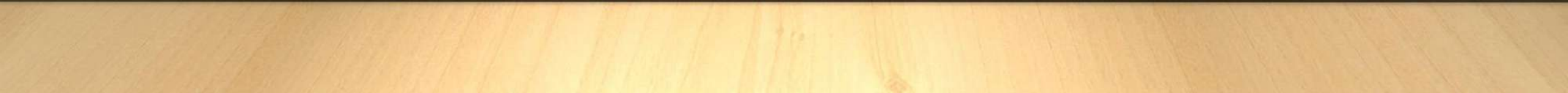
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Research Methodology

BY DR. SHAANIKA

13 APRIL 2023





Lesson Objectives

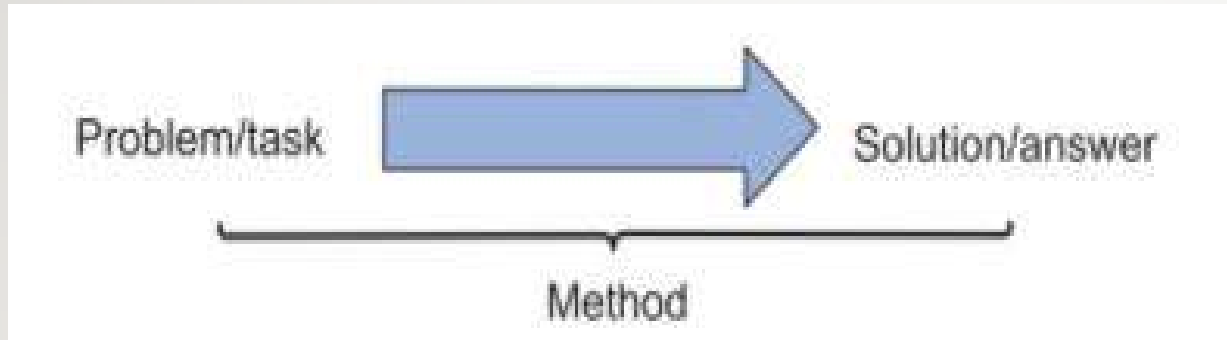
- Understand the need for methodology in research
- Explain the differences between quantitative, qualitative and mixed methods research
- Be able to select a methodology for your study

What is methodology

- A way in which we approach problems and seek answers.
 - Methodology :covers data collection techniques, data collection process and data analysis.
- A carrier to the desired destination.
- Selected in alignment with research aims, objectives and questions.

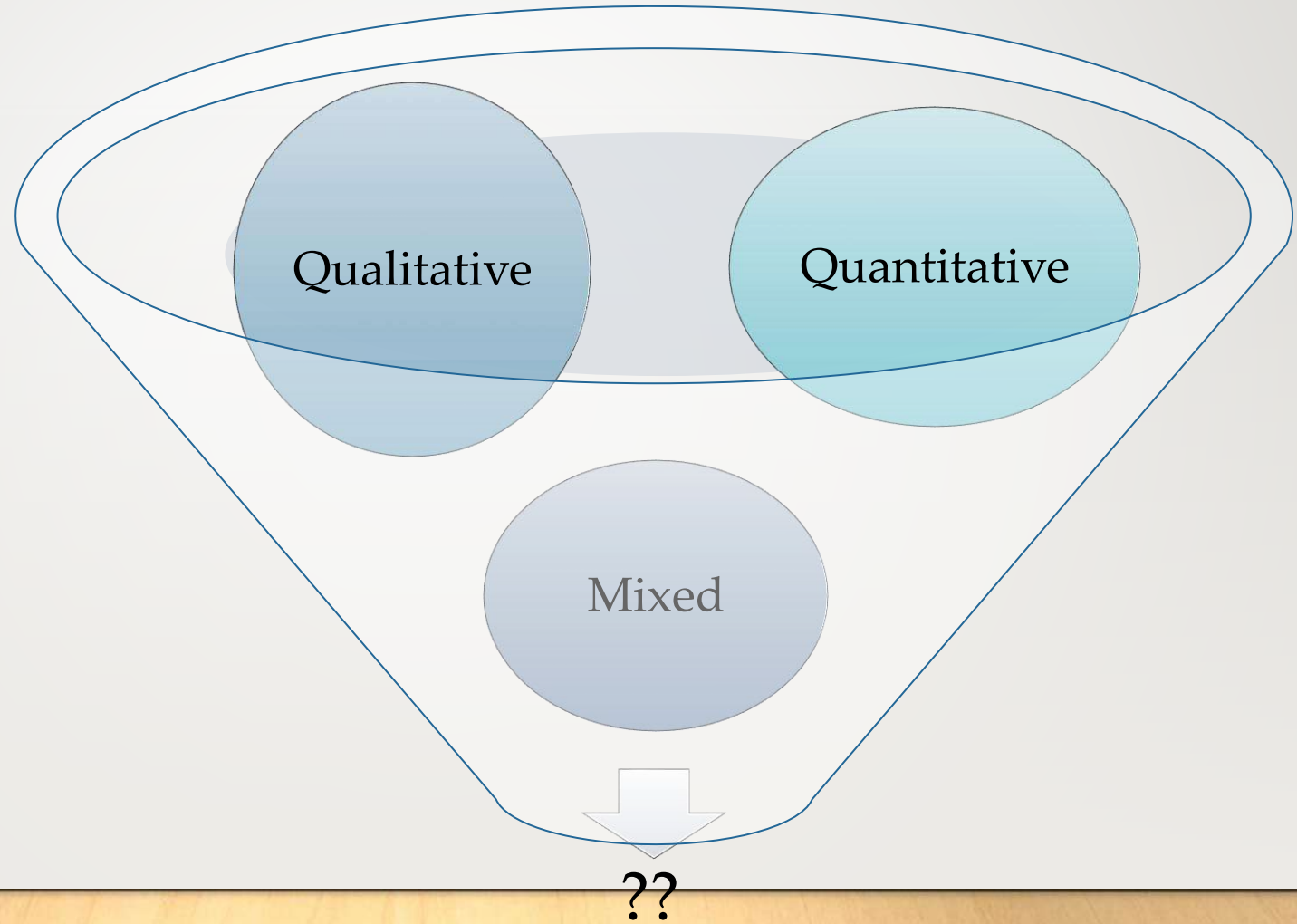
What is research methodology ?

- A vehicle undertaken to achieve study objectives.



- Logic or ways of reasoning to arrive at solutions.
- Rules of communication, i.e. to explain how the findings have been achieved (Ghauri, Grønhaug & Strange, 2020).

Methodological choice



Qualitative research method

- A form of social inquiry -explore and uncover explanations, rather than deduce them from measurements.
- Study participants' meanings and the relationships between them.
- Natural- study participants in their natural environment
- Subjective-Interpretive -each individual experience the world from their own point of view

Quantitative research method

- Main focus is counts and measurements of things
- Tends to be confirmatory
- Often begins with hypothesis and variables that needs to be tested and the results obtained are used for generalisation
- Reality should be studied objectively by the researchers who should put a distance between themselves.

Mixed research method

- Social phenomena are so complex, different kinds of methods are needed to best understand these complexities.
- Combines the use of quantitative and qualitative.
- Integration of subjectivity and objectivity.
- An objective need to be achieved by applying both methods.

Mixed Methods Designs

Research Design	Process	Purpose	Level of Interaction	Priority	Example from Literature
Convergent parallel (concurrent)	Qual Quant	To obtain different but complementary data to answer a single research question	Data collected & analysed independently	Equal	Peters and Cotton (2013) collected both mailed surveys and undertook unstructured interviews with women with physical disability in order to gain a broad understanding of the barriers and facilitating factors associated with in accessing and experiencing breast and cervical cancer screening services.
Sequential explanatory	QUANT → qual	Qualitative data are collected to explain the quantitative findings	Quant data frames qualitative data collection	Quantitative dominant	Pfaff et al. (2014) used a mailed survey to measure perceived confidence in interprofessional collaboration amongst new graduate nurses. Following analysis of the survey data they conducted interviews with 16 new graduate nurses to explain and expand upon the quantitative findings.
Sequential exploratory	QUAL → quant	Quant data builds on qualitative findings to provide generalizability	Qualitative data frames quant data collection	Qualitative dominant	Hamshire, Willgoss and Wibberley (2013) conducted a series of interviews with nursing students to explore their experiences and expectations of their nursing course. These interview findings informed the development of an online survey that was administered to 1080 students across nine UK universities.
Embedded / Nested	Qual (quant) Or Quant (qual)	To obtain different data to answer a complementary research question	Embedded dataset provides answers to a complementary research question.	Can be either Qualitative or Quantitative dominant	Kinser et al. (2013) conducted a randomised controlled trial of an 8 week yoga intervention for women with major depression. Outcomes included depression severity, stress, anxiety and rumination. Qualitative interviews were embedded in the trial to explore the feasibility and acceptability of the intervention.

Methodologic choice

- A closer look at your objectives.
- What is the best way to resolve your problem: explanations or measurements , experiments?.
- Look at the potency/weakness of each method in relation to your research problem.
- The methods used will influence the type of data collection techniques.

Exercise : Identify type of research

- Performance evaluations of two algorithms (memory usage, processing time, etc)
- Users perceptions about a particular movie. (Visual quality, acceptance, interesting, etc)
- To understand the perceptions of using new mobile applications to disseminate information related to COVID-19
- Study the performance of two systems producing a certain product e.g Bottle milk production



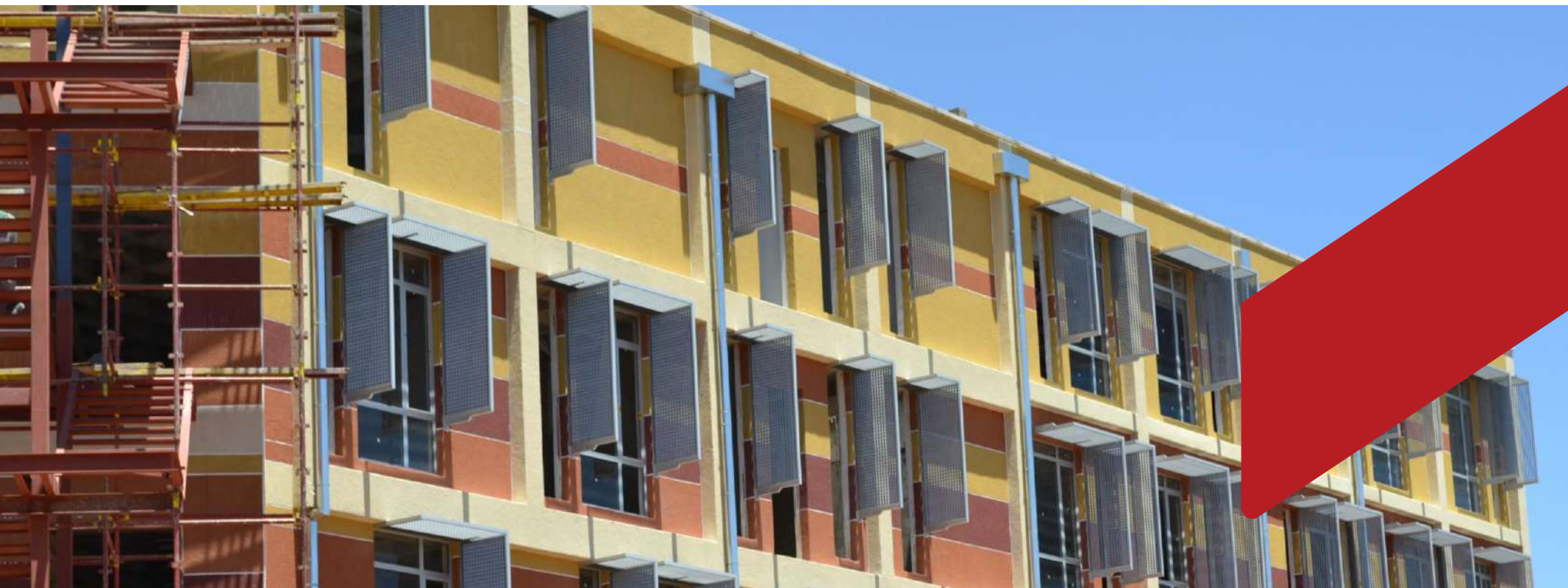
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Faculty of Computing and Informatics

RIT812s: Research Methodology

Dr Irja Shaanika
Dept. of Informatics

Date: February 2023





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What is Research?





*“Research is an **original contribution** to the **existing stock of knowledge** making for its advancement. It is the **pursuit of truth** with the help of study, observation, comparison and experiment. In short, the search for knowledge through objective and **systematic** method of finding solution to a problem is research.”*



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What is Scientific Research

*“If a research contributes
to a **body of science**, and follows the **scientific method**.”*



And what is Methodology ?

*“Methodology is a research strategy that translates **ontological** and **epistemological** principles into guidelines that show **how** research is to be conducted and principles, procedures, and practices that **govern** research.”*



The Functions of Research

The main function of research is to improve research procedures through the refinement and extension of knowledge.

The improvement is associated with various aspects:

1. Research encourages scientific and inductive thinking, besides promoting the development of logical habits of thinking and organisation.
2. The function of research is to make a decision concerning the refinement or extension of knowledge in this particular area.
3. Research assumes significant role in the formation of policies, for the government and business. Research also tries to solve various operational and planning problems.



The Functions of Research.....

4. Another function of research is to aid administrators to improve the Social systems. Research is equally important to social scientists for analysing social relationships and seeking explanations to various social problems.

5. Research helps in carrier development.



Classification of Research

1. Basic level(Basic research is designed to add an organized body of scientific knowledge and does not necessarily produce results of immediate practical value).
2. Applied level (to solve an immediate practical problem and the goal of adding to scientific knowledge is secondary).



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Faculty of Computing and Informatics

Thank You.

Ethics in Research

By Dr Shaanika





Table of Content

- Ethics: the concept
- Stakeholders in research
- Ethical issue to consider concerning
 - Research Participants
 - Collecting Information
 - Seeking informed consent
 - Providing incentives



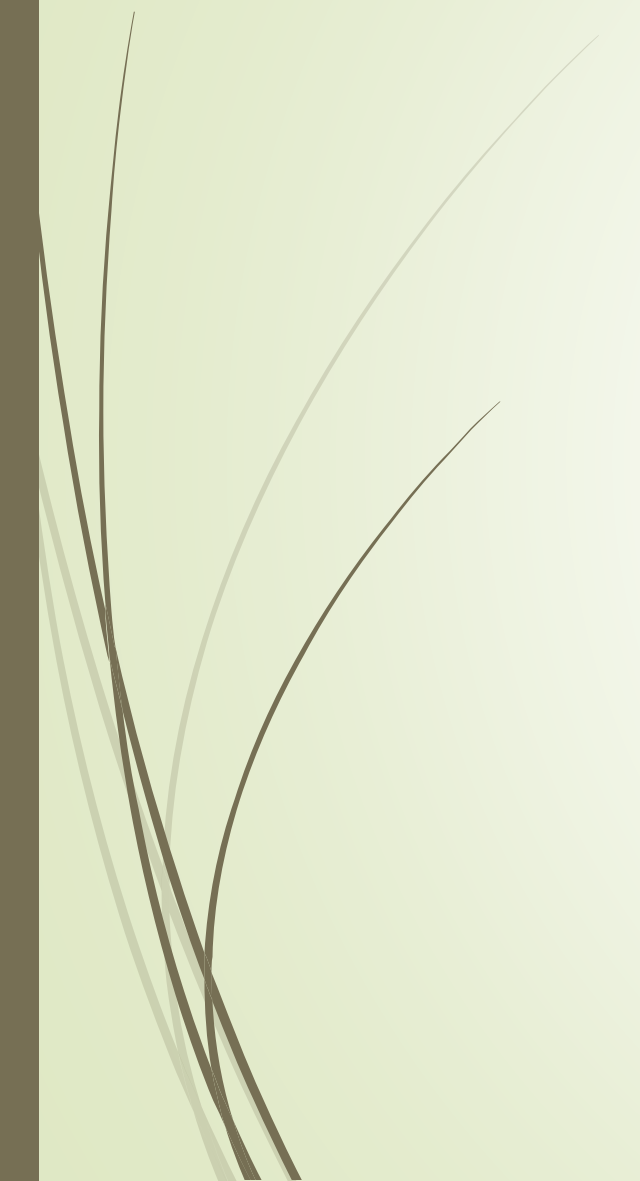
Ethics: the concept

- Babbie (2007) describes ethics as conforming to standards of conducts in accordance with professionalism or environmental culture.
- Ethical considerations in research reflect significant *moral concerns about practice and responsibilities* of researchers in the environments that they intend to conduct their studies.
- Ethical Standard (factors to consider)
 - Participant's consent, freewill to enroll in the interview process
 - Confidentiality and anonymity(how will it be maintained)
 - Data retention -protection





Why Do ethics Matter

- Crucial for maintaining research integrity.
 - Research misconduct can have dire consequences.
 - Following ethics protects both the researcher and participants.
- 



University Research Policy

- See document attached on ELearning



Stakeholders in research

- NUST
- Researcher
- Supervisors
- Organisations(Cases)
- Participants



Research Participants

- From study population –
 - From whom the information required to find answers to your research question is obtained.
 - Decide specifically and clearly who constitutes your study population, in order to select the appropriate respondents.



Collecting Information

- Data collection tools(interviews, questionnaires, observations)
- Informed consent
 - Organisation
 - Individual participants



Seeking informed consent

- A copy of the Approved proposal signed by both supervisor and student.
- A copy of the questionnaire/interview(if any)/ data collection protocol/tools
- Filled in Ethical Clearance Application Form
- Completed Informed Consent form
- Letter Requesting permission from target partner/institution.

Plagiarism

Common Types of Plagiarism

- ✓ **Complete plagiarism:** Submit another authors work in your name
- ✓ **Source-based plagiarism:** Reference an incorrect or non-existent source
- ✓ **Direct plagiarism:** Copy text from another document word-to-word
- ✓ **Self or auto plagiarism:** Reuse a major part of your own work without attribution
- ✓ **Paraphrasing plagiarism:** Make minor changes and use others' writing
- ✓ **Mosaic plagiarism:** Interlay someone else's phrases or text within your work.
- ✓ **Accidental plagiarism:** Unintentional paraphrasing or copying due to neglect
- ✓ **Inaccurate authorship:** Authorship instead of acknowledgment to contributors and vice versa

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Thank You

➡ Any Questions





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Department of Computer Science

Research Methodology

ASSUMPTIONS,

LIMITATIONS AND,

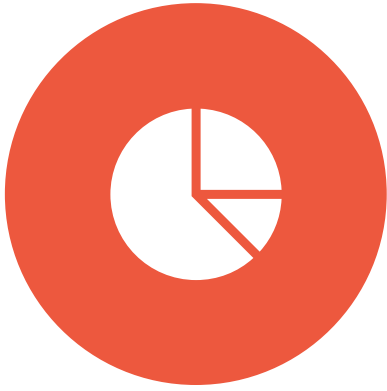
DELINIATIONS IN THE RESEARCH PROCESS

ETHICAL CONSIDERATIONS
CHAPTER OUTLINES

Why are we here?

During the process of your thesis writing, you might suddenly realize that your research has innate flaws, of which some you have control of, and others you don't. Being able to recognize and accurately describe these problems is what makes you a distinctive researcher.

What are they?



LIMITATIONS.



**DELINEATION
S.**



ASSUMPTIONS.

Limitations?

- ❖ These are potential **weaknesses** of the study that are **out** of the researcher's control.
- ❖ They are difficult to contain.
- ❖ In most cases, they are closely associated with the chosen research design, funding constraints, or other factors.
- ❖ Do not worry about limitations because limitations affect almost all research projects, as well as most things in life.
- ❖ However, they should be **clearly acknowledged** as they may affect the study design, results and ultimately, conclusions.

Common examples of limitations

❖ Access

If your study depends having access to organisations, people, or documents, and for some reasons, this access is denied, or limited in one way or the other, the reason for this need to be stated and described.

❖ Time

Another common limitation is time. If your study was limited to a certain amount of time, your results are affected and are depended on conditions occurring during that time and by the operations of society during that time period (e.g., economy, social trends).

❖ Lack of prior research studies on the topic

Citing prior research forms the basis of your literature review, and helps lay a foundation for understanding your problem. A limitation like this one can serve as an opportunity to describe the need for further research.

❖ Resources (e.g. finance)

❖ Participant dropout,

❖ Limitations of instrument used, etc.

Why should I state my limitations?

- ❖ **Always admit and identify the study's limitations**

Not mentioning them means you are giving your supervisor an opportunity to point them out on your behalf, and this can lead to failure as it will be assumed that you have willingly ignored them

- ❖ **They are an opportunity for you, as a researcher to make suggestions for future research.**

- ❖ **All studies have limitations**

- ❖ **The discussion of Your limitations should therefore be related and restricted to the problem under investigation.**

Delineations?

- ❖ This are the boundaries that you, the researcher, sets for your own study.
- ❖ They define the scope of your study, and unlike the limitations, they are in your control.
- ❖ *Delimitations are set so that the study's aims and objectives do not become impossible to achieve.*

Listed below are examples of delimitations:

- ❖ Research objectives,
- ❖ Research questions,
- ❖ Methodology,
- ❖ Variables and populations chosen as targets to study.

Things to consider on Delineations

- ❖ When you are stating your delimitations, clearly inform the reader why you chose a certain aspect of this particular subject/problem and not the other.
- ❖ The answer might simply be that you were curious about the topic and/or wanted to improve it.
- ❖ In any case, you should clearly list the other options available and the reasons why you did not choose these options immediately after you list your delimitations.
- ❖ Limit your delimitations to the things that the reader might reasonably expect you to do, but you, for clearly explained reasons, have decided not to do/cover
- ❖ Remember that delineations are not weaknesses, and you don't have to apologize for them. Good, strong research projects have clear boundaries.

Assumptions

- ❖ Assumptions are things that are accepted as true, or at least plausible, by researchers and anyone who will read your thesis/paper.
- ❖ In other words, any scholar reading your paper will assume that certain aspects of your study is true given your population, statistical test, research design, or other delimitations.
- ❖ They can't be controlled, but they need to exist for your study to be credible, or essential.

For Example:

If you are conducting a survey, you need to assume that people will answer truthfully. Or, if you are choosing a certain sample, you need to assume that this sample is a true representative of the population that you wish to make inferences to.

Summary on the DLA

	Delimitations	Limitations	Assumptions
<i>Purpose</i>	To determine the parameters/scope of the study	To determine the weakness associated to decisions made in the study	To make a study relevant
<i>Control</i>	There is a level of control over where to draw the boundaries	Difficult to control but some can be addressed in the study	Can't be totally controlled but they should exist to make the study relevant

Ethical considerations

Research Ethics Clearance is an instrument to measure the ethical fulfilment of a research process. Approval from the Ethics Committee must be obtained before research begins. In other words, Research Clearance Ethics is a reference for researchers in upholding the values of integrity, honesty, and fairness in conducting research.

The main purpose of obtaining ethical clearance is to:

- protect research subjects/respondents from physical harm (threats),
- psychological harm (stress, regret),
- social harm (stigma, hatred from society) and most importantly,
- legal consequences (sued) as result of participating in a research.

Who needs ethical clearance?

Researchers can apply for an ethical clearance if their research meets the following criteria:

- All research involving humans must go through a process of ethical clearance. Excluded from this is research that uses secondary data, literature reviews or published data.
- Research that uses a review of confidential materials (medical records from hospital/health clinic) must go through an ethical clearance process (although it still requires permission from the issuing institution).
- Likewise, research that uses information from closed/non-public media, such as statistical reviews from an institution (employees, clients, patients, users, service providers, service records, financial records, company records) must go through ethical clearance.
- Studies that use additional methods, which deals directly with humans such as interviews, Focus Group Discussions (FGD) and others, still require an ethical clearance process, although the main method used is a review of publicly available material.

What is needed?

Supervisor + student has to validate that the following documents have been fully completed before sharing documents with the Ethical Clearance committee.

Checklist:

1. A copy of the Approved Proposal signed by both supervisor and student.
2. A copy of the Questionnaires/ Interview (if any)/ data collection protocol/tool.
3. A filled-in Ethical Clearance Application Form
4. A written/completed Informed Consent Form (ICF) - **Information Sheet, Certificate of Consent**
5. A copy of Draft letter requesting Permission from target partner/institution (alternatively, a permission letter from them)

Thesis Outline

The final thesis will have the following chapters

Chapter One: The Introduction and background information

Chapter Two: Literature review

Chapter Three: Methodology - Design of proposed framework/system

Chapter Four: The controller design and its classifiers

Chapter Five: Findings, discussion and contributions

Chapter Six: Conclusion and Future work

References

Marwat, K. (2015). *Limitations, delimitations, assumptions and keywords of research*. [PowerPoint slide]. Comsats Wah Campus. Retrieved from <https://www.slideshare.net/kamalshah3975/limitationsdelimitaions-assumptions-and-keywords>

Simon, M. K. (2011). *Dissertation and scholarly research: Recipes for success* (2011 Ed.). Seattle, WA, Dissertation Success, LLC. <http://dissertationrecipes.com/wp-content/uploads/2011/04/AssumptionslimitationsdelimitationsX.pdf>

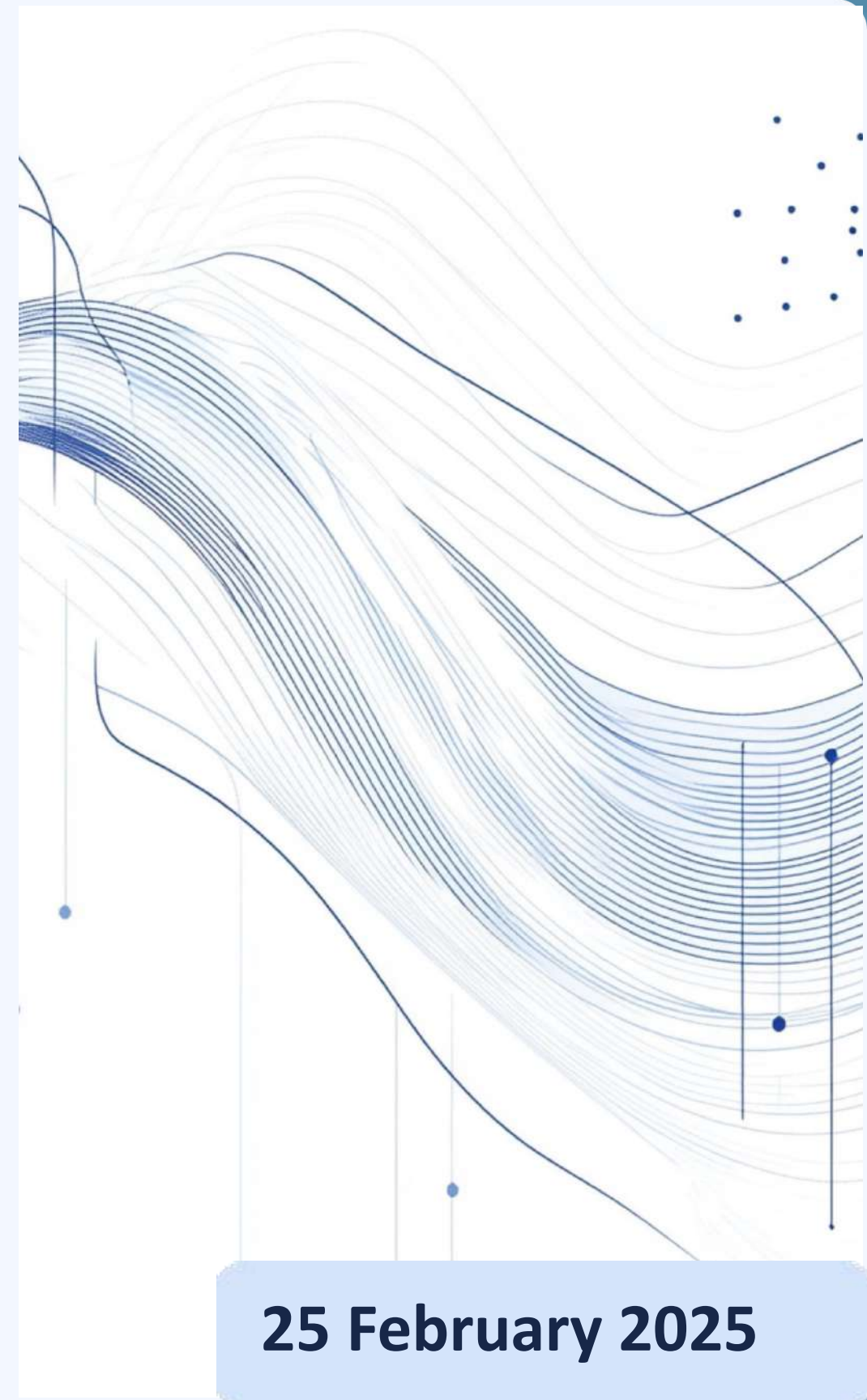
Thank you

Any Questions?

Creating an Effective Problem Statement

Prof Jude Osakwe

25 February 2025



CONTENTS

1. Understanding the Purpose

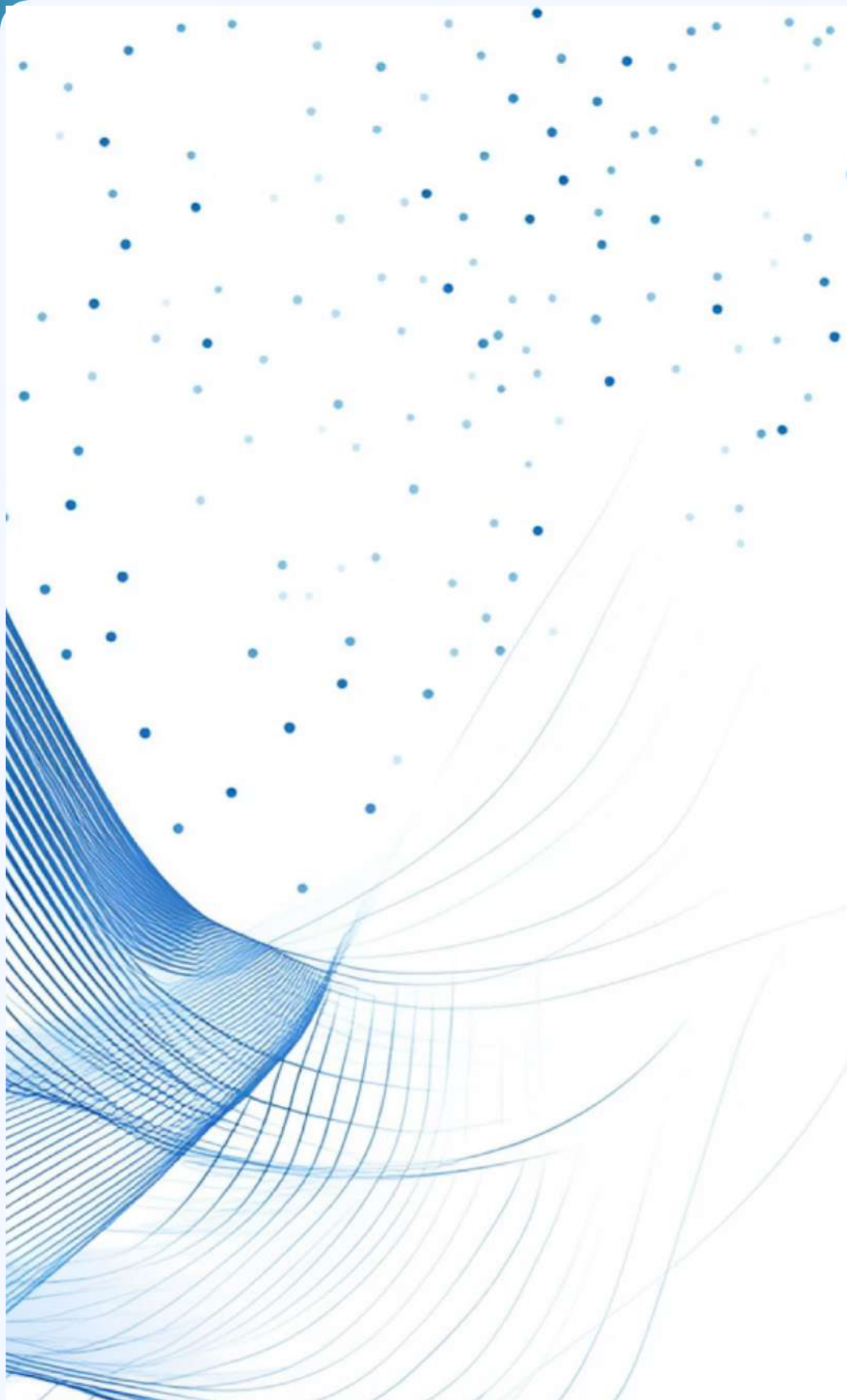
2. Steps to Writing a Problem Statement

3. Putting It All Together: A Template

4. Tips for Writing a Strong Problem Statement
Statement



1. Understanding the Purpose



Importance of a Problem Statement

1

Clear Definition of the Issue

- Clearly define the issue at hand.
- Ensure immediate understanding of the problem.

2

Justification for a Solution

- Explain the significance of the problem.
- Highlight the importance of solving it.

3

Focus for Efforts

- Provide a clear target for efforts.
- Ensure alignment on the goal.

4

Support Statements with References

2. Steps to Writing a Problem Statement Statement



Describe the Ideal Situation (The "What Should Be")

Vision of the Desired State

- Paint a picture of the optimal scenario.
- Use specific and concrete language.

Establishing a Baseline

- Quantify the ideal situation with metrics or benchmarks.
- Help measure progress later.

Describe the Current Reality (The "What Is")



Articulation of the Problem

- Clearly describe the current situation.
- Focus on observable effects or symptoms.

Objective and Factual Description

- Base description on data and evidence.
- Avoid opinions and use statistics.



Providing Context

- Explain relevant background or circumstances.
- Include information about the organization or industry.





Explain the Consequences (The "So What")

Articulating the Impact

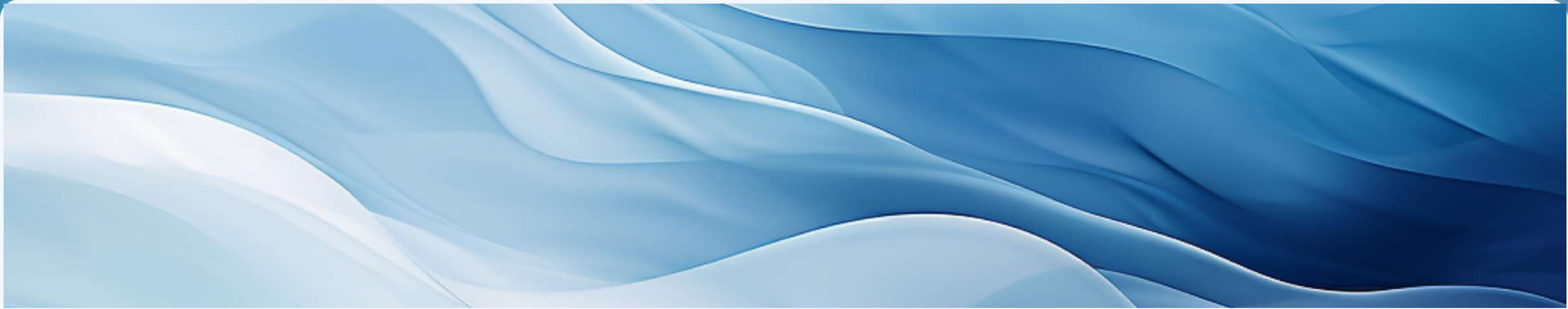
- Describe the negative consequences consequences of the problem.
- Identify who is affected and how.

Quantifying the Impact

- Use numbers to illustrate the impact.
- Provide specific examples.

Long-term Effects

- Consider immediate and future implications.
- Relate consequences to organizational goals.



Suggest a Solution (Optional, but often helpful - The "What Could Be")

Outlining a Potential Direction

- Briefly state what needs to be done.
- Focus on the desired outcome.

Keeping it Broad

- Avoid specificity about the solution.
- Encourage brainstorming and exploration.

Reiterating Benefits

- Summarize the positive impact of solving the problem.

3. Putting It All Together: A Template



3. Putting It All Together: A Template

1

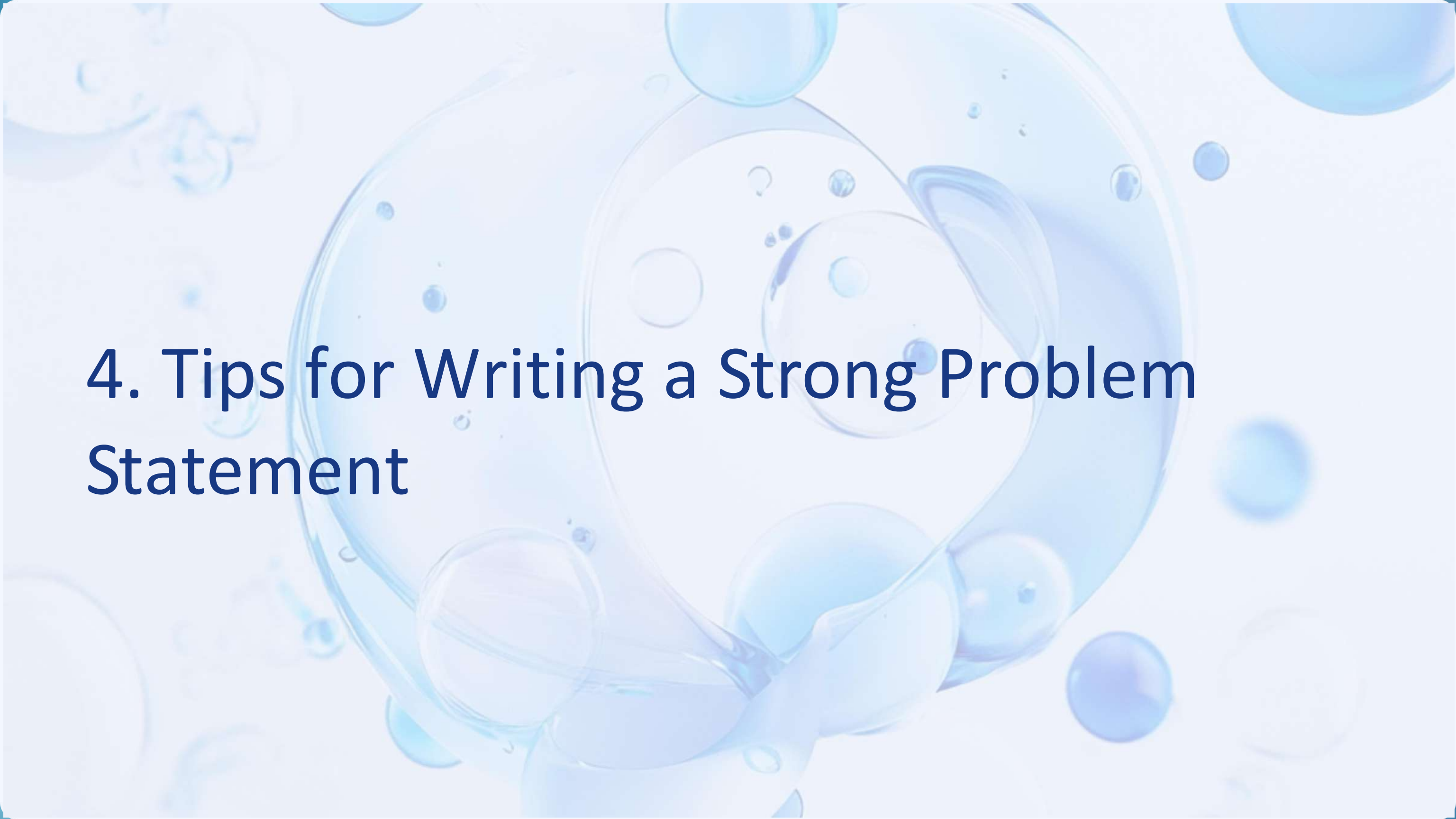
Structure of the Problem Statement

- Ideal: "Ideally, [desired state/situation]..."
- Reality: "However, currently [problem/issue]..."
- Consequences: "As a result, [negative consequences/impact]..."
- Suggestion (Optional): "To address this, we need to [general solution direction] in order to [desired outcome/benefit]..."

2

Example of a Problem Statement

- Ideal: "Ideally, our customers should be able to easily resolve their inquiries through our online chat system within 5 minutes..."
- Reality: "However, currently, the average wait time for customers using our online chat system is 15 minutes..."
- Consequences: "As a result, customer satisfaction scores related to online chat have decreased by 25%..."
- Suggestion: "To address this, we need to improve the efficiency and effectiveness of our online chat system..."



4. Tips for Writing a Strong Problem Statement

4. Tips for Writing a Strong Problem Statement(1)

1

Conciseness and Clarity

- Use precise language and avoid jargon.
- Keep it short while conveying necessary information.

2

Focus on One Problem

- Break down multiple issues into separate statements.

3

Specific and Measurable

- Use data and metrics to support claims.



4. Tips for Writing a Strong Problem Statement(2)

1

Avoid Blame

- Focus on the problem, not assigning fault.

2

Get Feedback

- Share with others for input and understanding.

3

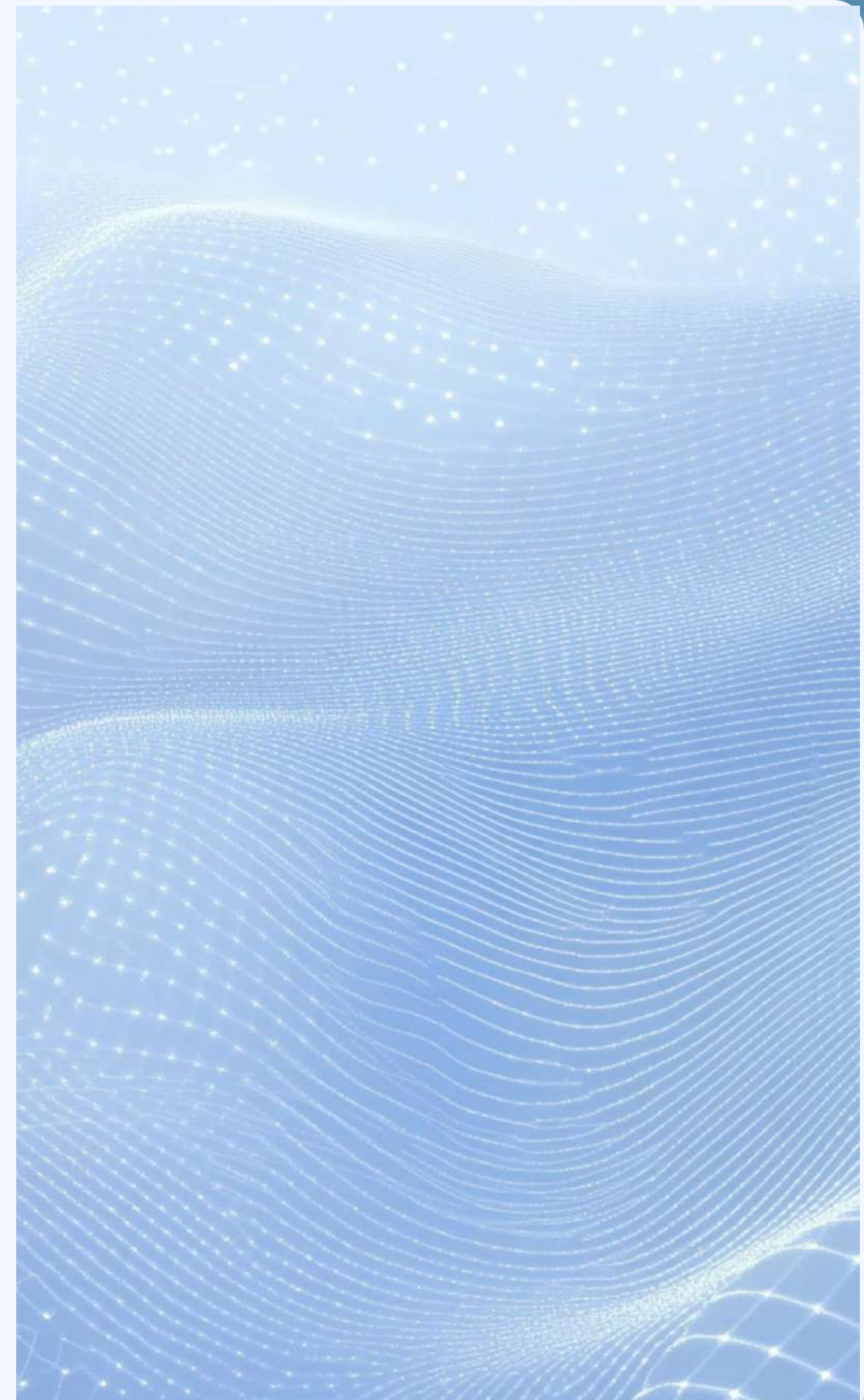
Revise as Needed

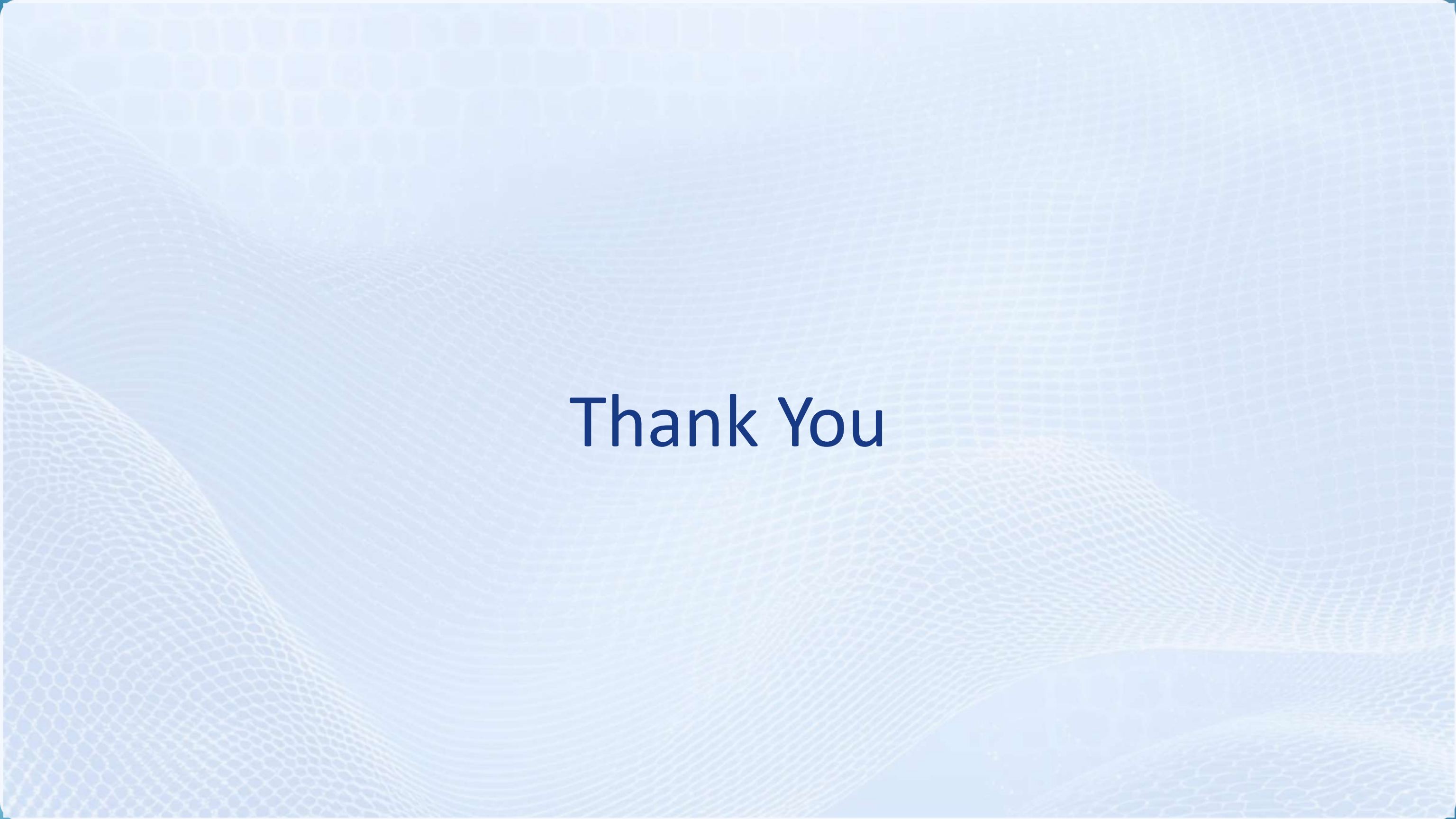
- Be prepared to refine and update the statement.

4

Consider the Audience

- Tailor language and framing for the intended readers.





Thank You



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Faculty of Computing and Informatics

2024

Basics of research?



Meaning of Research

Research in common parlance refers to a search for knowledge. Research is an art of scientific investigation. Dictionary definition of research is a careful investigation or inquiry specially through search for new facts in any branch of knowledge. Some people consider research as a movement from the known to the unknown.



What is research?

The systematic investigation into and study of materials, sources, etc, in order to establish facts and reach new conclusions. b. an endeavour to discover new or collate old facts etc by the scientific study of a subject or by a course of critical investigation



What is research?

Research is what we do when we have a question or a problem we want to resolve

We may already think we know the answer to our question already

We may think the answer is obvious, common sense even

But until we have subjected our problem to rigorous scientific scrutiny, our 'knowledge' remains little more than guesswork or at best, intuition.



What is research?

First priority is to formulate your question

Then figure out how you are going to answer it

- How have others answered it?
- How does your proposal fit in with what others have done?
- How will you know when you have answered it?

Then you can present your answer



OBJECTIVES OF RESEARCH

The purpose of research is to discover answers to questions through the application of scientific procedures. The main aim of research is to find out the truth which is hidden and which has not been discovered as yet. Each research study has its own specific purpose.



Characteristics of Research

Research is directed towards the solution of a problem.

Based upon observable experience or empirical evidence.

Demands accurate observation and description.

Involves gathering new data from primary sources or using existing data for a new purpose.

Requires expertise

Objective and logical

Involves the quest for answers to unsolved problems.

Requires courage.

Should be carefully recorded and reported.



Research Categories

Basic Research

- Conducted to verify the acceptability of a given theory or to know more about certain concepts.
- Also known as fundamental research, „theoretical“ research or „pure“ research.

Applied Research

- conducted when a decision must be made about a specific real-life problem.
 - It is of two types:
- ▮ Applied Research can further be divided into two groups:
 - ▮ Problem Solving Research
 - ▮ Problem Oriented Research



Research Process

The research process involves several key stages, including problem identification, literature review, data collection, analysis and reporting. Each stage is essential for producing valid and reliable research outcomes



Quantitative Research

Utilises statistical analysis to quantify observations and generalise results. Emphasizes objectivity and control in data collection and analysis.



Qualitative Research

Focuses on understanding human behavior and experiences through in-depth observations and interviews. Values subjective interpretation and context.



Mixed Methods Research

Combines quantitative and qualitative approaches to provide a comprehensive understanding of a research problem. Offers diverse perspectives and insights.



Experimental Research

Employs controlled experiments to establish cause-and-effect relationships. Requires strict adherence to predefined protocols and conditions.





Survey Research

Utilises Questionnaires and interviews to gather data from a sample population. Provides insights into attitudes, behaviors, and opinions.



Ethnographic Research

Involves immersion in a culture to understand social phenomena.
Requires prolonged fieldwork and participant observation.

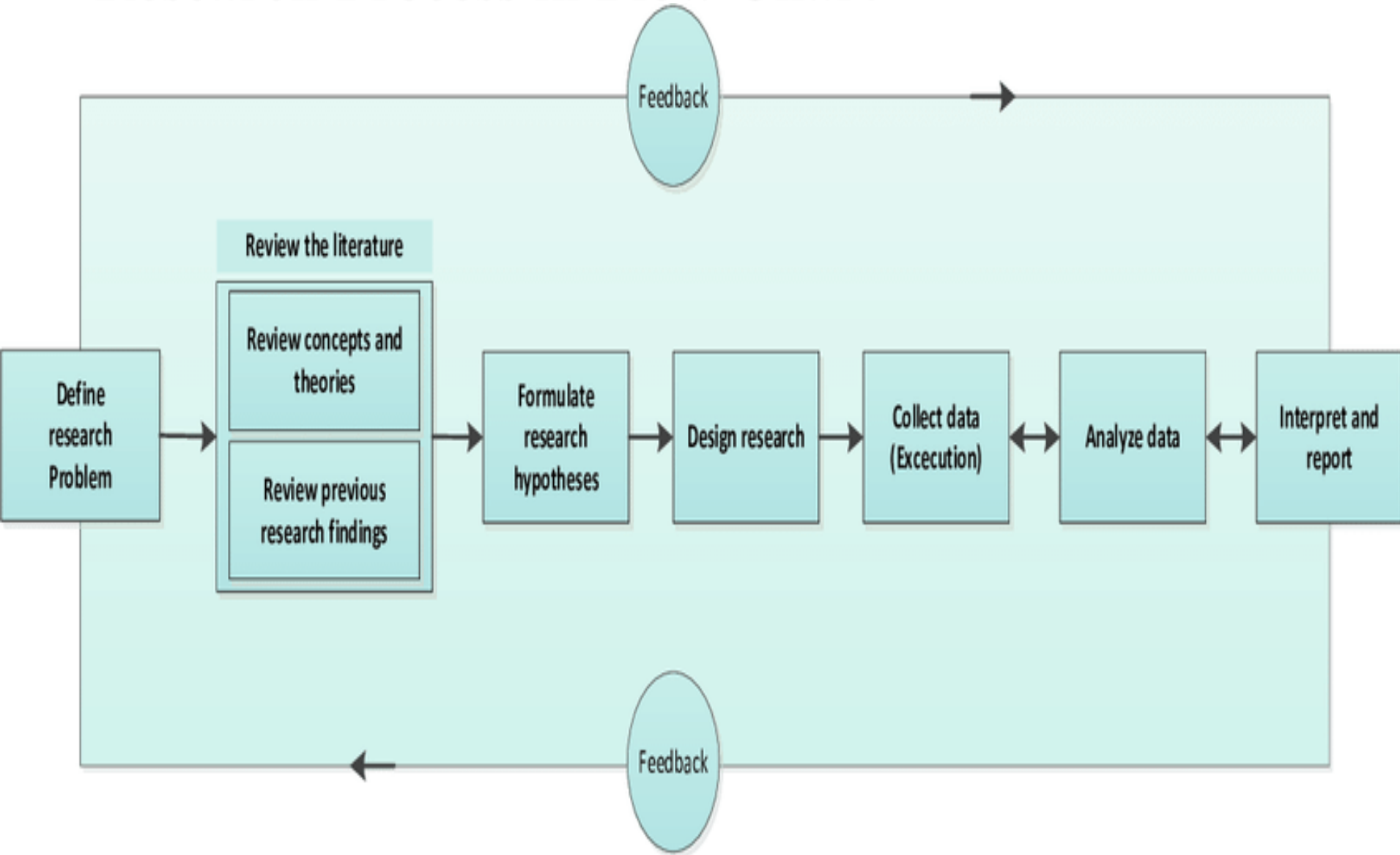


Meta-Analysis

Synthesizes findings from multiple studies to draw overarching conclusions. Provides a comprehensive overview of existing research.



Research Process in Flow Chart





Steps involved in Research Process in Research Methodology

Formulating the research problem: There are two types of research problems, viz., those which relate to states of nature and those which relate to relationships between variables. At the very outset the researcher must single out the problem he wants to study, i.e., he must decide the general area of interest or aspect of a subject-matter that he would like to inquire into.

Extensive literature survey: Once the problem is formulated, a brief summary of it should be written down. It is compulsory for a research writing a thesis to write synopsis of the topic and submit to his/her supervisor. At this juncture the researcher should undertake extensive literature survey connected with the problem.



Research Design

A **research design** is a strategy for answering your research question using empirical data. Creating a research design means making decisions about:

- Your overall research objectives and approach
- Whether you'll rely on Primary or secondary research
- Your sampling methods or criteria for selecting subjects
- Your data collection methods
- The procedures you will follow to collect data
- Your data analysis methods

A well-planned research design helps ensure that your methods match your research objectives and that you use the right kind of analysis for your data.



Determining sample design: All the items under consideration in any field of inquiry constitute „universe“ or „population“. A complete enumeration of all the items in the „population“ is known as a census inquiry. It can be presumed that in such an inquiry when all the items are covered no element of chance is left and highest accuracy is obtained. But in practice this may not be true.

Collecting the data: In dealing with any real life problem it is often found that data at hand are inadequate, and hence, it becomes necessary to collect data that are appropriate. There are several ways of collecting the appropriate data which differ considerably in context of money costs, time and other resources at the disposal of the researcher.



Execution of the project: Execution of the project is a very important step in the research process. If the execution of the project proceeds on correct lines, the data to be collected would be adequate and dependable. The researcher should see that the project is executed in a systematic manner and in time.

Analysis of data: After the data have been collected, the researcher turns to the task of analyzing them. The analysis of data requires a number of closely related operations such as establishment of categories, the application of these categories to raw data through coding, tabulation and then drawing statistical inferences. The unwieldy data should necessarily be condensed into a few manageable groups and tables for further analysis. Thus, researcher should classify the raw data into some purposeful and usable categories.



The main text of the report should have the following parts:

Introduction: It should contain a clear statement of the objective of the research and an explanation of the methodology adopted in accomplishing the research. The scope of the study along with various limitations should as well be stated in this part.

Summary of findings: After introduction there would appear a statement of finding sand recommendations in non-technical language. If the findings are extensive, they should be summarized.

Main report: The main body of the report should be presented in logical sequence and broken-down into readily identifiable sections.

Conclusion: Towards the end of the main text, researcher should again put down the results of his research clearly and precisely. In fact, it is the final summing up.

At the end of the report, appendices should be enlisted in respect of all technical data. Bibliography, i.e., list of books, journals, reports, etc., consulted, should also be given in the end.

Index should also be given specially in a published research report.



Ethical Considerations

Ethical considerations are paramount in research to ensure the protection of participants and the integrity of the study. Adhering to ethical guidelines and obtaining informed consent are fundamental principles.

THANK YOU



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OF SCIENCE AND TECHNOLOGY**

-RESEARCH PROPOSAL TEMPLATE-

PROPOSED TITLE

(Type the proposed title in capital letters, the title should be in the middle of the page)

by

Name(s) & Surname

Student Number

Submitted in fulfillment of the requirements for the degree

XXXXXXXXXXXXXXXXXXXXXXX

in the

(Department of XXXXXXXXX)

at

NAMIBIA UNIVERSITY OF SCIENCE AND TECHNOLOGY

Supervisor:

(Title, initials and surname of supervisor)

Co-supervisor:

(Title, initials and surname of supervisor)

Date of submission:

dd/mm/yyyy

METADATA

TITLE: (your title)

STUDENT NAME:

SUPERVISOR:

CO-SUPERVISOR:

DEPARTMENT:

QUALIFICATION:

SPECIALISATION:

CLUSTER:

KEYWORDS:

TYPE OF RESEARCH: Applied Research

METHODOLOGY:

STATUS: Proposal

DOCUMENT DATE: (date of submission)

SPONSOR (Cluster/ Research lab):

1. Topic:
2. Problem Statement
3. Aim
4. Objectives
5. Research Questions

Supervisor Signature:

NUST & FCI LOGO (here)

-RESEARCH PROPOSAL TEMPLATE-

PROPOSED TITLE

(Type the proposed title in capital letters, the title should be in the middle of the page)

by

Name(s) & Surname

Student Number

Submitted in fulfillment of the requirements for the degree

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in the

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(Title, initials and surname of supervisor)

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METADATA

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DEPARTMENT:

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SPECIALISATION:

CLUSTER:

KEYWORDS:

TYPE OF RESEARCH: Applied Research

METHODOLOGY:

STATUS: Proposal

DOCUMENT DATE: (date of submission)

SPONSOR (Cluster/ Research lab):

DECLARATION

I, ...(name (s))... , hereby declare that the work contained in the proposal for the degree.....project, entitled:

“.....(title of the project).....”,

is my own original work and that I have not previously in its entirety or in part submitted it at any university or other higher education institution for the award of a degree.

I further declare that I will fully acknowledge any sources of information I will use for the research in accordance with the Institution rules.

Signature: _____ Date: _____

SIGNATURE OF THE SUPERVISOR

I, _____, herewith declare that I accept this proposal for my supervision

Signature: _____ Date: _____

INSTRUCTIONS:

- ✓ The body of your proposal must address the sections listed below; however you are free to adapt to the type and nature of your investigation.
- ✓ Use Calibri font 12; line spacing 1.5; justify-alignment (except if intended)
- ✓ Page numbering: use “Page X of Y”, or “x/y” at bottom of page
- ✓ Avoid use of “first person singular” like “I”, “me”, etc.
- ✓ The length of the proposal write-up should be between 10 to 15 pages (up to 20 for PhD), including everything.
- ✓ Remove all instructions given in this template in your final write-up.

PROPOSED TITLE

Type the proposed title in capital letters, the title should be in the middle of the page; an unambiguous title that conveys the focus of the research

ABSTRACT

(short summary of the study in about 200 -250 words; the content should include: context, problem, methods/solutions, significance, (future..)); (This is the last section of your proposal to be written)

(Also see Appendix A for more insight)

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1 INTRODUCTION/BACKGROUND

Note that a research proposal is always future directed. It is an “architects plan” which explains what the researcher intends to do. It is generally written in the future tense. Here you give the context or background and introduction to the research.

2 PROBLEM STATEMENT

(Main statement; may be followed by related questions) (and/ or) Hypothesis (assumptions) or / and the type of envisaged solution)

Divide this part into three:

- Motivate the reason / need for the research by, for example, explaining the dilemma and problem. What will you be investigating in order to come to a conclusion about the problem that you have identified.
- Immediately after this one should state any HYPOTHESIS, IF ANY, if it refers to basic statement related to the problem; otherwise if the hypothesis is related to the type of suggested solution then it will be stated after “Research Objectives or Research Questions

3 RESEARCH OBJECTIVE(S)

(Main objective and sub-objectives as major tasks to be done which will bring you to, or help you reach a given milestone)

- Here you also have to mention possible research objectives (what do you want to achieve with this study). Usually these are 3-5 steps or milestones or sub-objectives you will follow to reach the purpose. Remember when readers get to your conclusion of your study they will go back to this part to see whether your stated purpose or objectives have been achieved.
- (alternatively) **RESEARCH QUESTIONS**: Main and sub-questions. Each (sub) Research question should be answered when at least one Research (sub) objective has been met.

4 PRELIMINARY LITERATURE REVIEW and or Theoretical Underpinnings/ Framework

(Review of existing state of knowledge (understanding, methodologies, solutions, projections...) on the main questions/ issues of the research that your Keywords and or practices (refer to the main variables of the problem) encapsulates.

- **More than 60% should deal with the problem focus** (one or some of this: problem domain, the process, methodology, solution, significance); and **less than 40%** on the remaining aspect.
- Advised to use sub-sections corresponding to knowledge/ issues encapsulated in the main questions/ or objectives of the study
- The section should end with a short preliminary conclusion, and link to your initially indicated problem statement).

Normally your full literature review (in the final Thesis) must adequately cover the following issues:

- Conceptual and operational definitions of the key constructs mentioned in the title and hypotheses
- A discussion of relevant previous research or conceptual work on the constructs
- A discussion of possible relationships between key constructs or expected differences between groups on the key constructs included in the study
- A discussion of possible measurement approaches and scales for each of the key constructs included in the study
- Demonstrating your ability to critically review and logically present the materials in a coherent manner.

It must provide enough theoretical support for the hypotheses formulated, if any, for testing.

Use relevant, current and comprehensive sources from authority in the field

5 IMPORTANCE / SIGNIFICANCE/BENEFITS OF THE STUDY/VALUE

State why this study is important, and to whom it may have any benefits/values. Why is your work worth doing? There is usually a theoretical (oftentimes academic) and a practical significance (to the social, human, industry context, etc.) which needs attention/explanation.

6 RESEARCH METHODOLOGY (Approach to the solution)

This section should consist of the following:

- Your Methodology (blueprint view of the inquiry)
- The research Design: Name and discuss the overall approach you will use to test your thesis statement or research questions. Qualitative or quantitative or triangulation, back this with references. Here you can also highlight the strength and weaknesses of the design.
- Your research Methods: Each design has certain methods, indicate the methodology in detail, how you apply it and justify this to the reader. Sub-section here should be research/data collection instruments, participants and data analysis.

6.1 Research/data collection instruments

Describe the instruments you will use for data collection, what you expect them to deliver and why you believe they will get you reliable data. Also indicate the reliability and validity of the instruments. How do you know the instruments will get the data you want and that this data will measure what you say it should. Will the data be sufficient for your purposes?

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Which sampling technique did you apply for selecting your participants? Why only these and what will their feedback provide/mean to assist you to answer the research questions? Provide a detailed table of how many participants from which levels in the company/business you will include in the study.

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(see [Appendix B](#) for a more hierarchical presentation of the same.)

- Limitations of the method: All methods have limitations. To identify what limitations your method have, brainstorm how else you might have tackled your problem.

7 DELINEATIONS AND LIMITATIONS

7.1 Delineation

covers you back and explains exactly what you are responsible for and by detailing what you are not responsible for and why. This is the scope of your work.

7.2 Limitations

are those things which you will not cover because of money, time and also has to do with the method you have chosen and its limitations (scope, context, resources/time/finance/skills, etc.)

[and (optional) **Risks** (impact of constraints, and ethics issues)] ; (It is advised at proposal level you make use of bulleting items)

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(Matching methods/tasks with sub-objectives over time; Gantt-chart can be used; landscape paper format too)

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Chapter One: Introduction

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Next Chapter: Research methodology

Next Chapter: Findings/results and discussion

Next Chapter: Significance of the findings (=> contribution)

Next Chapter: Conclusions, recommendations and future works

REFERENCES / BIBLIOGRAPHY

The list of references at the end of your report contains details on all materials consulted (books, proceedings, journal articles, e-mail messages and web pages) and that have been cited in the text. Please make sure that you use the correct referencing method in the text as well as in the list of references at the end of your report.

Bibliography may be chosen if the list of sources should include materials used but could not be cited in the text (check with your supervisor)

Please use the **APA method** of referencing as explained in the CITATION guide as provided by NUST Library and from the FCI research Coordinators.

APPENDIX A: ABSTRACT CONTENT

Item in abstract	Example
Problem specification	The lack of <i>[state what is missing that causes the problem you have identified]</i> remains an open problem.
Main ways of attacking the problem(<i>show knowledge of area</i>)	The most common approaches to <i>[your problem area]</i> are based on <i>[state broad underlying assumption/theory]</i> . These approaches <i>[list main features]</i> . While such approaches have the advantage of <i>[say what they are good at]</i> , they cannot <i>[identify limitations]</i> .
Recent approaches (<i>show problem hasn't been solved recently</i>)	In recent years, <i>[alternative 'leading edge' type of approach/system/model/theory]</i> has gained increasing support, because <i>[say why it is getting popular]</i> .
Specific recent approach that you are going to develop further, and its limitations (<i>establish small research gap</i>)	<i>[Specify one form if the new approach]</i> is one form of approach/system/model/theory that <i>[say what this form is particularly good at]</i> . However, attempts at implementing/trialing the <i>[specific approach]</i> have exposed limitations. <i>[describe some different attempts and their respective limitations]</i> .
Aim of the thesis. Main features of proposed solution.	This thesis is to design/develop and test a new <i>[approach/system/model/theory]</i> , which <i>[summarise its main features]</i> . It will do this by <i>[indicate methodology/methods]</i> .
Expected benefits/contribution. How these will be evaluated.	It is expected that this approach/system/model/theory will overcome the limitations of <i>[say what you expect it will better than existing approaches]</i> . The new approach/system/model/theory will be evaluated by <i>[say how the benefits will be demonstrated]</i> .

(Source: Muyingi et al. (unpublished). "The complete research guidebook full of things your supervisor should have told you")

APPENDIX B: METHODOLOGY

Methodology can be restructured in Hierarchy:

1-The research Paradigm (philosophical view (ontology, epistemology, axiology, and rhetoric) of knowledge and its relationship with the researcher; e.g. Positivism, Interpretivism, Pragmatism

2-The Main Approach / Methodology : High-level map/ Set of Principles and Focal Approach (Descriptive; Explanatory; Exploratory, Theory testing/building/ constructive; Comparative; Longitudinal; Action Research; Participatory; Blended methodologies)

3- The Main Design or Argumentation Flow or Problem Solving Approach or Strategy (Deductive/ Inductive/ etc.)

4-The Methods (How to apply the approach: (Data type: qualitative or Quantitative or Mixed); and (Procedures)

5-Tools/Techniques/Instruments: (how to materialise the methods):

a)-The instruments/ tools to Collect Data

b)-The instruments/ tools to Analyse Data

c)-The instruments/ tools to Evaluate/ test Findings)

APPENDIX C: TECHNICAL CARE CHECKLIST FOR FINAL RESEARCH PROPOSAL

Please check that your final research proposal complies with the specific requirements set out below. Students often lose unnecessary marks because of sloppy technical care. Ask yourself the following questions ...

TECHNICAL CARE	✓ or x
1. PRELIMINARIES (Before the main body text starts on page 1)	
Is the layout of and information supplied on the title page correct?	
Have I included an Abstract (on a separate page) before the Table of Contents?	
Does the Abstract provide a <u>complete</u> summary of the whole research proposal?	
Is the Table of Contents complete and correct?	
Is the List of Figures complete and correct?	
Is the List of Tables complete and correct?	
Do the wordings of headings correspond meticulously with the wording in the Table of Contents?	
Do the headings numbers in the text correspond meticulously with the numbers in the Table of Contents?	
Do the figure / table captions listed in the List of Tables and List of Figures correspond a 100% with the captions used in the text?	
Have all tables and figures been supplied with correct captions (situated <u>above</u> the table or figure)?	
Have all tables and figures been supplied with correct and complete source references where appropriate (situated <u>below</u> the table or figure)?	
Have all tables and figures been numbered correctly? (Figures and tables are numbered independently starting from 1. Do not include section numbers in the numbering of tables / figures)	
Are the preliminary pages (i.e. Executive Summary, Table of Contents, List of Figures and List of Tables) numbered in Roman numerals: (i), (ii), (iii), (iv)?	
2. TEXT	
Are all pages in the body of the text numbered correctly with Arabic figures (1, 2, 3, etc. out of total page)?	
Have I used only one method of emphasizing (<i>italics</i> , or bold or <u>underlining</u>) throughout the report?	

Is there a line open between paragraphs?	
Are the paragraphs in my report not perhaps too long? Can't I split long paragraphs to improve readability?	
Have I made any "sweeping" or unsubstantiated statements, such as "there is no literature available on this topic" or "this research will contribute to the body of knowledge"?	
Have I used an impersonal, objective writing style (One symptom of a subjective writing style is that the researcher refers to him-/herself directly. References to "I", "we" and even "the researcher" should be avoided.)	
3. REFERENCING TECHNIQUES	
Are all sources (books, journals, e-mail messages, web pages) that have been cited in the text, also been included in the List of References at the end of the report and vice versa?	
Is the list of references strictly in the required order of the referencing style you are using ?	
Have I used the correct revised (abridged) Harvard referencing method in the text, as well as in the list of references?	
Have I correctly indicated the sources of all tables / figures taken or adapted from the literature? (Such references are provided <u>below</u> the table or figure)	
4. APPENDICES	
Do all the appendices have appropriate descriptive titles?	
Are all the appendices numbered up to the last page?	
Have all the appendices been tagged / "flagged" for easy cross-referencing?	
Are all the appendices included in the Table of Contents with their correct page numbers?	
5. GENERAL TECHNICAL CARE	
<i>Have I checked my proposal for spelling and grammatical errors?</i>	
<i>Have I used capital letters and abbreviations correctly?</i>	
Have I consistently rounded off all numeric values to three (3) decimals?	
Has text of the whole document been "justified" (i.e. aligned evenly along both the left and right margins to form a square box)?	
Are all headings numbered correctly?	
Headings in capital letters (first order headings) should not be underlined.	
There should be no full stops at the end of headings.	
Have I used an A4 paper setting?	

Have all pages printed correctly?	
Have I consistently used one and a half (1½) line spacing in the body of the text?	
Is the font size of the body text correct (12 pt)?	
Have I consistently used the same font type (i.e. Ariel or Times New Roman) and font size (12 pt) for the body text?	
Is the left margin set to at least 2 cm?	

Postgraduate Research Progress Report Template (one page)

Faculty of Computing and Informatics (FCI)

Year under Review: (tick-off)	One	Two	Three	Four	Five	Six	Seven	Eight
Semester End: (tick-off)								
Student Name, Surname								
Student Number								
Degree								
Research Title								
1. Research Problem in brief (Problem Statement)								
2. Research Objectives (use bullet points)		•						
3. Evaluation	Activities Completed/ Achievements? What outputs have been Delivered? (use bullet points)							
	Status (If No, please provide details: How far from schedule, budget or objectives? What actions are you taking to resolve the delays? What support is required?)	On schedule?						
		On budget?						
		On target?						
	Activities in Progress (use bullet points)							
	Activities Outstanding (use bullet points)							
4. Adjustments Were there any adjustments to the scope? (use bullet points, If yes, please provide details: What was changed? How will it impact your schedule, budget or objectives?)								
5. Challenges to Date (use bullet points)								

6. Risks (use bullet points)	
7. Students Comments	
8. Supervisor Comments	
Student's Signature	
Supervisor's Name and Signature	

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**NAMIBIA UNIVERSITY
OF SCIENCE AND TECHNOLOGY**

-RESEARCH PROPOSAL TEMPLATE-

PROPOSED TITLE

(Type the proposed title in capital letters, the title should be in the middle of the page)

by

Name(s) & Surname

Student Number

Submitted in fulfillment of the requirements for the degree

XXXXXXXXXXXXXXXXXXXXXXX

in the

(Department of XXXXXXXXX)

at

NAMIBIA UNIVERSITY OF SCIENCE AND TECHNOLOGY

Supervisor:

(Title, initials and surname of supervisor)

Co-supervisor:

(Title, initials and surname of supervisor)

Date of submission:

dd/mm/yyyy

METADATA

TITLE: (your title)

STUDENT NAME:

SUPERVISOR:

CO-SUPERVISOR:

DEPARTMENT:

QUALIFICATION:

SPECIALISATION:

CLUSTER:

KEYWORDS:

TYPE OF RESEARCH: Applied Research

METHODOLOGY:

STATUS: Proposal

DOCUMENT DATE: (date of submission)

SPONSOR (Cluster/ Research lab):

1. Topic:
2. Problem Statement
3. Aim
4. Objectives
5. Research Questions

Supervisor Signature:



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 - b. Participants Population
 - c. Data Analysis
9. Delineations and Limitations.
 - a. Delineations
 - b. Limitations
10. Assumptions,
11. Risk/ Feasibility Analysis And Ethical Considerations
12. Work Plan
13. Chapter Outline/ Overview

14. Reference List

NUST & FCI LOGO (here)

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DECLARATION

I, ...(name (s))... , hereby declare that the work contained in the proposal for the degree.....project, entitled:

“.....(title of the project).....”,

is my own original work and that I have not previously in its entirety or in part submitted it at any university or other higher education institution for the award of a degree.

I further declare that I will fully acknowledge any sources of information I will use for the research in accordance with the Institution rules.

Signature: _____ Date: _____

SIGNATURE OF THE SUPERVISOR

I, _____, herewith declare that I accept this proposal for my supervision

Signature: _____ Date: _____

INSTRUCTIONS:

- ✓ The body of your proposal must address the sections listed below; however you are free to adapt to the type and nature of your investigation.
- ✓ Use Calibri font 12; line spacing 1.5; justify-alignment (except if intended)
- ✓ Page numbering: use “Page X of Y”, or “x/y” at bottom of page
- ✓ Avoid use of “first person singular” like “I”, “me”, etc.
- ✓ The length of the proposal write-up should be between 10 to 15 pages (up to 20 for PhD), including everything.
- ✓ Remove all instructions given in this template in your final write-up.

PROPOSED TITLE

Type the proposed title in capital letters, the title should be in the middle of the page; an unambiguous title that conveys the focus of the research

ABSTRACT

(short summary of the study in about 200 -250 words; the content should include: context, problem, methods/solutions, significance, (future..)); (This is the last section of your proposal to be written)

(Also see Appendix A for more insight)

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3 PRELIMINARY LITERATURE REVIEW AND OR THEORETICAL UNDERPINNINGS/ FRAMEWORK	2
4 IMPORTANCE / SIGNIFICANCE/BENEFITS OF THE STUDY/VALUE	2
5 RESEARCH METHODOLOGY (APPROACH TO THE SOLUTION)	2
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1 INTRODUCTION/BACKGROUND

Note that a research proposal is always future directed. It is an “architects plan” which explains what the researcher intends to do. It is generally written in the future tense. Here you give the context or background and introduction to the research.

2 PROBLEM STATEMENT

(Main statement; may be followed by related questions) (and/ or) Hypothesis (assumptions) or / and the type of envisaged solution)

Divide this part into three:

- Motivate the reason / need for the research by, for example, explaining the dilemma and problem. What will you be investigating in order to come to a conclusion about the problem that you have identified.
- Immediately after this one should state any HYPOTHESIS, IF ANY, if it refers to basic statement related to the problem; otherwise if the hypothesis is related to the type of suggested solution then it will be stated after “Research Objectives or Research Questions

3 RESEARCH OBJECTIVE(S)

(Main objective and sub-objectives as major tasks to be done which will bring you to, or help you reach a given milestone)

- Here you also have to mention possible research objectives (what do you want to achieve with this study). Usually these are 3-5 steps or milestones or sub-objectives you will follow to reach the purpose. Remember when readers get to your conclusion of your study they will go back to this part to see whether your stated purpose or objectives have been achieved.
- (alternatively) **RESEARCH QUESTIONS**: Main and sub-questions. Each (sub) Research question should be answered when at least one Research (sub) objective has been met.

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Next Chapter: Research methodology

Next Chapter: Findings/results and discussion

Next Chapter: Significance of the findings (=> contribution)

Next Chapter: Conclusions, recommendations and future works

REFERENCES / BIBLIOGRAPHY

The list of references at the end of your report contains details on all materials consulted (books, proceedings, journal articles, e-mail messages and web pages) and that have been cited in the text. Please make sure that you use the correct referencing method in the text as well as in the list of references at the end of your report.

Bibliography may be chosen if the list of sources should include materials used but could not be cited in the text (check with your supervisor)

Please use the **APA method** of referencing as explained in the CITATION guide as provided by NUST Library and from the FCI research Coordinators.

APPENDIX A: ABSTRACT CONTENT

Item in abstract	Example
Problem specification	The lack of <i>[state what is missing that causes the problem you have identified]</i> remains an open problem.
Main ways of attacking the problem(<i>show knowledge of area</i>)	The most common approaches to <i>[your problem area]</i> are based on <i>[state broad underlying assumption/theory]</i> . These approaches <i>[list main features]</i> . While such approaches have the advantage of <i>[say what they are good at]</i> , they cannot <i>[identify limitations]</i> .
Recent approaches (<i>show problem hasn't been solved recently</i>)	In recent years, <i>[alternative 'leading edge' type of approach/system/model/theory]</i> has gained increasing support, because <i>[say why it is getting popular]</i> .
Specific recent approach that you are going to develop further, and its limitations (<i>establish small research gap</i>)	<i>[Specify one form if the new approach]</i> is one form of approach/system/model/theory that <i>[say what this form is particularly good at]</i> . However, attempts at implementing/trialing the <i>[specific approach]</i> have exposed limitations. <i>[describe some different attempts and their respective limitations]</i> .
Aim of the thesis. Main features of proposed solution.	This thesis is to design/develop and test a new <i>[approach/system/model/theory]</i> , which <i>[summarise its main features]</i> . It will do this by <i>[indicate methodology/methods]</i> .
Expected benefits/contribution. How these will be evaluated.	It is expected that this approach/system/model/theory will overcome the limitations of <i>[say what you expect it will better than existing approaches]</i> . The new approach/system/model/theory will be evaluated by <i>[say how the benefits will be demonstrated]</i> .

(Source: Muyingi et al. (unpublished). "The complete research guidebook full of things your supervisor should have told you")

APPENDIX B: METHODOLOGY

Methodology can be restructured in Hierarchy:

1-The research Paradigm (philosophical view (ontology, epistemology, axiology, and rhetoric) of knowledge and its relationship with the researcher; e.g. Positivism, Interpretivism, Pragmatism

2-The Main Approach / Methodology : High-level map/ Set of Principles and Focal Approach (Descriptive; Explanatory; Exploratory, Theory testing/building/ constructive; Comparative; Longitudinal; Action Research; Participatory; Blended methodologies)

3- The Main Design or Argumentation Flow or Problem Solving Approach or Strategy (Deductive/ Inductive/ etc.)

4-The Methods (How to apply the approach: (Data type: qualitative or Quantitative or Mixed); and (Procedures)

5-Tools/Techniques/Instruments: (how to materialise the methods):

a)-The instruments/ tools to Collect Data

b)-The instruments/ tools to Analyse Data

c)-The instruments/ tools to Evaluate/ test Findings)

APPENDIX C: TECHNICAL CARE CHECKLIST FOR FINAL RESEARCH PROPOSAL

Please check that your final research proposal complies with the specific requirements set out below. Students often lose unnecessary marks because of sloppy technical care. Ask yourself the following questions ...

TECHNICAL CARE	✓ or x
1. PRELIMINARIES (Before the main body text starts on page 1)	
Is the layout of and information supplied on the title page correct?	
Have I included an Abstract (on a separate page) before the Table of Contents?	
Does the Abstract provide a <u>complete</u> summary of the whole research proposal?	
Is the Table of Contents complete and correct?	
Is the List of Figures complete and correct?	
Is the List of Tables complete and correct?	
Do the wordings of headings correspond meticulously with the wording in the Table of Contents?	
Do the headings numbers in the text correspond meticulously with the numbers in the Table of Contents?	
Do the figure / table captions listed in the List of Tables and List of Figures correspond a 100% with the captions used in the text?	
Have all tables and figures been supplied with correct captions (situated <u>above</u> the table or figure)?	
Have all tables and figures been supplied with correct and complete source references where appropriate (situated <u>below</u> the table or figure)?	
Have all tables and figures been numbered correctly? (Figures and tables are numbered independently starting from 1. Do not include section numbers in the numbering of tables / figures)	
Are the preliminary pages (i.e. Executive Summary, Table of Contents, List of Figures and List of Tables) numbered in Roman numerals: (i), (ii), (iii), (iv)?	
2. TEXT	
Are all pages in the body of the text numbered correctly with Arabic figures (1, 2, 3, etc. out of total page)?	
Have I used only one method of emphasizing (<i>italics</i> , or bold or <u>underlining</u>) throughout the report?	

Is there a line open between paragraphs?	
Are the paragraphs in my report not perhaps too long? Can't I split long paragraphs to improve readability?	
Have I made any "sweeping" or unsubstantiated statements, such as "there is no literature available on this topic" or "this research will contribute to the body of knowledge"?	
Have I used an impersonal, objective writing style (One symptom of a subjective writing style is that the researcher refers to him-/herself directly. References to "I", "we" and even "the researcher" should be avoided.)	
3. REFERENCING TECHNIQUES	
Are all sources (books, journals, e-mail messages, web pages) that have been cited in the text, also been included in the List of References at the end of the report and vice versa?	
Is the list of references strictly in the required order of the referencing style you are using ?	
Have I used the correct revised (abridged) Harvard referencing method in the text, as well as in the list of references?	
Have I correctly indicated the sources of all tables / figures taken or adapted from the literature? (Such references are provided <u>below</u> the table or figure)	
4. APPENDICES	
Do all the appendices have appropriate descriptive titles?	
Are all the appendices numbered up to the last page?	
Have all the appendices been tagged / "flagged" for easy cross-referencing?	
Are all the appendices included in the Table of Contents with their correct page numbers?	
5. GENERAL TECHNICAL CARE	
<i>Have I checked my proposal for spelling and grammatical errors?</i>	
<i>Have I used capital letters and abbreviations correctly?</i>	
Have I consistently rounded off all numeric values to three (3) decimals?	
Has text of the whole document been "justified" (i.e. aligned evenly along both the left and right margins to form a square box)?	
Are all headings numbered correctly?	
Headings in capital letters (first order headings) should not be underlined.	
There should be no full stops at the end of headings.	
Have I used an A4 paper setting?	

Have all pages printed correctly?	
Have I consistently used one and a half (1½) line spacing in the body of the text?	
Is the font size of the body text correct (12 pt)?	
Have I consistently used the same font type (i.e. Ariel or Times New Roman) and font size (12 pt) for the body text?	
Is the left margin set to at least 2 cm?	